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CMA REVIEW

PART 1: FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS



by

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and

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CMA REVIEW

PART 1: FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS



Part 1 – Financial Planning, Performance, and Analytics

A. External Financial Reporting Decisions (15% - Levels A, B, and C)

1. *Financial statements*

- a. Balance sheet (1.2)
- b. Income statement (1.3)
- c. Statement of changes in equity (1.4)
- d. Statement of cash flows (1.5)
- e. Integrated reporting (SU 6)

2. *Recognition, measurement, valuation, and disclosure*

- a. Asset valuation (SUs 2-3)
- b. Valuation of liabilities (SU 4)
- c. Equity transactions (1.4)
- d. Revenue recognition (5.1-5.2)
- e. Income measurement (1.3, 5.3)
- f. Major differences between U.S. GAAP and IFRS (SUs 1-5, Appendix B)

STUDY UNIT ONE

EXTERNAL FINANCIAL STATEMENTS

1.1	Concepts of Financial Accounting	1
1.2	Statement of Financial Position (Balance Sheet)	5
1.3	Income Statement and Statement of Comprehensive Income	9
1.4	Statement of Changes in Equity and Equity Transactions	15
1.5	Statement of Cash Flows	22

This study unit is the **first of six** on **external financial reporting decisions**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The six study units are

Study Unit 1: External Financial Statements

Study Unit 2: Measurement, Valuation, and Disclosure: Assets -- Short-Term Items

Study Unit 3: Measurement, Valuation, and Disclosure: Assets -- Long-Term Items

Study Unit 4: Measurement, Valuation, and Disclosure: Liabilities

Study Unit 5: Revenue and Impairment Recognition

Study Unit 6: Integrated Reporting

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

1.1 CONCEPTS OF FINANCIAL ACCOUNTING

1. The Objective of General-Purpose Financial Reporting

- a. The objective of general-purpose financial reporting is to report financial information that is **useful in making decisions** about providing resources to the reporting entity.
- b. The information reported relates to the entity's economic resources and claims to them (financial position) and to changes in those resources and claims.
 - 1) Information about economic resources and claims helps to evaluate liquidity, solvency, financing needs, and the probability of obtaining financing.
- c. Users need to differentiate between changes in economic resources and claims arising from (1) the entity's performance (income statement) and (2) other events and transactions, such as issuing debt and equity (balance sheet).
 - 1) Information about financial performance is useful for
 - a) Understanding the return on economic resources, its variability, and its components;
 - b) Evaluating management; and
 - c) Predicting future returns.

- d. For general-purpose financial statements to be useful to external parties, they **must be prepared in conformity with accounting principles that are generally accepted in the United States (GAAP)**.



SUCCESS TIP

The CMA exam also tests some knowledge of International Financial Reporting Standards (IFRS). When international standards diverge significantly from U.S. GAAP, the differences are highlighted in the outline. A list of additional IFRS differences can be found in Appendix B. If an exam question does not distinguish between GAAP and IFRS, use GAAP.

- e. Financial accounting differs from management accounting.
- 1) Management accounting assists management decision making, planning, and control. Management accounting information is therefore primarily directed to specific internal users, and it ordinarily need not follow GAAP.

2. Users of Financial Statements

- a. Users may directly or indirectly have an economic interest in a specific business. Users with direct interests usually invest in or manage the business, and users with indirect interests advise, influence, or represent users with direct interests.
 - 1) Users with direct interests include
 - a) Investors or potential investors
 - b) Suppliers and creditors
 - c) Employees
 - d) Management
 - 2) Users having indirect interests include
 - a) Financial advisors and analysts
 - b) Stock markets or exchanges
 - c) Regulatory authorities
- b. **External users** use financial statements to determine whether doing business with the firm will be beneficial.
 - 1) Investors need information to decide whether to increase, decrease, or obtain an investment in a firm.
 - 2) Creditors need information to determine whether to extend credit and under what terms.
 - 3) Financial advisors and analysts need financial statements to help investors evaluate particular investments.
 - 4) Stock exchanges need financial statements to evaluate whether to accept a firm's stock for listing or whether to suspend the stock's trading.
 - 5) Regulatory agencies may need financial statements to evaluate the firm's conformity with regulations and to determine price levels in regulated industries.
- c. **Internal users** use financial statements to make decisions affecting the operations of the business. These users include management, employees, and the board of directors.
 - 1) Management needs financial statements to assess financial strengths and deficiencies, to evaluate performance results and past decisions, and to plan for future financial goals and steps toward accomplishing them.
 - 2) Employees want financial information to negotiate wages and fringe benefits based on the increased productivity and value they provide to a profitable firm.

3. Features of Financial Statements



SUCCESS TIP

Note disclosures are no longer tested on the CMA exam. However, we continue to provide limited coverage about note disclosures in the outline because they are an integral part of financial statements.

- a. Financial statements are the primary means of communicating financial information to external parties.
 - 1) Additional information is provided by financial statement notes, supplementary information (such as management's discussion and analysis), and other disclosures. Information typically disclosed in notes is essential to understanding the financial statements.
 - a) The notes are considered part of the basic financial statements. They amplify or explain information recognized in the statements and are an integral part of statements prepared in accordance with GAAP.
 - i) The first footnote accompanying any set of complete financial statements is generally one describing significant accounting policies, such as the use of estimates and assumptions, and policies relating to, among others, revenue recognition and allocation of asset (tangible and intangible) costs to current and future periods.
 - ii) Financial statement notes should not be used to correct improper presentation.
- b. A full set of financial statements includes the following statements:
 - 1) Statement of financial position (also called a balance sheet)
 - 2) Income statement
 - 3) Statement of comprehensive income
 - 4) Statement of changes in equity
 - 5) Statement of cash flows
- c. To be useful, information presented in the financial statements must be relevant and faithfully represented.
 - 1) To enhance the usefulness, the information should be comparable with similar information for
 - a) Other entities and
 - b) The same entity for another period or date.
 - 2) Thus, **comparability** allows users to understand similarities and differences.
- d. Financial statements are prepared under the **going-concern assumption**, which means that the entity is assumed to continue operating indefinitely.
 - 1) As a result, liquidation values are not important. It is assumed that the entity is not going to be liquidated in the near future.

4. Financial Statement Relationships

- a. Financial statements complement each other. They describe different aspects of the same transactions, and more than one statement is necessary to provide information for a specific economic decision.
- b. The components (elements) of one statement relate to those of other statements. Among the relationships are those listed below.
 - 1) Net income or loss from the statement of income is reported and accumulated in the retained earnings account, a component of the equity section of the statement of financial position.
 - 2) The components of cash and equivalents from the statement of financial position are reconciled with the corresponding items in the statement of cash flows.
 - 3) Items of equity from the statement of financial position are reconciled with the beginning balances on the statement of changes in equity.
 - 4) Ending inventories are reported in current assets on the statement of financial position and are reflected in the calculation of cost of goods sold on the statement of income.
 - 5) Amortization and depreciation reported in the statement of income also are reflected in asset and liability balances in the statement of financial position.

5. Accrual Basis of Accounting

- a. Financial statements are prepared under the accrual basis of accounting. Accrual accounting records the financial effects of transactions and other events and circumstances when they occur rather than when their associated cash is paid or received.
 - 1) Revenues are recognized in the period in which they were earned even if the cash will be received in a future period.
 - 2) Expenses are recognized in the period in which they were incurred even if the cash will be paid in a future period.



**AUTHOR'S
NOTE**

Under the **cash basis**, revenues are recognized when cash is received and expenses are recognized when cash is paid. Under GAAP, financial statements cannot be prepared under the cash basis of accounting.

1.2 STATEMENT OF FINANCIAL POSITION (BALANCE SHEET)

1. Overview

- a. The statement of financial position, also called the balance sheet, reports the amounts of assets (items of value), liabilities (debt), and equity (net worth) and their relationships at a moment in time, such as at the end of the fiscal year.
 - 1) It helps users assess liquidity, financial flexibility, the efficiency with which assets are used, capital structure, and risk.
- b. The **basic accounting equation** presents a perfect balance between the entity's resources and its capital structure.
 - 1) The entity's resources consist of the assets the entity deploys in its attempts to earn a return.
 - 2) The capital structure consists of the amounts contributed by creditors (liabilities) and investors (equity).

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

- c. The equation is based on the **proprietary theory**. According to this theory, equity in an enterprise is what remains after the economic obligations of the enterprise are deducted from its economic resources.

2. Elements of the Balance Sheet

- a. **Assets** are resources controlled by the entity as a result of past events. They represent probable future economic benefits to the entity.
 - 1) Examples include inventory; accounts receivable; investments; and property, plant, and equipment.
- b. **Liabilities** are present obligations of the entity arising from past events. Their settlement is expected to result in an outflow of economic benefits from the entity.
 - 1) Examples include loans payable, bonds issued by the entity, and accounts payable.
- c. **Equity** is the residual interest in the assets of the entity after subtracting all its liabilities.
 - 1) Examples include a company's common stock, preferred stock, and retained earnings.
 - 2) Equity is affected not only by operations but also by transactions with owners, such as dividends and contributions.
 - a) **Investments by owners** are increases in equity of a business entity. They result from transfers of something of value to increase ownership interests. Assets are the most commonly transferred item, but services also can be exchanged for equity interests.
 - b) **Distributions to owners** are decreases in equity. They result from transferring assets to owners. A distribution to owners decreases the ownership interest in the company.
- d. Assets and liabilities are separated in the statement of financial position into **current** and **noncurrent** categories.
 - 1) Assets are generally reported in order of liquidity.

- e. Some variation of the following classifications is used by most entities:

Assets

Current assets:

Cash
 Certain investments
 Accounts and notes receivable
 Inventories
 Prepaid expenses

Noncurrent assets:

Certain investments and funds
 Property, plant, and equipment (PPE)
 Intangible assets
 Other noncurrent assets

Liabilities

Current liabilities:

Accounts payable
 Current notes payable
 Current maturities of noncurrent liabilities

Noncurrent liabilities:

Noncurrent notes payable
 Bonds payable

Equity

Investments by owners
 Retained earnings
 Accumulated other comprehensive income
 Noncontrolling interest in a consolidated entity

3. Current and Noncurrent Assets

- a. An asset is classified as **current** on the statement of financial position if it is expected to be realized in cash or sold or consumed within the entity's operating cycle or 1 year, **whichever is longer**.
- b. The following are the major categories of current assets: (1) cash and cash equivalents; (2) certain individual trading, available-for-sale, and held-to-maturity debt securities; (3) receivables; (4) inventories; and (5) prepaid expenses, etc.
 - 1) Prepaid expenses are valued on the balance sheet at the cost less the expired or used portion.
- c. **Noncurrent assets** are those not qualifying as current.
- d. The following are the major categories of noncurrent assets:
 - 1) **Investments and funds** include nonoperating items intended to be held beyond the longer of 1 year or the operating cycle. The following assets are typically included:
 - a) Investments in securities made to control or influence another entity and other noncurrent securities.
 - b) Certain available-for-sale and held-to-maturity debt securities may be noncurrent.
 - c) Funds restricted as to withdrawal or use for other than current operations, for example, to (1) retire long-term debt, (2) satisfy pension obligations, or (3) pay for the acquisition or construction of noncurrent assets.
 - d) A lessor's net investment in a sales-type lease.
 - 2) **Property, plant, and equipment (PPE)** are tangible operating items recorded at cost and reported net of any accumulated depreciation. They include
 - a) Land and natural resources subject to depletion, e.g., oil and gas
 - b) Buildings, equipment, furniture, fixtures, leasehold improvements, land improvements, a lessee's right-of-use assets held under finance and operating leases, noncurrent assets under construction, and other depreciable assets
 - 3) **Intangible assets** are nonfinancial assets without physical substance. Examples are patents and goodwill.

4. Current and Noncurrent Liabilities

- a. **Current liabilities** are expected to be settled or liquidated in the ordinary course of business during the longer of the next year or the operating cycle.
 - 1) Generally speaking, current liabilities are expected to be settled or liquidated within 1 year from the balance sheet date.
- b. The following are the major categories of current liabilities:
 - 1) **Trade payables** for items entering into the operating cycle, e.g., for materials and supplies used in producing goods or services for sale.
 - 2) **Other payables** arising from operations, such as accrued wages, salaries, rentals, royalties, and taxes.
 - 3) **Unearned revenues** arising from collections in advance of delivering goods or performing services, e.g., ticket sales revenue.
 - 4) **Other obligations** expected to be liquidated in the ordinary course of business during the longer of the next year or the operating cycle. These include
 - a) Short-term notes given to acquire capital assets
 - b) Payments required under sinking-fund provisions
 - c) Payments on the current portion of serial bonds or other noncurrent debt
 - d) Long-term obligations that are or will become callable by the creditor because of the debtor's violation of a provision of the debt agreement at the balance sheet date
 - e) Possible obligations for warranties (guarantees) and estimated returns
 - f) Short-term leases if the lessee elects to recognize a right-of-use asset and a lease liability
- c. Current liabilities **do not include** short-term debt if an entity
 - 1) Intends to refinance them on a noncurrent basis and
 - 2) Demonstrates an ability to do so.
 - a) The ability to refinance may be demonstrated by entering into a **refinancing agreement** before the balance sheet is issued.
- d. **Noncurrent liabilities** are those not qualifying as current. The noncurrent portions of the following items are reported in this section of the balance sheet:
 - 1) Noncurrent notes and bonds
 - 2) A lessee's liabilities under finance and operating leases
 - 3) Most postretirement benefit obligations
 - 4) Deferred tax liabilities arising from interperiod tax allocation
 - 5) Obligations under product or service warranty agreements
 - 6) Deferred revenue

5. Equity

- a. Any recognized transaction that does not have equal and offsetting effects on total assets and total liabilities changes equity. The following are the major items of equity:
 - 1) **Capital contributions by owners** (par value of common and preferred stock issued and additional paid-in capital).
 - a) Additional paid-in (contributed) capital is the amount received in excess of par value at the time stock was sold.
 - 2) **Retained earnings** are the accumulated net income not yet distributed to owners. Dividends can be paid when retained earnings has a credit balance. A payment in excess of this balance is a return of capital, not a dividend.
 - a) Occasionally, the board of directors restricts retained earnings to prevent payment of dividends. The usual reason is that the board plans to reinvest the earnings in the business.
 - 3) **Treasury stock** is the firm's own stock that has been repurchased.
 - a) It is reflected in shareholders' equity as a contra account (which reduces the balance of a related account). Thus, it is not an asset.
 - 4) **Accumulated other comprehensive income** items not included in net income.

6. Balance Sheet Elements Are Permanent Accounts

- a. Assets, liabilities, and equity are recorded in **permanent (real) accounts**. Their balances at the end of one accounting period (the balance sheet date) are carried forward as the beginning balances of the next accounting period.

7. Major Balance Sheet Note Disclosures

- a. Note disclosures and schedules specifically related to the balance sheet include
 - 1) Investment securities
 - 2) Maturity patterns of bond issues
 - 3) Significant uncertainties, such as pending litigation
 - 4) Details of capital stock issues

8. Limitations of the Balance Sheet

- a. The balance sheet shows a company's financial position at a single point in time; accounts may vary significantly a few days before or after the publication of the balance sheet.
- b. Many balance sheet items, such as fixed assets, are recorded at historical costs, which may not equal their fair value.
- c. The preparation of the balance sheet requires estimates and management judgment.
- d. The balance sheet omits many items that cannot be recorded objectively but have financial value to the company.

1.3 INCOME STATEMENT AND STATEMENT OF COMPREHENSIVE INCOME

1. Income Statement Elements

- a. The income statement reports the results of an entity's operations over a period of time, such as a year.

The Income Equation

$$\text{Income (Loss)} = \text{Revenues} + \text{Gains} - \text{Expenses} - \text{Losses}$$

- b. The following are the elements of an income statement:
 - 1) **Revenues** are inflows or other enhancements of assets or settlements of liabilities (or both) from delivering or producing goods, providing services, or other activities that qualify as ongoing major or central operations.
 - 2) **Gains** are increases in equity (or net assets) other than from revenues or investments by owners.
 - 3) **Expenses** are outflows or other usage of assets or incurrences of liabilities (or both) from delivering or producing goods, providing services, or other activities that qualify as ongoing major or central operations.
 - 4) **Losses** are decreases (which may or may not occur) in equity (or net assets) other than from expenses or distributions to owners.
 - c. All transactions affecting the net change in equity during the period are included in income except
 - 1) Transactions with owners
 - 2) Prior-period adjustments (such as error correction or a change in accounting principle)
 - 3) Items reported initially in other comprehensive income
 - 4) Transfers to and from appropriated retained earnings
 - d. Revenues, expenses, gains, and losses are recorded in **temporary (nominal) accounts** because they record the transactions, events, and other circumstances during a period of time. These accounts are closed (reduced to zero) at the end of each accounting period, and their balances are transferred to real accounts.
 - 1) For example, income or loss for the period (a nominal account) is **closed to retained earnings (a real account)** at the end of the reporting period.
2. **Gross profit** is the net difference between sales revenue and cost of goods sold.
 - a. Gross profit margin as a percentage of sales is calculated as follows:

$$\text{Gross profit margin (\%)} = \text{Gross profit} \div \text{Sales}$$

3. Typical Items of Cost and Expense

a. The **expense recognition principles** are associating cause and effect, systematic and rational allocation, and immediate recognition.

1) **Matching** is essentially synonymous with **associating cause and effect**. Under the **matching principle**, the recognition of expense and related revenue occurs in the same accounting period.

a) Such a direct relationship is found when the cost of goods sold is recognized in the same period as the revenue from the sale of the goods.

b. Cost of Goods Sold

1) For a **retailer**, cost of goods sold is calculated based on changes in inventory:

Beginning inventory	\$10,000
Plus: Net purchases	14,000
Plus: Freight-in	1,000
Goods available for sale	<u>\$25,000</u>
Minus: Ending inventory	(5,000)
Cost of goods sold	<u><u>\$20,000</u></u>

2) For a **manufacturer**, cost of goods sold is calculated as follows:

Beginning direct materials inventory	\$3,000	
Purchases during the period	3,000	
Ending direct materials inventory	<u>(1,000)</u>	
Direct materials used in production		\$5,000
Direct labor costs		5,000
Manufacturing overhead costs (Fixed + Variable)		<u>4,000</u>
Total manufacturing costs		\$14,000
Beginning work-in-process inventory		5,000
Ending work-in-process inventory		<u>(4,000)</u>
Cost of goods manufactured		\$15,000
Beginning finished goods inventory		6,000
Ending finished goods inventory		<u>(11,000)</u>
Cost of goods sold		<u><u>\$10,000</u></u>

c. Share-Based Payments and Employee Benefits

1) Common share-based payment arrangements between employers and employees include

- Call options that give employees the right to purchase an entity's shares in exchange for their services,
- Share appreciation rights that entitle employees to cash payments calculated by reference to increases in the market price of an entity's shares, and
- Share ownership plans that distribute shares to employees in exchange for their services.

2) ASC 718, *Compensation—Stock Compensation*, requires all entities to recognize compensation expense in an amount equal to the fair value of share-based payments (e.g., stock options and restricted stock) granted to employees.



IFRS 2, *Share-based Payment*, contains more stringent criteria for determining whether an employee share purchase plan is compensatory. As a result, some employee share purchase plans for which IFRS 2 requires recognition of compensation cost will not be considered to give rise to compensation cost under U.S. GAAP.

IFRS 2 applies the same measurement requirements to employee share options regardless of whether the issuer is a public or a nonpublic entity. U.S. GAAP require that a nonpublic entity account for its options and similar equity instruments based on their fair value unless it is not practicable to estimate the expected volatility of the entity's share price. In that situation, the entity is required to measure its equity share options and similar instruments at a value using the historical volatility of an appropriate industry sector index.

In tax jurisdictions such as the United States, where the time value of share options generally is not deductible for tax purposes, IFRS 2 requires that no deferred tax asset be recognized for the compensation cost related to the time value component of the fair value of an award.

U.S. GAAP require a portfolio approach in determining excess tax benefits of equity awards in paid-in capital available to offset write-offs of deferred tax assets, whereas IFRS 2 requires an individual instrument approach. Thus, some write-offs of deferred tax assets that will be recognized in paid-in capital under U.S. GAAP will be recognized in determining net income under IFRS 2.

d. Other Expenses

- 1) General and administrative expenses are incurred for the benefit of the enterprise as a whole and are not related wholly to a specific function, e.g., selling or manufacturing.
 - a) They include accounting, legal, and other fees for professional services; officers' salaries; insurance; wages of office staff; miscellaneous supplies; and office occupancy costs.
- 2) Selling expenses are those incurred in selling or marketing.
 - a) Examples include sales representatives' salaries, commissions, and traveling expenses; advertising; sales department salaries and expenses, including rent; and credit and collection costs.
 - b) Shipping costs are also often classified as selling costs.

- e. Interest expense is recognized based on the passage of time. In the case of bonds, notes, and finance leases, the effective interest method is used.

4. Income Statement Formats

- a. The **single-step** income statement provides one grouping for revenue items and one for expense items. The single step is the one subtraction necessary to arrive at net income.
- b. The **multiple-step** income statement presents operating revenues and expenses in a section separate from nonoperating items.
 - 1) The most common way to present the income statement is the condensed format of the multiple-step income statement, which includes only the section totals.

EXAMPLE 1-1 Multiple-Step Income Statement	
Net sales	\$ 200,000
Cost of goods sold	<u>(150,000)</u>
Gross profit	\$ 50,000
Selling expenses	(6,000)
Administrative expenses	<u>(5,000)</u>
Income from operations	\$ 39,000
Other revenues and gains	3,500
Other expenses and losses	<u>(2,500)</u>
Income before taxes	\$ 40,000
Income taxes	<u>(16,000)</u>
Net income	<u>\$ 24,000</u>

5. Reporting Irregular Items

a. Discontinued Operation

- 1) When an entity reports a **discontinued operation**, it must be presented in a separate section between **income from continuing operations and net income**.
 - a) Because these items are reported after the presentation of income taxes, they must be shown net of tax.
 - b) The term “continuing operations” is used only when a discontinued operation is reported.
- 2) **Discontinued operations**, if reported, may have two components:
 - a) Gain or loss from operations of the component that has been disposed of or is classified as held for sale from the first day of the reporting period until the date of disposal (or the end of the reporting period if it is classified as held for sale)
 - b) Gain or loss on the disposal of this component

EXAMPLE 1-2		Discontinued Operations
	Net sales	\$ 200,000
	Cost of goods sold	(150,000)
	Gross profit	\$ 50,000
	Selling expenses	(6,000)
	Administrative expenses	(5,000)
	Income from operations	\$ 39,000
	Other revenues and gains	3,500
	Other expenses and losses	(2,500)
	Income before taxes	\$ 40,000
	Income taxes	(16,000)
	Income from continuing operations	\$ 24,000
	Discontinued operations	
	Gain from operations of component (including loss on disposal of \$5,000)	\$10,000
	Income taxes	(4,000)
	Gain from discontinued operations	\$ 6,000
	Net income	\$ 30,000

b. Extraordinary Items

- 1) FASB Accounting Standards Update (ASU) 2015-01, *Income Statement—Extraordinary and Unusual Items*, removed the concept of extraordinary items from U.S. GAAP for both public and private companies.
 - a) The separate, net-of-tax presentation is not allowed. An entity is required to separately present items that are of an unusual nature and/or occur infrequently on a pre-tax basis within income from continuing operations.
 - b) The standard took effect for fiscal years beginning after December 15, 2015.

6. Major Income Statement Note Disclosures

- a. Note disclosures and schedules specifically related to the income statement include the following:
 - 1) Earnings per share
 - 2) Depreciation schedules
 - 3) Components of income tax expense
 - 4) Components of pension expense

7. Limitations of the Income Statement

- a. The income statement does not always show all items of income and expense. Some of the items are reported on a statement of other comprehensive income and not included in the calculation of net income.
- b. The financial statements report accrual-basis results for the period. The company may recognize revenue and report net income before any cash was actually received.
 - 1) For example, the data from the income statement itself is not sufficient enough for assessing liquidity. This statement must be viewed in conjunction with other financial statements, such as the balance sheet and statement of cash flows.
- c. The preparation of the income statement requires estimates and management judgment.

8. Statement of Comprehensive Income

- a. **Comprehensive income** includes all changes in equity (net assets) of a business during a period except those from investments by and distributions to owners.
 - 1) It consists of
 - a) **Net income or loss** (the bottom line of the income statement) and
 - b) **Other comprehensive income (OCI).**
 - 2) Certain income items are excluded from the calculation of net income and instead are included in comprehensive income. The following are the major items included in other comprehensive income:
 - a) The effective portion of a gain or loss on a hedging instrument in a cash flow hedge
 - b) Unrealized holding gains and losses due to changes in the fair value of available-for-sale debt securities
 - c) Translation gains and losses for financial statements of foreign operations
 - d) Certain amounts associated with accounting for defined benefit postretirement plans
- b. All items of comprehensive income are recognized for the period in either
 - 1) **One continuous financial statement** that has two sections, net income and OCI, or
 - 2) **Two separate but consecutive statements.**
 - a) The first statement (the income statement) presents the components of net income and total net income.
 - b) The second statement (the statement of OCI) is presented immediately after the first. It presents a total of OCI with its components and a total of comprehensive income.

EXAMPLE 1-3		Separate Statement of Comprehensive Income	
Net income			\$70,000
Other comprehensive income (net of tax):			
Loss on defined benefit postretirement plans	\$(15,000)		
Gains on foreign currency translation	6,000		
Gains on remeasuring available-for-sale securities	4,000		
Effective portion of losses on cash flow hedges	<u>(3,000)</u>		
Other comprehensive income (loss)			<u>(8,000)</u>
Total comprehensive income			<u>\$62,000</u>

1.4 STATEMENT OF CHANGES IN EQUITY AND EQUITY TRANSACTIONS

1. Statement of Changes in Equity

- a. A statement of changes in equity presents a reconciliation for the accounting period of the beginning balance for each component of equity to the ending balance.
- b. Each change is disclosed separately in the statement. The following are the common changes in the equity component balances during the accounting period:
 - 1) Net income (loss) for the period, which increases (decreases) the retained earnings balance.
 - 2) Distributions to owners (dividends paid), which decreases the retained earnings balance.
 - 3) Issuance of common stock, which increases the common stock balance. If the amount paid for the stock is above the par value of stock, the balance of additional paid-in capital is increased for the difference.
 - 4) Total change in other comprehensive income during the period.
 - 5) Repurchase of common stock (treasury stock) decreases the total shareholders' equity.

EXAMPLE 1-4 Statement of Changes in Equity						
	Year Ended December 31, Year 1 (in millions)					
	Total Equity	Retained Earnings	Accumulated Other Comprehensive Income	Common Stock	Additional Paid-in Capital	Treasury Stock
Beginning balance	\$500	\$350	\$100	\$40	\$ 30	\$(20)
Net income for the period	110	110				
OCI for the period	25		25			
Common stock issued	90			10	80	
Dividends declared	(60)	(60)				
Repurchase of common stock	(15)					(15)
Ending balance	<u>\$650</u>	<u>\$400</u>	<u>\$125</u>	<u>\$50</u>	<u>\$ 110</u>	<u>\$(35)</u>

2. Statement of Retained Earnings

a. A statement of retained earnings reconciles the beginning and ending balances of the account. This statement is reported as part of the statement of changes in equity in a separate column.

1) The following is a common example of retained earnings reconciliation:

Retained earnings beginning balance
+ Net income (loss) for the period
– Dividends distributed during the period
+ Positive (negative) prior-period adjustments
<u>Retained earnings ending balance</u>

2) Prior-period adjustments include the cumulative effect on the income statement of **changes in accounting principle** (e.g., change in inventory valuation method) and **corrections of prior-period financial statement errors**.

a) These items require **retrospective application** (i.e., adjustment of the carrying amounts of assets, liabilities, and retained earnings at the beginning of the first period reported for the cumulative effect of the new principle or the error on the prior periods).

b) Thus, correction of prior-period errors and the cumulative effect of changes in accounting principle **must not** be included in the calculation of current-period net income.

3) However, **changes in accounting estimate** (e.g., a change in the percentage of credit sales for estimation of bad debt expense) require **prospective application** (i.e., the effect of a change in accounting estimate is accounted for only in the period of change and any future periods affected).

3. Common and Preferred Stock

a. The most widely used classes of stock are common and preferred. The following basic terminology is related to stock:

1) Stock authorized is the maximum amount of stock that a corporation is legally allowed to issue.

2) Stock issued is the amount of stock authorized that has actually been issued by the corporation.

3) Stock outstanding is the amount of stock issued that has been purchased and is held by shareholders.

b. The **common shareholders** are the owners of the firm. They have voting rights, and they select the firm's board of directors and vote on resolutions. Common shareholders are not entitled to dividends unless so declared by the board of directors. A firm may choose not to declare any.

1) Common shareholders are entitled to receive **liquidating distributions** only after all other claims have been satisfied, including those of preferred shareholders.

2) Common shareholders ordinarily have **preemptive rights**.

a) Preemptive rights give current common shareholders the right to purchase any additional stock issuances in proportion to their ownership percentages. This way the preemptive rights safeguard a common shareholder's proportionate interest in the firm.

- c. **Preferred stock** has features of debt and equity. It is classified as an equity instrument and presented in the equity section of the firm's balance sheet. Preferred stock has a fixed charge, but payment of dividends is not an obligation. The payment of dividends is at the firm's discretion. Preferred shareholders tend not to have voting rights.
- 1) Preferred shareholders have the right to receive
 - a) Dividends at a specified fixed rate before common shareholders may receive any and
 - b) Distributions before common shareholders, but after creditors, in the event of firm bankruptcy (liquidation).
 - 2) The following are common features of preferred stock:
 - a) **Cumulative preferred stock** accumulates unpaid dividends (called dividends in arrears). Dividends in arrears must be paid before any common dividends can be paid.
 - b) Holders of **convertible preferred stock** have the right to convert the stock into shares of another class (usually common stock) at a predetermined ratio.

4. Equity Transactions – Issuance of Stock

- a. Cash is increased (debited), the appropriate stock account is increased (credited) for the total par value of stock issued, and additional paid-in capital (paid-in capital in excess of par) is increased (credited) for the difference.

EXAMPLE 1-5 Issuance of Stock

A company issued 50,000 shares of its \$1 par-value common stock. The market price of the stock was \$17 per share on the day of issue.

Cash (50,000 shares × \$17 market price)	\$850,000	
Common stock (50,000 shares × \$1 par value)		\$ 50,000
Additional paid-in capital -- common (difference)		800,000

The balances of cash, common stock, and additional paid-in capital were increased by \$850,000, \$50,000, and \$800,000, respectively.

- b. Direct costs of issuing stock (underwriting, legal, accounting, tax, registration, etc.) must not be recognized as an expense. Instead, they reduce the net proceeds received and additional paid-in capital.

5. Equity Transactions – Treasury Stock

- a. Treasury stock is reported either at cost (as an unallocated reduction of total equity) or at par (as a direct reduction of the relevant contributed capital account).
 - 1) When treasury stock is recorded at cost, treasury stock is debited for the full cost of the purchase.
 - 2) Under the par value method, treasury stock is debited for the par value of the stock.
 - 3) Debits also will be made to additional paid-in capital or retained earnings.
- b. **Acquisition Price > Original Issue Price**
 - 1) Under the par-value method, if the acquisition price is greater than the original issue price, the difference is a reduction of retained earnings or a credit balance in paid-in capital (PiC) from treasury stock transactions.

EXAMPLE 1-6 Treasury Stock -- Cost Method vs. Par-Value Method		
Parvenu reacquired 5,000 shares of its \$1 par-value common stock for \$20 per share. This stock had originally been issued at \$17 per share. Parvenu had no prior treasury stock transactions.		
<u>Cost Method:</u>		
Treasury stock (5,000 shares × \$20 market price)	\$100,000	
Cash (payment)		\$100,000
<u>Par-Value Method:</u>		
Treasury stock (5,000 shares × \$1 par value)	\$ 5,000	
Additional paid-in capital -- common (5,000 shares × \$16 original issue excess)	80,000	
Retained earnings (difference)	15,000	
Cash (payment)		\$100,000

- c. **Acquisition Price < Original Issue Price**
 - 1) Under the par-value method, if the acquisition price is less than the original issue price, additional paid-in capital is decreased (debited) for the difference between the acquisition price and the par value of the stock reacquired.

EXAMPLE 1-7 Treasury Stock -- Acquisition Price < Issue Price		
Parvenu reacquired 5,000 shares of its \$1 par-value common stock for \$10 per share.		
<u>Cost Method:</u>		
Treasury stock (5,000 shares × \$10 market price)	\$ 50,000	
Cash (payment)		\$50,000
<u>Par-Value Method:</u>		
Treasury stock (5,000 shares × \$1 par value)	\$ 5,000	
Additional paid-in capital -- common (difference)	45,000	
Cash (payment)		\$50,000

- 2) However, if the acquisition price is less than the par value of stock reacquired, the difference is an increase in PiC from treasury stock transactions. (Gains and losses are not recognized on transactions in an entity's own stock.)
- d. Treasury stock has no voting rights and receives no cash dividends or distributions in liquidation.

6. Equity Transactions – Cash Dividend

- a. On the **date of declaration**, the board of directors formally approves a dividend. A declaration of a dividend **decreases** (debits) the **retained earnings** account.
- b. All holders of the stock on the **date of record** are legally entitled to receive the dividend.
- c. The date of payment is the date on which the dividend is paid.

EXAMPLE 1-8		Cash Dividend	
On September 12, a company's board of directors declared a \$3 per-share dividend to be paid on October 15 to all holders of common stock. On the date of declaration, 40,000 shares of common stock were outstanding.			
<u>September 12 -- Declaration date</u>		<u>October 15 -- Payment date</u>	
Retained earnings (40,000 × \$3)	\$120,000	Dividends payable	\$120,000
Dividends payable	\$120,000	Cash	\$120,000

7. Equity Transactions – Property Dividend

- a. When an entity declares a dividend consisting of tangible property,
 - 1) First, the property is **remeasured to fair value** as of the date of declaration, and any gain or loss on the remeasurement is recognized in the statement of income.
 - 2) Second, the carrying amount of retained earnings is decreased for the fair value of the property to be distributed.
 - 3) Third, the property is distributed as a dividend.

EXAMPLE 1-9

Property Dividend

On August 1, a company's board of directors declared a property dividend (land) to be distributed on December 1 to holders of common stock. On August 1, the carrying amount of the land to be distributed is \$50,000 and its fair value is \$80,000. The journal entries to record the declaration and distribution of the property dividend are as follows:

August 1 -- Declaration date

Land (\$80,000 – \$50,000)	\$30,000	Retained earnings	\$80,000
Gain on land remeasurement	\$30,000	Property dividend payable	\$80,000

December 1 -- Payment date

Property dividend payable	\$80,000
Land	\$80,000

8. Equity Transactions – Stock Dividend and Stock Split

- a. A stock dividend involves no distribution of cash or other property. Stock dividends are accounted for as a reclassification of different equity accounts, not as liabilities.
 - 1) The recipient does not recognize income. It has the same proportionate interest in the entity and the same total carrying amount as before the stock dividend.
- b. The accounting for stock dividends depends on the percentage of new shares to be issued.
 - 1) An issuance of shares **less than 20% to 25%** of the previously outstanding common shares should be recognized as a **stock dividend**.
 - 2) An issuance of **more than 20% to 25%** of the previously outstanding common shares should be recognized as a **stock split in the form of a dividend**.
- c. In accounting for a **stock dividend**, the **fair value** of the additional shares issued is reclassified from retained earnings to common stock (at par value) and the difference to additional paid-in capital.

EXAMPLE 1-10 Stock Dividend

On May 1, a company's board of directors declared and paid a 10% stock dividend on the 45,000 shares of common stock outstanding (\$1 par value). The stock was trading for \$15 per share at the declaration date.

May 1 -- Declaration and payment date

Retained earnings [(45,000 shares × 10%) × \$15 market price]	\$67,500	
Common stock [(45,000 shares × 10%) × \$1 par value]		\$ 4,500
Additional paid-in capital (difference)		63,000

- d. For a stock dividend that is accounted for as a **stock split in the form of a dividend**, the **par value** of the additional shares issued is reclassified from retained earnings to common stock.

EXAMPLE 1-11 Stock Split

Continuing from Example 1-10, assume that a 40% stock split in the form of a dividend was declared.

May 1 -- Declaration and payment date

Retained earnings [(45,000 shares × 40%) × \$1 par value]	\$18,000	
Common stock		\$18,000

- e. **Stock splits** are issuances of shares that do not affect any aggregate par value of shares issued and outstanding or total equity. Stock split reduces the par value of each stock and increases the number of shares outstanding.
 - 1) No entry is made, and no transfer from retained earnings occurs.

9. Major Statement of Changes in Equity Note Disclosures

- a. Note disclosures and schedules specifically related to the statement of changes in equity include
 - 1) Class of stock (including related rights and privileges)
 - 2) Share-based payment compensation
 - 3) Dividends
 - 4) Retained earnings
 - 5) Stockholders' equity

10. Limitations of the Statement of Changes in Equity

- a. The financial statements report accrual-basis results for the period. The company may recognize revenue and/or expense and report net income before any cash was actually received and/or paid.
 - 1) Hence, the data for retained earnings is not sufficient for assessing the amount actually available to be reinvested in the company or to pay debt.
- b. The statement of changes in equity illustrates a company's equity based on a specific time period; equity may vary significantly a few days before or after publication of the statement.

1.5 STATEMENT OF CASH FLOWS

1. Overview

- a. The **primary purpose** of the statement of cash flows is to provide relevant information about the cash receipts and cash payments of an entity during the period.
 - 1) To achieve this purpose, the statement should provide information about cash inflows and outflows from the operating, investing, and financing activities of an entity. This is the accepted order of presentation.
 - 2) The statement of cash flows should help users assess the entity's ability to generate positive future net cash flows (liquidity), its ability to meet obligations (solvency), and its financial flexibility.
- b. The statement of cash flows explains the change in cash and cash equivalents during the period. It reconciles the period's beginning balance of cash and cash equivalents with the ending balance.

EXAMPLE 1-12 Summary of Cash Flow Statement

The following is an example of the summarized format of the statement of cash flows (only the headings). The amounts of cash and cash equivalents at the beginning and end of the year are taken from the balance sheet.

Entity A's Statement of Cash Flows for the Year Ended December 31, Year 1

Net cash provided by (used in) operating activities	\$20,000
Net cash provided by (used in) investing activities	(5,000)
Net cash provided by (used in) financing activities	9,000
Net increase (decrease) in cash and cash equivalents during the year	\$24,000
Cash and cash equivalents at beginning of year (January 1, Year 1)	6,000
Cash and cash equivalents at end of year (December 31, Year 1)	<u>\$30,000</u>

2. Operating Activities

- a. Operating activities are all transactions and other events that are not financing or investing activities.
 - 1) Cash flows from operating activities are primarily derived from the principal revenue-producing activities of the entity. They generally result from transactions and other events that enter into the determination of net income.
- b. The following are examples of **cash inflows** from operating activities:
 - 1) Cash receipts from the sale of goods and services (including collections of accounts receivable)
 - 2) Cash receipts from royalties, fees, commissions, trading debt securities, and other revenue
 - 3) Cash received in the form of **interest** or **dividends**
- c. The following are examples of **cash outflows** from operating activities:
 - 1) Cash payments to suppliers for goods and services
 - 2) Cash payments to employees
 - 3) Cash payments to government for taxes, duties, fines, and other fees or penalties
 - 4) Payments of **interest on debt**

- d. The two acceptable methods of presentation of cash flows from operating activities are the direct and the indirect methods.
 - 1) The only difference between these two methods is their presentation of **net cash flows from operating activities**.
 - a) The total cash flows from operating, investing, and financing activities are the same regardless of which method is used.
 - 2) The CMA exam requires candidates to know how to prepare the statement of cash flows using the indirect method.

3. Investing Activities

- a. Cash flows from investing activities represent the extent to which expenditures have been made for resources intended to generate future income and cash flows.
- b. The following are examples of **cash outflows (and inflows)** from investing activities:
 - 1) Cash payments to acquire (cash receipts from sale of) property, plant, and equipment; intangible assets; and other long-lived assets
 - 2) Cash payments to acquire (cash receipts from sale and maturity of) equity and debt instruments (such as held-to-maturity securities and available-for-sale debt securities) of other entities for investing purposes
 - 3) Cash advances and loans made to other parties (cash receipts from repayment of advances and loans made to other parties)

4. Financing Activities

- a. Cash flows from financing activities generally involve the cash effects of transactions and other events that relate to the issuance, settlement, or reacquisition of the entity's debt and equity instruments.
- b. The following are examples of **cash inflows** from financing activities:
 - 1) Cash proceeds from issuing shares and other equity instruments (obtaining resources from owners).
 - 2) Cash proceeds from issuing loans, notes, bonds, and other short-term or long-term borrowings.
- c. The following are examples of **cash outflows** from financing activities:
 - 1) Cash repayments of amounts borrowed
 - 2) Payments of cash dividends
 - 3) Cash payments to acquire or redeem the entity's own shares
 - 4) Cash payments by a lessee for a reduction of the outstanding liability relating to a finance lease

5. Major Statement of Cash Flows Note Disclosures

- a. Information about all **noncash investing and financing activities** (i.e., investing and financing activities that affect recognized assets or liabilities **but not cash flows**) must be disclosed in the notes.
 - 1) The following are examples of noncash investing and financing activities:
 - a) Conversion of debt to equity
 - b) Acquisition of assets either by assuming directly related liabilities or by a lessee's recognition of a finance or operating lease
 - c) Exchange of a noncash asset or liability for another

6. Indirect Method of Presenting Operating Cash Flows

- a. Under the indirect method (also called the reconciliation method), the net cash flow from operating activities is determined by adjusting the net income of a business for the effect of the following:
- 1) **Noncash revenue and expenses** that were **included in net income**, such as depreciation and amortization expenses, impairment losses, undistributed earnings of equity-method investments, and amortization of discount and premium on bonds
 - 2) Items **included in net income** whose cash effects relate to **investing or financing cash flows**, such as gains or losses on sales of property and equipment (related to investing activities) and gain or losses on extinguishment of debt (related to financing activities)
 - 3) All **deferrals** of past operating cash flows, such as changes during the period in inventory and deferred income
 - 4) All **accruals** of expected future operating cash flows, such as changes during the period in accounts receivable and accounts payable



AUTHOR'S NOTE

The net income for the period as it is reported in the income statement was calculated using the accrual method of accounting. Therefore, adjustments must be made to reach the amount of cash flow from operating activities.

- b. The reconciliation of net income to net cash flow from operating activities must disclose all major classes of reconciling items. At a minimum, this disclosure reports changes in
- 1) Accounts receivable and accounts payable related to operating activities and
 - 2) Inventories.

The following rules will help reconcile net income to net cash flow from operating activities under the indirect method:

Increase in current operating liabilities	Added to net income
Decrease in current operating assets	Added to net income
Increase in current operating assets	Subtracted from net income
Decrease in current operating liabilities	Subtracted from net income
Noncash losses and expenses included in net income	Added to net income
Losses and expenses whose cash effects are related to investing or financing cash flows	Added to net income
Noncash gains and revenues included in net income	Subtracted from net income
Gains and revenues whose cash effects are related to investing or financing cash flows	Subtracted from net income



SUCCESS TIP

EXAMPLE 1-13 Statement of Cash Flows -- Indirect Method

Dice Corp's consolidated balance sheet accounts as of December 31, Year 6 and Year 5, are presented below. Information relating to Year 6 activities is to the left.

Information Relating to Year 6 Activities

		December 31	
		Year 6	Year 5
- Cash dividends of \$240,000 were declared and paid by Dice in Year 6. - The accounts receivable balances at the beginning and end of Year 6 were net of allowances for credit losses of \$50,000 and \$60,000, respectively. Dice wrote off \$40,000 of accounts receivable during Year 6. - Current investments consist of Treasury bills maturing on 6/30/Year 7. They were acquired for cash on December 31, Year 6. - Equipment costing \$400,000 and having a carrying amount of \$140,000 was sold on January 1, Year 6, for \$150,000 in cash. Additional plant assets were purchased in Year 6 for cash. - Dice accounts for its interest in Thrice Corp. under the equity method. Its equity in Thrice's Year 6 earnings was \$25,000. During Year 6, Dice received a \$10,000 cash dividend from Thrice. At the end of Year 6, Dice sold part of its investment in Thrice for \$135,000 in cash. Significant influence over Thrice was not lost as a result of the sale. - The provision for Year 6 income taxes was \$210,000. 10,000 shares of common stock were issued in Year 6 for \$22 a share.	<u>Assets</u>		
	Cash	\$ 195,000	\$ 100,000
	Current investments	300,000	0
	Accounts receivable (net)	480,000	510,000
	Inventory	680,000	600,000
	Prepaid expenses	15,000	20,000
	Equity method investment	215,000	300,000
	Plant assets	1,730,000	1,000,000
	Accumulated depreciation	(480,000)	(450,000)
	Goodwill	90,000	100,000
	Total assets	\$ 3,225,000	\$ 2,180,000
	<u>Liabilities and Equity</u>		
	Accounts payable	\$ 825,000	\$ 720,000
	Interest payable	15,000	10,000
	Income tax payable	20,000	30,000
	Current debt	325,000	0
	Deferred taxes	250,000	300,000
	Common stock, \$10 par	800,000	700,000
	Additional paid-in capital	370,000	250,000
	Retained earnings	620,000	170,000
	Total liabilities and equity	\$ 3,225,000	\$ 2,180,000

-- Continued on next page --

EXAMPLE 1-13 -- Continued

The following computations are necessary to determine the net cash flows from operating, investing, and financing activities:

- a. **Net income.** The starting point for presenting the net cash flow from operating activities is net income (loss) for the period. Net income can be calculated as follows:

Ending retained earnings	\$620,000
Dividends declared	240,000
Beginning retained earnings	(170,000)
Year 6, net income	<u><u>\$690,000</u></u>

- 1) The \$240,000 of cash dividends paid in Year 6 are a cash outflow from financing activities.
- b. **Accounts receivable.** The easiest way to determine the reconciling adjustment for accounts receivable is to calculate the change in their net amount (Accounts receivable – Allowance for credit losses). Net accounts receivable are current operating assets. A decrease in net accounts receivable of **\$30,000** (\$510,000 – \$480,000) is **added** to net income to determine the net cash flow from operating activities.
- c. **Plant assets.** The items that affect the presentation of cash flow from operating activities are depreciation expense, gain or loss on disposal, and impairment loss.

- 1) **Depreciation expense** for Year 6 can be calculated as follows:

Ending accumulated depreciation	\$480,000
Accumulated depreciation on items sold	260,000
Beginning accumulated depreciation	(450,000)
Depreciation expense	<u><u>\$290,000</u></u>

Depreciation expense is a noncash expense included in net income. Thus, **\$290,000** is **added** to net income in determining the net cash flows from operating activities.

- 2) The **gain on disposal** of the equipment is **\$10,000** (\$150,000 cash received – \$140,000 carrying amount). The cash effect is related to investing activities. Thus, it is **subtracted** from net income to determine the net cash flow from operating activities.
- 3) The \$150,000 of cash proceeds from sale of equipment is a cash inflow from investing activities.
- 4) Plant assets purchased in Year 6 (cash outflow from investing activities) can be calculated as follows:

Ending plant assets at cost	\$1,730,000
Plant assets sold at cost	400,000
Beginning plant assets at cost	(1,000,000)
Plant assets purchased	<u><u>\$1,130,000</u></u>

NOTE: The following equation may be useful for deriving the required information if the data given in the question are for the carrying amount (Cost – Accumulated depreciation) of the PPE item.

Beginning carrying amount	\$XXX
Purchases during the period	XXX
Depreciation expense	(XXX)
Disposals during the period	(XXX)
Ending carrying amount	<u><u>\$XXX</u></u>

-- Continued on next page --

EXAMPLE 1-13 -- Continued

- d. **Equity-method investment.** The carrying amount of the equity-method investment sold can be calculated as follows:

Beginning carrying amount	\$300,000
Equity in Thrice's current year earnings	25,000
Dividends received from Thrice	(10,000)
Ending carrying amount	<u>(215,000)</u>
Carrying amount of investment sold	<u>\$100,000</u>

- 1) **The gain on sale of the investment is \$35,000** (\$135,000 cash received – \$100,000 carrying amount). The cash effect is related to investing activities. Thus, it is **subtracted** from net income in determining the net cash flows from operating activities.
 - 2) **Undistributed earnings on equity-method investment.** Under the equity method, the investor's share of the investee's earnings is debited to the investment account and credited to income. A cash dividend from the investee is a return of an investment that results in a debit to cash and a credit to the investment. The undistributed earnings on the equity-method investments equal **\$15,000** (\$25,000 share in earnings – \$10,000 dividends received). This amount is a noncash revenue included in net income. Thus, it is **subtracted** from net income in determining the net cash flow from operating activities.
 - 3) The cash received on the sale of the investment of \$135,000 is a cash inflow from investing activities.
- e. **Goodwill.** Goodwill is not amortized. Thus, the **\$10,000** decrease in the amount of goodwill (\$100,000 beginning balance – \$90,000 ending balance) must be a result of impairment. A loss on impairment of goodwill is a noncash loss included in net income. Thus, it is **added** to net income in determining the net cash flow from operating activities.
- f. **Current investments.** The purchase of current investments for \$300,000 is a cash outflow from investing activities.
- g. **Common stock.** The proceeds from issuing common stock were \$220,000 (10,000 × \$22). This cash inflow from financing activities equals the sum of the increases in the common stock and additional paid-in capital accounts.
- h. **Inventory** is a current operating asset. Inventory increased by **\$80,000** (\$680,000 – \$600,000). This amount is **subtracted** from net income in determining the net cash flow from operating activities.
- i. **Prepaid expenses** are current operating assets. Prepaid expenses decreased by **\$5,000** (\$15,000 – \$20,000). This amount is **added** to net income in determining the net cash flow from operating activities.
- j. **Accounts payable** is a current operating liability. Accounts payable increased by **\$105,000** (\$825,000 – \$720,000). This amount is **added** to net income in determining the net cash flow from operating activities.
- k. **Interest payable** is a current operating liability. Interest payable increased by **\$5,000** (\$15,000 – \$10,000). This amount is **added** to net income in determining the net cash flow from operating activities.
- l. **Income tax payable** is a current operating liability. Income tax payable decreased by **\$10,000** (\$20,000 – \$30,000). This amount is **subtracted** from net income in determining the net cash flow from operating activities.
- m. **Current debt.** The issuance of \$325,000 of current debt (\$325,000 – \$0) is a cash inflow from financing activities.
- n. The **deferred tax liability** decreased by **\$50,000** (\$250,000 – \$300,000). The decrease in the deferred tax liability increases net income by decreasing income tax expense. This decrease is a noncash item included in net income. Thus, it is **subtracted** from net income in determining the net cash flow from operating activities.

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EXAMPLE 1-13 -- Continued

Dice Corp.
Consolidated Statement of Cash Flows -- Indirect Method
for the Year Ended December 31, Year 6

Cash flows from operating activities:

Net income for Year 6	\$ 690,000	a.
Decrease in accounts receivable	30,000	b.
Depreciation expense	290,000	c1.
Gain on disposal of equipment	(10,000)	c2.
Gain on sale of investment	(35,000)	d1.
Undistributed earnings of equity-method investment	(15,000)	d2.
Loss on impairment of goodwill	10,000	e.
Increase in inventories	(80,000)	h.
Decrease in prepaid expenses	5,000	i.
Increase in accounts payable	105,000	j.
Increase in interest payable	5,000	k.
Decrease in income tax payable	(10,000)	l.
Decrease in deferred tax liability	(50,000)	n.
Net cash provided by operating activities		<u>\$ 935,000</u>

Cash flows from investing activities:

Proceeds from sale of equipment	\$ 150,000	c3.
Purchases of plant assets	(1,130,000)	c4.
Proceeds from sale of equity-method investment	135,000	d3.
Purchases of current investments	(300,000)	f.
Net cash used in investing activities		<u>\$(1,145,000)</u>

Cash flows from financing activities:

Dividends paid	\$ (240,000)	a1.
Proceeds from issuing common stock	220,000	g.
Proceeds from current debt	325,000	m.
Net cash provided by financing activities		<u>\$ 305,000</u>

Net increase in cash	\$ 95,000
Cash, beginning of year	100,000
Cash, end of year	<u>\$ 195,000</u>

7. Direct Method of Presenting Operating Cash Flows

- a. Under the direct method, the entity presents major classes of actual gross operating cash receipts and payments and their sum (net cash flow from operating activities). At a minimum, the following must be presented:
 - 1) Cash collected from customers
 - 2) Interest and dividends received
 - 3) Other operating cash receipts, if any
 - 4) Cash paid to employees and other suppliers of goods and services
 - 5) Interest paid
 - 6) Income taxes paid
 - 7) Other operating cash payments, if any
- b. If the direct method is used, the reconciliation of net income to net cash flow from operating activities (the operating section of the indirect method format) must be provided in a separate schedule.
 - 1) Most entities apply the indirect method because the reconciliation must be prepared regardless of the method chosen.

8. Limitations of the Statement of Cash Flows

- a. A cash flow statement is not sufficient for forecasting the profitability of a firm as noncash items are not included in the calculation of cash flow from operating activities.
- b. A cash flow statement may not represent the true liquid position of an entity.
 - 1) Hence, decisions regarding large expenditures could be based on misconceived information when decisions are based only on the statement of cash flows.
 - 2) This statement must be viewed in conjunction with other financial statements, such as the balance sheet and the income statement.
- c. Information can be manipulated in the statement of cash flows. For instance, management can schedule vendor payments to occur after year end to increase net cash flows reported on the statement of cash flows.

STUDY UNIT TWO

MEASUREMENT, VALUATION, AND DISCLOSURE: ASSETS -- SHORT-TERM ITEMS

2.1	<i>Accounts Receivable</i>	2
2.2	<i>Inventory -- Fundamentals</i>	8
2.3	<i>Inventory -- Cost Flow Methods</i>	14
2.4	<i>Measurement of Inventory Subsequent to Initial Recognition</i>	18

This study unit is the **second of six** on **external financial reporting decisions**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The six study units are

Study Unit 1: External Financial Statements

Study Unit 2: Measurement, Valuation, and Disclosure: Assets -- Short-Term Items

Study Unit 3: Measurement, Valuation, and Disclosure: Assets -- Long-Term Items

Study Unit 4: Measurement, Valuation, and Disclosure: Liabilities

Study Unit 5: Revenue and Impairment Recognition

Study Unit 6: Integrated Reporting

If you are interested in reviewing more introductory or background material, go to www.gleim.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Gleim outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

2.1 ACCOUNTS RECEIVABLE

1. Overview

- a. Accounts receivable (**trade receivables**) are amounts owed to an entity by its customers resulting from credit sales in the ordinary course of business that are due in customary trade terms.
 - 1) All other receivables are non-trade receivables.
- b. The recording of a receivable, which often coincides with revenue recognition, is consistent with the accrual method of accounting.
- c. Receivables should be separated into **current and noncurrent** portions. Most of the entity's accounts receivable are classified as current assets because they are expected to be collected within 1 year or the entity's normal operating cycle.
 - 1) **Current** accounts receivable are reported in the balance sheet at **net realizable value (NRV)**, i.e., net of allowance for credit losses (uncollectible accounts), allowance for sales returns, and billing adjustments.
 - 2) **Noncurrent** receivables are measured at net present value of future cash flows expected to be collected.
- d. The principal measurement issue for accounts receivable is the estimation of the **allowance for credit loss** and calculation of **credit loss expenses** for the period.
- e. The direct write-off method expenses bad debts when they are determined to be uncollectible.
 - 1) It is **not acceptable under GAAP** because it does not match revenue and expense when the receivable and the write-off are recorded in different periods.
 - 2) But this method is used for tax purposes.

2. Allowance for Customers' Right of Sales Return

- a. When a right of return exists, an entity may recognize revenue from a sale of goods at the time of sale only if the amount of future returns can be reliably estimated. A provision must be made for the return of merchandise because of product defects, customer dissatisfaction, etc.
- b. To be consistent with the **matching principle** (recognition of revenue and related expense in the same accounting period), the revenue from the sale of goods and the expense for the estimated sales returns must be recognized on the same date.
 - 1) Accordingly, an **allowance** for sales returns should be established.

EXAMPLE 2-1 Allowance for Sales Returns

A company has \$500,000 of sales in July, its first month of operations. Management estimates that total returns will be 1% of sales.

Recognition of revenue from sale		Recognition of allowance for sales returns	
Cash/accounts receivable	\$500,000	Sales returns (contra revenue)	\$5,000
Sales	\$500,000	Allowance for sales returns (contra asset)	\$5,000

3. Allowance Method (Required under GAAP)

- a. The allowance method attempts to match **credit loss expense** with the **related revenue**. This method is **required under GAAP**.

- 1) The periodic journal entry to record credit loss expense is

Credit loss expense	\$XXX	
Allowance for credit losses		\$XXX

- 2) As specific accounts receivable are written off, they are charged to the allowance account.

Allowance for credit losses	\$XXX	
Accounts receivable		\$XXX

- 3) Thus, the **write-off** of a particular account has no effect on expenses.

- a) Write-offs do not affect the carrying amount of the net accounts receivable balance because the reductions of gross accounts receivable and the allowance are the same.

- 4) In the balance sheet, the carrying amount of accounts receivable is reported at the net amount expected to be collected.

$$\begin{array}{l} \text{Gross accounts} \\ \text{receivable} \end{array} - \begin{array}{l} \text{Allowance for} \\ \text{credit losses} \end{array} = \begin{array}{l} \text{Carrying amount of} \\ \text{accounts receivable} \end{array}$$

- b. Under the allowance method, the two approaches to calculating the amount charged to credit loss expense are the income statement approach and the balance sheet approach.



**AUTHOR'S
NOTE**

The direct write-off method (i.e., expense bad debts only when they are determined to be uncollectible) is **not acceptable under GAAP**. This method does not match revenues and expenses. Occasionally, a small company will use the direct write-off method under the assumption that the difference between it and the allowance method is immaterial. In other words, the materiality principle is used as an excuse to violate the matching principle.

4. Income Statement Approach (Percentage of Sales)

- a. Under the income statement approach, the entity estimates the periodic credit loss expense as a percentage of sales on credit. The journal entry is

Credit loss expense	\$XXX	
Allowance for credit losses		\$XXX

- b. The ending balance of the allowance for credit losses is determined after all the activity in the allowance account is recorded during the period.

EXAMPLE 2-2 Income Statement Approach

Midburg Co. has the following unadjusted account balances at year end:

Cash	\$ 85,000	Dr.
Accounts receivable	100,000	Dr.
Allowance for credit losses	2,000	Cr.
Sales on credit	500,000	Cr.

Based on its experience, Midburg expects credit losses to average 2% of credit sales. Thus, the estimated credit loss expense is \$10,000 ($\$500,000 \times 2\%$). The year-end adjusting entry is

Credit loss expense	\$10,000	
Allowance for credit losses		\$10,000

The year-end adjusted balance of allowance for credit losses is a \$12,000 credit ($\$10,000 + \$2,000$).

<u>Balance sheet presentation</u>	
Accounts receivable	\$100,000
(Allowance for credit losses)	(12,000)
Accounts receivable, net	<u>\$ 88,000</u>

5. Balance Sheet Approach (Percentage of Receivables)

- a. Under this approach, the ending balance of the allowance for credit losses is a percentage of the ending balance of accounts receivable.
- 1) Credit loss expense reflects the adjustment of the allowance to its correct ending balance.

EXAMPLE 2-3 Balance Sheet Approach

Using the data from Example 2-2, assume that, based on Midburg's experience, 6% of accounts receivable are determined to be uncollectible. Thus, the ending balance of the allowance for credit losses is \$6,000 ($\$100,000 \times 6\%$). Because the allowance currently has a balance of \$2,000, the following journal entry is required:

Credit loss expense ($\$6,000 - \$2,000$)	\$4,000	
Allowance for credit losses		\$4,000

<u>Balance sheet presentation</u>	
Accounts receivable	\$100,000
(Allowance for credit losses)	(6,000)
Accounts receivable, net	<u>\$ 94,000</u>

- b. An entity rarely experiences a single rate of uncollectibility on all its accounts. For this reason, entities using the balance sheet approach to estimate expected credit losses for accounts receivable generally prepare an **aging schedule**.

EXAMPLE 2-4 Balance Sheet Approach with Aging Schedule

Midburg prepares the following aging schedule of its accounts receivable:

Balance Range	Less than 30 Days	31-60 Days	61-90 Days	Over 90 Days	Total Balances
\$0 - \$100	\$ 5,000	\$ 200	\$ 100	\$ 100	\$ 5,400
\$100 - \$1,000	8,000	3,800			11,800
\$1,000 - \$5,000	20,000	2,000	1,900		23,900
\$5,000 - \$10,000	38,000		8,000	900	46,900
Over \$10,000		12,000			12,000
Totals	<u>\$ 71,000</u>	<u>\$18,000</u>	<u>\$10,000</u>	<u>\$1,000</u>	<u>\$ 100,000</u>

Midburg then applies an appropriate percentage to each stratum based on experience.

Aging Intervals	Balance	Estimated Uncollectible	Ending Allowance
Less than 30 days	\$ 71,000	2%	\$1,420
30-60 days	18,000	12%	2,160
61-90 days	10,000	15%	1,500
Over 90 days	1,000	20%	200
Total	<u>\$100,000</u>		<u>\$5,280</u>

Because the allowance currently has a balance of \$2,000, the following journal entry is required to establish the proper measurement:

Credit loss expense (\$5,280 – \$2,000)	\$3,280	
Allowance for credit losses		\$3,280
<u>Balance sheet presentation</u>		
Accounts receivable	\$100,000	
(Allowance for credit losses)		(5,280)
Accounts receivable, net		<u>\$ 94,720</u>

6. Collection of Accounts Previously Written Off

- a. Occasionally, a customer pays on an account previously written off.

Cash	\$XXX
Allowance for credit losses	\$XXX

- 1) Credit loss expense is not affected when an account receivable is written off or when an account previously written off becomes collectible.
- b. The following equation illustrates the reconciliation of the beginning and ending balances of gross accounts receivable (accounts receivable before adjustment for allowance for credit losses):

Beginning accounts receivable
+ Credit sales during the period
– Cash collected on credit sales during the period
– Accounts receivable written-off during the period
= <u>Ending accounts receivable</u>

- c. The following equation illustrates the reconciliation of the beginning and ending balances of the allowance for credit losses:

Beginning allowance for credit losses
+ Credit loss expense recognized for the period
– Accounts receivable written off
+ Collection of accounts receivable previously written off
= <u>Ending allowance for credit losses</u>

- 1) Under the **income statement approach**, credit loss expense is a percentage of sales on credit, and the ending balance of the allowance is calculated using the equation above.
- 2) Under the **balance sheet approach**, the ending balance of the allowance is a percentage of the ending balance of accounts receivable, and credit loss expense is calculated using the equation above.

7. Factoring of Accounts Receivable

- a. **Factoring** is a transfer of receivables to a third party (a factor) who assumes the responsibility of collection.
- b. Factoring discounts receivables on a **nonrecourse**, notification basis. Thus, payments by the debtors on the transferred assets are made to the factor. If the transferor (seller) surrenders control, **the transaction is a sale**.
- 1) If a sale is **without recourse**, the transferee (credit agency) assumes the risks and receives the rewards of collection. This sale is final, and the seller has no further liabilities to the transferee. Accordingly, the receivables are no longer reported on the seller's books.
- 2) If a sale is **with recourse**, the transferor (seller) may be required to make payments to the transferee or to buy back receivables in specified circumstances.
- a) In this circumstance, the transfer might not qualify as a sale. The parties account for the transaction as a secured borrowing with a pledge of noncash collateral.
- b) Accordingly, the receivables are still on the seller's books and it must recognize a liability for the amount of cash received.

EXAMPLE 2-5 Factoring of Accounts Receivable

A factor charges a 2% fee plus an interest rate of 18% on all cash advanced to a transferor of accounts receivable. Monthly sales are \$100,000, and the factor advances 90% of the receivables submitted after deducting the 2% fee and the interest. Credit terms are net 60 days. What is the cost to the transferor of this arrangement?

Amount of receivables submitted	\$100,000
Minus: 10% reserve	(10,000)
Minus: 2% factor's fee	<u>(2,000)</u>
Amount accruing to the transferor	\$ 88,000
Minus: 18% interest for 60 days	<u>(2,640) [\$88,000 × 18% × (60 ÷ 360)]</u>
Amount to be received immediately	<u><u>\$ 85,360</u></u>

The transferor also will receive the \$10,000 reserve at the end of the 60-day period if it has not been absorbed by sales returns and allowances. Thus, the total cost to the transferor to factor the receivables for the month is \$4,640 (\$2,000 factor fee + interest of \$2,640). Assuming that the factor has approved the customers' credit in advance (the sale is without recourse), the transferor will not absorb any bad debts.

The journal entry to record the transaction is

Cash	\$85,360
Due from factor	10,000
Loss on sale of receivables	2,000
Prepaid interest	2,640
Accounts receivable	\$100,000

- c. The main reasons for factoring transactions are as follows:
- 1) A factor usually receives a high financing fee plus a fee for collection. Furthermore, the factor often operates more efficiently than its clients because of the specialized nature of its services.
 - 2) An entity (seller) that uses a factor tries to speed up its collections. Also, it can eliminate its credit department and accounts receivable staff. In addition, bad debts are eliminated from the financial statements.
 - a) These reductions in costs can offset the fee charged by the factor.
- d. Credit card sales are a common form of factoring. The retailer benefits by prompt receipt of cash and avoidance of bad debts and other costs. In return, the credit card company charges a fee.

2.2 INVENTORY -- FUNDAMENTALS

1. Overview

- a. Inventory is the total of tangible personal property
 - 1) Held for sale in the ordinary course of business,
 - 2) In the form of work-in-process to be completed and sold in the ordinary course of business, or
 - 3) To be used up currently in producing goods or services for sale.
 - a) Inventory does not include long-term assets subject to depreciation.
- b. Inventories are generally classified as current assets in the financial statements. They are expected to be realized in cash or sold or consumed during the normal operating cycle of the business.
 - 1) Long-term assets either subject to depreciation or retired from regular use and held for sale are not classified as inventories.
- c. The inventories of a **retailer** (trading entity) consist of goods purchased to be resold without substantial modification.
- d. The inventories of a **manufacturer** consist of (1) goods to be consumed in production (materials), (2) goods in the process of production (work-in-process), and (3) finished goods.
- e. The **gross profit** for the period is measured and presented in the income statement as the difference between the amount of sales and the amount of cost of goods sold.
 - 1) The calculation of cost of goods sold for a retailer and a manufacturer is discussed in Study Unit 1, Subunit 3.

2. Cost Basis of Inventory -- Initial Measurement

- a. The **cost of inventory** includes all costs incurred in bringing the inventories to their existing location and ready-to-use condition.
- b. The **cost of purchased inventories** includes
 - 1) The price paid or consideration given to acquire the inventory (net of trade discounts, rebates, and other similar items);
 - 2) Import duties and other unrecoverable taxes; and
 - 3) Handling, insurance, freight-in, and other costs directly attributable to (a) acquiring finished goods and materials and (b) bringing them to their present location and condition (salable or usable condition).
 - 4) The following is the calculation of net purchases:

$$\begin{array}{r}
 \text{Purchases} \\
 - \text{Returns and allowances} \\
 - \text{Purchase discounts} \\
 + \text{Transportation-in} \\
 \hline
 = \text{Net purchases}
 \end{array}$$

- c. The **cost of manufactured inventories** (work-in-process and finished goods) includes the costs of direct materials used and conversion costs. Conversion costs consist of (1) direct labor costs and (2) manufacturing overhead costs.

3. Inventory Accounting Systems

- a. A **perpetual inventory system** updates inventory accounts after each purchase or sale. This system is generally more suitable for entities that sell relatively expensive and heterogeneous items and requires continuous monitoring of inventory and cost of goods sold accounts. Automobile dealers are an example. Under this system,
 - 1) Purchases and other items related to inventory costing are charged directly to inventory.
 - 2) Inventory and cost of goods sold are adjusted as sales occur.
- b. An **advantage** of the perpetual inventory system is that the amount of inventory on hand and the cost of goods sold can be determined at any time. A **disadvantage** of the perpetual inventory system is that the bookkeeping is more complex and expensive.
 - 1) A perpetual system is more often used when individual inventory items are of high value because a perpetual system provides better internal control.
- c. In the **periodic inventory system**, inventory and cost of goods sold are updated at specific intervals, such as quarterly or annually, based on the results of a **physical count**.
 - 1) Bookkeeping is simpler under this system. Thus, entities with relatively inexpensive and homogeneous items, such as grain dealers, that have no need to continuously monitor their inventory and cost of goods sold generally use this system.
 - 2) Under the periodic system,
 - a) Goods bought from suppliers and other items related to inventory costs are tracked during the period in a separate temporary account (**purchases**).
 - b) The beginning inventory balance remains unchanged until the end of the period when the purchases account is closed.
 - c) Changes in inventory and cost of goods sold are recorded **only at the end of the period**, based on the physical count.

EXAMPLE 2-6 Perpetual System vs. Periodic System

Entity A's January 1, Year 1, inventory consists of 1,000 units with a cost of \$5 per unit. The following are Entity A's Year 2 transactions:

April 1: Sold 600 inventory units for \$4,800 in cash.

May 1: Purchased 250 inventory units for \$5 in cash per unit.

The year-end result of the physical count was 650 inventory units. The following are Entity A's journal entries under the perpetual and periodic systems:

Perpetual System		Periodic System	
Inventory sale April 1:			
Cash	\$4,800	Cash	\$4,800
Sales	\$4,800	Sales	\$4,800
Cost of goods sold (600 × \$5)	\$3,000		
Inventory	\$3,000		
Inventory purchase May 1:			
Inventory (250 × \$5)	\$1,250	Purchases	\$1,250
Cash	\$1,250	Cash	\$1,250
After the physical count on December 31:			
No journal entry is needed because the physical count equals the amount of inventory on the books (1,000 – 600 + 250 = 650).		Inventory (year-end) (650 × \$5)	\$3,250
		Cost of goods sold (difference)	3,000
		Inventory (beginning)	\$5,000
		Purchases	1,250
Beginning inventory (1,000 × \$5)	\$5,000		
Purchases of inventory during the period	1,250		
Ending inventory	(3,250)		
Cost of goods sold	<u>\$3,000</u>		

The perpetual and periodic systems have the same result. However, under the periodic system, the amounts of inventory and cost of goods sold are updated only at the end of the period after the physical count.

4. Inventory Period-End Physical Count

a. An annual period-end **inventory physical count** is necessary under both the perpetual and periodic inventory accounting systems. The amount of inventory reported in the annual financial statements should be based on a physical count.

- 1) Under the **perpetual system**, a physical count helps to detect (a) misstatements in the records and (b) thefts of inventory.
 - a) The differences between the physical count and the inventory in the books (inventory shortages and overages) are recognized in the cost of goods sold account or as a separate line item in the current period income statement.
- 2) Under the **periodic system**, the amounts of inventory and cost of goods sold can be determined based only on the results of a physical count.
 - a) Thus, the amount of cost of goods sold for the period includes both (1) inventory cost of goods sold and (2) inventory shortages and overages.

- b. For a physical count to be accurate, the entity must count all items considered to be inventory and eliminate all items that are not. Items to be counted as inventory include the following:
- 1) **Goods in transit** – Items in transit are inventories that on the physical count date
 - (a) are not on the entity's premises and are on the way to the desired location and
 - (b) whose legal title is held by the entity; i.e., the entity bears the risk of loss on inventory in transit. The following are the most common shipping terms:
 - a) **FOB shipping point** (sometimes called FOB Factory) – Legal title and risk of loss pass to the buyer when the seller delivers the goods to the carrier. The **buyer** must include the goods in inventory during shipping.
 - b) **FOB destination** – Legal title and risk of loss pass to the buyer when the seller delivers the goods to a specified destination. The **seller** must include the goods in inventory during shipping.
 - 2) **Goods out on consignment** – A consignment sale is an arrangement between the owner of goods (consignor) and the sales agent (consignee). Consigned goods are not sold but rather transferred to an agent for possible sale. The consignor records sales only when the goods are sold to third parties by the consignee.
 - a) Goods out on consignment are included in the **consignor's** inventory at cost. Costs of transporting the goods to the consignee are inventoriable costs, not selling expenses.
 - b) The **consignee** never records the consigned goods as an asset.

EXAMPLE 2-7 Goods Out on Consignment

Entity A's December 31, Year 1, warehouse inventory physical count results in the amount of inventory of \$50,000. The following is additional information regarding Entity A's year-end inventory:

- During the year, Entity A consigned goods with a total cost of \$60,000 to Entity B (the consignee). The annual statement that was sent from Entity B to Entity A states that 60% of the consignment goods were sold for \$42,000.
- Merchandise costing \$40,000 shipped FOB shipping point from a vendor on December 29, Year 1, and was received by Entity A on January 4, Year 2.
- Merchandise costing \$70,000 shipped FOB destination from a vendor on December 30, Year 1, and was received by Entity A on January 5, Year 2.
- The goods billed to the customer FOB destination on December 27, Year 1, had a cost of \$25,000. The goods were shipped by Entity A on December 28, Year 1, and were received by the customer on January 3, Year 2.

In the December 31, Year 1, balance sheet, Entity A reports an inventory amount of \$139,000. This amount consists of

Warehouse physical inventory count	\$ 50,000
Goods held on consignment ($\$60,000 \times 40\%$)	24,000
Merchandise shipped FOB shipping point (title and risk of loss passed to Entity A on December 29, Year 1, at the time of shipment)	40,000
Goods shipped FOB destination to customer (title and the risk of loss will pass to the customer only on January 3, Year 2)	25,000
December 31, Year 1, inventory balance	<u>\$139,000</u>

5. Inventory Estimation

- a. An estimate of inventory may be needed when an exact count is not feasible, e.g., for interim reporting purposes or when inventory records have been destroyed. The gross profit method may be used for inventory estimation, even when the inventory may have been stolen or destroyed by fire.

- 1) **Gross profit margin** (gross profit percentage) equals gross profit divided by sales.

$$\text{Gross profit margin (\%)} = \text{Gross profit} \div \text{Sales}$$

EXAMPLE 2-8

Inventory Estimation

A retailer needs to estimate ending inventory for quarterly reporting purposes. The firm’s best estimate of the gross percentage is its historical rate of 25%. The following additional information is available:

	Net sales	\$1,000,000	
	Purchases	300,000	
	Beginning inventory	800,000	
Beginning inventory			\$ 800,000
Purchases			<u>300,000</u>
Goods available for sale			\$1,100,000
Sales	\$1,000,000		
Gross profit (Sales × Gross profit margin)	<u>(250,000)</u>		
Cost of goods sold [Sales × (1 – Gross profit margin)]			<u>(750,000)</u>
Ending inventory			<u><u>\$ 350,000</u></u>

6. Inventory Errors

- a. These errors may have a material effect on current assets, working capital (Current assets – Current liabilities), cost of goods sold, net income, and equity. A common error is inappropriate timing of the recognition of transactions.
- 1) If a purchase on account is not recorded and the goods are not included in ending inventory, cost of goods sold (BI + Purchases – EI) and net income are unaffected. But current assets and current liabilities are understated.
 - 2) If purchases and beginning inventory are properly recorded but items are excluded from ending inventory, cost of goods sold is overstated. Net income, inventory, retained earnings, working capital, and the current ratio are understated.
- b. Errors arising from recording transactions in the wrong period may reverse in the subsequent period.
- 1) If ending inventory is overstated, the overstatement of net income will be offset by the understatement in the following year that results from the overstatement of beginning inventory.

- c. An **overstatement error in year-end inventory** of the current year affects the financial statements of 2 different years.

1) The **first year's** effects may be depicted as follows:

Balance Sheet Effects		Income Statement Effects	
		Net sales revenue	\$XXX,XXX
		Beginning inventory	\$XXX,XXX
		Plus: Cost of goods purchased/produced	XXX,XXX
		Goods available for sale	XXX,XXX
Current assets:		Minus: Ending inventory	Overstated ↑
Ending inventory	Overstated ↑	Cost of goods sold	Understated ↓
Equity:		Gross profit	Overstated ↑
Retained earnings, 12/31/Yr 0	\$XXX,XXX	Expenses	(XX,XXX)
Plus: Net income	Overstated ↑	Net income	Overstated ↑
Retained earnings, 12/31/Yr 1	Overstated ↑		

Figure 2-1

2) At the end of the **second year**, retained earnings is correctly stated as follows:

Balance Sheet Effects		Income Statement Effects	
		Net sales revenue	\$XXX,XXX
		Beginning inventory	Overstated ↑
		Plus: Cost of goods purchased/produced	XXX,XXX
		Goods available for sale	Overstated ↑
Current assets:		Minus: Ending inventory	\$(XXX,XXX)
Ending inventory	\$XXX,XXX	Cost of goods sold	Overstated ↑
Equity:		Gross profit	Understated ↓
Retained earnings, 12/31/Yr 1	Overstated ↑	Expenses	(XX,XXX)
Plus: Net income	Understated ↓	Net income	Understated ↓
Retained earnings, 12/31/Yr 2	<u>\$XXX,XXX</u>		

Figure 2-2

2.3 INVENTORY -- COST FLOW METHODS

1. Specific Identification Method

- a. **Specific identification** requires determining which specific items are sold and therefore reflects the actual physical flow of goods. This system is appropriate for (1) items that are not ordinarily interchangeable and (2) items that are segregated for a specific project. It can be used for blocks of investment securities or special inventory items, such as automobiles or heavy equipment. Any item that has a serial number on it would be a candidate for the specific identification method.
 - 1) Specific identification is the most accurate method because it identifies each item of inventory.
 - 2) However, it requires detailed records and may not be feasible or cost effective.
- b. When the inventory items purchased or produced are identical and interchangeable, specific identification is not appropriate.
 - 1) In such circumstances, several assumptions about the flow of cost, such as **average**, **FIFO**, or **LIFO**, may be appropriate for the measurement of periodic income. The method selected should be the one that, under the circumstances, most clearly reflects periodic income.

2. Average Method

- a. The average method assumes that goods are indistinguishable and are therefore measured at an average of the costs incurred. The average may be calculated on the periodic basis or as each additional purchase occurs (perpetual basis).
 - 1) The **moving-average method** is used under the **perpetual** inventory accounting system. It requires determination of a new weighted-average inventory cost after each purchase. This cost is used for every sale until the next purchase.

EXAMPLE 2-9 Moving-Average Method

The following data relate to Entity A's Year 1 activities:

Date	Transaction	Number of units	Purchase price per unit (\$)	Sale price per unit (\$)
January 1	Beginning balance	100	20	
March 1	Purchase	20	32	
April 1	Sale	70		40
June 1	Purchase	30	14	
October 1	Sale	40		24

Under the **moving-average method**, the year-end inventory and Year 1 cost of goods sold are calculated as follows:

Date	Activity	Units	Price	Cost of inventory purchased/sold	Inventory total balance	On-hand units	Cost per unit
Jan. 1	Beg. bal.	100	\$20		\$2,000 (100 × 20)	100	\$20
Mar. 1	Purchase	20	\$32	\$640 = 20 × \$32	\$2,640 (2,000 + 640)	120	\$22 (\$2,640 ÷ 120)
Apr. 1	Sale	70	\$22	(\$1,540) = 70 × \$22	\$1,100 (2,640 – 1,540)	50	\$22 (\$1,100 ÷ 50)
Jun. 1	Purchase	30	\$14	\$420 = 30 × \$14	\$1,520 (1,100 + 420)	80	\$19 (\$1,520 ÷ 80)
Oct. 1	Sale	40	\$19	(\$760) = 40 × \$19	\$760 (1,520 – 760)	40	\$19 (\$760 ÷ 40)

The cost of **inventory** on December 31, Year 1, is **\$760**. The Year 1 **cost of goods sold** is **\$2,300**.

Beginning inventory	\$2,000
Purchases (\$640 + \$420)	1,060
Ending inventory	(760)
Cost of goods sold (\$1,540 + \$760)	\$2,300

- 2) The **weighted-average method** is used under the **periodic** inventory accounting system. The average cost is determined only at the end of the period. The weighted-average cost per unit is used to determine the ending inventory and the cost of goods sold for the period. It is calculated as follows:

EXAMPLE 2-10 Weighted-Average Method

Under the **weighted-average method**, Entity A's ending inventory and Year 1 cost of goods sold are determined as follows:

First, the weighted-average cost per unit is calculated.

$$\frac{\text{Cost of beginning inventory} + \text{Cost of purchases during the period}}{\text{Units in beginning inventory} + \text{Number of units purchased}} = \frac{\$2,000 + \$1,060}{100 + 20 + 30} = \$20.40$$

Second, the ending inventory and Year 1 cost of goods sold are calculated using the weighted-average cost per unit (WACPU).

Beginning inventory	\$2,000	
Purchases	1,060	
Ending inventory	(816)	(40 × \$20.40) = (WACPU × Units in ending inventory)
Cost of goods sold	<u>\$2,244</u>	(110 × \$20.40) = (WACPU × Units sold during the period)

3. First-in, First-out (FIFO)

- This method assumes that the first goods purchased are the first sold. Thus, ending inventory consists of the latest purchases.
- Cost of goods sold includes the earliest goods purchased.
- Under the FIFO method, year-end inventory and cost of goods sold for the period are **the same** regardless of whether the perpetual or the periodic inventory accounting system is used.

EXAMPLE 2-11 FIFO

The number of units in Entity A's ending inventory is 40. Under the FIFO method, the cost of these units is the cost of the **latest purchases (\$740)**.

<u>Date of purchase</u>	<u>Units</u>	<u>Price per unit</u>	<u>Total cost</u>
June 1, Year 1	30	\$14	\$420
March 1, Year 1	10	\$32	\$320
Ending inventory	<u>40</u>		<u>\$740</u>

The Year 1 cost of goods sold is **\$2,320**.

Beginning inventory	\$2,000
Purchases (\$640 + \$420)	1,060
Ending inventory	(740)
Cost of goods sold	<u>\$2,320</u>

NOTE: The results are the same under the periodic and perpetual systems.

4. Last-in, First-out (LIFO)

a. The LIFO (last-in, first-out) method assumes the newest items of inventory are sold first. Thus, the items remaining in inventory are the oldest. Under the LIFO method, the perpetual and the periodic inventory accounting systems may result in different values for year-end inventory and cost of goods sold.

- 1) Under the **periodic** inventory accounting system, the calculation of inventory and cost of goods sold are made at the end of the period.

EXAMPLE 2-12 LIFO Periodic

The number of units in Entity A's ending inventory is 40. Under the LIFO method, the cost of those units is the cost of the **earliest purchases** (beginning inventory) of **\$800** (40 units × \$20). The Year 1 cost of goods sold is **\$2,260**.

Beginning inventory	\$2,000
Purchases (\$640 + \$420)	1,060
Ending inventory	(800)
Cost of goods sold	<u>\$2,260</u>

- 2) Under the **perpetual** inventory accounting system, cost of goods sold is calculated every time a sale occurs and consists of the most recent (latest) purchases.

b. Note Disclosure

- 1) If a method other than LIFO is used to account for the flow of inventory, the excess of the total amount under such method over the total LIFO amount will be presented in the notes.

EXAMPLE 2-13 LIFO Perpetual

Date	Activity	Units	Cost per unit	Cost of inventory purchased/sold	Inventory total balance	Number of units
Jan. 1	Beg. bal.	100	\$20		100 × \$20 = \$2,000	100
Mar. 1	Purchase	20	\$32	20 × \$32 = \$640	Jan. 1, layer 100 × \$20 = \$2,000 Mar. 1, layer 20 × \$32 = 640 \$2,640	120
Apr. 1	Sale	70		20 × \$32 = \$ 640 50 × \$20 = 1,000 \$(1,640)	Jan. 1, layer 50 × \$20 = \$1,000	50
Jun. 1	Purchase	30	\$14	30 × \$14 = \$420	Jan. 1, layer 50 × \$20 = \$1,000 Jun. 1, layer 30 × \$14 = 420 \$1,420	80
Oct. 1	Sale	40		30 × \$14 = \$ 420 10 × \$20 = 200 \$(620)	Jan. 1, layer 40 × \$20 = \$800	40

Entity A's cost of ending **inventory** is **\$800**, and the Year 1 **cost of goods sold** is **\$2,260** (\$1,640 + \$620).

NOTE: The results of the LIFO method under the perpetual and periodic systems are the same in this example but may differ in other situations.



IFRS
DIFFERENCE

LIFO is not permitted.

5. Retail Inventory Method

- a. The retail inventory method is a very effective means of estimating inventory value. It is used for
 - 1) Interim and annual financial reporting in accordance with GAAP,
 - 2) Federal income tax purposes, and
 - 3) Verifying year-end inventory and cost of goods sold data, e.g., as an analytical procedure by an independent auditor.

6. Cost Flow Methods -- Comparison

- a. The cost flow model selected should be the one that most clearly reflects periodic income.

EXAMPLE 2-14 Cost Flow Methods		
Follows are Entity A's varying results under each of the five cost flow methods:		
	Ending Inventory	Cost of Goods Sold
Moving average	\$760	\$2,300
Weighted average	816	2,244
FIFO	740	2,320
LIFO periodic	800	2,260
LIFO perpetual	800	2,260

- b. An advantage of FIFO is that ending inventory approximates current replacement cost.
 - 1) A disadvantage is that current revenues are matched with older costs.
- c. Under LIFO, management can affect net income with an end-of-period purchase that immediately alters cost of goods sold.
 - 1) A last-minute FIFO purchase included in the ending inventory has no such effect.
- d. Under LIFO, if fewer units are purchased than sold,
 - 1) The beginning inventory is partially or fully liquidated and
 - 2) Old costs are matched against current revenues in the year's income statement.
- e. In a time of **rising prices** (inflation), use of the LIFO method results in the lowest year-end inventory, the highest cost of goods sold, and the lowest gross profit. LIFO assumes that the oldest (and therefore the lowest-priced) goods purchased are in year-end inventory, and that cost of goods sold consists of the latest (and therefore the highest-priced) goods purchased.
 - 1) The results for the FIFO method are the opposite of those for the LIFO method.

During a Period of Inflation	Ending Inventory	Cost of Goods Sold	Gross Profit (Net Income)
LIFO	Lowest	Highest	Lowest
FIFO	Highest	Lowest	Highest

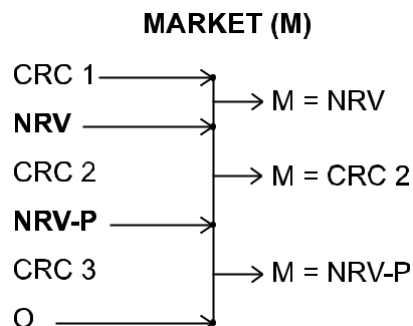
2.4 MEASUREMENT OF INVENTORY SUBSEQUENT TO INITIAL RECOGNITION

1. Statement of Rule

- a. The subsequent measurement of inventory depends on the cost method used.
 - 1) Inventory accounted for using LIFO or the retail inventory method is measured at the **lower of cost or market**.
 - 2) Inventory accounted for using any other cost method (e.g., **FIFO or average cost**) is measured at the **lower of cost or net realizable value**.
- b. The **loss on write-down** of inventory to market or net realizable value (NRV) generally is presented as a component of cost of goods sold. However, if the amount of loss is material, it should be presented as a separate line item in the current-period income statement.
 - 1) A write-down of inventory below its cost may result from damage, deterioration, obsolescence, changes in price levels, changes in demand, etc.
- c. A **reversal of a write-down** of inventory recognized in the annual financial statements is **prohibited** in subsequent periods.
 - 1) Once inventory is written down below cost, the reduced amount is the new cost basis.
- d. Depending on the nature of the inventory, the rules for write-down below cost may be applied either directly to each item or to the total of the inventory (or in some cases, to the total of each major category).
 - 1) The method should be the one that most clearly reflects periodic income.

2. Measurement of Inventory at the Lower of Cost or Market (LCM)

- a. Inventory accounted for using the **LIFO or retail inventory method** must be written down to **market** if its utility is no longer as great as its cost.
 - 1) The excess of cost over market is recognized as a loss on write-down in the income statement.
- b. **Market** is the current cost to replace inventory, subject to certain limitations. Market should not (1) exceed a **ceiling** equal to NRV or (2) be less than a **floor** equal to NRV reduced by an allowance for an approximately **normal profit margin**.
 - 1) NRV is the estimated selling price in the ordinary course of business minus reasonably predictable costs of completion, disposal, and transportation.
 - 2) Thus, **current replacement cost (CRC)** is not to be greater than NRV or less than NRV minus a normal profit (NRV – P).



EXAMPLE 2-15 Calculating LCM

The following information is related to a company's year-end inventories:

Cost per inventory unit	Item A	Item B	Item C
Estimated selling price	\$80	\$70	\$44
Minus: Cost of completion	(20)	--	(3)
Minus: Cost of disposal	(6)	(5)	(2)
NRV (ceiling)	\$54	\$65	\$39
Minus: Normal profit margin	(3)	(7)	(4)
NRV – NPM (floor)	\$51	\$58	\$35
Current replacement cost (CRC)	\$53	\$55	\$40
(a) Market	\$53 Ceiling > CRC > Floor	\$58 Floor > CRC	\$39 CRC > Ceiling
(b) Historical cost per unit	\$50	\$60	\$45
Lower of cost (b) or market (a)	\$50 Cost < Market	\$58 Market < Cost	\$39 Market < Cost

- c. The excess of **cost over market** is recognized as a loss on write-down in the income statement.

EXAMPLE 2-16 LIFO -- LCM

Lala Co. accounts for its inventory using the **LIFO** cost method. The following is its inventory information at the end of the fiscal year:

Historical cost	\$100,000
Current replacement cost	82,000
Net realizable value (NRV)	90,000
Normal profit margin	5,000

Under the LIFO method, inventory is measured at the lower of cost or market (current replacement cost subject to certain limitations). Market cannot be higher than NRV (\$90,000) or lower than NRV reduced by a normal profit margin (\$90,000 – \$5,000 = \$85,000). Thus, market is \$85,000. (The current replacement cost of \$82,000 is below the floor.) Because market is lower than cost, the inventory is reported in the balance sheet at market of \$85,000. The write-down of inventory of \$15,000 (\$100,000 – \$85,000) is recognized as a loss in the income statement. The journal entry is as follows:

Loss from inventory write-down	\$15,000
Inventory	\$15,000



IFRS
DIFFERENCE

Inventories are measured at the lower of cost or net realizable value (NRV) regardless of the cost method used. NRV is the estimated selling price less the estimated costs of completion and disposal. NRV is assessed each period. Accordingly, a write-down **may be reversed** but not above original cost. The write-down and reversal are recognized in profit or loss.

Using the data from Example 2-16, NRV (\$90,000) is lower than cost (\$100,000). Thus, the inventory is reported in the statement of financial position at its NRV (\$90,000). The write-down of inventory of \$10,000 (\$100,000 – \$90,000) is recognized in profit or loss. The journal entry is as follows:

Loss from inventory write-down	\$10,000
Inventory	\$10,000

3. Measurement of Inventory at the Lower of Cost or NRV (LCNRV)

- a. Inventory measured using any method other than LIFO or retail (e.g., **FIFO or average cost**), must be measured at the **lower of cost or net realizable value**.
 - 1) **NRV** is the estimated selling price in the ordinary course of business minus reasonably predictable costs of completion, disposal, and transportation.
 - 2) The excess of **cost over NRV** is recognized as a loss on write-down in the income statement.

EXAMPLE 2-17 FIFO -- LCNRV

Using the data from Example 2-16, assume that Lala Co. accounts for its inventory using the **FIFO** cost method.

Under the FIFO method (or any other method except for LIFO or retail), inventory is measured at the **lower of cost or net realizable value**. NRV of \$90,000 is lower than cost of \$100,000. Thus, a loss on write-down to NRV of \$10,000 is recognized. The journal entry is as follows:

Loss from inventory write-down	\$10,000	
Inventory		\$10,000

NOTE: The same loss on write-down is recognized under IFRS.

4. Inventory Measurement at Interim Dates

- a. A write-down of inventory below cost (to market for LIFO and retail and to NRV for all other methods) may be **deferred in the interim financial statements** if no loss is reasonably anticipated for the year.
 - 1) But inventory losses from a **nontemporary** decline below cost must be recognized at the interim date.
 - 2) If the loss is recovered in another quarter, it is recognized as a gain and treated as a change in estimate. The amount recovered is limited to the losses previously recognized.



IFRS
DIFFERENCE

Under IFRS, each interim period is viewed as a discrete (individually separate) reporting period. Accordingly, the accounting treatment for inventory measurement in the interim statements is the same as in the annual statements. For an interim period, an inventory loss from a write-down to NRV must be recognized even if no loss is reasonably expected for the year.

STUDY UNIT THREE

MEASUREMENT, VALUATION, AND DISCLOSURE: ASSETS -- LONG-TERM ITEMS

3.1	<i>Investments in Equity Securities</i>	1
3.2	<i>Equity Method</i>	4
3.3	<i>Investments in Debt Securities</i>	5
3.4	<i>Property, Plant, and Equipment</i>	11
3.5	<i>Intangible Assets</i>	17

This study unit is the **third of six** on **external financial reporting decisions**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The six study units are

Study Unit 1: External Financial Statements

Study Unit 2: Measurement, Valuation, and Disclosure: Assets -- Short-Term Items

Study Unit 3: Measurement, Valuation, and Disclosure: Assets -- Long-Term Items

Study Unit 4: Measurement, Valuation, and Disclosure: Liabilities

Study Unit 5: Revenue and Impairment Recognition

Study Unit 6: Integrated Reporting

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

3.1 INVESTMENTS IN EQUITY SECURITIES

1. Overview

- a. An **equity security** is an **ownership** interest in an entity (e.g., common stock or preferred stock) or a right to acquire or dispose of such an interest (e.g., warrants or call options).
 - 1) Convertible debt securities are not equity interests in the issuer.
- b. This subunit applies to all investments in equity securities **except** for investments
 - 1) Accounted for under the equity method
 - 2) In consolidated subsidiaries
 - 3) For which the entity has elected the fair value option (FVO)

- c. The accounting method for an investment in voting stock depends on the presumed influence that the investor has over the investee. The presumed influence usually is determined based on the ownership interest held. The table to the right depicts the three possibilities.

Percentage Ownership	Presumed Influence	Accounting Method
100%	Control	Consolidation
50%	Significant	Equity Method or FVO
20%	Little or none	Fair Value Measurement
0%		

Figure 3-1

2. Measurement of an Investment in Equity Securities

- The investment is measured at **fair value** at each balance sheet date.
- Unrealized holding gains and losses** on the remeasurement of the investment to fair value are reported in the **income statement** (net income) at each subsequent reporting date.
- Dividends received from investments in equity securities are reported as dividend income in the income statement.
- Cash flows from purchases and sales of equity securities are classified in the statement of cash flows based on the nature and purpose for which the securities were acquired.

EXAMPLE 3-1 Investment in Equity Securities

On November 1, Year 1, Abi Co. purchased 200 shares of Gail Co.'s common stock at fair value. This investment is less than 1% of the ownership interests in Gail Co. The following are the fair values per share of Gail common stock at the relevant dates:

Date	Fair Value
November 1, Year 1	\$100
December 31, Year 1	90
December 31, Year 2	115

November 1, Year 1

Investment in equity securities (200 × \$100)	\$20,000
Cash	\$20,000

This investment in equity securities of Gail Co. is reported at fair value through net income on each balance sheet date.

December 31, Year 1

Unrealized holding loss	[200 × (\$90 – \$100)]	\$2,000
Investment in equity securities		\$2,000

In Abi's December 31, Year 1, balance sheet, the investment in equity securities of Gail Co. is reported at its fair value of \$18,000 (200 × \$90). In the Year 1 income statement, a loss of \$2,000 is recognized.

December 31, Year 2

Investment in equity securities	[200 × (\$115 – \$90)]	\$5,000
Unrealized holding gain		\$5,000

In Abi's December 31, Year 2, balance sheet, the investment in equity securities of Gail Co. is reported at its fair value of \$23,000 (200 × \$115). In the Year 2 income statement, a gain of \$5,000 is recognized.

3. Measurement Alternative for Investment in Equity Securities without a Readily Determinable Fair Value

a. Measurement Alternative

- 1) An entity may elect a measurement alternative for an investment in equity securities without a readily determinable fair value.
 - a) This alternative is **cost minus impairment** (if any), **plus or minus** changes resulting from observable price changes for the identical or a similar investment of the same issuer.
- 2) If the measurement alternative is selected, it must be applied until the investment has a readily determinable fair value.
 - a) The entity must reassess at each reporting period whether the fair value of an equity investment is readily determinable.
 - b) When the fair value of an equity investment is readily determinable, the investment is measured at fair value through net income.

b. Impairment Test

- 1) A qualitative assessment of whether an investment is impaired must be performed at each reporting date. An investment is impaired if the fair value of the investment is lower than its carrying amount.
 - a) A **qualitative** assessment may consider many impairment indicators, such as significant deterioration in earnings performance, credit rating, or asset quality.
- 2) If the qualitative assessment indicates potential impairment, the entity must estimate the fair value of the investment and perform a **quantitative** impairment test.
 - a) The carrying amount of the investment is compared with its fair value. An impairment loss is recognized in the income statement (net income) for the excess of the carrying amount over the fair value.

$$\text{Impairment loss} = \text{Carrying amount} - \text{Fair value}$$

c. Observable Price Changes

- 1) To identify observable price changes, a reasonable effort should be made to identify relevant transactions by the same issuer that occurred on or before the balance sheet date. Accordingly, an entity does not need to make an exhaustive search for all observable price changes.

d. Similar Investment of the Same Issuer

- 1) Different rights and obligations of the securities should be considered when identifying whether a security issued by the same issuer is similar to the equity investment.

4. Decision Tree: Classification and Measurement of an Investment in Equity Securities

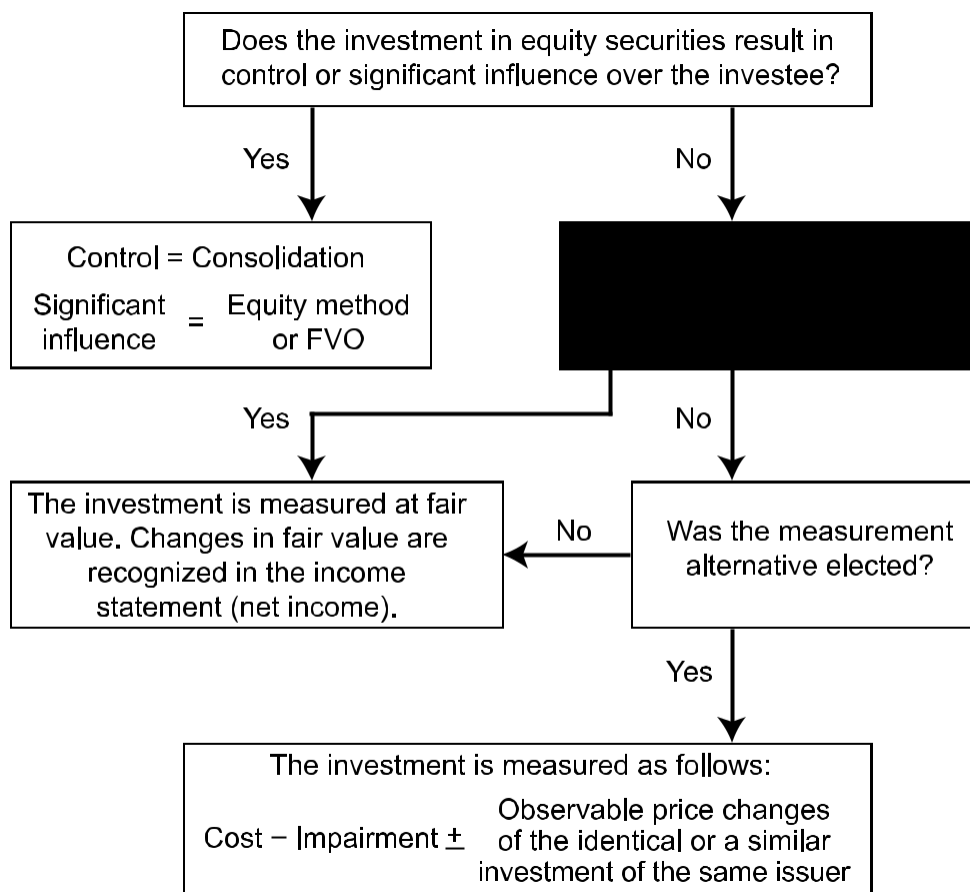


Figure 3-2

3.2 EQUITY METHOD

1. Significant Influence

- a. An investment in voting stock that enables the investor to exercise significant influence over the investee should be accounted for by the **equity method** (assuming no FVO election).
- b. Significant influence is presumed to exist when the investor holds between 20% and 50% of the investee's voting interests (shares of common stock).
 - 1) The amount of preferred stock held by the investor is irrelevant. Preferred stock usually is non-voting.

2. Application of the Equity Method

- a. An equity method investment is initially recognized at cost.
- b. Under the equity method, the investor recognizes in income its **share of the investee's earnings or losses** in the periods for which they are reported by the investee. The journal entries are

<u>Investee reported net income for the period</u>		<u>Investee reported net loss for the period</u>	
Investment in X Co.	\$XXX	Loss -- Share of X Co. losses	\$XXX
Revenue -- Share of X Co. earnings	\$XXX	Investment in X Co.	\$XXX

- 1) An investor recognizes increases in earnings and the **investment account** for its share of the investee's net income for the period.
- 2) An investor recognizes a **loss** and a **decrease in the investment account** for its share of the investee's net loss for the period.
 - a) The investor's share of the investee's earnings or losses is recognized only for the portion of the year that the investment was held under the equity method.
- c. Dividends from the investee are treated as a return of an investment. They have no effect on the investor's income.
 - 1) The investor's share of dividends distributed by the investee increases cash and **decreases the investment**. The journal entry is

Cash or dividend receivable	\$XXX	
Investment in X Co.		\$XXX
- d. If an investor can **no longer be presumed** to exercise significant influence (for example, due to a decrease in the level of ownership), it ceases to account for the investment using the equity method.

3.3 INVESTMENTS IN DEBT SECURITIES

1. Overview

- a. A **debt security** represents a **creditor** relationship with the issuer.
 - 1) In addition to the common forms of debt, this category includes
 - a) Mandatorily redeemable preferred stock (stock that must be redeemed by the issuer),
 - b) Preferred stock redeemable at the investor's option, and
 - c) Collateralized mortgage obligations.
 - 2) Leases, options, financial futures contracts, and forward contracts are not debt securities.
- b. This subunit applies to all investments in debt securities.

- c. Debt securities are classified at acquisition into one of **three categories**. The classification is reassessed at each reporting date.

Category	Criteria
Held-to-maturity	Debt securities that the reporting entity has the positive intent and ability to hold to maturity
Trading	Debt securities intended to be sold in the near term
Available-for-sale	Debt securities not classified as held-to-maturity or trading

2. Held-to-Maturity Securities -- Amortized Cost

- a. An investment in a debt security is classified as held-to-maturity when the holder has both the **positive intent** and the **ability** to hold the security until its maturity date.
- 1) The investor may intend to hold the security for an indefinite period. Also, the possibility may exist that it will be sold before maturity to supply needed cash, avoid interest rate risk, etc. In these cases, the security cannot be classified as held-to-maturity.
 - 2) If a sale before maturity takes place, the security still can be deemed to have been held-to-maturity if
 - a) Sale is near enough to the maturity or call date (e.g., within 3 months) so that interest rate risk (change in the market rate) does not have a significant effect on fair value, or
 - b) Sale is after collection of 85% or more of the principal.
- b. Held-to-maturity securities are reported at **amortized cost**.
- 1) The purchase of held-to-maturity securities is recorded as follows:
- | | | |
|-----------------------------|-------|-------|
| Held-to-maturity securities | \$XXX | |
| Cash | | \$XXX |
- c. **Presentation -- balance sheet.** Held-to-maturity securities are presented **net of any unamortized premium or discount**. No valuation account is used.
- 1) Amortization of any discount (premium) is reported by a debit (credit) to held-to-maturity securities and a credit (debit) to interest income.
 - 2) Individual securities are presented as current or noncurrent.
- d. **Presentation -- income statement.** Realized gains and losses and interest income (including amortization of premium or discount) are included in earnings.
- e. **Presentation -- cash flow statement.** Cash flows are from investing activities.

3. Trading Securities -- Fair Value through Net Income

- a. Trading securities are bought and held primarily for sale in the near term. They are purchased and sold frequently.
- 1) Each trading security is initially recorded at **cost** (including brokerage commissions and taxes).
- | | | |
|--------------------|-------|-------|
| Trading securities | \$XXX | |
| Cash | | \$XXX |
- 2) At each balance sheet date, trading securities are **remeasured at fair value**.

- b. **Unrealized holding gains and losses** on trading securities are reported in the **income statement** (net income). A holding gain or loss is the net change in fair value during the period, not including recognized dividends or interest not received.
- 1) To retain historical cost in the accounts while reporting changes in the carrying amount from changes in fair value, a valuation allowance may be established.
 - a) For example, the entry below debits an allowance for an increase in the fair value of trading securities.
- | | | |
|--|-------|-------|
| Securities fair value adjustment (trading) | \$XXX | |
| Unrealized holding gain | | \$XXX |
- c. **Presentation -- balance sheet.** The balances of the securities and valuation allowances are netted. One amount is displayed for fair value.
- 1) Trading securities are current assets.
- d. **Presentation -- income statement.** Unrealized and realized holding gains and losses, dividends, and interest income (including premium or discount amortization) are included in earnings (income statement).
- e. **Presentation -- cash flow statement.** Classification of cash flows depends on the nature of the securities and the purpose of their acquisition. They are typically considered to be from operating activities.
- f. The accounting for (1) trading securities and (2) financial assets under the fair value option is similar.
- 1) Under both methods, the investment is measured in the balance sheet at fair value.
 - 2) Unrealized holding gains or losses on the remeasurement to fair value then are recognized in earnings (income statement).

EXAMPLE 3-2 Trading Securities

On October 1, Year 1, Maverick Co. purchased 1,000 shares of Larson Co. mandatorily redeemable preferred stock (i.e., debt securities) at fair value. Maverick classified this investment as trading securities. On March 1, Year 2, Maverick sold all of its investment at fair value. The following are the fair values per share of Larson mandatorily redeemable preferred stock at the relevant dates:

Date	Fair Value
October 1, Year 1	\$15
December 31, Year 1	14
March 1, Year 2	21

October 1, Year 1

Trading securities (1,000 × \$15)	\$15,000	
Cash		\$15,000

December 31, Year 1

At each balance sheet date, trading securities are remeasured at fair value. Unrealized holding gains and losses are reported in earnings.

Unrealized holding loss [1,000 × (\$15 – \$14)]	\$1,000	
Securities fair value adjustment (trading)		\$1,000

In Maverick's December 31, Year 1, balance sheet, the investment in Larson is reported in the current assets section as trading securities. It is measured at year-end fair value of \$14,000 (1,000 × \$14).

March 1, Year 2

Cash (1,000 × \$21)	\$21,000	
Trading securities		\$14,000
Gain on disposal of trading securities		7,000

4. Available-for-Sale Securities -- Fair Value through OCI

- a. Securities that are not classified as held-to-maturity or trading are considered available-for-sale. The initial acquisition is recorded at **cost** by a debit to available-for-sale securities and a credit to cash.

Available-for-sale securities	\$XXX	
Cash		\$XXX

- b. At each balance sheet date, available-for-sale securities are **remeasured at fair value**.
- c. **Unrealized holding gains and losses** resulting from the remeasurement to fair value are reported in **other comprehensive income (OCI)**.

Unrealized holding loss -- OCI	\$XXX	
Securities fair value adjustment (available-for-sale)		\$XXX

- 1) Tax effects are debited or credited directly to OCI.
 - 2) Amortization of any discount (premium) is reported by a debit (credit) to available-for-sale securities or an allowance and a credit (debit) to interest income.
 - 3) Receipt of cash dividends is recorded by a debit to cash and a credit to dividend income.
 - 4) All or part of unrealized gains and losses for an available-for-sale security designated and qualifying as the hedged item in a fair value hedge are recognized in earnings.
- d. **Presentation -- balance sheet.** The balances of the securities and valuation allowances are netted. One amount is displayed for fair value.
- 1) Individual securities are presented as current or noncurrent.
 - 2) In the equity section, unrealized holding gains and losses are reported in accumulated OCI (the real account to which OCI is closed).
- e. **Presentation -- income statement.** Realized gains and losses, dividends, and interest income (including premium or discount amortization) are included in earnings.
- f. **Presentation -- statement of comprehensive income.** Unrealized holding gains and losses for the period are included in comprehensive income.
- 1) Reclassification adjustments also must be made for each component of OCI. Their purpose is to avoid double counting when an item included in net income also was included in OCI for the same or a prior period.
 - a) For example, if a gain on available-for-sale securities is realized in the current period, the prior-period recognition of an unrealized holding gain must be eliminated by debiting OCI and crediting a gain.
- g. **Presentation -- cash flow statement.** Cash flows are from investing activities.

EXAMPLE 3-3 Available-for-Sale Securities

On April 1, Year 1, Maverick Co. purchased 1,000 shares of White Co. mandatorily redeemable preferred stock at fair value. Maverick classified this investment as available-for-sale securities. On May 1, Year 3, Maverick sold all of its investment at fair value. The following are the fair values per share of White mandatorily redeemable preferred stock at the relevant dates:

Date	Fair Value
April 1, Year 1	\$25
December 31, Year 1	32
December 31, Year 2	27
May 1, Year 3	31

The changes in the fair value of White preferred shares are temporary.

April 1, Year 1, Journal Entry

Available-for-sale securities (1,000 × \$25)	\$25,000	
Cash		\$25,000

December 31, Year 1, Journal Entry

At each balance sheet date, available-for-sale securities are remeasured at fair value. Unrealized holding gains and losses are included in OCI.

Available-for-sale securities fair value adjustment [1,000 × (\$32 – \$25)]	\$7,000	
Unrealized holding gain (OCI item)		\$7,000

Presentation in Maverick's December 31, Year 1, financial statements:

Balance sheet: Assets section -- Available-for-sale securities (1,000 × \$32)	\$32,000
Equity section -- Accumulated OCI	7,000
Statement of comprehensive income -- Unrealized holding gain (OCI)	7,000

December 31, Year 2, Journal Entry

Unrealized holding loss [1,000 × (\$27 – \$32)]	\$5,000	
Available-for-sale securities fair value adjustment		\$5,000

Presentation in Maverick's December 31, Year 2, financial statements:

Balance sheet: Assets section -- Available-for-sale securities (1,000 × \$27)	\$27,000
Equity section -- Accumulated OCI (\$7,000 – \$5,000)	2,000
Statement of comprehensive income -- Unrealized holding loss (OCI)	5,000

May 1, Year 3, Journal Entry

Cash (1,000 × \$31)	\$31,000	
Accumulated OCI	2,000	
Available-for-sale securities		\$27,000
Realized gain on disposal of available-for-sale securities		6,000

Presentation in Maverick's December 31, Year 3, financial statements:

Income statement: Realized gain on disposal on AFS securities	\$6,000
Statement of OCI: Reclassification of holding gains from accumulated OCI	(2,000)
Total comprehensive income for Year 3 (\$31,000 – \$27,000)	<u>\$4,000</u>

5. Impairment

- a. Unrealized changes in fair value are recognized in earnings (income statement) if they represent **permanent declines**. The **amortized cost basis** is used to calculate any impairment.
 - 1) The amortized cost basis differs from fair value, which equals the cost basis plus or minus the net unrealized holding gain or loss.
- b. If a decline in fair value of an individual **held-to-maturity** or **available-for-sale** security below its amortized cost basis is permanent, the amortized cost basis is **written down to fair value** as a new cost basis.
- c. The impairment is a **realized loss** included in **earnings** (income statement).
 - 1) The new cost basis is not affected by recoveries in fair value.
 - 2) Subsequent changes in fair value of available-for-sale securities, except for other-than-temporary declines, are included in OCI.

6. Transfers between Categories

- a. Transfers between categories are accounted for at **transfer-date fair value**. The following describes the treatment of **unrealized holding gains and losses** at that date:
 - 1) **From trading to any category**. Amounts already recognized in earnings are not reversed.
 - 2) **To trading from any category**. Amounts not already recognized in earnings are recognized in earnings.
 - 3) **To available-for-sale from held-to-maturity**. Amounts are recognized in OCI.
 - 4) **To held-to-maturity from available-for-sale**. Amounts recognized in OCI are not reversed but are amortized in the same way as a premium or discount.
 - a) Fair value accounting may result in recognition of a premium or discount when a debt security is transferred to the held-to-maturity category.
 - 5) Transfers **from held-to-maturity** or **into or from trading** should be rare.

b. Summary of Transfers

From	To	Earnings Recognition
Trading	Any category	Already recognized, not reversed
Any category	Trading	If not already recognized
Held-to-maturity	Available-for-sale	Unrealized gain (loss) recognized in OCI
Available-for-sale	Held-to-maturity	Amounts in OCI not reversed but are amortized in same way as premium (discount)

3.4 PROPERTY, PLANT, AND EQUIPMENT

1. Overview

- a. Property, plant, and equipment (PPE), also called fixed assets, are tangible property expected to benefit the entity for **more than 1 year**. They are held for the production or supply of goods or services, rental to others, or administrative purposes.

2. PPE -- Initial Measurement

- a. PPE are initially measured at historical cost, which consists of all the costs necessarily incurred to bring the asset to the condition and location necessary for its intended use. The historical (initial) cost includes
 - 1) The net purchase price (minus trade discounts and rebates, plus purchase taxes and import duties).
 - 2) The directly attributable costs of bringing the asset to the location and condition needed for its intended operation, such as architects' and engineers' fees, site preparation, delivery and handling, installation, assembly, and testing.
 - 3) The interest (borrowing costs) attributable to the acquisition, construction, or production of PPE.
- b. In the case of a constructed building, historical cost does not include site preparation costs (e.g., the costs of clearing, draining, filling, and leveling the land). Instead, such costs are costs of the attached land, not of the building to be constructed on the land.



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DIFFERENCE

Historical cost includes the cost of borrowing (i.e., interest) that is directly attributable to the acquisition, construction, or production of PPE **less** any investment income earned on the temporary investment of such borrowing.

3. PPE -- Measurement Subsequent to Initial Recognition

- a. The carrying amount of an item of PPE is the amount at which it is presented in the balance sheet. This amount is equal to the historical cost minus accumulated depreciation and impairment losses.

$$\begin{array}{r}
 \text{Historical or initial cost} \\
 - \text{Accumulated depreciation} \\
 - \text{Impairment losses} \\
 \hline
 = \text{Asset's carrying amount} \\
 \hline
 \hline
 \end{array}$$

- b. The accounting issue related to expenditures for PPE subsequent to initial recognition is to determine whether they should be
 - 1) Capitalized at cost and depreciated in future periods (a capital expenditure) or
 - 2) Recognized as an expense as incurred (a revenue expenditure).

- c. **Capital expenditures** provide additional benefits by improving the quality of services rendered by the asset, extending its useful life, or increasing its output. These expenditures are capitalized at cost.
- 1) An improvement expenditure must be capitalized if it (a) results in a betterment to the unit of property, (b) adapts the unit of property to a new or different use, or (c) results in a restoration of the unit of property.
 - a) An expenditure is a **betterment** if it ameliorates a condition or defect that existed before acquisition of the property or arose during the production of the property; is for a material addition to the property; or increases the property's productivity, efficiency, strength, etc.
 - b) An expenditure is an **adaptation** to a new or different use if it adapts the unit of property to a use inconsistent with the taxpayer's intended ordinary use at the time the taxpayer originally placed the property into service.
 - c) An expenditure is a **restoration** if it
 - i) Restores a basis that has been taken into account,
 - ii) Returns the unit of property to working order from a state of nonfunctional disrepair,
 - iii) Results in a rebuilding of the unit of property to a like-new condition after the end of the property's alternative depreciation system class life, or
 - iv) Replaces a major component or substantial structural part of the unit of property.
- d. **Revenue expenditures** (expenses) maintain an asset's normal service capacity. These costs are recurring, not expected to benefit future periods, and expensed as incurred.
- 1) Routine, minor expenditures made to maintain the operating efficiency of PPE are ordinarily expensed as incurred.

4. PPE -- Depreciation

- a. **Depreciation** is the process of systematically and rationally allocating the depreciable base of a tangible capital asset over its expected useful life. The periodic depreciation expense is recognized in the income statement. Accumulated depreciation is a contra-asset account.
- 1) The debit is to depreciation expense, and the credit is to accumulated depreciation. The journal entry is

Depreciation expense	\$XXX	
Accumulated depreciation		\$XXX

- b. The asset's **depreciable base** (the amount to be allocated) is calculated as follows:

Depreciable base = Historical cost – Salvage value – Recognized impairment loss

- c. **Estimated useful life** is an estimated period over which services or economic benefits are expected to be obtained from the use of the asset.
- d. **Salvage value** (residual value) is the amount that the entity expects to obtain from disposal of the asset at the end of the asset's useful life.
- e. **Land** has an indefinite useful life and therefore must not be depreciated. Thus, the depreciable base of property that consists of land or a building is the depreciable base of the building only.

EXAMPLE 3-4 Depreciable Base

Jayhawk Co. recently acquired a robot to be used in its fully automated factory for a purchase price of \$850,000. Jayhawk spent another \$150,000 installing and testing the robot. The company estimates that the robot will have a 5-year useful life and can be sold at the end of that time for \$100,000.

The depreciable base for this asset is calculated as follows:

Purchase price	\$ 850,000
Installation and testing	<u>150,000</u>
Historical cost	\$1,000,000
Estimated salvage value	<u>(100,000)</u>
Depreciable base	<u>\$ 900,000</u>

- f. The **depreciation method** chosen should reflect the pattern in which economic benefits (or services) from the assets are expected to be received. The chosen method allocates the cost of the asset as equitably as possible to the periods during which services (or economic benefits) are obtained from the use of the asset.



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DIFFERENCE

Each part of an item with a cost significant to the total cost must be depreciated separately. But an entity may separately depreciate parts that are not significant.

5. Depreciation Methods

- a. **Straight-line (S-L)** depreciation is the simplest method because an equal amount of depreciation is charged to each period of the asset's useful life.

- 1) The easiest way to calculate straight-line depreciation is to divide the depreciable base by the estimated useful life.

$$\text{Periodic depreciation expense} = \text{Depreciable base} \div \text{Estimated useful life}$$

EXAMPLE 3-5 Depreciation Expense -- Straight-Line

If Jayhawk applies the straight-line method, depreciation expense over the life of the asset will be calculated as follows:

	Depreciable Base	Divided: Estimated Useful Life	Equals: Depreciation Expense	Accumulated Depreciation	Carrying Amount, End of Year
Year 1:	\$900,000	5	\$180,000	\$180,000	\$820,000
Year 2:	900,000	5	180,000	360,000	640,000
Year 3:	900,000	5	180,000	540,000	460,000
Year 4:	900,000	5	180,000	720,000	280,000
Year 5:	900,000	5	<u>180,000</u>	900,000	100,000
Total			<u>\$900,000</u>		

The straight-line percentage for Jayhawk's new robot is 20% (100% ÷ 5-year estimated useful life).

- 2) **Depreciation for a fractional period.** Because an asset is most likely to be acquired or disposed of in the middle of a period, the calculation often involves a fraction of a period. Time-based methods most often compute depreciation to the nearest month of a partial year. However, other conventions also are permitted.

EXAMPLE 3-6 Depreciation for a Fractional Period

Assume that Jayhawk purchased the robot on October 1, Year 1. Depreciation expense recognized in Year 1 is \$45,000 [$\$180,000$ depreciation expense for a full year $\times (3 \div 12)$]. Annual depreciation expense recognized in Years 2 through 5 is \$180,000. Depreciation expense recognized in Year 6 is \$135,000 [$\$180,000 \times (9 \div 12)$].

- b. Usage-centered activity methods calculate depreciation as a function of an asset's use rather than the time it has been held.
- 1) The **units-of-output method** allocates cost based on production. As production varies, so will the depreciation expense.

EXAMPLE 3-7 Units-of-Output Method

On the date of purchase, Jayhawk anticipated that the robot would produce 8,000 units of product over its 5-year life. In actuality, the robot produced the following:

Year 1	Year 2	Year 3	Year 4	Year 5	Total
2,300 units	2,000 units	1,800 units	1,200 units	700 units	8,000 units

Depreciation expense over the life of the asset will be calculated as follows:

	Depreciable Base	Times: Units-of- Production Fraction	Equals: Depreciation Expense	Accumulated Depreciation	Carrying Amount, Year End
Year 1:	\$900,000	(2,300 $\div$ 8,000)	\$258,750	\$258,750	\$741,250
Year 2:	900,000	(2,000 $\div$ 8,000)	225,000	483,750	516,250
Year 3:	900,000	(1,800 $\div$ 8,000)	202,500	686,250	313,750
Year 4:	900,000	(1,200 $\div$ 8,000)	135,000	821,250	178,750
Year 5:	900,000	(700 $\div$ 8,000)	78,750	900,000	100,000
Total			<u>\$900,000</u>		

- c. **Accelerated depreciation methods** are time-based. They result in decreasing depreciation charges over the life of the asset. The two major time-based methods are declining balance and sum-of-the-years'-digits.
- 1) **Declining balance** determines depreciation expense by multiplying the carrying amount (not the depreciable base equal to cost minus salvage value) at the beginning of each period by some percentage (e.g., 200% or 150%) of the straight-line rate of depreciation.

Periodic depreciation expense = Carrying amount $\times$ Declining-balance percentage

- a) The carrying amount decreases by the depreciation recognized. The result is the use of a constant rate against a declining balance.
- b) Salvage value is ignored in determining the carrying amount, but the asset is not depreciated below salvage value.

EXAMPLE 3-8 Declining Balance Method

If Jayhawk applies double-declining-balance (DDB) depreciation to the robot, the declining-balance percentage will be 40% (20% straight-line rate $\times$ 2). Depreciation expense over the life of the asset will be calculated as follows:

	Carrying Amount, <u>First of Year</u>	Times: <u>DDB Rate</u>	Equals: Depreciation <u>Expense</u>	Accumulated <u>Depreciation</u>	Carrying Amount, <u>End of Year</u>
Year 1:	\$1,000,000	40%	\$400,000	\$400,000	\$600,000
Year 2:	600,000	40%	240,000	640,000	360,000
Year 3:	360,000	40%	144,000	784,000	216,000
Year 4:	216,000	40%	86,400	870,400	129,600
Year 5:	129,600	40%	29,600*	900,000	100,000
			<u>\$900,000</u>		

*Year 5 depreciation expense is \$29,600 because the carrying amount cannot be less than salvage value.

- 2) The **sum-of-the-years'-digits (SYD)** depreciation method multiplies not the carrying amount but the constant depreciable base (cost minus salvage value) by a declining fraction. It is a declining rate, declining-charge method.

$$= \text{Depreciable base} \times \frac{\text{Remaining life}}{\text{Sum of years' digits}}$$

EXAMPLE 3-9 SYD Method

If Jayhawk applies sum-of-the-years'-digits depreciation, the denominator of the SYD fraction is 15 (1 + 2 + 3 + 4 + 5). Depreciation expense over the life of the asset will be calculated as follows:

	Depreciable Base	SYD Fraction	Depreciation Expense	Accumulated Depreciation	Carrying Amount, Year End
Year 1:	\$900,000	(5 ÷ 15)	\$300,000	\$300,000	\$700,000
Year 2:	900,000	(4 ÷ 15)	240,000	540,000	460,000
Year 3:	900,000	(3 ÷ 15)	180,000	720,000	280,000
Year 4:	900,000	(2 ÷ 15)	120,000	840,000	160,000
Year 5:	900,000	(1 ÷ 15)	60,000	900,000	100,000
			<u>\$900,000</u>		

- d. **Group and composite depreciation** methods apply **straight-line** accounting to a collection of assets depreciated as if they were a single asset.

- 1) The composite method applies to groups of **dissimilar assets** with varying useful lives, and the group method applies to **similar assets**. They provide an efficient way to account for large numbers of depreciable assets. They also result in the offsetting of under- and overstated depreciation estimates.



IFRS
DIFFERENCE

An entity may choose either the **cost model** (as under U.S. GAAP) or the **revaluation model** as its accounting policy. It must apply that policy to an entire class of PPE. A class is a grouping of assets of similar nature and use in an entity's operations, for example, land, office equipment, or motor vehicles.

An item of PPE whose fair value can be reliably measured may be carried at a **revalued amount** equal to fair value at the revaluation date (minus subsequent accumulated depreciation and impairment losses).

Revaluation is needed whenever fair value and the asset's carrying amount differ materially. Accumulated depreciation is restated proportionately or eliminated.

A revaluation increase must be recognized in other comprehensive income and accumulated in equity as **revaluation surplus**. But the increase must be recognized in profit or loss to the extent it reverses a decrease of the same asset that was recognized in profit or loss.

A revaluation decrease must be recognized in profit or loss. But the decrease must be recognized in other comprehensive income to the extent of any credit in revaluation surplus for the same asset.



IFRS
DIFFERENCE

Under **IAS 40, Investment Property**, **investment property** is property (land, building, part of a building, or both) held by the owner or by the lessee under a finance lease to earn rental income or for capital appreciation or both. Investment property may be accounted for according to (1) the **cost model** and carried at historical cost minus accumulated depreciation and impairment losses (as under U.S. GAAP) or (2) the **fair value model**. If the fair value model is chosen as the accounting policy, all of the entity's investment property must be measured at **fair value** at the end of the reporting period. A **gain or loss** from a change in the fair value of investment property must be recognized in profit or loss for the period in which it arises. Investment property that is accounted for according to the fair value model is not depreciated.

3.5 INTANGIBLE ASSETS

1. Initial Recognition

- a. An intangible asset is an identifiable, nonmonetary asset that lacks physical substance. Examples of intangible assets include licenses, patents, copyrights, franchises, and trademarks.
- b. **Externally acquired intangible assets** (other than goodwill) are initially recorded at acquisition cost plus any additional costs, such as legal fees.
- c. **Internally developed** intangible assets (other than goodwill) are recorded initially at the amount of the additional costs other than those for research and development (e.g., legal fees).
- d. **Research and development (R&D)** costs must be **expensed as incurred** (not capitalized) except under certain limited scenarios.

EXAMPLE 3-10 Internally Developed Patent

A company invested \$200,000 and \$300,000 in the research phase and the development phase, respectively, of an internally developed patent. In addition, the company paid \$10,000 and \$15,000 for patent registration fees and legal fees, respectively.

The patent will be recorded at the amount of the incidental costs of \$25,000 (\$10,000 patent registration fees + \$15,000 legal fees). The amounts paid for research and development must be expensed as incurred and are never capitalized to the cost of the asset.



IFRS
DIFFERENCE

Intangible assets may be accounted for under either the **cost model** (as under U.S. GAAP) or the **revaluation model** (described on the previous page). The revaluation model can be applied only if the intangible asset is traded in an active market.



IFRS
DIFFERENCE

Research costs must be expensed as incurred.

Development costs may result in **recognition of an intangible asset** if the entity can demonstrate the (1) technical feasibility of completion of the asset, (2) intent to complete, (3) ability to use or sell the asset, (4) way in which it will generate probable future economic benefits, (5) availability of resources to complete and use or sell the asset, and (6) ability to measure reliably expenditures attributable to the asset.

Using the data from Example 3-10, assume that all the criteria mentioned above were met. Under IFRS, the patent will be recorded at the amount of \$325,000 (\$300,000 development costs + \$25,000 incidental costs). The research costs of \$200,000 are expensed as incurred and are never capitalized to the cost of the asset.

2. Intangible Assets with Finite Useful Lives

- a. An intangible asset with a **finite useful life** (an amortized intangible asset) to the reporting entity is **amortized** over that useful life.
- 1) The debit is to amortization expense, and the credit is to intangible asset. The journal entry is

Amortization expense	\$XXX	
Intangible asset		\$XXX

- b. The **amortizable amount** equals the amount of cost initially assigned minus the residual value.

$$\text{Amortizable amount} = \text{Historical (initial) cost} - \text{Residual value}$$

- c. The amortization methods are similar to the depreciation methods described in item 5. in Subunit 3.4.
- d. The carrying amount of an intangible asset with a finite useful life equals its historical cost minus accumulated amortization and any impairment losses.
- e. The **impairment test** for an intangible asset with a finite useful life (an amortized intangible asset) is the **two-step impairment test** for long-lived assets described in Study Unit 5, Subunit 3.



IFRS
DIFFERENCE

An impairment loss for an asset (except goodwill) may be reversed if a change in the estimates used to measure the recoverable amount has occurred. The test for impairment of assets other than goodwill has one step: determine whether an asset's carrying amount is greater than its recoverable amount (greater of fair value minus costs to sell or value in use). The impairment test for long-lived assets is described in Study Unit 5, Subunit 3.

3. Intangible Assets with Indefinite Useful Lives

- a. An intangible asset with an **indefinite useful life** is **not amortized**. The carrying amount of an intangible asset with an indefinite useful life equals its historical cost minus any impairment losses.
 - 1) Goodwill is an intangible asset with an indefinite useful life. However, the accounting treatment of goodwill differs from that for other intangible assets. Thus, it is described separately in item 4. on the next page.
- b. **Impairment test.** An intangible asset with an indefinite useful life (a **nonamortized** intangible asset) must be reviewed for impairment at least annually. It is tested more often if events or changes in circumstances suggest that the asset may be impaired.
 - 1) An entity may elect to perform a **qualitative assessment** to determine whether a **quantitative impairment test** is needed. The entity also may directly perform the quantitative test.
 - a) **Qualitative assessment.** After the assessment of qualitative factors, the entity may determine that it is more likely than not (probability > 50%) that an indefinite-lived intangible asset is not impaired.
 - i) In this case, the quantitative impairment test is not required.
 - ii) If potential impairment is found, the quantitative impairment test must be performed.
 - b) **Quantitative impairment test.** The carrying amount of an asset is compared with its fair value. If the carrying amount exceeds the fair value, the asset is impaired, and the excess is the recognized impairment loss.

Determination of an Impairment Loss

1. Review for impairment

2. Impairment loss = Carrying amount – Fair value

- 2) This impairment loss is **nonreversible**, so the adjusted carrying amount is the new accounting basis.



IFRS
DIFFERENCE

A one-step quantitative impairment test is performed. No qualitative assessment exists.

4. Goodwill

- a. Goodwill is recognized only in a business combination. It is an intangible asset reflecting the future economic benefits arising from those assets acquired in the combination that are not individually identified and separately recognized.
- 1) Goodwill has an indefinite useful life. Thus, it must not be amortized subsequent to its initial recognition and is instead periodically tested for impairment.
 - 2) The parent presents any goodwill recognized in its consolidated balance sheet as one amount under noncurrent assets.
 - 3) Internally generated goodwill must not be recognized in the financial statements.
 - 4) Goodwill recognized equals the excess of a) over b) below:
 - a) The sum of the acquisition-date **fair value** of the consideration transferred, any NCI recognized, and any previously held equity interest in the acquiree
 - b) The acquisition-date fair value of identifiable assets acquired and liabilities assumed (fair value of net assets acquired)

EXAMPLE 3-11 Calculation of Goodwill

Entity C acquired 80% of the outstanding common stock of Entity D for \$192,000. Entity D's acquisition-date fair values of identifiable assets and liabilities were \$350,000 and \$140,000, respectively. The acquisition-date fair value of NCI was \$48,000. The goodwill is calculated as follows:

Consideration transferred		\$192,000
Noncontrolling interest		48,000
Acquisition-date fair value of identifiable net assets (Assets – Liabilities) acquired:		
Assets	\$350,000	
Liabilities	<u>(140,000)</u>	<u>(210,000)</u>
Goodwill		<u>\$ 30,000</u>

- 5) If b) exceeds a) above, a **gain from a bargain purchase** must be recognized in the parent's consolidated statement of income.

EXAMPLE 3-12 Calculation of Goodwill

Using the information provided in Example 3-11, assume that the acquisition-date fair value of the identifiable assets acquired was \$400,000 (instead of \$350,000). The gain from the bargain purchase is calculated as follows:

Acquisition-date fair value of identifiable net assets (Assets – Liabilities) acquired:		
Assets	\$400,000	
Liabilities	<u>(140,000)</u>	\$260,000
Consideration transferred		(192,000)
Noncontrolling interest		<u>(48,000)</u>
Gain from bargain purchase		<u>\$ 20,000</u>

- b. **Impairment test.** Goodwill is tested for impairment at the reporting-unit level. All goodwill is assigned to the reporting units that will benefit from the business combination. It is tested for impairment each year at the same time.
- c. As in the case of an intangible asset with an indefinite useful life, an entity may elect to perform a **qualitative assessment** to determine whether the quantitative impairment test is needed.
- d. If the qualitative assessment indicates that **potential impairment exists**, the quantitative impairment test below is performed to determine whether goodwill is impaired.
 - 1) The **fair value of the reporting unit is compared with its carrying amount, including goodwill.**
 - 2) A goodwill impairment loss is recognized for the amount that the carrying amount of a reporting unit, including goodwill, exceeds its fair value. The loss is limited to the total amount of goodwill allocated to that reporting unit. The journal entry is as follows:

Impairment loss	\$XXX	
Goodwill		\$XXX
 - 3) The goodwill impairment loss is recognized in the income statement and cannot be reversed in subsequent periods.

5. Patents

- a. The **amortization period** for a patent is the **shorter** of
 - 1) Its **useful life** or
 - 2) The **legal life** remaining after acquisition or the moment the application was filed.
- a) The useful life may be substantially shorter than the legal life because of changes in consumer tastes, delays in marketing the product or service, and development of substitutes or improvements.
- b. The accounting treatment of the costs of the **legal defense of a patent** depends upon the outcome of the litigation.
 - 1) The costs of **successful litigation** are **capitalized** to the cost of the patent because they will benefit future periods. They are amortized over the shorter of
 - a) The remaining legal life or
 - b) The estimated useful life of the patent.
 - 2) The costs of **unsuccessful litigation** (damages, attorneys' fees, etc.) are **expensed** as incurred.
 - a) An unsuccessful suit also may indicate that the unamortized cost of the patent has no value, or lower value, and impairment loss might be recognized.

STUDY UNIT FOUR

MEASUREMENT, VALUATION, AND DISCLOSURE: LIABILITIES

4.1	<i>Different Types of Expenses and Liabilities</i>	1
4.2	<i>Leases</i>	6
4.3	<i>Income Taxes</i>	14
4.4	<i>Accounting for Bonds and Noncurrent Notes Payable</i>	20

This study unit is the **fourth of six** on **external financial reporting decisions**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The six study units are

Study Unit 1: External Financial Statements

Study Unit 2: Measurement, Valuation, and Disclosure: Assets -- Short-Term Items

Study Unit 3: Measurement, Valuation, and Disclosure: Assets -- Long-Term Items

Study Unit 4: Measurement, Valuation, and Disclosure: Liabilities

Study Unit 5: Revenue and Impairment Recognition

Study Unit 6: Integrated Reporting

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

4.1 DIFFERENT TYPES OF EXPENSES AND LIABILITIES

1. Purchase Commitment

- a. A commitment to acquire goods in the future is not recorded at the time of the agreement. The goods are recognized as inventory when they are received.
- b. But a **loss** is recognized on a firm's noncancelable purchase commitment if the current market price of the goods is less than the commitment price.
 - 1) An example of such an agreement is a **take-or-pay contract** that requires one party to purchase a certain number of goods from the other party or else pay a penalty.

- c. Material losses expected on purchase commitments are measured in the same way as inventory losses, recognized, and separately disclosed.
- 1) The reason for current loss recognition is the same as that for inventory. A decrease (not an increase) in its future benefits should be recognized when it occurs even if the contract is unperformed on both sides.
 - 2) The entry is

Unrealized holding loss -- earnings	\$XXX
Liability -- purchase commitment	\$XXX
 - 3) The nature and the term of the contract (obligation) must be described in the notes to the financial statements.

EXAMPLE 4-1 Purchase Commitment

During the year, the Lisbon Company signed a noncancelable contract to purchase 2,000 pounds of materials at \$64 per pound during the forthcoming year. On December 31, the market price of the materials is \$52 per pound, and the selling price of the finished product is expected to decline accordingly. The financial statements prepared for the year should report a loss of \$24,000 [$2,000 \times (\$64 - \$52)$] in the income statement.

GAAP require recognition in the income statement of a material loss on a purchase commitment as if the inventory were already owned. Losses on firm purchase commitments are measured in the same way as inventory losses. If the cost is \$128,000 and the market price is \$104,000, a \$24,000 loss should be disclosed.

2. Warranty Liability

- a. A warranty is a **written guarantee** of the integrity of a product or service. The seller also agrees to repair or replace a product, refund all or part of the price, or provide additional service.
 - 1) A warranty is customarily offered for a limited time, such as 2 years.
 - 2) It may or may not be separable.
- b. A warranty that provides a customer assurance that a product will function as expected in accordance with agreed-upon specifications is an **assurance-type warranty**. A standard one-year computer warranty against manufacturing defects is an example.
 - 1) If a customer does **not** have the option to purchase a warranty separately (or if the warranty is required by law), it is an assurance-type warranty.
- c. A warranty that provides a service in addition to assurance is a **service-type warranty**. An example is a warranty against customer-inflicted damage (such as to a smartphone).
 - 1) A warranty that can be **purchased separately** by the customer is a service-type warranty.
 - 2) Service-type warranties are more likely to have longer coverage periods than assurance-type warranties.
- d. **Accounting.** An **assurance-type** warranty is **not** a separate performance obligation in a contract. No transaction price is allocated to the warranty.
- e. An assurance-type warranty creates a **loss contingency**. **Accrual accounting** should be used if
 - 1) Warranty expense is probable,
 - 2) The amount can be reasonably estimated, and
 - 3) The amount is material.

f. A warranty liability is recognized when revenue is recognized **at the time of sale**.

- 1) Even if the warranty extends beyond the period of sale, the **entire liability** (expense) for expected costs is recognized at the time of sale. It is not prorated over the annual periods covered by the warranty.

Beginning warranty liability	
Warranty expense recognized in the current period	
Minus: Warranty payments in the current period	
<u>Ending warranty liability</u>	

- 2) Actual payments for warranty costs reduce the warranty liability and do not affect warranty expense.

- a) If warranty payments for the period exceed the warranty liability, the excess is recognized as warranty expense.

- 3) The following are **accrual-basis** entries for warranty expense estimated as a percentage of sales for an assurance-type warranty:

- a) The following entry records a sale of a product:

Cash or accounts receivable	\$XXX	
Sales revenue		\$XXX

- b) The following entry records related warranty expense recognized on the day of sale:

Warranty expense	\$XXX	
Estimated warranty liability		\$XXX

- c) The following entry records actual warranty expenditures paid in the current period:

Estimated warranty liability	\$XXX	
Cash		\$XXX

EXAMPLE 4-2 Assurance-Type Warranty

In Year 1, a company began selling a product under a 2-year warranty. The estimated warranty costs are 3% of sales in the year of sale and 5% in the following year. Sales and actual warranty payments for Year 1 and Year 2 are as follows:

	<u>Sales</u>	<u>Warranty Payments</u>
Year 1	\$300,000	\$ 5,000
Year 2	500,000	37,000

In Year 1, warranty expense of \$24,000 [$\$300,000 \times (3\% + 5\%)$] is recognized. The warranty liability of \$19,000 ($\$24,000 - \$5,000$) is reported on the December 31, Year 1, balance sheet.

In Year 2, warranty expense of \$40,000 [$\$500,000 \times (3\% + 5\%)$] is recognized. The warranty liability of \$22,000 is reported on the December 31, Year 2, balance sheet.

Beginning warranty liability (1/1/Year 2)	\$19,000
Warranty expense recognized in Year 2	40,000
Warranty payments in Year 2	<u>(37,000)</u>
Ending warranty liability (12/31/Year 2)	<u>\$22,000</u>

g. **Accounting.** A **service-type warranty** is a **separate performance obligation in a contract**. Thus, a portion of the total transaction price is allocated to the service-type warranty.

- 1) The total transaction price is allocated to the service-type warranty and the related product sold based on their estimated standalone selling prices.
- 2) At contract inception, the consideration received for the service-type warranty is accounted for as an advance payment, and a contract liability is recognized. The following entry is recorded on a sale of a product with a service-type warranty:

Cash	\$XXX (total transaction price)
Revenue	\$XXX (transaction price allocated to the product)
Contract liability	XXX (transaction price allocated to the service-type warranty)

- 3) Revenue from a service-type warranty is recognized **over time** (i.e., over the coverage period). The pattern of revenue recognized from a service-type warranty depends on the way the warranty performance obligation is satisfied.
 - a) If warranty service is provided continuously over the warranty period, revenue is recognized on the **straight-line basis** over the coverage period.
 - i) If warranty service costs are **not** incurred on a straight-line basis, revenue recognition over the contract's term is proportionate to the estimated service costs.
 - b) The following entry records revenue from a service-type warranty:

Contract liability	\$XXX
Revenue	\$XXX

- c) The following entry records costs when an entity pays claims under the warranty:

Warranty expense	\$XXX
Cash	\$XXX

h. **Accounting.** If an assurance-type warranty and a service-type warranty **cannot** be separated, they are accounted for as a single performance obligation (as a service-type warranty).

3. Contingencies -- Recognition and Reporting

- a. A contingency is “an existing condition, situation, or set of circumstances **involving uncertainty** as to possible gain (a gain contingency) or loss (a loss contingency) to an enterprise that will ultimately be resolved when one or more future events occur or fail to occur.”
- b. A contingency may be
 - 1) Probable. Future events are likely to occur.
 - 2) Reasonably possible. The chance of occurrence is more than remote but less than probable.
 - 3) Remote. The chance of occurrence is slight.
- c. A **loss contingency** must be accrued (recognition of a liability and related loss contingency) when the following two conditions are met:
 - 1) It is **probable** that, at a balance sheet date, an asset has been impaired or a liability has been incurred.
 - 2) The amount of the loss can be **reasonably estimated**.
 - a) The amount with the better estimate within a **range of loss** must be accrued.
 - b) If no amount within that range appears to be a better estimate than any other, the **minimum** should be accrued.
 - c) Disclosure of the nature of the accrual and, in some cases, the amount or the range of loss may be required to prevent the financial statements from being misleading.
- d. If at least one condition is not met but the probability of loss is at least **reasonably possible**, the nature of the contingency must be **disclosed** in the notes to the financial statements.
- e. Loss contingencies with a **remote** probability ordinarily are **not disclosed**.
- f. Gain contingencies are **recognized only when realized**. A gain contingency must be adequately disclosed in the notes.
 - 1) For example, an award of damages in a lawsuit is not realized if it is being appealed.

4.2 LEASES

1. Definition of a Lease

- a. A lease is a long-term contract in which the owner of property (the lessor) allows another party (the lessee) to use the property for a stated period in exchange for a stated payment.
- b. The primary issue is whether the lease agreement transfers substantially all the benefits and risks of ownership of the asset to the lessee.

2. Lease Classification Test

- a. A lease is classified as a **finance lease by the lessee** and as a **sales-type lease by the lessor** if, at lease commencement, **at least one** of the following **five criteria** is met:
 - 1) The lease **transfers ownership** of the leased asset to the lessee by the end of the lease term.
 - 2) The lease includes an **option to purchase** (commonly referred to as a bargain purchase option) the leased asset that the lessee is reasonably certain to exercise.
 - 3) The lease term is for the major part of the remaining **economic life** of the leased asset.
 - a) A lease term of **75%** or more of the remaining economic life of the leased asset generally is considered to be a major part of its remaining economic life.
 - b) This criterion is inapplicable if the beginning of the lease term is at or near the end of the economic life of the leased asset.
 - i) This period generally is considered to be the last 25% of the leased asset's total economic life.
 - 4) The present value of the sum of (a) the **lease payments** and (b) any **residual value guaranteed by the lessee** equals or exceeds substantially all of the **fair value** of the leased asset.
 - a) A present value of **90%** or more of the fair value of the leased asset generally is considered to be substantially all of its fair value.
 - 5) The leased asset is so specialized that it is expected to have **no alternative use** to the lessor at the end of the lease term.
- b. When none of the five classification criteria described above are met, the lease is classified as
 - 1) An **operating lease** by the **lessee**.
 - 2) An **operating lease** or a **direct financing lease** by the **lessor**.
- c. When the lease is not a sales-type lease, the lessor reports a **direct financing lease** only when
 - 1) The present value of the sum of (a) the lease payments and (b) any residual value guaranteed by the lessee **or any other third party** equals or exceeds substantially all of the fair value of the leased asset, and
 - 2) It is probable that the lease payments and any residual value guarantee will be collected.

- d. If the lease is **not** a direct financing lease, it is classified as an **operating lease** by the lessor.
- 1) Classification of a lease as a direct financing lease is rare. It happens only when the lease includes residual value guaranteed by a third party other than the lessee that results in meeting the “substantially all of the fair value” classification criterion (the 90% of the fair value of the leased asset criterion).
- a) Thus, a lessor classifies a lease without residual value guaranteed by a third party (not the lessee) as either
- A sales-type lease or
 - An operating lease.
- b) In most cases, when none of the five classification criteria are met (when the lease is not a sales-type lease), the **lessor classifies the lease as an operating lease**.

Decision Tree: Classification of the Lease by the Lessor

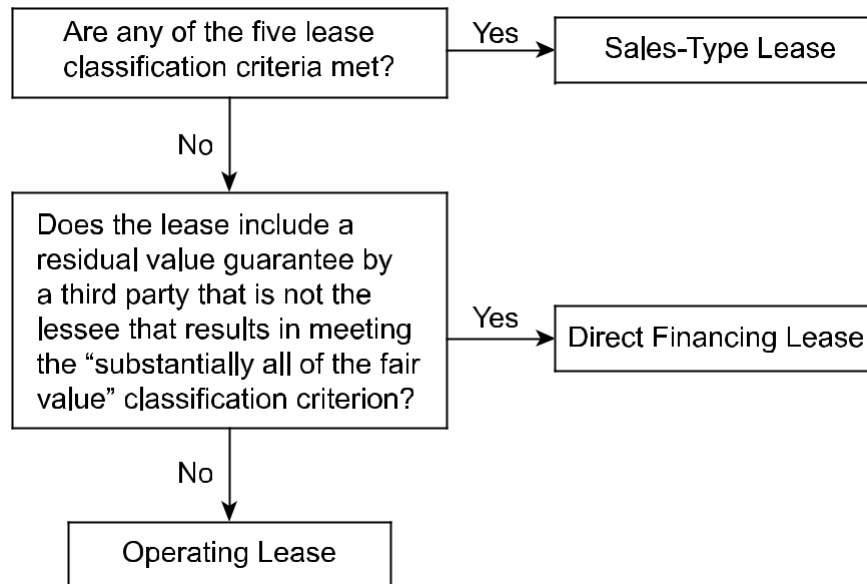


Figure 4-1

- 2) Thus, a lease qualifies as an operating lease when it cannot qualify as any other type of lease.

- e. A **short-term lease** is a lease that, at the commencement date, has a lease term of **12 months or less** and does not include a purchase option that the lessee is reasonably certain to exercise.
- 1) As an accounting policy for **short-term leases**, a **lessee** may elect **not** to recognize the **right-of-use asset** and **lease liability**.
 - 2) Under this short-term lease exception, the lessee recognizes lease payments as rent expense on the straight-line basis over the full lease term.
 - a) The lessee records the following journal entry:

Rent expense	\$XXX	
Cash or rent payable		\$XXX
 - 3) A short-term lease is a form of off-balance-sheet financing. The lessee has the right to use the leased asset, but neither the asset nor a liability for future lease payments is recorded in its financial statements.
 - a) Lessees may prefer to account for a lease as a short-term lease instead of as a finance lease to avoid recognition in the financial statements of
 - i) A liability for future lease payments,
 - ii) Interest expense, and
 - iii) Depreciation of the leased asset.



IFRS
DIFFERENCE

An operating lease is not a classification option for the lessee. The lessee accounts for all leases, except for "short-term" leases and "low-value" leases, similarly to a finance lease under U.S. GAAP.

The lessee may elect to account for a lease of low-value assets (assets with value of \$5,000 or less) the same way as for "short-term" leases. The lessee then recognizes lease payments as an expense on a straight-line basis over the lease term.

3. Lessee Accounting -- Initial Measurement

- a. **General rule.** For **finance and operating leases**, a **lessee** must recognize a **lease liability** and a **right-of-use asset** at the lease commencement date.
 - 1) Finance and operating leases result in the **same accounting** for
 - a) Initial recognition and measurement of the lease liability,
 - b) Initial recognition and measurement of the right-of-use asset, and
 - c) Subsequent measurement of the lease liability.
 - 2) The accounting for subsequent measurement of a right-of-use asset differs under finance and operating leases.
- b. **Lease liability.** At the lease commencement date, a **lease liability** is measured at the **present value of the lease payments** to be made over the lease term.
 - 1) The lease payments are discounted using the discount rate for the lease.
 - a) It is the rate implicit in the lease, if known to the lessee.
 - b) If not, it is the lessee's incremental borrowing rate.

- 2) The lease payments used to calculate the lease liability depend on the specific terms of each lease contract.
- If the lease includes a purchase option that the lessee is reasonably certain to exercise, the lease payments consist of the following:
 - Rental payments
 - Exercise price of the purchase option
 - If no purchase option exists, the lease payments may have the following three components:
 - Rental payments
 - Any penalties for terminating the lease (nonrenewal penalties)
 - Amounts probable of being owed by the lessee under residual value guarantees



AUTHOR'S
NOTE

For the “**substantially all of the fair value**” lease classification criterion, the present value of the **full amount** of the residual value guaranteed by the lessee is included in the test. However, in measuring the lease liability, only the amounts **probable of being owed** by the lessee under residual value guarantees are included.

EXAMPLE 4-3 Residual Value Guaranteed by the Lessee

In a lease contract, a lessee guarantees that the residual value of the leased property at the end of the lease term is at least \$50,000. At the lease commencement date, the lessee estimates that the value of the property at the end of the lease term will be only \$40,000. Thus, it is probable that the lessee will owe \$10,000 to the lessor under the residual value guarantee.

- To test the “substantially all of the fair value” lease classification criterion, the present value of the full amount of the \$50,000 residual value guaranteed by the lessee is included.
- In the measurement of the lease liability, only the present value of the amount that is probable of being owed by the lessee under the residual value guarantee of \$10,000 is included.

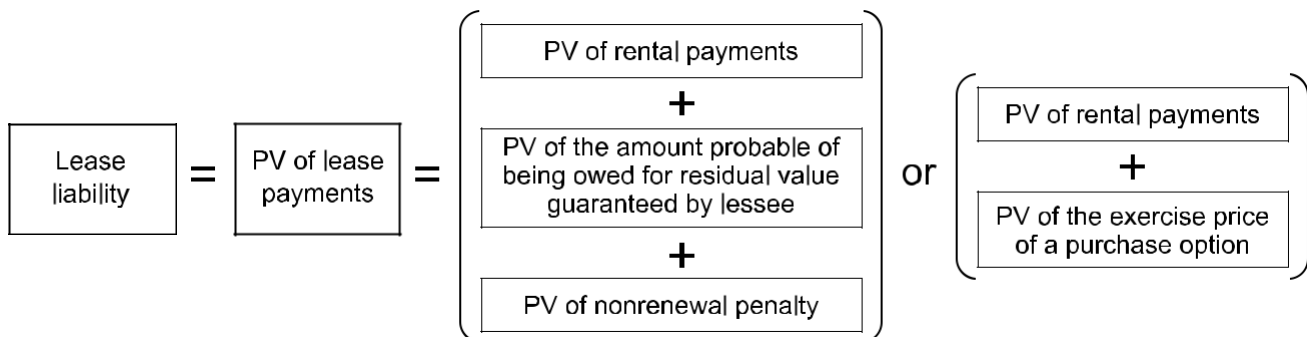


Figure 4-2

- 3) In a balance sheet, the total lease liability is allocated between current and noncurrent portions. The current portion at a balance sheet date is the reduction of the lease liability in the forthcoming year.

- c. **Right-of-use asset.** At the lease commencement date, a right-of-use asset is measured at the amount at which the lease liability was recognized plus initial direct costs incurred by the lessee.

- 1) When no initial direct costs were incurred by the lessee, a **right-of-use asset equals the lease liability** recognized. The following journal entry is recorded by the lessee:

Right-of-use asset	\$XXX	
Lease liability		\$XXX

- 2) Subsequent to initial recognition, the right-of-use asset is reported in the balance sheet at cost minus accumulated amortization and any impairment losses.

EXAMPLE 4-4 Finance Lease

On January 1, Year 1, Cottle, Inc., entered into a 3-year lease of a machine from Crimson, LLC. Cottle must pay Crimson three annual payments of \$100,000 starting on December 31, Year 1. The machine's useful life from the lease commencement date is 5 years. The lease allows Cottle the option to purchase the machine at the end of the lease term for \$15,000. Cottle is reasonably certain to exercise this purchase option. Cottle's incremental borrowing rate is 15%, but the rate implicit in the lease is 10%, which is known to Cottle.

- The present value factor for an ordinary annuity at 10% for 3 periods is 2.48685, and the present value of \$1 at 10% for 3 periods is 0.7513.
- The present value factor for an ordinary annuity at 15% for 3 periods is 2.28323, and the present value of \$1 at 15% for 3 periods is 0.65752.

The lease is a **finance lease** because it meets the lease classification criterion of including a purchase option that the lessee is reasonably certain to exercise. The rate implicit in the lease of 10% is used to calculate the present value of the lease payments because Cottle knows this rate.

PV of rental payments (\$100,000 × 2.48685)	\$248,685
PV of purchase option (\$15,000 × 0.7513)	11,270
PV of lease payments	<u>\$259,955</u>
Right-of-use asset	\$259,955
Lease liability	\$259,955

EXAMPLE 4-5 Operating Lease -- Lessee

Using the scenario in Example 4-4, assume that (1) Cottle concludes that the contract is an **operating lease**, (2) the lease does not include a purchase option, (3) the rental payments are \$100,000 at the end of Years 1 and 2 and \$160,000 at the end of Year 3, and (4) the rate implicit in the lease is not known to Cottle.

Because Cottle does not know the rate implicit in the lease, it uses its **incremental borrowing rate** of 15% to calculate the present value of lease payments.

The PV of the rental payments is \$267,774 [(\$100,000 × 2.28323) + (\$60,000 × 0.65752)].

Right-of-use asset	\$267,774
Lease liability	\$267,774

4. Lessee Accounting for Finance Leases -- Subsequent Measurement

a. **Interest expense and lease liability amortization.** Each periodic lease payment made by the lessee has two components: **interest expense** and the **reduction of the lease liability**.

- 1) If the first periodic lease payment is made at the **commencement date** of the lease, its only component is the reduction of the lease liability. No interest expense is recognized for the first payment because no time has elapsed between the lease commencement date and the payment.
- 2) **Interest expense** is calculated using the effective interest method (also known as the effective-rate method or the interest method). It is calculated as the carrying amount of the lease liability at the beginning of the period times the discount rate of the lease.

Interest expense = Lease liability at the beginning of the period × Discount rate

- 3) The **reduction of the lease liability** is the excess of the periodic lease payment over the interest expense recognized during the period.

Reduction of lease liability = Periodic lease payment – Interest expense

EXAMPLE 4-6 Interest Expense and Amortization of Lease Liability

In Example 4-4, the lease was classified as a **finance lease**. Cottle prepares the following amortization schedule and records the following journal entries:

Date	Period Beginning Lease Liability		Effective Interest Rate		Interest Expense		Cash Payment		Reduction of Lease Liability		Period Ending Lease Liability
1/1/Yr 1	\$259,955	×	10%	=	\$25,995	–	\$100,000	=	\$ 74,005		\$185,950
1/1/Yr 2	185,950	×	10%	=	18,595	–	100,000	=	81,405		104,545
1/1/Yr 3	104,545	×	10%	=	10,455	–	115,000	=	104,545		0

The last payment on 12/31/Yr 3 is \$115,000 (\$100,000 annual rental payment + \$15,000 exercise price of the option to purchase the machine).

12/31/Yr 1

Interest expense	\$25,995
Lease liability	74,005
Cash	\$100,000

On December 31, Year 1, the carrying amount of the lease liability is **\$185,950**. In its 12/31/Yr 1 balance sheet, Cottle reports a current lease liability of \$81,405 and a noncurrent lease liability of \$104,545.

12/31/Yr 2

Interest expense	\$18,595
Lease liability	81,405
Cash	\$100,000

On December 31, Year 2, the carrying amount of the lease liability is **\$104,545**. This entire amount is reported as a current liability in the 12/31/Yr 2 balance sheet because Cottle expects to pay the entire amount in Year 3.

12/31/Yr 3

Interest expense	\$ 10,455
Lease liability	104,545
Cash	\$115,000 (\$100,000 annual payment + \$15,000 option price)

b. **Amortization of a right-of-use asset.** A lessee amortizes the right-of-use asset on a **straight-line basis**.

- 1) The right-of-use asset is amortized over the shorter of (a) its **useful life** or (b) the **lease term**.
 - a) However, if, at the end of the lease term, (1) the ownership of the leased asset is transferred to the lessee, or (2) the lessee is reasonably certain to exercise the purchase option, the amortization period is the **useful life of the leased asset**.

Lease Classification Criterion Satisfied	Amortization Period of the ROU Asset
Criterion 1 - Transfer of ownership	Useful life of the leased asset
Criterion 2 - Exercise of purchase option	Useful life of the leased asset
Criterion 3 - Major part of the remaining economic life	Shorter of ROU asset's useful life or lease term
Criterion 4 - Substantially all of the fair value	Shorter of ROU asset's useful life or lease term
Criterion 5 - No alternative use to the lessor	Shorter of ROU asset's useful life or lease term

EXAMPLE 4-7	Amortization of Right-of-Use Asset over Useful Life		
In Example 4-4 on page 10, Cottle is reasonably certain to exercise the option to purchase the machine. The right-of-use asset therefore is amortized over the useful life of the machine of 5 years. Annual amortization expense of \$51,991 ($\$259,955 \div 5$ years) is recognized by Cottle. The journal entry is			
	Amortization expense	\$51,991	
	Right-of-use asset		\$51,991
In its 12/31/Yr 1 balance sheet, Cottle reports the right-of-use asset at \$207,964 ($\$259,955$ initial cost – \$51,991 accumulated amortization).			

5. Lessee Accounting for Operating Leases -- Subsequent Measurement

a. **Finance leases vs. operating leases.** As noted previously, accounting for finance leases and operating leases is **the same** for (1) initial recognition and measurement of the lease liability and the right-of-use asset and (2) subsequent measurement of the lease liability.

- 1) The following are the **differences** in accounting for finance and operating leases:
 - a) Subsequent accounting for (amortization of) the right-of-use asset
 - b) Income statement presentation of interest expense and amortization of the right-of-use asset
 - c) Statement of cash flow classification of cash lease payments

- b. **Recognition of lease expense in operating leases.** A **single (equal) lease expense** is recognized in each period. It is calculated so that the total undiscounted lease payments are allocated over the lease term on a **straight-line basis**.

Single periodic lease expense = Total undiscounted lease payments (\$) ÷ Lease term (years)

- 1) Initial direct costs incurred by the lessee are included in the total undiscounted lease payments. Thus, they are recognized in the single periodic lease expense on a straight-line basis over the lease term.
- 2) The single periodic lease expense has two components: (a) interest expense on the lease liability and (b) amortization of the right-of-use asset.

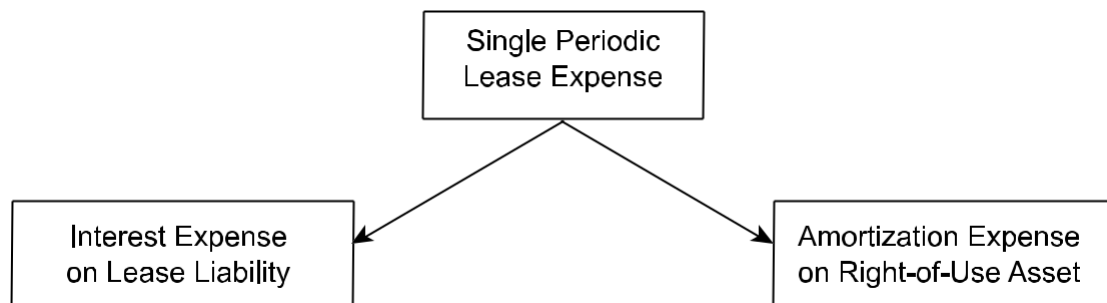


Figure 4-3

- 3) In the **income statement**, a single amount for the total lease expense for the period is reported in income from continuing operations.
 - a) Thus, interest expense for the lease liability and amortization expense for the right-of-use asset are **not reported separately**.
- c. **Balance sheet presentation. Finance lease liabilities and operating lease liabilities** must **not** be presented together in the same line item. They are presented in the balance sheet or disclosed in the notes, separately from each other and separately from other liabilities.
 - 1) **Finance lease right-of-use assets and operating lease right-of-use assets** must **not** be presented together in the same line item. They are presented in the balance sheet or disclosed in the notes, separately from each other and separately from other assets.
- d. **Statement of cash flows presentation.** Payments for operating leases (repayment of the lease liability and interest expense on the lease liability) are cash outflows from **operating activities**.
 - 1) Payments for **short-term leases** are cash outflows from **operating activities**.

4.3 INCOME TAXES

1. Objectives

- a. The objectives of accounting for income taxes are to recognize the following:
 - 1) The amount of taxes currently payable or refundable
 - 2) Deferred tax liabilities and assets for the future tax consequences of events that have been recognized in the financial statements or tax returns
- b. To achieve these objectives, an entity uses the asset-and-liability approach to account for (1) income taxes currently payable or deductible and (2) deferred taxes.

2. Basic Definitions

- a. **Income tax expense or benefit** is the sum of (1) current tax expense or benefit and (2) deferred tax expense or benefit.
- b. **Current tax expense or benefit** is the amount of taxes paid or payable (or refundable) for the year as determined by applying the enacted tax law to the taxable income or excess of deductions over revenues for that year.
 - 1) This amount is the **income tax payable** for the period.
 - 2) The approach is sometimes called income-tax-basis accounting, which is an other comprehensive basis of accounting (not GAAP). Nontaxable revenues and nondeductible expenses are included in the income statement.
- c. **Taxable income** is the income calculated under the tax code. Taxable income equals pretax accounting income adjusted for permanent and temporary tax differences.
- d. **Deferred tax expense or benefit** is the net change during the year in an entity's deferred tax amounts.
- e. A **deferred tax liability (or asset)** records the deferred tax consequences of taxable (or deductible) temporary differences.
 - 1) A deferred tax liability or asset is recognized for the estimated future tax effects of temporary differences and carryforwards.
 - 2) A deferred tax amount is measured using the enacted tax rate(s) expected to apply when the liability or asset is expected to be settled or realized.

3. Intraproduct Tax Allocation

- a. Intraproduct tax allocation **is required**. Income tax expense (benefit) is allocated to
 - 1) Continuing operations,
 - 2) Discontinued operations,
 - 3) Other comprehensive income, and
 - 4) Items debited or credited directly to equity.
- b. Intraproduct tax allocation is needed because items included in determining taxable income may be presented in different sections of the income statement. A tax on tax benefit should be shown in the same section as the related income or expense.

4. Interperiod Tax Allocation

- a. Amounts in the entity's income tax return for a year include the tax consequences of most items recognized in the financial statements for the same year.
 - 1) But significant exceptions may exist because of differences between the GAAP-basis of accounting and the income-tax-basis of calculating taxable income.
- b. **Temporary differences** result when the GAAP basis and the tax basis of an asset or liability differ. Differences in the two bases arise when items of income and expense are recognized in different periods under GAAP and under the tax code. The effect is that a taxable or deductible amount will occur in future years when the asset is recovered or the liability is settled.
 - 1) Tax consequences of some items may be recognized in **tax returns** for a year different from that in which their **financial-statement effects** are recognized. The following are examples:
 - a) Different depreciation methods may be used for tax purposes and in the financial statements.
 - i) Accelerated depreciation is allowed for tax purposes for certain assets, but they may be depreciated using the straight-line method in the financial statements.
 - b) Expenses for warranty liability are recognized in the financial statements on the date of the sale under the accrual method of accounting.
 - i) For tax purposes, warranty expenses are recognized under the cash basis when actual payments of warranty costs are made.
 - c) Bad debt expense is recognized in the financial statements under the allowance method in accordance with the income-statement or balance-sheet approach.
 - i) For tax purposes, bad debt expense is recognized when the debts are determined to be uncollectible using the direct write-off method.
- c. A **permanent difference** is an event that is recognized either in pretax financial income or in taxable income **but never in both**. It does not result in a deferred tax amount. The following are examples:
 - 1) Payments of fines or penalties are recognized as an expense in the financial statements but are never deducted in the tax return.
 - 2) Interest on state or municipal bonds is recognized as income in the financial statements but not in taxable income for tax purposes.
- d. When tax consequences and financial-statement effects differ, income taxes currently payable or refundable also may differ from income tax expense or benefit.
 - 1) The accounting for these differences is **interperiod tax allocation**.

5. Deferred Tax Assets and Liabilities

- a. Deferred tax assets and liabilities result from temporary differences, not permanent differences.
- b. Taxable temporary differences result in future taxable amounts and deferred tax liabilities (DTL).



Figure 4-4

- 1) DTLs arise when **revenues or gains** are recognized under GAAP before they are included in taxable income.
 - a) An example is income recognized under the equity method for financial statement purposes and at the time of distribution in taxable income.
- 2) DTLs also result when **expenses or losses** are deductible for tax purposes before they are recognized under GAAP.
 - a) An example is accelerated tax depreciation of property.

$$\text{DTL} = \text{Future taxable amount} \times \text{Tax rate}$$

EXAMPLE 4-8 DTL

On January 1, Year 1, Luxor buys a piece of equipment for \$100,000. It will be depreciated over its 5-year useful life on the straight-line basis for financial reporting (20% per year). The following accelerated percentages will be used for tax purposes: 59%, 16.5%, 9.5%, 8%, and 7% for Years 1-5, respectively.

Year 1 excess of GAAP income over taxable income \$100,000 × (59% – 20%)	<u>\$39,000</u>
Future taxable amount in Year 2 (16.5% – 20%)	\$ 3,500
Future taxable amount in Year 3 (9.5% – 20%)	10,500
Future taxable amount in Year 4 (8% – 20%)	12,000
Future taxable amount in Year 5 (7% – 20%)	<u>13,000</u>
Total future taxable amount	\$39,000
Enacted tax rate	× 40%
Year 1 deferred tax liability	<u>\$15,600</u>

NOTE: On December 31, Year 1, the reported amount of the equipment in the financial statements of \$80,000 (\$100,000 – \$20,000) exceeds its tax basis of \$41,000 (\$100,000 – \$59,000) by \$39,000. This taxable temporary difference results in recognition of a deferred tax liability of \$15,600 (\$39,000 × 40% enacted tax rate for future years).

- c. Deductible temporary differences result in future deductible amounts and deferred tax assets (DTA).



Figure 4-5

- 1) DTAs result when **revenues or gains** are included in taxable income before they are recognized under GAAP.
 - a) Examples are unearned revenues such as rent and subscriptions received in advance.
- 2) DTAs also result when **expenses or losses** are recognized under GAAP before they are deductible for tax purposes.
 - a) Examples are bad debt expense recognized under the allowance method and warranty costs.

$$\text{DTA} = \text{Future deductible amount} \times \text{Tax rate}$$

EXAMPLE 4-9 DTA

In Year 1, Luxor accrued \$70,000 of warranty costs for financial reporting purposes. From past experience, it expects these costs to be incurred in Years 2 through 5 as follows: \$5,000, \$15,000, \$40,000, \$10,000.

Year 1 excess of taxable income over GAAP income	<u>\$70,000</u>
Future deductible amount in Year 2	\$ 5,000
Future deductible amount in Year 3	15,000
Future deductible amount in Year 4	40,000
Future deductible amount in Year 5	<u>10,000</u>
Total future deductible amount	\$70,000
Enacted tax rate	× 40%
Year 1 deferred tax asset	<u>\$28,000</u>

6. Calculating Tax Expense or Benefit

a. Income tax expense or benefit reported on the income statement is the sum of the current component and the deferred component.

- 1) **Current tax expense or benefit** is the amount of taxes paid or payable (or refundable) for the year based on the enacted tax law.

Taxable income

Current tax expense or benefit = (or excess of deductions over revenue) × Tax rate

- 2) **Deferred tax expense or benefit** is the net change during the year in an entity's deferred tax amounts.

Changes in DTL balances ± Changes in DTA balances

b. Current income tax expense is recorded as follows:

Income tax expense -- current	\$XXX
Income tax payable	\$XXX

c. Deferred income tax expense or benefit is recognized for the **net change** during the year in the deferred tax amounts (DTL and DTA) and recorded as follows:

If the DTL balance increased during the year:

Income tax expense -- deferred	\$XXX
Deferred tax liability	\$XXX

If the DTL balance decreased during the year:

Deferred tax liability	\$XXX
Income tax expense -- deferred	\$XXX

If the DTA balance increased during the year:

Deferred tax asset	\$XXX
Income tax expense -- deferred	\$XXX

If the DTA balance decreased during the year:

Income tax expense -- deferred	\$XXX
Deferred tax asset	\$XXX

EXAMPLE 4-10 Calculating Tax Expense or Benefit

Lucas Company had the following deferred tax balances for the year just ended. The deferred tax asset is fully realizable. The company's taxable income was \$1,000,000 for the year. The enacted tax rate is 40%.

	Beginning Balance	Ending Balance
Deferred tax asset	\$ 9,000	\$17,000
Deferred tax liability	13,000	23,000

Lucas calculates income tax expense for the year as follows:

- Current tax expense is \$400,000 ($\$1,000,000 \times 40\%$).
- Deferred tax expense is the net change in the deferred tax liability and asset balances for the year. The DTL balance increased by \$10,000 ($\$23,000 - \$13,000$), and the DTA balance increased by \$8,000 ($\$17,000 - \$9,000$). Thus, the net DTL increase is \$2,000 ($\$10,000 - \$8,000$).

Lucas records the following entry:

Income tax expense -- current	\$400,000	
Income tax expense -- deferred	2,000	
Deferred tax asset	8,000	
Income tax payable		\$400,000
Deferred tax liability		10,000

4.4 ACCOUNTING FOR BONDS AND NONCURRENT NOTES PAYABLE

1. Nature of Bonds

- a. A bond is a formal contract to pay an amount of money (face amount) at the maturity date plus interest at the stated rate at specific intervals.

EXAMPLE 4-11 Nature of Bonds

At the beginning of the year, a company issues 200 8%, 5-year, \$5,000 bonds. Annual cash interest payments will be made at the end of each year. The total face amount of bonds issued is \$1,000,000 (200 bonds $\times$ \$5,000 face amount), and the annual interest payment is \$80,000 (\$1,000,000 face amount $\times$ 8% stated rate).

- b. The proceeds received from the investors on the day the bonds are sold equal the present value of the sum of the future cash flows expected to be received from the bonds. These proceeds equal
- 1) The present value of the face amount plus
 - 2) The present value of the annuity of interest payments.
- c. The bonds are recognized in the financial statements as the amount of proceeds paid for them, i.e., the face amount of the bonds plus any premium or minus any discount.
- 1) They are recorded as a debt in the issuer's financial statements and as an investment in the investors' financial statements.
- d. Some theoreticians have argued that bonds should be reported on the balance sheet at their fair value, which is the amount they would sell for in the open market. For example, if a \$1,000 bond could be bought at 98% of its face value, an application of fair-value accounting would result in the issuer showing a liability of only \$980.

2. Bond Issuance

- a. The cash proceeds from the sale of bonds can be equal to, less than, or greater than the face amount of the bonds depending on the relationship of the bonds' stated rate of interest to the market rate of interest on the date the bonds are sold.
- 1) If the stated rate is equal to the market rate, the cash proceeds equal the face amount of the bonds, and the bonds are sold at **par**.
 - 2) If the stated rate is greater than the current market rate, the cash proceeds are greater than the face amount, and the bonds are sold at a **premium**.
 - 3) If the stated rate is lower than the current market rate, the cash proceeds are lower than the face amount, and the bonds are sold at a **discount**.
- b. The current market rate of interest is used to discount the cash flows expected to be received by the investor (paid by the issuer) from the bonds.

EXAMPLE 4-12 Bond Issuance

The following calculation uses the data from Example 4-11 and the following present value factors:

	At 6%	At 10%
Present value of 1 for 5 periods	0.747	0.621
Present value of ordinary annuity of 1 for 5 periods	4.212	3.791

- (1) Assume that the market interest rate was 6% on the date the bonds were issued.

Present value of face amount (\$1,000,000 $\times$ 0.747)	\$ 747,000
Present value of cash interest (\$80,000 $\times$ 4.212)	336,960
Cash proceeds from bonds issue	<u>\$1,083,960</u>

-- Continued on next page --

EXAMPLE 4-12 -- Continued

The amount of premium is \$83,960 (\$1,083,960 proceeds – \$1,000,000 face amount).

(2) Assume that the market interest rate was 10% on the date the bonds were issued.

Present value of face amount (\$1,000,000 × 0.621)	\$621,000
Present value of cash interest (\$80,000 × 3.791)	303,280
Cash proceeds from bonds issue	<u>\$924,280</u>

The amount of discount is \$75,720 (\$1,000,000 face amount – \$924,280 proceeds).

3. Amortization of Premium or Discount

- a. Bond premium or discount must be amortized over the life of the bonds using the **effective-interest method** (the market interest rate on the date the bond was sold). Under this method, interest expense changes every period and equals the following:

$$\text{Annual interest expense} = \text{Carrying amount of the bond} \times \text{Effective interest rate at the beginning of the period}$$

- b. The annual interest expense consists of the cash interest paid plus the effect of amortization of premium or discount.
- 1) When the bond is issued at a premium, annual interest expense equals cash interest paid minus the amount of premium amortized.
 - 2) When the bond is issued at a discount, annual interest expense equals cash interest paid plus the amount of discount amortized.
 - 3) The carrying amount of bonds as they are presented in the financial statements equals the face amount plus the premium (or minus the discount).

EXAMPLE 4-13 Amortization of Premium and Discount

Using the data from Example 4-12, the following interest expense will be recognized by the company in the first 2 years of the bonds:

(1) When market interest rate was 6% and bonds were issued at premium:

Year	A					B		
	Beginning Carrying Amount of Bonds	Rate				Premium Amortized	Remaining Premium	
1	\$1,083,960	× 6%	=	\$65,038	– \$80,000	= \$14,962	\$68,998	\$1,068,998
2	1,068,998	× 6%	=	64,140	– 80,000	= 15,860	53,138	1,053,138

(2) When market interest rate was 10% and bonds were issued at discount:

Year	A					B		
	Beginning Carrying Amount of Bonds	Rate				Discount Amortized	Remaining Discount	
1	\$924,280	× 10%	=	\$92,428	– \$80,000	= \$12,428	\$63,292	\$936,708
2	936,708	× 10%	=	93,671	– 80,000	= 13,671	49,621	950,379

- 4) At the maturity date, the discount or premium is fully amortized, and the carrying amount of the bonds equals the face amount.

4. The following are different types of bonds:

a. Maturity Pattern

- 1) A **term bond** has a single maturity date at the end of its term. The examples in this study unit pertain to a regular term bond.
- 2) A **serial bond** matures in stated amounts at regular intervals. Investors can choose the maturity that suits their financial needs.

b. Repayment Provisions

- 1) **Income bonds** pay interest contingent on the issuing firm's profitability.
- 2) **Revenue bonds** are issued by governments and are payable from specific revenue sources.

c. Securitization

- 1) **Mortgage bonds** are backed by specific assets, usually real estate.
- 2) **Debentures** are not secured by specific collateral. Instead, they are secured by the full faith and credit of the issuing firm.
- 3) **Guaranty bonds** are guaranteed by a third party, e.g., the parent of the subsidiary that issued the bonds.
- 4) **Collateral trust bonds** are secured by a financial asset, such as stock or other bonds.
- 5) **Equipment trust bonds** are secured by a mortgage on movable equipment, such as airplanes or railroad cars.

d. Valuation

- 1) **Variable (or floating) rate bonds** pay interest that is dependent on market conditions.
- 2) **Zero-coupon or deep-discount bonds** bear no stated rate of interest and thus involve no periodic cash payments. The interest component consists entirely of the bond's discount. The discount is amortized annually on a straight-line basis, but no periodic interest payments are required. The cash outflow for interest does not occur until the bond itself is retired.
- 3) **Commodity-backed bonds** are payable at prices related to a commodity such as gold.

e. Redemption Provisions

- 1) **Callable bonds** (also called redeemable bonds) may be repurchased (redeemed) by the issuer at a specified price before maturity.
 - a) During a period of falling interest rates, the issuer can replace high-interest debt with low-interest debt.
- 2) **Convertible bonds** may be converted into equity securities of the issuer at the option of the holder (buyer) under specified conditions. Convertible bonds generally pay a lower yield than comparable nonconvertible bonds.

5. Noncurrent Notes Payable

- a. Some notes require one principal payment at the end of the note's term plus periodic interest payments during the note's term (like a term bond).

EXAMPLE 4-14 Note -- One Principal Payment

An entity agrees to give, in return for merchandise, a 3-year, \$100,000 note bearing 8% interest paid annually. The effective interest rate is 6%. Because the note's stated rate exceeds the effective rate, the note will be issued at a premium.

The entity records the note at the present value of (1) a single payment of \$100,000 in 3 years and (2) three interest payments of \$8,000 each. These payments are discounted at the effective rate (five decimal places are used for increased accuracy).

Present value of principal ($\$100,000 \times 0.83962$)	\$ 83,962
Present value of interest ($\$8,000 \times 2.67301$)	21,384
Present value of note	<u>\$105,346</u>

The entry to record the note is

Inventory	\$105,346	
Premium on note payable		\$ 5,346
Note payable		100,000

- b. Other notes require equal periodic principal payments plus interest. Each periodic payment includes an equal amount of return of principal and an amount of interest accrued on the beginning carrying amount.

EXAMPLE 4-15 Note -- Periodic Principal Payments

On January 1, Year 1, Shark Co. borrowed \$120,000 on a 10% note payable to Bank. Three equal annual principal payments of \$40,000 plus interest are paid beginning December 31, Year 1.

Year	Beginning Carrying Amount (a)	Interest Rate	Interest Payment/Expense (b)	Principal Payment (c)	Total Payment (b) + (c)	Ending Carrying Amount (a) - (c)
1	\$120,000	10%	\$12,000	\$40,000	\$52,000	\$80,000
2	80,000	10%	8,000	40,000	48,000	40,000
3	40,000	10%	4,000	40,000	44,000	0

- c. A third type of note requires equal periodic cash payments. Each payment includes a principal component (i.e., return of principal) and an interest component.

EXAMPLE 4-16 Note -- Equal Periodic Payments

On January 1, Year 1, Star Co. borrowed \$120,000 on a 10% note payable to Bank. Three equal annual payments of \$48,254 are paid beginning December 31, Year 1.

Year	Beginning Carrying Amount (a)	Interest Rate	Interest Payment/ Expense (b)	Total Payment (c)	Principal Payment (c) - (b) = (d)	Ending Carrying Amount (a) - (d)
1	\$120,000	10%	\$12,000	\$48,254	\$36,254	\$83,746
2	83,746	10%	8,375	48,254	39,879	43,867
3	43,867	10%	4,387	48,254	43,867	0

NOTE: The proceeds from the note are equal to the present value of the cash payments associated with the note. The equal annual payment (\$48,254) multiplied by the present value of an ordinary annuity of \$1 at 10% for three periods (2.48685) equals the proceeds from the note of \$120,000.

STUDY UNIT FIVE

REVENUE AND IMPAIRMENT RECOGNITION

5.1	<i>Revenue from Contracts with Customers</i>	2
5.2	<i>Recognition of Revenue over Time</i>	9
5.3	<i>Impairment and Disposal of Long-Lived Assets</i>	13

This study unit is the **fifth of six** on **external financial reporting decisions**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The six study units are

Study Unit 1: External Financial Statements

Study Unit 2: Measurement, Valuation, and Disclosure: Assets -- Short-Term Items

Study Unit 3: Measurement, Valuation, and Disclosure: Assets -- Long-Term Items

Study Unit 4: Measurement, Valuation, and Disclosure: Liabilities

Study Unit 5: Revenue and Impairment Recognition

Study Unit 6: Integrated Reporting

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

5.1 REVENUE FROM CONTRACTS WITH CUSTOMERS

1. Overview

- a. The **core principle** is that an entity recognizes revenue for the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity expects to be entitled in the exchange.
- b. Below is the **five-step model** for recognizing revenue from contracts with customers.

Step 1:	Identify the contract(s) with a customer.
Step 2:	Identify the performance obligations in the contract.
Step 3:	Determine the transaction price.
Step 4:	Allocate the transaction price to the performance obligations in the contract.
Step 5:	Recognize revenue when (or as) a performance obligation is satisfied.

2. Step 1: Identify the Contract with a Customer

- a. A contract is an agreement between two or more parties that creates enforceable rights and obligations.
- b. Revenue is recognized for a contract with a customer if **all** of the following criteria are met:
 - 1) The contract was approved by the parties.
 - 2) The contract has commercial substance.
 - 3) Each party's rights regarding (a) goods or services to be transferred and (b) the payment terms can be identified.
 - 4) It is probable that the entity will collect substantially all of the consideration to which it is entitled according to the contract.
 - a) **Probable** means that the future event is likely to occur.
- c. If the criteria described above are not met (e.g., if collectibility cannot be reliably estimated), the consideration received is recognized as a liability, and **no revenue is recognized** until the criteria are met.
 - 1) However, even when the criteria described above are not met, revenue in the amount of **nonrefundable consideration** received from the customer is recognized if at least one of the following has occurred:
 - a) The contract has been terminated.
 - b) Control over the goods or services was transferred to the customer and the entity has stopped transferring (and has no obligation to transfer) additional goods or services to the customer.
 - c) The entity (1) has no obligation to transfer goods or services and (2) has received substantially all consideration from the customer.

- d. A **contract modification** exists when the parties approve a change in the scope or price of a contract.
 - 1) It is accounted for as a **separate contract** if the following conditions are met:
 - a) The scope of the contract increases because of the addition of promised goods or services that are distinct, and
 - b) The price of the contract increases by a consideration amount that reflects the entity's standalone selling prices of the additional promised goods or services.

3. Step 2: Identify the Performance Obligations in the Contract

- a. A **performance obligation** is a promise in a contract with a customer to transfer to the customer (1) a good or service that is distinct or (2) a series of distinct goods or services that are substantially the same and have the same pattern of transfer to the customer.
- b. Promised goods or services are **distinct** if
 - 1) The customer can benefit from them either on their own or together with other resources that are readily available (**capable of being distinct**), and
 - 2) The entity's promise to transfer them to the customer is separately identifiable from other promises in the contract (**distinct within the context of the contract**). A **separately identifiable good or service**
 - a) Does not significantly modify or customize another good or service promised in the contract.
 - b) Is not highly dependent on, or highly interrelated with, other goods or services promised in the contract.
- c. A contract may include a customer option to acquire **additional goods or services** for free or at a discount (e.g., coupon, sales incentives, etc.). If the option provides a **material right** to the customer, the result is a separate performance obligation in the contract.

4. Step 3: Determine the Transaction Price

- a. The **transaction price** is the amount of consideration to which an entity expects to be entitled in exchange for transferring promised goods or services to a customer.
 - 1) It excludes amounts collected on behalf of third parties (e.g., sales taxes).
 - 2) Any consideration payable to the customer, such as coupons, credits, or vouchers, reduces the transaction price.
 - 3) To determine the transaction price, an entity should consider the effects of the **time value of money** and **variable consideration**.
- b. The revenue recognized must reflect the price that a customer would have paid for the promised goods or services if the cash payment had been made when they were transferred to the customer (i.e., the cash selling price).
 - 1) Thus, the transaction price is adjusted for the effect of the time value of money when the contract includes a **significant financing component**.
 - 2) The following factors should be considered in assessing whether a contract includes a significant financing component:
 - a) The difference between (1) the cash selling price of the promised goods or services and (2) the amount of consideration to be received
 - b) The combined effect of (1) the expected time between the payment and the delivery of the promised goods or services and (2) market interest rates
- c. The transaction price should **not** be adjusted for the effect of the time value of money if
 - 1) The time between the payment and the delivery of the promised goods or services to the customer is **1 year or less**
 - 2) The customer paid in advance and the transfer of goods or services is at the discretion of the customer
 - a) An example is a bill-and-hold contract in which the seller provides storage services for goods it sold to the buyer.
 - 3) A substantial amount of the consideration promised is **variable** and its amount or timing varies with future circumstances that are **not** within the control of the entity or the customer
 - a) An example is consideration in the form of a sales-based royalty.
- d. Interest income or expense is recognized using the **effective interest method**.
 - 1) It must be presented in the income statement **separately** from revenue from contracts with customers.

EXAMPLE 5-1 Significant Financing Component

On January 1, Year 1, BIF Co. sold and transferred a machine to a customer for \$583,200 that is payable on December 31, Year 2. Other customers pay \$500,000 upon delivery of the same machine at contract inception. The cost of the machine to BIF is \$400,000. BIF determined that the contract includes a significant financing component because of the difference between the consideration (\$583,200) and the cash selling price (\$500,000). The contract includes an implicit interest rate of 8%. The following entries are recorded by BIF:

January 1, Year 1

Accounts receivable	\$500,000	Cost of goods sold	\$400,000
Revenue	\$500,000	Machine inventory	\$400,000

December 31, Year 1

Accounts receivable ($\$500,000 \times 8\%$)	\$40,000
Interest income	\$40,000

December 31, Year 2

Accounts receivable ($\$540,000 \times 8\%$)	\$43,200	Cash	\$583,200
Interest income	\$43,200	Accounts receivable	\$583,200

EXAMPLE 5-2 Significant Financing Component -- Advance Payment

On January 1, Year 1, Eva Co. received a payment of \$100,000 for delivering a machine to a customer at the end of Year 2. The cost of the machine to Eva is \$70,000. Eva determined that (1) the contract includes a significant financing component and (2) a financing rate of 10% is an appropriate discount rate. The following entries are recorded by Eva:

January 1, Year 1

Cash	\$100,000
Contract liability	\$100,000

December 31, Year 1

Interest expense ($\$100,000 \times 10\%$)	\$10,000
Contract liability	\$10,000

December 31, Year 2

Interest expense ($\$110,000 \times 10\%$)	\$11,000	Contract liability	\$121,000
Contract liability	\$11,000	Revenue	\$121,000
Cost of goods sold	70,000		
Machine inventory	70,000		

e. Variable Consideration

- 1) If a contract includes a variable amount, an entity must estimate the consideration to which it will be entitled in exchange for transferring the promised goods or services to a **customer**. For example, the contract price may vary because of the following:
 - a) Refunds due to a right of return provided to customers
 - b) Sales incentives
 - c) Prompt payment discounts
 - d) Volume discounts
 - e) Other uncertainties in contract price based on the occurrence or nonoccurrence of some future event
- 2) Variable consideration is **estimated** using one of the following methods:
 - a) The **expected value** is the sum of probability-weighted amounts in the range of possible consideration amounts. This method may provide an appropriate estimate if an entity has many contracts with similar characteristics. Expected value is discussed in Study Unit 12, Subunit 3.
 - b) The **most likely amount** is the single most likely amount in a range of possible consideration amounts. This method may provide an appropriate estimate if the contract has only two possible outcomes; e.g., a construction entity either will receive a performance bonus for finishing construction on time or will not.

- 3) The estimated transaction price must be updated at the **end** of each reporting period.
- 4) **Constraint**
 - a) Revenue from variable consideration is recognized only to the extent that it is **probable** that a **significant reversal** will **not** occur when the uncertainty associated with the variable consideration is subsequently resolved.
- 5) A **volume discount** offered as an incentive to increase future sales requires the customer to purchase a specified quantity of goods or services to receive a discount. The discount may be applied (a) **prospectively** on additional goods purchased in the future or (b) **retrospectively** on all goods purchased to date.
 - a) A **prospective volume discount** that provides a **material right** to the customer is accounted for as a separate performance obligation in the contract.
 - b) **Retrospective volume discounts** are accounted for as **variable consideration**. The uncertainty of the contract price for current goods sold is based on the occurrence or nonoccurrence of some future event (i.e., whether the customer completes the specified volume of purchase).

EXAMPLE 5-3 Retrospective Volume Discount

Barashka Co. manufactures wool coats. On October 1, Year 1, Barashka entered into a 2-year contract with a customer to sell coats for \$200 per unit. Based on the contract, if the customer purchased more than 3,000 coats over the contract period, the contract price per coat would be retroactively reduced to \$150. The cost per coat to Barashka is \$80. Barashka determined that the contract has no significant financing component.

The retrospective volume discount is variable consideration. In Year 1, the customer purchased with cash 100 coats, and Barashka estimated that the customer's purchases would not exceed 3,000 during the contract period. Thus, based on the most likely amount method, the contract price per coat was \$200.

The following entries were recorded by Barashka in Year 1:

Cash (100 × \$200)	\$20,000	Cost of goods sold (100 × \$80)	\$8,000	
Sales revenue		Inventory of coats		\$8,000
			\$20,000	

The winter in Year 2 was colder than expected. The customer purchased an additional 2,200 coats. Accordingly, Barashka estimated that the customer would purchase more than 3,000 coats over the contract period. The price per coat therefore was retrospectively reduced to \$150, and Year 2 revenue is calculated as a cumulative catch up adjustment. Year 2 revenue of \$325,000 was the difference between total revenue that should be recognized for Year 1 and Year 2 of \$345,000 [(2,200 + 100) × \$150] minus revenue recognized in Year 1 of \$20,000. A contract liability is recognized for the excess of consideration received over the amount of revenue recognized. It equals the future amount of goods to be transferred to the customer for which the consideration was already received.

The following entries were recorded by Barashka in Year 2:

Cash (2,200 × \$200)	\$440,000	Cost of goods sold (2,200 × \$80)	\$176,000	
Sales revenue		Inventory of coats		\$176,000
Contract liability			115,000	
			\$325,000	

As expected, the customer purchased an additional 1,100 coats in Year 3. Accordingly, 3,400 (100 + 2,200 + 1,100) coats were purchased during the contract period. In Year 3, the customer retroactively received the discount on all the coats previously purchased, paying cash of \$50,000 [(3,400 × \$150) – (\$440,000 + \$20,000)].

The following entries were recorded by Barashka in Year 3:

Cash	\$ 50,000	Cost of goods sold (1,100 × \$80)	\$88,000	
Contract liability	115,000	Inventory of coats		\$88,000
Sales revenue (1,100 × \$150)			\$165,000	

5. Step 4: Allocate the Transaction Price to the Performance Obligations in the Contract

- a. After separate performance obligations are identified and the total transaction price is determined, the transaction price is allocated to performance obligations on the basis of relative standalone selling prices.
- b. A **standalone selling price** is the price at which an entity would sell a promised good or service separately to a customer.
 - 1) The best evidence of a standalone selling price is the **observable price** of a good or service when it is (a) sold separately (b) in similar circumstances and (c) to similar customers (e.g., the list price of a good or service).
- c. If the standalone price is **not directly observable**, it must be estimated. The following are suitable approaches:
 - 1) **Adjusted market assessment.** An entity evaluates the market in which it sells goods or services and estimates the price that a customer in that market would be willing to pay for them.
 - a) For example, the prices of competitors for similar goods or services adjusted for the entity's costs and margins are estimates of standalone selling prices.
 - 2) **Expected cost plus an appropriate margin.** An entity forecasts its expected costs of satisfying a performance obligation and adds an appropriate margin for that cost.
 - 3) **Residual.** An entity estimates the standalone selling price by reference to the total transaction price minus the sum of the observable standalone selling prices of other goods or services promised in the contract.
 - a) The residual approach may be used only in limited circumstances.

EXAMPLE 5-4 Allocation of the Contract Price

A company entered into a contract with a customer to sell a machine and provide 3 years of maintenance services for it. The total consideration is \$200,000. The company determined that the machine and the maintenance services are distinct performance obligations. The company regularly sells machines separately at a directly observable standalone selling price of \$160,000. But it does not sell maintenance services on a standalone basis. Based on the expected cost plus an appropriate margin approach, the estimated standalone selling price for 3 years of maintenance services was \$90,000. The transaction price is allocated to each performance obligation in the contract using relative standalone selling prices.

<u>Performance Obligation</u>	<u>Standalone Selling Price</u>	<u>Allocation of the Contract Price</u>
Machine	\$160,000	\$128,000 = $(\$160,000 \div \$250,000) \times \$200,000$
Maintenance Services	90,000	72,000 = $(\$90,000 \div \$250,000) \times \$200,000$
Total	<u>\$250,000</u>	<u>\$200,000</u>

6. Step 5: Recognize Revenue when (or as) a Performance Obligation Is Satisfied

- a. An entity recognizes revenue when (or as) it satisfies a performance obligation by transferring a promised good or service (an asset) to a **customer**.
 - 1) An **asset** is transferred when (or as) the customer obtains control of that asset.
- b. **Control** of an asset is transferred when the customer
 - 1) Has the ability to direct the use of the asset and
 - 2) Obtains substantially all of the remaining benefits (potential cash flows) from the asset.
- c. A performance obligation can be satisfied either over time or at a point in time.
 - 1) Recognizing revenue **over time** requires transfer of the control of goods or services to a customer over time and therefore satisfaction of a performance obligation over time. **One** of the following criteria must be met:
 - a) The customer **simultaneously** receives and consumes the benefits provided by the entity's performance as the entity performs. For example, cleaning services are provided to a customer's offices every day throughout the accounting period.
 - b) The entity's performance **creates or enhances an asset** that the customer controls as the asset is created or enhanced. For example, a construction company erects a building on the customer's land.
 - c) The asset created has **no alternative use** to the entity, and the entity has an enforceable **right** to payment for the performance completed to date. For example, an aerospace company contracts to build a satellite designed for the unique needs of a specific customer.
 - i) An entity does not have an alternative use for an asset if the entity is restricted contractually or limited practically from directing the asset for another use.
 - 2) The accounting for contracts in which revenue is recognized over time is described in Subunit 5.2.
 - 3) If a performance obligation is **not satisfied over time**, an entity satisfies the performance obligation **at a point in time**.
 - a) Revenue is recognized at a point in time when the customer obtains **control** over the promised asset. The following indicators of the transfer of control should be considered:
 - i) The entity has a present right to payment for the asset.
 - ii) The customer has legal title to the asset.
 - iii) The entity has transferred physical possession of the asset.
 - iv) The customer has the significant risks and rewards of ownership of the asset.
 - v) The customer has accepted the asset.

7. Balance Sheet Presentation

- a. A **contract liability** is recognized for an entity's obligation to transfer goods or services to a customer for which the entity has received consideration from the customer.
 - 1) Deposits and other advance payments by the customer, such as sales of gift certificates, are recognized as contract liabilities.
- b. A **contract asset** is recognized for an entity's right to consideration in exchange for goods or services that the entity has transferred to a customer.
 - 1) However, the entity must have an **unconditional** right to the consideration to recognize a **receivable**.
 - 2) A right to consideration is unconditional if only the passage of time is required before payment of that consideration is due.
- c. Contract assets and contract liabilities resulting from different contracts must not be presented net in the statement of financial position.

5.2 RECOGNITION OF REVENUE OVER TIME

1. For each performance obligation satisfied over time, an entity must recognize revenue over time. For this purpose, the entity measures the **progress toward complete satisfaction** using the **output method** or the **input method**.
 - a. To determine the appropriate method, an entity must consider the nature of the good or service that it promised to transfer to the customer.
 - b. The chosen method should describe the entity's performance in transferring control of the promised asset to the customer.
2. At the end of each reporting period, the progress toward complete satisfaction of the performance obligation must be **remeasured** and updated for any changes in the outcome of the performance obligation.
 - a. Such changes must be accounted for prospectively as a **change in accounting estimate**.

3. The **input method** recognizes revenue on the basis of (a) the entity's inputs to the satisfaction of the performance obligation relative to (b) the total expected inputs to the satisfaction of that performance obligation.
 - a. Examples of input include
 - 1) Costs incurred,
 - 2) Labor hours expended,
 - 3) Resources consumed,
 - 4) Time elapsed, or
 - 5) Machine hours used.
 - b. In long-term construction contracts, **costs incurred relative to total estimated costs** often are used to measure the progress toward completion. This method is the **cost-to-cost** method (formerly known as the percentage-of-completion method).
 - 1) Only costs that contribute to progress in satisfying the performance obligation are used in the cost-to-cost method. Thus, the following costs must not be included in measuring the progress:
 - a) Costs incurred that relate to significant inefficiencies in the entity's performance (e.g., abnormal amounts of wasted materials or labor) that were not chargeable to the customer under the contract
 - b) General and administrative costs not directly related to the contract
 - c) Selling and marketing costs
 - c. When an entity's inputs are incurred evenly over time, recognition of revenue on a straight-line basis may be appropriate.

EXAMPLE 5-5 Cost-to-Cost Method

On January 1, Year 1, a contractor agrees to build on the customer's land a bridge that is expected to be completed at the end of Year 3. The promised bridge is a single performance obligation to be satisfied over time. The contractor determines that the progress toward completion of the bridge is reasonably measurable using the input method based on costs incurred. The contract price is \$2,000,000, and expected total costs of the project are \$1,200,000.

	<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>
Costs incurred during each year	\$300,000	\$600,000	\$550,000
Costs expected in the future	900,000	600,000	

Year 1

By the end of Year 1, 25% [$\$300,000 \div (\$300,000 + \$900,000)$] of the total expected costs have been incurred. Using the input method based on costs incurred, the contractor recognizes 25% of the total expected revenue ($\$2,000,000$ contract price $\times$ 25% = \$500,000) and cost of goods sold ($\$1,200,000 \times 25\%$ = \$300,000). The difference between these amounts is the gross profit for Year 1.

Revenue	\$500,000
Cost of goods sold	<u>(300,000)</u>
Gross profit -- Year 1	<u>\$200,000*</u>

* The gross profit in Year 1 of \$200,000 also may be calculated as total expected gross profit from the project of \$800,000 ($\$2,000,000 - \$1,200,000$) times the progress toward completion of the contract of 25%.

Year 2

By the end of Year 2, total costs incurred are \$900,000 ($\$300,000 + \$600,000$). Given that \$600,000 is expected to be incurred in the future, the total expected cost is \$1,500,000 ($\$900,000 + \$600,000$). The change in the total cost of the contract must be accounted for prospectively. By the end of Year 2, 60% ($\$900,000 \div \$1,500,000$) of expected costs have been incurred. Thus, \$1,200,000 ($\$2,000,000 \times 60\%$) of cumulative revenue and \$900,000 ($\$1,500,000 \times 60\%$) of cumulative cost of goods sold should be recognized for Years 1 and 2. Because \$500,000 of revenue and \$300,000 of cost of goods sold were recognized in Year 1, revenue of \$700,000 ($\$1,200,000$ cumulative revenue $- \$500,000$) and cost of goods sold of \$600,000 ($\$900,000$ cumulative cost of goods sold $- \$300,000$) are recognized in Year 2.

Revenue	\$700,000
Cost of goods sold	<u>(600,000)</u>
Gross profit -- Year 2	<u>\$100,000*</u>

* The gross profit in Year 2 of \$100,000 also may be calculated as the cumulative gross profit for Years 1 and 2 of \$300,000 [$(\$2,000,000 - \$1,500,000) \times 60\%$] minus the gross profit recognized in Year 1 of \$200,000.

Year 3

At the end of Year 3, the project is completed, and the total costs incurred for the contract are \$1,450,000 ($\$300,000 + \$600,000 + \$550,000$). Given \$1,200,000 of cumulative revenue and \$900,000 of cumulative cost of goods sold for Years 1 and 2, \$800,000 ($\$2,000,000$ contract price $- \$1,200,000$) of revenue and \$550,000 ($\$1,450,000$ total costs $- \$900,000$) of cost of goods sold are recognized in Year 3.

Revenue	\$800,000
Cost of goods sold	<u>(550,000)</u>
Gross profit -- Year 3	<u>\$250,000</u>

NOTE: (1) The total gross profit from the project of \$550,000 ($\$200,000 + \$100,000 + \$250,000$) equals the contract price of \$2,000,000 minus the total costs incurred of \$1,450,000. (2) When progress toward completion is measured using the cost-to-cost method, as in the example above, the cost of goods sold recognized for the period equals the costs incurred during that period.

4. The **output method** recognizes revenue based on direct measurement of (a) the value of goods or services transferred to the customer to date relative to (b) the remaining goods or services promised under the contract.
- However, revenue might not be recognized until all performance obligations are satisfied by transfer of control of the goods or services to the buyer. The effect then is that revenue is recognized at a point in time, not over time. The result is the same as under what was formerly known as the completed-contract method.
 - Examples of output methods include (1) appraisals of results achieved, (2) milestones reached, (3) units produced, and (4) units delivered.
 - An entity may have a right to consideration from a customer in an amount corresponding directly with the value to the customer of performance to date. Using a practical expedient, revenue may be recognized at the amounts to which the entity has a **right to invoice the customer**.

EXAMPLE 5-6 Output Method -- Practical Expedient

A law firm enters into a contract to provide consulting services to a customer for a 1-year period for a fixed amount per hour of service provided. Because the customer simultaneously receives and consumes the benefits provided by the law firm's performance as it performs, revenue is recognized over time. Under the practical expedient, the law firm may recognize revenue that it has a right to bill to the customer.

5. An entity recognizes revenue for a performance obligation satisfied over time only if progress toward complete satisfaction of the performance obligation can be reasonably measured.
- However, revenue can be recognized to the **extent of the cost incurred** (zero profit margin) when
 - An entity is not able to reasonably measure the outcome of a performance obligation or its progress toward satisfaction of that obligation, but
 - An entity expects to recover the costs incurred in satisfying the performance obligation.

EXAMPLE 5-7 Revenue Recognition -- Extent of Costs

On January 1, Year 1, Sadik Co. agrees to build on the customer's land a bridge that is expected to be completed at the end of Year 3. The contract price is \$2 million. The promised bridge is a single performance obligation to be satisfied over time. Because Sadik has no experience with this type of contract, it cannot reasonably determine the total expected costs of the project. Accordingly, by the end of Year 1, progress toward completion of the bridge is not reasonably determinable. In Year 1, \$300,000 of costs were incurred and paid by Sadik. However, the contract specified that Sadik has an enforceable right to payment of the costs incurred. Sadik therefore expects to recover these costs.

In Year 1, revenue and cost of goods sold are recognized at the amount of costs incurred of \$300,000, and no gross profit is recognized. The following entries are recorded by Sadik in Year 1:

Accounts receivable	\$300,000	Cost of goods sold	\$300,000
Revenue		Cash	\$300,000
	\$300,000		

6. As soon as an **estimated loss** on any project becomes apparent, it must be recognized in full, regardless of the methods used.

5.3 IMPAIRMENT AND DISPOSAL OF LONG-LIVED ASSETS

1. Two-Step Impairment Test

- a. This two-step impairment test is applied to fixed assets and to intangible assets with finite useful lives.
- b. Testing for impairment occurs when events or changes in circumstances indicate that the carrying amount of the asset may not be recoverable, for example, when
 - 1) The market price has decreased significantly or
 - 2) The use or physical condition of the asset has changed significantly and adversely.
- c. The test for impairment is a **two-step test**:
 - 1) **Recoverability test.** The carrying amount of a long-lived asset to be held and used is not recoverable if it exceeds the sum of the **undiscounted** future cash flows expected from the use and disposition of the asset.
 - 2) If the carrying amount is not recoverable, an impairment loss is recognized. It equals the excess of the carrying amount of the asset over its fair value.
 - a) An impairment loss is recognized immediately in income from continuing operations.
 - b) The journal entry to recognize the impairment loss is

Impairment loss	\$XXX
Accumulated depreciation	\$XXX

Determination of an Impairment Loss

1. Events or changes in circumstances indicate a possible loss
2. Carrying amount of an asset > Sum of undiscounted cash flows
3. Impairment loss = Carrying amount – Fair value

- d. The carrying amount of a long-lived asset adjusted for an impairment loss is its new cost basis. A previously recognized impairment loss **must not be reversed**.

EXAMPLE 5-8 Two-Step Impairment Test

On December 31, Year 2, a machine's carrying amount after the recognition of annual depreciation expense is \$160,000 (\$200,000 historical cost – \$40,000 accumulated depreciation). On that date, as a result of low demand for the company's products, management concludes that the carrying amount of the machine may not be recoverable. Management estimates that the undiscounted future cash flows over the remaining useful life of the machine will be \$150,000. On that date, the machine's estimated fair value is \$136,000.

The carrying amount of the machine exceeds the undiscounted future cash flows expected from the machine (\$160,000 > \$150,000). Thus, the carrying amount is deemed to not be recoverable, and the amount of impairment loss recognized is the excess of the machine's carrying amount over its fair value (\$160,000 – \$136,000 = \$24,000).

In the Year 2 income statement, **\$24,000** is recognized as an impairment loss.

The carrying amount of the machine as it is reported in the balance sheet on December 31, Year 2, is calculated as follows:

Historical (initial) cost	\$200,000
Accumulated depreciation	(40,000)
Impairment losses	(24,000)
Asset's carrying amount	<u>\$136,000</u>

The entity assesses at each reporting date whether an indication of impairment exists. Given such an indication, IFRS requires a **one-step impairment test**.

- The carrying amount of an asset is compared with its recoverable amount. An impairment loss is recognized equal to the excess of the carrying amount over the recoverable amount.



IFRS
DIFFERENCE

Determination of an Impairment Loss

1. Reporting-date assessment indicates a possible loss

2. Impairment loss = Carrying amount – Recoverable amount

The **recoverable amount** is the greater of an asset's (1) fair value minus cost to sell or (2) value in use. Value in use of the asset is the present value of its expected cash flows.

An impairment loss on an asset (besides goodwill) **may be reversed** in a subsequent period if a change in the estimates used to measure the recoverable amount has occurred. The reversal of an impairment loss is recognized immediately in profit or loss as income from continued operations.

2. Disposal of Long-Lived Assets

a. When an item of PPE is sold, the gain or loss on disposal is the difference between the net proceeds and the carrying amount of the asset.

- Depreciation (if any) is recognized to the date of sale, the carrying amount is removed from the books, the proceeds are recorded, and any gain or loss is recognized.

EXAMPLE 5-9 Disposal of a Long-Lived Asset

A company sold a machine with a carrying amount of \$100,000 (\$180,000 historical cost – \$80,000 accumulated depreciation) for \$135,000 in cash. The gain on disposal recognized is \$35,000 (\$135,000 – \$100,000). The journal entry is

Cash	\$135,000	
Accumulated depreciation	80,000	
Machine		\$180,000
Gain on disposal		35,000

If the machine were sold for \$90,000 in cash, the loss on disposal recognized would be \$10,000 (\$90,000 – \$100,000). The journal entry would be

Cash	\$90,000	
Accumulated depreciation	80,000	
Loss on disposal	10,000	
Machine		\$180,000

STUDY UNIT SIX

INTEGRATED REPORTING

6.1	<i>Integrated Reporting, Integrated Thinking, and the Integrated Report</i>	2
6.2	<i>Fundamental Concepts of Integrated Reporting</i>	4
6.3	<i>Guiding Principles and Content Elements of the Integrated Report</i>	8
6.4	<i>Adoption of Integrated Reporting</i>	11

Integrated Reporting

Many organizations have begun to report more than just their financial state. As investors, regulators, and society have become more demanding, especially in the wake of the 2008 financial crisis, the quest for not only financial information but also nonfinancial information has grown.

In the past, additional information was presented in separately published sustainability reports or corporate social responsibility (CSR) reports, but many organizations worldwide are now shifting toward an integrated report. An **integrated report** is designed to provide a holistic view of the organization by connecting financial and nonfinancial information to the long-term value-creation possibilities of the organization.

This study unit addresses most of the elements from the International Integrated Reporting Council's (IIRC's) International Integrated Reporting Framework as well as the issues organizations might face while switching to integrated reporting.

This study unit is the **sixth of six** on **external financial reporting decisions**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The six study units are

Study Unit 1: External Financial Statements

Study Unit 2: Measurement, Valuation, and Disclosure: Assets -- Short-Term Items

Study Unit 3: Measurement, Valuation, and Disclosure: Assets -- Long-Term Items

Study Unit 4: Measurement, Valuation, and Disclosure: Liabilities

Study Unit 5: Revenue and Impairment Recognition

Study Unit 6: Integrated Reporting

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.



SUCCESS TIP

CMA candidates should be prepared to answer questions about integrated reporting, integrated thinking, and the integrated report, as well as the relationships between these concepts. Integrated reporting may be tested in either or both the multiple-choice and essay sections. Like all other topics, integrated reporting is eligible to be tested at all three levels of difficulty, requiring candidates to

1. Recall aspects of integrated reporting and integrated thinking;
2. Evaluate and explain the fundamental concepts of value creation, the six capitals, and the value-creation process; and
3. Understand the different elements of an integrated report and the benefits and challenges it may take to adopt integrated reporting.

6.1 INTEGRATED REPORTING, INTEGRATED THINKING, AND THE INTEGRATED REPORT

BACKGROUND 6-1 Integrated Reporting

Traditionally, the main goal of corporate reporting was disclosure of the organization's financial information to increase transparency and accountability. However, organizations have come to own more intangible assets, from only 17% in 1975 to more than 80% currently. To represent this value accurately, organizations began to provide more nonfinancial information, often in the form of separate CSR reports. However, such information was usually scattered and incomplete and often lacked consistency and comparability.

Integrated reporting is a relatively new concept. It gained momentum with the creation of the International Integrated Reporting Council (IIRC) in 2010. This global entity, composed of regulators, organizations, accounting firms, and standards setters, issued the International <IR> (Integrated Reporting) Framework in 2013. This principles-based framework gives guidance to organizations preparing integrated reports. The objective is to bring together capital allocation and corporate activity to create financial well-being and sustainable expansion through the use of integrated thinking and integrated reporting.

Although the <IR> Framework primarily focuses on providers of financial capital in the private sector, its mission is for IR to become the worldwide corporate reporting standard for both private and public companies.

1. Integrated Reporting

- a. As defined in the glossary of the <IR> Framework, integrated reporting is “a process founded on integrated thinking that results in a periodic integrated report by an organization about value creation over time and related communications regarding aspects of value creation.”
 - 1) This definition of IR requires an understanding of three major concepts:
 - a) Integrated thinking,
 - b) The integrated report, and
 - c) Value creation.
 - 2) These three concepts will be explained later in this study unit.
- b. Through IR, an organization is able to explain how strategy, corporate governance, employee and management performance, and prospects lead to value creation over time.
 - 1) In the context of the entity's external environment, value is created by providing a combination of financial and nonfinancial information.
 - 2) IR enables an organization to tell its own unique story of value creation.
- c. The IR process is based on integrated thinking within the organization and the <IR> Framework.
- d. These concepts interact. IR is not a static concept but a process that continuously improves reporting, internal decision making, and integrated thinking.

2. Integrated Thinking

- a. Integrated thinking is a process of decision making, management, and reporting. It is based on the connectivity and interdependencies among the organization-specific factors that affect the organization's ability to create value over time.
- b. Integrated thinking is a prerequisite to IR. Understanding the influences financial and nonfinancial factors have on each other is necessary to (1) report in an integrated manner about the performance of the organization and (2) make well-informed decisions for long-term value creation.
- c. Through integrated thinking, an organization learns how the interdependencies among its financial and nonfinancial aspects interact and affect value creation. Thus, integrated thinking is the basis for the externally communicated integrated report.
- d. Integrated thinking cannot be done in a stand-alone department. Different departments must work together to measure and report the value creation of the organization for both itself and its environment.

3. Integrated Report

- a. The product of IR and integrated thinking is the integrated report. In the context of the organization's external environment, an integrated report concisely communicates how strategy, governance, performance, and prospects lead to value creation over the short, medium, and long term.
- b. Integrated reports should be prepared in accordance with the <IR> Framework published in 2013 by the IIRC.
- c. An integrated report provides a holistic view of the organization for all relevant stakeholders.
- d. An integrated report is **not** a mere combination of the annual report with a separate sustainability or corporate social responsibility report. It should be a designated, identifiable communication.

4. Purpose of the Integrated Report

- a. The main goal of an organization's integrated report is to give providers of financial capital more information about how the entity creates value over a period of time.
- b. An integrated report also benefits other stakeholders who are interested in the entity's effectiveness in creating value, such as employees, customers, suppliers, communities in which the entity is located, and assorted regulators.

6.2 FUNDAMENTAL CONCEPTS OF INTEGRATED REPORTING

Overview

- a. The main purpose of IR is to show how an organization creates value over time. An entity's value-creation process is influenced by
 - 1) The external environment in which the organization operates,
 - 2) Relationships with stakeholders, and
 - 3) Various resources (e.g., the six capitals defined below).
- b. These topics are addressed in the integrated report based on the three required fundamental concepts:
 - 1) Value creation,
 - 2) The six capitals, and
 - 3) The value-creation process.
- c. By focusing on the fundamental concepts defined by the <IR> Framework, an organization provides not only financial information but also context about its operations and goals over the short, medium, and long term.

1. Fundamental Concept 1 – Value Creation

- a. When an organization sells a product, it creates value for itself (in the form of revenues) and indirectly for the customer (the form of which depends on the nature of the good or service sold). This process affects not only the financial aspect of the organization but also its reputation and its relationship with its stakeholders.
- b. In selling products to customers, an organization is operating within a social environment, and the outcomes of these interactions affect the organization's **social license to operate**.
- c. When the value created by the organization for itself or for others is **material**, it should be incorporated in the integrated report.
- d. Externalities also need to be included in the integrated report. **Externalities** are the positive and/or negative effects on the capitals not owned by the organization itself.

2. Fundamental Concept 2 – Capitals

- a. Value creation depends on different sources, defined as the six capitals:
 - 1) Financial capital
 - 2) Natural capital
 - 3) Manufactured capital
 - 4) Intellectual capital
 - 5) Human capital
 - 6) Social and relationship capital
- b. The six capitals are affected by the regular course of business; i.e., they are increased, decreased, or transformed by the activities and outputs of the organization. For example,
 - 1) By selling a product and making a profit, financial capital increases.
 - 2) By having a toxic substance leak in a plant, natural (environmental) capital decreases.
 - 3) By filing a patent on a new product, intellectual capital increases.

- c. Not all capitals are equally important for a given organization. Also, sometimes one capital may need to decrease in order to make another capital increase.
 - 1) Value can be created over different capitals and is not necessarily measured in financial capital.
 - 2) Value creation in the sense of IR is not only seen as an increase of the overall stock of capitals, but also as preservation of this overall value.
- d. **Description and Interpretation of the Capitals**
 - 1) **Financial capital** is the available pool of funds.
 - a) It is the liquid capital available through financing for production of goods or provision of services (e.g., equity, debt, investments, or operational activities).
 - 2) **Natural capital** consists of the renewable and nonrenewable environmental resources and processes that provide goods or services that support the organization's prosperity.
 - a) These include land, water, air, minerals, biodiversity, and ecosystems.
 - 3) **Manufactured capital** consists of manufactured tangible objects.
 - a) It is the physical capital (not natural capital) available for the production of goods or provision of services.
 - i) Examples include property, plant, and equipment and infrastructures, such as roads and ports.
 - b) Manufactured capital is not about ownership but about availability of the tangible objects.
 - i) For example, the closure of a port a company relies on due to political irregularities decreases its manufactured capital, even though the port is not owned by the company itself.
 - 4) **Intellectual capital** is the intangible knowledge of the organization. It includes intellectual property, e.g., software, patents, copyrights, and licenses, and organizational capital, e.g., systems, procedures, and protocols.
 - a) Employees' competencies are not intellectual capital. They can be transferred to another organization.
 - 5) **Human capital** refers to employees' competencies, abilities, and experience, including
 - a) Training and development;
 - b) Ability and willingness to understand and implement the organization's strategy;
 - c) Loyalty to the organization and motivation for the job; and
 - d) Support for governance, risk management, and ethics.
 - 6) **Social and relationship capital** is the relationship of the organization with the environment in which it operates. This includes
 - a) Networks between and among the organization, stakeholders, institutions, communities, and shareholders
 - b) Brand recognition and the external reputation of the organization
 - c) Key stakeholders' willingness to share information, maintain the values and goals of the organization, and keep in touch with the organization
 - d) The organization's social license to operate

e. Capitals in the Integrated Report

- 1) All six capitals need not be present in every integrated report.
- 2) Some capitals may not be relevant for a particular organization.
- 3) Capitals need not be identified by the same name in every integrated report.
- 4) Organizations are allowed to show the change in value of the capitals by combining them or by using different names.
- 5) An integrated report need not be structured using the six capitals.
 - a) The capitals merely show how and through which vehicles value is created for the organization.
- 6) The capitals ensure that organizations at least think about all the different forms of capital affected by their operations.

Examples of Organizations' Reporting Metrics for Types of Capital

Capital	Organization	Example of Metric in IR
Financial capital	Prudential Financial (Financial services, North America)	Return on equity
	General Electric (Technology, North America)	Cash from operating activities
Natural capital	Coca-Cola Hellenic Bottling Company (Food and beverages, Europe)	Water resources used
	PostNL (Postal company, Europe)	CO ₂ emissions per package delivered
Manufactured capital	Royal Bafokeng Platinum, Ltd. (Mining, Africa)	Mining infrastructure above and below ground
	Transnet (Consumer services, Africa)	Multi-cargo ports, rolling stock
Intellectual capital	Philips (Technology, Europe)	Investments in R&D
Human capital	Sasol (Oil and gas, Africa)	Skilled and experienced employees
	CIMA (Professional services, Europe)	Number of new students attracted
	Smithfield (Consumer goods, North America)	Employee injury rates
Social and relationship capital	Schiphol Group (Airport, Europe)	Network of destinations
	Clorox (Consumer goods, North America)	Number of employee volunteer hours

3. Fundamental Concept 3 – The Value-Creation Process

- a. The value-creation process consists of the following elements:
 - 1) **Inputs** are resources and relationships (capitals) used in business activities.
 - a) The organization obtains inputs from the external environment.
 - 2) The **business model** transforms inputs through business activities into outputs and outcomes.
 - a) A business model is sensitive to changes in quality, cost, and availability of inputs. It may or may not be able to adapt quickly to changes.
 - 3) **Business activities** are the core of the business model that transforms the inputs (capitals) into outputs.
 - 4) **Outputs** are the results of the organization's business activities.
 - a) Outputs may be products, services, by-products, or waste.
 - 5) **Outcomes** are the internal and external effects of business activities and outputs on the capitals.
 - a) Outcomes relate directly to the external environment and are not necessarily controlled by the organization.
 - b) Outcomes determine the status for new inputs. In this sense, the value-creation process is circular.



AUTHOR'S NOTE

The way the organization shows the three fundamental concepts in the integrated report is not fixed. Organizations can either show all of the capitals and depict the whole value-creation process or they can stress certain details.

6.3 GUIDING PRINCIPLES AND CONTENT ELEMENTS OF THE INTEGRATED REPORT

1. Overview of Guiding Principles

- a. Preparing an integrated report involves certain challenges. The Guiding Principles are designed by the IIRC to help preparers of integrated reports present information.
- b. The Guiding Principles are guidelines and cannot always be applied strictly.
- c. Judgment is needed in applying these guidelines because they sometimes conflict (e.g., conciseness and completeness).

2. Guiding Principles

a. Strategic Focus and Future Orientation

- 1) The report should indicate how the strategy affects the ability to create value in the short, medium, and long term; how the strategy affects the capitals; and how the availability, quality, and affordability of capitals contribute to creating value.

b. Connectivity of Information

- 1) The integrated report should show a holistic picture of the interrelatedness of different aspects of the organization.
- 2) The report should explain how these factors affect the ability of the organization to create value over time.

c. Stakeholder Relationships

- 1) The key stakeholders and the quality of their relationship with the organization should be reported.

d. Materiality

- 1) The organization should disclose information that has a substantial effect on its ability to create value over time.

e. Conciseness

- 1) An integrated report should be clear and concise.
- 2) A judgment must be made as to what information should be included in the report.

f. Reliability and Completeness

- 1) An integrated report should be free of material errors.
- 2) An integrated report must include all material information, whether it is positive or negative.

g. Consistency and Comparability

- 1) The material information in an integrated report should be consistent over time.
- 2) Changes and improvements made should be explained.
- 3) Comparability among organizations is difficult due to the application of the <IR> Framework to the specific situation of each organization.

3. Overview of Content Elements

- a. Each integrated report should contain the nine Content Elements.
 - 1) Each of the Content Elements is explained in the <IR> Framework and includes one question that should be answered in the integrated report.
 - 2) Answering questions rather than checking off a list of requirements makes the content of the integrated report adaptable for different organizations in different situations.

4. Relationships among Content Elements

a. The Content Elements are interrelated.

- 1) The information gathered from answering the questions should be presented in a logical way.

5. The Content Elements and the questions posed include the following:

a. **Organizational Overview and External Environment**

- 1) *What does the organization do, and what are the circumstances under which it operates?*
 - a) An integrated report identifies the market in which the organization operates, its mission and vision, competitive landscape, key quantitative information, stakeholder needs, external environment changes, and reactions to external environment changes.

b. **Governance**

- 1) *How does the organization's governance structure support its ability to create value in the short, medium, and long term?*
 - a) An integrated report provides insight about an organization's leadership structure, diversity, strategic decision-making processes, values and ethics, and linkage between remuneration and value creation.

c. **Business Model**

- 1) *What is the organization's business model?*
 - a) An organization's business model is part of its value-creation process and is usually supported by a narrative, identification of critical stakeholder and other (e.g., materials) dependencies, and a diagram or flowchart.

d. **Risks and Opportunities**

- 1) *What are the specific risks and opportunities that affect the organization's ability to create value over the short, medium, and long term, and how is the organization dealing with them?*
 - a) Risks and opportunities may be internal, external, or a combination. An assessment of materiality should be made in the event the identified risk actually occurs.

e. **Strategy and Resource Allocation**

- 1) *Where does the organization want to go, and how does it intend to get there?*
 - a) Strategic objectives, resource allocation plans, and measurement of goal achievement are often included.
 - b) This Content Element is closely linked to the business model and external environment.

f. **Performance**

- 1) *To what extent has the organization achieved its strategic objectives for the period, and what are its outcomes in terms of effects on the capitals?*
 - a) Qualitative and quantitative measures should be used.
 - b) Linkages between historical performance and future plans are important.
 - c) Connections between financial performance and performance of other capitals can be made.

g. Outlook

1) *What challenges and uncertainties is the organization likely to encounter in pursuing its strategy, and what are the potential implications for its business model and future performance?*

- a) An analysis is usually made of expectations about the future, possible changes and responses, and the organization's ability to react to those changes.

h. Basis of Preparation and Presentation

1) *How does the organization determine what matters to include in the integrated report, and how are such matters quantified or evaluated?*

- a) An overview of the materiality determination process (often shown as a materiality matrix) is provided.
- b) The organization identifies the reporting standards and other frameworks used in its preparation.

i. General Reporting Guidance

1) This item in the Content Elements is not based on a question. It provides information about essential issues that should be considered during the preparation process, such as materiality, capital disclosures, and an explanation of the business model.

The Value-Creation Process Described by the <IR> Framework

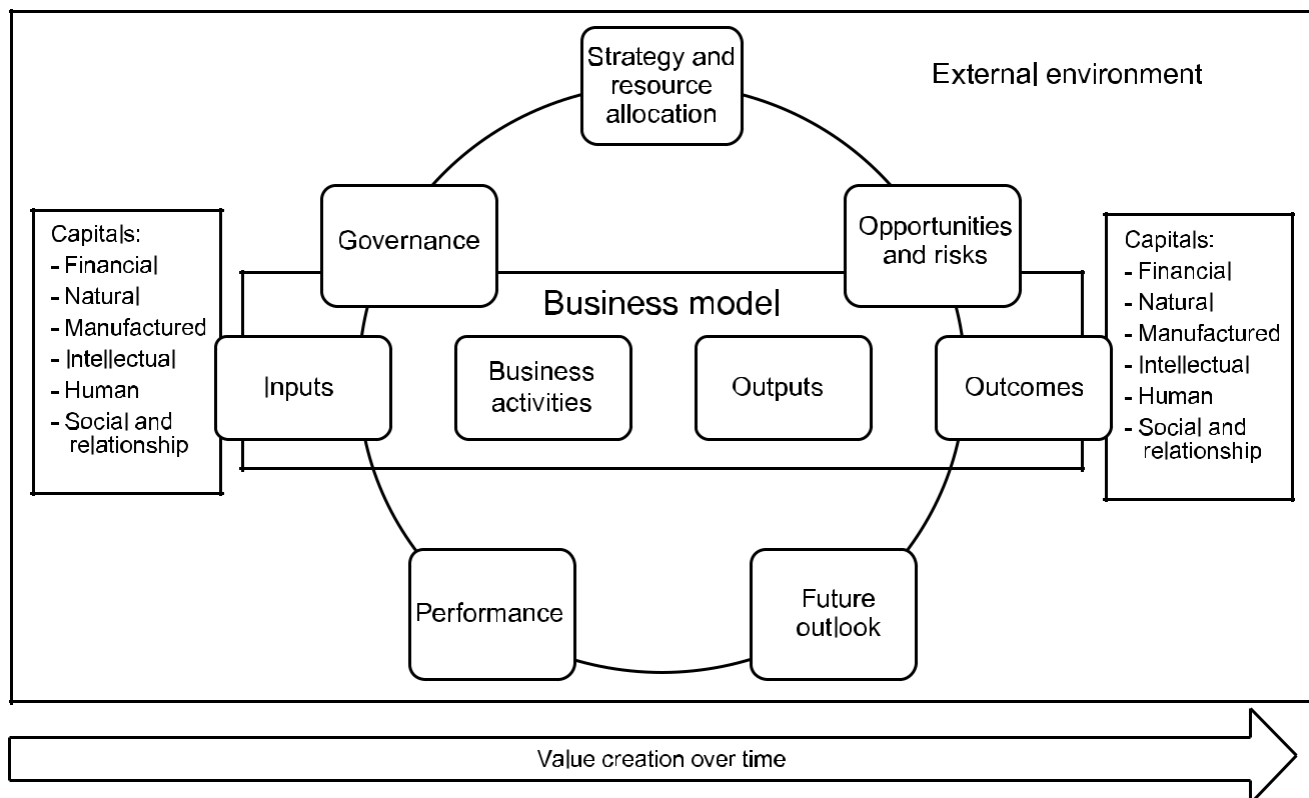


Figure 6-1

6.4 ADOPTION OF INTEGRATED REPORTING

1. Challenges

- a. An organization must be aware of the following concepts and challenges:
 - 1) **Materiality.** Because IR follows a multi-capital approach, a provider needs a materiality filter. For example, what financial capital information does a user of the report want to know? A successful start to integrated reporting requires a proper materiality definition process.
 - 2) **Data quality.** Because material matters are communicated in the report, the quality of nonfinancial data relied on is crucial. Internal controls for nonfinancial data must ensure quality.
 - 3) **Assurance.** Although validating integrated reports and, more specifically, nonfinancial data may be difficult, it enhances the quality and comparability of integrated reports.
 - 4) **Lack of standards.** No universally accepted standards exist for measuring and reporting value creation and nonfinancial information, making it difficult to communicate this kind of information. Adoption of the <IR> Framework, as well as other frameworks, such as the Global Reporting Initiative (GRI) or Sustainability Accounting Standards Board (SASB), can help overcome this problem.
 - 5) **Tone at the top.** Possibly the most important aspect in the success of IR is the tone at the top. To adopt IR, support from the CEO and the board of directors is absolutely necessary.

2. Building the Business Case for Integrated Reporting

- a. The adoption of IR has many challenges and benefits. Stressing these benefits can help build the business case for integrated reporting worldwide.
 - 1) The benefits of IR include
 - a) Linking financial and nonfinancial information, which provides more clarity about the value-creation process
 - b) Better decision making and resource allocation
 - c) Better relationships with stakeholders
 - d) More employee engagement
 - e) Lower reputational risk
 - f) More committed customers
 - g) Better measurement and internal control systems for nonfinancial information
 - h) Breaking down silos within the organization by requiring different departments to work together to produce an integrated report
 - i) Lower costs of, for example, debt and equity over the long term
 - 2) IR incurs costs. When building the business case for IR, management needs to address the costs associated with
 - a) Collecting and analyzing new types of data;
 - b) Setting up new data sets;
 - c) Hiring people with the proper analytical skills;
 - d) Setting up new internal control systems;
 - e) Assurance on the integrated report; and
 - f) Disclosing true (but sometimes negative or competitive) information.

APPENDIX B

IFRS DIFFERENCES

The following appendix provides additional testable differences between IFRS and U.S. GAAP.

Topic	IFRS	U.S. GAAP
<u>Investment in debt securities</u>		
The classification and measurement of investments in debt securities contingent on the investor's business model for managing them and the nature of cash flows from them	The investment is measured at amortized cost when the only business purpose is to collect contractual cash flows solely from principal and interest and the fair value option was not elected. The investment is measured at fair value through profit or loss when (1) the only business purpose is to collect contractual cash flows solely from principal and interest and the fair value option was elected; (2) the contractual cash flows are not solely from principal and interest; (3) the business purpose is to collect contractual cash flows and sell the investment, the cash flows are solely from principal and interest, and the fair value option was elected; or (4) the business purpose is something other than to collect contractual cash flows and sell the investment, the contractual cash flows are not solely from principal and interest, and the fair value option was elected. The investment is measured at fair value through other comprehensive income (OCI) when the business purpose is to collect contractual cash flows and sell the investment, the cash flows are solely from principal and interest, and the fair value option was not elected.	Held-to-maturity securities are reported at amortized cost. Trading securities are initially recorded at cost and remeasured at each balance sheet date at fair value through profit or loss. Available-for-sale securities are initially recorded at cost and remeasured at each balance sheet date at fair value through OCI.
<u>Investment in equity securities</u>		
Accounting for investment in equity securities that does not result in significant influence or control over the investee	The investment is measured at fair value through profit or loss (as under U.S. GAAP). The investor may irrevocably elect at initial recognition to measure the investment that is not held for trading at fair value through OCI. The holding gains and losses accumulated in OCI are never reclassified to profit or loss.	The investment is measured at fair value through net income. When the fair value is not readily determinable, the investment may be measured at cost minus impairment (if any), plus or minus changes resulting from observable price changes for the identical or a similar investment of the same issuer.

Topic	IFRS	U.S. GAAP
<u>Equity method of accounting</u>		
Accounting method for significant influence investments (20%-50% ownership but not control)	The equity method is applied for such investments.	The investor may choose either the (1) fair value option (FVO) or (2) equity method to account for such investments.
<u>Intangible assets</u>		
Test for impairment – Goodwill	A qualitative assessment is not an option. The test for impairment of a cash-generating unit (CGU) to which goodwill has been allocated is whether the carrying amount of the CGU (including allocated goodwill) exceeds its recoverable amount. Thus, the test has one step. An impairment loss for a CGU is allocated first to reduce allocated goodwill to zero and then pro rata to other assets of the CGU.	An entity may first perform a qualitative assessment to determine whether it is necessary to perform the quantitative impairment test. The fair value of a reporting unit is compared with its carrying amount, including goodwill. A goodwill impairment loss is recognized for the amount that the carrying amount of a reporting unit, including goodwill, exceeds its fair value. The loss is limited to the total amount of goodwill allocated to that reporting unit.
<u>Leases</u>		
Accounting by lessor	The lessor classifies a lease as a finance lease (similar to sales-type lease under U.S. GAAP) or an operating lease. A direct financing lease is not a classification option for the lessor.	The lessor classifies a lease as a sales-type lease, an operating lease, or a direct financing lease.
<u>Noncontrolling interest (NCI)</u>		
Measurement of NCI on the business combination date	A NCI may be measured at (1) fair value or (2) a proportionate share of the fair value of the acquiree's identifiable net assets.	A NCI is measured at its acquisition-date fair value.
<u>Statement of cash flows</u>		
Classification of dividends paid	Dividends paid are classified as cash outflows either from (1) operating activities or (2) financing activities.	Dividends paid are cash outflows from financing activities.
Classification of interest paid	Interest paid is classified as a cash outflow either from (1) operating activities or (2) financing activities.	Interest paid is a cash outflow from operating activities.
Classification of interest and dividends received	Interest and dividends received are classified as cash inflows either from (1) operating activities or (2) investing activities.	Interest and dividends received are cash inflows from operating activities.

GLEIM®

**2021
EDITION**

CMA REVIEW

PART 1:

FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS



B. Planning, Budgeting, and Forecasting (20% - Levels A, B, and C)

1. *Strategic planning*

- a. Analysis of external and internal factors affecting strategy (12.4)
- b. Long-term mission and goals (12.4)
- c. Alignment of tactics with long-term strategic goals (12.4-12.5, 13.1, 16.4)
- d. Strategic planning models and analytical techniques (12.4-12.5, 16.4)
- e. Characteristics of a successful strategic planning process (12.4-12.5, 16.4)

2. *Budgeting concepts*

- a. Operations and performance goals (13.1)
- b. Characteristics of a successful budget process (13.1)
- c. Resource allocation (SU 13)
- d. Other budgeting concepts (13.2-13.4, SU 14)

3. ***Forecasting techniques***

- a. Regression analysis (12.1)
- b. Learning curve analysis (12.2)
- c. Expected value (12.3)

4. ***Budgeting methodologies***

- a. Annual business plans (master budgets) (13.3)
- b. Project budgeting (13.4, SU 14)
- c. Activity-based budgeting (13.4)
- d. Zero-based budgeting (13.4)
- e. Continuous (rolling) budgets (13.4)
- f. Flexible budgeting (15.2)

5. ***Annual profit plan and supporting schedules***

- a. Operational budgets (14.1-14.3)
- b. Financial budgets (14.4-14.6)
- c. Capital budgets (13.3, 14.4)

6. ***Top-level planning and analysis***

- a. Pro forma income (14.6)
- b. Financial statement projections (14.6)
- c. Cash flow projections (14.5-14.6)

STUDY UNIT TWELVE

ANALYSIS, FORECASTING, AND STRATEGY

12.1	Correlation and Regression	1
12.2	Learning Curve Analysis	5
12.3	Expected Value and Sensitivity Analysis	8
12.4	Strategic Management	13
12.5	Strategic Planning	21

This study unit is the **first of three** on **planning, budgeting, and forecasting**. The relative weight assigned to this major topic in Part 1 of the exam is **20%**. The three study units are

Study Unit 12: Analysis, Forecasting, and Strategy

Study Unit 13: Budgeting -- Concepts and Methodologies

Study Unit 14: Budgeting -- Calculations and Pro Forma Financial Statements

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

12.1 CORRELATION AND REGRESSION

1. Forecasting Methods

- a. Forecasts are the basis for business plans and budgets. Forecasts are used to project product demand, inventory levels, cash flow, etc.
 - 1) **Qualitative methods** of forecasting rely on the manager's experience and intuition.
 - 2) **Quantitative methods** use mathematical models and graphs.
 - a) When some factor in the organization's environment is plotted on the x-axis, the technique is causal relationship forecasting.
 - b) When time periods are plotted on the x-axis, the technique is time-series analysis.

2. Correlation Analysis

- a. Correlation analysis is the foundation of any quantitative method of forecasting.
 - 1) Correlation is the strength of the linear relationship between two variables, expressed mathematically in terms of the **coefficient of correlation (r)**. It can be graphically depicted by plotting the values for the variables on a graph in the form of a scatter diagram.
 - a) The value of r ranges from 1 (perfect direct relationship) to -1 (perfect inverse relationship).
 - i) The more the scatter pattern resembles a straight line, the greater the absolute value of r .

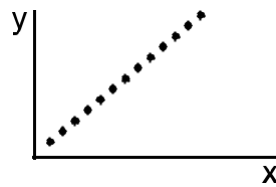
b) **Perfect direct relationship ($r = 1$)**

Figure 12-1

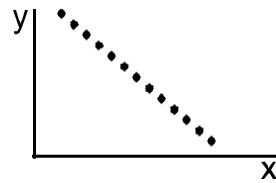
c) **Perfect inverse relationship ($r = -1$)**

Figure 12-2

d) **Strong direct relationship ($r = 0.7$)**

Figure 12-3

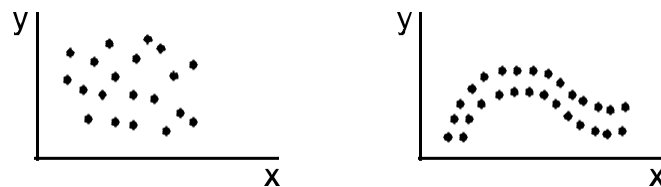
e) **No linear relationship ($r = 0$)**

Figure 12-4

- i) Note from the right-hand graph of the pair above that a coefficient of correlation of zero does not mean the variables have no relationship, only that any relationship cannot be expressed as a linear equation.
- 2) The **coefficient of determination (r^2)**, or the coefficient of correlation squared, is a measure of how good the fit between the two variables is.
 - a) Mathematically, the coefficient of determination is the proportion of the total variation in the dependent variable that is accounted for by the independent variable.

EXAMPLE 12-1 Coefficient of Determination

A car dealership determines that new car sales are a function of disposable income with a coefficient of correlation of .8.

This is equivalent to stating that 64% ($.8^2$) of the variation of new car sales from the average can be explained by changes in disposable income.

3. Regression Analysis



SUCCESS TIP

CMA candidates should be able to demonstrate an understanding of the measures associated with simple regression as well as calculate the result of the equation. They should also be able to identify when it is appropriate to use multiple regression. Therefore, candidates should memorize and fully understand the formulas for both simple and multiple regression analysis.

- a. Regression analysis, also called least-squares analysis, is the process of deriving the linear equation that describes the relationship between two variables with a nonzero coefficient of correlation.
 - 1) **Simple regression** is used given one independent variable.
 - a) The simple regression equation is the algebraic formula for a straight line:

$$y = a + bx$$

If: y = the dependent variable
 a = the y-intercept
 b = the slope of the regression line
 (i.e., the regression coefficient)
 x = the independent variable
 - b) The best straight line that fits a set of data points is derived using calculus to minimize the sum of the squares of the vertical distances of each point to the line (hence the name least-squares method).

EXAMPLE 12-2 Simple Regression

A firm has collected observations on advertising expenditures and annual sales.

Advertising (\$000s)	Sales (\$000,000s)
71	26.3
31	13.9
50	19.8
60	22.9
35	15.1

Solving with the least-squares method reveals that expected sales equal \$4.2 million plus 311.741 times the advertising expenditure.

$$y = \$4,200,000 + 311.741x$$

The observations are graphed as follows:

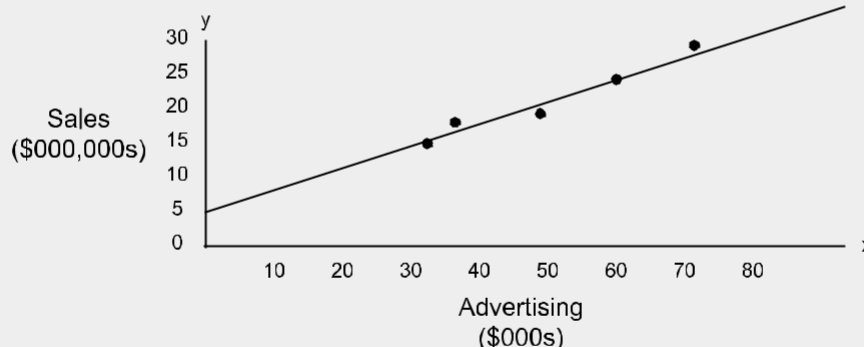


Figure 12-5

The firm can now project the amount it will have to spend on advertising to generate \$32,000,000 in sales.

$$\begin{aligned}
 y &= \$4,200,000 + 311.741x \\
 \$32,000,000 &= \$4,200,000 + 311.741x \\
 311.741x &= \$27,800,000 \\
 x &= \$89,177
 \end{aligned}$$

2) **Multiple regression** is used given more than one independent variable.

- a) Example 12-2 relating advertising to sales is clearly unrealistic. Sales are dependent upon more than just advertising expenditures.
- b) Multiple regression allows a firm to identify many factors (independent variables) and to weight each one according to its influence on the overall outcome.

$$y = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + \text{etc.}$$

b. Benefits and Limitations of Regression Analysis

1) **Assumptions** of the linear regression model include the following.

- a) The linear relationship established for x and y is only valid across the **relevant range** (i.e., the range within which per-unit variable costs remain constant and fixed costs are not changeable). The user must identify the relevant range and ensure that (s)he does not project the relationship beyond it.
 - i) A negative y-intercept in the simple regression equation usually indicates that it is outside the relevant range.
- b) Regression analysis assumes that **past relationships** can be validly projected into the future. Economists call this the *ceteris paribus* assumption.
 - i) Thus, a limitation of the regression method is that it can only be used when cost patterns remain unchanged from prior periods.
- c) The values of y around the regression line for a given value of x have a normal distribution and a mean of zero.
 - i) Moreover, the variances of the subpopulations of y are constant for different values of x, i.e., **homoscedasticity** or **constant variance**.
 - ii) The values of y also must be statistically independent.

2) Regression analysis is particularly valuable for budgeting and cost accounting purposes.

- a) Regression analysis is almost a necessity for computing the fixed and variable portions of mixed costs for flexible budgeting.
 - i) The y-intercept is the fixed portion and the slope of the regression line is the variable portion.

3) **Regression does not determine causality.**

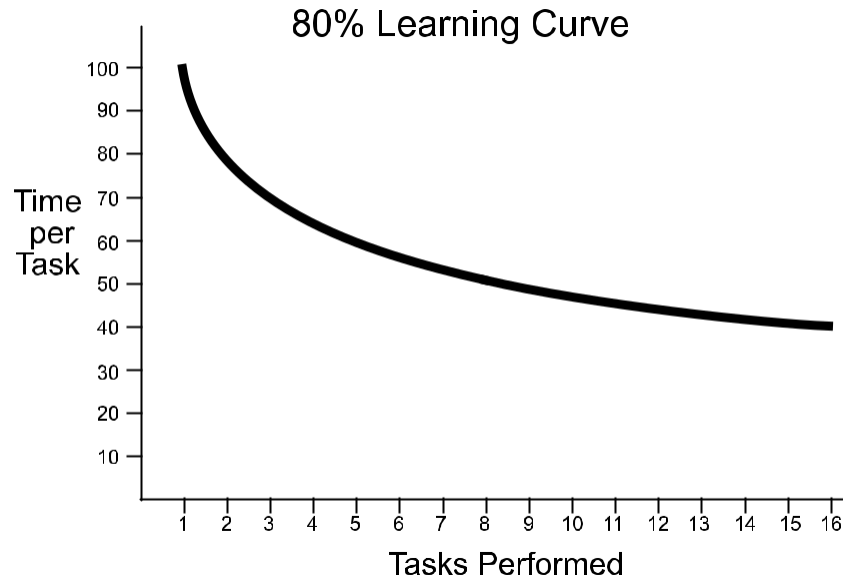
- a) Although x and y move together, the apparent relationship may be caused by some other factor.
 - i) For instance, car wash sales volume and sunny weather are strongly correlated, but car wash sales do not cause sunny weather.

12.2 LEARNING CURVE ANALYSIS

1. Learning Curves

a. Learning curve analysis reflects the increased rate at which people perform tasks as they gain experience.

- 1) The time required to perform a given task becomes progressively shorter during the early stages of production.



- 2) The curve is usually expressed as a percentage of reduced time each time **cumulative production doubles**.

- a) When determining whether cumulative production doubles, it is important to look at units produced as well as batch numbers.

EXAMPLE 12-3 Learning Curve

A firm produces units in batches of 100 and experiences an 80% learning curve.

<u>Batch</u>	<u>Cumulative Units Produced</u>	<u>Doubling</u>
1	100	
2	200	1 st doubling from 100 to 200
3	300	
4	400	2 nd doubling from 200 to 400
5	500	
6	600	
7	700	
8	800	3 rd doubling from 400 to 800

- 3) In practice, the most common percentage used is 80%. However, on the exam, be prepared to see a variety of percentages.

EXAMPLE 12-4 Learning Curves of 70%, 80%, and 90%

The following table illustrates learning curves for a product whose first unit takes 100 minutes to produce:

Batch	Cumulative Units Produced	70% Cumulative Average Time per Unit	80% Cumulative Average Time per Unit	90% Cumulative Average Time per Unit
1 (Unit 1)	1	100	100	100
2 (Unit 2)	2	70 (100 × 70%)	80 (100 × 80%)	90 (100 × 90%)
3 (Units 3-4)	4	49 (70 × 70%)	64 (80 × 80%)	81 (90 × 90%)
4 (Units 5-8)	8	34.3 (49 × 70%)	51.2 (64 × 80%)	72.9 (81 × 90%)
5 (Units 9-16)	16	24.01 (34.3 × 70%)	40.96 (51.2 × 80%)	65.61 (72.9 × 90%)

The time listed in the cumulative average time per unit column is an average of all the units produced up to that point.

Note that with an 80% learning curve upon completion of the final batch (units 9-16), the average had come down to 40.96 minutes per unit.

For it to reach this level from the 51.2 minutes it had reached at the end of the fourth batch (units 5-8), the average of the units in the fifth batch alone must have been 30.72 minutes [(40.96 minutes × 2) – 51.2 minutes].

With more sophisticated quantitative techniques, a more accurate average can be calculated of the units within each “batch.”



SUCCESS TIP

CMA candidates need to be alert as to the nature of the question being asked. Sometimes the question might ask, “What is the average time per unit after two units?” From the table in Example 12-4, with an 80% learning curve, you can see that the answer is 80. Alternatively, sometimes the question asks, “What is the time to produce the second unit?” The answer would be 60. Since the first unit took 100 minutes and the average for the two units is 80 minutes (a total of 160), the second unit must have taken only 60 minutes.

2. Application

- a. **Two methods** of applying learning curve analysis are in common use.
- 1) The **cumulative average-time learning model** projects the reduction in the cumulative average time it takes to complete a certain number of tasks.
 - 2) The **incremental unit-time learning model** projects the reduction in the incremental time it takes to complete the last task.

EXAMPLE 12-5 Learning Curve -- Cumulative Average-Time vs. Incremental Unit-Time

A firm determines that 100 minutes of labor are required to complete one unit of product. Assuming an 80% learning curve, the following table illustrates the difference between the two methods.

Learning Curve 80% at Each Doubling		Cumulative Average-Time Model		Incremental Unit-Time Model	
(A)	(B)	(A) × (B)		□ (B)	□ (B) / (A)
Unit Produced	Cumulative Average Time per Unit	Cumulative Total Time	Time Spent on Most Recent Unit	Incremental Unit Total Time	Average Time Spent on Most Recent Unit
1	100.00	100.00	100.00	100.00	100.00
2	80.00	160.00	60.00	180.00	90.00
3	70.21	210.63	50.63	250.21	83.40
4	64.00	256.00	45.37	314.21	78.55



**AUTHOR'S
NOTE**

CMA candidates will not need to know how to calculate units within a batch. You should know how to calculate the learning curves for 1, 2, and 4 unit(s) produced.

- 3) The difference between the two methods is clear in the way each calculates total time. Most CMA questions have historically used the cumulative-average-time method, and it is often called the “traditional” learning curve model.

3. Limitations

- a. The limitation of the learning curve in practice is the difficulty in knowing the shape of the learning curve.
- 1) There is no question that the learning curve effect exists, but companies typically do not know what percentage they should use in calculations until after it is too late to use the information effectively. As a result, many companies simply assume an 80% learning curve and make decisions based on those results.

12.3 EXPECTED VALUE AND SENSITIVITY ANALYSIS

1. Expected Value

- a. Expected value is a means of associating a dollar amount with each of the possible outcomes of a **probability distribution**. It should be distinguished from a **deterministic** approach, which assumes that a value is known with certainty.
 - 1) The outcome yielding the highest expected monetary value (which may or may not be the most likely one) is the optimal alternative.
 - a) The **decision** alternative is under the manager's control.
 - b) The **state of nature** is the future event whose outcome the manager is attempting to predict.
 - c) The **payoff** is the financial result of the combination of the manager's decision and the actual state of nature.
 - 2) The expected value of an event is calculated by multiplying the probability of each outcome by its payoff and summing the products.

EXAMPLE 12-6 Expected Value

An investor is considering the purchase of two identically priced pieces of property. The value of the properties will change if a road, currently planned by the state, is built.

The following are estimates that road construction will occur:

<u>Future State of Nature (SN)</u>	<u>Event</u>	<u>Probability</u>
SN 1	No road is ever built.	.1
SN 2	A road is built this year.	.2
SN 3	A road is built more than 1 year from now.	.7

The following are estimates of the values of the properties under each of the three possible events:

<u>Property</u>	<u>SN 1</u>	<u>SN 2</u>	<u>SN 3</u>
Bivens Tract	\$10,000	\$40,000	\$35,000
Newnan Tract	\$20,000	\$50,000	\$30,000

The expected value of each property is determined by multiplying the probability of each state of nature by the value under that state of nature and adding all of the products.

		<u>Expected Value</u>
Bivens Tract:	$.1(\$10,000) + .2(\$40,000) + .7(\$35,000)$	<u>= \$33,500</u>
Newnan Tract:	$.1(\$20,000) + .2(\$50,000) + .7(\$30,000)$	<u>= \$33,000</u>

Thus, Bivens Tract is the better investment.

- a) The calculation shown in Example 12-6 is often referred to as a payoff table.

- 3) The difficult aspect of constructing a payoff table is the determination of all possible outcomes of decisions and their probabilities. Thus, a probability distribution must be established.
 - a) The assigned probabilities may reflect prior experience with similar decisions, the results of research, or highly subjective estimates.
- 4) It is important to capture all values when constructing a payoff table. Otherwise, results may be inaccurate.
 - a) For example, unsold items may be sold for scrap or may require storage costs.
 - b) Failure to account for these and other factors will skew results.

EXAMPLE 12-7 Expected Value -- Scrap Option

A vendor can order 10, 20, or 30 hats. The probabilities of demand are as follows:

	Demand	Probability
10	60%	
20	30%	
30	10%	

Each hat can be sold for \$5. Unsold hats can be sold as scrap for \$1. The expected profit or loss if the ability to sell unsold hats as scrap is included when calculating the payoff table is as follows:

Decision	States of Nature			Expected Value
	Demand = 10	Demand = 20	Demand = 30	Totals
10 hats	\$50	\$ 50	\$ 50	\$50
20 hats	60	100	100	76
30 hats	70	110	150	90

$$\$50 = [.6 \times \$50] + [.3 \times \$50] + [.1 \times \$50]$$

$$\$76 = [.6 \times \$60] + [.3 \times \$100] + [.1 \times \$100]$$

$$\$90 = [.6 \times \$70] + [.3 \times \$110] + [.1 \times \$150]$$

The expected profit or loss if the ability to sell unsold hats as scrap is **not** included when calculating the payoff table is as follows:

Decision	States of Nature			Expected Value
	Demand = 10	Demand = 20	Demand = 30	Totals
10 hats	\$50	\$ 50	\$ 50	\$50
20 hats	50	100	100	70
30 hats	50	100	150	75

$$\$50 = [.6 \times \$50] + [.3 \times \$50] + [.1 \times \$50]$$

$$\$70 = [.6 \times \$50] + [.3 \times \$100] + [.1 \times \$100]$$

$$\$75 = [.6 \times \$50] + [.3 \times \$100] + [.1 \times \$150]$$

In this example, failing to include the option to sell unsold hats for \$1 results in much more similar expected values across the different decisions. Even though the result is the same, the goal is to provide the most accurate and useful information for decision-making purposes.

- 5) The expected value criterion is likely to be adopted by a decision maker who is risk neutral. However, other circumstances may cause the decision maker to be risk averse or even risk seeking.

EXAMPLE 12-8 Expected Value -- Risk Aversion

A dealer in luxury yachts may order 0, 1, or 2 yachts for this season's inventory. The dealer projects demand for the season as follows:

<u>Demand</u>	<u>Probability</u>
0 yachts	10%
1 yacht	50%
2 yachts	40%

The cost of carrying each excess yacht is \$50,000, and the gain for each yacht sold is \$200,000. The profit or loss resulting from each combination of decision and outcome is thus as follows:

<u>Decision</u>	<u>States of Nature</u>			<u>Expected Value</u>
	<u>Demand = 0</u>	<u>Demand = 1</u>	<u>Demand = 2</u>	<u>Totals</u>
Stock 0 yachts	\$ 0	\$ 0	\$ 0	\$ 0
Stock 1 yacht	(50,000)	200,000	200,000	175,000
Stock 2 yachts	(100,000)	150,000	400,000	225,000
\$175,000	= [.1 × -\$50,000] + [.5 × \$200,000] + [.4 × \$200,000]			
\$225,000	= [.1 × -\$100,000] + [.5 × \$150,000] + [.4 × \$400,000]			

In this example, a risk-averse decision maker may not wish to accept the risk of losing \$100,000 by ordering two yachts.

- 6) The **benefit** of expected value analysis is that it allows a manager to apply scientific management techniques to applications that would otherwise be guesswork.
- a) Although exact probabilities may not be known, the use of expected value analysis forces managers to evaluate decisions in a more organized manner. At the least, managers are forced to think of all of the possibilities that could happen with each decision.
- 7) A **limitation** of expected value is that it is based on repetitive trials, whereas in reality, most business decisions involve only one trial.

EXAMPLE 12-9 Expected Value -- Impossible Value

A company wishes to launch a communications satellite. The probability of launch failure is .2, and the value of the satellite if the launch fails is \$0. The probability of a successful launch is .8, and the value of the satellite would then be \$25,000,000. The expected value is calculated as follows:

$$.2(\$0) + .8(\$25,000,000) = \$20,000,000$$

But \$20,000,000 is not a possible value for a single satellite; either it flies for \$25,000,000 or it crashes for \$0.

2. Perfect Information

a. Perfect information is the certain knowledge of which state of nature will occur.

- 1) The **expected value of perfect information (EVPI)** is the additional expected value that could be obtained if a decision maker knew ahead of time which state of nature would occur.

EXAMPLE 12-10 Expected Value with Perfect Information

The yacht dealer from Example 12-8 would maximize profits if (s)he were able to determine exactly what all potential customers intended to do for the season. Since the yacht dealer has perfect information, the values used in the payoff table are based on the known demand; i.e., if demand will be 0 yachts, the dealer knows this and purchases 0 yachts, if demand will be 1 yacht, the dealer knows this and only purchases 1 yacht, etc.

Decision	States of Nature			Expected Value with Perfect Information
	Demand = 0	Demand = 1	Demand = 2	Totals
Stock 0 yachts	\$ 0	\$ 0	\$ 0	\$ 0
Stock 1 yacht	(50,000)	200,000	200,000	100,000
Stock 2 yachts	(100,000)	150,000	400,000	160,000
				<u>\$260,000</u>

The profit that could be obtained with this perfect knowledge of the market is calculated as follows:

$$\$260,000 = (.1 \times \$0) + (.5 \times \$200,000) + (.4 \times \$400,000)$$

The difference between this amount and the best choice without perfect information is the EVPI.

Expected value with perfect information	\$260,000
Expected value without perfect information	(225,000)
Expected value of perfect information	<u>\$ 35,000</u>

The dealer is therefore not willing to pay more than \$35,000 for perfect information about future demand.

- 2) The maximum anyone should be willing to pay for perfect information is the expected value of perfect information.

3. Sensitivity Analysis

a. Sensitivity analysis is the process of evaluating the effect of changes in variables on the optimum solution. Such analysis helps management analyze alternatives and estimate the consequences of possible prediction errors. Sensitivity analysis reveals how sensitive expected value calculations are to the accuracy of the initial estimates.

- 1) Sensitivity analysis is thus useful in determining whether expending additional resources to obtain better forecasts is justified.
 - a) If a change in the probabilities assigned to the various states of nature results in large changes in the expected values, the decision maker is justified in expending more effort to make better predictions about the outcomes.
 - b) The benefit of sensitivity analysis is that managers can see the effect of changed assumptions on the final objective.
 - i) For example, in a capital budgeting situation, a proposed investment might promise a return of \$10,000 per year and a rate of return of 15%. The \$10,000 is based on an estimate. What management needs to know is how acceptable the investment would be if the return was only \$6,000 per year.

EXAMPLE 12-11 Expected Value -- Probability Combinations

The yacht dealer in the expected value computation illustrated in Example 12-8 is testing different combinations of probabilities. All three of the scenarios depicted here yield the same decision (stock two yachts for the season):

Decision Alternatives	States of Nature	Payoff	Original		First Alternative		Second Alternative	
			Probability	Expected Value	Probability	Expected Value	Probability	Expected Value
Stock 0 Yachts	Demand = 0	\$ 0	0.1	\$ 0	0.5	\$ 0	0.333	\$ 0
	Demand = 1	0	0.5	0	0.1	0	0.333	0
	Demand = 2	0	0.4	0	0.4	0	0.333	0
				<u>\$ 0</u>		<u>\$ 0</u>		<u>\$ 0</u>
Stock 1 Yacht	Demand = 0	(50,000)	0.1	(5,000)	0.5	(25,000)	0.333	(16,650)
	Demand = 1	200,000	0.5	100,000	0.1	20,000	0.333	66,600
	Demand = 2	200,000	0.4	80,000	0.4	80,000	0.333	66,600
				<u>\$ 175,000</u>		<u>\$ 75,000</u>		<u>\$116,550</u>
Stock 2 Yachts	Demand = 0	(100,000)	0.1	(10,000)	0.5	(50,000)	0.333	(33,300)
	Demand = 1	150,000	0.5	75,000	0.1	15,000	0.333	49,950
	Demand = 2	400,000	0.4	160,000	0.4	160,000	0.333	133,200
				<u>\$ 225,000</u>		<u>\$ 125,000</u>		<u>\$149,850</u>

-- Continued on next page --

EXAMPLE 12-11 -- Continued

However, the following combination indicates that only one yacht should be stocked:

<u>Decision Alternatives</u>	<u>States of Nature</u>	<u>Payoff</u>	<u>Third Alternative</u>	
			<u>Probability</u>	<u>Expected Value</u>
Stock 0 Yachts	Demand = 0	\$ 0	0.1	\$ 0
	Demand = 1	0	0.8	0
	Demand = 2	0	0.1	0
				<u>\$ 0</u>
Stock 1 Yacht	Demand = 0	(50,000)	0.1	\$ (5,000)
	Demand = 1	200,000	0.8	160,000
	Demand = 2	200,000	0.1	20,000
				<u>\$175,000</u>
Stock 2 Yachts	Demand = 0	(100,000)	0.1	\$ (10,000)
	Demand = 1	150,000	0.8	120,000
	Demand = 2	400,000	0.1	40,000
				<u>\$150,000</u>

Clearly, the more accurately the dealer is able to anticipate demand, the more profit (s)he will make. In this case, the dealer considers it worthwhile to expend further resources gathering more data about market conditions for yachts.

- 2) A trial-and-error method inherent in sensitivity analysis is facilitated by the use of computer software.
- 3) A major use of sensitivity analysis is in capital budgeting, where small changes in prevailing interest rates or payoff amounts can significantly affect the profitability of a project.

12.4 STRATEGIC MANAGEMENT

1. Strategic Management

- a. Strategic management sets overall objectives for an entity and guides the process of reaching those objectives. It is the responsibility of upper management.
 - 1) Strategic planning is the design and implementation of the specific steps and processes necessary to reach the overall objectives.
- b. Strategic management and strategic planning are thus closely linked. By their nature, strategic management and strategic planning have a long-term planning horizon.

2. Steps in the Strategic Management Process

a. Strategic management is a five-stage process:

- 1) The board of directors drafts the organization's mission statement.
- 2) The organization performs a situational analysis, also called a SWOT analysis.
- 3) Based on the results of the situational analysis, upper management develops strategies describing how the mission will be achieved.
- 4) Strategic plans are implemented through the execution of component plans at each level of the entity.
- 5) Strategic controls and feedback are used to monitor progress, isolate problems, and take corrective action. Over the long term, feedback is the basis for adjusting the original mission and objectives.

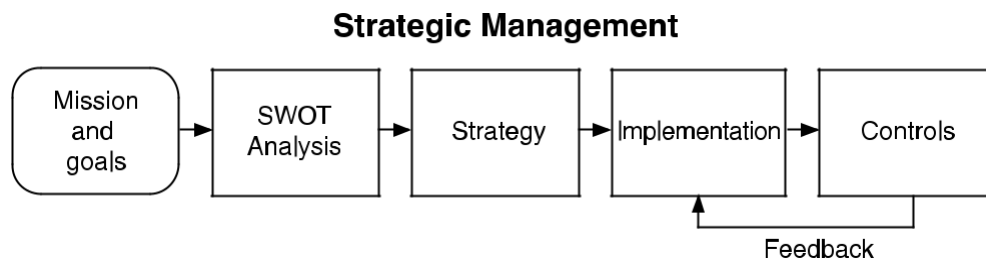


Figure 12-7

b. The **mission statement** summarizes the entity's reason for existing. It provides the framework for formulation of the company's strategies.

- 1) Missions tend to be most effective when they consist of a single sentence.
 - a) For example, the mission of Starbucks Coffee Company is "to inspire and nurture the human spirit – one person, one cup and one neighborhood at a time."
- 2) Missions tend to be stated in general terms. Setting specific objectives in the mission statement can limit an entity's ability to respond to a changing marketplace.
- 3) A mission statement and a vision statement serve different purposes for a company but are sometimes confused with each other. A mission statement describes what a company wants to achieve now, while a **vision statement** outlines what a company wants to achieve in the future.
 - a) The mission statement concentrates on the present; it defines the customers and critical processes, and it explains the desired level of performance.
 - b) The vision statement focuses on the future; it is a source of inspiration and motivation.
 - i) It often describes not just the future of the organization, but the future of the industry or society in which the organization operates.
 - ii) It answers the question, "Where do we want to be in the future?"
 - c) A mission statement describes how a company will achieve its vision. It defines the entity's purpose and primary objectives and answers the question, "What do we do, and what makes us different?"

- c. The **situational analysis** is most often called a SWOT analysis because it identifies the entity's strengths, weaknesses, opportunities, and threats.
 - 1) **Strengths and weaknesses** are usually identified by considering the entity's capabilities and resources (its internal environment).
 - a) What the entity does particularly well or has in greater abundance are its core competencies. But many entities may have the same core competencies (cutting-edge IT, efficient distribution, etc.).
 - b) An entity gains a competitive advantage in the marketplace by developing one or more distinctive competencies, i.e., competencies that are unlike those of its competitors.
 - 2) **Opportunities and threats** exist in the entity's external environment. They are identified by considering
 - a) Macroenvironment factors (economic, demographic, political, legal, social, cultural, and technical) and
 - b) Microenvironment factors (suppliers, customers, distributors, competitors, and other competitive factors in the industry).
 - 3) The SWOT analysis tends to highlight the basic factors of cost, quality, and the speed of product development and delivery.
- d. Based on the results of the situational analysis, upper management develops a **group of strategies** describing how the mission will be achieved. The strategies answer such questions as
 - 1) "Which lines of business will we be in?"
 - 2) "How do we penetrate and compete in the international marketplace?"
 - 3) "How will this line of business reach its objectives that contribute to achievement of the overall entity's mission?"
 - 4) "How do we perform each strategic business unit's basic processes (materials handling, assembly, shipping, human resources, customer relations, etc.) as efficiently as possible?"
- e. **Implementing** the chosen strategies involves every employee at every level of the entity. Incentive systems and employee performance evaluations must be designed so that they encourage employees to focus their efforts on achieving the entity's objectives.
 - 1) This approach requires communication among senior managers, who devise strategies; middle managers, who supervise and evaluate employees; and human resources managers, who must approve evaluation and compensation plans.
- f. As plans are executed at each organizational level, **strategic controls and feedback** allow management to determine the degree of progress toward the stated objectives.
 - 1) For controls to be effective, standards against which performance can be measured must be established. Then, the results of actual performance must be measured against the standards and reported to the appropriate managers.
 - a) If performance is unsatisfactory, managers take corrective action.
 - 2) Results are sent to higher-level management for continual refinement of the strategies.

3. Porter's Five Competitive Forces

- a. Business theorist Michael E. Porter has developed a model of the structure of industries and competition. It includes an analysis of the five competitive forces that determine long-term profitability as measured by long-term return on investment.
 - 1) This analysis includes an evaluation of the basic economic and technical characteristics that determine the strength of each force and the attractiveness of the industry. The competitive forces are depicted in the diagram below and discussed in detail on the following pages.

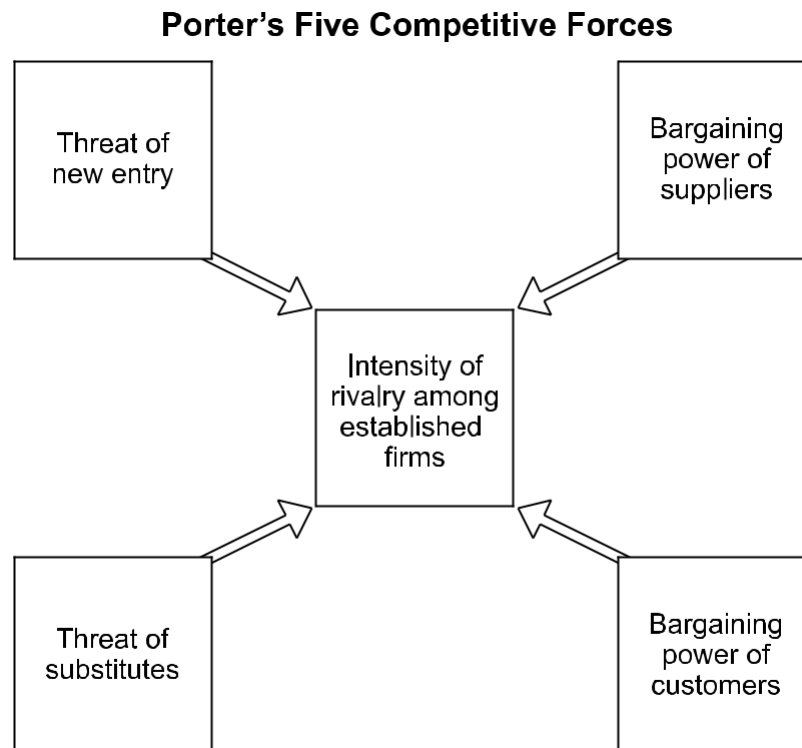


Figure 12-8

- b. **Rivalry among existing firms** will be intense when an industry contains many strong competitors. Price-cutting, large advertising budgets, and frequent introduction of new products are typical. The intensity of rivalry and the threat of entry vary with the following factors:
 - 1) The stage of the industry life cycle, e.g., rapid growth, growth, maturity, decline, or rapid decline
 - a) Thus, growth is preferable to decline. In a declining or even a stable industry, a firm's growth must come from winning other firms' customers, thereby strengthening competition.
 - 2) The distinctions among products (product differentiation) and the costs of switching from one competitor's product to another
 - a) Less differentiation tends to heighten competition based on price, with price cutting leading to lower profits. But high costs of switching suppliers weaken competition.

- 3) Whether fixed costs are high in relation to variable costs
 - a) High fixed costs indicate that rivalry will be intense. The greater the cost to generate a given amount of sales revenues, the greater the investment intensity and the need to operate at or near capacity. Hence, price cutting to sustain demand is typical.
- 4) Capacity expansion
 - a) If the size of the expansion must be large to achieve economies of scale, competition will be more intense. The need for large-scale expansion to achieve production efficiency may result in an excess of industry capacity over demand.
- c. The prospects of long-term profitability depend on the industry's **barriers to entry**.
 - 1) Factors that increase the threat of entry are the following:
 - a) Economies of scale (and learning curve effects) are not significant.
 - b) Brand identity of existing products is weak.
 - c) Costs of switching suppliers are low.
 - d) Existing firms do not have the cost advantages of vertical integration.
 - e) Product differences are few.
 - f) Access to existing suppliers is not blocked, and distribution channels are willing to accept new products.
 - g) Capital requirements are low.
 - h) Exit barriers are low.
 - i) The government's policy is to encourage new entrants.
 - 2) The most favorable industry condition is one in which entry barriers are high and exit barriers are low.
 - a) When the threat of new entrants is minimal and exit is not difficult, returns are high, and risk is reduced in the event of poor performance.
 - b) Low entry barriers keep long-term profitability low because new firms can enter the industry, increasing competition and lowering prices and the market shares of existing firms.
- d. The **threat of substitutes** limits price increases and profit margins. The greater the threat, the less attractive the industry is to potential entrants.
 - 1) Substitutes are types (not brands) of goods and services that have the same purposes, for example, plastic and metal or minivans and SUVs. Hence, a change in the price of one such product (service) causes a change in the demand for its substitutes.
 - 2) Structural considerations affecting the threat of substitutes are
 - a) Relative prices,
 - b) Costs of switching to a substitute, and
 - c) Customers' inclination to use a substitute.

- e. As the **threat of buyers' bargaining power** increases, the appeal of an industry to potential entrants decreases. Buyers seek lower prices, better quality, and more services. Moreover, they use their purchasing power to obtain better terms, possibly through a bidding process. Thus, buyers affect competition.
 - 1) Buyers' bargaining power varies with the following factors:
 - a) When purchasing power is concentrated in a few buyers or when buyers are well organized, their bargaining power is greater.
 - i) This effect is reinforced when sellers are in a capital-intensive industry.
 - b) High (low) switching costs decrease (increase) buyers' bargaining power.
 - c) The threat of backward (upstream) vertical integration, that is, the acquisition of a supply capacity, increases buyers' bargaining power.
 - d) Buyers are most likely to bargain aggressively when their profit margins are low and a supplier's product accounts for a substantial amount of their costs.
 - e) Buyers are in a stronger position when the supplier's product is undifferentiated.
 - f) The more important the supplier's product is to buyers, the less bargaining power they have.
- f. As the **threat of suppliers' bargaining power** increases, the appeal of an industry to potential entrants decreases. Accordingly, suppliers affect competition through pricing and the manipulation of the quantity supplied.
 - 1) Suppliers' bargaining power is greater when
 - a) Switching costs are substantial.
 - b) Prices of substitutes are high.
 - c) They can threaten forward (downstream) vertical integration.
 - d) They provide something that is a significant input to the value added by the buyer.
 - e) Their industry is concentrated, or they are organized.
 - 2) Suppliers' best responses are to develop favorable, mutually beneficial relationships with suppliers or to diversify their sources of supply.

4. Generic Competitive Analysis (Strategies) Model

- a. Strategies with a Broad Competitive Scope
 - 1) **Cost leadership** is the generic strategy of entities that seek competitive advantage through lower costs and that have a broad competitive scope.
 - 2) **Differentiation** is the generic strategy of entities that seek competitive advantage through providing a unique product and that have a broad competitive scope.
- b. Strategies with a Narrow Competitive Scope
 - 1) **Cost focus** is the generic strategy of entities that seek competitive advantage through lower costs and that have a narrow competitive scope (a regional or smaller market).
 - 2) **Focused differentiation** is the generic strategy of entities that seek competitive advantage through providing a unique product and that have a narrow competitive scope (a regional or smaller market).

Competitive Scope	Competitive Advantage	
	Low Cost	Product Uniqueness
Broad (Industry-wide)	Cost Leadership Strategy	Differentiation Strategy
Narrow (Market segment)	Focused Strategy: Cost	Focused Strategy: Differentiation

Figure 12-9

5. The Growth-Share Matrix

- Since a large firm may be viewed as a portfolio of investments in the form of strategic business units (SBUs), techniques of portfolio analysis have been developed to aid management in making decisions about resource allocation, new business startups and acquisitions, downsizing, and divestitures.
- One of the models most frequently used for competitive analysis was created by the Boston Consulting Group (BCG). This model, the growth-share matrix, has two variables. The market growth rate (MGR) is on the vertical axis, and the firm's relative market share (RMS) is on the horizontal axis.

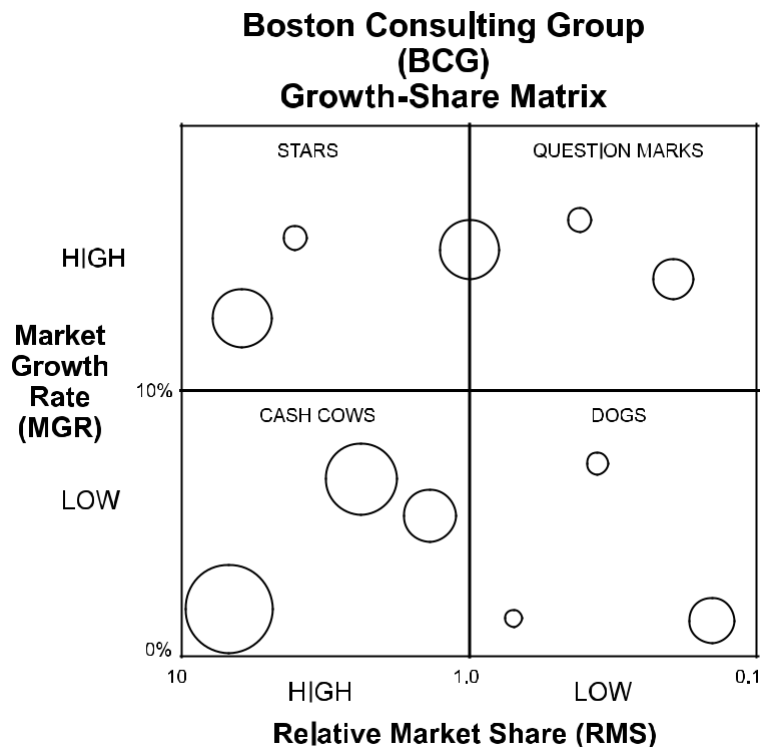


Figure 12-10

- The annual MGR is stated in constant units of the currency used in the measurement. It reflects the maturity and attractiveness of the market and the relative need for cash to finance expansion.
- The RMS reflects the SBU's competitive position in the market segment. It equals the SBU's absolute market share divided by that of its leading competitor.

- 3) The growth-share matrix has four quadrants. The firm's SBUs are commonly represented in their appropriate quadrants by circles. The size of a circle is directly proportional to the SBU's sales volume.
 - a) **Dogs** (low RMS, low MGR) are weak competitors in low-growth markets. Their net cash flow (plus or minus) is modest.
 - b) **Question marks** (low RMS, high MGR) are weak competitors and poor cash generators in high-growth markets. They need large amounts of cash not only to finance growth and compete in the market, but also to increase RMS.
 - i) If RMS increases significantly, a question mark may become a star. If not, it becomes a dog.
 - c) **Cash cows** (high RMS, low MGR) are strong competitors and cash generators. A cash cow ordinarily enjoys high profit margins and economies of scale. Financing for expansion is not needed, so the SBU's excess cash can be used for investments in other SBUs. However, marketing and R&D expenses should not necessarily be slashed excessively.
 - i) Maximizing net cash inflow might precipitate a premature decline from cash cow to dog.
 - d) **Stars** (high RMS, high MGR) are strong competitors in high growth markets. Such an SBU is profitable but needs large amounts of cash for expansion, R&D, and meeting competitors' attacks. Net cash flow (plus or minus) is modest.
- 4) A portfolio of SBUs should not have too many dogs and question marks or too few cash cows and stars.
- 5) Each SBU should have objectives, a strategy should be formulated to achieve those objectives, and a budget should be allocated.
 - a) A **hold strategy** is used for strong cash cows.
 - b) A **build strategy** is necessary for a question mark with the potential to be a star.
 - c) A **harvest strategy** maximizes short-term net cash inflow. Harvesting means zero-budgeting R&D, reducing marketing costs, not replacing facilities, etc. This strategy is used for weak cash cows and possibly question marks and dogs.
 - d) A **divest strategy** is normally used for question marks and dogs that reduce the firm's profitability. The proceeds of sale or liquidation are then invested more favorably.
 - i) A harvest strategy may undermine a future divestiture by decreasing the fair value of the SBU.
- 6) The life cycle of a successful SBU is reflected by its movement within the growth-share matrix.
 - a) The progression is from question mark to star, cash cow, and dog. Accordingly, a firm should consider an SBU's current status and its probable progression when formulating a strategy.
- 7) A serious mistake is to not tailor objectives (e.g., rates of return or growth) to the circumstances of each SBU.

- 8) Cash cows should not be underfunded because premature decline is risked. However, overfunding cash cows means less investment in SBUs with greater growth prospects.
 - a) A large investment in a dog with little likelihood of a turnaround is also a typical mistake.
 - b) A firm should not have too many question marks. Results are excess risk and underfunded SBUs.
- 9) According to Kotler, managers need to be aware of the limitations inherent in the use of a matrix (*Principles of Marketing*, 14th edition). Managers may find it difficult to measure market share and growth or even define SBUs. Thus, BCG's growth-share matrix may have limited strategic value.

12.5 STRATEGIC PLANNING

1. Planning

- a. Planning is the determination of what is to be done and of how, when, where, and by whom it is to be done. Plans serve to direct the activities that all organizational members must undertake and successfully perform to move the organization from where it is to where it wants to be (accomplishment of its objectives).
 - 1) However, no transactions occur during the planning cycle that must be recorded in the general ledger.
- b. Planning must be completed before undertaking any other managerial function.
 - 1) Forecasting is the basis of planning because it projects the future.
- c. Planning establishes the means to reach organizational ends (objectives).
 - 1) This means-end relationship extends throughout the organizational hierarchy and ties together the parts of the organization so that the various means all focus on the same end.
 - 2) One organizational level's ends provide the next higher level's means.

2. The Planning Process

- a. Long-range (strategic) planning includes **strategic budgeting**. A strategic budget describes the long-term position and objectives of an entity within its environment.
 - 1) Generally, timeframes range from 3 to 5 years, depending on what is best for the company; longer timeframes are not uncommon. Such planning is difficult because of uncertainty about future events and conditions.
 - 2) Thus, strategic plans tend to be general and exclude operational detail.
 - 3) An entity must complete its strategic plan before any specific budgeting can begin. The strategic plan states the means by which an entity expects to achieve its stated mission.
- b. Strategic planning embodies the concerns of senior management. It is based on a strategic analysis that includes the following:
 - 1) Identifying and specifying organizational objectives.
 - 2) Evaluating the strengths (competitive advantages) and weaknesses of the organization and its competitors.
 - 3) Assessing risk levels.
 - 4) Identifying and forecasting the effects of external (environmental) factors relevant to the organization. For example, market trends, changes in technology, international competition, and social change may provide opportunities, impose limitations, or represent threats.
 - a) Forecasting is the basis of planning because it projects the future. A variety of quantitative methods are used in forecasting.
 - 5) Deriving the best strategy for reaching the objectives, given the organization's strengths and weaknesses and the relevant future trends.
 - 6) Capital budgeting, a planning process for choosing and financing long-term projects and programs.
 - 7) Capacity planning, an element of planning closely related to capital budgeting that includes, among other things, consideration of business combinations or divestitures.
 - 8) **Contingency planning** involves having alternative strategies in place (i.e., disaster recovery plans or business continuity plans) as a way of preparing for unexpected emergencies or other disruptions to operations. The primary purpose for a contingency plan is to ensure operations continue with minimal interruption, losses, or damages.
 - 9) **Scenario planning** involves projecting plausible future events (i.e., economic conditions, political environment, scientific and technological developments, etc.) and developing alternative strategies for each one of them to ensure operations continue.
- c. Strategic plans are translated into measurable and achievable intermediate and operational plans. Thus, intermediate and operational plans must be consistent with, and contribute to achieving, strategic objectives.

3. Premises

- a. Premises are the underlying assumptions about the expected environment in which the strategic plan will be carried out. Thus, the next step in planning is premising, or the generation of planning assumptions. Premises should be limited to those crucial to the success of the plans.
- b. Managers should ask, "What internal and external factors would influence the actions planned for this organization (division, department, program)?" Premises must be considered at all levels of the organization.
 - 1) Thus, capital budgeting plans should be premised on assumptions (forecasts) about economic cycles, price movements, etc.
 - 2) The inventory department's plans might be premised on stability of parts prices or on forecasts that prices will rise.

EXAMPLE 12-12	Premises
• The general economy will suffer an 11% decline next year. • Our closest competitor's new model will provide greater competition for potential sales. • Union negotiations will result in a general wage increase of 8%. • Over the next 5 years, the cost of our raw materials will increase by 30%. • The elasticity of demand for the company's products is 1.2.	

4. Organizational Objectives

- a. Organizations may have multiple objectives that are contradictory.
 - 1) The objective of maximizing profit and the objective of growth could be mutually exclusive within a given year. Maximizing short-term profit might hamper or preclude future growth.
 - 2) Conflict among an organization's objectives is common.
- b. Objectives vary with the organization's type and stage of development.

5. Management Objectives

- a. The primary task of management is to carry on operations effectively and efficiently.
 - 1) **Effectiveness** is the degree to which the objective is accomplished. **Efficiency** is maximizing the output for a given quantity of input.
 - 2) Effectiveness is sometimes called "doing the right things," and efficiency is known as "doing things right."
 - a) Trade-offs are frequently made between efficiency and effectiveness.
- b. Objectives should be clearly and specifically stated, communicated to all concerned parties, and accepted by those affected.

6. Means-End Hierarchy

- a. Objectives should be established at the top and retranslated in more specific terms as they are communicated downward in the means-end hierarchy.

EXAMPLE 12-13	Means-End Hierarchy
• An entity has a socioeconomic purpose, such as providing food. • The entity's mission is the accomplishment of its socioeconomic purpose through the production of breakfast cereal. • The entity develops long-range or strategic objectives with regard to profitability, growth, or survival. • A more specific overall objective might be to provide investors with an adequate return on their investment. • Divisional objectives can be developed, e.g., to increase the sales of a certain kind of cereal. • Departmental objectives are developed, e.g., to reduce waste in the packaging department. • Low-level managers and supervisors then develop personal performance and development objectives.	

7. Policies, Procedures, and Rules

- a. After premises and objectives are formulated, the next step in the planning process is the development of policies, procedures, and rules. These elements are necessary at all levels of the organization and overlap both in definition and in practice.
 - 1) Intermediate and operational plans are translated into policies, procedures, and rules, which are standing plans for repetitive situations.
- b. Policies and procedures provide **feedforward control** because they anticipate and prevent problems and provide guidance on how an activity should be performed to best ensure that an objective is achieved.
 - 1) **Policies** are general statements that guide thinking and action in decision making. Policies may be explicitly published by, or implied by the actions of, management.
 - a) A strong organizational culture means that the organization's key values are intensely held and widely shared. In this case, the need for formal written policies is minimized.
 - 2) **Procedures** are specific directives that define how work is to be done.
 - 3) **Rules** are specific, detailed guides that restrict behavior.

8. Management by Objectives (MBO)

- a. MBO identifies relationships between an individual's job objectives (ends) and the immediate superior's objectives (ends). Thus, the subordinate can understand how his or her job is the means by which the superior's job is accomplished.
- b. MBO is a behavioral, communications-oriented, responsibility approach to management and employee self-direction. It is a comprehensive management approach and therefore is relevant to planning and control.
- c. MBO is based on the philosophy that employees
 - 1) Want to work hard if they know what is expected
 - 2) Like to understand what their jobs actually entail
 - 3) Are capable of self-direction and self-motivation

d. MBO requires

- 1) Senior management participation and commitment to the program. These managers must
 - a) Determine the overall direction and objectives for the organization
 - b) Communicate these effectively in operational or measurable terms
 - c) Coordinate subordinates' objectives with overall objectives
 - d) Follow up at the end of the MBO cycle period to reward performance and review problems
- 2) Integration of objectives for all subunits into a compatible, balanced system directed toward accomplishment of the overall objectives.
- 3) Provisions for regular periodic reporting of performance toward attainment of the objectives.
- 4) Free and honest communication between supervisor and subordinate.
- 5) A commitment to taking the ideas of subordinates seriously on the part of supervisors.
- 6) An organizational climate that encourages mutual trust and respect.

- e. Steps necessary to implement an MBO program include establishing objectives and action plans (the planning steps) and periodic review and final appraisal (the control steps).

9. **Goal Congruence**

- a. Goal congruence ensures the harmonization of objectives, procedures for achieving these objectives, operational planning, and strategic planning are working as a whole in order to realize the company goals.
- 1) Planning activities must be designed to encourage goal congruence at various levels of management.
 - 2) Periodic (i.e., monthly) analysis of actual results compared to budget assists with identifying whether a company is on or off course to achieving long-term strategic goals.
 - a) Tactics can be assessed and revised wherever necessary in order to get the company back on course to realizing company goals.
- b. Characteristics of successful strategic plans include but are not limited to
- 1) Clarity of purpose and realistic goals
 - 2) Monitoring, measurement, and feedback
 - 3) Discipline and commitment
 - 4) Leadership

STUDY UNIT THIRTEEN

BUDGETING -- CONCEPTS AND METHODOLOGIES

13.1	<i>Roles of Budgets and the Budgeting Process</i>	2
13.2	<i>Budgeting and Standard Costs</i>	11
13.3	<i>The Master Budget</i>	13
13.4	<i>Budget Methodologies</i>	17

Planning, Budgeting, and Forecasting

A budget is a realistic plan for the future that is expressed in quantitative terms. A budget is many tools in one—it is a planning tool, a control tool, a communication tool, and a motivational tool. As such, the area of budgeting, as tested on the CMA exam, is a composite of theory and calculations. Some of the calculations have many steps, making budgeting problems among the most-missed questions on the exam. However, budgeting should not be viewed as a difficult area; the concepts are easy, but you need to pay close attention to detail as you work numerical questions.

This study unit is the **second of three on planning, budgeting, and forecasting**. The relative weight assigned to this major topic in Part 1 of the exam is **20%**. The three study units are

Study Unit 12: Analysis, Forecasting, and Strategy

Study Unit 13: Budgeting -- Concepts and Methodologies

Study Unit 14: Budgeting -- Calculations and Pro Forma Financial Statements

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.



Multiple-choice questions can also help you study for the essay questions. As you read each multiple-choice question, cover up the answer explanations and write a response as if it were an essay. If your response is similar to the answer explanation, you know you are on the right track. This exercise will also help your overall comprehension, which will translate into higher scores on your essay questions.

13.1 ROLES OF BUDGETS AND THE BUDGETING PROCESS

1. The Budget as a Tool

- a. The budget is a **planning** tool.
 - 1) A budget is a written plan for the future that forces management to evaluate the assumptions and the objectives identified in the budgetary process.
 - 2) Companies that prepare budgets anticipate problems before they occur.
 - a) **EXAMPLE:** If a company runs out of critical raw material, it may have to shut down. At best, it will incur extremely high freight costs to have the needed materials rushed in. The company with a budget will have anticipated the shortage and planned around it.
 - 3) A firm that has no goals may not always make the best decisions. A firm with a goal in the form of a budget will be able to plan.
- b. The budget is a **control** tool.
 - 1) A budget helps a firm control costs by setting cost guidelines.
 - 2) Guidelines reveal the efficient or inefficient use of company resources.
 - a) These guidelines should reflect careful analysis that may begin, but should not end with consideration of prior period results.
 - 3) A manager is less apt to spend money for things that are not needed if (s)he knows that all costs will be compared with the budget.
 - a) (S)he will be accountable if controllable costs exceed budgeted amounts.
 - 4) Budgets can also reveal the progress of highly effective managers. Consequently, employees should not view budgets negatively. A budget is just as likely to provide a boost to a manager's career as it is to be detrimental.
 - 5) Managers can also use a budget as a personal self-evaluation tool.
 - 6) Budgetary slack (i.e., underestimating revenues or overestimating expenses), however, must be avoided if a budget is to have its desired effects. The natural tendency of a manager is to negotiate for a less stringent measure of performance so as to avoid unfavorable variances from expectations.
 - 7) For the budgetary process to serve effectively as a control function, it must be integrated with the accounting system and the organizational structure. Such integration enhances control by transmitting data and assigning variances to the proper organizational subunits.

- c. The budget is a **motivational** tool.
 - 1) A budget helps motivate employees to do a good job.
 - a) Employees are particularly motivated if they help prepare the budget.
 - b) A manager who is asked to prepare a budget for his or her department will work hard to stay within the budget.
 - 2) A budget must be seen as realistic by employees before it can become a good motivational tool.
 - 3) Unfortunately, the budget is not always viewed in a positive manner. Some managers view a budget as a restriction.
 - 4) Employees are more apt to have a positive feeling toward a budget if some degree of flexibility is allowed.
- d. The budget is a means of **communication**.
 - 1) A budget can help tell employees what goals the firm is attempting to accomplish.
 - 2) A budget functions as an aid to planning, coordination, and control. Thus, a budget helps management allocate resources efficiently and ensure that subunit objectives are consistent with those of other subunits and of the organization.
 - a) For the budget to function in these roles, senior management must be involved with the process. This involvement does not extend to dictating exact numerical contents of the budget because senior management lacks detailed knowledge of daily operations.
 - 3) If the firm does not have an overall budget, each department might think the firm has different goals.
 - 4) For example, the sales department may want to keep as much inventory as possible so that no sales will be lost, but the company treasurer may want to keep the inventory as low as possible in order to conserve cash reserves. If the budget specifies the amount of inventory, all employees can work toward the same objectives.
- e. The budget is a **goal congruence** tool.
 - 1) The budget notifies all departments and all levels of responsibility what the company's goals are. Everyone is working toward the same end objective.
 - 2) For example, the strategic plan and the capital budget must be congruent with the operational planning in the master budget.

2. The Budget as a Formal Quantification of Management's Plans

- a. Corporations have goals for market share, profitability, growth, dividend payout, etc. Not-for-profit organizations also have goals, such as increased number of free meals served, lowered recidivism rate among offenders, etc.
 - 1) These goals cannot be achieved without careful planning about the allocation of resources and the expected results.
- b. A budget lays out in specific terms an organization's expectations about the consumption of resources and the resulting outcomes.
- c. An organizational budget requires a significant commitment of internal resources. The most important factor in ensuring its success is for senior managers to demonstrate that they take the project seriously and consider it vital to the organization's future.

3. Budgeting's Role in the Overall Planning and Evaluation Process

- a. **Planning** is the process by which an organization sets specific goals for itself and sets about pursuing those goals. Planning is an organization's response to the saying "If you don't know where you're going, any path will take you there."
 - 1) The starting point for any organization's planning process is the formulation of its **mission statement**. The mission statement, formulated by the board and senior management, embodies the organization's reason for existing.
 - a) **EXAMPLE:** Increase shareholder value by providing global telecommunications services.
 - 2) Next, the organization draws up its **strategic plan** containing the means by which the firm expects to fulfill its stated mission.
 - a) To a great extent, the strategy is made up of **long-term objectives**, a set of specific, measurable goals.
 - b) **EXAMPLE:** Hold a 35% market share of U.S. cell phone users within 5 years.
 - 3) Once the long-term objectives are in place, the **priorities** of the organization will be clear.
 - a) Awareness of priorities is crucial for the **allocation of limited resources**.
 - b) **EXAMPLE:** How many cell towers, each of which require the outlay of construction and maintenance costs, will provide the optimum amount of coverage.
 - 4) **Short-term objectives** flow directly from the priorities.
 - a) **EXAMPLE:** Determine the appropriate number of cell towers needed and where they can feasibly be placed in the Metro Atlanta region.
 - b) The planning process coordinates the efficient allocation of organizational resources.
- b. To **evaluate progress** toward success in each of these stages, quantification is necessary. This is the role of the various types of budgets.
 - 1) Not all quantification is in monetary terms. To extend the previous example, although cell towers obviously have a dollar cost, they must be simply counted as well.
 - 2) **Comparing actual results to the budget** allows the organization as a whole to evaluate its performance and managers to do the same on an individual level.
 - a) The budget is therefore a strategic control.

4. Participation in the Budget Process

- a. Participation in the budget preparation process is up and down the organization.
 - 1) The budget process begins with the mission statement formulated by the **board of directors**.
 - 2) **Senior management** translates the mission statement into a strategic plan with measurable, realizable goals.

- 3) A **budget committee/department** composed of top management is formed to draft the budget calendar and budget manual. The budget committee/department also reviews and approves the departmental budgets submitted by operating managers.
 - a) A budget director's primary responsibility is to compile the budget and manage the budget process.
 - 4) **Middle and lower management** receive their budget instructions, draw up their departmental budgets in conformity with the guidelines, and submit them to the budget committee.
- b. **Top-down (authoritative) budgeting** is imposed by upper management and therefore has less chance of acceptance by those on whom the budget is imposed.
- 1) This approach has the advantage of ensuring consistency across functional areas. It is also far less complex and time-consuming than coordinating input from middle and lower levels.
 - 2) Even having a budget set by an outside consulting firm is usually considered a top-down approach because the firm was hired by top management. The consultant may not fully understand the company's processes, and employees who were not fully involved in the process may react negatively.
- c. **Bottom-up (participative) budgeting** is characterized by general guidance from the highest levels of management, followed by extensive input from middle and lower management. Because of this level of participation within the company, there is usually a greater chance of acceptance and optimal decision making.
- 1) Disadvantages of participative standards setting include its time and money costs. In addition, the quality of participation is affected by the goals, values, beliefs, and expectations of those involved.

	Advantages	Disadvantages
Top-down budgeting	• Ensures consistency across all functional areas • Is far less complex and time-consuming than coordinating input from the middle and lower levels	• An imposed budget is much less likely to promote a sense of commitment
Bottom-up budgeting	• Encourages employees to have a sense of ownership of the output of the process, resulting in acceptance of, and commitment to, objectives expressed in the budget • Enables employees to relate performance to rewards or penalties • Provides a broader information base (middle- and lower-level managers often are far better informed about operational realities than senior managers)	• Higher costs in time and money • Quality of participation is affected by the objectives, values, beliefs, and expectations of those involved • Creation of budgetary slack

- d. Participation in developing a budget may result in a **padding** of the budget, also known as budgetary slack.
- 1) **Budgetary slack** is the excess of resources budgeted over the resources necessary to achieve organizational goals. This practice results in the underestimation of revenues and overestimation of expenses. This must be avoided if a budget is to have its desired effects.
 - a) The natural tendency of a manager is to negotiate for a less stringent measure of performance so as to avoid unfavorable variances from expectations.
 - 2) Management may create slack by overestimating costs and underestimating revenues.
 - a) A firm may decrease slack by emphasizing the consideration of all variables, holding in-depth reviews during budget development, and allowing for flexibility in making additional budget changes.
 - b) A manager who expects his or her request to be reduced may inflate the amount.
 - c) If a budget is to be used as a performance evaluator, a manager asked for an estimate may provide one that is easily attained.
 - 3) The existence of slack can have both positive and negative effects on the budgeting process. The existence of slack can reduce the planning benefits of a budget, since the budget may not be entirely accurate.
 - a) For example, a cash budget might show that \$500,000 needs to be borrowed this month, whereas that amount is not really needed because managers were just being cautious.
 - b) Alternatively, the lack of slack may discourage managers from implementing new programs or might cause managers to avoid routine maintenance when the budget does not show funds available in a particular period.

5. Time Frames for Budgets

- a. Each phase of the organization's planning cycle has its own budget with an appropriate **time frame**.
 - 1) **Strategic** plans and budgets most concern senior managers and have time frames of up to 10 years or more.
 - 2) **Intermediate** plans and budgets most concern middle managers and have time frames of up to 2 years.
 - 3) **Operational** plans and budgets most concern lower-level managers and generally have time frames of 1 month to 1 year.

6. Budgeting's Role in Formulating and Controlling Short-Term Objectives

- a. A company's goal of increasing market share, making a steady dividend payout, etc., is generally achieved through the completion of incremental steps.
- b. The budget lays out the specific revenue targets and expense limitations for each functional area and department of the organization on a month-by-month basis.
 - 1) A budget cannot simply be a lump-sum total for a year. Incremental goals are achieved each month or week. This is especially true in seasonal businesses, such as agricultural supply.
- c. A **static** budget is for one level of activity, whereas a flexible budget is a plan for varying levels of activity.
 - 1) The static budget is easier and cheaper to prepare. But the flexible budget is more useful, particularly when actual activity is not the planned level.
 - 2) If a company uses an unchanged master budget throughout the year for comparison with actual results, it is probably using a static budget.

7. Budgeting's Role in Measuring Performance against Established Goals

- a. One of the most important reasons for adopting a budget is to provide guideposts for the assessment of success or failure on the part of individual managers and functional areas.
- b. As the fiscal year progresses, revenues, expenses, and other metrics can be compared to the budget to determine where organizational performance is meeting, lagging, or exceeding expectations.

8. Budgeting's Role in Monitoring and Controlling Expenditures

- a. The initial budget is a planning tool. To monitor how actual performance compares with the budget, budget reports are produced periodically during the year.
 - 1) The difference between actual performance and a budgeted amount is called a **variance**. Analysis of variances reveals the efficient or inefficient use of company resources (variance analysis is covered in Study Unit 15).

9. Role of Budgeting Process in Facilitating Communication among Organizational Units and Enhancing Coordination of Organizational Activities

- a. On a detailed level, the budget informs employees at all levels what objectives the firm is attempting to accomplish.
 - 1) If the firm does not have an overall budget, each department tends to pursue its own objectives without regard to what is good for the firm as a whole. Thus, a budget promotes goal congruence.
- b. The concrete nature of a budget facilitates coordination of the activities of a firm. An example is the purchasing of raw materials.
 - 1) Materials are needed prior to production, but the proper quantity to buy cannot be determined until the projected level of output is established.
 - a) Thus, a production budget (in units) is a prerequisite to the preparation of a materials purchases budget.
 - 2) Similarly, a direct labor budget is based on how many units are to be produced and how fast the workers are.
 - a) Labor standards are also complex in that they must consider the impact of the learning curve on productivity.

10. Characteristics of a Successful Budgeting Process

- a. **Sufficient lead time.** For a budget to be useful, it must be finalized when the fiscal year begins. This often calls for months of preparation, since the overall goals and baseline assumptions must be announced before functional areas and individual departments can begin formulating their numbers.
 - 1) The preparation of a complete organizational budget usually takes several months. A firm with a calendar year end may start the budget process in September, anticipating its completion by the first of December.
 - 2) The **budget planning calendar** is the schedule of activities for the development and adoption of the budget. It includes a list of dates indicating when specific information is to be provided to others by each information source.
 - a) Because all of the individual departmental budgets are based on forecasts prepared by others and the budgets of other departments, it is essential to have a planning calendar to integrate the entire process.
 - 3) The budget department is responsible for compiling the budget and managing the budget process. However, the budget director and department are not responsible for actually developing the estimates on which the budget is based.
- b. **Budget manual.** Everyone involved in preparing the budget at all levels must be educated on the detailed procedures for preparing and submitting their part of the overall budget.
 - 1) Because of the number of component departments, budgets must be prepared in a standard format.
 - a) In addition, all concerned must be informed of the ultimate goals that are being pursued and the baseline assumptions that have been laid down. A budget may, for example, begin with a blanket mandate to raise revenues by 6.5% or to cut expenses across all departments by 2%.
 - 2) Distribution instructions are vital because of the interdependencies of a master budget.
 - a) One department's budget may be dependent on another's, and functional areas must be aggregated from their constituent department budgets. The distribution instructions coordinate these interdependencies.
- c. **Acceptance at all levels.** Participative budgeting has a much greater chance of acceptance by those affected and thus of achieving ultimate success than does a budget that is imposed from above.
 - 1) The support of top management is crucial to the budgeting efforts. This is because the single most important factor in ensuring the success of a budget process is for senior management to demonstrate that it takes the project seriously and considers it vital to the organization's future.

- d. Best practice guidelines for the budget process include but are not limited to
 - 1) Automating the process in order to permit more time analyzing and implementing changes instead of spending time manually creating the budget,
 - 2) Setting realistic goals and obtaining assistance from all levels of personnel to help gain support (i.e., “buy in”) from personnel,
 - 3) Considering “what if” scenarios,
 - 4) Updating budgets regularly to reflect current conditions, and
 - 5) Using an iterative budgeting process that reflects changes suggested by operating or senior managers.

11. Effects of External Factors on the Budgeting Process

- a. Decisions about a firm’s strategy, and in turn about its budget, are dependent upon **general economic conditions** and their expected trends as well as the availability of financial resources.
 - 1) For instance, if the economy is entering a period of lower demand, a manufacturer will not project increased sales. If costs are not changeable, the company may budget losses for the short term to hold on to market share.
- b. **Industry situation** includes the company’s current market share, governmental regulatory measures, the labor market, and the activities of competitors.
 - 1) For instance, if input costs are rising in a firm’s industry, the budget must reflect that reality; profit margins and cash flows will not be the same as in prior years. Also, a company in or near bankruptcy will face a different financial situation than would the market leader.

12. The Concept of Controllability

- a. Controllability is a key concept in the use of budgets and other standards to evaluate performance. Controllability is the extent to which a manager can influence activities and related revenues and costs.
- b. Controllable costs are those that are under the discretion of a particular manager. Noncontrollable costs are those to which another level of the organization has committed, removing the manager’s discretion.
- c. Controllability can be difficult to isolate because few costs or revenues are under the sole influence of one manager. Also, separating the effects of current management’s decisions from those of former management is difficult.
 - 1) If responsibility exceeds the extent to which a manager can influence an activity, the result may be reduced morale, a decline in managerial effort, and poor performance.
 - 2) The principle of controllability must be kept in mind when the budget is used as the basis for managerial evaluation.

13. Revisions to the Budget

- a. Often an organization will find that the assumptions under which the budget was prepared undergo significant change during the year. A policy must be in place to accommodate revisions to the budget resulting from these changes.
 - 1) Accommodation of change is a key characteristic of successful budgeting. If such a policy is not in place, managers can come to believe they are being held to a budget that is no longer possible to achieve, and morale can suffer.
- b. Information gained during the year as actual results and variances are reported can be used to help the company take corrective action. These steps make up a control loop:
 - 1) Establishing standards of performance (the budget)
 - 2) Measuring actual performance
 - 3) Analyzing and comparing performance with standards
 - 4) Devising and implementing corrective actions
 - 5) Reviewing and revising the standards

14. Criticisms and Limitations of the Budgeting Process

- a. Lower-level managers and other workers often feel that management makes across-the-board cuts when early budget iterations show that planned expenses are too high.
 - 1) The reason may be that management either does not take the time to examine specific areas of the budget closely or is unwilling to make tough decisions about where cuts need to be made.
- b. Budgeting is usually based on a year's time span and the component months and quarters of that year, which is an artificial time horizon. However, innovation and rapid responses should not be limited by an artificial deadline imposed by a budget.
- c. A budget is supposed to be an active document and should be used regularly with budget-to-actual comparisons being analyzed monthly, or even weekly. Unfortunately, some managers develop a budget and then wait until the end of the year to make evaluations. The use of a rolling, or continuous, budget helps avoid this problem.
- d. Many companies are reluctant to change a budget. Even the best budget needs to be changed occasionally to respond to changes in the organization's operating environment.

13.2 BUDGETING AND STANDARD COSTS

1. The Use of Cost Standards

- a. Standard costs are **predetermined expectations** about how much a unit of input, a unit of output, or a given activity should cost.
 - 1) The use of standard costs in budgeting allows the standard-cost system to alert management when the actual costs of production differ significantly from the standard.
- b. A standard cost is not just an average of past costs but an objectively determined estimate of what a cost should be. Standards may be based on accounting, engineering, or statistical quality control studies.
 - 1) Because of the impact of fixed costs in most businesses, a standard costing system is usually not effective unless the company also has a flexible budgeting system (flexible budgeting systems are covered in Study Unit 7, Subunit 4, and Study Unit 15, Subunit 2).

2. Developing Standards

- a. **Activity analysis** identifies, describes, and evaluates the activities that go into producing a particular output. Determining the resources and steps that go into the production process aids in the development of standard costs.
 - 1) Each operation requires its own unique set of inputs and preparations. Activity analysis describes what these inputs are and who performs these preparations.
 - a) Inputs include the amounts and kinds of equipment, facilities, materials, and labor. Engineering analysis, cost accounting, time-and-motion study, and other approaches may be useful.
 - 2) **Historical data** may be used to set standards by firms that lack the resources to engage in the complex task of activity analysis.
- b. For **direct materials**, there is often a direct relationship between unit price and quality. In establishing its cost standards, a manufacturer must decide whether it will use an input that is
 - 1) Cheaper per unit but will ultimately result in using more because of low quality or
 - 2) More expensive per unit but will ultimately result in using less because of lower waste and spoilage.
- c. For **direct labor**, the complexity of the production process and the restrictions on pay scales imposed by union agreements have the most impact on formulating cost standards. Human resources also must be consulted to help project the costs of benefits.
- d. Standards can be set using the **authoritative** (top-down) approach or the **participative** (bottom-up) approach.
 - 1) A form of the bottom-up approach that involves line managers and their supervisors, accountants, engineers, and other interested employees before standards are accepted by top management is also called the **team development approach**.

3. Ideal vs. Practical Standards

- a. **Ideal (theoretical) standards** are standard costs that are set for production under optimal conditions. For this reason, they are also called perfection or maximum efficiency standards.
 - 1) They are based on the work of the most skilled workers with no allowance for waste, spoilage, machine breakdowns, or other downtime.
 - 2) Often called “tight” standards, they can have positive behavioral implications if workers are motivated to strive for excellence. However, they are not widely used because they can have negative behavioral effects if the standards are perceived as impossible to attain.
 - 3) Ideal standards have been adopted by some companies that apply continuous improvement and other total quality management principles.
 - 4) Ideal standards are ordinarily replaced by currently attainable standards for cash budgeting, product costing, and budgeting departmental performance. Otherwise, accurate financial planning will be impossible.
- b. **Practical (currently attainable) standards** are defined as the performance that is expected to be achieved by reasonably well-trained workers with an allowance for normal spoilage, waste, and downtime.
 - 1) An alternative interpretation is that practical standards represent possible but difficult-to-attain results.
 - 2) Compared to ideal standards, practical standards serve as a better motivating target for manufacturing personnel.

**SUCCESS TIP**

This study unit provides a brief overview of the master budget process. CMA candidates are expected to be able to prepare budgets based on relevant information (i.e., standard costs, discussed in Subunit 13.2). Take care to know the budget cycle and, especially, the order in which each budget is prepared.

2. Master Budget

- a. The master budget, also called the **comprehensive budget** or **annual profit plan**, encompasses the organization's **operating and financial plans** for a specified period (ordinarily a year or single operating cycle).
- b. The importance of carefully drafting the budget calendar is that it includes specific dates when each department must submit its budget to the next designated department.
 - 1) For example, the production budget cannot be prepared until after the sales budget. The direct materials budget and the direct labor budget cannot be prepared until after the production budget has been completed.

3. Operating Budget

- a. In the operating budget, the emphasis is on obtaining and using current resources.
 - 1) Sales budget
 - 2) Production budget
 - 3) Direct materials budget
 - 4) Direct labor budget
 - 5) Employee fringe benefits budget
 - 6) Manufacturing overhead budget
 - 7) Ending finished goods inventory budget
 - 8) Cost of goods sold budget
 - 9) Nonmanufacturing budget
 - a) Research and development budget
 - b) Design budget
 - c) Marketing budget
 - d) Distribution budget
 - e) Customer service budget
 - f) Administrative budget
 - 10) Pro forma income statement

4. Financial Budget

- a. In the financial budget, the emphasis is on obtaining the funds needed to purchase operating assets. It contains the
 - 1) Capital budget (completed before operating budget is begun)
 - 2) Projected cash disbursement schedule
 - 3) Projected cash collection schedule
 - 4) Cash budget
 - 5) Pro forma balance sheet
 - 6) Pro forma statement of cash flows

5. Sales Budget

- a. The sales budget is the first budget prepared because sales volume affects production and purchasing levels, operating expenses, and cash flows.
 - 1) Thus, expectations about sales drive the entire budget process.
- b. Once a firm can estimate sales, the next step is to decide how much to produce or purchase.
 - 1) Sales are usually budgeted by product or department. The sales budget also establishes targets for sales personnel.
 - 2) Sales credit policies can have a large effect on the sales budget.
 - 3) If sales volume is seasonal, certain items in the budget also must be coordinated seasonally. Examples are production volume, finished goods inventory, and sales volume.

NOTE: A more detailed discussion of sales budgets can be found in Study Unit 14, Subunit 1.

6. Production Budgets

- a. Production budgets (for manufacturing firms) are based on sales in units (not dollars) plus or minus desired inventory buildup or reduction.
 - 1) They are prepared for each department and each item. Production budgets are usually stated in units instead of dollars.
- b. When the production budget has been completed, it is used to prepare three additional budgets:
 - 1) Raw materials purchases, which is similar to the purchases budget of a merchandising firm
 - 2) Direct labor budget, which includes hours, wage rates, and total dollars
 - 3) Factory overhead budget, which is similar to a department expenses budget

NOTE: A more detailed discussion of production budgets can be found in Study Unit 14, Subunit 2, and a more detailed discussion of direct labor and factory overhead budgets can be found in Study Unit 14, Subunit 3.

7. Purchases Budget

- a. The purchases budget may be prepared on a monthly or even a weekly basis.
 - 1) Purchases should be planned to prevent stockouts and minimize carrying costs.
 - 2) Attempts should be made to determine potential purchasing terms.

8. Expense Budgets

- a. Expense budgets are prepared by department heads using the sales budget as a basis.
 - 1) Expense budgets are based on prior years' costs and adjusted for anticipated changes in prices, wages, and sales volume estimates.

9. Capital Budget

- a. Equipment purchases (capital expenditures) are technically not part of the operating budget, but they must be incorporated into the preparation of the cash budget and pro forma financial statements.
- b. Capital budgets may be prepared more than a year in advance to allow sufficient time to
 - 1) Plan financing of major expenditures for equipment or buildings or
 - 2) Receive custom orders of specialized equipment, buildings, etc.

NOTE: A more detailed discussion of capital budgets can be found in Study Unit 14, Subunit 4.

10. Cash Budget

- a. The cash budget is usually the last to be prepared and is often the most important part of a company's budget program.
 - 1) Almost all organizations, regardless of size, prepare a cash budget. It is particularly important for organizations operating in seasonal industries.
 - 2) An organization must have adequate cash at all times. Even with plenty of other assets, an organization with a temporary shortage of cash can be driven into bankruptcy. Proper planning can keep an entity from financial embarrassment.
- b. A cash budget details projected cash receipts and disbursements. It cannot be prepared until the other budgets have been completed.
 - 1) Cash budgeting facilitates loans and other financing. A bank is more likely to lend money to a firm if the money will not be needed immediately.

NOTE: A more detailed discussion of cash budgets can be found in Study Unit 14, Subunit 5.

13.4 BUDGET METHODOLOGIES



SUCCESS TIP

CMA candidates must know how to both apply and select a specific budgeting method. They may also be required to explain a specific action, such as why a method should be selected. All topics are eligible to be tested on the multiple-choice section, the essay section, or both.

1. Project Budget

- a. A project budget consists of all the costs expected to attach to a particular project, such as the design of a new airliner or the building of a single ship.
 - 1) While the project is obviously part of the company's overall line of business, the costs and profits associated with it are significant enough to be tracked separately.
- b. A project will typically use resources from many parts of the organization, e.g., design, engineering, production, marketing, accounting, and human resources.
 - 1) All of these aspects of the project budget must align with those of the firm's master budget.

EXAMPLE 13-1 Project Budget					
Function	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	Totals
Design	\$ 800,000	\$ 200,000	\$ --	\$ --	\$1,000,000
Engineering	500,000	1,200,000	400,000	--	2,100,000
Production	--	2,100,000	1,500,000	1,500,000	5,100,000
Marketing	--	100,000	200,000	200,000	500,000
Accounting	100,000	100,000	100,000	100,000	400,000
Human Resources	20,000	20,000	20,000	20,000	80,000
Totals	\$1,420,000	\$3,720,000	\$2,220,000	\$1,820,000	\$9,180,000

2. Activity-Based Budgeting

- a. Activity-based budgeting applies activity-based costing principles (Study Unit 8, Subunit 3) to budgeting. Its greatest effect is on the application of **indirect costs**.
 - 1) A traditional budgeting system accumulates all indirect costs into a single pool and allocates them to products based on a cost driver such as volume or machine hours.

EXAMPLE 13-2 Traditional Budgeting					
A manufacturer produces two valves, a simple one and a complex one. It has budgeted production costs for the upcoming year using a traditional budgeting system:					
Cost Category	Simple Valve		Complex Valve		Total
	Total	Per Unit	Total	Per Unit	
Direct materials	\$ 450,000	\$ 9.00	\$270,000	\$27.00	\$ 720,000
Direct labor	240,000	4.80	78,000	7.80	318,000
Total direct costs	\$ 690,000	\$13.80	\$348,000	\$34.80	\$1,038,000
Allocated indirect costs	720,000	14.40	234,000	23.40	954,000
Total manufacturing costs	\$1,410,000	\$28.20	\$582,000	\$58.20	\$1,992,000

- b. Activity-based budgeting involves defining the activities that drive indirect costs.
- 1) A cost pool is established for each activity, and a cost driver is identified for each pool.
 - a) The key to successful activity-based budgeting is selecting a driver for each pool that has a direct cause-and-effect relationship with the level of activity in that pool.
 - 2) The budgeted cost for each pool is determined by multiplying the demand for the activity by the estimated cost of a unit of the activity.

EXAMPLE 13-3 Indirect-Cost Assignment

The manufacturer in Example 13-2 has designed an indirect-cost assignment system based on the following pools and drivers:

Indirect cost pool	Driver
Product design	Engineering hours
Production setup	Number of batches
Machining	Machine hours
Inspection & testing	Number of valves
Customer maintenance	Salesperson hours

- c. Since activity-based budgeting employs multiple indirect cost pools, it provides far greater detail regarding indirect costs than traditional functional or spending-category budgeting (which only employs a single pool).

EXAMPLE 13-4 Activity-Based Budgeting

Note that while the amounts of indirect costs assigned to the two products in Examples 13-2 and 13-3 are different, indirect costs in total are (necessarily) the same as under the traditional system:

Cost Category	Estimated Driver Level	Cost per Unit of Driver	Simple Valve		Complex Valve		Total
			Total	Per Unit	Total	Per Unit	
Direct materials			\$ 450,000	\$ 9.00	\$270,000	\$27.00	\$ 720,000
Direct labor			240,000	4.80	78,000	7.80	318,000
Total direct costs			\$ 690,000	\$13.80	\$348,000	\$34.80	\$1,038,000
Indirect cost assignment:							
Product design:							
Simple valve	1,000 ×	\$23.75 =	\$ 23,750	\$ 0.48			
Complex valve	2,800 ×	23.75 =			\$ 66,500	\$ 6.65	\$ 90,250
Production setup:							
Simple valve	219 ×	21.00 =	4,600	0.09			
Complex valve	20 ×	21.00 =			420	0.04	5,020
Machining:							
Simple valve	2,000 ×	3.25 =	6,500	0.13			
Complex valve	19,000 ×	3.25 =			61,750	6.18	68,250
Inspection & testing:							
Simple valve	50,000 ×	12.50 =	625,000	12.50			
Complex valve	10,000 ×	12.50 =			125,000	12.50	750,000
Customer maintenance:							
Simple valve	1,500 ×	17.60 =	26,400	0.53			
Complex valve	800 ×	17.60 =			14,080	1.41	40,480
Total indirect costs			\$ 686,250	\$13.73	\$267,750	\$26.78	\$ 954,000
Total manufacturing costs			\$1,375,850	\$27.53	\$615,750	\$61.58	\$1,992,000*

*\$1,992,000 is not equal to the sum of \$1,375,850 and \$615,750 due to rounding differences.

3. Zero-Based Budgeting

- a. Zero-based budgeting (ZBB) is a budget and planning process in which each manager must justify his or her department's entire budget every budget cycle.

BACKGROUND 13-1 Zero-Based Budgeting

The concept of ZBB originated in the U.S. Department of Agriculture in the early 1960s but was abandoned. Texas Instruments Corporation began using it in the late 1960s and early 1970s, as did the state of Georgia under Governor Jimmy Carter. Carter also tried to introduce the concept into the federal budget system when he served as president (1977–1980).

- b. ZBB differs from the traditional concept of **incremental budgeting**, in which the current year's budget is simply adjusted to allow for changes planned for the coming year.
- 1) The managerial advantage of incremental budgeting is that the manager has to put forth less effort to justify changes in the budget.
- c. Under ZBB, a manager must build the budget every year from a base of zero. All expenditures must be justified regardless of variance from previous years.
- 1) The objective is to encourage periodic reexamination of all costs in the hope that some can be reduced or eliminated.
- d. ZBB begins with the deepest budgetary units of the entity.
- 1) It requires determination of objectives, operations, and costs for each activity and the alternative means of carrying out that activity.
 - 2) Different levels of service (work effort) are evaluated for each activity, measures of work and performance are established, and activities are ranked according to their importance to the entity.
 - 3) For each budgetary unit, a decision package is prepared that describes various levels of service that may be provided, including at least one level of service lower than the current one.
 - a) Accordingly, ZBB requires managers to justify each expenditure for each budget period and to review each cost component from a cost-benefit perspective.
- e. The major limitation of ZBB is that it requires more time and effort to prepare than a traditional budget.

4. Continuous (Rolling) Budgeting

- a. A continuous (rolling) budget is one that is revised on a regular (continuous) basis.
 - 1) Typically, a company continuously extends such a budget for an additional month or quarter in accordance with new data as the current month or quarter ends.
 - 2) For example, if the budget cycle is 1 year, a budget for the next quarter will be available continuously as each quarter ends.

EXAMPLE 13-5 Rolling Budget					
<u>Product Line</u>	<u>Fiscal Year 1</u>				<u>Four</u>
	<u>1st</u>	<u>2nd</u>	<u>3rd</u>	<u>4th</u>	<u>Quarter</u>
	<u>Quarter</u>	<u>Quarter</u>	<u>Quarter</u>	<u>Quarter</u>	<u>Totals</u>
Feed	\$ 12,000	\$ 10,000	\$ 10,000	\$ 14,000	\$ 46,000
Animal health	2,000	1,200	800	1,000	5,000
Fertilizer	80,000	75,000	20,000	10,000	185,000
Crop protectants	74,000	76,000	41,000	11,000	202,000
Petroleum	120,000	20,000	14,000	100,000	254,000
Farm supplies	15,000	45,000	55,000	20,000	135,000
Totals	\$303,000	\$227,200	\$140,800	\$156,000	\$827,000

<u>Product Line</u>	<u>Fiscal Year 1</u>			<u>Fiscal</u>	<u>Four</u>
	<u>2nd</u>	<u>3rd</u>	<u>4th</u>	<u>Year 2</u>	<u>Quarter</u>
	<u>Quarter</u>	<u>Quarter</u>	<u>Quarter</u>	<u>1st</u>	<u>Totals</u>
Feed	\$ 10,000	\$ 10,000	\$ 14,000	\$ 9,000	\$ 43,000
Animal health	1,200	800	1,000	2,200	5,200
Fertilizer	75,000	20,000	10,000	90,000	195,000
Crop protectants	76,000	41,000	11,000	90,000	218,000
Petroleum	20,000	14,000	100,000	85,000	219,000
Farm supplies	45,000	55,000	20,000	10,000	130,000
Totals	\$227,200	\$140,800	\$156,000	\$286,200	\$810,200

- b. The principal advantage of a rolling budget is that it requires managers to always be thinking ahead.
 - 1) The disadvantage is the amount of time managers must constantly spend on budget preparation.

STUDY UNIT FOURTEEN

BUDGETING -- CALCULATIONS AND PRO FORMA FINANCIAL STATEMENTS

14.1	<i>Sales Forecasts and the Sales Budget</i>	1
14.2	<i>Operating Budget Calculations -- Production and Direct Materials</i>	4
14.3	<i>Operating Budget Calculations -- Others</i>	5
14.4	<i>Projecting Cash Collections</i>	10
14.5	<i>The Cash Budget</i>	11
14.6	<i>Pro Forma Financial Statements</i>	14

This study unit is the **third of three** on **planning, budgeting, and forecasting**. The relative weight assigned to this major topic in Part 1 of the exam is **20%**. The three study units are

Study Unit 12: Analysis, Forecasting, and Strategy

Study Unit 13: Budgeting -- Concepts and Methodologies

Study Unit 14: Budgeting -- Calculations and Pro Forma Financial Statements

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

14.1 SALES FORECASTS AND THE SALES BUDGET

1. Sales Forecasts

- a. The sales forecast starts by looking back at historical trends and seeks to determine a pattern so that next year's sales can be predicted.
 - 1) Managers will also forecast sales many years into the future in order to plan more effectively.
- b. One of the most effective ways to do this is to plot sales on a graph and use regression analysis (covered in Study Unit 12, Subunit 1) to forecast next year's sales.
 - 1) Using either the regression analysis or sales per year, the average annual growth rate (AAGR) in sales can be determined.

Growth rate (GR) = (Ending value – Beginning value) ÷ Beginning value

AAGR = (GR₁ + GR₂ + GR₃ + . . . GR_x) ÷ Number of periods

- 2) Based on other factors, managers may choose to revise the initial AAGR up or down.
- a) Other factors may include macroeconomic factors, production capabilities, competitors' production capabilities, new product introductions, markets opening or closing, inflation, advertising, promotional discounts, etc.

EXAMPLE 14-1 Preliminary Forecast

MoeCo had the following sales for Years 1-5:

<u>Year</u>	<u>Sales (in millions)</u>
1	\$1,146
2	\$1,318
3	\$1,214
4	\$1,362
5	\$1,589
6	--

MoeCo: Sales Forecast
(in millions)

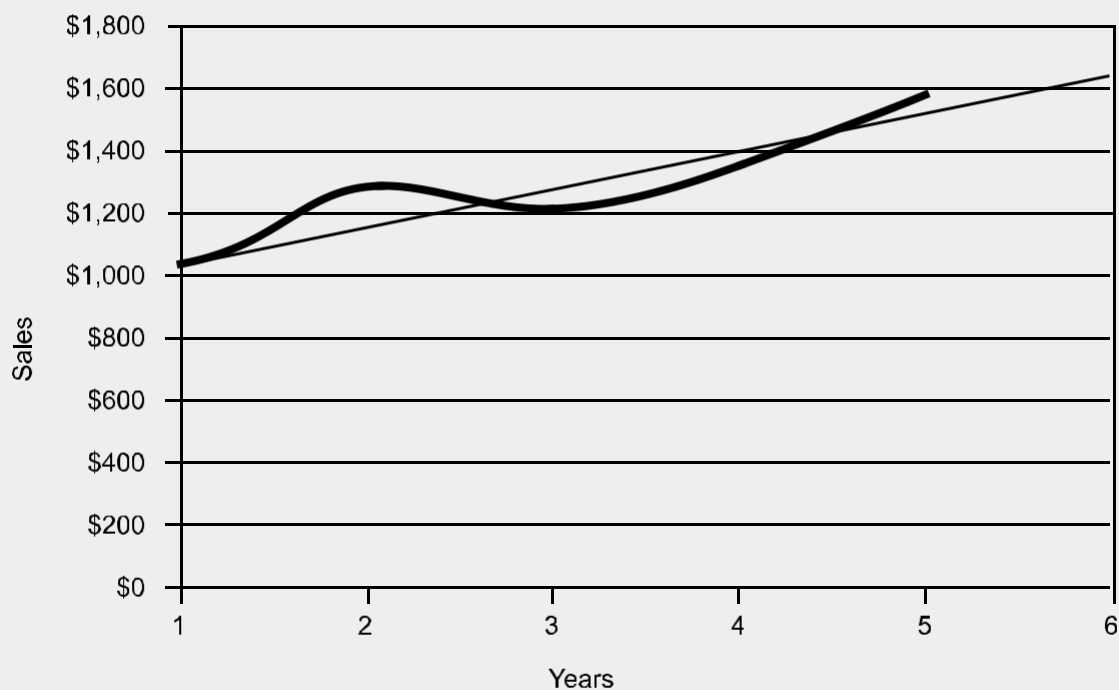


Figure 14-1

Using this information, a preliminary forecast for Year 6 sales can be developed. MoeCo's AAGR can be calculated as follows:

$$\begin{array}{rcl}
 (\$1,318 - \$1,146) \div \$1,146 & = & 15.0\% \\
 (\$1,214 - \$1,318) \div \$1,318 & = & -7.9\% \\
 (\$1,362 - \$1,214) \div \$1,214 & = & 12.2\% \\
 (\$1,589 - \$1,362) \div \$1,362 & = & 16.7\% \\
 & & \underline{36.0\%} \\
 \text{Divided by: number of periods} & & \div 4 \\
 \text{AAGR} & & \underline{\underline{9.0\%}}
 \end{array}$$

-- Continued on next page --

EXAMPLE 14-1 -- Continued

Because MoeCo plans to cut its sales budget by 25% for Year 6, MoeCo anticipates that sales for Year 6 will rise by only 7.8%, not 9.0%.

MoeCo forecasts sales increasing as follows:

<u>Year 6</u>	<u>Year 7</u>	<u>Year 8</u>	<u>Year 9</u>	<u>Year 10</u>	<u>Year 11</u>
7.8%	7.8%	7.8%	7.0%	7.0%	6.5%

MoeCo forecasts that the rate of increase in sales will decline after Year 8 due to factors such as increased competition in the marketplace and lower customer demand.

2. Sales Budget

- a. The sales budget, also called the revenue budget, is the starting point for the massive cycle that produces the annual profit plan (i.e., the master budget).
- b. The sales budget is an outgrowth of the sales forecast. The sales forecast distills recent sales trends, overall conditions in the economy and industry, market research, activities of competitors, and credit and pricing policies.
 - 1) For example,
 - a) A company may determine that demand is highly elastic for its mature products and that growth will come only from new product introductions and from cost savings on existing products.
 - b) At the same time, the company determines that a tight monetary policy on the Fed's part must cause the firm to tighten its credit standards.
 - c) Simultaneously, a competitor that the firm knows is a low-cost producer is also considering moving into the markets that the budgeting company is considering.
 - 2) All of these factors must be taken into account when forming expectations about product sales for the coming budget cycle.
- c. The sales budget must specify both projected unit sales and dollar revenues.

EXAMPLE 14-2 Sales Budget

The demand for this firm's product is elastic, so the price cut in June is expected to boost sales.

	<u>April</u>	<u>May</u>	<u>June</u>	<u>2nd Quarter Totals</u>	<u>Ref.</u>
Projected sales in units	1,000	1,200	1,800	4,000	SB1
Selling price	× \$400	× \$400	× \$380		
Projected total sales	<u>\$400,000</u>	<u>\$480,000</u>	<u>\$684,000</u>	<u>\$1,564,000</u>	SB2

14.2 OPERATING BUDGET CALCULATIONS -- PRODUCTION AND DIRECT MATERIALS

1. Production Budget

- a. The production budget follows directly from the sales budget. To minimize finished goods carrying costs and obsolescence, the levels of production are dependent upon the projections contained in the sales budget.
 - 1) The production budget is concerned with **units only**. Product pricing is not a consideration since the goal is purely to plan output and inventory levels and the necessary manufacturing activity.

EXAMPLE 14-3 Production Budget						
	Source	April	May	June	2nd Quarter Totals	Ref.
Projected sales in units	SB1	1,000	1,200	1,800		
Add: Desired ending inventory (10% of next month's sales)		120	180	200		
Total needed		1,120	1,380	2,000		
Less: Beginning inventory		(100)	(120)	(180)		
Units to be produced		1,020	1,260	1,820	4,100	PB

2. Direct Materials Budget

- a. The direct materials budget follows directly from the production budget. It is concerned with both units and input prices.
 - 1) To minimize raw materials carrying costs and obsolescence, the purchasing of inputs is tied closely to the projections contained in the production budget.

EXAMPLE 14-4 Two Direct Materials Budgets						
Note that in the third month, the process is expected to experience improved efficiency with regard to Raw Material A. A price break on Raw Material B is expected.						
Raw Material A	Source	April	May	June	2nd Quarter Totals	Ref.
Units to be produced	PB	1,020	1,260	1,820		
Raw material per finished product		× 4	× 4	× 3		DMB1
Total units needed for production		4,080	5,040	5,460		
Raw material cost per unit		× \$12	× \$12	× \$12		DMB2
Cost of units used in production		<u>\$48,960</u>	<u>\$60,480</u>	<u>\$65,520</u>	<u>\$174,960</u>	DMB3
Add: Desired units in ending inventory (20% of next month's need)		1,008	1,092	1,600		
Total needs		5,088	6,132	7,060		
Less: Beginning inventory		(400)	(1,008)	(1,092)		
Raw material to be purchased		4,688	5,124	5,968		
Raw material cost per unit		× \$12	× \$12	× \$12		
Cost of raw material to be purchased		<u>\$56,256</u>	<u>\$61,488</u>	<u>\$71,616</u>		DMB4

-- Continued on next page --

EXAMPLE 14-4 -- Continued

Raw Material B	Source	April	May	June	2nd Quarter Totals	Ref.
Units to be produced	PB	1,020	1,260	1,820		
Raw material per finished product		× 2	× 2	× 2		DMB5
Total units needed for production		2,040	2,520	3,640		
Raw material cost per unit		× \$10	× \$10	× \$8		DMB6
Cost of units used in production		<u>\$20,400</u>	<u>\$25,200</u>	<u>\$29,120</u>	<u>\$74,720</u>	DMB7
Add: Desired units in ending inventory (20% of next month's need)		504	728	900		
Total needs		2,544	3,248	4,540		
Less: Beginning inventory		(200)	(504)	(728)		
Raw material to be purchased		2,344	2,744	3,812		
Raw material cost per unit		× \$10	× \$10	× \$8		
Cost of raw material to be purchased		<u>\$23,440</u>	<u>\$27,440</u>	<u>\$30,496</u>		DMB8

14.3 OPERATING BUDGET CALCULATIONS -- OTHERS**1. Direct Labor Budget**

- a. The direct labor budget depends on wage rates, amounts and types of production, numbers and skill levels of employees to be hired, etc.

EXAMPLE 14-5 Direct Labor Budget

No new efficiencies are expected, and the wage rate is set by contract with the union.

	Source	April	May	June	2nd Quarter Totals	Ref.
Units to be produced	PB	1,020	1,260	1,820		
Direct labor hours per unit		× 2	× 2	× 2		DLB1
Projected total direct labor hours		2,040	2,520	3,640		DLB2
Direct labor cost per hour		×\$18.64	×\$18.64	×\$18.64		
Total projected direct labor cost*		<u>\$38,026</u>	<u>\$46,973</u>	<u>\$67,850</u>	<u>\$152,849</u>	DLB3

*NOTE: For the remaining calculations, please round to the nearest whole number.

2. Employee Fringe Benefits

- a. The **cost of fringe benefits** must be derived once the cost of wages has been determined.

EXAMPLE 14-6 Employee Fringe Benefit Projection						
	Source	April	May	June	2nd Quarter Totals	Ref.
Projected direct labor wages	DLB3	\$38,026	\$46,973	\$67,850	\$152,849	
Employer FICA match (7.65%)		2,909	3,593	5,191	11,693	
Health insurance (12.1%)		4,601	5,684	8,210	18,495	
Life insurance (5%)		1,901	2,349	3,393	7,643	
Pension matching (4%)		1,521	1,879	2,714	6,114	
Total projected direct labor cost		\$48,958	\$60,478	\$87,358	\$196,794	DLB4
The full per-hour cost of labor can now be determined. This will be used in determining the costs embedded in units remaining in ending finished goods inventory.						
Since a first-in, first-out (FIFO) assumption is used for all inventories and only units produced in June are expected to remain at the end of June, the calculation is only necessary for June's data.						
Total projected direct labor cost	÷	Total projected direct labor hours	=	Full direct labor cost per hour	Ref.	
\$87,358	÷	3,640	=	\$24	DLB5	

- b. Whether employee fringes are included in direct labor costs or treated as overhead, the effect on cost of goods sold is the same. Both ways include the amounts in variable manufacturing costs.

3. Variable Overhead

- a. The **manufacturing overhead budget** reflects the nature of overhead as a mixed cost, i.e., one that has a variable component and a fixed component.
- b. Variable overhead contains those elements that vary with the level of production.
- 1) Indirect materials
 - 2) Indirect labor
 - 3) Variable factory operating costs (e.g., electricity)
- c. Under GAAP, all manufacturing costs (direct materials, direct labor, variable overhead, and fixed overhead) must be recognized as product costs. This approach is **absorption costing** (full costing).

EXAMPLE 14-7 Variable Overhead Budget						
Note that variable overhead will be applied to finished goods on the basis of direct labor hours.						
	Source	April	May	June	2nd Quarter Totals	Ref.
Variable overhead	DLB2					
Projected total direct labor hours		2,040	2,520	3,640		
Variable OH rate per direct labor hour		× \$2	× \$2	× \$2		MOB1
Projected variable overhead		\$4,080	\$5,040	\$7,280	\$16,400	MOB2

4. Fixed Overhead

a. Fixed overhead contains those elements that remain the same regardless of the level of production.

- 1) Real estate taxes
- 2) Insurance
- 3) Depreciation

EXAMPLE 14-8 Fixed Overhead Budget

Note that fixed overhead will be applied based on the number of units produced.

Fixed overhead	Source	April	May	June	2nd Quarter Totals	Ref.
Projected fixed overhead		\$9,000	\$9,000	\$9,000	\$27,000	MOB3
Projected unit production	PB	÷1,020	÷1,260	÷1,820		
Fixed OH applied per unit		<u>\$ 8.82</u>	<u>\$ 7.14</u>	<u>\$ 4.95</u>		MOB4

5. Ending Finished Goods Inventory Budget

a. The ending finished goods inventory budget can be prepared now that the components of finished goods cost have been projected.

- 1) The end result will have a direct impact on the pro forma balance sheet. The higher the amount of costs capitalized in finished goods, the higher will be the firm's projected asset balance at year end.

EXAMPLE 14-9 Unit-Cost Calculation

Since a first-in, first-out (FIFO) assumption is used for all inventories and only units produced in June are expected to remain at the end of June, this calculation uses June's data.

	Source	Qty.	Source	Input cost	Cost per finished unit
Production costs in ending inventory:					
Direct materials -- raw material A	DMB1	3	DMB2	\$12.00	\$ 36.00
Direct materials -- raw material B	DMB5	2	DMB6	8.00	16.00
Direct labor	DLB1	2	DLB5	24.00	48.00
Variable overhead	DLB1	2	MOB1	2.00	4.00
Fixed overhead	--	1	MOB4	4.95	4.95
Finished goods cost					<u>\$108.95</u>

Now the total amount of cost embedded in ending inventory can be derived.

Total FIFO cost per finished unit	×	Projected units at June 30	=	Projected ending inventory	Ref.
\$108.95	×	200	=	\$21,790	EFGIB

6. Cost of Goods Sold Budget

a. The cost of goods sold budget combines the results of the projections for the three major inputs (materials, labor, overhead).

- 1) The result directly affects the pro forma income statement. Cost of goods sold is the single largest reduction to revenues for a manufacturer.

EXAMPLE 14-10 Cost of Goods Sold Budget for the Quarter			
	Source		Ref.
Beginning finished goods inventory		\$ 16,200	
Manufacturing costs:			
Direct materials used -- A	DMB3	\$174,960	
Direct materials used -- B	DMB7	74,720	
Direct labor employed	DLB4	196,794	
Variable overhead	MOB2	16,400	
Fixed overhead	MOB3	<u>27,000</u>	
Cost of goods manufactured		489,874	
Cost of goods available for sale		\$506,074	
Ending finished goods inventory	EFGIB	(21,790)	
Cost of goods sold		<u>\$484,284</u>	CGSB
Budgeted gross margin, the amount left over from sales revenue after the cost of the product, can now be calculated.			
	Source		
Sales	SB2	\$1,564,000	
Cost of goods sold	CGSB	<u>(484,284)</u>	
Gross margin		<u>\$1,079,716</u>	

- b. The calculation of gross margin is required for external financial reporting. Cost of goods sold for this purpose must be derived using absorption (full) costing, i.e., by including all manufacturing costs, both variable and fixed.
- c. For internal reporting, variable (direct) costing, which includes only variable costs in cost of goods sold, is more useful than absorption costing.

7. Variable Costing and Contribution Margin

- a. Contribution margin is the amount left over from sales after subtracting variable costs.
 - 1) Contribution margin is the amount available for "contributing" to the covering of fixed costs and providing a profit (a more detailed discussion of absorption and variable costing can be found in Study Unit 7, Subunit 5).
 - 2) Although it is impermissible for external financial reporting, contribution margin is more useful to management accountants because it more accurately reveals the change in profitability resulting from a given change in output.
- b. Cost of goods sold calculated on a variable-costing basis includes **only** those costs that vary directly with the level of production. The amount of sales left over after subtracting variable-basis cost of goods sold is contribution margin.
 - 1) Absorption costing (required for external reporting) includes certain amounts in cost of goods sold that do not vary directly with the level of production, such as fixed overhead costs.
 - 2) Because costs are accumulated so differently, inventory amounts as well as cost of goods sold are different under variable costing from what they are under absorption costing.
- c. The breakeven point is the level of production at which operating income equals zero, i.e., the level at which all fixed costs plus those variable costs incurred to that point have been covered. Every sales dollar beyond breakeven provides operating profit.

$$\text{Breakeven point} = \frac{\text{Total fixed costs}}{\text{Contribution margin per unit}}$$

EXAMPLE 14-11 Breakeven Point

A manufacturer has budgeted total fixed costs of \$1,240,000 and a budgeted contribution margin of \$6.80 per unit. The breakeven point for the budget period is 182,353 units ($\$1,240,000 \div \6.80).



**AUTHOR'S
NOTE**

Breakeven analysis, also called cost-volume-profit analysis, is tested in Part 2 of the CMA exam.

8. Nonmanufacturing Budget

- a. The nonmanufacturing budget consists of the individual budgets for R&D, design, marketing, distribution, customer service, and administrative costs.
 - 1) The development of separate R&D, design, marketing, distribution, customer service, and administrative budgets reflects a **value chain approach**.
 - 2) An alternative is to prepare a single **selling and administrative budget** for nonproduction costs.
- b. The variable and fixed portions of selling and administrative costs (S&A) must be treated separately.
 - 1) Some S&A costs vary directly and proportionately with the level of sales. For example, as more product is sold, sales representatives must travel more miles and serve more customers.
 - 2) Other S&A expenses, such as sales support staff, are fixed; they must be paid no matter the level of sales.
 - 3) As the variable portion of S&A costs increases, contribution margin, i.e., the amount available for covering fixed costs, is decreased.

EXAMPLE 14-12 Nonmanufacturing Costs Budget

Note the separate treatment of the variable and fixed portions.

	Source	April	May	June	2nd Quarter Totals	Ref.
Variable nonmanufacturing costs:						
Projected sales in units	SB1	1,000	1,200	1,800		
Variable S&A expenses (\$3 per unit sold)		$\times \$3$	$\times \$3$	$\times \$3$		
Total variable nonmanufacturing costs		\$ 3,000	\$ 3,600	\$ 5,400	\$ 12,000	
Fixed nonmanufacturing costs:						
Research and development		\$ 8,000	\$ 8,000	\$ 8,000	\$ 24,000	
Design		4,000	4,000	4,000	12,000	
Marketing		7,000	7,000	7,000	21,000	
Distribution		10,000	10,000	10,000	30,000	
Customer service		11,000	11,000	11,000	33,000	
Administrative		50,000	50,000	50,000	150,000	
Total fixed nonmanufacturing costs		\$90,000	\$90,000	\$90,000	\$270,000	
Total nonmanufacturing costs		\$93,000	\$93,600	\$95,400	\$282,000	NMB

- c. Note that management can make tradeoffs among elements of selling and administrative expenses that can affect contribution margin.
 - 1) For example, use of fixed advertising expense will increase contribution margin, while the same sales level might be reached using variable sales commissions, a method that would reduce contribution margin.

9. Pro Forma Operating Income

- a. If the projected level of operating income is insufficient, the various components of the operating budget can be adjusted.

EXAMPLE 14-13		Pro Forma Operating Income	
		Source	
Sales		SB2	\$1,564,000
Cost of goods sold		CGSB	(484,284)
Gross margin			\$1,079,716
Nonmanufacturing costs		NMB	(282,000)
Operating income			\$ 797,716

14.4 PROJECTING CASH COLLECTIONS

1. Capital Budget

- a. The capital budget concerns financing of major expenditures for long-term assets and must therefore have a multi-year perspective. Productive machinery must be acquired to enable the company to achieve its projected levels of output.
- b. Procedures for ranking projects according to their risk and return characteristics are necessary because every organization has finite resources.



**AUTHOR'S
NOTE**

These procedures (net present value, internal rate of return, payback method, etc.) are tested in Part 2 of the CMA exam.

- c. The capital budget has a direct impact on the cash budget and the pro forma financial statements.
 - 1) A capital expenditures budget is a listing of all desired expenditures for long-lived assets. It will have two or more columns.
 - a) The first column is for the upcoming budget year and will be incorporated into the operating budget.

- b) The second and any subsequent columns are for future years. These latter columns are not incorporated into the current operating budget, but serve as a starting point for subsequent years' capital expenditures budgets.
- 2) Principal and interest on debt acquired to finance capital purchases require regular cash outflows. The acquired debt also appears in the liabilities section of the pro forma balance sheet.
- 3) At the same time, the output produced by the new productive assets generates regular cash inflows. In addition, the new assets themselves appear in the assets section of the pro forma balance sheet.

2. Cash Collections Schedule

- a. The projected cash collections schedule is used to estimate the inflows of cash from customer payments.

EXAMPLE 14-14 Cash Collections Schedule						
Note the assumption that 5% of sales will prove to be uncollectible.						
Projected February sales		\$180,000				
Projected March sales		\$220,000				
	<u>Source</u>	<u>April</u>	<u>May</u>	<u>June</u>	<u>Ref.</u>	
Projected sales	SB2	\$400,000	\$480,000	\$684,000		
Cash collections from sales:						
From 2nd prior month sales (30%)		54,000	66,000	120,000		
From prior month sales (50%)		110,000	200,000	240,000		
From current month sales (15%)		60,000	72,000	102,600		
Total cash collections from sales		\$224,000	\$338,000	\$462,600		PCC S

14.5 THE CASH BUDGET

1. Cash Budget Preparation

- a. The cash budget combines the results of the operating budget with the cash collection and disbursement schedules to produce a comprehensive view of the sources and uses of cash.
 - 1) The cash budget is used to plan outside financing activities. For example, if the budget shows a cash deficit at some future date, the firm can plan ahead to borrow the necessary funds or sell stock.
 - 2) The cash budget also is used to plan the investing of excess cash.
 - 3) Dividend policy can also be planned using the cash budget. Dividend payment dates should correspond to a time when the firm has excess cash.

- b. Cash budgets are prepared not only for annual and quarterly periods but also for monthly and weekly periods.
- 1) They are particularly important for organizations operating in seasonal industries.
 - 2) The factors needed to prepare a cash forecast include all other elements of the budget preparation process, plus consideration of collection policies, bad debt estimates, changes in the economy, and anticipation of non-routine transactions.
 - 3) Factors should be considered for short-term and long-term cash forecasting (i.e., nonroutine property purchases and sales).
- c. **Credit and purchasing policies** have a direct impact on the cash budget. Loose credit policies toward customers' credit result in delayed cash receipts. Taking advantage of purchase discounts results in accelerated cash outlays.
- d. The cash budget consists of a cash disbursements schedule for raw materials.

EXAMPLE 14-15		Materials Cash Disbursements Schedule				
March raw materials purchases -- A		\$45,000				
March raw materials purchases -- B		\$17,000				
	Source	April	May	June	Ref.	
Projected raw materials cost -- A	DMB4	\$56,256	\$61,488	\$71,616		
Cash payments for purchases of A:						
For prior month purchases (40%)		18,000	22,502	24,595		
For current month purchases (60%)		33,754	36,893	42,970		
Total cash disbursements for A		\$51,754	\$59,395	\$67,565		PCDS1
Projected raw materials cost -- B	DMB8	\$23,440	\$27,440	\$30,496		
Cash payments for purchases of B:						
For prior month purchases (40%)		6,800	9,376	10,976		
For current month purchases (60%)		14,064	16,464	18,298		
Total cash disbursements for B		\$20,864	\$25,840	\$29,274		PCDS2

EXAMPLE 14-16

Cash Budget

The last section deals with the anticipated handling of a cash deficit.

	Source	April	May	June
Beginning cash balance		\$ 50,000	\$100,000	\$130,767
Cash collections from sales	PCCS	224,000	338,000	462,600
Cash available for disbursement		\$274,000	\$438,000	\$593,367
Cash disbursements:				
For raw material A	PCDS1	\$ 51,754	\$ 59,395	\$ 67,565
For raw material B	PCDS2	20,864	25,840	29,274
For direct labor	DLB4	48,958	60,478	87,358
For variable overhead	MOB2	4,080	5,040	7,280
For fixed overhead	MOB3	9,000	9,000	9,000
For nonmanufacturing costs	NMB	93,000	93,600	95,400
For equipment purchases	Cap. Budg.	0	0	30,000
Total disbursements		\$227,656	\$253,353	\$325,877
Surplus of cash available over disbursements		\$ 46,344	\$184,647	\$267,490
Desired ending cash balance		(100,000)	(100,000)	(100,000)
Surplus (deficiency) of cash		\$ (53,656)	\$ 84,647	\$167,490

-- Continued on next page --

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EXAMPLE 14-16 -- Continued

	<u>April</u>	<u>May</u>	<u>June</u>
Financing:			
Borrowings	\$ 53,656	\$ 0	\$ 0
Repayments:			
Principal	0	(53,656)	0
Interest	0	(224)	0
Net financing	<u>\$ 53,656</u>	<u>\$(53,880)</u>	<u>\$ 0</u>
Ending cash balance	<u>\$100,000</u>	<u>\$130,767</u>	<u>\$267,490</u>

EXAMPLE 14-17 Cash Budget

Following is a cash budget for a company that had budgeted sales of \$9,000 for January, \$9,700 for February, and \$13,950 for March.

Cash Budget				
For Quarter Ending March 31				
	<u>January</u>	<u>February</u>	<u>March</u>	<u>Total</u>
Beginning cash balance	\$ 80	\$ 20	\$ 1,957	\$ 80
Receipts:				
Collection from sales*	<u>6,800</u>	<u>9,350</u>	<u>11,825</u>	<u>27,975</u>
Total cash available	<u>\$6,880</u>	<u>\$9,370</u>	<u>\$13,782</u>	<u>\$28,055</u>
Payments:				
Purchases**	\$3,150	\$2,760	\$ 3,960	\$ 9,870
Sales salaries	1,350	1,455	2,093	4,898
Supplies	360	388	558	1,306
Utilities	120	110	100	330
Administrative salaries	1,800	1,800	1,800	5,400
Advertising	80	80	80	240
Equipment purchases	<u>0</u>	<u>820</u>	<u>3,000</u>	<u>3,820</u>
Total payments	<u>\$6,860</u>	<u>\$7,413</u>	<u>\$11,591</u>	<u>\$25,864</u>
Ending balance	<u>\$ 20</u>	<u>\$1,957</u>	<u>\$ 2,191</u>	<u>\$ 2,191</u>

* Sales are 50% cash sales and 50% on credit (net 30 days). Thus, 50% of each month's sales are collected in the month of the sale and 50% are collected in the following month. For example, the February collections were calculated as follows:

50% of January sales	\$4,500
50% of February sales	<u>4,850</u>
	<u>\$9,350</u>

** Purchases are on terms of net 30 days. Thus, purchases are paid for in the month following the purchase. The amount paid in February (\$2,760) was the total purchases for January.

14.6 PRO FORMA FINANCIAL STATEMENTS

1. Pro Forma Statement of Income

- a. **Pro forma** is a Latin phrase meaning “according to form.” It can be loosely translated to mean “as if.” Financial statements are referred to as pro forma when they reflect projected, rather than actual, results.
- 1) The pro forma income statement is used to decide whether the budgeted activities will result in an acceptable level of income. If the initial pro forma income shows a loss or an unacceptable level of income, adjustments can be made to the component parts of the master budget.
 - 2) Other strategic objectives can also be observed from the pro forma income statement, such as a target gross margin percentage and the interest coverage ratio (times interest earned). The adequacy of earnings per share can also be observed from the pro forma income statement.

EXAMPLE 14-18 Pro Forma Income Statement		
Manufacturing Company		
Pro Forma Statement of Income		
2nd Quarter		
Sales		\$1,564,000
Beginning finished goods inventory	\$ 16,200	
Add: Cost of goods manufactured	<u>489,874</u>	
Goods available for sale	<u>\$506,074</u>	
Less: Ending finished goods inventory	<u>(21,790)</u>	
Cost of goods sold		<u>(484,284)</u>
Gross margin		\$1,079,716
Less: Selling and administrative expenses		<u>(282,000)</u>
Operating income		\$ 797,716
Add: Other revenues and gains		15,000
Less: Other expenses and losses		<u>(10,000)</u>
Earnings before interest and taxes		\$ 802,716
Less: Interest expense		<u>(224)</u>
Earnings before income taxes		\$ 802,492
Less: Income taxes (40%)		<u>(320,997)</u>
Net income		<u>\$ 481,495</u>
Basic earnings per share for 2nd quarter		
(5,000,000 common shares issued and outstanding)		\$0.096

- b. Revenue and cost assumptions can be changed and their effects on pro forma net income observed.

EXAMPLE 14-19 Pro Forma Income Statement -- Change in Gross Margin

The company has projected gross margin to be 69% of sales (\$1,079,716 ÷ \$1,564,000). If gross margin is changed to 75% of sales (\$1,564,000 × .75 = \$1,173,000), the pro forma income statement will reflect a different bottom line.

Manufacturing Company	
Pro Forma Statement of Income	
2nd Quarter	
Sales	\$1,564,000
Cost of goods sold	(391,000)
Gross margin	\$1,173,000
Less: Selling and administrative expenses	(282,000)
Operating income	\$ 891,000
Add: Other revenues and gains	15,000
Less: Other expenses and losses	(10,000)
Earnings before interest and taxes	\$ 896,000
Less: Interest expense	(224)
Earnings before income taxes	\$ 895,776
Less: Income taxes (40%)	(358,310)
Net income	\$ 537,466
Basic earnings per share for 2nd quarter	
(5,000,000 common shares issued and outstanding)	\$0.107

2. Percentage of Sales Method

- a. After sales are forecasted, future financial statements must be forecasted. The most common method is the percent of sales method.
 - 1) Under this method, many items on the income statement and balance sheets are assumed to increase proportionately to sales.
 - 2) Other items may be based off historical data (i.e., interest expense may remain constant due to contracts previously entered into) or be based off forecasted net sales (i.e., cost of goods sold will be 60% of net sales).
- b. The first financial statement forecasted is generally the income statement.

EXAMPLE 14-20 Percentage of Sales Method

LisaCo's Actual Year 5 and Projected Year 6 Income Statement
(millions of dollars)

	Actual Year 5	Forecast Basis	Year 6 Forecast
Sales	\$4,000	1.11 × Year 6 sales	\$4,440
Cost of goods sold	(3,200)	80% of net sales	(3,552)
Gross margin	\$ 800		\$ 888
Selling and administrative expenses	(300)	1.11 × Year 6 sales	(333)
Operating income	\$ 500		\$ 555
Other revenue	200	Same as last year	200
Earnings before interest and taxes (EBIT)	\$ 700		\$ 755
Interest	(100)	Same as last year	(100)
Earnings before taxes (EBT)	\$ 600		\$ 655
Taxes (40%)	(240)		(262)
Net income	\$ 360		\$ 393

Notes

- Sales and selling and administrative expenses are expected to increase by 11% in the next year.
- Cost of goods sold is expected to remain at 80% of net sales.
- Other revenue and interest expense are expected to remain constant.

3. Pro Forma Balance Sheet

- a. The pro forma balance sheet is prepared using the cash and capital budgets and the pro forma income statement.
 - 1) The pro forma balance sheet is the beginning-of-the-period balance sheet updated for projected changes in cash, receivables, payables, inventory, etc.
 - 2) If the balance sheet indicates that a contractual agreement may be violated, the budgeting process must be repeated.
 - a) For example, some loan agreements require that owners' equity be maintained at some percentage of total debt or that current assets be maintained at a given multiple of current liabilities.

4. Pro Forma Statement of Cash Flows

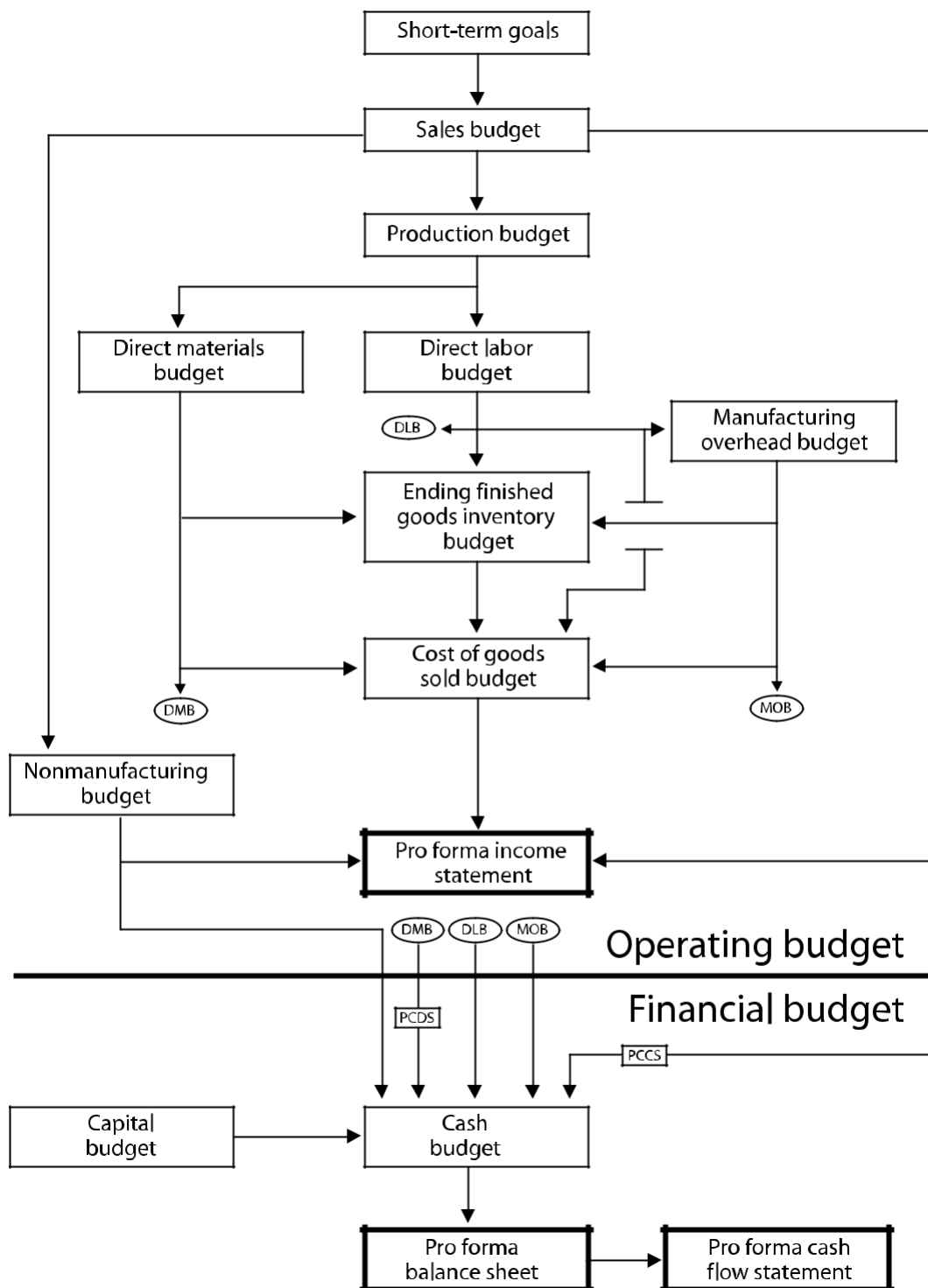
- a. The pro forma statement of cash flows is normally the last statement prepared.
- b. The pro forma statement of cash flows is intended to help users evaluate the firm's liquidity, solvency, and financial flexibility.
- c. The pro forma statement of cash flows classifies cash receipts and disbursements depending on whether they are from operating, investing, or financing activities.
 - 1) The **direct** presentation reports the major classes of gross cash operating receipts and payments and the difference between them.
 - 2) The **indirect** presentation reconciles net income with net operating cash flow. Under GAAP, this reconciliation must be disclosed, regardless of which presentation is chosen.
 - a) The reconciliation requires balance sheet data, such as the changes in accounts receivable, accounts payable, and inventory, as well as net income.
- d. All the pro forma statements are interrelated (articulated), e.g., the pro forma cash flow statement will include anticipated borrowing. The interest on this borrowing will appear in the pro forma income statement.

5. Financial Projections and Ratio Analysis

- a. Pro forma financial statements are of interest to parties outside the organization as well as inside.
 - 1) Banks and stock analysts in particular want to know what the firm believes its results will be.
- b. Projections help the bank assess whether the company anticipates satisfying the requirements of debt covenants.
 - 1) Typically, a firm's financing agreement with its bank requires that its debt ratio remain below a certain threshold and that its coverage ratios remain above a threshold.
 - a) The debt ratio is the portion of the firm's capital structure that consists of debt, i.e., total liabilities divided by total assets.
 - b) The most common coverage ratio is times interest earned, i.e., earnings before interest and taxes divided by interest expense.
 - 2) Projection of satisfactory levels of these ratios provide the bank some assurance that the firm will remain solvent for the foreseeable future.
- c. Earnings per share (EPS) is probably the most heavily relied-upon performance measure used by investors.
 - 1) EPS states the amount of current-period earnings that can be associated with a single share of a corporation's common stock.
 - a) EPS is only calculated for common stock because common shareholders are the residual owners of a corporation.
 - 2) Of the two versions of EPS required for external financial reporting (basic and diluted), only basic is needed on Part 1 of the CMA exam.

$$\text{Basic EPS} = \frac{\text{Net income} - \text{Preferred dividend}}{\text{Weighted-average number of common shares outstanding}}$$

- a) The claims of preferred shareholders must be satisfied before those of the residual owners. Thus, amounts associated with preferred stock are not available to the common shareholders and must be removed from the numerator.
- b) The number of shares outstanding is weighted by the number of months the shares are outstanding.



GLEIM®

**2021
EDITION**

CMA REVIEW

PART 1: FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS



C. Performance Management (20% - Levels A, B, and C)

1. *Cost and variance measures*

- a. Comparison of actual to planned results (SU 15)
- b. Use of flexible budgets to analyze performance (15.2)
- c. Management by exception (15.1)
- d. Use of standard cost systems (7.4, 13.2, 15.1)
- e. Analysis of variation from standard cost expectations (15.3-15.8)

2. *Responsibility centers and reporting segments*

- a. Types of responsibility centers (16.1)
- b. Transfer pricing (16.5)
- c. Reporting of organizational segments (16.2-16.3)

3. *Performance measures*

- a. Product profitability analysis (16.2)
- b. Business unit profitability analysis (16.2)
- c. Customer profitability analysis (16.2)
- d. Return on investment (16.3)
- e. Residual income (16.3)
- f. Investment base issues (16.3)
- g. Key performance indicators (KPIs) (16.4)
- h. Balanced scorecard (16.4)

STUDY UNIT FIFTEEN

COST AND VARIANCE MEASURES

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Performance Management

Performance reporting is a major topic on the CMA exam. Factors to be analyzed for control and performance evaluation include revenues, costs, profits, and investment in assets. Variance analysis based on flexible budgets and standard costs is heavily tested, as is responsibility accounting for revenue, cost, investment, and profit centers. The balanced scorecard (explained in Study Unit 16) is included in this discussion.

This study unit is the **first of two** on **performance management**. The relative weight assigned to this major topic in Part 1 of the exam is **20%**. The two study units are

Study Unit 15: Cost and Variance Measures

Study Unit 16: Responsibility Accounting and Performance Measures

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

15.1 VARIANCE ANALYSIS OVERVIEW

1. Uses of a Budget

- a. A budget communicates to employees the organization's operational and strategic objectives. The budget quantifies the operational steps that ultimately lead to the achievement of strategic objectives.
- b. A performance evaluation system must be used to monitor progress toward the budget's objectives.
- c. Feedback should be timely so that managers can take corrective action.
 - 1) Information sent to top management is ordinarily more highly aggregated and less timely than that communicated to managers at operational levels.
 - a) Top managers are concerned with the organization's overall financial results and long-term prospects and are responsible for the strategic planning function.
 - 2) Lower-level reports contain more quantitative information of an operational nature, e.g., production data.

2. Variance Analysis

- a. Variance analysis is the basis of any performance evaluation system using a budget.
- b. Variances are the differences between the amounts budgeted and the amounts actually incurred (or earned in the case of revenues).
 - 1) A primary reason for calculating variances is to notify management whenever an unusual event has occurred.
 - 2) Management uses this information to improve future performance forecasts and take early corrective action (e.g., changing a business process).
- c. Variances may be favorable or unfavorable.
 - 1) Whether a variance is favorable or unfavorable depends on how it affects income. A **favorable variance** increases income, and an **unfavorable variance** decreases income. Favorable variances are labeled with an F and unfavorable variances are labeled with a U.
 - 2) On the revenue side,
 - a) A **favorable variance** occurs when actual revenues exceed budgeted revenues.
 - b) An **unfavorable variance** occurs when actual revenues are less than budgeted revenues.
 - 3) On the cost side,
 - a) A **favorable variance** occurs when actual costs are less than budgeted costs.
 - b) An **unfavorable variance** occurs when actual costs exceed budgeted costs.

EXAMPLE 15-1 Unfavorable Variance

Under efficient conditions, a worker should complete one unit of product per hour. If workers are normally paid \$6 per hour, the standard labor cost per unit is \$6 per unit. If the actual per-unit amounts for a 1-week period were 1.1 hours at \$6.25 per hour, or \$6.88 per unit, the variance is \$.88 per unit. The variance is unfavorable because the actual cost exceeded the standard cost.

- d. The significance of variances depends not only on their amount but also on their direction, frequency, and trend.
 - 1) Persistent variances may indicate that standards need to be reevaluated.
- e. Variance analysis enables **management by exception**, the practice of giving attention primarily to significant deviations from expectations (whether favorable or unfavorable).
 - 1) Concentrating on operations that are not performing within expected limits is likely to yield the best ratio of benefits to costs.

3. Assignment of Responsibility

- a. A crucial part of variance analysis is the **assignment of responsibility**. The performance measures on which managers are judged should be directly related to the factors that drive the element being measured, e.g., cost drivers and revenue drivers.
- b. The goal is to assign responsibility for variances to those most likely to have information that will enable management to find solutions. A manager who does not control an activity may nevertheless be the individual who is best informed about it.
 - 1) The constructive approach is to promote learning and continuous improvement in manufacturing operations, not to assign blame. However, variance analysis may be useful in evaluating managers' performance.

4. Overview of Variances



AUTHOR'S
NOTE

The variances beginning below are covered in greater detail throughout this study unit. CMA candidates should be prepared to calculate and understand these variances.

a. Static Budget Variance

- 1) The beginning of variance analysis is the static budget variance.
- 2) The static budget variance measures the difference between the static (master) budget amount and the actual results. It is the total variance to be explained.

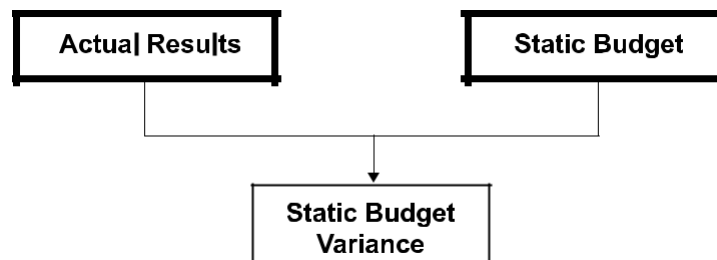


Figure 15-1

b. Flexible Budget Variance and Sales-Volume Variance

- 1) The static budget variance consists of a flexible budget variance and a sales-volume variance.
- 2) The **flexible budget variance** is the difference between the actual results and the budgeted amount for the **actual activity level**. It may be analyzed in terms of variances related to selling prices, input costs, and input quantities.
 - a) A **flexible budget** must be based on per-unit variable costs (costs that vary with activity) and total fixed costs (costs incurred regardless of activity). Thus, it consists of the costs that should have been incurred given the actual level of production.
 - i) Actual production is actual output, but variable costs are measured using the standard level of inputs (e.g., direct labor hours, machine hours, etc.).
 - b) **Standard costs** alert management to cost problems, permit management by exception, and may increase the efficiency of employees who participate in setting standards. Standards costs also facilitate flexible budgeting, performance evaluation, and setting employee goals.
 - c) Standard costs should be established for direct materials, direct labor, and overhead. These standards can then be used to calculate variances.
 - i) Standard costs exclude inefficiencies and reflect expected changes. Their emphasis is on current performance and potential improvement.
- 3) The **sales-volume variance** is the difference between the flexible budget and static budget amounts if selling prices and costs are constant.
- 4) These concepts are explained more fully in Study Unit 7, Subunit 4, item 3.

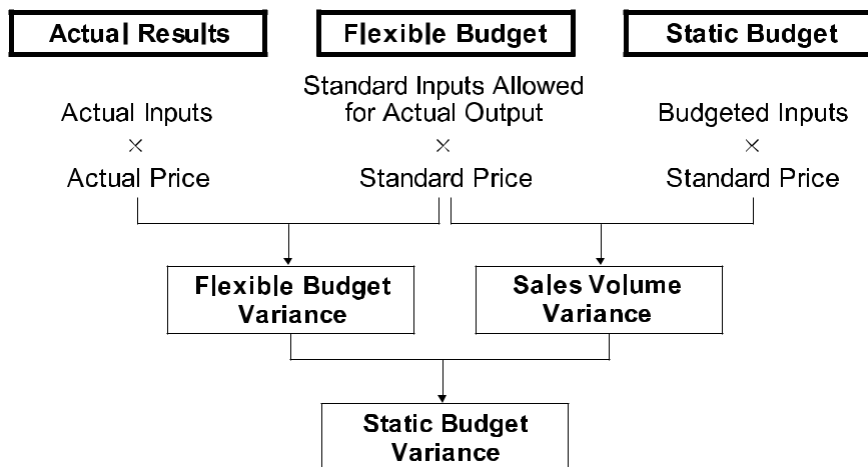


Figure 15-2

c. **Components of the Flexible Budget Variance**

- 1) The flexible budget variance consists of the following variances, which are depicted in Figure 15-3.

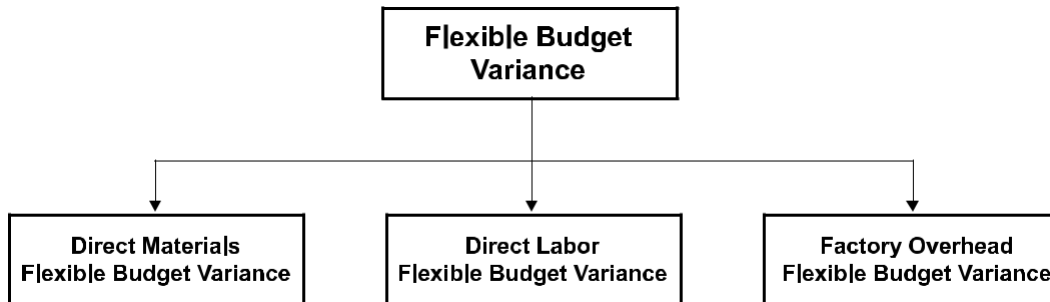


Figure 15-3

- 2) A **direct materials variance** includes a
- a) Price variance
 - b) Quantity or usage variance (an efficiency variance for direct materials)
 - i) When a product has more than one input, the (a) materials mix variance and (b) materials yield variance can be calculated.
- 3) A **direct labor variance** includes a(n)
- a) Rate variance (a price variance for direct labor)
 - b) Efficiency variance
 - i) When labor rates vary, the (a) labor mix variance and (b) labor yield variance can be calculated.
- 4) **Factory overhead variances** have variable and fixed components. A four-way analysis includes two variable and two fixed components:
- a) Variable overhead spending variance
 - b) Variable overhead efficiency variance
 - c) Fixed overhead spending variance (also known as a budget variance)
 - d) Fixed overhead production-volume

variance d. **Components of the Sales-Volume Variance**

- 1) When more than one product is made, the sales volume variance consists of the following variances:
- a) **Sales yield variance**
 - b) **Sales mix variance**

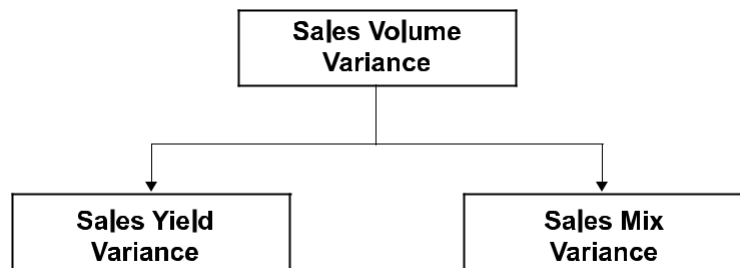


Figure 15-4

NOTE: Each of the variances introduced on the previous pages and above will be examined in detail in subsequent subunits.

5. Advantages and Disadvantages of Variance Analysis

- a. Advantages of variance analysis include that it
 - 1) Identifies areas where actual results are different from budget and management can take appropriate action
 - 2) Provides a basis for investigation that promotes an understanding of operations
 - 3) Promotes accountability among personnel
 - 4) Evaluates performance of personnel
 - 5) Identifies budget estimates requiring revision
- b. Disadvantages of variance analysis include that
 - 1) The length of time it takes to identify a problem may be too long for management to effectively take action
 - 2) There is a focus on the past without looking at the future

15.2 STATIC AND FLEXIBLE BUDGET VARIANCES

1. The Static Budget Variance

- a. The static budget and actual results of the period are required to calculate the static budget variance.
- b. A **static budget** is prepared before the budget period begins and is not changed. It is based only on the output planned at the beginning of the budget period (i.e., for only one level of activity).
 - 1) The static budget reflects management's best estimates of, for example, sales, production, input prices, labor and overhead costs, and selling and administrative costs.
- c. **Standard prices (SP)** are determined for the underlying cost drivers. They are budgeted unit costs established to improve productivity and efficiency.
 - 1) A **cost driver** is used to assign costs to a cost object. It is a measure of activity, such as direct labor hours or machine hours, that has a cause-and-effect relationship with total cost.
 - 2) A standard price is not an average of past costs but an objectively determined estimate of what a cost should be. It may be based on accounting, engineering, or statistical quality control studies.
 - a) A standard price can be compared to "par" on a golf course.
 - 3) Standard prices must be kept **current** to provide relevant information.
 - a) If prices have changed considerably for a particular material, a variance is always reported if the standard price is not changed. Much of the usefulness of standard prices is lost if a large variance is always expected.

- d. The **standard quantity (SQ)** is an objectively determined estimate of the quantity expected to be produced.
- 1) **Ideal standards** are standard costs under optimal conditions. They are based on the work of the most skilled workers with no allowance for waste, spoilage, machine breakdowns, or other downtime.
 - a) These **tight standards** are not in wide use because they are difficult to attain.
 - 2) **Practical standards** are defined as the performance that is reasonably expected to be achieved with an allowance for normal spoilage, waste, and downtime.
 - a) Practical standards represent possible but difficult-to-attain results; thus, practical standards are useful in determining standard quantity.



SUCCESS TIP

It is important to note that on the CMA exam, you may encounter slightly different names for the terms used in this outline. As a CMA candidate, it is important to be able to understand and apply variance analysis regardless of the terminology used.

Standard Price	Standard Quantity
Standard Cost	Standard Materials
Budgeted Price	Budgeted Quantity
Budgeted Cost	Budgeted Materials

- e. Once all standard prices and standard quantities are determined, the static budget can be calculated.

$$\text{Static budget} = \text{Standard quantity} \times \text{Standard price} = \text{SQ} \times \text{SP}$$

- f. The **actual results** are prepared after the budget period ends.
- 1) The actual results reflect the revenues actually earned and the costs actually incurred.
 - 2) The drivers are therefore adjusted to the actual amounts.

$$\text{Actual results} = \text{Actual quantity} \times \text{Actual price} = \text{AQ} \times \text{AP}$$

- g. The **static budget variance** is the difference between the static budget and the actual results for the period.
- 1) The static budget variance provides useful information, but it does not explain the cause of the variance.

$$\text{Static budget variance} = \text{Actual results} - \text{Static budget} = (\text{AQ} \times \text{AP}) - (\text{SQ} \times \text{SP})$$

EXAMPLE 15-2 Total Static Budget Variance

At the end of the current period, ChowDown, Inc., a pet food manufacturer, prepared the following analysis of the static budget variance:

	AQ×AP Actual Results	Static Budget Variances	SQ×SP Static Budget
Units sold	16,500	(1,500)U	18,000
Revenue	\$ 1,947,000	\$ (213,000)U	\$ 2,160,000
Variable costs:			
Direct materials	750,750	160,500 F	911,250
Direct labor	321,750	(24,750)U	297,000
Variable manufacturing overhead	214,500	(41,700)U	172,800
Total variable costs	<u>\$ 1,287,000</u>	<u>\$ 94,050 F</u>	<u>\$ 1,381,050</u>
Contribution margin	\$ 660,000	\$(118,950) U	\$ 778,950
Fixed overhead	429,000	21,000 F	450,000
Operating income	<u>\$ 231,000</u>	<u>\$ (97,950)U</u>	<u>\$ 328,950</u>
Total Static Budget Variance			
\$ (97,950)U			

- 2) Interpretation of the static budget variance can sometimes be misleading. For example, the \$160,500 variance on direct materials in Example 15-2 is labeled as favorable because actual cost was less than standard. However, if the variance was mostly the result of production being lower than planned, the favorable variance resulted regardless of the actions of the manager responsible. In fact, (s)he may have been the cause of production delays, which led to the use of fewer materials.
- 3) A flexible budget using standard costs helps management determine how much of the static budget variance arose from inaccurate forecasts of output sold and variations in the effectiveness and efficiency of actual output.

2. Flexible Budget and Sales-Volume Variances

- a. Flexible budget variances and sales-volume variances are based on flexible budgets.
- b. A **flexible budget** adjusts for changes in the volume of activity. It can be adapted to any level of production. Budgeted revenues and costs are based on the actual quantities and standard costs.

$$\text{Flexible budget} = \text{Actual quantity} \times \text{Standard price} = \text{AQ} \times \text{SP}$$

- c. **Flexible budget variances** result from variations in the efficiency and effectiveness of producing actual output. They are the differences between actual results and flexible budget amounts.

$$\begin{aligned} \text{Flexible budget variance} &= \text{Actual results} - \text{Flexible budget} \\ &= (\text{AQ} \times \text{AP}) - (\text{AQ} \times \text{SP}) \\ &= \text{AQ} \times (\text{AP} - \text{SP}) \end{aligned}$$

- d. **Sales-volume variances** result from inaccurate forecasts of output sold. They are the differences between flexible budget amounts and static budget amounts.

$$\begin{aligned}\text{Sales-volume variance} &= \text{Flexible budget} - \text{Static budget} \\ &= (\text{AQ} \times \text{SP}) - (\text{SQ} \times \text{SP}) \\ &= \text{SP} \times (\text{AQ} - \text{SQ})\end{aligned}$$

**SUCCESS TIP**

The questions in this study unit use formulas written in different ways. For example, the flexible budget variance can be written in the following formats: $\text{AQ} \times (\text{AP} - \text{SP})$, $\text{AQ} \times (\text{SP} - \text{AP})$, $(\text{AP} - \text{SP}) \times \text{AQ}$, or $(\text{SP} - \text{AP}) \times \text{AQ}$. In each case, the absolute value is derived. The effect on income is then determined to decide whether the variance is favorable or unfavorable. Understanding the variances and understanding what each question is looking for will help you avoid becoming confused by the formulas being presented in different formats.

EXAMPLE 15-3 Total Static Budget Variance -- Components

At the end of the current period, ChowDown, Inc., prepared the following analysis of the total static budget variance:

	AQ×AP Actual Results (AR)	AR–FB Flexible Budget Variances	AQ×SP Flexible Budget (FB)	FB–SB Sales Volume Variances	SQ×SP Static Budget (SB)
Units sold	<u>16,500</u>		<u>16,500</u>	<u>(1,500) U</u>	<u>18,000</u>
Revenue	\$1,947,000	\$(33,000) U	\$1,980,000	\$(180,000) U	\$2,160,000
Variable costs:					
Direct materials	750,750	84,563 F	835,313	75,937 F	911,250
Direct labor	321,750	(49,500) U	272,250	24,750 F	297,000
Variable non-mfg. overhead	<u>214,500</u>	<u>(56,100) U</u>	<u>158,400</u>	<u>14,400 F</u>	<u>172,800</u>
Total variable costs	<u>\$1,287,000</u>	<u>\$(21,037) U</u>	<u>\$1,265,963</u>	<u>\$115,087 F</u>	<u>\$1,381,050</u>
Contribution margin	660,000	(54,037) U	714,037	(64,913) U	778,950
Fixed overhead	<u>429,000</u>	<u>21,000 F</u>	<u>450,000</u>	<u>0</u>	<u>450,000</u>
Operating income	<u>\$231,000</u>	<u>\$(33,037) U</u>	<u>\$264,037</u>	<u>\$(64,913) U</u>	<u>\$328,950</u>
	Total Flexible Budget Variance \$(33,037) U		Total Sales Volume Variance \$(64,913) U		
	Total Static Budget Variance \$(97,950) U				

15.3 DIRECT MATERIALS VARIANCES



SUCCESS TIP

CMA candidates must be able to calculate and analyze variances, which involves identifying causes and recommending corrective actions. Typically, a favorable variance is seen as desirable while an unfavorable one is undesirable, but this is not always the case. When taking the exam, it is easy to automatically see a favorable variance as desirable; however, you always need to analyze the variances, determine their causes, and evaluate whether they are desirable or not before recommending a course of action.

1. The **direct materials flexible budget variance** has price and quantity (efficiency) components. These two sources of the total variance can be isolated.
 - a. Part of the total variance is attributed to using more or less materials than planned (the quantity component).
 - b. Part is attributed to the price of materials being different than what was planned (the price component).

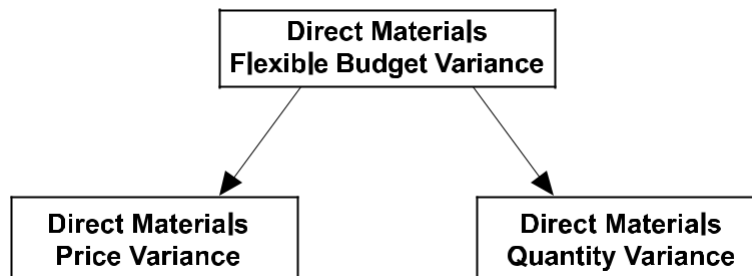


Figure 15-5

2. Direct Materials Price Variance

- a. The direct materials price variance equals the actual quantity of total input times the difference between the standard price and the actual price.

(Actual units produced × Actual materials per unit) × (Actual price – Standard price)

- b. The price variance may be isolated at the time of purchase or when materials are transferred to work-in-process.
 - 1) A **purchase price variance** is a nonmanufacturing variance. It measures the difference between (a) the amount paid for **all** units of materials purchased during a specific period and (b) the amount expected to be paid. This formula can be used to calculate variances at the earliest time possible, which is at the time goods are purchased.

- c. Once a variance has been calculated, the next step is to analyze the reasons for the variance. In general, a favorable variance is desirable and an unfavorable variance is undesirable. But that is not always true.
- 1) An **unfavorable direct materials price variance** means that the actual price paid for materials is higher than the estimated standard price. This is usually undesirable.
 - a) The purchasing function may be responsible under the assumption that it bought materials that cost too much. Thus, an unfavorable variance is not the fault of the production departments.
 - b) However, the unfavorable variance may simply indicate that prices in the industry have risen, and standard prices should be updated.
 - 2) A **favorable direct materials price variance** may mean that the purchasing function performed well by finding a lower-cost source for the materials.
 - a) However, the favorable variance may simply indicate that prices in the industry have fallen, and standard prices should be updated.
 - b) Another possibility is that the lower price may be attributable to lower-quality materials. An analysis must be made to determine whether the lower-quality materials result in a lower-quality product or the use of excessive quantities in production. In these cases, the purchasing function is at fault.
 - 3) This variance is generally considered the responsibility of the purchasing manager and purchasing department.

3. Direct Materials Quantity Variance

- a. The direct materials quantity variance (an efficiency or usage variance) equals the budgeted price times the difference between the actual quantity of total input and the budgeted quantity of total input.

$$\text{Standard price} \times \left[\left(\begin{array}{cc} \text{Actual units} & \text{Actual materials} \\ \text{produced} & \times \text{per unit} \end{array} \right) - \left(\begin{array}{cc} \text{Actual units} & \text{Standard} \\ \text{produced} & \times \text{materials per unit} \end{array} \right) \right]$$

- 1) When calculating the quantity variance, the standard quantity equals actual units produced times the standard materials per unit. This equals the materials that should have been used given the actual level of production.
- 2) The actual cost of the materials is ignored because the variance isolates the effect that would have occurred given no price variance.

- b. Once a variance has been calculated, the next step is to analyze the reasons for the variance. In general, a favorable variance is desirable and an unfavorable variance is undesirable. But that is not always true.
- 1) An **unfavorable materials quantity variance** is usually the responsibility of the production department and may indicate an excessive use of materials. It also may indicate theft of materials or other waste or shrinkage.
 - a) However, the excessive use might be attributable to using lower-quality materials that were purchased at a lower price.
 - b) An alternative explanation is that excessive usage may have been caused by using unskilled (and lower cost) labor. Thus, a favorable labor rate variance may have contributed to an unfavorable materials quantity variance.
 - 2) A **favorable materials quantity variance** may indicate that workers have been unusually efficient, for example, by reducing normal spoilage.
 - a) However, the reduced usage may indicate that they are producing lower-quality products with less than the standard quantity of materials.
 - b) Accordingly, a favorable variance is not always desirable. It may be as bad as, or worse than, an unfavorable variance. It may suggest that costs have been reduced at the expense of product quality.
 - 3) This variance is generally considered the responsibility of the production manager and production department.

EXAMPLE 15-4 Direct Materials -- Flexible Budget Variance

ChowDown, Inc., estimated output for the period of 18,000 units. However, actual output was 16,500 units. Standard direct materials per unit were estimated at 7.5, but the actual usage was 6.5 per unit. The standard price was budgeted at \$6.75, but the actual price was \$7.00. The direct materials variances are calculated as follows:

Direct Materials Variance	Units	×	Materials per Unit	=	Quantity of Total Input	×	Price	=	Total
Static budget direct materials	18,000		7.5		135,000		\$6.75		\$911,250
Flexible budget direct materials	16,500		7.5		123,750		\$6.75		\$835,313
Actual direct materials	16,500		6.5		107,250		\$7.00		\$750,750
Direct materials price variance	107,250 × (\$7.00 – \$6.75)				\$ (26,813) U				
Direct materials quantity variance	\$6.75 × (107,250 – 123,750)				111,375 F				
Direct materials flexible budget variance					<u>\$ 84,562 F</u>		(\$1 rounding difference)		

4. Service Organizations

- a. In a service organization, the direct materials variances are usually immaterial compared with the direct labor variances. The reason is that only a relatively small investment is made in direct materials because these organizations tend to be labor intensive.

15.4 DIRECT LABOR VARIANCES

1. The direct labor variance is similar to the direct materials variance. The total direct labor flexible budget variance consists of the rate (price) variance and the efficiency (quantity) variance.

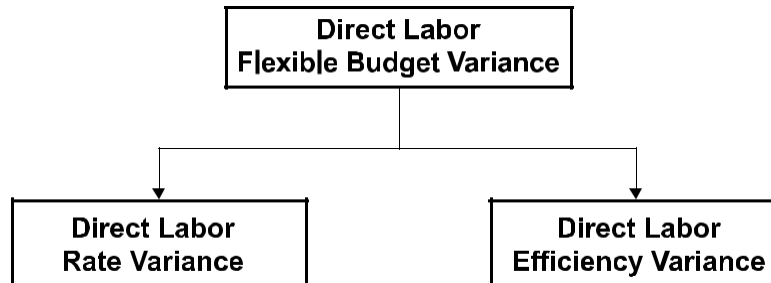


Figure 15-6

2. Direct Labor Rate Variance

- a. This variance equals the actual quantity of total input times the difference between the actual price and the standard price.

(Actual units produced × Actual labor hours per unit) × (Actual price – Standard price)

- b. Once a variance has been calculated, the next step is to analyze the reasons for the variance. In general, a favorable variance is desirable and an unfavorable variance is undesirable. But that is not always true.
 - 1) An **unfavorable labor rate variance** is usually caused by assigning skilled workers to a production process when the standard price calculation assumed that unskilled (lower paid) workers could complete the job.
 - a) However, the greater efficiency of skilled workers might result in a favorable labor efficiency variance to offset the unfavorable labor rate variance.
 - b) Another explanation for an unfavorable labor rate variance is that a new union contract resulted in a higher wage to workers, in which case the standard prices should be updated.
 - 2) A **favorable labor rate variance** is usually caused by assigning workers with less skill to a job. This may be desirable when they are qualified for the job.
 - a) However, the favorable rate variance may be offset by an unfavorable efficiency variance or lower-quality products.

3. Direct Labor Efficiency Variance

- a. This variance equals the standard price times the difference between the actual quantity of total input and the standard quantity of total input.

$$\text{Standard price} \times \left[\left(\frac{\text{Actual units produced}}{\text{Actual labor hours per unit}} \right) - \left(\frac{\text{Actual units produced}}{\text{Standard labor hours per unit}} \right) \right]$$

- b. Once a variance has been calculated, the next step is to analyze the reasons for the variance. In general, a favorable variance is desirable and an unfavorable variance is undesirable. But that is not always true.
- 1) An **unfavorable labor efficiency variance** means that workers are spending too much time on a production process, which is normally undesirable.
 - a) However, it may be caused by using workers with less skill than anticipated, in which case the labor rate variance may be favorable.
 - b) Another explanation is the use of low-quality materials that require extra time in the production process, in which case, the material price variance may be favorable.
 - 2) A **favorable labor efficiency variance** is almost always desirable. It means that employees are working efficiently and have been able to complete production in fewer hours than anticipated.
 - a) It is considered a production department efficiency.

EXAMPLE 15-5 Direct Labor -- Flexible Budget Variance

ChowDown, Inc., estimated output for the period of 18,000 units. However, actual output was 16,500 units. Standard direct labor was estimated at 3 hours per unit, but the actual usage was 3.25 hours per unit. The standard price was \$5.50, but the actual price was \$6.00. The direct labor variances are calculated as follows:

	Units	×	Labor Hours per Unit	=	Quantity of Total Input ×	Price =	Total
Static budget direct labor	18,000		3		54,000	\$5.50	\$297,000
Flexible budget direct labor	16,500		3		49,500	\$5.50	\$272,250
Actual direct labor	16,500		3.25		53,625	\$6.00	\$321,750
Direct labor rate variance	53,625 × (\$6.00 – \$5.50)				\$26,813U		
Direct labor efficiency variance	\$5.50 × (53,625 – 49,500)				22,688U		
Direct labor flexible budget variance					<u>\$49,501 U</u>		(\$1 rounding difference)

4. Service Organizations

- a. In a service organization, the direct labor variances are usually more relevant than the direct materials variances. A relatively large investment is made in direct labor and only a minor investment in direct materials.

15.5 MIX AND YIELD VARIANCES

1. In some production processes, inputs are **substitutable**; for example, a baker of pecan pies may use pecans from Florida instead of Georgia or higher-skilled labor instead of lower-skilled labor.
 - a. Given substitutable inputs, the direct materials quantity variance consists of the **materials mix variance** and the **materials yield variance**.
 - 1) A favorable (unfavorable) materials mix variance means that more (less) lower-priced materials were used than budgeted.
 - 2) A favorable (unfavorable) materials yield variance means that less (more) materials than budgeted were used to produce the output.

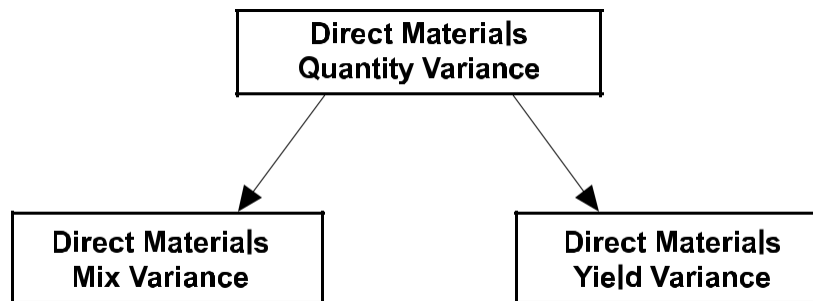


Figure 15-7

- b. Given substitutable inputs, the direct labor efficiency variance consists of the **labor mix variance** and the **labor yield variance**.
 - 1) A favorable (unfavorable) labor mix variance means that low-paid employees worked more (less) hours and higher-paid employees worked less (more) hours than budgeted.
 - 2) A favorable (unfavorable) labor yield variance means less (more) hours than budgeted were used to produce the output.
 - c. The sum of the mix and yield variance is the efficiency variance.

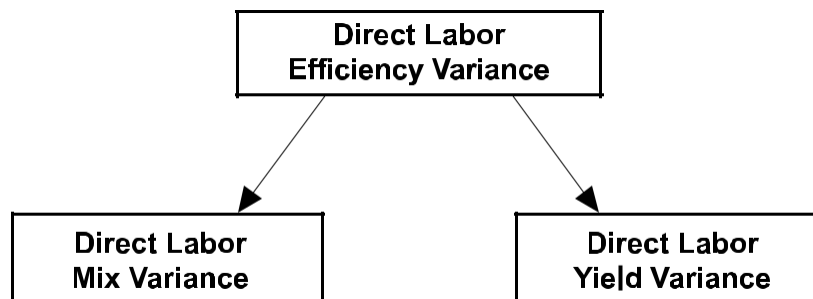


Figure 15-8

2. To calculate these variances, the entity must determine the weighted-average standard price using
- The standard mix of inputs at standard prices (**SMSP**) and
 - The actual mix of inputs at standard prices (**AMSP**).

EXAMPLE 15-6 SMSP vs. AMSP

A retail store budgets its employee hours for the upcoming month and calculates its weighted-average standard price of wages using the standard mix (SMSP) as follows:

	Budgeted Hours (SM)		Standard Wage (SP)		Subtotals
Managers	200	×	\$22	=	\$ 4,400
Sales associates	800	×	14	=	11,200
Warehouse	600	×	8	=	4,800
Totals	<u>1,600</u>				<u>\$20,400</u>

$$\text{SMSP} = \$20,400 \div 1,600 \text{ hours} = \$12.75 \text{ per hour}$$

After month end, the store employs the actual hours worked to calculate the weighted-average standard price using the actual mix (AMSP) as follows:

	Actual Hours (AM)		Standard Wage (SP)		Subtotals
Managers	220	×	\$22	=	\$ 4,840
Sales associates	800	×	14	=	11,200
Warehouse	480	×	8	=	3,840
Totals	<u>1,500</u>				<u>\$19,880</u>

$$\text{AMSP} = \$19,880 \div 1,500 \text{ hours} = \$13.2533 \text{ per hour}$$

3. The **mix variance** measures the relative use of higher-priced and lower-priced inputs in the production process based on standard input prices and **actual total quantity (ATQ)** of inputs. It isolates the effect of using the actual mix instead of the standard mix.

$$\text{Mix variance} = \text{ATQ} \times (\text{SMSP} - \text{AMSP})$$

EXAMPLE 15-7 Labor Mix Variance

The store calculates its labor mix variance as follows:

$$\begin{aligned}
 \text{Labor mix variance} &= \text{ATQ} \times (\text{SMSP} - \text{AMSP}) \\
 &= 1,500 \text{ hours} \times (\$12.75 - \$13.2533) \\
 &= 1,500 \text{ hours} \times -\$0.5033 \\
 &= \$755 \text{ unfavorable}
 \end{aligned}$$

This variance was unfavorable because higher-paid managers worked more hours and lower-paid warehouse employees worked fewer hours than budgeted.

4. The **yield variance** isolates the effect of the difference between the ATQ of inputs and the **STQ (standard total quantity)**. The calculation is based on standard input prices and the standard mix.

$$\text{Yield variance} = (\text{STQ} - \text{ATQ}) \times \text{SMSP}$$

EXAMPLE 15-8 Labor Yield Variance

The store calculates its labor yield variance as follows:

$$\begin{aligned} \text{Labor yield variance} &= (\text{STQ} - \text{ATQ}) \times \text{SMSP} \\ &= (1,600 \text{ hours} - 1,500 \text{ hours}) \times \$12.75 \\ &= 100 \text{ hours} \times \$12.75 \\ &= \$1,275 \text{ favorable} \end{aligned}$$

The variance was favorable because fewer hours than budgeted were used to produce the output. The sum of the mix and yield variances is the efficiency variance (\$755 U + \$1,275 F = \$520 F).

5. The same formulas can be applied to the mix and yield variances for direct materials.

15.6 OVERHEAD VARIANCES

1. The total overhead variance consists of four variances. Two are calculated for variable overhead and two for fixed overhead.

2. Variable Overhead

- a. The total **variable overhead variance** is the flexible-budget variance. It is the difference between actual variable overhead and the amount applied based on the budgeted application rate and the standard input allowed for the actual quantity.

$$\text{Actual variable overhead} - \left[\overset{\text{Budgeted}}{\text{application rate}} \times \left(\overset{\text{Standard cost}}{\text{driver per unit} \times \text{Actual units}} \right) \right]$$

- b. Variable overhead includes the following:

- 1) The **spending variance** is the difference between (a) actual variable overhead and (b) the product of the budgeted application rate and the actual amount of the allocation base.

$$\left(\overset{\text{Actual}}{\text{allocation base}} \times \overset{\text{Actual}}{\text{application rate}} \right) - \left(\overset{\text{Actual}}{\text{allocation base}} \times \overset{\text{Budgeted}}{\text{application rate}} \right)$$

$$\text{Actual allocation base} \times \left(\overset{\text{Actual}}{\text{application rate}} - \overset{\text{Budgeted}}{\text{application rate}} \right)$$

- a) The variable overhead spending variance is favorable or unfavorable if production spending is less or more, respectively, than the standard.

- 2) The **efficiency variance** is the budgeted application rate times the difference between (a) the actual allocation base and (b) the standard input allowed for the actual quantity.

$$\text{Budgeted application rate} \times \left[\overset{\text{Actual}}{\text{allocation base}} - \left(\overset{\text{Standard cost}}{\text{driver per unit}} \times \text{Actual units} \right) \right]$$

- a) Variable overhead applied equals the flexible-budget amount for the actual output level. The reason is that unit variable costs are assumed to be constant within the relevant range.

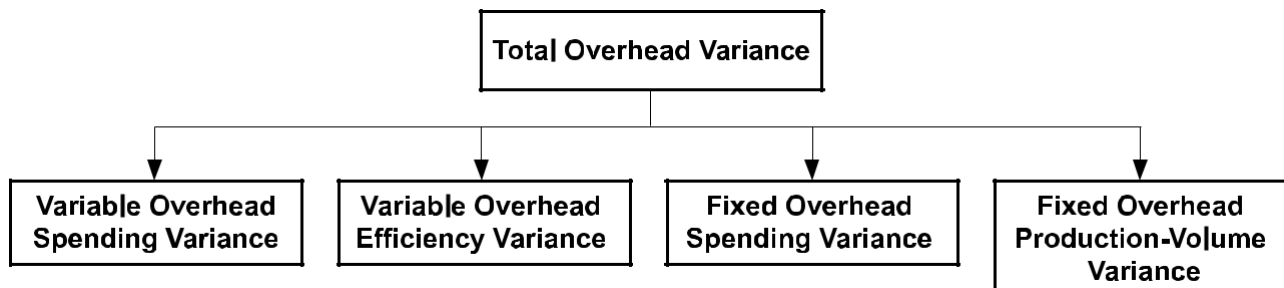


Figure 15-9

- b) If variable overhead is applied on the basis of output, not inputs, no efficiency variance arises.
- c) The variable overhead efficiency variance is related to the labor efficiency variance if overhead is applied to production on the basis of direct labor hours.
- i) For example, if the labor efficiency variance is unfavorable, the overhead efficiency variance also is unfavorable because it is based on the same number of input hours.

Variable Overhead Variances

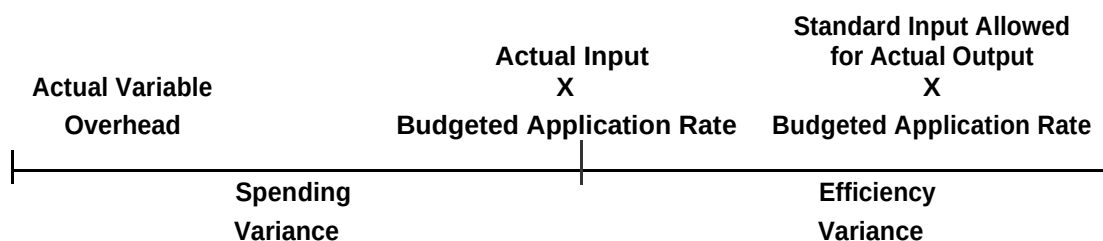


Figure 15-10

EXAMPLE 15-9 Variable Overhead Variances

ChowDown, Inc., estimated output for the period of 18,000 units. However, actual output was 16,500 units. Standard processing time is 1.2 machine hours per unit. But actual usage was 1.3 machine hours per unit. The standard application rate was \$8.00 per machine hour, but the actual rate was \$10.00 per machine hour. The variable overhead variances are calculated as follows:

Variable Overhead Variances	Units ×	Cost Driver per Unit	=	Allocation Base	×	Application Rate	=	Total
Static budget variable overhead	18,000	1.2		21,600		\$ 8.00		\$172,800
Flexible budget variance overhead	16,500	1.2		19,800		\$ 8.00		\$158,400
Actual variable overhead	16,500	1.3		21,450		\$ 10.00		\$214,500
Variable overhead spending variance				21,450 × (\$10.00 – \$8.00)		\$42,900 U		
Variable overhead efficiency variance				\$8.00 × (21,450 – 19,800)		13,200 U		
Total variable overhead variances						<u>\$56,100 U</u>		

3. Fixed Overhead

- a. The **total fixed overhead variance** is the difference between (1) actual fixed overhead and (2) the product of the budgeted application rate and the standard input allowed for the actual output.

$$\text{Actual fixed overhead} - \left[\text{Budgeted application rate} \times \left(\text{Standard cost driver per unit} \times \text{Actual units} \right) \right]$$

- b. Fixed overhead includes the following:

- 1) The fixed overhead **spending variance** (also known as budget variance) is the easiest to calculate because it is the difference between (a) actual fixed overhead and (b) the amount budgeted, regardless of actual and budgeted production levels.

Actual fixed overhead – Budgeted fixed overhead

- a) This variance is the same as the fixed overhead flexible-budget variance. The reason is that the static budget lump-sum of fixed overhead is also the flexible budget amount over the relevant range of output. Moreover, the efficiency of production does not affect the fixed overhead variances.
- b) Thus, an efficiency variance is calculated for variable, but not fixed, overhead.
- c) The fixed overhead variance is simply attributable to more or less spending by the production function. Whether the difference is justified should be investigated.

EXAMPLE 15-10

If the budgeted fixed overhead is \$450,000 for 18,000 units, with a standard cost of \$25 per unit, and the actual cost is \$429,000 for 16,500 units, the fixed overhead spending variance is \$21,000 favorable. This was calculated by deducting the \$429,000 actual cost from the \$450,000 budgeted. The difference in units produced is not considered. The firm spent \$21,000 less on fixed costs than was anticipated.

- 2) The **production-volume variance** (idle capacity variance or denominator-level variance) is the difference between (a) budgeted fixed overhead and (b) the product of the budgeted application rate and the standard input allowed for the actual output.

$$\text{Budgeted fixed overhead} - \left[\overset{\text{Budgeted}}{\text{application rate}} \times \left(\overset{\text{Standard cost}}{\text{driver per unit}} \times \text{Actual units} \right) \right]$$

- a) This variance results when production capacity differs from capacity usage. A favorable (unfavorable) variance occurs when overhead applied is more (less) than budgeted fixed costs. For example, the variance is favorable when actual production exceeds planned production.
- b) The production-volume variance is typically not the fault of the production function. The sales staff often is blamed, or rewarded, for a volume variance.
 - i) If sales are greater than expected, production increases, and the variance may be favorable.
 - ii) An unfavorable volume variance may be caused by low sales (the fault of the sales staff) or by a production shutdown, perhaps due to a labor strike, power failure, or natural disaster. In these cases, the variance is attributable to actions of the general administration of the entity or to uncontrollable external factors.

Fixed Overhead Variances

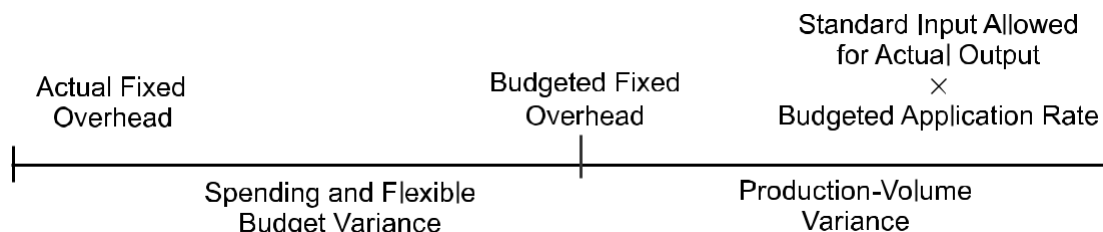


Figure 15-11

EXAMPLE 15-11 Fixed Overhead Variances

ChowDown, Inc., estimates output for the period of 18,000 units. However, actual output was 16,500 units. Fixed overhead is applied at \$25 per unit sold. But actual cost per unit was \$26. The fixed overhead variances are calculated below:

Fixed Overhead Variances	Units	×	Cost Driver per Unit	×	Application Rate	=	Total
Static and flexible budget fixed overhead	18,000		1		\$25.00		\$450,000
Applied fixed overhead	16,500		1		\$25.00		\$412,500
Actual fixed overhead	16,500		1		\$26.00		\$429,000
Fixed overhead spending variance	\$429,000 – \$450,000						\$21,000 F
Fixed overhead production-volume variance	\$450,000 – [\$25 × (1 × 16,500)]						(37,500) U

NOTE: The production-volume variance is unfavorable because it indicates that fixed overhead was underallocated to actual output. The variance is the amount of fixed overhead incurred for unused production capacity.

4. Analysis of Overhead Variances

- a. **Four-way overhead variance analysis** includes all four components of the total overhead variance.

Four-Way Analysis	Spending Variance	Efficiency Variance	Production-Volume Variance
Variable OH	\$(42,900) U	\$(13,200) U	--
Fixed OH	\$ 21,000 F	--	\$(37,500) U

- b. **Three-way overhead variance analysis** combines the variable and fixed spending variances and reports the other two variances separately.

Three-Way Analysis	Spending Variance	Efficiency Variance	Production-Volume Variance
Total OH	\$(21,900) U	\$(13,200) U	\$(37,500) U

- c. **Two-way overhead variance analysis** combines the spending and efficiency variances into one flexible-budget variance and reports the production-volume variance separately.

Two-Way Analysis	Flexible-Budget Variance	Production-Volume Variance
Total OH	\$(35,100) U	\$(37,500) U

- 1) The flexible-budget variance in two-way analysis also is called the **controllable variance**. It is the portion of the total variance not attributable to the production-volume variance.

15.7 COMPREHENSIVE EXAMPLE



SUCCESS TIP

As a CMA candidate, you will be expected to know and understand the different variance formulas. Please note that you may need to solve more than one formula to get the values you need to input into the formula that will give you your final answer. Take the time to memorize them but also to understand them. You may be given different elements of the variance formulas and be expected to solve for the unknown variable. Remember to take into account the way that a favorable/unfavorable variance affects your answer. We have provided a summary below, and it may be useful to create notecards for these formulas to aid in retention. If asked to find total overhead variance, it may be useful to convert the fixed and variable overhead to a per unit basis, add together, and then multiply by the actual rate base.

When you arrive at the exam, you will first be presented with a tutorial before your time for the exam begins. However, by using the Glem CMA Review Course and CMA Test Prep, you will already be familiar with the information provided in the tutorial. Take these few minutes to write any formulas or variances you can recall on your scrap paper. (Don't forget to pay attention to the time, as once the time for the tutorial is up, your exam will begin.) This way, when you are presented with questions requiring you to calculate variances, you will be able to look at your scrap paper rather than trying to recall from memory during a time when you are stressed. This will help you keep the formulas straight and manage your time on the exam.

1. Summary of Variance Formulas

a. Direct Materials Variances

- 1) Flexible budget direct materials variances consist of a price variance and a quantity variance.

a) Direct materials **price variance**

$$\left(\frac{\text{Actual units}}{\text{produced}} \times \frac{\text{Actual materials}}{\text{per unit}} \right) \times \left(\frac{\text{Actual}}{\text{price}} - \frac{\text{Standard}}{\text{price}} \right)$$

b) Direct materials **quantity variance**

$$\text{Standard price} \times \left[\left(\frac{\text{Actual units}}{\text{produced}} \times \frac{\text{Actual materials}}{\text{per unit}} \right) - \left(\frac{\text{Actual units}}{\text{produced}} \times \frac{\text{Standard}}{\text{materials per unit}} \right) \right]$$

- c) A favorable materials quantity variance indicates the use of less than the standard quantity of materials. A favorable quantity variance may therefore result from unusual efficiency or the production of lower-quality products. An unfavorable materials quantity variance is usually caused by waste, shrinkage, or theft.

b. **Direct Labor Variances**

- 1) Flexible budget direct labor variance consists of a price (rate) variance and a quantity (efficiency) variance:

- a) Direct labor **rate variance**

$$\left(\frac{\text{Actual units}}{\text{produced}} \times \frac{\text{Actual labor}}{\text{hours per unit}} \right) \times \left(\text{Actual price} - \text{Standard price} \right)$$

- b) Direct labor **efficiency variance**

$$\text{Standard price} \times \left[\left(\frac{\text{Actual units}}{\text{produced}} \times \frac{\text{Actual labor}}{\text{hours per unit}} \right) - \left(\frac{\text{Actual units}}{\text{produced}} \times \frac{\text{Standard labor}}{\text{hours per unit}} \right) \right]$$

- c) A favorable labor efficiency variance indicates the use of less than the standard number of labor hours. A favorable variance may therefore result from unusual efficiency or the production of lower-quality products. An unfavorable labor efficiency variance may be caused by production delays resulting from materials shortages, inferior materials, or excessive work breaks.

- c. The quantity variance for materials and the efficiency variance for labor can be further subdivided into **mix and yield variances**:

- 1) Mix variance

$$\text{ATQ} \times (\text{SMSP} - \text{AMSP})$$

- 2) Yield variance

$$(\text{STQ} - \text{ATQ}) \times \text{SMSP}$$

- d. A manufacturer's total **overhead variance** consists of variable and fixed portions.

- 1) The **variable overhead variance** consists of a spending variance and an efficiency variance.
 2) The **fixed overhead variance** consists of a spending variance and a production-volume variance.

2. At the end of the current period, the following analysis of the flexible budget variance was prepared:

	<u>Actual Amounts</u>	<u>Standard Amounts</u>
Units	7,000	6,000
Selling Price	\$65.00	\$63.00
Materials Per Unit	1.50	2
Materials Price	\$7.00	\$6.00
Labor Hours Per Unit	3.50	3
Labor Price	\$5.50	\$5.00
Machine Hours Per Unit	2.00	1.50
VOH Application Rate	\$4/Machine Hour	\$3/Machine Hour
FOH Application Rate	\$8/Machine Hour	\$7.50/Machine Hour

	AQ×AP	AR–FB	AQ×SP	FB–SB	SQ×SP
	Actual Results (AR)	Flexible Budget	Flexible Budget (FB)	Sales Volume	Static Budget (SB)
		Variances		Variances	
Units sold	<u>7,000</u>		<u>7,000</u>	<u>1,000 F</u>	<u>6,000</u>
Revenue	\$ 455,000	\$ 14,000 F	\$441,000	\$ 63,000 F	\$378,000
Variable costs:					
Direct materials	73,500	10,500 F	84,000	(12,000) U	72,000
Direct labor	134,750	(29,750) U	105,000	(15,000) U	90,000
Variable non-mfg. overhead	<u>56,000</u>	<u>(24,500) U</u>	<u>31,500</u>	<u>(4,500) U</u>	<u>27,000</u>
Total variable costs	<u>\$ 264,250</u>	<u>\$(43,750) U</u>	<u>\$220,500</u>	<u>\$(31,500) U</u>	<u>\$189,000</u>
Contribution margin	190,750	(29,750) U	220,500	31,500 F	189,000
Fixed overhead	<u>112,000</u>	<u>(33,250) U</u>	<u>78,750</u>	<u>(11,250) U</u>	<u>67,500</u>
Operating income	<u>\$ 78,750</u>	<u>\$(63,000) U</u>	<u>\$141,750</u>	<u>\$ 20,250 F</u>	<u>\$121,500</u>
		Total Flexible Budget Variance		Total Sales Volume Variance	
		\$63,000 U		\$20,250 F	
		Total Static Budget Variance			
		\$42,750 U			

Variances

MATERIALS VARIANCES

Price Variance

$$\begin{aligned}
 &= (\text{Actual Units} \times \text{Actual Materials/Unit}) \times (\text{Actual Price} - \text{Std Price}) \\
 &= (7,000 \times 1.50) \times (\$7.00 - \$6.00) \\
 &= 10,500 \times \$1.00 \\
 &= \$ (10,500) \text{ U}
 \end{aligned}$$

Quantity Variance

$$\begin{aligned}
 &= \text{Std Price} \times [(\text{Actual Units} \times \text{Actual Materials/Unit}) - (\text{Actual Units} \times \text{Std Materials/Unit})] \\
 &= \$6.00 \times [(7,000 \times 1.50) - (7,000 \times 2.00)] \\
 &= \$6.00 \times (10,500 - 14,000) \\
 &= \$21,000 \text{ F}
 \end{aligned}$$

Direct Materials Price Variance **\$ (10,500) U**

Direct Materials Quantity Variance **\$ 21,000 F**

Direct Materials Flexible Budget Variance **\$ 10,500 F**

LABOR VARIANCES

Rate Variance

$$\begin{aligned}
 &= (\text{Actual Units} \times \text{Actual Labor Hrs/Unit}) \times (\text{Actual Price} - \text{Std Price}) \\
 &= (7,000 \times 3.50) \times (\$5.50 - \$5.00) \\
 &= 24,500 \times \$0.50 \\
 &= \$ (12,250) \text{ U}
 \end{aligned}$$

Efficiency Variance

$$\begin{aligned}
 &= \text{Std Price} \times [(\text{Actual Units} \times \text{Actual Labor Hrs/Unit}) - (\text{Actual Units} \times \text{Std Labor Hrs/Unit})] \\
 &= \$5.00 \times [(7,000 \times 3.50) - (7,000 \times 3.00)] \\
 &= \$5.00 \times (24,500 - 21,000) \\
 &= \$ (17,500) \text{ U}
 \end{aligned}$$

Direct Labor Rate Variance **\$ (12,250) U**

Direct Labor Efficiency Variance **\$ (17,500) U**

Direct Labor Flexible Budget Variance **\$ (29,750) U**

VARIABLE OVERHEAD VARIANCESSpending Variance

$$\begin{aligned}
 &= \text{Actual Allocation Base} \times (\text{Actual Application Rate} - \text{Budgeted Application Rate}) \\
 &= 14,000 \times (\$4.00 - \$3.00) \\
 &= \$14,000) \text{ U}
 \end{aligned}$$

Efficiency Variance

$$\begin{aligned}
 &= \text{Budgeted Application Rate} \times [\text{Actual Allocation Base} - (\text{Std Cost Driver/Unit} \times \text{Actual Units})] \\
 &= \$3.00 \times [14,000 - (1.50^* \times 7,000)] \\
 &= \$3.00 \times (14,000 - 10,500) \\
 &= \$10,500) \text{ U}
 \end{aligned}$$

*The standard cost driver/unit is the standard input allowed for actual output. In this example, it is the \$3 per machine hour (standard input) ÷ 2 machine hours per unit (actual output).

$$1.5 = \$3 \text{ per machine hour} \div 2 \text{ machine hours per unit}$$

Variable Overhead Spending Variance	<u>\$(14,000) U</u>
Variable Overhead Efficiency Variance	<u>\$(10,500) U</u>
Variable Overhead Flexible Budget Variance	<u>\$(24,500) U</u>

FIXED OVERHEAD VARIANCESSpending Variance

$$\begin{aligned}
 &= \text{Actual Fixed Overhead} - \text{Budgeted Fixed Overhead} \\
 &= \$112,000 - 67,500 \\
 &= \$ (44,500) \text{ U}
 \end{aligned}$$

Production Volume Variance

$$\begin{aligned}
 &= \text{Budgeted Fixed Overhead} - [\text{Budgeted Application Rate} \times (\text{Std Cost Driver/Unit} \times \text{Actual Units})] \\
 &= \$67,500 - [\$7.50 \times (1.50 \times 7,000)] \\
 &= \$67,500 - (\$7.50 \times 10,500) \\
 &= \$11,250 \text{ F}
 \end{aligned}$$

Fixed Overhead Spending Variance	<u>\$(44,500) U</u>
Fixed Overhead Production Volume Variance	<u>\$ 11,250 F</u>
Total Fixed Overhead Variance	<u>\$(33,250) U</u>

	Actual Output at Actual Input and Cost	AQ×SP Actual Output at Standard Input and Cost	Variance
Materials	\$ 73,500	\$ 84,000	\$ 10,500 F
Labor	134,750	105,000	(29,750) U
Variable overhead	56,000	31,500	(24,500) U
Fixed overhead	112,000	78,750	(33,250) U
	<u>376,250</u>	<u>299,250</u>	
Net unfavorable variance		77,000	(77,000) U
	<u><u>\$376,250</u></u>	<u><u>\$376,250</u></u>	

15.8 SALES VARIANCES

1. Single Product Sales Variances

- a. Variance analysis is useful for evaluating not only the production function but also the selling function.
 - 1) If sales differ from the amount budgeted, the difference could be attributable to either the **sales price variance** or the **sales volume variance** (sum of the sales quantity and mix variances).
 - 2) The analysis of these variances concentrates on **contribution margins** because fixed costs are assumed to be constant.
- b. For a single product, the **sales price variance** is the change in the contribution margin attributable solely to the change in selling price (holding quantity constant).

$$\text{Sales price variance} = (\text{AP} - \text{SP}) \times \text{AQ}$$

- c. For a single product, the **sales volume variance** is the change in the contribution margin attributable solely to the difference between the actual and standard unit sales.

$$\text{Sales volume variance} = (\text{AQ} - \text{SQ}) \times \text{SCM}$$

- 1) For a single product, the sales mix variance is zero.

EXAMPLE 15-12 Budgeted vs. Actual Results

A firm has budgeted sales of 10,000 units of its sole product at \$17 per unit. Variable costs are expected to be \$10 per unit, and fixed costs are budgeted at \$50,000. The following compares budgeted and actual results:

	Budget Computation	Budget Amount	Actual Computation	Actual Amount
Sales	10,000 units × \$17 per unit	\$170,000	11,000 units × \$16 per unit	\$176,000
Variable costs	10,000 units × \$10 per unit	(100,000)	11,000 units × \$10 per unit	(110,000)
Contribution margin		\$ 70,000		\$ 66,000
Fixed costs		(50,000)		(50,000)
Operating income		<u>\$ 20,000</u>		<u>\$ 16,000</u>
Unit contribution margin (UCM)	\$70,000 ÷ 10,000 units	\$7	\$66,000 ÷ 11,000 units	\$6

Although sales were greater than budgeted, the actual contribution margin (ACM) is less than the standard contribution margin (SCM). The discrepancy can be analyzed in terms of the sales price variance and the sales volume variance.

$$\begin{aligned} \text{Sales price variance} &= (\text{AP} - \text{SP}) \times \text{AQ} \\ &= (\$16 - \$17) \times 11,000 \\ &= \$ (11,000) \text{ U} \end{aligned}$$

$$\begin{aligned} \text{Sales volume variance} &= (\text{AQ} - \text{SQ}) \times \text{SCM} \\ &= (11,000 - 10,000) \times \$7 \\ &= \$7,000 \text{ F} \end{aligned}$$

The sales price variance (\$11,000 U) plus the sales volume variance (\$7,000 F) equals the total change in the contribution margin (\$4,000 U).

- d. A similar analysis may be done for **cost of goods sold**.
 - 1) The average production cost per unit is used instead of the average unit selling price, but the quantities for unit production are the same.
 - 2) Accordingly, the overall variation in gross profit is the sum of the variation in revenue plus the variation in cost of goods sold.

2. Multiproduct Sales Variances

- a. For two or more products, the multiproduct sales variances reflect not only the change in total unit sales but also the change in the sales mix.
- 1) The **multiproduct sales price variance** may be calculated as in the single-product case for each product. The results are then added.
 - a) An alternative is to multiply the actual total units sold times the difference between the following:
 - i) The weighted-average price based on actual units sold at actual unit prices
 - ii) The weighted-average price based on actual units sold at standard prices
 - 2) The **multiproduct sales volume variance** may be calculated as in the single-product case for each product. The results are then added.
 - a) An alternative is to determine the difference between the following:
 - i) Actual total unit sales times the standard weighted-average UCM for the actual mix
 - ii) Standard total unit sales times the standard weighted-average UCM for the standard mix
 - 3) The multiproduct sales volume variance consists of the sales quantity and sales mix variances.
 - a) The **sales quantity variance** is the difference between (1) the standard contribution margin based on actual unit sales and (2) the standard contribution margin based on standard unit sales.
 - i) One way to calculate this variance is to multiply the standard UCM (SUCM) for each product times the difference between (a) its standard percentage of actual total unit sales and (b) the standard unit sales of the product. The results for each product are then added together. The equation to calculate this variance for each product is

$$\text{Sales quantity variance} = \text{SUCM} \times [(\text{Total AQ} \times \text{Standard mix \%}) - \text{Standard unit sales}]$$

- ii) An alternative is to multiply the standard weighted-average UCM based on the standard mix times the difference between (a) total actual unit sales and (b) the total standard unit sales.

- iii) The components of the sales quantity variance are the market size variance and the market share variance.

- The **market size variance** is the change in contribution margin caused by a change in the market size.

$$\text{Market size variance} = \text{Budgeted market share percentage} \times \left(\begin{array}{c} \text{Actual market size} \\ \text{in units} \end{array} - \begin{array}{c} \text{Budgeted market size} \\ \text{in units} \end{array} \right) \times \text{Budgeted weighted-average UCM}$$

- The **market share variance** is the amount of contribution margin gained (forgone) because of a change in the market share.

$$\text{Market share variance} = \left(\begin{array}{c} \text{Actual market share} \\ \text{percentage} \end{array} - \begin{array}{c} \text{Budgeted market share} \\ \text{percentage} \end{array} \right) \times \begin{array}{c} \text{Actual market size} \\ \text{in units} \end{array} \times \text{Budgeted weighted-average UCM}$$

- b) The **sales mix variance** is the difference between (1) the standard contribution margin for the actual mix and actual total unit sales and (2) the standard contribution margin for the standard mix and actual total unit sales.
- i) One way to calculate this variance is to multiply the SUCM for each product times the difference between (a) actual unit sales of the product and (b) its standard percentage of actual total unit sales. The results for each product are then added together. The equation to calculate this variance for each product is

$$\text{Sales mix variance} = \text{SUCM} \times [\text{AQ} - (\text{Total AQ} \times \text{Standard mix \%})]$$

- ii) An alternative is to multiply total actual unit sales times the difference between (a) the standard weighted-average UCM for the standard mix and (b) the standard weighted-average UCM for the actual mix.

EXAMPLE 15-13 Contribution Margin and Sales Volumes Variances

	Plastic	Metal	Total
Standard selling price per unit	\$6.00	\$10.00	
Standard variable cost per unit	3.00	7.50	
Standard contribution margin per unit	<u>\$3.00</u>	<u>\$ 2.50</u>	
Standard unit sales	300	200	500
Standard mix percentage	60%	40%	100%
Actual units sold	260	260	520
Actual selling price per unit	\$6.00	\$9.50	

As shown below (000 omitted), the **total contribution margin variance** was \$100 unfavorable (\$130 unfavorable sales price variance – \$30 favorable sales volume variance).

Sales price variance:

Plastic 260 × (\$6.00 – \$6.00)	\$ 0	
Metal 260 × (\$10 – \$9.50)	<u>(130)</u>	\$130 unfavorable

Sales volume variance:

Plastic (260 – 300) × \$3.00	\$(120)	
Metal (260 – 200) × \$2.50	<u>150</u>	\$ 30 favorable
Total contribution margin variance		<u><u>\$100</u></u> unfavorable

The sales volume variance consists of the following:

Sales quantity variance:

Plastic [(520 × .6) – 300] × \$3.00	\$ 36	
Metal [(520 × .4) – 200] × \$2.50	<u>20</u>	\$ 56 favorable

Sales mix variance:

Plastic [260 – (520 × .6)] × \$3.00	\$(156)	
Metal [260 – (520 × .4)] × \$2.50	<u>130</u>	\$ 26 unfavorable

Sales volume variance

\$ 30 favorable

STUDY UNIT SIXTEEN

RESPONSIBILITY ACCOUNTING AND PERFORMANCE MEASURES

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16.2	Performance Measures -- Cost, Revenue, and Profit Centers (Contribution Margin Reporting)	6
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This study unit is the **second of two** on **performance management**. The relative weight assigned to this major topic in Part 1 of the exam is **20%**. The two study units are

Study Unit 15: Cost and Variance Measures

Study Unit 16: Responsibility Accounting and Performance Measures

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

16.1 RESPONSIBILITY CENTERS

1. Decision Making and Decentralization

- a. The primary distinction between centralized and decentralized organizations is in the degree of freedom of decision making by managers at many levels.
 - 1) In a **centralized** organization, decision making is consolidated so that activities throughout the organization may be more effectively coordinated from the top.
 - 2) In a **decentralized** organization, decision making is at as low a level as possible. The premise is that the local manager can make more informed decisions than a manager farther from the decision.

2. Responsibility Centers

- a. A decentralized organization is divided into **responsibility centers** (also called **strategic business units**, or SBUs) to facilitate local decision making and to provide a basis for measuring the performance of subunits.
 - 1) Four types of responsibility centers are generally recognized: cost, revenue, profit, and investment.
 - 2) The main characteristic of a responsibility center is the authority to make decisions affecting the major determinants of the type of center.
- b. A **cost center** (e.g., a maintenance department) is responsible for costs only.
 - 1) Cost drivers are the relevant performance measures.
 - 2) A disadvantage of a cost center is the potential for cost shifting, for example, replacement of variable costs for which a manager is responsible with fixed costs for which (s)he is not.
 - a) Another disadvantage is that long-term issues may be disregarded when the emphasis is on, for example, annual cost amounts.
 - b) Yet another issue is allocation of service department costs to cost centers.
 - 3) **Service centers** exist primarily and sometimes solely to provide specialized support to other organizational subunits. They are usually operated as cost centers.
- c. A **revenue center** (e.g., a sales department) is responsible for revenues only.
 - 1) Revenue drivers are the relevant performance measures. They are factors that influence unit sales, such as changes in prices and products, customer service, marketing efforts, and delivery terms.
- d. A **profit center** (e.g., an appliance department in a retail store) is responsible for both revenues and expenses.
- e. An **investment center** (e.g., a branch office) is responsible for revenues, expenses, and invested capital. Thus, compared to the other types of responsibility centers, it is most like an independent business.
 - 1) The advantage of an investment center is that it permits an evaluation of performance that can be compared with that of other responsibility centers or other potential investments on a return on investment basis, i.e., on the basis of the effectiveness of asset usage.

3. Performance Measures and Manager Motivation

- a. Each responsibility center is structured such that a logical group of operations is under the direction of a single manager.
 - 1) Measures are designed for every responsibility center to monitor performance.

- b. **Controllability.** The performance measures on which the manager's incentive package are based must be, as far as practicable, under the manager's direct responsibility and authority.
 - 1) "Controllable" factors are those factors that a manager can influence in a given time period.
 - a) Inevitably, some costs, especially **common costs** such as the costs of central administrative functions, cannot be traced to particular activities or responsibility centers.
 - 2) Controllable cost is not synonymous with variable cost. Often this classification is particular to the level of the organization.
 - a) For instance, the fixed cost of depreciation may not be a controllable cost of the manager of a revenue center but is controllable by the division vice president to which that manager reports.
- c. **Goal congruence.** These performance measures must be designed such that the manager's pursuit of them ties directly to accomplishment of the organization's overall goals.
 - 1) Suboptimization results when segments of the organization pursue goals that are in that segment's own best interests rather than those of the organization as a whole.
- d. Along with the responsibility, a manager must be granted sufficient authority to control those factors on which his or her incentive package is based.

4. Allocating Common Costs

a. Issues

- 1) Common costs are the costs of products, activities, facilities, services, or operations shared by two or more cost objects.
 - a) The difficulty with common costs is that they are **indirect costs** whose allocation may be arbitrary.
- 2) A direct **cause-and-effect** relationship between a common cost and the actions of the cost object to which it is allocated is desirable.
 - a) Such a relationship promotes acceptance of the allocation by managers who perceive the fairness of the procedure, but identification of cause and effect may not be feasible.
 - b) A common method of allocation is to use budgeted rates and standard hours allowed for actual output for variable costs and budgeted rates and capacity available for fixed costs.
- 3) An alternative allocation criterion is the **benefit received**.
 - a) For example, advertising costs that do not relate to particular products may increase sales of all products.
 - b) Allocation based on the increase in sales by organizational subunits is likely to be accepted as equitable despite the absence of clear cause-and-effect relationships.
- 4) Allocating costs to foster competition may also be appropriate.
 - a) Care must be taken to ensure this competition is healthy for organizational dynamics.

b. Headquarters Costs

- 1) A persistent problem in large organizations is the treatment of the costs of headquarters and other central support costs. Such costs are frequently allocated.
- 2) Research has shown that central support costs are allocated to departments or divisions for the following reasons:
 - a) The allocation reminds managers that support costs exist and that the managers would incur these costs if their operations were independent.
 - b) The allocation reminds managers that profit center earnings must cover some amount of support costs.
 - c) Departments or divisions should be motivated to use central support services appropriately.
 - d) Managers who must bear the costs of central support services that they do not control may be encouraged to exert pressure on those who do. Thus, they may be able to restrain such costs indirectly.
 - i) However, department or division managers pressuring central managers is not a healthy organizational dynamic.

c. Effects of Arbitrary Allocations

- 1) Managers' morale may suffer when allocations depress operating results.
- 2) Dysfunctional conflict may arise among managers when costs controlled by one are allocated to others.
- 3) Resentment may result if cost allocation is perceived to be arbitrary or unfair.
 - a) For example, an allocation on an ability-to-bear basis, such as operating income, penalizes successful managers and rewards underachievers and may therefore have a demotivating effect.

d. Allocation Alternatives

- 1) If allocation is based on actual sales or contribution margin, responsibility centers that increase their sales (or contribution margin) will be charged with increased overhead.
- 2) If central administrative or other fixed costs are not allocated, responsibility centers might reach their revenue (or contribution margin) goals without covering all fixed costs (which is necessary to operate in the long run).
- 3) Allocation of overhead, however, is motivationally negative; central administrative or other fixed costs may appear noncontrollable and be unproductive.
- 4) A much preferred alternative is to budget a certain amount of contribution margin earned by each responsibility center to the central administration based on negotiation.
 - a) The intended result is for each unit to see itself as contributing to the success of the overall entity rather than carrying the weight (cost) of central administration.
 - b) Central administration can then make the decision whether to expand, divest, or close responsibility centers.

e. Calculations

- 1) Two specific approaches to common cost allocation are in general use.
 - a) Under the **stand-alone method**, the common cost is allocated to each cost object on a proportionate basis.
 - b) Under the **incremental method**, the cost objects are sorted in descending order by total traceable cost, and the common cost is allocated up to the amount of each.

EXAMPLE 16-1 Common Cost Allocation

The proportionate costs of servicing three customers are presented in the table below. The common cost of providing service to these customers is \$8,000.

	Cost of Servicing	%
Luciano	\$ 7,000	70%
Ratzinger	2,000	20%
Wojtyla	1,000	10%
Total	<u>\$10,000</u>	<u>100%</u>

Stand-Alone Method

	Total Cost to Be Allocated		Allocation %		Allocated Cost
Luciano	\$8,000	×	70%	=	\$5,600
Ratzinger	8,000	×	20%	=	1,600
Wojtyla	8,000	×	10%	=	800
Total			<u>100%</u>		<u>\$8,000</u>

Incremental Method

	Traceable Cost	Allocated Cost	Remaining Unallocated
To be allocated			\$8,000
Luciano	\$ 7,000	\$7,000	1,000
Ratzinger	2,000	1,000	0
Wojtyla	1,000	0	
Total	<u>\$10,000</u>	<u>\$8,000</u>	

16.2 PERFORMANCE MEASURES -- COST, REVENUE, AND PROFIT CENTERS (CONTRIBUTION MARGIN REPORTING)

1. Cost Centers and Revenue Centers

- a. Since managers of cost and revenue centers can influence only one type of factor, variance analysis (Study Unit 15) is the most appropriate performance measurement technique for these responsibility centers.
- b. To be effective, a performance measure should be based on a **cause-and-effect** relationship between the outcome (effect) being measured and a driver (cause) that is under the manager's control.
 - 1) An appropriate performance measure for a cost or revenue center might not even be financial. Examples might include number of invoices processed per hour or percentage of customer shipments correctly filled.

2. Profit Centers

- a. The **contribution margin approach** to reporting (in contrast to the financial reporting approach) is extremely useful in performance measurement for profit centers.
 - 1) The contribution margin approach isolates the effects of variable and fixed costs and thus highlights the effects of a manager's choices regarding improving the contribution margin.
 - 2) In addition to contribution margin and operating income, this approach can also be used to calculate multiple intermediate measures, as shown below:

Contribution Margin Income Statement

Sales		\$150,000
Variable production costs		(40,000)
Manufacturing contribution margin		\$110,000
Variable S&A expenses		(20,000)
Contribution margin		\$ 90,000
Controllable fixed costs:		
Fixed production costs	\$30,000	
Fixed S&A expenses	25,000	(55,000)
Short-run performance margin		\$ 35,000
Traceable fixed costs:		
Depreciation	\$10,000	
Insurance	5,000	(15,000)
Segment margin		\$ 20,000
Allocated common costs		(10,000)
Segment operating income		\$ 10,000

3. Segment Reporting

- a. A segment is a product line, geographical area, or other meaningful subunit of the organization.
 - 1) As the following examples illustrate, contribution margin reporting is extremely useful for manager decision making. In determining whether to drop a product or customer, the deciding factor is whether the product or customer is generating sufficient contribution margin to cover fixed costs.
- b. **Product profitability analysis** allows management to determine whether a product is providing any coverage of fixed costs.

EXAMPLE 16-2 Product Profitability Analysis

At first glance, a dairy operation appears to be comfortably profitable.

Sales	\$540,000
Variable costs	<u>312,000</u>
Contribution margin	\$228,000
Other traceable costs:	
Marketing	116,000
R&D	<u>18,000</u>
Product line margin	\$ 94,000
Fixed costs	<u>24,000</u>
Operating income	<u>\$ 70,000</u>

A product profitability analysis shows an entirely different picture. Two product lines are losing money, and one is not even covering its own variable costs.

	Milk	Cream	Cottage Cheese	Total
Sales	\$300,000	\$ 60,000	\$ 180,000	\$540,000
Variable costs	<u>110,000</u>	<u>62,000</u>	<u>140,000</u>	<u>312,000</u>
Contribution margin	\$190,000	\$ (2,000)	\$ 40,000	\$228,000
Other traceable costs:				
Marketing	66,000	10,000	40,000	116,000
R&D	<u>8,000</u>	<u>4,000</u>	<u>6,000</u>	<u>18,000</u>
Product line margin	<u>\$116,000</u>	<u>\$ (16,000)</u>	<u>\$ (6,000)</u>	\$ 94,000
Fixed costs				<u>24,000</u>
Operating income				<u>\$ 70,000</u>

- c. **Area office profitability analysis** performs the same function on the segment level.

EXAMPLE 16-3 Area Office Profitability Analysis

A geographic profitability analysis for a company that provides research services allows management to see which branch offices are the most profitable.

	Cartagena	Riyadh	Mumbai	Osaka	Total
Sales	\$1,200,000	\$800,000	\$2,000,000	\$4,600,000	\$8,600,000
Variable costs of sales	800,000	460,000	1,400,000	3,200,000	5,860,000
Other variable costs	256,000	176,000	320,000	544,000	1,296,000
Contribution margin	\$ 144,000	\$164,000	\$ 280,000	\$ 856,000	\$1,444,000
Traceable fixed costs	150,000	100,000	160,000	220,000	630,000
Area office margin	\$ (6,000)	\$ 64,000	\$ 120,000	\$ 636,000	\$ 814,000
Nontraceable fixed costs					200,000
Operating income					\$ 614,000

- d. **Customer profitability analysis** enables management to make decisions about whether to continue servicing a given customer.

EXAMPLE 16-4 Customer Profitability Analysis

At first, it might appear that the two unprofitable customers should be dropped.

	Gonzales	Abdullah	Patel	Kawanishi	Total
Sales	\$ 10,000	\$40,000	\$62,000	\$ 22,000	\$134,000
Cost of goods sold	7,200	26,000	41,000	18,100	92,300
Other relevant costs	1,000	2,200	4,400	4,100	11,700
Customer margin	\$ 1,800	\$11,800	\$16,600	\$ (200)	\$ 30,000
Allocated fixed costs	2,000	6,000	8,800	4,000	20,800
Operating income	\$ (200)	\$ 5,800	\$ 7,800	\$ (4,200)	\$ 9,200

Dropping Kawanishi makes sense. However, Gonzales is contributing to the coverage of fixed costs, which would have to be shifted to the other customers if Gonzales were dropped.

- e. Issues involved in determining product profitability, business unit profitability, and customer profitability include the following:
- 1) **Cost measurement** involves calculating correct costs and not undercosting (i.e., product consumes a high amount of resources but is reported as having low costs per unit) or overcosting (i.e., business unit consumes a low amount of resources but is reported as having high costs).
 - 2) **Cost allocation** is the assignment of costs to the appropriate product, business unit, and customer. Incorrect costs assignment will lead to overcosting and undercosting. Thus, providing erroneous feedback on the profitability of the applicable product, business unit, and/or customer will lead management to base decisions on inaccurate financial results.
 - 3) **Investment measurement** involves calculating the cost to expand production, develop new products, or enter new business markets through acquisition or internal development. This process involves (a) identifying investments, (b) determining the resources required, and (c) projecting the expected amounts and timing of returns. Hence, sound decisions regarding projected favorable returns on investments (i.e., ranking the identified investments) are contingent on accurate valuations of potential investment.

- 4) **Other measures** include but are not limited to (a) nonvalue-added costs and inefficient production activities, (b) selection and placement of value-added and efficient production activities, (c) response to customer requirements, and (d) high-quality products produced with less resources. Successful monitoring and management of these processes are ways to improve profitability and/or identify production activities, products, or customers requiring disposal. Nonfinancial measures are discussed in more detail in Study Unit 11, Subunit 2.

16.3 PERFORMANCE MEASURES -- INVESTMENT CENTERS

1. Purpose

- Performance measures for investment centers reveal how efficiently the manager is deploying capital to produce income for the organization.
- Thus, most performance measures relate the center's resources (balance sheet) to its income (income statement).
- Two widely used performance measures for an investment center are return on investment and residual income. These measures allow an investor to assess how effectively and efficiently management is using assets to obtain a return.

2. Return on Investment (ROI)

- ROI is the relationship between income and the investment in assets to generate such income.

$$\text{ROI} = \frac{\text{Income of business unit}}{\text{Assets of business unit}}$$

- Income means operating income unless otherwise noted. Operating income is known as earnings before interest and taxes (EBIT).

EXAMPLE 16-5 ROI Calculations

The ROI calculations for the branch offices displayed in Example 16-3 in Subunit 16.2 are shown below:

	Cartagena	Riyadh	Mumbai	Osaka
Income of business unit	\$ (6,000)	\$ 64,000	\$ 120,000	\$ 636,000
Total assets	121,000	825,000	1,015,000	9,900,000
Return on investment	(5.0%)	7.8%	11.8%	6.4%

Even though the managers of the Osaka branch generated by far the largest contribution, they were not as efficient in the deployment of the resources at their disposal as were the managers of the Riyadh or Mumbai branches.

This example illustrates the principle that the appraisal of individual performance must consider more factors than simple dollars.

- b. Benefits with the application of ROI include, but are not limited to, the following:
- 1) **Improve projects** by analyzing data collected to determine how the projects should change in order to achieve favorable returns.
 - 2) **Secure funding** through the use of positive ROI forecasts.
 - 3) **Comparative analysis** assists in making comparisons between different business units in terms of profitability and asset utilization.
 - 4) **Discontinue ineffective products or operations** by using ROI data to support that the product or operation does not add value.
- c. A limitation with the application of ROI is that an investment center with a high ROI may not accept a profitable investment even though the investment's return is higher than the center's target ROI.
- 1) **EXAMPLE:** An investment center has an 8% ROI, and its investors expect 2%. If the decision makers look only at current ROI, they will reject a project earning 6%, even though that return exceeds the target.

3. Residual Income

- a. Residual income is a variation of ROI that measures performance in dollar terms rather than as a percentage return.

$$\text{Residual income} = \text{Income of business unit} - (\text{Assets of business unit} \times \text{Required rate of return})$$

- 1) Income means operating income unless otherwise noted. Operating income is known as EBIT.
- 2) Residual income is a significant refinement of the ROI concept because it forces business unit managers to consider the **opportunity cost** of capital. Opportunity cost represents the return on the best alternative investment of similar risk that would have been generated if management had invested.

EXAMPLE 16-6 Residual Income Calculations

The residual income calculations for the branch offices displayed in Example 16-5 on the previous page are shown below:

	Cartagena	Riyadh
Income of business unit	\$ (6,000)	\$ 64,000
Total assets	\$121,000	\$825,000
Times: Target rate of return	x 6.0%	x 6.0%
Opportunity cost of capital	(7,260)	(49,500)
Residual income	<u><u>\$ (13,260)</u></u>	<u><u>\$ 14,500</u></u>
	Mumbai	Osaka
Income of business unit	\$120,000	\$636,000
Total assets	\$1,015,000	\$9,900,000
Times: Target rate of return	x 6.0%	x 6.0%
Opportunity cost of capital	(60,900)	(594,000)
Residual income	<u><u>\$ 59,100</u></u>	<u><u>\$ 42,000</u></u>

This calculation reveals that, by employing the most resources, the Osaka branch has by far the highest threshold to clear for profitability.

4. Comparability Issues with Investment Center Performance Measures

- a. Alternative income measurements include income of business unit, income of business unit adjusted for price level changes, cash flow, and EBIT.
- b. Invested capital may be defined in various ways, for example, as
 - 1) Total assets available
 - 2) Total assets employed, which excludes assets that are idle, such as vacant land
 - 3) Working capital plus other assets, which excludes current liabilities (i.e., capital provided by short-term creditors)
 - a) This investment base assumes that the manager controls short-term credit.
- c. Different attributes of financial information will also affect the elements of the investment base.
 - 1) Historical cost
 - 2) Replacement cost
 - 3) Market value
 - 4) Present value
- d. The comparability of performance measures may be affected by differences in the **accounting policies** used by different business units.
 - 1) For example, policies regarding depreciation, decisions to capitalize or expense, inventory flow assumptions, and revenue recognition can lead to comparability issues for performance measures.
 - 2) These differences may be heightened for the business units of a multinational enterprise.
- e. Issues other than accounting policy may also affect comparability.
 - 1) Differences in the tax systems in the jurisdictions where business units operate
 - 2) The presence of items that are of an unusual nature and/or occur infrequently
 - 3) Allocation of common costs
 - 4) The varying availability of resources
- f. Performance evaluation in multinational companies is impacted by the following:
 - 1) **Expropriation** is a foreign government's seizure (nationalization) of the assets of a business for a public purpose and for just compensation.
 - 2) **Repatriation** is conversion of funds held in a foreign country into another currency and remittance of these funds to another nation. A firm often must obtain permission from the currency exchange authorities to repatriate earnings and investments. Regulations in many nations encourage a reinvestment of earnings in the country.
 - 3) **Tariffs** are taxes imposed on imported goods. Tariffs can discourage consumption of foreign goods, raise revenue, or both.
 - 4) **Import quotas** set limits on the quantity of different products that can be imported.
 - 5) **Inflation risk** is the risk that purchasing power of the currency will decline.
 - 6) **Exchange rate risk** is the risk of loss because of fluctuation in the relative value of a foreign currency in which the investment is payable.
 - 7) **Political risk** is the probability of loss from actions of governments, such as changes in tax laws or environmental regulations or expropriation of assets.

5. ROI vs. Residual Income

- a. Residual income is often considered preferable to ROI because it deals in absolute dollars rather than percentages.
 - 1) A manager with a 10% ROI would be reluctant to invest in a project with only an 8% return because his or her average overall return would decline. This would be detrimental to the company as a whole if the cost of capital were only 5%.
 - 2) However, under the residual income method, the manager would invest in any project with a return greater than the cost of capital or the hurdle rate that (s)he has been assigned.
 - 3) Thus, overall, the company would be better off even though the individual manager's ROI declined.

6. Revenue and Expense Recognition Policies

- a. A company's revenue and expense recognition policies may affect the measurement of income and thus reduce comparability among business units.
 - 1) For example, a company that uses last-in, first-out (LIFO) for inventory valuation will often show lower inventories and higher costs than a company that uses the first-in, first-out (FIFO) methodology. As a result, the LIFO company could appear to have a lower rate of return than if it had used the FIFO method.
 - 2) Other ratios would also be impacted, such as inventory turnover and asset turnover (these activity ratios are covered in detail in CMA Part 2, Study Unit 3, Subunit 1).
- b. Thus, when comparing companies or units on the basis of either ROI or residual income, the analyst must be sure that both companies or units are using the same accounting policies in the determination of income.
- c. The sharing of assets by subunits within an organization may also affect measures of return.
 - 1) For instance, assets normally appear on the books of only one division, even though another division might have access to those assets. Therefore, the division that shares its assets with another division may find that it has a lower rate of return than the division that has access to the use of the assets.
- d. Similarly, a company or division that uses straight-line depreciation on its plant assets will have lower expenses in the early years of an asset's life than if an accelerated method were being used. Thus, the straight-line division would appear to be more profitable than the division using the accelerated method.
 - 1) Of course, the accelerated method may be preferred by top management because it results in a tax savings, but the implication of the ROI measure might be that the straight-line division is more profitable.
- e. International operations may not always be comparable to domestic divisions, since the complication of changes in foreign-currency exchange rates might make income comparisons difficult. Also, transfer pricing is complicated in the international arena, since differences in tax rates between countries may have a role in the selection of the transfer prices selected.
 - 1) For example, a company will set a transfer price at a level that will limit the profits in high-tax countries and shift that profit to the low-tax country.
 - 2) Similarly, high profits in a foreign country might not always be transferable to the home country; thus, to say that the foreign subsidiary is more profitable is meaningless if that profit cannot be enjoyed by the parent company.

7. Nonfinancial Measures

- a. Nonfinancial measures, such as the balanced scorecard, can be used to compare divisions and managers. The balanced scorecard is discussed in detail in Study Unit 16, Subunit 4.

16.4 THE BALANCED SCORECARD

1. Key Performance Indicators (KPIs)

- a. The trend in performance evaluation is the balanced scorecard approach to managing the implementation of the firm's strategy.
 - 1) The balanced scorecard is a tool for communicating the company's strategy throughout the organization, and connects the firm's critical success factors to measurements of its performance.
- b. KPIs are specific, measurable financial and nonfinancial elements of a firm's performance that are vital to its competitive advantage.
 - 1) Multiple measures of performance permit a determination as to whether a manager is achieving certain objectives at the expense of others that may be equally or more important.
 - a) For example, an improvement in operating results at the expense of new product development would be apparent using a balanced scorecard approach.
- c. The balanced scorecard is a goal congruence tool that informs managers about the financial and nonfinancial factors that are essential to have a competitive advantage. Those factors are defined as the **critical success factors**.
 - 1) Measures on the balanced scorecard may be internal or external, and short term or long term.
 - 2) The balanced scorecard facilitates **best practice analysis**. Best practice analysis determines a method of carrying on a business function or process that is considered to be superior to all other known methods.
 - a) A lesson learned from one area of a business can be passed on to another area of the business or between businesses.

2. SWOT Analysis

- a. A firm identifies its KPIs by means of a SWOT analysis.
 - 1) The firm's greatest strengths are its core competencies, which are the basis for its competitive advantages and strategy.
- b. Strengths and weaknesses are internal resources or a lack thereof, for example, technologically advanced products, a broad product mix, capable management, leadership in R&D, modern production facilities, and a strong marketing organization.
- c. **Opportunities and threats** arise from such externalities as government regulation, advances in technology, and demographic changes. They may be reflected in such competitive conditions as
 - 1) Raising or lowering of barriers to entry into the firm's industry by competitors
 - 2) Changes in the intensity of rivalry within the industry, for example, because of overcapacity or high exit barriers
 - 3) The relative availability of substitutes for the firm's products or services
 - 4) Bargaining power of customers, which tends to be greater when switching costs are low and products are not highly differentiated
 - 5) Bargaining power of suppliers, which tends to be higher when suppliers are few
- d. The SWOT analysis tends to highlight the basic factors of cost, quality, and the speed of product development and delivery.

3. PEST Analysis

- a. A PEST analysis is an analysis of certain factors in the external environment of an organization, which can affect its activities and performances.
 - 1) It is used to identify KPIs.
- b. PEST analysis includes
 - 1) **P**olitical factors,
 - 2) **E**conomic factors,
 - 3) **S**ocial factors, and
 - 4) **T**echnological factors.
- c. PEST analysis may also include environmental, legal, and ethical considerations, among others.

4. Measures

- a. Once the firm has identified its KPIs, it must establish specific measures for each KPI that are both relevant to the success of the firm and can be reliably stated.
 - 1) Thus, the balanced scorecard varies with the strategy adopted by the firm.
 - 2) For example, product differentiation or cost leadership either in a broad market or a narrowly focused market (a focus strategy). These measures provide a basis for implementing the firm's competitive strategy.
- b. The scorecard should include lagging indicators (such as output and financial measures) and leading indicators (such as many types of nonfinancial measures).
 - 1) The latter should be used only if they are predictors of ultimate financial performance.
- c. The scorecard should permit a determination of whether certain objectives are being achieved at the expense of others.
 - 1) For example, reduced spending on customer service may improve short-term financial results at a significant cost that is revealed by a long-term decline in customer satisfaction measures.
- d. By providing measures that are **nonfinancial as well as financial**, long term as well as short term, and internal as well as external, the balanced scorecard de-emphasizes short-term financial results and focuses attention on KPIs.
- e. An effective balanced scorecard requires a vast amount of data of many different types.
 - 1) For this reason, an enterprise resource planning (ERP) system is almost a necessity. An ERP system integrates information systems across the organization by creating one database linking all of the firm's applications.

5. Possible KPIs and Measures

- a. A typical balanced scorecard classifies objectives into one of four perspectives on the business, which are (1) financial, (2) customer satisfaction, (3) internal business processes, and (4) learning and growth.
- b. **Financial**

KPI	Measure
Sales	New Product Sales
FV of Firm's Stock	Price/Earnings Ratio
Profits	Return on Investment
Liquidity	Quick Ratio

- 1) Other measures may include sales, projected sales, accuracy of sales projections, stock prices, operating earnings, earnings trend, revenue growth, gross margin percentage, cost reductions, residual income, cash flow coverage and trends, turnover (assets, receivables, and inventory), and interest coverage.

c. **Customer Satisfaction**

KPI	Financial Measure	Nonfinancial Measure
Customer Satisfaction	Trends in dollar amounts of returns	Market share
Dealer and Distributor Relationships	Trends in dollar amounts of discounts taken	Lead time
Marketing and Selling Performance	Trends in dollar amounts of sales	Market research results
Prompt Delivery	Trend in delivery expenses	On-time delivery rate
Quality	Dollar amount of defects	Rate of defects

- 1) Other financial measures may include dollar amount of sales, dollar amount of returns, and warranty expense.
- 2) Other nonfinancial measures may include unit sales, trends in unit sales, trend in market share, number of returns, rate of returns, customer retention rate, number of defects, number of warranty claims, rate of warranty claims, survey results, coverage and strength of distribution channels, training of marketing people, service response time, and service effectiveness.

d. **Internal Business Processes**

KPI	Financial Measure	Nonfinancial Measure
Quality	Scrap costs	Rate of scrap and rework
Productivity	Change in company revenue/change in company costs	Units produced per machine hour
Flexibility of Response to Changing Conditions	Cost to repurpose machine for new use	Time to repurpose machine for new use
Operating Readiness	Set-up costs	Downtime
Safety	Dollar amount of injury claims	Number of injury claims

- 1) Other financial measures may include such things as quality costs and level of inventory carrying costs.
- 2) Other nonfinancial measures may include new products marketed, technological capabilities, survey results, field service reports, vendor defect rate, cycle time, labor and machine efficiency, setup time, scheduling effectiveness, capacity usage, maintenance, and accidents and their results.

e. Learning and Growth

KPI	Financial Measure	Nonfinancial Measure
Development of New Products	R&D costs	Number of new patents applied for
Promptness of Their Introduction	Lost revenue (from slow introduction of new product to market)	Length of time to bring a product to market
Human Resource Development	Recruiting costs	Personnel turnover
Morale	Orientation/team-building costs	Personnel complaints
Competence of Work Force	Training/retraining costs	Hours of training

- 1) Other financial measures may include financial and operating results.
- 2) Other nonfinancial measures may include number of design changes and copyrights registered, R&D personnel qualifications, actual versus planned shipping dates, skill set levels attained, personnel survey results, organizational learning, and industry leadership.

6. Development

- a. The active support and participation of senior management are essential.
 - 1) This involvement will in turn ensure the cooperation of lower-level managers in the identification of objectives, appropriate measures, targeted results, and methods of achieving the results.
- b. The scorecard should contain measures at the detail level that permit everyone to understand how his or her efforts affect the firm's results.
 - 1) The scorecard and the strategy it represents must be communicated to all managers and used as a basis for compensation decisions.

7. Functionality

- a. Each objective is associated with one or more measures that permit the organization to gauge progress toward the objective.
 - 1) In order for the balanced scorecard to be useful, the organization must be able to identify a cause-and-effect relationship between an action that could be taken (or avoided) and an effect on a KPI.
 - a) For example, if the organization increases its R&D budget, then the organization can increase the number of new patents applied for.
 - 2) Achievement of the objectives in each perspective makes it possible to achieve the objectives in the next higher perspective.
 - 3) This chaining of objectives and perspectives embodies the implementation of a **strategy map**.

- b. To achieve its objectives, the organization must establish the following:
- 1) Relevant criteria to measure outcomes.
 - a) This requires the use of **lagging indicators** which measure the results of actions.
 - 2) Relevant performance drivers
 - a) This requires the use of **leading indicators** which are used to determine how an outcome can be met.
- c. When measuring results and planning actions to take, the organization must stay focused on the cost-effectiveness of any action taken (or not taken).
- 1) Achieving goals should not become ends unto themselves.
 - a) Achievement of goals should be linked to the bottom line.
- d. The following are **problems in implementation** of the balanced scorecard approach:
- 1) Using too many measures, with a consequent loss of focus on KPIs
 - 2) Failing to evaluate personnel on nonfinancial as well as financial measures
 - 3) Including measures that will not have long-term financial benefits
 - 4) Not understanding that subjective measures (such as customer satisfaction) are imprecise
 - 5) Trying to achieve improvements in all areas at all times
 - 6) Not being aware that the hypothesized connection between nonfinancial measures and ultimate financial success may not continue to be true
- e. **Example Balanced Scorecard**

OBJECTIVES	PERFORMANCE MEASURES	TARGETS	OUTCOMES	CORRECTIVE INITIATIVES
PERSPECTIVE: Financial Performance				
Increase sales	Gross revenues	Increase 15%	Increase 3%	• Expand into new markets • Improve same-store sales
PERSPECTIVE: Customer Service				
Reduce returns	Number of returns	Decrease 10%	Decrease 2%	• Reduce number of defects • Determine customer needs prior to sale
PERSPECTIVE: Internal Business Processes				
Reduce scrap	Costs of scrap	Decrease 5%	Increase 4%	• Improve employee training • Seek higher quality materials
PERSPECTIVE: Learning and Growth				
Reduce personnel turnover	Length of time employed	Increase 50%	Increase 10%	• Improve hiring practices • Reevaluate compensation plan

16.5 TRANSFER PRICING

1. Purpose

- a. Transfer prices are the amounts charged by one segment of an organization for goods and services it provides to another segment of the same organization.
 - 1) The principal challenge is determining a price that motivates both the selling and the buying manager to pursue organizational **goal congruence**.
- b. In a decentralized system, each responsibility center theoretically may be completely separate.
 - 1) Thus, Division A should charge the same price to Division B as would be charged to an outside buyer.
 - 2) The reason for decentralization is to motivate managers, and the best interests of Division A may not be served by giving a special discount to Division B if the goods can be sold at the regular price to outside buyers. However, having A sell at a special price to B may be to the company's advantage.

2. Transfer Pricing Schemes

- a. Four basic methods of transfer price setting are in common use: variable cost, full cost, market price, and negotiated price.
- b. **Variable Cost**
 - 1) By allowing the buyer to purchase at the selling division's variable cost, unused production capacity will be utilized (this method should only be used when the selling division has **excess capacity**).
 - 2) However, there is no incentive for the selling division, since it will be producing the products at a loss (even though the company as a whole will benefit from the arrangement).
 - a) In practice, companies who wish to follow this philosophy actually adopt a negotiation policy wherein the transfer price will be something greater than variable costs but less than full costs. At least the seller would have a positive contribution margin if the price is set slightly above variable costs.
 - 3) The advantage of using variable costs is that the buyer is motivated to solve the company's excess capacity problem, even though that excess capacity is not in the buyer's division.

c. **Full (Absorption) Cost**

- 1) Full (absorption) cost includes materials, labor, and full allocation of manufacturing overhead.
- 2) The use of full (absorption) cost ensures that the selling division will not incur a loss and provides more incentive to the buying division to buy internally than does use of market price.
 - a) However, there is no motivation for the seller to control production costs since all costs can be passed along to the buying division.

d. **Market Price**

- 1) A market price is the best transfer price to use in many situations. For example, if the selling division is operating at **full capacity** and can sell all of its output at the market price, then there is no justification to use a lower price as the transfer price for intracompany transfers.
- 2) Alternatively, if the selling division is not producing at full capacity, the use of market prices for internal transfers is not justified. A lower price might be more motivational to either the buyer or the seller.

e. **Negotiation**

- 1) A negotiated price may result when organizational subunits are free to determine the prices at which they buy and sell internally. Hence, a transfer price may simply reflect the **best bargain** that the parties can strike between themselves.
- 2) The transfer price need not be based directly on particular market or cost information.
- 3) A negotiated price may be especially appropriate when market prices are subject to rapid fluctuation.

3. **Choice of Transfer Pricing Policy**

- a. The choice of a transfer pricing policy (which type of transfer price to use) is normally decided by top management at the corporate level. The decision typically includes consideration of multiple factors.
- b. Goal congruence factors
 - 1) The transfer price should promote the goals of the company as a whole.

c. Segmental performance factors

- 1) The segment making the transfer should be allowed to recover its incremental cost plus its opportunity cost of the transfer. The opportunity cost is the benefit forgone by not selling to an outsider.
 - a) For this purpose, the transfer should be at market price.
 - b) The selling manager should not lose income by selling within the company.
- 2) Properly allocating revenues and expenses through appropriate transfer pricing also facilitates evaluation of the performance of the various segments.

d. Negotiation factors

- 1) If the purchasing segment could purchase the product or service outside the company, it should be permitted to negotiate the transfer price.
- 2) The purchasing manager should not have to incur greater costs by purchasing within the company.

e. Capacity factors

- 1) If Division A has excess capacity, it should be used for producing products for Division B. If Division A is operating at full capacity and selling its products at the full market price, profitable work should not be abandoned to produce for Division B.

f. Cost structure factors

- 1) If Division A has excess capacity and an opportunity arises to sell to Division B at a price in excess of the variable cost, the work should be performed for Division B because a contribution to cover the fixed costs will result.

g. Tax factors

- 1) A wide range of tax issues on the interstate and international levels may arise, e.g., income taxes, sales taxes, value-added taxes, inventory and payroll taxes, and other governmental charges.
- 2) In the international context, exchange rate fluctuations, threats of expropriation, and limits on transfers of profits outside the host country are additional concerns.
 - a) Thus, because the best transfer price may be a low one because of the existence of tariffs or a high one because of the existence of foreign exchange controls, the effect may be to skew the performance statistics of management.
 - b) The high transfer price may result in foreign management appearing to show a lower return on investment than domestic management, but the ratio differences may be negated by the fact that a different transfer pricing formula is used.

Transfer Price Decision Tree

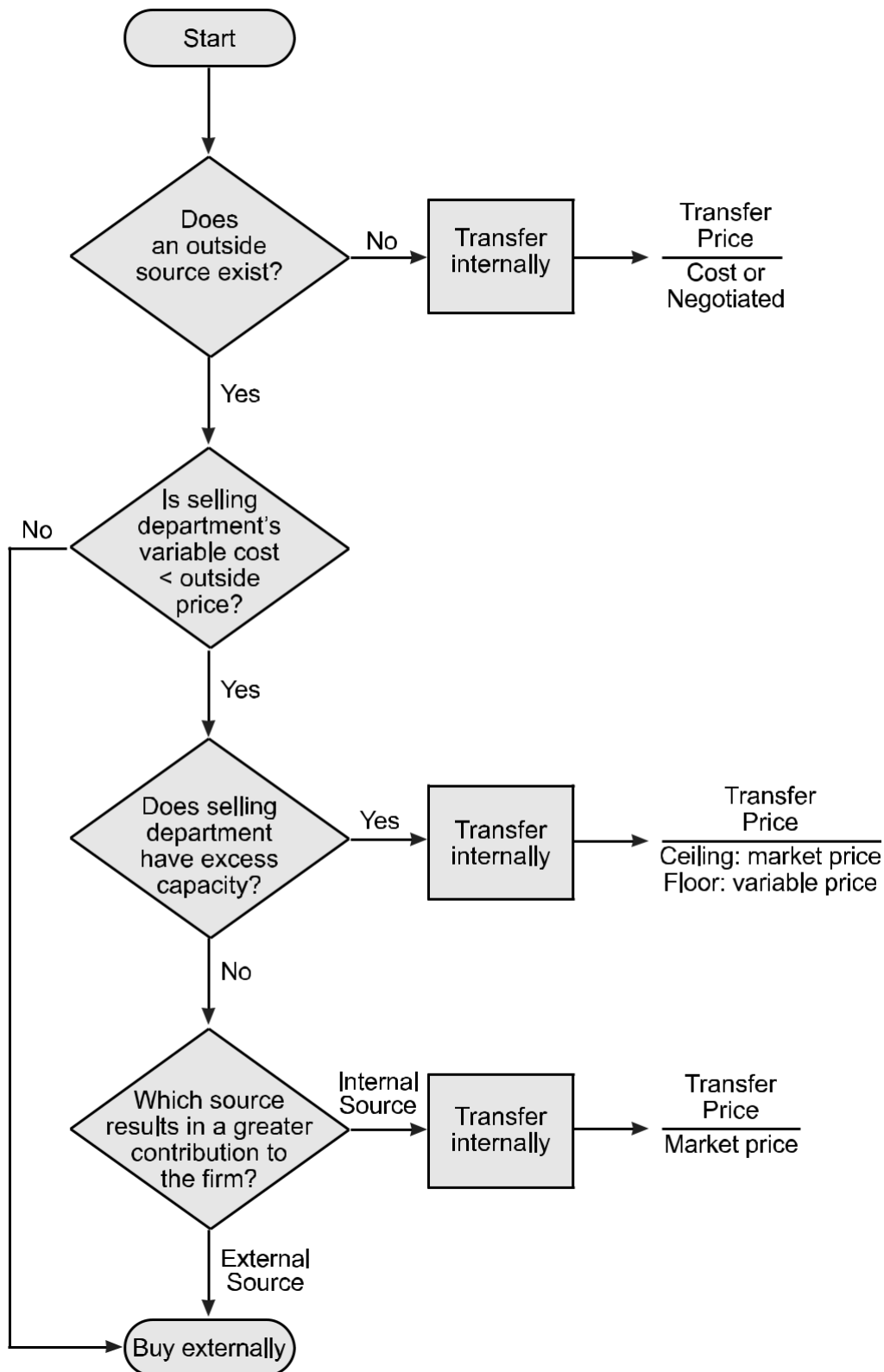


Figure 16-1

EXAMPLE 16-7 Transfer Pricing Decision Scenarios

Division A produces a small part at a cost of \$6 per unit. The regular selling price is \$10 per unit. If Division B can use the part in its production, the cost to the company (as a whole) will be \$6.

Division B has another supplier who will sell the item to B at \$9.50 per part. Division B wants to buy the \$9.50 part from the outside supplier instead of the \$10 part from Division A, but making the part for \$6 is in the company's best interest.

What amount should Division A charge Division B?

The answer is complicated by many factors. For example, if Division A has excess capacity, B should be charged a lower price. If it is operating at full capacity, B should be charged \$10.

Also consider what portion of Division A's costs is fixed. For example, if a competitor offered to sell the part to B at \$5 each, can Division A advantageously sell to B at a price lower than \$5? If Division A's \$6 total cost is composed of \$4 of variable costs and \$2 of fixed costs, it is beneficial for all concerned for A to sell to B at a price less than \$5. Even at a price of \$4.01, the parts would be providing a contribution margin to cover some of A's fixed costs.

4. Dual Pricing

- a. Under dual pricing, the selling and buying units record the transfer at different prices.
 - 1) For example, the seller could record the transfer to another segment as a sale at the usual market price that would be paid by an outsider. The buyer, however, would record a purchase at the variable cost of production.
 - 2) Each segment's reported performance is improved by the use of a dual-pricing scheme.
 - 3) The firm as a whole would benefit because variable costs would be used for decision-making purposes. In a sense, variable costs would be the relevant price for decision-making purposes, but the regular market price would be used for evaluation of production divisions.
- b. However, under a dual-pricing system, the profit for the company will be less than the sum of the profits of the individual segments. In effect, the seller is given a corporate subsidy.
 - 1) The dual-pricing system is rarely used because the incentive to control costs is reduced.
 - 2) The seller is assured of a high price, and the buyer is assured of an artificially low price. Thus, neither manager must exert much effort to show a profit on segmental performance reports.
 - 3) Also, an elimination entry must be recorded at the end of the period to reconcile the fact that sales were recorded at an amount different from the corresponding purchases.

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**2021
EDITION**

CMA REVIEW

PART 1:

FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS

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D. Cost Management (15% - Levels A, B, and C)

1. *Measurement concepts*

- a. Cost behavior and cost objects (7.1-7.3)
- b. Actual and normal costs (7.4, 9.2)
- c. Standard costs (7.4, 13.2)
- d. Absorption (full) costing (7.5)
- e. Variable (direct) costing (7.5)
- f. Joint and by-product costing (9.1)

2. *Costing systems*

- a. Job order costing (8.1)
- b. Process costing (8.2)
- c. Activity-based costing (8.3)
- d. Life-cycle costing (8.4)

3. *Overhead costs*

- a. Fixed and variable overhead expenses (9.2)
- b. Plant-wide vs. departmental overhead (9.2)
- c. Determination of allocation base (9.2)
- d. Allocation of service department costs (9.3)

4. *Supply chain management*

- a. Lean resource management techniques (10.1)
- b. Enterprise resource planning (ERP) (10.2)
- c. Theory of Constraints (10.3)
- d. Capacity management and analysis (10.4, 11.1)

5. *Business process improvement*

- a. Value chain analysis (11.1)
- b. Value-added concepts (11.1)
- c. Process analysis, redesign, and standardization (11.2)
- d. Activity-based management (8.3, 11.2)
- e. Continuous improvement concepts (8.3, 11.2)
- f. Best practice analysis (11.2)
- g. Cost of quality analysis (11.2)
- h. Efficient accounting processes (11.2)

STUDY UNIT SEVEN

COST MANAGEMENT CONCEPTS

7.1	<i>Cost Management Terminology</i>	2
7.2	<i>Cost Behavior and Relevant Range</i>	4
7.3	<i>Cost Classification</i>	9
7.4	<i>Costing Techniques</i>	12
7.5	<i>Absorption and Variable Costing</i>	15

Cost Management

Cost management is at the heart of the field of management accounting. Thus, the CMA exam places great emphasis on this area of study. The candidate will face many questions involving numerical calculations and others requiring a knowledge of cost terminology and the implications of cost management decisions. There is a strategic value in knowing cost information. For example, costs need to be known for pricing and overhead allocation purposes.

This study unit is the **first of five** on **cost management**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The five study units are

- Study Unit 7: Cost Management Concepts
- Study Unit 8: Cost Accumulation Systems
- Study Unit 9: Cost Allocation Techniques
- Study Unit 10: Supply Chain Management
- Study Unit 11: Business Process Improvement

If you are interested in reviewing more introductory or background material, go to www.GLEM.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following GLEM outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

7.1 COST MANAGEMENT TERMINOLOGY

1. Subdisciplines of Accounting

- a. **Financial accounting** is concerned principally with reporting to external users, usually through a set of financial statements produced in accordance with GAAP.
 - 1) Financial accounting thus has a historical focus.
- b. **Management accounting** is concerned principally with reporting to internal users. The management accountant's goal is to produce reports that improve organizational decision making.
 - 1) Management accounting is thus future-oriented.
- c. **Cost accounting** supports both financial and management accounting. Information about the cost of resources acquired and consumed by an organization underlies effective reporting for both internal and external users.

2. Basic Definitions

- a. A cost is defined by the IMA in two senses:
 - 1) "In management accounting, a measurement in monetary terms of the amount of resources used for some purpose. The term by itself is not operational. It becomes operational when modified by a term that defines the purpose, such as acquisition cost, incremental cost, or fixed cost."
 - 2) "In financial accounting, the sacrifice measured by the price paid or required to be paid to acquire goods or services. The term 'cost' is often used when referring to the valuation of a good or service acquired. When 'cost' is used in this sense, a cost is an asset. When the benefits of the acquisition (the goods or services) expire, the cost becomes an expense or loss."
- b. A **cost object** is any object to which costs can be attached.
 - 1) Examples are products, processes, employees, departments, and facilities.
 - 2) Costs are allocated to objects to assist management decision making on a variety of matters, including whether to expand a department, discontinue a product line, and make or buy a good.
- c. A **cost driver** is the basis used to assign costs to a cost object.
 - 1) Cost driver is defined by the IMA as "a measure of activity, such as direct labor hours, machine hours, beds occupied, computer time used, flight hours, miles driven, or contracts, that is a causal factor in the incurrence of cost to an entity."
 - 2) The key aspect of a cost driver is the existence of a direct cause-and-effect relationship between the quantity of the driver consumed and the amount of total cost. In other words, a cost driver is some event that causes costs to occur.

3. Direct vs. Indirect

- a. Costs can be classified by how they are assigned to cost objects.
 - 1) **Direct costs** are costs that can be associated with a particular cost object in an economically feasible way, i.e., they can be traced to that object.
 - a) Examples are the direct materials and direct labor inputs to a manufacturing process discussed in items 4.a.1) and 4.a.2) on the next page.

- 2) **Indirect costs** are costs that cannot be associated with a particular cost object in an economically feasible way and thus must be allocated to that object.
 - a) Examples are the indirect materials and indirect labor inputs to a manufacturing process discussed in items 4.a.3)a) and 4.a.3)b) below.
 - i) To simplify the allocation process, indirect costs are often collected in cost pools.
 - b) A **cost pool** is an account into which a variety of similar cost elements with a common cause are accumulated.
 - i) It is preferable for all the costs in a cost pool to have the same cost driver.
 - ii) Manufacturing overhead, as defined in item 4.a.3) below, is a commonly used cost pool into which various untraceable costs of the manufacturing process are accumulated prior to being allocated.
- 3) **Common costs** are another notable type of indirect cost. A common cost is a cost incurred for the benefit of more than one cost object.
 - a) The key to common costs is that, since they cannot be directly traced to a single cost object, they must be allocated using some systematic and rational basis.
 - b) An example is depreciation or rent on the headquarters building. This is a direct cost when treating the building as a whole, but is a common cost of the departments located in the building, and thus must be allocated when treating the individual departments.

4. Manufacturing vs. Nonmanufacturing

- a. The costs of manufacturing a product can be classified as one of three types:
 - 1) **Direct materials** are those tangible inputs to the manufacturing process that can practicably be traced to the product, e.g., sheet metal welded together for a piece of heavy equipment.
 - a) In addition to the purchase price, all costs of bringing raw materials to the production line, e.g., transportation-in, are included in the cost of direct materials.
 - 2) **Direct labor** is the cost of human labor that can practicably be traced to the product, e.g., the wages of the welder.
 - 3) **Manufacturing overhead** consists of all costs of manufacturing that are not direct materials or direct labor.
 - a) **Indirect materials** are tangible inputs to the manufacturing process that cannot practicably be traced to the product, e.g., the welding compound used to put together a piece of heavy equipment.
 - b) **Indirect labor** is the cost of human labor connected with the manufacturing process that cannot practicably be traced to the product, e.g., the wages of assembly line supervisors and janitorial staff.
 - c) **Factory operating costs**, also known as general overhead, include utilities, real estate taxes, insurance, depreciation on factory equipment, etc.

- b. Manufacturing costs are often grouped into the following classifications:
 - 1) **Prime cost** equals direct materials plus direct labor, i.e., those costs directly attributable to a product.
 - 2) **Conversion cost** equals direct labor plus manufacturing overhead, i.e., the costs of converting raw materials into the finished product.
- c. Operating a manufacturing concern requires the incurrence of nonmanufacturing costs.
 - 1) **Selling (marketing) expenses** are those costs incurred in getting the product from the factory to the consumer, e.g., sales personnel salaries, advertising, and product transportation.
 - 2) **Administrative expenses** are those costs incurred by a company not directly related to producing or marketing the product, e.g., executive salaries, depreciation on the headquarters building, and rent on a warehouse containing inventory.

5. Product vs. Period

- a. One of the most important classifications a management accountant can make is whether to capitalize a cost as part of finished goods inventory or to expense it as incurred.
 - 1) **Product costs** (also called inventoriable costs) are capitalized as part of finished goods inventory. They eventually become a component of cost of goods sold. Such costs include direct materials and direct labor. Product costs also include manufacturing overhead costs under various costing approaches, as discussed in Subunit 7.5.
 - 2) **Period costs** are expensed as incurred, i.e., they are not capitalized in finished goods inventory and are thus excluded from cost of goods sold.
 - a) The theory is that period costs are caused by the passage of time and would occur even if production was zero.

7.2 COST BEHAVIOR AND RELEVANT RANGE

1. Relevant Range

- a. The relevant range defines the limits within which per-unit variable costs remain constant and fixed costs are not changeable. It is synonymous with the short run.
- b. The relevant range is established by the efficiency of a company's current manufacturing plant, its agreements with labor unions and suppliers, etc.

2. Variable Costs

- a. **Variable costs** are a direct function of production volume. They increase when production grows and decrease when production shrinks. Raw materials and labor directly involved with production are common variable costs.

- b. **Variable cost per unit** remains constant in the short run regardless of the level of production.

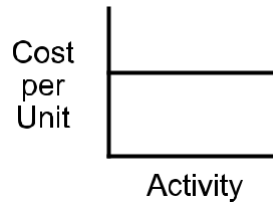


Figure 7-1

- c. **Variable costs in total**, on the other hand, vary directly and proportionally with changes in volume.

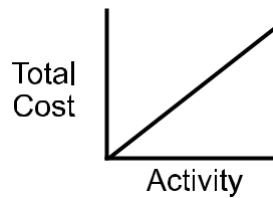


Figure 7-2

EXAMPLE 7-1 Variable Costs in Total

A company requires one unit of direct material to be used in each finished good it produces.

Number of Outputs Produced	Input Cost per Unit	Total Cost of Inputs
0	\$10	\$ 0
100	\$10	\$ 1,000
1,000	\$10	\$ 10,000
5,000	\$10	\$ 50,000
10,000	\$10	\$ 100,000

3. Fixed Costs

- a. **Fixed costs** remain constant regardless of production. Examples of fixed costs include rent, interest, insurance, and lease payments.
- b. **Fixed costs in total** remain unchanged in the short run regardless of production level, e.g., the amount paid for an assembly line is the same even if production is halted entirely.

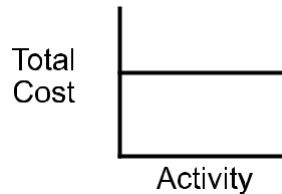


Figure 7-3

- c. **Fixed cost per unit**, on the other hand, varies indirectly with the activity level.

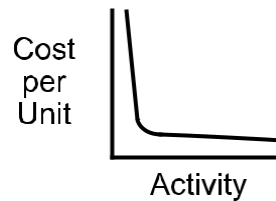


Figure 7-4

EXAMPLE 7-2 Fixed Cost per Unit

The historical cost of the assembly line is settled, but its cost per unit decreases as production increases.

Number of Outputs Produced	Cost of Assembly Line	Per Unit Cost of Assembly Line
1	\$1,000,000	\$ 1,000,000
100	\$1,000,000	\$ 10,000
1,000	\$1,000,000	\$ 1,000
5,000	\$1,000,000	\$ 200
10,000	\$1,000,000	\$ 100

4. Mixed (Semivariable) Costs

- a. Mixed (semivariable) costs **combine fixed and variable elements**, e.g., rental expense on a car that carries a flat fee per month plus an additional fee for each mile driven.

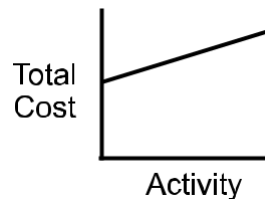


Figure 7-5

EXAMPLE 7-3 Mixed (Semivariable) Costs

The company rents a piece of machinery to make its production line more efficient. The rental is \$150,000 per year plus \$1 for every unit produced.

Number of Outputs Produced	Fixed Cost of Extra Machine	Variable Cost of Extra Machine	Total Cost of Extra Machine
0	\$150,000	\$ 0	\$150,000
100	\$150,000	\$ 100	\$150,100
1,000	\$150,000	\$ 1,000	\$151,000
5,000	\$150,000	\$ 5,000	\$155,000
10,000	\$150,000	\$ 10,000	\$160,000

- b. Two methods of estimating mixed costs are in general use:
- 1) The **regression (or scattergraph) method** is by far the more complex (and accurate) of the two methods. Regression analysis is covered in detail in Study Unit 12, Subunit 1.
 - 2) The **high-low method** is the less accurate but quicker of the two methods.
- c. The first step in applying the high-low method is to isolate the variable portion of the cost.
- 1) The difference in cost between the highest and lowest levels of activity for a group of periods is divided by the difference in the cost drivers (activity level) at the two levels (cost drivers are defined in item 2.c. in Subunit 7.1).

$$\text{Variable portion of mixed cost using high-low method} = \frac{\text{Cost at highest activity level} - \text{Cost at lowest activity level}}{\text{Driver at highest activity level} - \text{Driver at lowest activity level}}$$

- d. The fixed portion can now be calculated by inserting the appropriate values for either the high or low period in the range.
- e. The firm then uses the resulting information to project total cost at any level of activity.

EXAMPLE 7-4 Mixed Cost -- High-Low Method

A company has the following cost data:

Month	Machine Hours	Maintenance Costs
April	1,000	\$2,275
May	1,600	\$3,400
June	1,200	\$2,650
July	800	\$1,900
August	1,200	\$2,650
September	1,000	\$2,275
Variable portion = $\frac{\text{May cost} - \text{July cost}}{\text{May driver} - \text{July driver}} = \frac{\$3,400 - \$1,900}{1,600 - 800} = \frac{\$1,500}{800} = \$1.875 \text{ per machine hour}$		
Fixed portion = Total cost – Variable portion		
= \$1,900 – (800 machine hours × \$1.875 per hour)		
= \$1,900 – \$1,500		
= \$400		

An expenditure of 1,300 machine hours will generate a probable total cost of \$2,837.50 [\$400 + (1,300 × \$1.875)].

5. Linear vs. Nonlinear Cost Functions

- a. Four of the five costs described in this subunit are linear-cost functions; i.e., they change at a constant rate (or remain unchanged) over the short run.
- b. Fixed cost per unit, however, is an example of a nonlinear-cost function.
 - 1) Note that fixed cost per unit has an asymptotic character with respect to the x-axis, approaching it closely while never intersecting it (it does intersect the y-axis at the zero level of activity).
 - a) The function shows a high degree of variability over its range taken as a whole (item 3.c. on page 6).
 - 2) Another type of nonlinear-cost function is a step-cost function, one that is constant over small ranges of output but increases by steps (discrete amounts) as levels of activity increase.

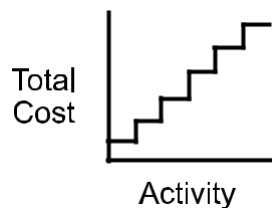


Figure 7-6

- a) Both fixed and variable costs can display step-cost characteristics.
 - i) If the steps are relatively narrow, these costs are usually treated as variable.
 - ii) If the steps are wide, they are more similar to fixed costs.
- 3) An example of a step cost would be the salary of production foremen. Operating at one shift per day might require one foreman, while two shifts would require two foremen.

6. Relevant Range and Marginal Cost

- a. Marginal cost is the cost incurred by a one-unit increase in the activity level of a particular cost driver.
 - 1) Necessarily then, marginal cost remains constant across the relevant range.
- b. Management accountants capture the concept of relevant range when they say that "All costs are variable in the long run."
 - 1) Investment in new, more productive equipment results in higher total fixed costs but may result in lower total and per-unit variable costs.

7.3 COST CLASSIFICATION

1. Controllable vs. Noncontrollable

- a. **Controllable costs** are those that are directly influenced at a given level of managerial authority within a given time period. Thus, they are costs that a particular manager has the authority to change.
 - 1) **Noncontrollable costs** are those that a particular manager does not have the authority to change.
- b. In other words, controllability is determined at different levels of the organization; it is not inherent in the nature of a given cost.
 - 1) For example, an outlay for new machinery may be controllable to the division vice president but noncontrollable to a plant manager or lower-level manager.

2. Avoidable vs. Committed

- a. **Avoidable costs** are those that may be eliminated by not engaging in an activity or by performing it more efficiently. An example is direct materials cost, which can be saved by ceasing production.
- b. **Committed costs** arise from holding property, plant, and equipment. Examples are insurance, real estate taxes, lease payments, and depreciation. They are by nature long-term and cannot be reduced by lowering the short-term level of production.

3. Incremental vs. Differential

- a. **Incremental cost** is the additional cost inherent in a given decision.
- b. **Differential cost** is the difference in total cost between two decisions.
- c. In practice, these two terms are often used interchangeably.

EXAMPLE 7-5 Incremental vs. Differential Cost

A company must choose between introducing two new product lines.

The incremental choice of the first option is the initial investment of \$1.5 million; the incremental choice of the second option is the initial investment of \$1.8 million.

The differential cost of the two choices is \$300,000.

4. Engineered vs. Discretionary

- a. **Engineered costs** are those having a direct, observable, quantifiable cause-and-effect relationship between the level of output and the quantity of resources consumed.
 - 1) Examples are direct materials and direct labor.
- b. **Discretionary costs** are those characterized by an uncertainty in the degree of causation between the level of output and the quantity of resources consumed. They tend to be the subject of a periodic (e.g., annual) outlay decision.
 - 1) Examples are advertising and R&D costs. Some routine maintenance costs might also fit this classification.

5. Outlay vs. Opportunity

- a. **Outlay costs** require actual cash disbursements. They are also called explicit, accounting, or out-of-pocket costs.
 - 1) An example is the tuition payment required to attend college.
- b. **Opportunity cost** is the maximum benefit forgone by using a scarce resource for a given purpose and not for the next-best alternative. It is also called implicit cost.
 - 1) An example is the wages foregone by attending college instead of working full-time.
- c. **Economic cost** is the sum of explicit and implicit costs.
- d. **Imputed costs** are those that should be involved in decision making even though no transaction has occurred that would be routinely recognized in the accounts. They are a type of opportunity cost.
 - 1) An example is the profit lost as a result of being unable to fill orders because the inventory level is too low.

6. Relevant vs. Sunk

- a. **Relevant costs** are those future costs that will vary depending on the action taken. All other costs are assumed to be constant and thus have no effect on (are irrelevant to) the decision.
 - 1) An example is tuition that must be spent to attend a fourth year of college.
- b. **Sunk costs** are costs either already paid or irrevocably committed to incur. Because they are unavoidable and will therefore not vary with the option chosen, they are not relevant to future decisions.
 - 1) An example is 3 years of tuition already spent. The previous 3 years of tuition make no difference in the decision to attend a fourth year.
- c. **Historical cost** is the actual (explicit) price paid for an asset. Financial accountants rely heavily on it for balance sheet reporting.
 - 1) Because historical cost is a sunk cost, however, management accountants often find other (implicit) costs to be more useful in decision making.

7. Normal vs. Abnormal Spoilage

- a. **Normal spoilage** is the spoilage that occurs under normal operating conditions. It is essentially uncontrollable in the short run.
 - 1) Since normal spoilage is expected under efficient operations, it is treated as a **product cost**; that is, it is absorbed into the cost of the good output.
- b. **Abnormal spoilage** is spoilage that is not expected to occur under normal, efficient operating conditions. Thus, it is generally thought to be more controllable than normal spoilage. The cost of abnormal spoilage should be separately identified and reported to management.
 - 1) Abnormal spoilage is typically treated as a **period cost** (a loss) because of its unusual nature.

8. Rework, Scrap, and Waste

- a. **Rework** consists of end products that do not meet standards of salability but can be brought to salable condition with additional effort.
 - 1) The decision to rework or discard is based on whether the marginal revenue to be gained from selling the reworked units exceeds the marginal cost of performing the rework.
- b. **Scrap** consists of raw material left over from the production cycle but still usable for purposes other than those for which it was originally intended.
 - 1) Scrap may be used for a different production process or may be sold to outside customers, usually for a nominal amount.
- c. **Waste** consists of raw material left over from the production cycle for which there is no further use.
 - 1) Waste is not salable at any price and must be discarded.

9. Other Costs

- a. **Carrying costs** are the costs of storing or holding inventory. Examples include the cost of capital, insurance, warehousing, breakage, and obsolescence.
- b. **Transferred-in costs** are those incurred in a preceding department and received in a subsequent department in a multi-departmental production setting.
- c. **Value-adding costs** are the costs of activities that cannot be eliminated without reducing the quality, responsiveness, or quantity of the output required by a customer or the organization.
- d. **Stockout costs** are the opportunity costs of missing a customer order. These can also include the costs of expediting a special shipment necessitated by insufficient inventory on hand.

10. Manufacturing Capacity

- a. **Normal capacity** is the long-term average level of activity that will approximate demand over a period that includes seasonal, cyclical, and trend variations.
 - 1) Deviations in a given year will be offset in subsequent years.
- b. **Practical capacity** is the maximum level at which output is produced efficiently. It allows for unavoidable delays in production for maintenance, holidays, etc.
- c. **Theoretical (ideal) capacity** is the maximum capacity assuming continuous operations with no holidays, downtime, etc.

7.4 COSTING TECHNIQUES

1. Actual vs. Normal Costing

- a. **Actual costing** is the recording of product costs based on actual (1) cost of materials, (2) cost of labor, and (3) overhead incurred. Actual costing is the most accurate method of accumulating costs in a cost accounting system due to it not including any budget-based amounts. However, it is also the least timely and most volatile method.
 - 1) After the end of the production period, all actual costs incurred for a cost object are totaled; indirect costs are allocated.
 - 2) Because per-unit costs depend on the level of production in a period, large fluctuations arise from period to period. This volatility can lead to the reporting of misleading financial information.
- b. **Normal costing** charges actual direct materials and direct labor to a specific product or a production department but applies overhead on the basis of a budgeted rate. This compensates for the fluctuations in unit cost inherent in actual costing.
 - 1) There is usually a difference between budgeted overhead and actual overhead. If the difference is immaterial, it should be allocated to cost of goods sold. If the difference is material, it should be prorated between cost of goods sold, work-in-process, and finished goods inventories.
- c. **Extended normal costing** extends the use of normalized rates to direct material and direct labor, so that all three major input categories use normalized rates.

2. Cost Accumulation Systems

- a. **Job-order costing** is appropriate when producing products with individual characteristics or when identifiable groupings are possible. Study Unit 8, Subunit 1, has a detailed explanation.
 - 1) Costs are attached to specific "jobs." Each job will result in a single, identifiable end product.
 - 2) Examples are any industry that generates custom-built products, such as shipbuilding or a sign shop.
- b. **Process costing** is used when similar products are mass produced on a continuous basis. Study Unit 8, Subunit 2, has a detailed explanation.
 - 1) Costs are attached to specific departments or phases of production. Examples are fence and candy manufacturing.
 - 2) Since costs are attached to streams of products rather than individual items, process costing involves calculating an average cost for all units. The two widely used methods are weighted-average and first-in, first-out (FIFO).
 - 3) Some units remain unfinished at the end of the period. For each department to adequately account for the costs attached to its unfinished units, the units, called work-in-process, must be restated in terms of equivalent units of production (EUP).

- c. **Activity-based costing (ABC)** first assigns resource costs to activities. These activity costs are then assigned to physical goods. Study Unit 8, Subunit 3, has a detailed explanation.
- 1) ABC is a response to the distortions of product cost information brought about by peanut-butter costing, which is the inaccurate averaging or spreading of costs like peanut butter over products or service units that use different amounts of resources.
 - a) A major cause of the problems associated with peanut-butter, or traditional, costing is the significant increase in indirect costs brought about by the increasing use of technology.
 - i) The fixed costs associated with technology are harder to correlate to specific units of product than are the variable costs associated with direct labor.
 - 2) The difference between traditional (that is, volume-based) costing systems and ABC can be summarized as follows:
 - a) Under volume-based systems, a single pool collects all indirect costs and the total cost in the pool is then allocated to production.
 - b) Under ABC, by contrast, every activity that bears on the production process has its own cost pool. The costs in each pool are assigned based on a cost driver specific to the activity.
 - i) For example, with a single pool, all costs may be assigned based on a single driver, such as machine hours or direct-labor hours.
 - ii) Under ABC, a dozen different cost pools might be assigned on the basis of a dozen different cost drivers. The increasing use of computers in the accounting process has made it easier to implement ABC.
- d. **Life-cycle costing** emphasizes the need to price products to cover all the costs incurred over the lifespan of a product, not just the immediate costs of production. Study Unit 8, Subunit 4, has a detailed explanation.
- 1) Costs incurred before production, such as R&D and product design, are referred to as **upstream costs**.
 - 2) Costs incurred after production, such as marketing and customer service, are called **downstream costs**.

3. Standard Costing, Flexible Budgeting, and Variance Analysis

- a. **Standard costing** is a system designed to alert management when the actual costs of production differ significantly from budgeted (“standard”) costs.
 - 1) Standard costs are predetermined, attainable unit costs. A standard cost is not just an average of past costs, but an objectively determined estimate of what a cost should be. The cost estimate excludes past inefficiencies and takes into account expected future changes.
 - 2) Utilizing standard costs tends to simplify recordkeeping.
 - 3) The costs of using standard costing are low relative to using actual costing.
 - 4) Standard costs can be used with both job-order and process-costing systems.
 - 5) There is usually a difference between budgeted overhead and actual overhead. If the difference is immaterial, it should be allocated to cost of goods sold. If the difference is material, it should be prorated between cost of goods sold, work-in-process, and finished goods inventories.
- b. **Flexible budgeting** is the calculation of the quantity and cost of inputs that should have been consumed given the achieved level of production.
 - 1) Flexible budgeting supplements the **static budget**, which is the company’s best projection of the resource consumption and levels of output that will be achieved for an upcoming period.
- c. The static and flexible budgets are compared to the actual results and the differences are calculated. These differences are referred to as variances. Variances are covered in detail in Study Unit 15.
 - 1) Variance analysis enables **management by exception**, the practice of giving attention primarily to significant deviations from expectations (whether favorable or unfavorable).

7.5 ABSORPTION AND VARIABLE COSTING

1. Two Ways of Treating Fixed Production Costs: Period vs. Product Costs

- a. Recall that product costs are capitalized costs because they become a component of cost of goods sold.
- b. **Period costs** are expensed as incurred, i.e., they are not capitalized in finished goods inventory and are thus excluded from cost of goods sold.
- c. For **internal purposes**, decision making is improved by treating fixed overhead as a period cost so that only costs that are variable in the short run are included in the cost of the product.
 - 1) Fixed overhead costs are considered period costs and deducted in the period in which they are incurred.
 - 2) This practice is termed **variable, or direct, costing**. Variable costing is the preferred term because it concisely describes what is happening—namely that product costs are based only on variable costs.
- d. For **external reporting purposes**, the cost of a product must include **all** the costs of manufacturing it: direct labor, direct materials, and all factory overhead (both fixed and variable).
 - 1) This method is commonly known as **absorption, full costing, or full absorption costing**.

2. Absorption Costing

- a. Under absorption costing, the fixed portion of manufacturing overhead is “absorbed” into the cost of each unit of product.
 - 1) Product cost thus includes **all manufacturing costs**, both fixed and variable.
 - 2) Absorption-basis cost of goods sold is subtracted from sales to arrive at gross margin.
 - 3) Total selling and administrative (S&A) expenses (i.e., both fixed and variable) are then subtracted from gross margin to arrive at operating income.
 - 4) The inventoried cost of the product thus includes all production costs, whether variable or fixed. This technique is required for external financial reporting and for income tax purposes.
 - 5) **Gross margin** (also called gross profit) is the net difference between sales revenue and absorption cost of goods sold. It represents the amount available to cover S&A expenses.
- b. This method is required for external reporting purposes and for tax purposes.

3. Variable costing

- a. **Variable costs** are a direct function of production volume. They increase when production grows and decrease when production shrinks. Raw materials and labor directly involved with production are common variable costs.
- b. This method is more appropriate for internal reporting. It is sometimes called **direct costing**, although variable costing is the more accurate term (the phrase “direct costing” is considered misleading because it implies traceability).
 - 1) Product cost includes only the variable portion of manufacturing costs.
 - 2) Fixed manufacturing costs are considered period costs and are thus expensed as incurred. This technique is not allowed for external financial reporting but is very useful for internal decision making. It also stops management from manipulating income by overproducing during the period.
 - 3) Variable-basis cost of goods sold and the variable portion of S&A expenses are subtracted from sales to arrive at **contribution margin**.
 - a) This amount (Sales – Total variable costs) is an important element of the variable costing income statement because it is the amount available for covering fixed costs (both manufacturing and S&A).
 - b) Contribution margin is an important metric internally but is generally considered irrelevant to external financial statement users.
 - c) Manufacturing contribution margin considers only the actual costs of manufacturing (i.e., direct materials, direct labor, and variable manufacturing overhead) to be product costs, i.e., inventoriable.

EXAMPLE 7-6 Contribution Margin Income Statement

Company X sells 20,000 units in the current year at a sales price of \$60 per unit. X has the following costs:

Variable costs per unit		Total fixed costs	
Direct materials	\$8	Overhead	\$200,000
Direct labor	5	Selling and administrative	150,000
Overhead	2		
Selling	1		

Company X's contribution margin formatted income statement is prepared as follows:

Sales (\$60 × 20,000)		\$1,200,000
Less: Variable costs		
Direct materials (\$8 × 20,000)	\$160,000	
Direct labor (\$5 × 20,000)	100,000	
Overhead (\$2 × 20,000)	40,000	
Selling (\$1 × 20,000)	20,000	(320,000)
Contribution margin		\$ 880,000
Less: Fixed costs		
Overhead	\$200,000	
Selling and administrative	150,000	(350,000)
Operating income		<u>\$ 530,000</u>

4. Variable vs. Absorption Costing

- The accounting for variable production costs and fixed S&A expenses is identical under the two methods.
- The difference lies in the varying treatment of fixed production costs and presentation of variable S&A expenses.
- Absorption and variable costing income statements can be illustrated as follows:

Absorption Costing	Variable Costing
Sales	Sales
– Cost of goods sold:	– Variable expenses:
Direct materials	Direct materials
Direct labor	Direct labor
Variable overhead	Variable overhead
Fixed overhead	Variable S&A expenses
= <u>Gross margin</u>	= <u>Contribution margin</u>
– <u>Total S&A expenses</u>	– <u>Fixed expenses:</u>
= <u><u>Operating income</u></u>	Fixed overhead
	Fixed S&A expenses
	= <u><u>Operating income</u></u>

- Note that ending finished goods inventory will differ between the two methods due to the different treatment of fixed production costs.

- This leads to a difference in cost of goods sold and operating income.

EXAMPLE 7-7

Absorption vs. Variable Costing

		Absorption Costing (Required for ext. rptg.)	Variable Costing (For internal reporting only)
Sales		\$100,000	\$100,000
Beg. finished goods inventory		\$10,000	\$10,000
Product	Add: Variable production costs	20,000 (a)	20,000 (a)
Costs	Add: Fixed production costs	30,000 (b)	-
Goods available for sale		\$60,000	\$30,000
Minus: End. finished goods inventory		(35,000)	(25,000)
Cost of goods sold		\$(25,000)	\$(5,000)
Period Costs	Minus: Variable S&A expenses	-	(10,000) (c)
	Gross margin (abs.) / Contribution margin (var.)	\$75,000	\$85,000
	Minus: Fixed production costs	-	(30,000) (b)
	Minus: Variable S&A expenses	(10,000) (c)	-
	Minus: Fixed S&A expenses	(20,000) (d)	(20,000) (d)
Operating income		<u>\$45,000</u>	<u>\$35,000</u>

Legend

Cost Component

(a) Variable production costs

(b) Fixed production costs

(c) Variable selling and administrative expenses

(d) Fixed selling and administrative expenses

The \$10,000 difference in operating income (\$45,000 – \$35,000) is the difference between the ending inventory values (\$35,000 – \$25,000). In essence, the absorption method carries 33.33% of the fixed overhead costs (\$30,000 × 33.33% = \$10,000) on the balance sheet as an asset because 33.33% of the month's production is still in inventory.

e. The following table summarizes product and period costs under both methods:

	Absorption Costing (Required under GAAP)	Variable Costing (For Internal Reporting Only)
Product Costs (Included in Cost of Goods Sold)	Variable production costs	
	Fixed production costs	
Period Costs (Excluded from Cost of Goods Sold)		Fixed production costs
	Variable S&A expenses	
	Fixed S&A expenses	

5. Justification for Variable Costing

- a. Under variable costing, fixed manufacturing overhead cost is considered a cost of maintaining capacity, not a cost of production.
 - 1) To illustrate, a company has a fixed rental expense of \$10,000 per month on its factory building. That cost will be \$10,000 regardless of whether there is any production.
 - a) If the company produces zero units, the cost will be \$10,000; if the company produces 10,000 units, the cost will be \$10,000.
 - 2) Therefore, the \$10,000 is not viewed as a cost of production and is not added to the cost of the inventories produced. The \$10,000 is a cost of maintaining a certain level of production capacity.
- b. To emphasize, variable costing is used only for internal decision-making purposes; it is not permitted for external financial reporting or for tax calculation.
 - 1) The main advantage of the variable costing method is that income cannot be manipulated by management action, but management can manipulate income when using the absorption method. Additional advantages of variable costing are discussed in item 8., on page 22.

EXAMPLE 7-8 Absorption Basis vs. Variable Basis

During its first month in business, a firm produced 100 units and sold 80 while incurring the following costs:

Direct materials	\$1,000
Direct labor	2,000
Variable overhead	<u>1,500</u>
Manufacturing costs used in variable costing	<u>\$4,500</u>
Fixed overhead	<u>3,000</u>
Manufacturing costs used in absorption costing	<u><u>\$7,500</u></u>

The effects on the financial statements of using one method rather than the other can be seen in these calculations:

	Absorption Basis	Variable Basis
Manufacturing costs	\$7,500	\$4,500
Divided by: Units produced	÷ 100	÷ 100
Per-unit cost	\$ 75	\$ 45
Times: Ending inventory	× 20	× 20
Value of ending inventory	<u><u>\$1,500</u></u>	<u><u>\$ 900</u></u>

-- Continued on next page --

EXAMPLE 7-8 -- Continued

The \$600 difference (\$1,500 – \$900) in inventory value represents the fixed overhead that is charged to the absorption basis inventory. Because 20% of the units produced are still in inventory, 20% of the \$3,000 of fixed costs, or \$600, is still in inventory. Under the variable basis, all fixed costs are expensed.

The per-unit selling price of the finished goods was \$100, and the company incurred \$200 of variable selling and administrative expenses and \$600 of fixed selling and administrative expenses.

The following are partial income statements prepared using the two methods:

		Absorption Costing (Required under GAAP)	Variable Costing (For Internal Reporting Only)
		\$ 8,000	\$ 8,000
Sales			
Beginning finished goods inventory		\$ 0	\$ 0
Product	Plus: Variable production costs	4,500 (a)	4,500 (a)
Costs	Plus: Fixed production costs	3,000 (b)	
Goods available for sale		\$7,500	\$4,500
Less: Ending finished goods inventory		<u>(1,500)</u>	<u>(900)</u>
Cost of goods sold		<u>\$(6,000)</u>	<u>\$(3,600)</u>
Less: Variable S&A expenses			(200) (c)
Gross margin (abs.) / Contribution margin (var.)		<u>\$ 2,000</u>	<u>\$ 4,200</u>
Period	Less: Fixed production costs		(3,000) (b)
Costs	Less: Variable S&A expenses	(200) (c)	
Less: Fixed S&A expenses		(600) (d)	(600) (d)
Operating income		<u>\$ 1,200</u>	<u>\$ 600</u>

The \$600 difference in operating income (\$1,200 – \$600) is the difference between the two ending inventory values (\$1,500 – \$900). In essence, the absorption method carries 20% of the fixed overhead costs (\$3,000 × 20% = \$600) on the balance sheet as an asset because 20% of the month's production (100 available – 80 sold = 20 on hand) is still in inventory.

This calculation is for illustrative purposes only. The difference in operating income is exactly the difference in ending inventory only when beginning inventory is \$0.

6. Effects on Operating Income

- a. As production and sales levels change, the two methods have varying effects on operating income.
 - 1) When everything produced during a period is sold that period, the two methods report the same operating income.
 - a) Total fixed costs budgeted for the period are charged to sales revenue in the period under both methods.
 - 2) When production and sales are not equal for a period, the two methods report different operating income.
- b. When production exceeds sales, ending inventory expands.
 - 1) Under absorption costing, some fixed costs are embedded in ending inventory whereas, under variable costing, all costs have been expensed.
 - 2) **Therefore, when production exceeds sales, operating income is higher under absorption costing than it would be under variable costing.**
- c. When production is less than sales, ending inventory contracts.
 - 1) Under absorption costing, fixed costs embedded in beginning inventory get expensed whereas, under variable costing, only the current period's fixed costs are expenses.
 - 2) **Therefore, when production is less than sales, operating income is higher under variable costing than it would be under absorption costing.**

- d. Many companies prefer variable costing for internal reporting because of the perverse incentive inherent in absorption costing.
- 1) Whenever production exceeds sales, fewer fixed costs are expensed under the absorption basis, and operating income always increases.
 - 2) A production manager can thus **increase absorption-basis operating income merely by increasing production**, whether there is any customer demand for the additional product or not.
 - a) The company must also deal with the increased carrying costs resulting from swelling inventory levels.
 - 3) This practice, called producing for inventory, can be effectively discouraged by using variable costing for performance reporting and consequent bonus calculation.

EXAMPLE 7-9 Extended Example of Absorption and Variable Operating Income

A company has the following sales and cost data:

	Year 1	Year 2	Year 3
Production in units	40,000	50,000	0
Sales in units	30,000	30,000	30,000
Ending inventory in units (Wgt. Avg.)	10,000	30,000	0
Unit sales price	\$ 1.00		
Unit variable cost	0.50		
Fixed manufacturing costs	4,000	per year	
Variable S&A expenses	0.03	per unit	
Fixed S&A expenses	1,000	per year	

Compare the 3-year income statements prepared under the two methods:

Absorption Costing (Required for external reporting)				Variable Costing (For internal reporting only)			
	Year 1	Year 2	Year 3		Year 1	Year 2	Year 3
Sales	\$30,000	\$30,000	\$30,000	Sales	\$30,000	\$30,000	\$30,000
Beginning inventory	\$ 0	\$ 6,000	\$17,500	Beginning inventory	\$ 0	\$ 5,000	\$15,000
Variable mfg. costs	20,000	25,000	0	Variable mfg. costs	20,000	25,000	0
Fixed mfg. costs	4,000	4,000	4,000				
Goods available for sale	\$24,000	\$35,000	\$21,500	Goods available for sale	\$20,000	\$30,000	\$15,000
Less: Ending inventory*	(6,000)	(17,500)	0	Less: Ending inventory	(5,000)	(15,000)	0
Absorption COGS	\$18,000	\$17,500	\$21,500	Variable COGS	\$15,000	\$15,000	\$15,000
				Variable S&A exps.	(900)	(900)	(900)
Gross margin	\$12,000	\$12,500	\$ 8,500	Contribution margin	\$14,100	\$14,100	\$14,100
				Fixed mfg. costs	(4,000)	(4,000)	(4,000)
Variable S&A expenses	(900)	(900)	(900)				
Fixed S&A expenses	(1,000)	(1,000)	(1,000)	Fixed S&A expenses	(1,000)	(1,000)	(1,000)
Operating income	\$10,100	\$10,600	\$ 6,600	Operating income	\$ 9,100	\$ 9,100	\$ 9,100

* Ending inventory is calculated on the weighted-average basis. The use of FIFO would result in slightly different numbers in Year 2 under the absorption method, but the impact would be the same.

-- Continued on next page --

EXAMPLE 7-9 -- Continued

Note that, assuming zero inventory at the beginning of Year 1 and at the end of Year 3, the total operating income for the 3-year period is the same under either costing method.

	Absorption Costing	Variable Costing
Year 1	\$10,100	\$ 9,100
Year 2	10,600	9,100
Year 3	6,600	9,100
3-Year Total	<u>\$27,300</u>	<u>\$ 27,300</u>

Absorption costing shows a higher operating income than variable costing in Years 1 and 2 because fixed overhead has been capitalized and does not get expensed until Year 3. Variable costing, on the other hand, treats fixed overhead as an expense of the period in which the cost is incurred. In Year 2, despite the same cash flow, there is a \$1,500 difference between the final operating income figures. There is an even greater difference in Year 3.

If fixed costs increase relative to variable costs, the differences become more dramatic (here, 50% of the selling price is variable manufacturing cost, and fixed overhead is no more than 20% of the variable manufacturing cost).

From an internal point of view, a manager can manipulate absorption income by changing production levels. But, with variable costing, a manager cannot manipulate it simply by changing production levels.

Note that, under the absorption method, management was able to show higher incomes in Years 1 and 2 by overproducing. If the manager was being given a bonus for a higher level of income, (s)he could obtain the bonus by producing more units than could be sold. As a result, some fixed costs would be added to the balance sheet as inventories. Thus, the income statement and balance sheet both look good, despite the fact that the production manager has done a bad thing: (S)he has produced excessive inventories that require the company to incur storage and financing costs. Spoilage may also be a result.

7. Summary of Effects on Income and Ending Inventory

- The value of ending inventory is **never** higher under variable costing than it is under absorption costing because fixed manufacturing costs are not included in inventory under variable costing.
- Income and inventory levels will differ whenever sales and production differ.
 - Income will be higher or lower under variable costing depending upon whether inventories are increased during the period or liquidated.
 - If inventories increase during a period, the variable costing method will show a lower income because all fixed costs are being subtracted on the income statement, while under the absorption method, some fixed costs are being capitalized as inventories.
 - Variable costing will show a higher income in periods when inventories decline because the absorption method forces the subtraction of current period fixed costs included in inventory sold, plus some fixed costs incurred (and capitalized) in prior periods.
- Under variable costing, profits always move in the same direction as sales volume. Profits reported under absorption costing behave erratically and sometimes move in the opposite direction from sales trends.
- In the long run, the two methods will report the same total profits if sales equal production. The inequalities between production and sales are usually minor over an extended period.

8. Benefits of Variable Costing

- a. Although the use of variable costing for financial statements is prohibited, most agree about its superiority for internal reporting. It is far better suited than absorption costing to the needs of management.
 - 1) Management requires a knowledge of cost behavior under various operating conditions. For planning and control, management is more concerned with treating fixed and variable costs separately than with calculating full costs.
 - 2) Full costs are usually of dubious value because they contain arbitrary allocations of fixed cost.
- b. Under the variable costing method, a production manager can manipulate income levels by deferring costs but not by overproducing. Given the same cost structure every year, the income levels will be based on sales, not the level of production.
- c. Under variable costing, the cost data for profit planning and decision making are readily available from accounting records and statements. Reference to auxiliary records and supplementary analyses is not necessary.
 - 1) For example, cost-volume-profit relationships and the effects of changes in sales volume on net income can easily be computed from the income statement prepared under the variable costing concept, but not from the conventional absorption cost income statement based on the same data.
- d. Profits and losses reported under variable costing have a relationship to sales revenue and are not affected by inventory or production variations.
- e. Absorption cost income statements may show decreases in profits when sales are rising and increases in profits when sales are decreasing, which may be confusing to management. Attempts at explanation by means of volume variances often compound rather than clarify the confusion.
- f. When variable costing is used, the favorable margin between selling prices and variable cost should provide a constant reminder of profits forgone because of lack of sales volume. A favorable margin justifies a higher production level.
- g. The full impact of fixed costs on net income, partially hidden in inventory values under absorption costing, is emphasized by the presentation of costs on an income statement prepared under variable costing.
- h. Proponents of variable costing maintain that fixed factory overhead is more closely correlated to capacity to produce than to the production of individual units.

9. Further Aspects of Variable Costing

- a. Variable costing is also preferred over absorption costing for studies of relative profitability of products, territories, and other segments of a business. It concentrates on the contribution that each segment makes to the recovery of fixed costs that will not be altered by decisions to make and sell. Under variable costing procedures,
 - 1) The marginal income concept leads to better pricing decisions, which are a principal advantage of variable costing.
 - 2) The impact of fixed costs on net income is emphasized by showing the total amount of such costs separately in financial reports.
 - 3) Out-of-pocket expenditures required to manufacture products conform closely with the valuation of inventory.
 - 4) The relationship between profit and the major factors of selling price, sales mix, sales volume, and variable manufacturing and nonmanufacturing costs is measured in terms of a single index of profitability.
 - a) This profitability index, expressed as a positive amount or as a ratio, facilitates the analysis of cost-volume-profit relationships, compares the effects of two or more contemplated courses of action, and aids in answering many questions that arise in profit planning.
 - 5) Inventory changes have no effect on the breakeven computations.
 - 6) Marginal income amounts facilitate appraisal of products, territories, and other business segments without having the results hidden or obscured by allocated joint fixed costs.
 - 7) Questions regarding whether a particular part should be made or bought can be more effectively answered if only variable costs are used.
 - a) Management must consider whether to charge the product being made with variable costs only or to charge a percentage of fixed costs as well.
 - b) Management must also consider whether the making of the part will require additional fixed costs and a decrease in normal production.
 - 8) Disinvestment decisions are facilitated because whether a product or department is recouping its variable costs can be determined.
 - a) If the variable costs are being covered, operating a department at an apparent loss may be profitable.
 - 9) Management is better able to judge the differences between departments if certain fixed costs are omitted from the statements instead of being allocated arbitrarily.
 - 10) Costs are guided by sales.
 - a) Under variable costing, cost of goods sold will vary directly with sales volume, and the influence of production on gross profit is avoided.
 - b) Variable costing also eliminates the possible difficulties of having to explain over- or underapplied factory overhead to higher management.

STUDY UNIT EIGHT

COST ACCUMULATION SYSTEMS

8.1	<i>Job-Order Costing</i>	2
8.2	<i>Process Costing</i>	6
8.3	<i>Activity-Based Costing</i>	15
8.4	<i>Life-Cycle Costing</i>	24

This study unit is the **second of five** on **cost management**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The five study units are

- Study Unit 7: Cost Management Concepts
- Study Unit 8: Cost Accumulation Systems
- Study Unit 9: Cost Allocation Techniques
- Study Unit 10: Supply Chain Management
- Study Unit 11: Business Process Improvement

The strategic value of cost information includes financial information about costs and revenues as well as relevant nonfinancial information about productivity and market share. Competitive organizations are concerned with effectively managing costs and pricing, product and manufacturing improvements, and product quality. The management accountant needs to make solid strategic decisions, which include but are not limited to,

- 1) Choice of products and manufacturing methods,
- 2) Replacing equipment,
- 3) Managing cash flow,
- 4) Inefficient operations,
- 5) Budgeting raw materials and other production costs,
- 6) Pricing, and
- 7) Accurate accounting for inventory and other financial statement items.

If you are interested in reviewing more introductory or background material, go to www.GLEM.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following GLEM outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

8.1 JOB-ORDER COSTING

1. Use of Job-Order Costing

- a. Job-order costing is concerned with accumulating costs by specific job.
 - 1) This method is appropriate when producing products with individual characteristics (e.g., yachts), or when identifiable groupings are possible (e.g., jewelry).
 - 2) Units (jobs) should be dissimilar enough to warrant the special recordkeeping required by job-order costing.
 - 3) Products are usually custom made for a specific customer.

2. Steps in Job-Order Costing

- a. The first step in the process is the receipt of a sales order from a customer requesting a product or special group of products.
- b. The sales order is approved, and a production order is issued.
- c. Costs are recorded by classification, such as direct materials, direct labor, and manufacturing overhead, on a job-cost sheet (may be manual or electronic) that is specifically prepared for each job.
 - 1) The physical inputs required for the production process are obtained from suppliers. The journal entry to record the acquisition of inventory would be

Raw materials	\$XXX	
Accounts payable		\$XXX

- 2) Under job-order costing, direct materials and direct labor are charged based on the amounts actually applied to each job.

- a) Materials requisition forms request direct materials to be pulled from the warehouse and sent to the production line.

Work-in-process – Job 1015	\$XXX	
Raw materials		\$XXX

- b) Time tickets track the direct labor that workers expend on various jobs.

Work-in-process – Job 1015	\$XXX	
Wages payable		\$XXX

- 3) Under job-order costing, the third component, manufacturing overhead, is charged using an estimated rate.
 - a) The application of an estimated overhead rate is necessary under job-order costing because the outputs are customized and the processes vary from period to period.
 - b) As indirect costs are paid throughout the year, they are collected in the manufacturing overhead control account.
 - i) Note that work-in-process is not affected when actual overhead costs are incurred.
 - ii) The debits are made to a manufacturing overhead control account, not work-in-process.

Manufacturing overhead control	\$XXX	
Property taxes payable		\$XXX
Manufacturing overhead control	\$XXX	
Prepaid insurance		\$XXX
Manufacturing overhead control	\$XXX	
Accumulated depreciation – factory equipment		\$XXX

- c) Overhead costs are applied to (“absorbed” by) each job based on a predetermined overhead application rate for the year (such as \$5 per direct labor hour, or machine hour, etc., or based on an activity-based costing system). If activity-based costing is used, the procedure for overhead applications is the same as above, except that multiple rates based on multiple cost drivers will be used (such as \$5 per direct labor hour, plus \$2 per machine hour, plus \$1 per material requisition used).
- At the beginning of the year, an estimate is made of the total amount that will be spent for manufacturing overhead during that year.
 - An estimate is also made of the total quantity of allocation base, such as direct labor hours or machine hours, that will be required for manufacturing overhead during that year.

Estimated total manufacturing overhead

Application rate = Estimated total quantity of allocation base

- The application rate is calculated by dividing the estimate of the total amount that will be spent for manufacturing overhead during that year by the estimated total amount of allocation base.
- The amount applied equals the number of units of the allocation base used during the period times the application rate.

Amount overhead applied = Total quantity of allocation base × Application rate

The credit is to manufacturing overhead applied, a contra-account for manufacturing overhead control.

Work-in-process – Job 1015	\$XXX	
Manufacturing overhead applied		\$XXX

- By tracking the amounts applied to the various jobs in a separate account, the actual amounts spent on overhead are preserved in the balance of the overhead control account.
 - In addition, the firm can determine at any time how precise its estimate of overhead costs for the period was by comparing the balances in the two accounts. The closer they are (in absolute value terms), the better the estimate was.
- d) At the end of the period, the overhead control and applied accounts are netted to examine any differences, which are called variances.
- If the result is a credit, overhead was overapplied for the period. If the result is a debit, overhead was underapplied.
 - If the difference is immaterial, it can be closed directly to cost of goods sold. If the difference is material, it should be allocated based on the relative values of work-in-process, finished goods, and cost of goods sold.

- 4) The amounts from the input documents are accumulated on job-cost sheets. These serve as a subsidiary ledger page for each job.
- The total of all job-cost sheets for jobs in progress will equal the balance in the general ledger work-in-process inventory account.
 - Once the job is completed, but before it is delivered to the customer, the job-cost sheet serves as the subsidiary ledger for the finished goods inventory account.
- d. When a job order is completed, all the costs are transferred to finished goods.
- | | | |
|----------------------------|---------|---------|
| Finished goods | \$X,XXX | |
| Work-in-process – Job 1015 | | \$X,XXX |
- e. When the output is sold, the appropriate portion of the cost is transferred to cost of goods sold.
- | | | |
|--------------------|---------|---------|
| Cost of goods sold | \$X,XXX | |
| Finished goods | | \$X,XXX |

3. Job-Order Cost Flow Diagram

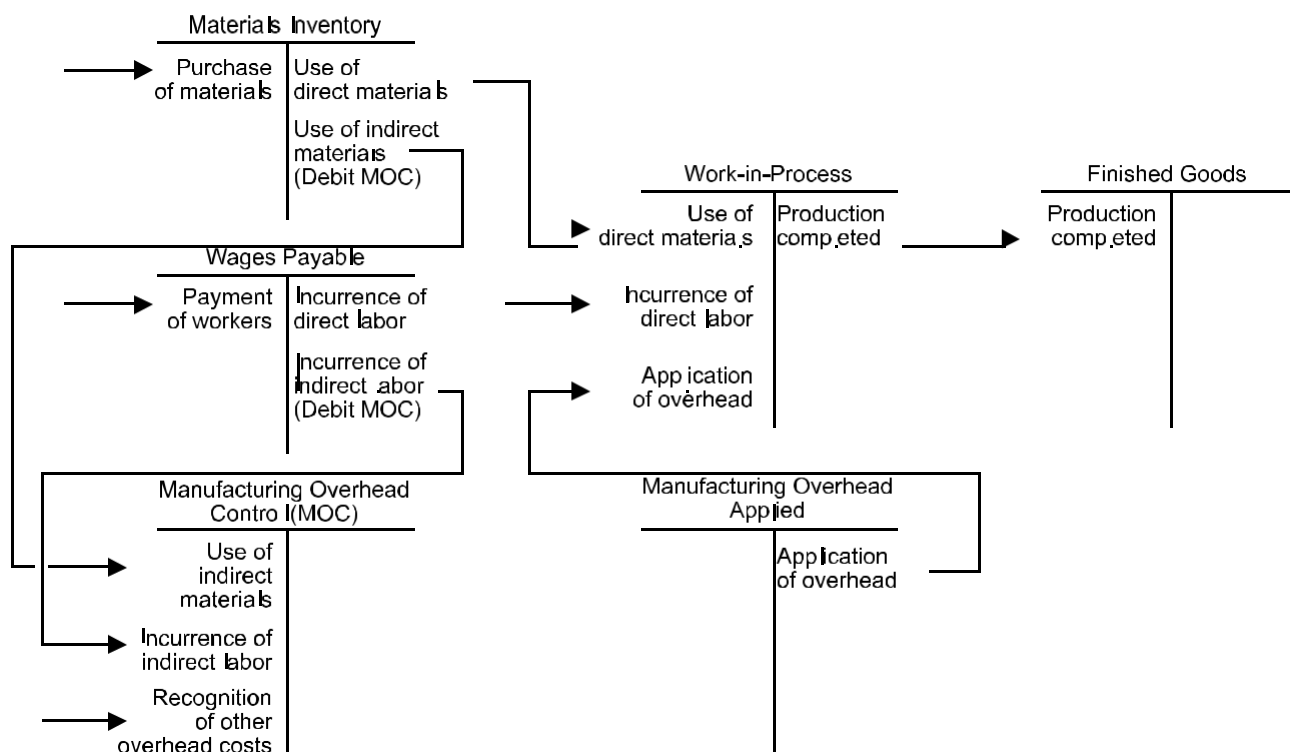


Figure 8-1



SUCCESS TIP

CMA candidates will be expected to understand the proper accounting procedure for normal and abnormal spoilage under both job-order costing and process costing. In job-order costing, normal spoilage is treated as a product cost while abnormal spoilage is treated as a period cost. It is important to understand not only that they are treated differently, but why. When answering questions pertaining to spoilage, pay attention to the question stem, what system is being used, and whether the product can be sold or not.

4. Spoilage, Rework, and Scrap

a. Output that does not meet the quality standards for salability is considered spoilage.

1) **Normal spoilage** is the amount expected in the ordinary course of production.

- a) Because management understands that good units cannot be produced without some risk of spoilage, the cost of the normal spoilage is included in the cost of the good units produced (i.e., treated as a product cost).
- b) This is accomplished by allowing the net cost of the spoilage to remain in the work-in-process account of the job that generated it.
 - i) If the normal spoilage is worthless, it should be discarded. No journal entry will be made.
 - ii) If the normal spoilage can be sold, its value should **not** be included in the cost of the good units produced. The entry is

Spoiled inventory (at fair market value)	\$XXX	
Work-in-process – Job 1015		\$XXX

2) **Abnormal spoilage** is spoilage over and above the amount expected in the ordinary course of production.

- a) Abnormal spoilage is **not** treated as a manufacturing cost and is **not** included in the cost of the good units produced (i.e., treated as a period cost).
- b) Instead, abnormal spoilage costs are expensed in the period they occurred or the loss may be accrued. This is accomplished by charging a loss account for the net cost of the spoilage.
 - i) If the abnormal spoilage is worthless, it should be discarded. The entry is

Loss from abnormal spoilage (costs up to point of inspection)	\$XXX	
Work-in-process – Job 1015		\$XXX

ii) If the abnormal spoilage can be sold, it should be sold. The entry is

Spoiled inventory	\$XXX	
Loss from abnormal spoilage (difference)	XXX	
Work-in-process – Job 1015 (costs up to point of inspection)		\$XXX

b. **Rework** consists of unacceptable units that can be repaired and sold as acceptable units.

1) Normal rework may be attributable to a job or to all jobs. It is debited to

- a) A job (job costing) or
- b) Manufacturing overhead and allocated (job and process costing).

2) Abnormal rework is debited to a loss (job and process costing).

c. **Scrap** is leftover material with no attached cost and low sales value. It may be recognized as revenue (debit cash, credit revenue) at the time of sale.

1) Material scrap revenue attributable to (a) a job or (b) all jobs (or a process) is credited to work-in-process or manufacturing overhead, respectively.

8.2 PROCESS COSTING

1. Use of Process Costing

- a. Process cost accounting is used to assign costs to inventoriable goods or services. It is applicable to relatively homogeneous products that are mass produced on a continuous basis (e.g., petroleum products, thread, computer monitors).
 - 1) Assigning an exact amount of materials, labor, and indirect costs to thousands, or even millions, of individual end products is simply not cost-effective. For this reason, process costing involves averaging the costs of production and allocating them to work-in-process and finished goods.

2. Accumulation of Costs

- a. The accumulation of costs under a process costing system is **by department rather than by project**. There will normally be a work-in-process inventory account for each department.
 - 1) This reflects the continuous, homogeneous nature of the manufacturing process.
- b. As in job-order costing, the physical inputs required for the production process are obtained from suppliers.

Raw materials	\$XXX
Accounts payable	\$XXX

- c. **Direct materials** actually used by the first department in the process (Department A) are added to work-in-process for that department.

Work-in-process – Department A	\$XXX
Raw materials	\$XXX

- d. **Conversion costs**, which include direct labor and manufacturing overhead used by the first department, are added to work-in-process for that department.
 - 1) Actual amounts are used.
 - 2) Standard costs are applied at a later stage for purposes of variance analysis.

Work-in-process – Department A	\$XXX
Wages payable (direct and indirect labor)	\$XXX
Manufacturing supplies (indirect materials)	XXX
Property taxes payable	XXX
Prepaid insurance	XXX
Accumulated depreciation – factory equipment	XXX

- e. The products can move from one department to the next (from Department A to Department B).

Work-in-process – Department B	\$XXX	
Work-in-process – Department A		\$XXX

- f. The second department (Department B) can add more direct materials and conversion costs.

Work-in-process – Department B	\$XXX	
Raw materials		\$XXX

Work-in-process – Department B	\$XXX	
Wages payable (direct and indirect labor)		\$XXX
Manufacturing supplies (indirect materials)		XXX
Property taxes payable		XXX
Prepaid insurance		XXX
Accumulated depreciation – factory equipment		XXX

- g. If a standard costing system is used, standard costs are applied before transfer from work-in-process to finished goods.

- 1) The differences are accumulated in a direct materials variance and a conversion costs variance account.

- h. When processing is finished in the last department, all the costs are transferred to finished goods.

Finished goods	\$XXX	
Work-in-process – Department B		\$XXX

- i. As products are sold, the costs are transferred to cost of goods sold.

Cost of goods sold	\$XXX	
Finished goods		\$XXX

3. Process Costing Cost Flow Diagram

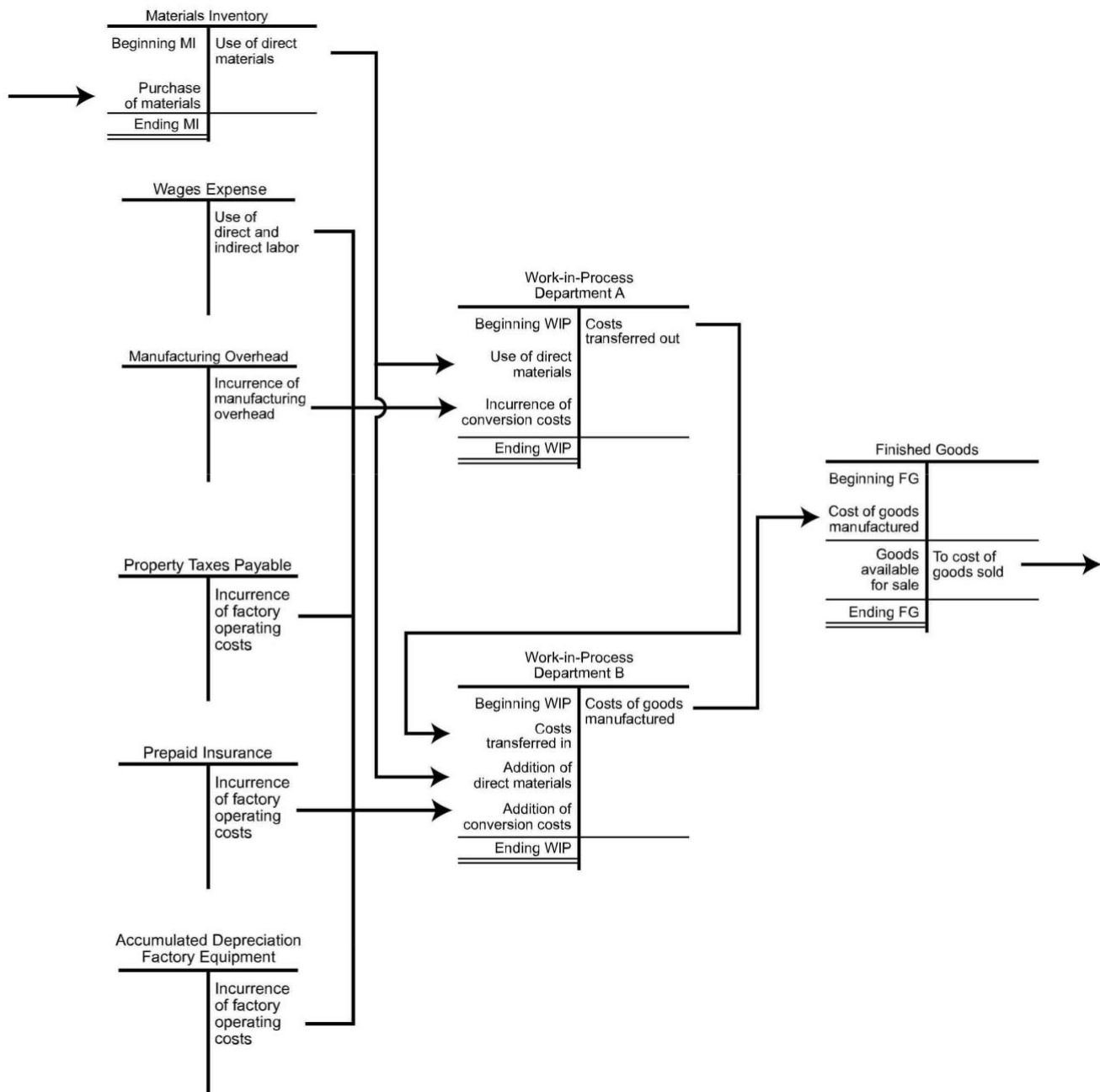


Figure 8-2

4. Equivalent Units of Production – Basics

a. Some units remain unfinished at the end of the period. For each department to account adequately for costs attached to its unfinished units, the units must be restated in terms of equivalent units of production (EUP).

- 1) EUP are the number of complete units that could have been produced using the inputs consumed during the period.

EXAMPLE 8-1 EUP

One thousand work-in-process units 80% completed for direct materials and 60% for conversion costs represent 800 EUP of direct materials ($1,000 \times 80\%$) and 600 EUP of conversion costs ($1,000 \times 60\%$).

- 2) Cost-per-unit can be calculated using EUP (cost per EUP).
- b. In all EUP calculations, three populations of units must be accounted for:
 - 1) Units in beginning work-in-process (beginning WIP)
 - a) Units in beginning WIP can be calculated as follows:

$$\begin{array}{r}
 \text{Units transferred out} \\
 + \text{Ending WIP} \\
 - \text{Units started} \\
 \hline
 \text{Beginning WIP} \\
 \hline
 \end{array}$$

- 2) Units started and completed during the current period
 - a) Units started and completed can be calculated as follows:

$$\begin{array}{r}
 \text{Units transferred out} \\
 - \text{Beginning WIP} \\
 \hline
 \text{Units started and completed} \\
 \hline
 \end{array}
 \quad \text{OR} \quad
 \begin{array}{r}
 \text{Units started} \\
 - \text{Ending WIP} \\
 \hline
 \text{Units started and completed} \\
 \hline
 \end{array}$$

- 3) Units in ending work-in-process (ending WIP)
 - a) Units in ending WIP can be calculated as follows:

$$\begin{array}{r}
 \text{Beginning WIP} \\
 + \text{Units started} \\
 - \text{Units transferred out} \\
 \hline
 \text{Ending WIP} \\
 \hline
 \end{array}$$

- b) These units have not been completed during the period.

EXAMPLE 8-2 Beginning WIP, Units Started and Completed, and Ending WIP

A company's beginning inventory consists of 20,000 units that were 40% complete. At the end of the month, work-in-process inventory consisted of 15,000 units that were 30% complete. During the period, the company transferred 150,000 units to finished goods.

Three EUP populations to account for:

Beginning WIP	20,000 units
Units started and completed	130,000 units (150,000 transferred out – 20,000 beginning WIP)
Ending WIP	15,000 units

EXAMPLE 8-3 Beginning WIP, Units Started and Completed, and Ending WIP

A company's beginning work-in-process inventory included 7,000 units that were 60% complete. During the period, 100,000 units started production. By the end of the period, work-in-process inventory totaled 18,000 units that were 70% complete.

Three EUP populations to account for:

Beginning WIP	7,000 units
Units started and completed	82,000 units (100,000 started – 18,000 ending WIP)
Ending WIP	18,000 units

- c. There are two methods for calculating EUP:
- 1) Under the **weighted-average method**, the beginning WIP is treated as if it is started and completed during the current period.
 - 2) Under the **first-in, first-out (FIFO) method**, work done in the current period on units in beginning WIP are included in the calculation.
 - a) For example, units in beginning WIP that are 40% complete at the beginning of the period will require an additional 60% of work in the current period for those units to be completed. The 60% of work done in the current period is included in EUP, while the 40% done in the previous period is not.

5. Equivalent Units of Production – Materials

- a. Materials can be added at the beginning. Refer to Section A. of the Extended Example on page 13 for an illustration.
 - 1) Units in beginning WIP are already 100% complete with respect to direct materials; however, whether or not they produce current-period EUP depends on the method used to calculate EUP.
 - 2) Beginning WIP -- Weighted-average method
 - a) Units in beginning WIP are treated as 100% complete and **will produce EUP in the current period** even though materials were added in the previous period.
 - 3) Beginning WIP -- FIFO method
 - a) Units in beginning WIP **will not produce EUP in the current period** since materials were added in the previous period.
 - 4) Units started and completed are always 100% complete with respect to direct materials.
 - 5) Units in ending WIP are 100% complete with respect to direct materials when materials are all added at the beginning of the production process.
 - a) All materials will be treated as being added during the current period.
- b. Materials can be added evenly throughout the process. Refer to Section B. of the Extended Example on page 13 for an illustration.
 - 1) It is important to note both the method used and the completion percentage.
 - 2) Beginning WIP -- Weighted-average method
 - a) Units in beginning WIP are treated as 100% complete and **will produce EUP in the current period** even though materials were added in the previous period.
 - 3) Beginning WIP -- FIFO method
 - a) Since we are looking for the amount completed in the current period, we subtract the completion percentage from 100%. This will give us the percentage completed in the current period.
 - b) For example, if there are 1,000 units in beginning WIP that are 30% complete with respect to materials, 70% (100% – 30%) will be completed in the current period.
 - i) EUP from beginning WIP will be 700 units (1,000 units × 70%).

- 4) Units started and completed are always 100% complete with respect to direct materials.
- 5) Units in ending WIP will be complete based on the completion percentage.
 - a) For example, if there are 1,000 units in ending WIP that are 60% complete with respect to materials, 60% were completed in the current period.
 - i) EUP from ending WIP will be 600 units ($1,000 \text{ units} \times 60\%$).
- c. Materials added at a specific time during production:
 - 1) Whether or not materials are included in the current period's calculation of EUP depends on when materials would have been added and whether or not that point was reached.
 - 2) Beginning WIP
 - a) If the point at which materials are added occurs during the current period, beginning WIP will produce EUP.
 - b) If the point at which materials are added occurred during the previous period, beginning WIP will not produce EUP for the current period.
 - c) For example, assume that materials are added at 50% completion. If beginning WIP is 20% complete, materials have not been added yet. All materials will be added in the current period, and the units will count 100% toward EUP.
 - i) If beginning WIP is 60% complete, materials were added in the prior period and no more materials will be added in the current period. Therefore, these materials will not produce EUP in the current period.
 - 3) Units started and completed are always 100% complete with respect to direct materials.
 - 4) Ending WIP
 - a) If the point at which materials are added occurred during the current period, ending WIP will produce EUP for the current period.
 - b) If the point at which materials are added occurs during the next period, ending WIP will not produce EUP for the current period.
 - c) For example, assume that materials are added at 50% completion. If ending WIP is 20% complete, materials have not been added yet. Therefore, these units will not produce EUP in the current period.
 - i) If ending WIP is 60% complete, materials were added during the current period and these units will produce EUP in the current period.

6. Equivalent Units of Production – Conversion Costs

- a. Since conversion costs are generally added throughout the process, these calculations are similar to those in item 5.b. beginning on page 10. Refer to Section C. of the Extended Example on page 14 for an illustration.
- b. It is important to note both the method used and the completion percentage.
 - 1) Beginning WIP -- Weighted-average method
 - a) Units in beginning WIP are treated as 100% complete and **will produce EUP in the current period** even though conversion costs were added in the previous period.
 - 2) Beginning WIP -- FIFO method
 - a) Since we are looking for the amount completed in the current period, we subtract the completion percentage from 100%. This will give us the percentage completed in the current period.
 - b) For example, if there are 1,000 units in beginning WIP that are 20% complete with respect to conversion costs, 80% (100% – 20%) will be completed in the current period.
 - i) EUP from beginning WIP will be 800 units (1,000 units × 80%).
 - 3) Units started and completed are always 100% complete with respect to conversion costs.
 - 4) Units in ending WIP will be complete based on the completion percentage.
 - a) For example, if there are 1,000 units in ending WIP that are 10% complete with respect to conversion costs, 10% were completed in the current period.
 - i) EUP from ending WIP will be 100 units (1,000 units × 10%).

EXAMPLE 8-4 Extended Example

Quantity Schedule

	Units	Completed for Direct Materials	Completed for Conversion Costs
Beginning work-in-process	2,000	80%	40%
Units started during period	8,000		
Units to account for	10,000		
Units transferred to next department	9,000		
Ending work-in-process	1,000	90%	70%
Units accounted for	10,000		

In all EUP calculations, three populations of units must be accounted for:

- Beginning WIP: 2,000 (given)
- Units started and completed in the current period

9,000 Units transferred out	8,000 Units started
– 2,000 Beginning WIP	– 1,000 Ending WIP
<u>7,000 Units started and completed</u>	<u>7,000 Units started and completed</u>
- Ending WIP: 1,000 (given)

-- Continued on next page --

EXAMPLE 8-4 -- Continued**Section A.****EUP for direct materials – added at beginning of process:**

When direct materials are added at the beginning of the process

- Under **weighted-average**, prior period and current period costs are both considered. Thus, direct materials costs embedded in beginning WIP, as well as those added during the period, are used in the calculation.
- Under **FIFO**, only current period activity is considered. Since all DM were added in the previous period, no DM costs are added in the current period.
- Under both methods, ending WIP is by definition 100% complete with respect to materials. Notice that the ending inventory is only 70% complete with respect to conversion costs.

	Weighted Average		FIFO	
Beginning WIP	2,000 units × 100% (a)	= 2,000	2,000 units × 0% (c)	= 0
Started and completed	7,000 units × 100%	= 7,000	7,000 units × 100%	= 7,000
Ending WIP	1,000 units × 100% (b)	= 1,000	1,000 units × 100% (d)	= 1,000
Totals	<u>10,000</u>	<u>10,000</u>	<u>10,000</u>	<u>8,000</u>

(a) Always 100% (c) Needed to complete
(b) Degree of completeness(d) Degree of completeness

One way to remember the difference between weighted average and FIFO is to consider the theoretical implications of the calculation for beginning inventory. Under weighted average, costs will be the average for two periods, so the work done for both periods is considered—in this case, 2,000 units. Under FIFO, layers are considered. Since this year's layer was zero, that is the EUP.

Section B.**EUP for direct materials – added throughout process:**

When direct materials are added evenly throughout the process

- Under weighted-average, the treatment of beginning WIP is the same as that when materials are added at the beginning (i.e., all units treated as 100% complete and will produce EUP in the current period).
- Under FIFO, the costs incurred in the current period for beginning WIP are only those needed to complete them.
- Under both methods, ending WIP uses the percentage of completion.

Ending inventory is 90% complete, while beginning inventory is 80% complete.

	Weighted Average		FIFO	
Beginning WIP	2,000 units × 100% (a)	= 2,000	2,000 units × 20% (c)	= 400
Started and completed	7,000 units × 100%	= 7,000	7,000 units × 100%	= 7,000
Ending WIP	1,000 units × 90% (b)	= 900	1,000 units × 90% (d)	= 900
Totals	<u>10,000</u>	<u>9,900</u>	<u>10,000</u>	<u>8,300</u>

(a) Always 100% (c) Needed to complete
(b) Degree of completeness(d) Degree of completeness

Conversion costs by their nature are added evenly throughout the process. Based on the quantity schedule, ending inventory is 70% complete with respect to conversion costs.

-- Continued on next page --

EXAMPLE 8-4 -- Continued**Section C.****EUP for conversion costs:**

	Weighted Average	FIFO
Beginning WIP	2,000 units × 100% (a) = 2,000	2,000 units × 60% (c) = 1,200
Started and completed	7,000 units × 100% = 7,000	7,000 units × 100% = 7,000
Ending WIP	1,000 units × 70% (b) = 700	1,000 units × 70% (d) = 700
Totals	<u>10,000</u>	<u>8,900</u>

(a) Always 100% (c) Needed to complete
(b) Degree of completeness (d) Degree of completeness

Assume that the costs to be allocated are as follows:

	Direct Materials	Conversion Costs	Total
Beginning work-in-process	\$25,000	\$10,000	= \$ 35,000
Added during the month	55,000	50,000	= 105,000
			<u>\$140,000</u>

The per-unit costs under each of the two methods can now be derived.

- Under the weighted-average method, all direct materials and conversion costs are averaged in, both those incurred in the current period and those in beginning work-in-process. In other words, divide costs for both periods by all units worked on in both periods.
- Under the FIFO method, only the costs incurred in the current period are included in the calculation because only the work performed in the current period is included in EUP.

Per-unit costs – direct materials added at beginning of process:

Conversion costs are 70% complete on ending WIP.

	Weighted-average	FIFO
Direct materials:	$\frac{\$25,000 + \$55,000}{10,000 \text{ EUP}} = \8.000	$\frac{\$55,000}{8,000 \text{ EUP}} = \6.875
Conversion costs:	$\frac{\$10,000 + \$50,000}{9,700 \text{ EUP}} = \6.186	$\frac{\$50,000}{8,900 \text{ EUP}} = \5.618
Total per-unit cost	<u>\$14.186</u>	<u>\$12.493</u>

Per-unit costs – direct materials added throughout process:

	Weighted-average	FIFO
Direct materials:	$\frac{\$25,000 + \$55,000}{9,900 \text{ EUP}} = \8.081	$\frac{\$55,000}{8,300 \text{ EUP}} = \6.627
Conversion costs:	$\frac{\$10,000 + \$50,000}{9,700 \text{ EUP}} = \6.186	$\frac{\$50,000}{8,900 \text{ EUP}} = \5.618
Total per-unit cost	<u>\$14.267</u>	<u>\$12.245</u>

7. Spoilage in Process Costing

- a. As with job-order costing, the cost of a normal level of spoilage is left in cost of goods sold; abnormal spoilage is recognized separately as a loss.
- b. Recognizing the loss resulting from abnormal spoilage under process costing is a multi-step process.
 - 1) The manufacturer establishes inspection points, that is, the places in the production process where those goods not meeting specifications are pulled from the process. This is in contrast to job-order costing, in which a unit can be judged to be spoiled at any time.
 - a) The typical arrangement is to inspect units as they are being transferred from one department to the next. This way, each department has its own amount of spoilage, calculated using its own equivalent-unit costs.
 - 2) The loss is equal to the number of units of abnormal spoilage multiplied by the department's equivalent-units costs, whether weighted-average or FIFO.

Loss on abnormal spoilage	\$XXX	
Work-in-process – Department A		\$XXX

8.3 ACTIVITY-BASED COSTING

1. Use of Activity-Based Costing

- a. Activity-based costing (ABC) is a response to the significant increase in **indirect costs** resulting from the rapid advance of technology.
 - 1) ABC is a refinement of an existing cost accounting system (job-order or process) for use in internal reporting and management planning and control. It is also used for external reporting to the extent its product-costing function satisfies the relevant standard setter's requirements.
 - a) Under a traditional (volume-based) costing system, overhead is accumulated into a single cost pool and spread evenly across all end products.
 - b) Under ABC, indirect costs are attached to activities that are then rationally allocated to end products.
 - c) ABC may be used by manufacturing, service, or retailing entities.

2. Traditional (Volume-Based) Costing System

- a. The inaccurate averaging or spreading of indirect costs over products or service units that use different amounts of resources is sometimes called **peanut-butter costing**.
 - 1) Peanut-butter costing results in **product-cost cross-subsidization**, the condition in which the miscosting of one product causes the miscosting of other products.
 - 2) The peanut-butter effect of using a traditional (i.e., volume-based) costing system can be summarized as follows:
 - a) Direct labor and direct materials are traced to products or service units.
 - b) A single pool of indirect costs (overhead) is accumulated for a given organizational unit.
 - c) Indirect costs from the pool are assigned using an allocative (rather than a tracing) procedure, such as using a single overhead rate for an entire department, e.g., \$3 of overhead for every direct labor hour.
 - i) The effect is an averaging of costs that may result in significant inaccuracy when products or service units do not use similar amounts of resources.

EXAMPLE 8-5 Product-Cost Cross-Subsidization

A company produces two similar products. Both products require one unit of raw material and one hour of direct labor. Raw materials costs are \$14 per unit, and direct labor is \$70 per hour. During the month just ended, the company produced 1,000 units of Product A and 100 units of Product B. Manufacturing overhead for the month totaled \$20,000.

Using direct labor hours as the overhead allocation base, per-unit costs and profits are calculated as follows:

	Product A	Product B	Total
Raw materials	\$ 14,000	\$ 1,400	
Direct labor	70,000	7,000	
Overhead $\{ \$20,000 \times [1,000 \div (1,000 + 100)] \}$	18,182		
Overhead $\{ \$20,000 \times [100 \div (1,000 + 100)] \}$		1,818	
Total costs	<u>\$102,182</u>	<u>\$ 10,218</u>	<u>\$112,400</u>
Selling price	\$ 119.99	\$ 119.99	
Cost per unit	(102.18)	(102.18)	
Profit per unit	<u>\$ 17.81</u>	<u>\$ 17.81</u>	

The company's management accountants have determined that overhead consists almost entirely of production line setup costs, and that the two products require equal setup times. Allocating overhead on this basis yields vastly different results.

	Product A	Product B	Total
Raw materials	\$14,000	\$ 1,400	
Direct labor	70,000	7,000	
Overhead $(\$20,000 \times 50\%)$	10,000		
Overhead $(\$20,000 \times 50\%)$		10,000	
Total costs	<u>\$94,000</u>	<u>\$18,400</u>	<u>\$112,400</u>
Selling price	\$119.99	\$119.99	
Cost per unit	(94.00)	(184.00)	
Profit per unit	<u>\$ 25.99</u>	<u>\$ (64.01)</u>	

Rather than the comfortable profit the company believed it was making on both products using peanut-butter costing, it becomes clear that the company is losing money on every unit of Product B that it sells. The high-volume Product A has been heavily subsidizing the setup costs for the low-volume Product B.

- b. Example 8-5 on the previous page assumes a single component of overhead for clarity. In reality, overhead is made up of many components.

- 1) The peanut-butter effect of traditional overhead allocation is illustrated in the following diagram:

Overhead Allocation in a Traditional (Volume-Based) Cost Accumulation System

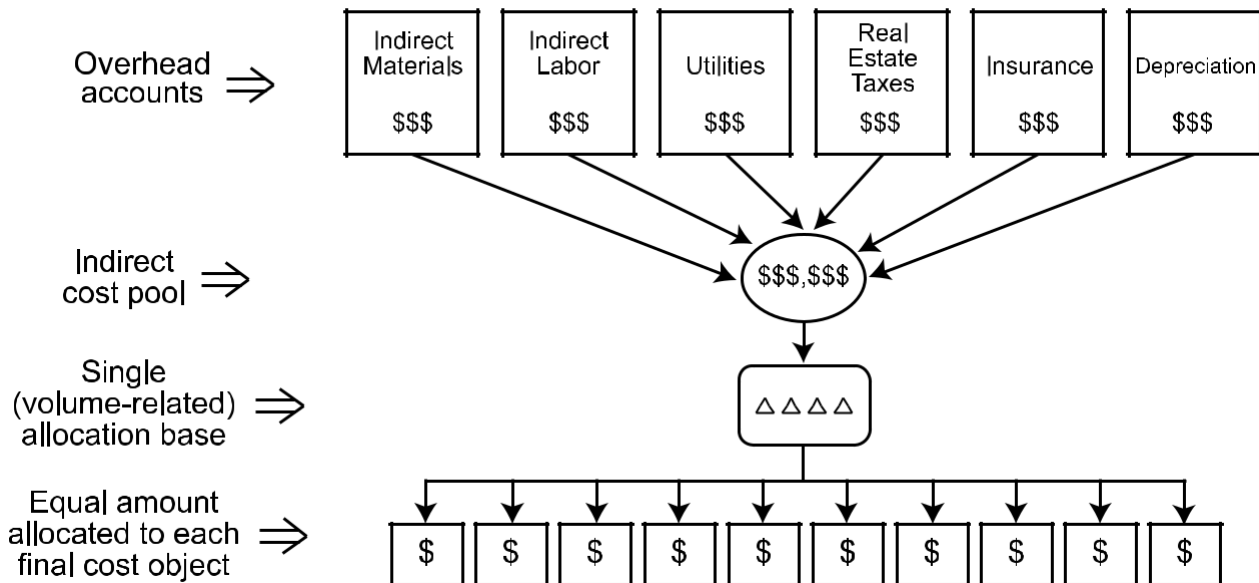


Figure 8-3

3. Traditional-Based Systems vs. Activity-Based Systems

- a. Traditional-based systems were appropriate when direct costs were the bulk of manufacturing costs. However, increased automation led to increased overhead. ABC was developed to address the increasing complexity of overhead costs.

- 1) **Traditional-based systems**, as illustrated above, involve

- Accumulating costs in general ledger accounts (utilities, taxes, etc.),
- Using a single cost pool to combine the costs in all the related accounts,
- Selecting a single driver to use for the entire indirect cost pool, and
- Allocating the indirect cost pool to final cost objects.

- 2) **Activity-based systems**, by contrast, involve

- Identifying organization activities that constitute overhead,
- Assigning the costs of resources consumed by the activities, and
- Assigning the costs of the activities to final cost objects, based on the activity that drives (causes) the costs.

4. Steps in Activity-Based Costing

a. Step 1 – Activity Analysis

- 1) An activity is a set of work actions undertaken within the entity, and a cost pool is established for each activity.
- 2) Activities are classified in a hierarchy according to the level of the production process at which they take place.
 - a) **Unit-level activities** are performed for each unit of output produced. Examples are using direct materials and using direct labor.
 - b) **Batch-level activities** occur for each group of outputs produced. Examples are materials ordering, materials handling, and production line setup.
 - c) **Product-sustaining** (or service-sustaining) **activities** support the production of a particular product (or service), irrespective of the level of production. Examples are product design, engineering changes, and testing.
 - d) **Facility-sustaining activities** concern overall operations and therefore cannot be traced to products at any point in the production process. Examples are accounting, human resources, maintenance of physical plant, and safety and security arrangements.

EXAMPLE 8-6 Identification of Activities

Fabulous Foundry uses a job-order system to accumulate costs for the custom pipe fittings of all sizes that it produces. Since the 1950s, Fabulous has accumulated overhead costs in six general ledger accounts (indirect materials, indirect labor, utilities, real estate taxes, insurance, and depreciation), combined them into a single indirect cost pool, and allocated the total to its products based on machine hours.

- At the time this system was established, overhead was a relatively small percentage of the foundry's total manufacturing costs.
- With increasing reliance on robots in the production process and computers for monitoring and control, overhead is now a greater percentage of the manufacturing costs while direct labor costs have shrunk.

To obtain better data about product costs, Fabulous has decided to refine its job-order costing system by switching to activity-based costing for the allocation of overhead.

- The foundry's management accountants conducted extensive interviews with production and sales personnel to determine how the incurrence of indirect costs can be viewed as activities that consume resources.
- The accountants identified five activities and created a cost pool for each to capture the incurrence of indirect costs:

<u>Activity</u>	<u>Hierarchy</u>
Product design	Product-sustaining
Production setup	Batch-level
Machining	Unit-level
Inspection & testing	Unit-level
Customer maintenance	Facility-sustaining

b. Step 2 – Assign Resource Drivers to Resource Costs

- 1) Identifying resource costs in ABC is more complex than it is in volume-based overhead allocation.
 - a) A separate accounting system may be necessary to track resource costs separately from the general ledger.
- 2) Once the resources have been identified, resource drivers are designated to allocate resource costs to the activity cost pools.
- 3) **Resource drivers** (causes) are measures of the resources consumed by an activity.

EXAMPLE 8-7 Resource Drivers

Fabulous Foundry's management accountants identified the following resources used by its indirect cost processes:

<u>Resource</u>	<u>Driver</u>
Computer processing	CPU cycles
Production line	Machine hours
Materials management	Hours worked
Accounting	Hours worked
Sales & marketing	Number of orders

c. Step 3 – Allocate Resource Costs to Activity Cost Pools

- 1) Once the resource drivers are determined, the dollar amount of resources per resource driver can be determined.
 - a) One method of doing this is by dividing the total dollar amount of a resource cost by the total amount of the resource driver used by the entire entity.
- 2) Costs of resources are then allocated to activity cost pools based on the amount of resource drivers used by each activity cost pool.

EXAMPLE 8-8 Allocation to Activity Pools

Fabulous Foundry's management accountants have determined that a total amount of \$1,000,000 of Materials Management was used over a total of 100,000 hours worked. Fabulous's management accountants will therefore allocate \$10 (\$1,000,000 ÷ 100,000 hours) to each activity pool for each hour of Materials Management worked for each cost pool.

Activity	Amount Allocated
Product Design	\$250,000 for 25,000 hours
Production Setup	\$270,000 for 27,000 hours
Machining	\$450,000 for 45,000 hours
Inspection & Testing	\$30,000 for 3,000 hours
Customer Maintenance	\$0 for 0 hours

- 3) This allocation is done for each resource. This is termed first-stage allocation.
 - a) Some cost activities may not be allocated resources if that cost activity did not use those resources.

d. **Step 4 – Allocate Activity Cost Pools to Final Cost Objects**

- 1) The final step in enacting an ABC system is allocating the activity cost pools to final cost objects. This is termed second-stage allocation.
- 2) Once the cost drivers are determined, the dollar amount of activity pool per activity driver can be determined.
 - a) One method of doing this is by dividing the total dollar amount assigned to an activity cost pool by the total amount of the activity driver used by the entire entity.
- 3) Costs are reassigned to final-stage (or, if intermediate cost objects are used, next-stage) cost objects on the basis of activity drivers.
 - a) **Activity drivers** are measures of the demands made on an activity by next-stage cost objects, such as the number of parts in a product used to measure an assembly activity.

EXAMPLE 8-9 Activity Drivers

Fabulous Foundry's management accountants have designated the following drivers to associate with their corresponding activities:

<u>Activity</u>	<u>Driver</u>
Product design	Number of products
Production setup	Number of setups
Machining	Number of units produced
Inspection & testing	Number of units produced
Customer maintenance	Number of orders

5. **Indirect Cost Assignment Diagram**

- a. The differences between traditional overhead allocation and ABC are illustrated in the following diagram:

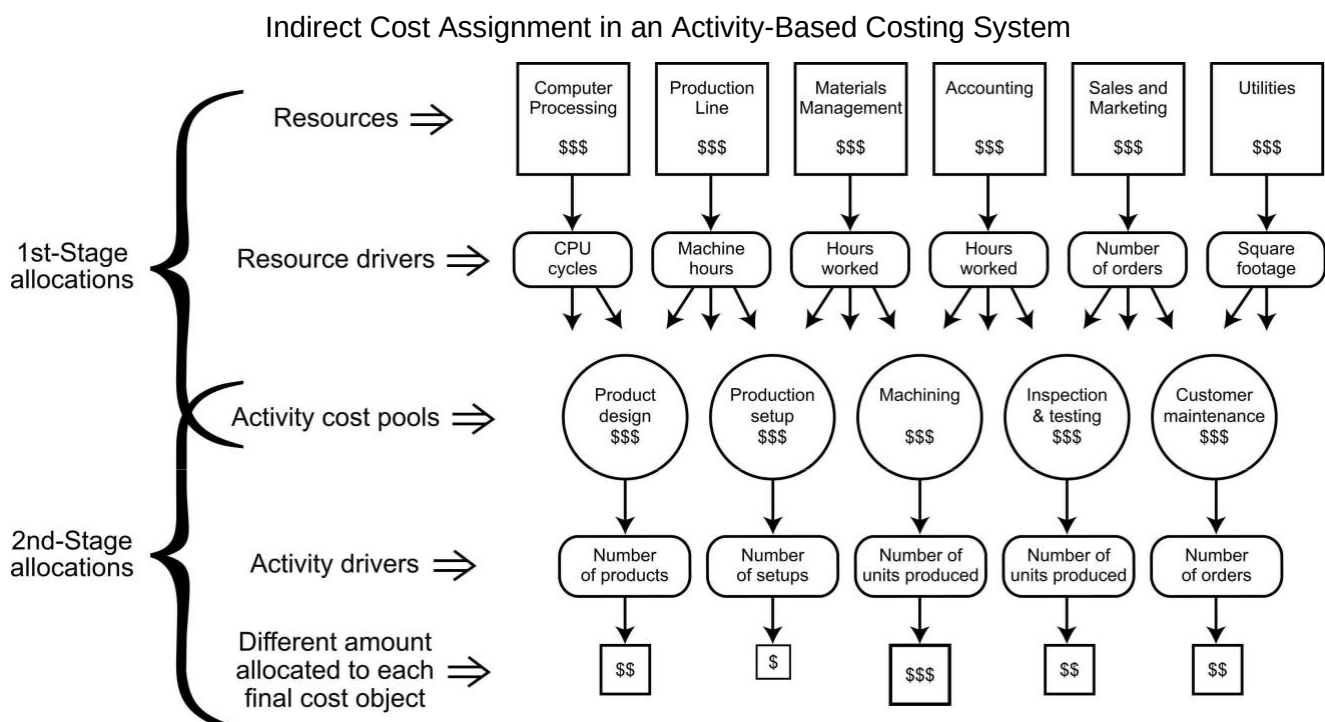


Figure 8-4

6. Process Value Analysis

- a. Design of an ABC system starts with process value analysis, a comprehensive understanding of how an organization generates its output.
 - 1) A process value analysis involves a determination of which activities that use resources are value-adding or nonvalue-adding and how the latter may be reduced or eliminated.
 - a) A **value-adding activity** contributes to customer satisfaction or meets a need of the entity. The perception is that it cannot be omitted without a loss of the quantity, quality, or responsiveness of output demanded by the entity or its customers.
 - b) A **nonvalue-adding activity** does not make such a contribution. It can be eliminated, reduced, or redesigned without impairing the quantity, quality, or responsiveness of the product or service desired by customers or the entity.
 - 2) The linkage of product costing and continuous improvement of processes is activity-based management (ABM). It encompasses driver analysis, activity analysis, and performance measurement.
- b. Using a four-level driver-analysis model, activities are grouped, and drivers are determined for the activities.
 - 1) Within each grouping of activities, the cost pools for activities that can use the same driver are combined into homogeneous cost pools.
 - a) In contrast, traditional systems assign costs largely on the basis of unit-level drivers.
 - 2) A difficulty in applying ABC is that, although the first three levels of activities (unit-, batch-, and product-level) pertain to specific products or services, facility-level activities do not.
 - a) Thus, facility-level costs are not accurately assignable to products or services. The theoretically sound solution may be to treat these costs as period costs.
 - b) Nevertheless, organizations that apply ABC ordinarily assign them to products or services to obtain a full-absorption cost suitable for external financial reporting in accordance with GAAP.

7. Cost Drivers

- a. Drivers (both resource and activity) must be chosen on the basis of a cause-and-effect relationship with the resource or activity cost being allocated, not simply a high positive correlation.
 - 1) A cost object may be a job, product, process, activity, service, or anything else for which a cost measure is desired.
 - 2) Intermediate cost objects receive temporary accumulations of costs as the cost pools move from their originating points to the final cost objects.
 - a) For example, work-in-process is an intermediate cost object, and finished salable goods are final cost objects.

8. Advantages and Disadvantages of Activity-Based Costing

- a. An advantage of ABC is that product costing is improved, making for better decision making.
 - 1) The process value analysis performed as part of ABC provides information for eliminating or reducing nonvalue-adding activities (e.g., scheduling production, moving components, waiting for the next operating step, inspecting output, or storing inventories).
 - a) The result is therefore not only more accurate cost assignments, especially of overhead, but also better cost control and more efficient operations.
 - 2) The real benefits of ABC occur when a company has a high level of fixed costs and produces a wide variety of products with widely varying levels of production.
- b. Disadvantages of ABC are its cost of implementation and the increased time and effort needed to (1) maintain a separate accounting system to capture resource costs and (2) design and implement drivers and cost pools.
 - 1) Initial costs are quite high, and continuing costs of application can also be significant. Thus, if a company has a low level of fixed costs, there is little to no advantage in using ABC as compared to a simple overhead application method (such as a fixed amount per direct labor hour).
 - 2) Similarly, if a company produces a single product, there is no advantage. Also, if volume levels among various products do not vary too much, there is little advantage.
 - 3) ABC-derived costs of products or services may not conform with GAAP; for example, ABC may assign research costs to products but not traditional product costs such as plant depreciation, insurance, or taxes.

9. Organizational Benefits

- a. Organizations most likely to benefit from using ABC are those with products or services that vary significantly in volume, diversity of activities, and complexity of operations; relatively high overhead costs; or operations that have undergone major technological or design changes.
 - 1) Although the previous illustrations have assumed a manufacturing operation, service organizations also can use ABC and receive most of the advantages. However, service organizations may have some difficulty in implementing ABC because they tend to have relatively high facility-level costs that are difficult to assign to specific service units.
 - a) Service organizations also engage in many nonuniform human activities for which information is not readily accumulated.
 - b) Moreover, output measurement is more problematic in service than in manufacturing entities.
 - c) Nevertheless, ABC has been adopted by various insurers, banks, railroads, and healthcare providers.

- b. Direct labor (hours or dollars) has long been the most common base for allocating overhead because of the simplicity of the calculation, but it is not always relevant.
 - 1) Companies now use dozens of different allocation bases depending on how activity affects overhead costs. One company reported that it used 37 different bases to allocate overhead, some of which were averages of several activities.
 - 2) In principle, a separate overhead account or subsidiary ledger account should be used for each type of overhead.
 - 3) In the past, “people were cheap and machines were expensive.” This meant that direct labor was ordinarily a larger component of total production cost than overhead and was the activity that drove (caused) overhead costs.
 - a) Due to the lowered cost of computers and robotics and the expansion of employee benefits (“machines are cheap and people are expensive”), overhead is more likely to be a large component of total production cost, with direct labor often a small percentage.
 - b) Most overhead costs vary in proportion to product diversity and the complexity of an operation.
 - i) Direct labor is not a cost driver for most overhead costs.
 - c) Allocating a very large cost (overhead) using a very small cost (direct labor) as a base is irrational.
 - i) A small change in direct labor on a product can make a significant difference in total production cost, an effect that may rest on an invalid assumption about the relationship of the cost and the allocation base.
 - 4) As previously noted, ABC is more useful when overhead costs are relatively high.
 - a) Also, the more diverse a company's line of products or services or the more significant the volume differences among its products or services, the more beneficial ABC will be.
- c. Companies use ABC because of its ability to solve costing problems that conventional cost accounting either creates or fails to address.
 - 1) These problems include suboptimal pricing, poor allocation of costs, and incorrect direction by management.
 - 2) For example, if overhead is allocated at 700% of direct labor, managers may try to reduce direct labor costs by \$1 to reduce the amount of overhead allocated by \$7.
 - a) But the better decision may be to ignore direct labor and concentrate on such cost-cutting efforts as eliminating setups, engineering changes, and movement of materials.

8.4 LIFE-CYCLE COSTING

1. Life-Cycle Approach

- a. A life-cycle approach to budgeting estimates a product's revenues and expenses over its entire sales life cycle.
 - 1) The product life cycle has five phases:
 - a) The **research and development** phase typically has no sales and high costs.
 - b) The **introduction** phase is characterized by few competitors. Profits are usually low in the introduction stage because of slow sales growth. Also, costs are high for sales promotion and relatively high for unit costs of production.
 - c) In the **growth** stage, the number of competitors increases but does not peak. The opportunity for cost reductions is at its maximum during the growth stage because production volume is increasing at a high rate. Thus, fixed costs are spread over more units of production.
 - d) In the **maturity** phase, sales growth declines and competitors are most numerous.
 - e) The number of competitors decreases in the **decline** stage.
 - 2) Accordingly, life-cycle costing takes a long-term view of the entire cost life cycle, also known as the value chain.
 - a) Costs incurred before production, such as R&D and product design, are referred to as **upstream costs**.
 - b) Costs incurred after production, such as marketing and customer service, are referred to as **downstream costs**.

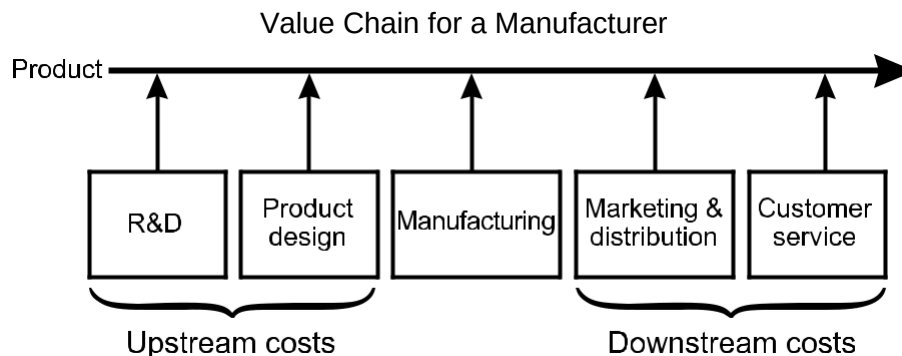


Figure 8-5

- 3) This information is important for pricing decisions because revenues must cover costs incurred in each stage of the value chain, not only production.

2. Potential Benefits

- a. Life-cycle costing emphasizes the relationships among costs incurred at different value-chain stages, for example, the effect of reduced design costs on future customer-service costs.
 - 1) Because it makes a distinction between incurring costs (actually using resources) and locking in (designing in) costs, life-cycle costing highlights the potential for cost reduction activities during the upstream phase of the value chain.
 - 2) It is in this phase that the greatest opportunity exists to minimize downstream costs. Indeed, it has been estimated that 90% or more of costs are committed (not incurred) before production begins.

3. Life-Cycle vs. Other Costing Methods

- a. Traditional approaches focus on cost control (as opposed to cost reduction) during production and treat pre- and postproduction (upstream and downstream) costs as period costs that are largely ignored in determining the profitability of specific products.
 - 1) Other costs that traditional methods ignore are the after-purchase costs (operating, support, repair, and disposal) incurred by customers.
 - 2) Accordingly, whole-life cost is a concept closely associated with life-cycle cost.
 - a) **Whole-life cost** equals the life-cycle cost plus after-purchase costs.
 - b) Attention to the reduction of all whole-life costs through analysis and management of all value-chain activities is a powerful competitive tool because of the potential for increasing customer satisfaction.
 - 3) Life-cycle and whole-life cost concepts are associated with **target costing** and **target pricing**.
 - a) A firm may determine that market conditions require that a product sell at a given target price. Target pricing is used in some industries to create a competitive advantage. It takes a customer-oriented approach to setting prices based on what customers will pay and what competitors will do. Thus, the target price is in effect a constraint that cannot be changed.
 - b) **Target costing** is the practice of calculating the price for a product by adding the desired unit profit margin to the total unit cost. It is an adjunct concept of target pricing.
 - i) For example, a furniture manufacturer might want to produce a sofa that sells for \$600 and has a \$200 profit margin. That means the cost can be no more than \$400.
 - ii) If the new-product development team says the expected cost will be more than \$400, then the product will not be manufactured because, under target costing, the product is not a “sofa” but a “\$600 sofa.”
 - c) The cost reduction objectives of life-cycle and whole-life cost management can therefore be determined using target costing.
 - 4) **Value engineering** is a means of reaching targeted cost levels.
 - a) Value engineering is a systematic approach to assessing all aspects of the value chain cost buildup for a product.
 - i) The purpose is to minimize costs without reducing customer satisfaction.
 - b) For this purpose, distinguishing between value-adding and nonvalue-adding activities is useful.
 - i) A **value-adding activity** contributes to customer value or satisfies a need of the entity. A **nonvalue-adding activity** does not make such a contribution.
 - ii) Accordingly, value engineering seeks to minimize nonvalue-adding activities and their costs by reducing the cost drivers of those activities.
 - iii) Value engineering also attempts to minimize the costs of value-adding activities by improving their efficiency.

4. Internal and External Reporting Effects

- a. For internal management accounting purposes, the costs (such as R&D) that result in marketable products represent a life-cycle investment and must be capitalized.
 - 1) The reporting system should also allow for capitalization and subsequent allocation of upstream costs for management accounting purposes.
- b. For external financial statement purposes, costs during the upstream phase must be expensed in the period incurred.
 - 1) As a result, organizations that focus on a product's life cycle must develop an accounting system consistent with GAAP for external financial reporting purposes.
 - 2) Essentially, life-cycle costing requires the accumulation of all costs over a product's lifetime, from inception of the idea to the abandonment of the product.
 - a) These costs are then allocated to production on an expected unit-of-output basis.
 - 3) The internal income statement for a product will report total sales for all periods, minus all expenses to date.
 - a) A risk reserve may be established as an account contra to the capitalized costs.
 - b) The risk reserve consists of any deferred product costs that might not be recovered if sales are less than planned.

5. Evaluating Management

- a. The overall advantage of life-cycle costing is that it provides a better measure for evaluating the performance of product managers.
 - 1) Traditional financial statements, however, might report that certain products were extremely profitable because upstream costs were expensed in previous periods.
 - 2) For example, if a substantial investment is made in the development of a new product but that product quickly becomes obsolete due to new technology, how worthwhile was the investment?
 - 3) Life-cycle costing combines all costs and revenues for all periods to provide a better view of a product's overall performance.

STUDY UNIT NINE

COST ALLOCATION TECHNIQUES

9.1	Joint Product and By-Product Costing	1
9.2	Overhead Allocation and Normal Costing	6
9.3	Allocating Service Department Costs	14

This study unit is the **third of five** on **cost management**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The five study units are

- Study Unit 7: Cost Management Concepts
- Study Unit 8: Cost Accumulation Systems
- Study Unit 9: Cost Allocation Techniques
- Study Unit 10: Supply Chain Management
- Study Unit 11: Business Process Improvement

If you are interested in reviewing more introductory or background material, go to www.GLEM.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following GLEM outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

9.1 JOINT PRODUCT AND BY-PRODUCT COSTING

1. Joint vs. Separable Costs

- a. Often a manufacturing process involves processing a single input up to the point at which multiple end products (joint products) become separately identifiable, called the **split-off point**.
 - 1) **Joint (common) costs** are those costs incurred before the split-off point; i.e., since they are not traceable to the end products, they must be allocated.
 - a) Joint costs include direct materials, direct labor, and manufacturing overhead. Because they are not separately identifiable, they must be allocated to the individual joint products.
 - b) For example, the cost of a tree would be a joint cost for a lumber yard.
 - 2) **Separable costs** are those incurred beyond the split-off point, i.e., once they can be identified with a particular joint product and allocated to a specific unit of output.
 - 3) **By-products** are products of relatively small total value that are produced simultaneously from a common manufacturing process with products of greater value and quantity (joint products).
 - a) For example, a lumber yard might have leftover lumber and sawdust that could be sold as by-products for a small amount.

- b. An example where joint costing is very important is petroleum refining.
- 1) Costs incurred in bringing crude oil to the fractionation process are joint costs. The fractionation process is the split-off point.
 - 2) Once the oil has been refined into its separately identifiable end products (asphalt, diesel fuel, kerosene, etc.), all further costs are separable costs.
 - 3) If selling costs are lower than disposal costs, the sludge left over after the high-value products have been processed may be sold as a cheap lubricant. It is considered a by-product.

2. Joint Cost Allocation

- a. Because joint costs cannot be traced to individual products, they must be allocated. The primary purpose for allocating common costs to joint products is to determine the inventory cost of joint products for financial reporting. The methods available for this allocation can be classified into two conceptual groups.
- 1) The **physical-measure-based approach** employs a physical measure, such as volume, weight, or a linear measure.
 - 2) **Market-based approaches** assign a proportionate amount of the total cost to each product on a monetary basis. They include the
 - a) Sales-value at split-off method
 - b) Estimated net realizable value (NRV) method
 - c) Constant-gross-margin percentage NRV method

3. Physical-Measure-Based Approach

- a. The **physical-unit method** (also called the physical quantities method) allocates joint production costs to each product based on their relative proportions of the measure selected.

EXAMPLE 9-1		Joint Costing -- Physical-Unit Method	
A refinery processes 1,000 barrels of crude oil and incurs \$100,000 of processing costs. The process results in the following outputs. Under the physical-unit method, the joint costs up to split-off are allocated as follows:			
Asphalt	$\$100,000 \times (300 \text{ barrels} \div 1,000 \text{ barrels})$	=	\$ 30,000
Fuel oil	$\$100,000 \times (300 \text{ barrels} \div 1,000 \text{ barrels})$	=	30,000
Diesel fuel	$\$100,000 \times (200 \text{ barrels} \div 1,000 \text{ barrels})$	=	20,000
Kerosene	$\$100,000 \times (100 \text{ barrels} \div 1,000 \text{ barrels})$	=	10,000
Gasoline	$\$100,000 \times (100 \text{ barrels} \div 1,000 \text{ barrels})$	=	<u>10,000</u>
Joint costs allocated			<u><u>\$100,000</u></u>

- b. The physical-unit method's simplicity makes it appealing, but it does not match costs with the individual products' revenue-generating potential.
- 1) Consequently, the physical-unit method is rarely beneficial. Advantages include (a) ease of use and (b) having objective criteria for the allocation of the joint costs.
 - 2) However, its limitations are that it treats low-value products that are large in size as if they were valuable. As a result, a large, low-value product might always show a loss, but a small, high-value product will always show a profit.

4. Market-Based Approaches

- a. These allocations are performed using the entire production run for an accounting period, not units sold. This is because the joint costs were incurred on all the units produced, not just those sold.

- b. The **sales-value at split-off method** (also called the gross market value method) is based on the relative sales values of the separate products at split-off.

EXAMPLE 9-2 Sales-Value at Split-Off Method

The refinery estimates that the five outputs can sell for the following prices at split-off:

Asphalt	300 barrels at \$ 60/barrel	= \$ 18,000
Fuel oil	300 barrels at \$ 180/barrel	= 54,000
Diesel fuel	200 barrels at \$ 160/barrel	= 32,000
Kerosene	100 barrels at \$ 80/barrel	= 8,000
Gasoline	100 barrels at \$ 180/barrel	= 18,000
Total sales value at split-off		<u>\$130,000</u>

The total expected sales value for the entire production run at split-off is thus \$130,000. Multiply the total joint costs to be allocated by the proportion of the total expected sales of each product:

Asphalt	$\$100,000 \times (\$ 18,000 \div \$130,000)$	= \$ 13,846
Fuel oil	$\$100,000 \times (\$ 54,000 \div \$130,000)$	= 41,539
Diesel fuel	$\$100,000 \times (\$ 32,000 \div \$130,000)$	= 24,615
Kerosene	$\$100,000 \times (\$ 8,000 \div \$130,000)$	= 6,154
Gasoline	$\$100,000 \times (\$ 18,000 \div \$130,000)$	= 13,846
Joint costs allocated		<u>\$100,000</u>

- c. The **estimated NRV method** also allocates joint costs based on the relative market values of the products.
- 1) The significant difference is that, under the estimated NRV method, all separable costs necessary to make the product salable are subtracted before the allocation is made.

EXAMPLE 9-3 Joint Costing -- Estimated NRV Method

The refinery estimates final sales prices as follows:

Asphalt	300 barrels at \$ 70/barrel	= \$21,000
Fuel oil	300 barrels at \$200/barrel	= 60,000
Diesel fuel	200 barrels at \$180/barrel	= 36,000
Kerosene	100 barrels at \$ 90/barrel	= 9,000
Gasoline	100 barrels at \$190/barrel	= 19,000

From these amounts, separable costs are subtracted (these costs are given):

Asphalt	$\$21,000 - \$1,000$	= \$ 20,000
Fuel oil	$\$60,000 - \$1,000$	= 59,000
Diesel fuel	$\$36,000 - \$1,000$	= 35,000
Kerosene	$\$ 9,000 - \$2,000$	= 7,000
Gasoline	$\$19,000 - \$2,000$	= 17,000
Total net realizable value		<u>\$138,000</u>

Multiply the total joint costs to be allocated by the proportion of the final expected sales of each product:

Asphalt	$\$100,000 \times (\$ 20,000 \div \$138,000)$	= \$ 14,493
Fuel oil	$\$100,000 \times (\$ 59,000 \div \$138,000)$	= 42,754
Diesel fuel	$\$100,000 \times (\$ 35,000 \div \$138,000)$	= 25,362
Kerosene	$\$100,000 \times (\$ 7,000 \div \$138,000)$	= 5,072
Gasoline	$\$100,000 \times (\$ 17,000 \div \$138,000)$	= 12,319
Joint costs allocated		<u>\$100,000</u>

- d. The **constant-gross-margin percentage NRV method** is based on allocating joint costs so that the gross-margin percentage is the same for every product.

- 1) The three steps under this method are
 - a) Determine the overall gross-margin percentage.
 - b) Subtract the appropriate gross margin from the final sales value of each product to calculate total costs for that product.
 - c) Subtract the separable costs to arrive at the joint cost amount.

EXAMPLE 9-4 Joint Costing -- Constant-Gross-Margin Percentage

The refinery uses the same calculation of expected final sales price as under the estimated NRV method:

Asphalt	300 barrels at \$	70/barrel	= \$	21,000
Fuel oil	300 barrels at \$	200/barrel	=	60,000
Diesel fuel	200 barrels at \$	180/barrel	=	36,000
Kerosene	100 barrels at \$	90/barrel	=	9,000
Gasoline	100 barrels at \$	190/barrel	=	19,000
Total of final sales prices				<u>\$145,000</u>

The final sales value for the entire production run is thus \$145,000. From this total, the joint costs and total separable costs are deducted to arrive at a total gross margin for all products:

$$\$145,000 - \$100,000 - \$7,000 = \$38,000$$

The gross margin percentage can then be derived:

$$\$38,000 \div \$145,000 = 26.21\%$$

Deduct gross margin from each product to arrive at a cost of goods sold:

Asphalt	\$ 21,000 – (\$ 21,000 × 26.21%)	=	\$15,497
Fuel oil	\$ 60,000 – (\$ 60,000 × 26.21%)	=	44,276
Diesel fuel	\$ 36,000 – (\$ 36,000 × 26.21%)	=	26,565
Kerosene	\$ 9,000 – (\$ 9,000 × 26.21%)	=	6,641
Gasoline	\$ 19,000 – (\$ 19,000 × 26.21%)	=	14,021

Deduct the separable costs from each product to arrive at the allocated joint costs:

Asphalt	\$ 15,497 – \$1,000	= \$	14,497
Fuel oil	\$ 44,276 – \$1,000	=	43,276
Diesel fuel	\$ 26,565 – \$1,000	=	25,565
Kerosene	\$ 6,641 – \$2,000	=	4,641
Gasoline	\$ 14,021 – \$2,000	=	<u>12,021</u>
Joint costs allocated			<u>\$100,000</u>

- e. The three market-based approaches are far superior to the physical-measure-based approach because they produce an allocation that yields a predictable, comparable level of profitability among the products.

5. Accounting for By-Products

- a. By-products are one or more products of relatively small total value that are produced simultaneously from a common manufacturing process with products of greater value and quantity. By-products, therefore, are defined as having a smaller market value (relative total sales value) than joint products.

- b. The first question that must be answered in regard to by-products is: Do the benefits of further processing and bringing them to market exceed the costs?

Selling price – Additional processing costs – Selling costs = Net realizable value

- 1) If the net realizable value is zero or negative, the by-products should be discarded as scrap.
- c. If the by-products are material, they are capitalized in a separate inventory account, as in this example:

Finished goods inventory – Asphalt (allocated costs)	\$XX,XXX
Finished goods inventory – Fuel oil (allocated costs)	XX,XXX
Finished goods inventory – Diesel fuel (allocated costs)	XX,XXX
Finished goods inventory – Kerosene (allocated costs)	XX,XXX
Finished goods inventory – Gasoline (allocated costs)	XX,XXX
By-product inventory – Sludge (estimated net realizable value)	X,XXX
Work-in-process (total manufacturing costs for period)	\$XXX,XXX

- 1) The amount capitalized is the entire estimated net realizable value of the by-products generated during the period.
- a) This treatment is justifiable when a ready market for the by-products is available.
- 2) By proportionally reducing the amounts capitalized for the major finished goods, this treatment of by-product inventory effectively reduces cost of goods sold. Thus, when the by-products are sold, the income statement effects have already been recognized.

Cash	\$X,XXX
By-product inventory – Sludge	\$X,XXX

- d. If the by-products are immaterial, they are not recognized until the time of sale.
- 1) The amount of miscellaneous revenue (or reduction to cost of goods sold) reported is the actual proceeds from the sale of the by-products.

Cash	\$X,XXX
Cost of goods sold/Miscellaneous revenue	\$X,XXX

- e. Regardless of the timing of their recognition in the accounts, by-products usually do not receive an allocation of joint costs because the cost of this accounting treatment ordinarily exceeds the benefit.

6. Sell-or-Process-Further Decisions

- a. The decision to sell or process further is made based on whether the incremental revenue to be gained by further processing exceeds the incremental cost thereof. This type of decision is tested on Part 2 of the CMA exam.
- 1) The joint cost of the product is irrelevant because it is a sunk cost.

9.2 OVERHEAD ALLOCATION AND NORMAL COSTING

1. Components of Manufacturing Overhead

- a. Manufacturing overhead consists of all costs of manufacturing that are not direct materials or direct labor.
- b. Indirect materials are tangible inputs to the manufacturing process that cannot practicably be traced to the product, e.g., the welding compound used to put together a piece of heavy equipment.
- c. Indirect labor is the cost of human labor connected with the manufacturing process that cannot practicably be traced to the product, e.g., the wages of assembly line supervisors and janitorial staff.
- d. Factory operating costs, such as utilities, real estate taxes, insurance, depreciation on factory equipment, etc.
 - 1) Overhead thus consists of all costs of manufacturing that are not direct materials or direct labor.

2. Variable and Fixed Components

- a. Unlike direct materials and direct labor, which are purely variable costs, overhead contains both variable and fixed components.
 - 1) **Variable overhead costs** include indirect materials, indirect labor, utilities, and depreciation expense under any method that ties depreciation to the level of output.
 - a) The **time frame** for budgeting variable overhead is the short run, i.e., the period within which per-unit variable costs remain constant and therefore predictable.
 - 2) **Fixed overhead costs** include real estate taxes, insurance, and depreciation expense under any method not related to the level of output. The costs of defective units, if normal, are also charged to overhead.
 - a) The **time frame** for budgeting fixed overhead is the long run. Such cost elements as real estate taxes and depreciation are determined by capital expenditures, i.e., those that by their nature span multiple years.
 - 3) A key factor in budgeting overhead is production volume. For example, variable overhead costs are fixed within a relevant range and may increase if output increases substantially. Another key factor is management's judgments about the nature and amounts of costs and estimates of output.

- b. Budgeted (estimated) overhead for the year is accumulated in two indirect cost pools (one for variable costs and one for fixed), which then are allocated to production using an approximate allocation base for each.

EXAMPLE 9-5 Budgeted Estimates of Total Overhead Costs

A manufacturer is preparing its budget for the upcoming year and has compiled the following estimates of total costs:

Cost Element	Estimated Variable Overhead	Estimated Fixed Overhead
Indirect materials	\$ 80,000	
Indirect labor	46,000	
Utilities	155,000	
Real estate taxes		\$ 81,000
Insurance		54,000
Straight-line depreciation		240,000
Totals	<u>\$281,000</u>	<u>\$375,000</u>

3. Selecting an Allocation Base

- a. The most important criterion in accurate cost allocations is using homogeneous cost pools. After creating cost pools, an allocation base is selected for each pool. The crucial quality of an allocation base is that it be a cost driver of the costs in the pool to be allocated.
- 1) Recall that a cost driver should capture a cause-and-effect relationship between the level of the driver and the level of the cost being allocated.
- b. In labor-intensive industries, direct labor hours or cost is an appropriate driver. In capital-intensive industries, machine hours is more appropriate. Still other bases may be used in an ABC system.
- 1) It is therefore possible that variable and fixed overhead will employ the same allocation base.
 - 2) Overhead is usually not allocated on the basis of units produced because of the lack of a cause-and-effect relationship.
- c. EXAMPLE: The variable elements of overhead vary directly with the level of production, so the company has chosen to use units of output as the allocation base for variable overhead. The fixed elements of overhead are related directly to the level of productive capacity, such as factory space and amount of machinery, so the company has decided that it will use machine hours as the allocation base for fixed overhead.

4. Calculating the Application Rate

- a. Once appropriate allocation bases have been selected, the predetermined overhead application rates are calculated.
 - 1) The estimates made of the total amounts of overhead will be the numerators.
 - 2) Estimates are then made of the total quantity of each allocation base that will be expended; these will be the denominators.
 - 3) The quotients are the application rates for that budget period.
- b. Estimates of the total quantity of each allocation base can be based off different capacity levels.
 - 1) **Normal capacity** is the long-term average level of activity that will approximate demand over a period that includes seasonal, cyclical, and trend variations. Deviations in a given year will be offset in subsequent years.
 - 2) **Practical capacity** is the maximum level at which output is produced efficiently. It allows for unavoidable delays in production for maintenance, holidays, etc. Practical capacity is based on realistic, attainable levels of production and input efficiency and is the most appropriate denominator level to use in selecting an overhead application rate.
 - 3) **Theoretical (ideal) capacity** is the maximum capacity assuming continuous operations with no holidays, downtime, etc.

EXAMPLE 9-6 Overhead Allocation Rates

The company's best projection for the upcoming year is that 1,110,000 units will be produced and 57,000 machine hours will be expended. The overhead allocation rates can thus be calculated as follows:

Variable overhead application rate:	$\$281,000 \div 1,110,000 \text{ units of output} = \0.253 per unit
Fixed overhead application rate:	$\$375,000 \div 57,000 \text{ machine hours} = \6.579 per hour

- c. A significant conceptual challenge is understanding the need to apply fixed overhead using an allocation base rather than simply to recognize one-twelfth of the estimated total every month.
 - 1) Since fixed costs are by their nature unchanging within the relevant range, using an allocation base at first appears to unnecessarily complicate the bookkeeping process. But, one way or another, fixed costs must be covered by selling products to customers.
 - 2) The advantage of applying fixed overhead at a predetermined rate is that an allocation base, even if only indirectly, reflects the level of productive activity.
 - a) If production is way down or way up in a particular month, using an allocation base will result in a fixed overhead production-volume variance, also called the denominator-level variance (discussed in more detail in Study Unit 15, Subunit 6).
 - b) The existence of a production-volume/denominator-level variance alerts management to the fact that fixed costs are being spread among fewer or more units, respectively, than anticipated.
- d. Under an **activity-based costing (ABC)** system, individual overhead costs are assigned based on the level of an associated activity rather than lumped in a single (or double) pool (discussed further in Study Unit 8, Subunit 3).

5. Departmental vs. Plant-Wide Rates

- a. All the examples of overhead application so far have employed a single plant-wide rate. This method has the benefit of simplicity.
 - 1) However, some production departments may be labor-intensive while others are machine-intensive. In these cases, the use of a single driver for applying overhead to every phase of the production results in the miscosting of products, less effective control, and less efficient operations.
 - a) Plant-wide rates in such circumstances may diminish the motivation of managers to achieve goals and hinder the appropriate evaluation of their performance.
 - 2) A more accurate method is the use of departmental rates.

EXAMPLE 9-7 Departmental Overhead Allocation Rates

A company is preparing its overhead budget for the coming year and has selected direct labor hours as the allocation base.

	<u>Budgeted Overhead</u>	<u>Allocation Base</u>	<u>Overhead Application Rate</u>
Department A	\$ 60,000		
Department B	40,000		
Total process	<u>\$ 100,000</u>	÷ 20,000	= \$5.00 per direct labor hour

A study by the company's management accountants reveals that Department A heavily employs direct labor while Department B is far more automated.

- Of the total direct labor hours budgeted for the year, 15,000 are projected for Department A and only 5,000 for Department B.
- At the same time, Department A is projected to consume 8,000 machine hours while Department B is projected to use 16,000.

Instead of applying a single plant-wide application rate, a more accurate allocation can be obtained by using a different allocation base for each production department.

	<u>Budgeted Overhead</u>	<u>Allocation Base</u>	<u>Overhead Application Rate</u>
Department A	\$60,000	÷ 15,000	= \$4.00 per direct labor hour
Department B	\$40,000	÷ 16,000	= \$2.50 per machine hour

- b. When indirect costs represent a large proportion of total production costs, ABC, which uses cost pools for all costs (not just overhead), may be the most appropriate cost accumulation system.

6. Time Frame for Calculating Application Rates

- a. Calculating new overhead application rates each interim period can result in misleading unit costs. Thus, overhead rates are normally calculated no more often than annually.
 - 1) During months of low production, per-unit overhead charges increase. The results are higher product costs during months of lower production and distortions in the financial statements.
 - 2) An annual application rate tends to smooth seasonal variability of overhead costs and activity levels.
 - 3) Example 9-8 illustrates this phenomenon.

EXAMPLE 9-8 Extended Example of Unit-Cost Distribution

A manufacturing firm divides its reporting year into 3 budget periods. The company is expecting the following units of production and sales during the upcoming year. Note that production is expected to fluctuate but sales are expected to be even.

	<u>Jan-Apr</u>	<u>May-Aug</u>	<u>Sep-Dec</u>	<u>Totals</u>
Production	10,000	6,000	8,000	24,000
Sales	7,000	7,000	7,000	21,000

Variable overhead costs are calculated at \$1 per unit:

	<u>Jan-Apr</u>	<u>May-Aug</u>	<u>Sep-Dec</u>	<u>Totals</u>
Variable overhead cost	\$10,000	\$ 6,000	\$ 8,000	\$24,000
Fixed overhead cost	20,000	20,000	20,000	60,000
Total overhead cost	<u>\$30,000</u>	<u>\$ 26,000</u>	<u>\$ 28,000</u>	<u>\$84,000</u>

For simplicity, a single overhead application rate is used in this example:

	<u>Jan-Apr</u>		<u>May-Aug</u>		<u>Sep-Dec</u>
Estimated total overhead	<u>\$30,000</u>	= \$3.00	<u>\$26,000</u>	= \$4.33	<u>\$28,000</u>
Estimated production	10,000		6,000		8,000
					= \$3.50

These fluctuations in the applied overhead rate will lead to fluctuations in unit cost:

	<u>Jan-Apr</u>	<u>May-Aug</u>	<u>Sep-Dec</u>
Direct materials	\$ 3.00	\$ 3.00	\$ 3.00
Direct labor	4.00	4.00	4.00
Manufacturing overhead	3.00	4.33	3.50
Total unit cost	<u>\$10.00</u>	<u>\$11.33</u>	<u>\$10.50</u>

-- Continued on next page --

EXAMPLE 9-8 -- Continued

The comparative income statements make clear the distorting effect:

	<u>Jan-Apr</u>	<u>May-Aug</u>	<u>Sep-Dec</u>	<u>Totals</u>
Sales:				
Produced in Jan-Apr	7,000	3,000		
Produced in May-Aug		4,000	2,000	
Produced in Sep-Dec			5,000	
Expected unit sales	<u>7,000</u>	<u>7,000</u>	<u>7,000</u>	
Expected selling price	<u>× \$12</u>	<u>× \$12</u>	<u>× \$12</u>	
Total expected sales	<u>\$84,000</u>	<u>\$84,000</u>	<u>\$84,000</u>	\$ 252,000
Cost of goods sold:				
From Jan-Apr	\$70,000	\$30,000		
From May-Aug		45,333	\$22,667	
From Sep-Dec			52,500	
Total expected COGS	<u>\$70,000</u>	<u>\$75,333</u>	<u>\$75,167</u>	\$ 220,500
Gross margin	<u>\$14,000</u>	<u>\$ 8,667</u>	<u>\$ 8,833</u>	<u>\$ 31,500</u>

Large fluctuations in gross margin are reported during a period when there was no change at all in the company's underlying cost structure, and sales were the same throughout.

- b. To prevent these distortions in the financial statements, **normal costing** derives a single overhead application rate by looking at the entire year (normal costing is defined in Study Unit 7, Subunit 4).

EXAMPLE 9-9 Extended Example of Normal Costing

Instead of using a different overhead application rate for each budget period, the company uses a single average figure for the entire year.

- The company expects to produce 24,000 units during the year, for an average of 8,000 units per budget period.
- Dividing the fixed overhead of \$20,000 for each budget period by 8,000 units yields a fixed overhead application rate of \$2.50.
- The new total overhead application rate per unit is thus \$3.50 (\$1.00 variable cost + \$2.50 fixed cost), and the new per-unit cost for all 3 budget periods is \$10.50 (\$3.00 direct materials + \$4.00 direct labor + \$3.50 overhead application rate).

The revised income statements prepared using a normalized overhead rate reveal the smoothing effect on gross margin:

	<u>Jan-Apr</u>	<u>May-Aug</u>	<u>Sep-Dec</u>	<u>Totals</u>
Sales:				
Produced in Jan-Apr	7,000	3,000		
Produced in May-Aug		4,000	2,000	
Produced in Sep-Dec			5,000	
Expected unit sales	<u>7,000</u>	<u>7,000</u>	<u>7,000</u>	
Expected selling price	<u>× \$12</u>	<u>× \$12</u>	<u>× \$12</u>	
Total expected sales	<u>\$84,000</u>	<u>\$84,000</u>	<u>\$84,000</u>	\$ 252,000
Cost of goods sold:				
From Jan-Apr	\$73,500	\$31,500		
From May-Aug		42,000	\$21,000	
From Sep-Dec			52,500	
Total expected COGS	<u>\$73,500</u>	<u>\$73,500</u>	<u>\$73,500</u>	\$ 220,500
Gross margin	<u>\$10,500</u>	<u>\$10,500</u>	<u>\$10,500</u>	<u>\$ 31,500</u>

- c. **Extended normal costing** applies a normalized rate to direct costs as well as to manufacturing overhead.
- 1) The following table summarizes the use of rates in the three costing methods described:

	Actual Costing	Normal Costing	Extended Normal Costing
Direct Materials	Actual	Actual	Budgeted
Direct Labor	Actual	Actual	Budgeted
Manufacturing Overhead	Actual	Budgeted	Budgeted

7. Recording Actual Overhead Costs

- a. During the budget period, actual overhead costs are accumulated in the control accounts as they are incurred.

EXAMPLE 9-10 Journal Entries for Actual Overhead Costs	
At the end of October, the company recorded the following journal entries to recognize actual overhead costs incurred during the month:	
Variable overhead control	\$22,050
Raw materials (withdrawals for indirect materials)	\$ 6,059
Wages payable (indirect labor)	4,120
Utilities payable (bill from utility provider)	11,871
Fixed overhead control	\$31,250
Real estate taxes (\$81,000 ÷ 12 months)	\$ 6,750
Insurance expense (\$54,000 ÷ 12 months)	4,500
Depreciation expense (\$240,000 ÷ 12 months)	20,000

8. Allocating Overhead to Work-in-Process

- a. At the end of the period, overhead is applied to work-in-process based on the actual level of the driver (e.g., Actual machine hours × Application rate).

EXAMPLE 9-11 Overhead Journal Entry	
During October, the company produced 91,000 units and expended 4,000 machine hours. The journal entry to apply overhead for October is as follows:	
Work-in-process	\$49,339
Variable overhead applied (91,000 units × \$0.253)	\$23,023
Fixed overhead applied (4,000 hours × \$6.579)	26,316

9. Over- and Underapplied Overhead

- a. Inevitably, the overhead amounts applied throughout the year will vary from the amount actually incurred, which is only determinable once the job is complete. This variance is called over- or underapplied overhead.
 - 1) Overapplied overhead (a credit balance in overhead applied) results when product costs are overstated because the
 - a) Activity level was higher than expected or
 - b) Actual overhead costs were lower than expected.
 - 2) Underapplied overhead (a debit balance in overhead applied) results when product costs are understated because the
 - a) Activity level was lower than expected or
 - b) Actual overhead costs were higher than expected.
- b. If the amount of over- or underapplied overhead is considered **immaterial**, it can be closed directly to cost of goods sold.

EXAMPLE 9-12 Immaterial Over- or Underapplied Overhead			
<u>If overapplied:</u>			
Variable overhead applied (balance)	\$23,023		
Cost of goods sold (difference)		\$	973
Variable overhead control (balance)			22,050
<u>If underapplied:</u>			
Fixed overhead applied (balance)	\$26,316		
Cost of goods sold (difference)	4,934		
Fixed overhead control (balance)		\$	31,250

- c. If the amount of over- or underapplied overhead is considered **material**, it should be allocated based on the relative values of work-in-process, finished goods, and cost of goods sold.

EXAMPLE 9-13 Material Over- or Underapplied Overhead			
Work-in-process, finished goods, and cost of goods sold bear a 20:20:60 cost relationship.			
<u>If overapplied:</u>			
Variable overhead applied (Balance)	\$23,023		
Work-in-process (Overapplied amount × Allocation %)		\$	195
Finished goods (Overapplied amount × Allocation %)			195
Cost of goods sold (Overapplied amount × Allocation %)			583
Variable overhead control (Balance)			22,050
<u>If underapplied:</u>			
Fixed overhead applied (Balance)	\$26,316		
Work-in-process (Underapplied amount × Allocation %)	987		
Finished goods (Underapplied amount × Allocation %)	987		
Cost of goods sold (Underapplied amount × Allocation %)	2,960		
Fixed overhead control (Balance)		\$	31,250

- 1) The most accurate method of allocating over- or underapplied overhead is based on the relative amounts of overhead in work-in-process, finished goods, and cost of goods sold.

9.3 ALLOCATING SERVICE DEPARTMENT COSTS

1. Reasons for Allocation

- a. Service (support) department costs are considered part of overhead (indirect costs). Thus, they cannot feasibly be traced to cost objects and must be allocated to the operating departments that use the services.
 - 1) When service departments also render services to each other, their costs may be allocated to each other before allocation to operating departments.
- b. Four criteria are used to allocate costs:
 - 1) Cause and effect should be used if possible because of its objectivity and acceptance by operating management.
 - 2) Benefits received is the most frequently used alternative when a cause-and-effect relationship cannot be determined.
 - a) However, it requires an assumption about the benefits of costs, for example, that advertising that promotes the company but not specific products was responsible for increased sales by the various divisions.
 - 3) Fairness is sometimes mentioned in government contracts but appears to be more of a goal than an objective allocation base. Management judgment is used to determine fairness.
 - 4) Ability to bear (based on profits) is usually unacceptable because of its dysfunctional effect on managerial motivation.
- c. The three methods of service department allocation in general use are the direct method, the step-down method, and the reciprocal method.
 - 1) With the direct and reciprocal methods, the order of allocation is irrelevant.
 - 2) However, under the step-down method, some service department cost is allocated to other service departments before allocation to the operating departments.
 - a) These first allocations change the proportions of the total constituted by each department.

2. Direct Method

a. The direct method is the simplest. Under the direct method, service department costs are allocated directly to the producing departments without regard for services rendered by service departments to each other.

- 1) Service department costs are allocated to production departments based on an allocation base appropriate to each service department's function.

EXAMPLE 9-14 Service Cost Allocation -- Direct Method

A company has the following service department costs and allocation bases:

<u>Service Department</u>	<u>Costs to Be Allocated</u>	<u>Allocation Base</u>
Information Technology	\$120,000	CPU cycles
Custodial Services	40,000	Floor space
Total	<u>\$160,000</u>	

The production departments have the following preallocation costs and allocation base amounts:

<u>Production Department</u>	<u>Preallocation Costs</u>	<u>CPU Cycles Used</u>	<u>%</u>	<u>Floor Space in Sq. Ft.</u>	<u>%</u>
Milling	\$300,000	60,000,000	62.5%	56,000	70.0%
Finishing	200,000	36,000,000	37.5%	24,000	30.0%
Totals	<u>\$500,000</u>	<u>96,000,000</u>	<u>100.0%</u>	<u>80,000</u>	<u>100.0%</u>

The direct method allocates the service department costs to the production departments as follows:

	<u>Service Departments</u>		<u>Production Departments</u>		
	<u>Information Technology</u>	<u>Custodial Services</u>	<u>Milling</u>	<u>Finishing</u>	<u>Total</u>
Totals before allocation	\$120,000	\$40,000	\$300,000	\$200,000	\$660,000
Allocate IT (62.5%, 37.5%)	(120,000)	—	75,000	45,000	0
Allocate Custodial (70.0%, 30.0%)	—	(40,000)	28,000	12,000	0
Totals after allocation	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$403,000</u>	<u>\$257,000</u>	<u>\$660,000</u>

3. Step (Step-Down) Method

a. Under the step (or step-down) method, some of the costs of services rendered by service departments are allocated to each other.

- 1) This method derives its name from the procedure involved. The service departments are allocated in order, from the one that provides the most service to other service departments down to the one that provides the least.

EXAMPLE 9-15 Service Cost Allocation -- Step Method

The services that each service department provides the other must be ascertained:

Service Department	Provided by IT		Provided by CS	
	CPU Cycles		Floor Space	
	Used	%	in Sq. Ft.	%
Information Technology	196,000,000	98.0%	20,000	80.0%
Custodial Services	4,000,000	2.0%	5,000	20.0%
Totals	200,000,000	100.0%	25,000	100.0%

Looking only at reciprocal service department activity, custodial services provides 80% of its services to information technology, but IT only provides 2% of its services to custodial. Thus, custodial will be allocated first.

The next step is to determine the relative proportions of the three departments that will receive the first allocation (the second allocation will only be distributed to the two production departments, whose allocation bases were determined under the direct method in Example 9-14).

Allocate Custodial Services:	Floor Space		Amount to	Departmental
	in Sq. Ft.	%	Be Allocated	Allocations
To Milling	56,000	56.0%	\$40,000	\$22,400
To Finishing	24,000	24.0%	40,000	9,600
To Information Technology	20,000	20.0%	40,000	8,000
Totals	100,000	100.0%		\$40,000

The step-down allocation is performed as follows:

	Service Departments		Production Departments		Total
	Custodial Services	Information Technology	Milling	Finishing	
Totals before allocation	\$ 40,000	\$120,000	\$300,000	\$200,000	\$660,000
Allocate Custodial	(40,000)	8,000	22,400	9,600	0
Totals after first allocation	\$ 0	\$128,000	\$322,400	\$209,600	\$660,000

Allocate IT:	CPU Cycles		Amount to	Departmental
	Used	%	Be Allocated	Allocations
To Milling	60,000,000	62.5%	\$128,000	\$ 80,000
To Finishing	36,000,000	37.5%	128,000	48,000
Totals	96,000,000	100.0%		\$128,000

	Information Technology		Production Departments		Total
	Technology		Milling	Finishing	
Totals after first allocation	\$128,000		\$322,400	\$209,600	\$660,000
Allocate IT	(128,000)		80,000	48,000	0
Totals after second allocation	\$ 0		\$402,400	\$257,600	\$660,000

4. Reciprocal Method

a. The reciprocal method is the most complex and the most theoretically sound of the three methods. It is also known as the simultaneous solution method, cross allocation method, matrix allocation method, or double distribution method.

- 1) Under the reciprocal method, services rendered by all service departments to each other are recognized.

EXAMPLE 9-16 Service Cost Allocation -- Reciprocal Method

The reciprocal method requires calculating the allocation base amounts for information technology; i.e., the service department that was not allocated to the other service department under the step method (Example 9-15).

Allocate Information Technology:	CPU Cycles	
	Used	%
To Milling	60,000,000	60.0%
To Finishing	36,000,000	36.0%
To Custodial Services	4,000,000	4.0%
Totals	<u>100,000,000</u>	<u>100.0%</u>

Use linear algebra to calculate fully reciprocated information technology costs (FRITC) and fully reciprocated custodial services costs (FRCSC):

$$\begin{aligned}\text{FRITC} &= \text{Preallocation IT costs} + (\text{FRCSC} \times \text{Portion of custodial effort used by IT}) \\ &= \$120,000 + (\text{FRCSC} \times 20\%)\end{aligned}$$

$$\begin{aligned}\text{FRCSC} &= \text{Preallocation custodial costs} + (\text{FRITC} \times \text{Portion of IT effort used by custodial}) \\ &= \$40,000 + (\text{FRITC} \times 4\%)\end{aligned}$$

These algebraic equations can be solved simultaneously.

$$\begin{aligned}\text{FRITC} &= \$120,000 + (\text{FRCSC} \times 20\%) \\ &= \$120,000 + \{[\$40,000 + (\text{FRITC} \times 4\%)] \times 20\%\} \\ &= \$120,000 + [(\$40,000 + .04\text{FRITC}) \times .2] \\ &= \$120,000 + \$8,000 + .008\text{FRITC} \\ .992\text{FRITC} &= \$128,000 \\ \text{FRITC} &= \$129,032 \\ \text{FRCSC} &= \$40,000 + (\text{FRITC} \times 4\%) \\ &= \$40,000 + (\$129,032 \times .04) \\ &= \$40,000 + \$5,161 \\ &= \$45,161\end{aligned}$$

The reciprocal allocation is performed as follows:

	Service Departments		Production Departments		
	Custodial Services	Information Technology	Milling	Finishing	Total
Totals before allocation	\$40,000	\$120,000	\$300,000	\$200,000	\$660,000
Allocate Custodial Services (20.0%, 56.0%, 24.0%)	(45,161)	9,032	25,290	10,839	0
Allocate Information Tech. (4.0%, 60.0%, 36.0%)	5,161	(129,032)	77,419	46,452	0
Totals after allocation	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$402,709</u>	<u>\$257,291</u>	<u>\$660,000</u>

5. Single-Rate vs. Dual-Rate Allocation

- a. The examples presented employed a single rate to allocate the costs of each support department. Some firms find that employing dual-rate allocation provides more useful information.

EXAMPLE 9-17 Dual-Rate Service Cost Allocation

The company has decided to allocate the IT department's costs using a dual-rate method, one rate for the costs of IT's investment in hardware and software (fixed) and another rate for the costs of services provided (variable).

- The IT department has determined that \$40,000 of its total allocable costs are associated with variable costs. These will henceforth be allocated using technician and programmer hours.
- The company's technicians and programmers worked a total of 1,600 hours on projects for the Milling and Finishing Departments during the period. Variable IT costs will thus be applied at the rate of \$25 per hour ($\$40,000 \div 1,600$).
- The remaining \$80,000 of allocable IT costs are associated with the department's investment in fixed plant. These costs will continue to be allocated using CPU cycles.
- Since the company's central computers consumed a total of 96 million CPU cycles doing processing for the Milling and Finishing Departments during the period, fixed IT costs will be applied at the rate of \$0.00083 per cycle ($\$80,000 \div 96,000,000$).

The dual-rate allocations will be made as follows:

<u>Allocate to Milling:</u>	<u>Driver Units Consumed</u>		<u>Application Rate</u>		<u>Totals</u>
Variable IT costs	640 hours	×	\$25.00	=	\$16,000
Fixed IT costs	60,000,000 cycles	×	\$0.00083	=	50,000
Total					<u>\$66,000</u>

<u>Allocate to Finishing:</u>	<u>Driver Units Consumed</u>		<u>Application Rate</u>		<u>Totals</u>
Variable IT costs	960 hours	×	\$25.00	=	\$24,000
Fixed IT costs	36,000,000 cycles	×	\$0.00083	=	30,000
Total					<u>\$54,000</u>

The total amount of IT department costs has been allocated ($\$66,000 + \$54,000 = \$120,000$).

- b. The dual-rate method can be used to refine a system currently using a single rate under any of the other methods (direct, step-down, or reciprocal).

STUDY UNIT TEN

SUPPLY CHAIN MANAGEMENT

10.1	<i>Just-in-Time Inventory and Lean Manufacturing (Lean Resource Management)</i>	1
10.2	<i>Enterprise Resource Planning and Outsourcing</i>	5
10.3	<i>Theory of Constraints and Throughput Costing</i>	12
10.4	<i>Capacity Management</i>	17

This study unit is the **fourth of five** on **cost management**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The five study units are

- Study Unit 7: Cost Management Concepts
- Study Unit 8: Cost Accumulation Systems
- Study Unit 9: Cost Allocation Techniques
- Study Unit 10: Supply Chain Management
- Study Unit 11: Business Process Improvement

If you are interested in reviewing more introductory or background material, go to www.GLEM.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following GLEM outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

10.1 JUST-IN-TIME INVENTORY AND LEAN MANUFACTURING (LEAN RESOURCE MANAGEMENT)

1. Overview

- a. Supply chain management involves a two-way exchange of information. For example, a customer may be able to track the progress of its order, and the supplier may be able to monitor the customer's inventory. Thus, the customer has better information about order availability, and the supplier knows when the customer's inventory needs replenishment.
- b. Modern inventory planning favors the **just-in-time (JIT)** model. Many companies have traditionally built parts and components for subsequent operations on a preset schedule. Such a schedule provides a cushion of inventory so that the next operation will always have parts to work with—a just-in-case method. In this respect, traditional inventory planning views inventory as an asset.
 - 1) In contrast, JIT views inventory as a liability and limits output to the demand of the subsequent operation. Reductions in inventory levels result in less money invested in idle assets; reduction of storage space requirements; and lower inventory taxes, pilferage, and obsolescence risks.

- c. JIT is a reaction to the trends of global competition and rapid technological progress that have resulted in shorter product life cycles and greater consumer demand for product diversity.
 - 1) High inventory levels often mask production problems because defective parts can be overlooked when plenty of good parts are available. If only enough parts are made for the subsequent operation, however, any defects will immediately halt production.
 - 2) The focus of quality control under JIT shifts from the discovery of defective parts to the prevention of quality problems, so zero machine breakdowns (achieved through preventive maintenance) and zero defects are ultimate goals. Higher quality and lower inventory go together.
- d. **Lean manufacturing** expands the concept of JIT. The central focus is on accomplishing more with fewer resources (labor, equipment, space, etc.), providing customers with what they want, and meeting their expectations.
 - 1) The five principles of lean manufacturing are described below. The phrases “lean resource management” or “the lean process” are sometimes used to imply that many of the techniques can be used by non-manufacturers.
 - a) **Value** involves identifying the features of the product or service that are valuable for the customer.
 - b) **Value stream** requires (1) examining every process within the production of a product, (2) identifying processes that add value, and (3) removing processes (if possible) that do not add value.
 - c) **Flow and pull** incorporates designing the production process to be capable of maximizing the flow of the product initiated by the pull of customer demand (i.e., only produce what the customer wants). This is where lean manufacturing and JIT principles coincide.
 - d) **Empowerment** provides each employee with the knowledge and authority to make valuable and timely decisions in order to (1) add customer value and (2) eliminate waste from the process.
 - e) **Perfection** focuses on making incremental improvement in each process with perfection as the goal.

2. Role of Kanban



SUCCESS TIP

The Japanese term “kanban” and JIT are often confused. For the purpose of this exam, CMA candidates will be required to understand the benefits of implementing JIT systems and how kanban is used in the process.

- a. JIT is a total system of purchasing, production, and inventory control. **Kanban** is one of the many elements in the JIT system and was developed by the Toyota Motor Corporation (kanban is not characteristic of Japanese industry as a whole).

- b. Kanban means ticket. Tickets (also described as cards or markers) control the flow of production or parts so that they are produced or obtained in the needed amounts at the needed times. A basic kanban system includes
 - 1) A withdrawal kanban that states the quantity that a later process should withdraw from its predecessor;
 - 2) A production kanban that states the output of the preceding process; and
 - 3) A vendor kanban that tells a vendor what, how much, where, and when to deliver.
- c. Kanban is essentially a visual workflow management system.
 - 1) When a worker sees a kanban, the card or ticket acts as authorization to release inventory to the next step. Work cannot move to the next stage until a kanban indicates the next stage is ready for it.
- d. U.S. companies have not been comfortable with controlling production using tickets on the production floor. **Computerized information systems** have been used for many years, and U.S. companies have been reluctant to give up their computers in favor of the essentially manual kanban system. Instead, U.S. companies have integrated their existing complex computerized planning systems with the JIT system.

3. Objectives

- a. Higher productivity, reduced order costs as well as carrying costs, faster and cheaper setups, shorter manufacturing cycle times, better due date performance, improved quality, and more flexible processes are objectives of JIT methods.
 - 1) The ultimate goal is increased competitiveness and higher profits.

4. Features

- a. JIT is a **pull system**; i.e., items are pulled through production by current demand, not pushed through by anticipated demand.
 - 1) One operation produces only what is needed by the next operation, and components and raw materials arrive just in time to be used.
- b. Demand-driven production allows **inventory levels to be minimized**. Counting, handling, and storing inventory are viewed as nonvalue-added.
 - 1) Indeed, carrying inventory is regarded as a symptom of correctable problems, such as poor quality, long cycle times, and lack of coordination with suppliers.
- c. Organizations that adopt JIT systems frequently experience a decrease in the number of suppliers.
 - 1) Consequently, organizations must exercise care to select suppliers that reliably deliver high-quality products.
 - 2) A decrease in suppliers typically results in an increase in the number of deliveries.
- d. Buyer-supplier relationships are facilitated by **electronic data interchange (EDI)**, a technology that allows the supplier access to the buyer's online inventory management system. Thus, electronic messages replace paper documents (purchase orders and sales invoices), and the production schedules and deliveries of the parties can be more readily coordinated.

5. Effects on Operations

- a. One consequence of the lower inventory levels associated with a JIT (lean) system is elimination of the need for certain internal controls.
 - 1) Frequent receipt of deliveries from suppliers often means less need for a sophisticated inventory control system and for control personnel.
- b. JIT also may eliminate central receiving areas, hard copy receiving reports, and storage areas.
- c. Manufacturing lead time is reduced.
 - 1) As a result, on-time delivery performance and response to changes in markets are enhanced, and production of customized goods in small lots becomes feasible.
- d. Production setup costs and times per lot are reduced. However, because smaller lots are used, total setup costs are not necessarily lower using a JIT production system.
- e. The quality of parts provided by suppliers is verified by use of statistical controls rather than inspection of incoming goods.
 - 1) Storage, counting, and inspecting are eliminated in an effort to perform only value-adding work.

6. Implementing JIT

- a. To implement a JIT inventory or lean production system, manufacturing is reorganized around what are called **manufacturing cells**.
 - 1) In a conventional plant layout, each department or function operates specialized machines that perform one task. All work moves from department to department.
 - 2) In a cellular layout, each cell is a miniature manufacturing plant. Cells are sets of machines, often grouped in semicircles, that produce a given product or product family.
- b. Each worker in a cell must be able to operate all machines and, possibly, to perform support tasks, such as setup activities, preventive maintenance, movement of work-in-process within the cell, and quality inspection.
 - 1) In such a pull system, workers might often be idle if they are not multi-skilled.
- c. A cellular organization requires workers to operate as effective teams, so employee empowerment is crucial in a JIT inventory or lean production system.
 - 1) Greater participation by employees is needed to achieve the objectives of continuous improvement and zero defects, so they may, for example, have the power to stop production to correct a problem, be consulted about changes in processes, or become involved in hiring co-workers. Thus, managers in such a system usually play more of a facilitating than a support role.

10.2 ENTERPRISE RESOURCE PLANNING AND OUTSOURCING

1. Overview

- a. Short-range (tactical or operational) plans must be converted into specific production targets for finished goods. The raw materials going into the creation of these end products must be carefully scheduled for delivery.
 - 1) The yearly/quarterly/monthly numbers and styles of finished goods called for in the demand forecasts included in the operational plans must be turned into specific dates for completion and availability for shipment to the customer. This is the task of the **master production schedule (MPS)**.

2. Materials Requirements Planning (MRP)

- a. MRP is a computerized system for moving materials through a production process according to a predetermined schedule.
- b. An MRP system enables a company to efficiently fulfill the requirements of the MPS by coordinating both the manufacture of component parts for finished goods and the arrival of the raw materials necessary to create the intermediate components.
 - 1) As computers were introduced into manufacturing, it was common for firms to have a production scheduling system and an inventory control system. MRP joins the two into a single application.
 - 2) The three overriding goals of MRP are the arrival of the right part, in the right quantity, at the right time.
- c. MRP is a **push system**; i.e., the demand for raw materials is driven by the forecasted demand for the final product, which can be programmed into the computer.
 - 1) For example, an automobile manufacturer need only tell the computer how many autos of each type are to be manufactured.
- d. MRP, in effect, creates schedules of when items of inventory will be needed in the production departments.
 - 1) If parts are not in stock, the system automatically generates a purchase order on the proper date (considering lead times) so that deliveries will arrive on time.
 - 2) The timing of deliveries is vital to avoid both production delays and a pileup of raw materials inventory that must be stored.
- e. The MRP system consults the **bill of materials (BOM)**, a record of which (and how many) subassemblies go into the finished product. The system then generates a complete list of every part and component needed.

EXAMPLE 10-1 MRP

A manufacturer has the following bill of materials for its main product AA115:

<u>Subunit</u>	<u>Quantity</u>
CM12	1
PR75	5

The bill of materials for the component subunits is as follows:

<u>Subunit</u>	<u>Contains</u>	<u>Quantity</u>
CM12	TT413	2
	XH511	3
PR75	LQ992	1

Current inventory quantities are as follows:

<u>Subunit</u>	<u>On Hand</u>
CM12	25
PR75	35
LQ992	50
TT413	30
XH511	40

The company has 20 units of finished AA115 in inventory and wishes to maintain this level throughout the year. Production of 40 units is scheduled for the upcoming month. The principal subunits that must be produced are calculated as

<u>Subunit</u>	<u>Quantity per Finished Product</u>		<u>Production Run</u>		<u>Quantity Needed</u>		<u>Quantity On Hand</u>		<u>To Be Built</u>
CM12	1	x	40	=	40	-	25	=	15
PR75	5	x	40	=	200	-	35	=	165

The parts that must be ordered from vendors can thus be calculated as follows:

<u>Subunit</u>	<u>Components</u>	<u>Component Quantity</u>		<u>Subunits To Be Built</u>		<u>Quantity Needed</u>		<u>Quantity On Hand</u>		<u>To Be Purchased</u>
CM12	TT413	2	x	15	=	30	-	30	=	0
	XH511	3	x	15	=	45	-	40	=	5
PR75	LQ992	1	x	165	=	165	-	50	=	115

The lead times required are

CM12	2 weeks
TT413	1 week
XH511	2 weeks
PR75	3 weeks
LQ992	1 week
AA115	1 week

-- Continued on next page --

EXAMPLE 10-1 -- Continued

Product	Action	Weeks					
		1	2	3	4	5	6
AA115	Due Date						40
	Place Order					40	
PR75	Due Date					165	
	Place Order		165				
LQ992	Due Date		115				
	Place Order	115					
CM12	Due Date					15	
	Place Order			15			
TT413	Due Date			0			
	Place Order		0				
XH511	Due Date			5			
	Place Order	5					

- f. When items of inventory will be needed is determined by their lead time and the lead times of other component items or procedures. **Lead time** is the amount of time between when a process starts and when it is completed.
- 1) Once it is known how many of a final product is necessary, the lead times of all the included components and processes can be used to create a schedule outlining when certain items and components must be ordered or produced so that the final product will be completed on time.
- g. Some benefits of MRP are
- 1) Reduced idle time
 - 2) Lower setup costs
 - 3) Lower inventory carrying costs
 - 4) Increased flexibility in responding to market changes

3. Manufacturing Resource Planning (MRP II)

- a. MRP II is a closed-loop manufacturing system that integrates all facets of a manufacturing business, including production, sales, inventories, schedules, and cash flows.
- 1) The same system is used for both the financial reporting and managing operations (both use the same transactions and numbers).
 - 2) Because manufacturing resource planning encompasses materials requirements planning, MRP is a component of an MRP II system.

4. Traditional Enterprise Resource Planning (ERP)

- a. MRP is a common function contained in ERP.
- 1) Although ERP and MRP are similar, they are not interchangeable since ERP includes functions not included in MRP.
 - 2) An ERP system allows a company to determine what hiring decisions might need to be made or whether a company should invest in new capital assets.
 - a) A company that merely needed to maintain inventory and materials levels would only need to implement an MRP system.

- b. The traditional ERP system is one in which subsystems share data and coordinate their activities. ERP is intended to integrate enterprise-wide information systems across the organization (accounting, supply chain management, treasury, etc.) by creating one database linked to all of the entity's applications.
- 1) Thus, if marketing receives an order, it can quickly verify that inventory is sufficient to notify shipping to process the order.
 - a) Otherwise, production is notified to manufacture more of the product, resulting in an automatic adjustment of output schedules.
 - b) If materials are inadequate for this purpose, the system issues a purchase order.
 - c) If more labor is needed, human resources is instructed to reassign or hire employees.
 - d) The foregoing business processes (and others) should interact seamlessly in an ERP system. Moreover, the current generation of ERP software also provides the capability for smooth (and instant) interaction with the business processes of external parties.
 - 2) The subsystems in a traditional ERP system are internal to the organization. Thus, they often are called **back-office functions**. The information produced is principally (but not exclusively) intended for internal use by the organization's managers.
 - 3) Because ERP software is costly and complex, it is usually installed only by the largest enterprises. However, mid-size organizations are increasingly likely to buy ERP software.
 - 4) The tremendous variety of forms that information systems can take and the diverse needs of users have led to the concept of information resources management (IRM), which takes a global view of the information holdings and needs of an organization. This view is promoted by the Information Resources Management Association of Hershey, PA (www.irma-international.org).
- c. Generally, separate financial and nonfinancial systems have the increased potential to experience
- 1) Different reporting results (requiring personnel to research discrepancies instead of actually enhancing operations)
 - 2) Inefficiencies
 - 3) A lack of standardized purchasing among departments (i.e., departments as a whole not taking advantage of quantity discounts)
- NOTE: This list is not all-inclusive.
- d. ERP connects all **functional financial and non-financial subsystems** (human resources, the financial accounting system, production, marketing, distribution, purchasing, receiving, order processing, shipping, etc.) and also connects the organization with its suppliers and customers.
- 1) Thus, ERP facilitates demand analysis and materials requirements planning.
 - 2) By decreasing lead times, it improves just-in-time inventory management.
 - 3) Even more importantly, ERP's coordination of all operating activities permits flexible responses to shifts in supply and demand.

- e. The **advantages** of developing a traditional ERP system are similar to those derived from business process reengineering.
 - 1) Using ERP software that reflects the best practices forces the linked subunits in the organization not only to redesign and improve their processes but also to conform to one standard.
 - 2) An organization may wish to undertake a reengineering project before choosing ERP software. The project should indicate what best practices already exist in the organization's processes. This approach may be preferable for a unique enterprise in a highly differentiated industry.
 - a) Carrying out a reengineering project before installing an ERP system defines what process changes are needed and which vendor software should be used.
 - b) If the organization is not especially unique, vendor software probably is already based on industry best practices.
 - 3) The processes reflected in the ERP software may differ from the organization's. In this case, the better policy is usually to change the organization's processes. Customizing the ERP software is expensive and difficult, and it may result in bugs and awkwardness when adopting upgrades.
 - a) Implementing an ERP system is likely to encounter significant resistance because of its comprehensiveness. Most employees will have to change ingrained habits and learn to use new technology. Thus, successful implementation requires effective change management.
- f. The **disadvantages** of traditional ERP are its extent and complexity, which make customization of the software difficult and costly.

5. Current Generation of ERP

- a. The current generation of ERP software (ERP II) has added **front-office functions**, which provide the capability for smooth (and instant) interaction with the business processes of **external** parties such as customers, suppliers, shareholders or other owners, creditors, and strategic allies (e.g., the members of a trading community or other business association). Accordingly, an ERP II system has the following interfaces with its back-office functions:
 - 1) **Supply-chain** management applications for an organization focus on relationships extending from its suppliers to its final customers. Issues addressed include distribution channels, warehousing and other logistical matters, routing of shipments, and sales forecasting.
 - a) In turn, one organization's supply chain is part of a linked chain of multiple organizations. This chain stretches from the producers of raw materials, to processors of those materials, to entities that make intermediate goods, to assemblers of final products, to wholesalers, to retailers, and lastly to consumers.
 - b) Supply chain management involves a two-way exchange of information. For example, a customer may be able to track the progress of its order, and the supplier may be able to monitor the customer's inventory. Thus, the customer has better information about order availability, and the supplier knows when the customer's inventory needs replenishment.
 - c) An advanced planning and scheduling system may be an element of a supply chain management application for a manufacturer. It controls the flow of material and components within the chain. Schedules are created given projected costs, lead times, and inventories.

- 2) **Customer relationship** management applications extend to customer service, finance-related matters, sales, and database creation and maintenance.
 - a) Integrated data is helpful in better understanding customer needs, such as product preference or location of retail outlets. Thus, the organization may be able to optimize its sales forecasts, product line, and inventory levels.
 - i) Business intelligence software is used to analyze customer data.
- 3) **Partner relationship** management applications connect the organization not only with such partners as customers and suppliers but also with owners, creditors, and strategic allies (for example, other members of a joint venture).
 - a) Collaborative business partnerships may arise between competitors or between different types of organizations, such as a manufacturer partnering with an environmental group. Special software may be helpful to the partners in sharing information, developing a common strategy, and measuring performance.

b. Advantages of a Current ERP System

- 1) Lower inventory costs
- 2) Better management of liquid assets
- 3) Reduced labor costs and greater productivity
- 4) Enhanced decision making
- 5) Elimination of data redundancy, centralization of data, and protection of data integrity
- 6) Avoidance of the costs of other means of addressing needed IT changes
- 7) Increased customer satisfaction
- 8) More rapid and flexible responses to changed circumstances
- 9) More effective supply chain management
- 10) Integration of global operations
- 11) Standardization and simplification of the decision-making process

c. Disadvantages of a Current ERP System

- 1) Losses from an unsuccessful implementation, e.g., sales declines
- 2) Purchasing hardware, software, and services
- 3) Data conversion from legacy systems to the new integrated system (although conversion software may help)
- 4) Training
- 5) Design of interfaces and customization
- 6) Software maintenance and upgrades
- 7) Salaries of employees working on the implementation

6. Challenges

- a. **Implementation** of ERP may take years and cost millions. Moreover, a poor implementation may cause the project to fail regardless of the quality of the software.
 - 1) However, more rapid and less costly implementation may be possible if no customization is done.
- b. The initial step is to do strategic planning and to organize a project team that is representative of affected employee groups.
- c. The second step is to choose ERP software and a consulting firm.
 - 1) One possibility is to choose the software before the consultants because the first decision may affect the second.
 - 2) Another option is to hire consultants to help with the selection of the software.
 - a) The organization then may hire other consultants to help with implementation.
- d. The third and longest step is preimplementation.
 - 1) The length of the process design phase is a function of the extent of
 - a) Reengineering
 - b) Customization of the software
 - 2) Data conversion may be delayed because all departments must agree on the meaning of every data field, i.e., what values will be considered valid for that field.
 - 3) The ERP system and its interfaces must be tested.
- e. Implementation ("going live") is not the final step. Follow-up is necessary to monitor the activities of employees who have had to change their routines. For example, a mistake caused by reverting to the old method of entering a sales order may have pervasive consequences in a new integrated system: a credit check, rescheduling of production, and ordering of materials.
- f. Training should be provided during implementation not only regarding technical matters but also to help employees understand the reasons for process changes. For example, the employees who enter sales orders should know what the effects will be throughout the system.
 - 1) Other change management techniques include effective communication to allay employee fears and the creation of user-friendly documents and interfaces.

7. Outsourcing

- a. Outsourcing is the management or day-to-day execution of an entire business function by a third-party service provider. Outsourced services may be provided on or off premises, in the same country, or in a separate country.
 - 1) Outsourcing enables a company to focus on its core business rather than having to be concerned with marginal activities. For example, payroll preparation is often outsourced because a company does not want to maintain a full-time staff to perform what is only a weekly or monthly activity.
- b. Business process outsourcing is the outsourcing of back office and front office functions typically performed by white-collar and clerical workers. Examples of these functions include data processing, accounting, human resources, and medical coding and transcription.
 - 1) **Insourcing** is the transfer of an outsourced function to an internal department of a company to be managed entirely by company employees. The term has also been used to describe a foreign company's locating of facilities in a host country where it employs local workers.
 - 2) **Cosourcing** is performance of a business function by both internal staff and external resources, such as consultants or outsourcing vendors, who have specialized knowledge of the business function.
- c. Benefits of outsourcing include reliable service, reduced costs, avoidance of the risk of obsolescence, and access to technology. Limitations include dependence on an outside party and loss of control over a necessary function.

10.3 THEORY OF CONSTRAINTS AND THROUGHPUT COSTING

1. Overview

- a. The theory of constraints (TOC) is a system to improve human thinking about problems. It has been greatly extended to include manufacturing operations.
- b. The basic premise of TOC as applied to business is that improving any process is best done not by trying to maximize efficiency in every part of the process, but by focusing on the slowest part of the process.
- c. The slowest part of the process is called the **constraint**.
 - 1) Constraints cause backup in the process because other processes need to "wait" for the constraint to finish before the next process can be completed.
 - 2) Increasing the efficiency of processes that are not constraints merely creates backup in the system.

- d. The **steps in a TOC analysis** are as follows (they are described in more detail under items 2. through 6. beginning below):

- 1) Identify the constraint.
- 2) Determine the most profitable product mix given the constraint.
- 3) Maximize the flow through the constraint.
- 4) Increase capacity at the constraint.
- 5) Redesign the manufacturing process for greater flexibility and speed.

2. Step 1 -- Identify the Constraint

- a. The bottleneck operation can usually be identified as the one where work-in-process backs up the most.
- b. A more sophisticated approach is to analyze available resources (number and skill level of employees, inventory levels, time spent in other phases of the process) and determine which phase has negative slack time, i.e., the phase without enough resources to keep up with input.

3. Step 2 -- Determine the Most Profitable Product Mix Given the Constraint

- a. A basic principle of TOC analysis is that **short-term** profit maximization requires maximizing the contribution margin through the constraint, called the **throughput margin** or throughput contribution. This short-term focus may be contrasted with the long-term focus of activity-based costing. It considers all costs as a basis for strategic pricing and profit planning.
 - 1) To make the most profitable use of the bottleneck operation, managers may need to produce the product with the highest throughput margin per unit, not the highest contribution margin per unit.
- b. **Throughput costing**, sometimes called **supervariable costing**, recognizes only **direct materials costs** as being truly variable and thus relevant to the calculation of throughput margin. All other manufacturing costs are ignored because they are considered fixed in the short run.
 - 1) Even direct labor is considered a fixed cost, which makes sense considering that many companies have union contracts or policies that involve employing laborers, or at least paying them, even when no work is available.

$$\text{Throughput margin} = \text{Sales} - \text{Direct materials}$$



**AUTHOR'S
NOTE**

Throughput margin is also called throughput contribution. Also, for external reporting purposes, absorption costing must be used. Therefore, an adjusting entry will be required to restate the inventory accounts for external reporting purposes.

EXAMPLE 10-2 Supervariable & Variable -- Absorption Costing

During its first month in business, a firm produced 100 units and sold 80 while incurring the following costs:

Direct materials	<u>\$1,000</u>
Manufacturing costs used in supervariable costing	<u>\$1,000</u>
Direct labor	2,000
Variable overhead	<u>1,500</u>
Manufacturing costs used in variable costing	<u>\$4,500</u>
Fixed overhead	<u>3,000</u>
Manufacturing costs used in absorption costing	<u><u>\$7,500</u></u>

- c. The different costing methods can result in drastically different costs per unit, which can affect the value of ending inventory.

EXAMPLE 10-3 Supervariable & Variable -- Absorption Costing

The following calculations use data from Example 10-2:

Cost per unit:

Absorption ($\$7,500 \div 100$ units)	\$75
Variable ($\$4,500 \div 100$ units)	45
Supervariable ($\$1,000 \div 100$ units)	10

Ending inventory:

Absorption ($\$75 \times 20$ units)	\$1,500
Variable ($\$45 \times 20$ units)	900
Supervariable ($\$10 \times 20$ units)	200

Note that, because throughput costing capitalizes so few costs as product costs, ending inventory and cost of goods sold are lower than under variable costing and much lower than under absorption costing.

Below is a comparison of the calculation of operating income under two of the methods. The units were sold at a price of \$100 each. The company incurred \$200 of variable selling and administrative expenses and \$600 of fixed selling and administrative expenses.

Note the drastic reduction in operating income resulting from the treatment of so many costs as period costs under throughput costing.

Variable Costing		Supervariable Costing	
Sales	\$8,000	Sales	\$8,000
Beginning inventory	\$ 0	Beginning inventory	\$ 0
Variable manufacturing costs	<u>4,500</u>	Direct materials costs	<u>1,000</u>
Goods available for sale	<u>\$4,500</u>	Goods available for sale	<u>\$1,000</u>
Less: Ending inventory	<u>(900)</u>	Less: Ending inventory	<u>(200)</u>
Variable cost of goods sold	(3,600)	Supervariable cost of goods sold	(800)
Variable S&A expenses	<u>(200)</u>		
Contribution margin	<u>\$4,200</u>	Throughput margin	<u>\$7,200</u>
		Direct labor	(2,000)
Fixed overhead	(3,000)	Variable overhead	(1,500)
Fixed S&A expenses	<u>(600)</u>	Variable S&A expenses	(200)
Operating income	<u><u>\$ 600</u></u>	Fixed overhead	(3,000)
		Fixed S&A expenses	<u>(600)</u>
		Operating loss	<u><u>\$ (100)</u></u>

- d. To determine the most profitable use of the bottleneck operation, a manager next calculates the throughput margin per unit of time spent in the constraint.
- Profitability is maximized by keeping the bottleneck operation busy with the product with the highest throughput margin per unit of time.

- e. When there are multiple products, the crucial factor in determining the optimal product mix is not which product is the most profitable in terms of absolute throughput margin, but which one has the highest **throughput margin per hour**.

EXAMPLE 10-4 Supervariable & Variable -- Absorption Costing

Using the data from Examples 10-2 and 10-3, under the current setup, the unit spends 5 hours in the manufacturing process.

Price	\$100
Less: Materials cost	<u>(10)</u>
Throughput margin	\$ 90
Divided by: Constraint time	<u>÷ 5</u>
Throughput margin per hour	<u>\$ 18</u>

- f. To derive the most profitable product mix given finite resources, customer demand must be taken into account.
- 1) The available time in the bottleneck operation is first devoted to the product with the highest throughput margin per hour, then to the other products in descending order until the company is unable to meet additional demand.

EXAMPLE 10-5 Throughput Margin

A firm sells 3 products, A, B, and C, with throughput margins per hour of \$18, \$15, and \$12, respectively. Product A spends 5 hours in the bottleneck, Product B spends 5 hours in the bottleneck, and Product C spends 4 hours in the bottleneck. The firm has determined it can sell 10 units of A, 8 units of B, and 6 units of C. The firm has a total of 100 hours available.

	Highest TM: A	2nd Highest TM: B	3rd Highest TM: C
Hours available	100	50	10
Hours needed to fill demand:			
Demand in units	10	8	6
Hours per unit in bottleneck	× 5 (50)	5 (40)	4 (24)
Hours remaining	<u>50</u>	<u>10</u>	<u>(14)</u>

The firm will forgo some sales of Product C to focus on products that are more profitable.

4. Step 3 -- Maximize the Flow Through the Constraint

- a. Production flow through a constraint is managed using the **drum-buffer-rope (DBR)** system.
 - 1) The drum (i.e., the beat to which a production process marches) is the bottleneck operation. The constraint sets the pace for the entire process.
 - 2) The buffer is a minimal amount of work-in-process input to the drum that is maintained to ensure that it is always in operation.
 - 3) The rope is the sequence of activities preceding and including the bottleneck operation that must be coordinated to avoid inventory buildup.

5. Step 4 -- Increase Capacity at the Constraint

- a. In the short run, TOC encourages a manager to make the best use of the bottleneck operation. The medium-term step for improving the process is to increase the bottleneck operation's capacity.

6. Step 5 -- Redesign the Manufacturing Process for Greater Flexibility and Speed

a. The long-term solution is to reengineer the entire process. The firm should take advantage of new technology, product lines requiring too much effort should be dropped, and remaining products should be redesigned to ease the manufacturing process.

- 1) Value engineering is useful for this purpose because it explicitly balances product cost and the needs of potential customers (product functions).

EXAMPLE 10-6 TOC Calculation

Step 1 -- Identify the Constraint

A company makes three products: an airborne radar unit, a seagoing sonar unit, and a ground sonar unit. Under the current setup, the hours spent by each product in the two phases of the manufacturing process are as follows:

Product	Assembly	Testing
Airborne Radar	3	4
Seagoing Sonar	8	10
Ground Sonar	5	5

The company has 150 hours available every month for testing. Under the current setup, therefore, the testing phase is the constraint.

Step 2 -- Determine the Most Profitable Product Mix Given the Constraint

The company calculates the throughput margin on each product and divides by the hours spent in testing:

	Airborne Radar	Seagoing Sonar	Ground Sonar
Price	\$200,000	\$600,000	\$300,000
Less: Materials costs	(100,000)	(400,000)	(250,000)
Throughput margin	\$100,000	\$200,000	\$ 50,000
Divided by: Constraint time	÷ 4	÷ 10	÷ 5
Throughput margin per hour	\$ 25,000	\$ 20,000	\$ 10,000

To derive the most profitable product mix given finite resources, customer demand must be taken into account. The company has determined that it can sell 12 units of airborne radar, 6 units of seagoing sonar, and 22 units of ground sonar per month.

The available time in the bottleneck operation is first devoted to the product with the highest throughput margin (TM), then in descending order until the company is unable to meet demand.

In the calculation below, the hours remaining after assignment to each product are the hours that can be devoted to the next product.

	Highest TM: Airborne Radar	2nd Highest TM: Seagoing Sonar	Lowest TM: Ground Sonar
Hours available	150	102	42
Hours needed to fulfill demand:			
Demand in units	12	6	22
Hours per unit in bottleneck	× 4 (48)	× 10 (60)	× 5 (110)
Hours remaining	<u>102</u>	<u>42</u>	<u>(68)</u>

Applying the principles of TOC, the company will forgo some sales of the ground sonar in favor of products that are more profitable given the current constraint.

-- Continued on next page --

EXAMPLE 10-6 -- Continued**Step 3 -- Maximize the Flow Through the Bottleneck Operation**

The company will apply a drum-buffer-rope system to ensure that the bottleneck operation stays busy on high-TM products while keeping work-in-process inventory to a minimum.

Step 4 -- Increase Capacity at the Bottleneck Operation

The company will hire and train more employees for the testing department.

Step 5 -- Redesign the Manufacturing Process for Greater Flexibility and Speed

The company will examine its markets and new manufacturing technology to determine which products it wants to continue selling, whether to add new ones, and whether to retool the production line.

10.4 CAPACITY MANAGEMENT**1. Capacity Levels**

- a. **Theoretical capacity** is the optimal level of output that can be completed with zero downtime and zero waste.
 - 1) Budgeting and planning based on theoretical capacity is not recommended because it is not reasonable to assume zero downtime (i.e., no maintenance) and zero waste.
- b. **Practical capacity** is the highest level of output that can reasonably be attained taking into account both planned and unplanned downtime (e.g., set-up costs, maintenance, breakdowns, etc.) and expected waste.
 - 1) Budgeting and planning based on practical capacity aligns the allocation of fixed costs with normal production activity. Any variance can be a signal for a change in demand for the product, an issue with production, etc.
- c. **Normal capacity** is the average level of output that can be completed over a period of time.

2. Capacity Planning

- a. Capacity planning is an element of strategic planning that is closely related to capital budgeting. The IMA's Statement on Management Accounting *Measuring the Cost of Capacity* (issued in March 1996) states that maximizing the value created within an organization starts with understanding the nature and capabilities of all of the company's resources. According to that statement, effective capacity cost management requires
 - 1) Investment Analysis
 - a) In the short run, optimizing capital decisions and the effective and flexible use of investments that have already been made
 - b) Helping minimize requirements for future investment
 - c) Providing useful costing information on current process costs versus those proposed in current or future investment proposals
 - 2) Capacity Assessment
 - a) Closing any gap between market demands and a firm's capabilities.
 - i) At times, the firm may have excess capabilities; at others, shortages may exist. These capabilities may be physical, human, technological, or financial.
 - b) Supporting the establishment of capacity usage measurements that identify the cost of capacity and its impact on business cycles and overall company performance.
 - c) Identifying the capacity required to meet strategic and operational objectives and to estimate current available capacity.
 - d) Detailing the opportunity cost of unused capacity and suggesting ways to account for that cost.
 - e) Creating a common language for, and understanding of, capacity cost management.
 - f) Capacity should be defined from several different perspectives. Managing the cost of that capacity starts when a product or process is first envisioned. It continues through the subsequent disposal of resources downstream.
 - 3) Manufacturing Process Assessment
 - a) Maximizing the value delivered to customers
 - b) Supporting effective matching of a firm's resources with current and future market opportunities
 - c) Eliminating waste in the short, intermediate, and long run
 - d) Supporting change efforts by providing pre-decision information and analysis on the potential resource and cost implications of a planned change
- b. Capacity planning is part of the capital budgeting process.
 - 1) Estimating capacity levels for future periods allows for the acquisition of more capacity when needed or disposal of capacity that is not expected to be utilized.
- c. Capacity level influences product costing, pricing decisions, and financial statements.
 - 1) Excess capacity has a cost. Having excess capacity means that a company will either have to charge higher prices for its products or report lower income on its financial statements.
- d. Similarly, producing at full capacity can have a cost in the form of opportunity costs. A company that could generate additional sales if it had more capacity needs to address whether the acquisition of additional capacity is warranted.

3. Capacity Expansion

- a. According to business strategy theorist Michael E. Porter, determining whether to expand capacity is a major strategic decision because of the capital required, the difficulty of forming accurate expectations, and the long timeframe of the lead times and the commitment. The key forecasting problems are long-term demand and behavior of competitors. The key strategic issue is avoidance of industry overcapacity.
 - 1) Capacity expansion is also referred to as market penetration because it involves increasing the amount of an existing product in an existing market.
- b. Undercapacity in a profitable industry tends to be a short-term issue. Profits ordinarily lure additional investors. Overcapacity tends to be a long-term problem because firms are more likely to compete intensely rather than reverse their expansion.
- c. The formal capital budgeting process entails predicting future cash flows related to the expansion project, discounting them at an appropriate interest rate, and determining whether the net present value is positive. This process permits comparison with other uses of the firm's resources.
 - 1) The apparent simplicity of this process is deceptive because it depends upon, among many other things, which expansion method is chosen, developments in technology, and profitability. The latter factor in turn depends on such uncertainties as total long-term demand and the expansion plans of rival firms.
- d. Porter's model of the decision process for capacity expansion has the following interrelated steps:
 - 1) The firm must identify the options in relation to their size, type, degree of vertical integration (if any), and possible response by competitors.
 - 2) The second step is to forecast demand, input costs, and technology developments. The firm must be aware that its technology may become obsolete or that future design changes to allow expansion may or may not be possible. Moreover, the expansion itself may put upward pressure on input prices.
 - 3) The next step is analysis of competitors to determine when each will expand. The difficulty is that forecasting their behavior depends on knowing their expectations. Another difficulty is that each competitor's actions potentially affect all other competitors' actions, with the industry leader being most influential.
 - 4) Using the foregoing information, the firm predicts total industry capacity and firms' market shares. These estimates, together with the expected demand, permit the firm to predict prices and cash flows.
 - 5) The final step is testing for inconsistencies.
- e. The extent of uncertainty about future demand is a crucial variable in determining the nature of industry expansion. For example, if uncertainty is great, firms willing to take greater risks because of their large cash resources or strategic stake in the industry will act first. Other firms will await events.
 - 1) When demand uncertainty is low, firms will tend to adopt a strategy of preemption, usually with strong market signals, to forestall competitors' expansion plans. Excess preemption leads to excess industry capacity because firms overestimate their competitive strengths, misunderstand market signals, or fail to accurately assess competitors' intentions.

STUDY UNIT ELEVEN

BUSINESS PROCESS IMPROVEMENT

11.1	Value-Chain Analysis	2
11.2	Other Process Improvement Tools	6

This study unit is the **fifth of five** on **cost management**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The five study units are

- Study Unit 7: Cost Management Concepts
- Study Unit 8: Cost Accumulation Systems
- Study Unit 9: Cost Allocation Techniques
- Study Unit 10: Supply Chain Management
- Study Unit 11: Business Process Improvement

If you are interested in reviewing more introductory or background material, go to www.GLEM.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following GLEM outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

11.1 VALUE-CHAIN ANALYSIS

1. Value and Customers' Perceptions

- a. To remain on the market, a product must provide value to the customer and a profit to the seller.
 - 1) Customers assign value to a product. The producer can affect the customers' perception of value by differentiating the product and lowering its price.
 - 2) The producer's profit is the difference between its costs and the price it charges for the product. Thus, by keeping costs low, the producer has more flexibility in pricing.
- b. The relationship of these three aspects of value creation can be graphically depicted as follows:

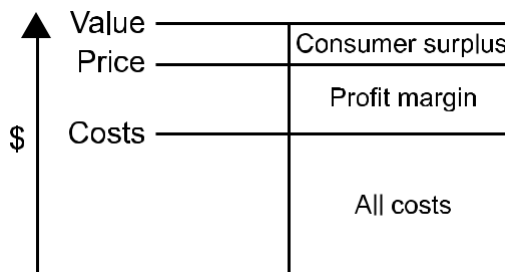


Figure 11-1

2. Value-Added and Nonvalue-Added Activities

- a. A value-added activity increases the value of a product or service to the customer. Examples include assembling the product and shipping it to the customer.
 - 1) In contrast to conventional financial statements, value-chain financial statements treat value-added activities as product costs (i.e., as inventoriable costs).
- b. A nonvalue-added activity does not increase the value of a product or service to the customer even if this activity is necessary. Examples include inspection for testing and inventory storage costs.
- c. A direct cost can be specifically associated with a single cost object. Direct costs usually are classified as either value-added or nonvalue-added.

3. The Value Chain

- a. The value chain is a model for depicting the way in which every function in a company adds value to the final product.
 - 1) The IMA's Statement on Management Accounting (SMA) *Value Chain Analysis for Assessing Competitive Advantage* (issued in March 1996) says, "The value chain approach for assessing competitive advantage is an integral part of the strategic planning process."
- b. A value chain depicts how costs and customer value accumulate along a chain of activities that lead to an end product or service.
 - 1) A value chain consists of the internal processes or activities a company performs: R&D, design, production, marketing, distribution, and customer service.

- c. Another view is that the value chain consists of all of the value-creating activities leading to the ultimate end-use product delivered into the final consumers' hands.
- 1) In other words, a value chain is a firm's overall chain of value-creating (value-added) processes.
 - 2) Customer retention is an important objective of value-adding and profit-maximizing processes because the customer base is a key intangible asset. Furthermore, high customer loyalty depends on the effectiveness of an organization's core business processes.
 - 3) Customer relationship management optimizes customer equity by managing information about individuals and their "touchpoints" for the purpose of maximizing customer loyalty. This process involves more than merely attracting customers and satisfying them.
- d. Primary activities deal with the product directly. Support activities lend aid to the primary activity functions. The value chain can be graphically depicted as follows:

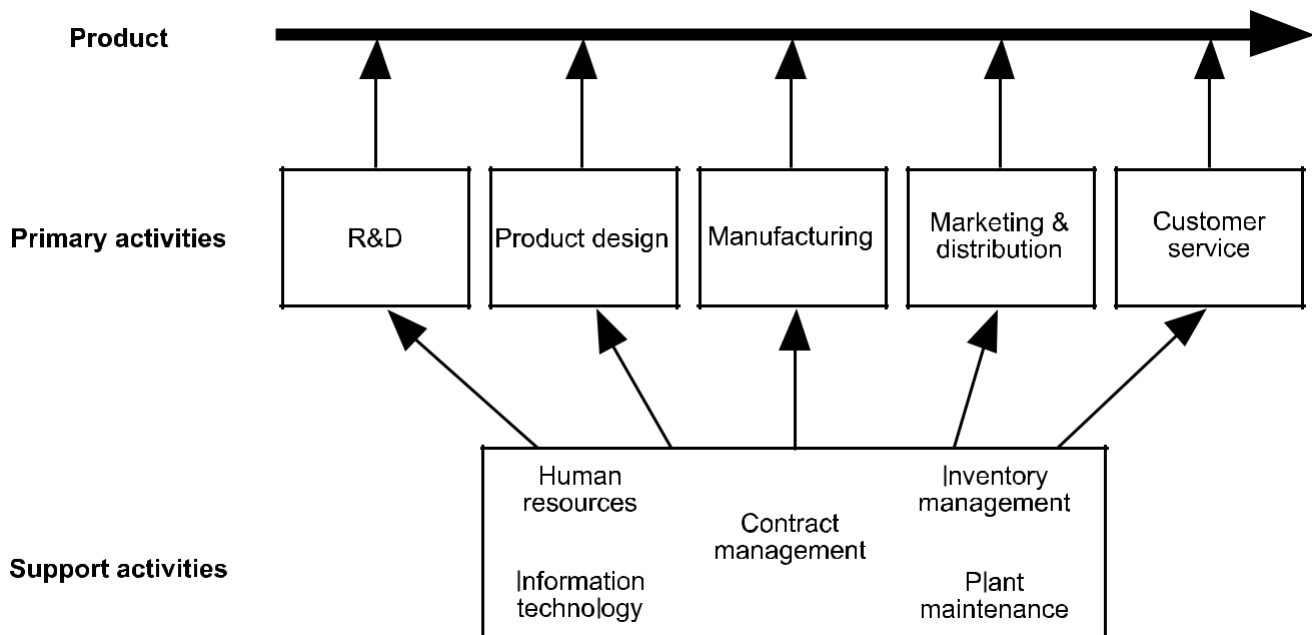


Figure 11-2

4. Value-Chain Analysis

- a. Value-chain analysis is a strategic analysis tool that allows a firm to focus on those activities that are consistent with its overall strategy.
- 1) Value-chain analysis allows a firm to decide which parts of the value chain it wants to occupy and how each activity then contributes to the firm's competitive advantage by adding customer value or by reducing costs.

- b. Because the value chain identifies and connects the organization's strategic activities, value-chain analysis improves the firm's knowledge of its relations with customers, suppliers, and competitors. It also facilitates the strategic determination of the phase(s) of the industry's value chain in which the firm should operate.
 - 1) The **first step** in a value-chain analysis is to identify the firm's value-creating activities.
 - 2) The **second step** is to determine how each value-creating activity can produce a competitive advantage for the firm. This step has multiple substeps.
 - a) Identify the firm's competitive advantage (e.g., cost reduction, product differentiation) so that the firm's position in the industry's value chain can be clarified.
 - b) Identify the ways in which the firm's value-creating activities can generate additional customer value.
 - c) Identify activities that are candidates for cost reduction or, in the case of non-core competencies, outsourcing.
 - d) Identify value-adding ways in which the firm's remaining activities can be linked.
- c. Value-chain analysis is a team effort. Management accountants need to collaborate with engineering, production, marketing, distribution, and customer service professionals to focus on the strengths, weaknesses, opportunities, and threats identified in the value-chain analysis results.
 - 1) Value-chain analysis offers an excellent opportunity to integrate strategic planning and management accounting to guide the firm to survival and growth.

5. The Supply Chain

- a. The supply chain is the flow of materials and services from their original sources to final consumers. Moreover, it usually encompasses more than one firm.
 - 1) Firms seeking to improve performance and reduce costs must analyze all phases of the supply chain as well as the value chain. Thus, a firm must reduce the cost of, and increase the value added by, its purchasing function.
- b. Purchasing is the management function that concerns the acquisition process. It includes choice of vendors, contract negotiation, the decision whether to purchase centrally or locally, and value analysis. The process is initiated by purchase requisitions issued by the production control function.
 - 1) Purchase requisitions ultimately result from insourcing vs. outsourcing (make vs. buy) decisions made when production processes were designed.
 - 2) The choice of vendors depends on price, quality, delivery performance, shipping costs, credit terms, and service. Purchasers with a competitive orientation and considerable economic power may be able to extract very favorable terms from vendors.
 - 3) Purchasers with a cooperative orientation adopt a longer-term approach: supply-chain coordination. The purchaser and the vendor are viewed as committed to a partnership involving joint efforts to improve quality.
 - a) For example, in the case of a major manufacturer and one of its suppliers, this orientation may include the purchaser's willingness to help develop the vendor's managerial, technical, and productive capacities. Thus, it tends to result in minimizing the number of vendors.

- c. Supply-chain analysis and coordination should extend to all parties in the chain, from initial sources of materials to retailers.
 - 1) Coordination has special relevance to inventory management. By sharing information among all parties, demand uncertainty is reduced at each level, with consequent decreases of inventory at each level, minimization of stockouts, and avoidance of overproduction and rush orders.
 - 2) For example, such cooperation counteracts what has been called the **bullwhip or backlash effect**. This phenomenon occurs when demand variability increases at each level of the supply chain.
 - 3) Retailers face only customer demand variability, but the manufacturer must cope with retailer demand variability that is greater than customer demand variability because retailers' purchases vary with additional factors, such as batching of orders and trade promotions.
 - 4) Similarly, the variability of manufacturer demands on suppliers may be greater than the variability of retailer demands on manufacturers.
- d. Value-chain and supply-chain analysis should be used to meet customer requirements for better performance regarding such critical success factors as
 - 1) Cost reduction,
 - 2) Efficiency,
 - 3) Continuous improvement of quality to meet customer needs and wants,
 - 4) Minimization or elimination of defects,
 - 5) Faster product development and customer response times, and
 - 6) Constant innovation.

6. Cost of Value Chain Analysis

- a. To achieve value chain analysis, a company will need value engineering.
- b. Value engineering is a means of reaching targeted cost levels. It is a systematic approach to assessing all aspects of the value chain cost buildup for a product. The purpose is to minimize costs without sacrificing customer satisfaction.
 - 1) Value engineering requires distinguishing between cost incurrence and **locked-in costs**.
- c. Cost incurrence is the actual use of resources, but locked-in (designed-in) costs will result in use of resources in the future as a result of past decisions. Thus, value engineering emphasizes controlling costs at the design stage, that is, before they are locked in.
- d. Life-cycle costing is sometimes used as a basis for cost planning and product pricing. Life-cycle costing estimates a product's revenues and expenses over its expected life cycle. The result is to highlight upstream and downstream costs in the cost-planning process that often receive insufficient attention. Emphasis is on the need to price products to cover all costs, not just production costs.

11.2 OTHER PROCESS IMPROVEMENT TOOLS

1. Process Analysis

- a. Process analysis is a means of linking a firm's internal processes to its overall strategy.
- b. Types of Processes
 - 1) Continuous, such as candy bars produced by machinery
 - 2) Batch, such as beer brewing
 - 3) Hybrid, in which both continuous and batch processes are used
 - 4) Make-to-stock, such as automobile assembly
 - 5) Make-to-order, such as deli sandwich making
- c. Process Interdependence
 - 1) The degree of interdependence among the stages in a process is referred to as "tightness."
 - 2) A tight process is one in which a breakdown in one stage brings the succeeding stages to a halt. This is characteristic of continuous processes that do not have buffer work-in-process inventories.
 - 3) A loose process is one in which subsequent stages can continue working after a breakdown in a previous stage. This is characteristic of batch processes and any others with extensive work-in-process inventories.
- d. Bottlenecks
 - 1) Very few processes run at precisely the same speed in every stage.
 - 2) One part of the process is almost always the slowest, referred to as the "bottleneck." If capacity is added at that point, the bottleneck simply shifts to the next slowest operation.
 - a) The theory of constraints (discussed in Study Unit 10, Subunit 3) was developed to deal with this challenge.
 - 3) The bottleneck issue only arises when demand for the firm's product is sufficient to absorb all of the output. When a production line is running at less than full capacity, bottlenecks can be avoided.

2. Process Value Analysis

- a. **Process value analysis** is a comprehensive understanding of how an organization generates its output. It involves a determination of which activities that use resources are value-adding or nonvalue-adding and how the latter may be reduced or eliminated.
 - 1) This linkage of product costing and continuous improvement of processes is **activity-based management (ABM)**. ABM redirects and improves the use of resources to increase the value created for customers and other stakeholders.
 - a) ABM identifies the key activities performed in an organization and focuses on understanding what causes costs to occur.
 - b) ABM uses include the following:
 - i) **Strategic analysis** explores various ways a company can create and sustain a competitive advantage in the marketplace.
 - ii) **Benchmarking** is a methodology that identifies an activity as the standard by which a similar activity will be judged.
 - iii) **Operations analysis** seeks to identify, measure, and improve current performance of key processes and operations within an entity.

- iv) **Profitability/pricing analysis** assists a company in analyzing the costs and benefits of products and processes and prelaunch analysis and improvement of product profitability.
 - v) **Process improvement** focuses on identifying the causes of variation, waste, and inefficiency. Process improvement includes both incremental and quantum change efforts that seek to increase the value created per resources consumed by an organization.
- 2) A **continuous improvement process (CIP)** is an ongoing effort involving management and workers to improve products, services, or processes.
- a) The purpose of CIP is the identification, reduction, and elimination of suboptimal processes to improve efficiency.
 - i) Ideal standards (discussed in Study Unit 13, Subunit 2, item 3.) are associated with production under optimal conditions.
 - b) CIP efforts can lead to “incremental” improvement over time or “breakthrough” improvement all at once.
 - c) Delivery (customer valued) processes are constantly evaluated and improved in light of their efficiency, effectiveness, and flexibility.
- 3) **Kaizen** is the Japanese word for the continuous pursuit of improvement in every aspect of organizational operations.
- a) For example, a budget prepared on the Kaizen principle projects costs based on future improvements. The possibility of such improvements must be determined, and the cost of implementation and the savings therefrom must be estimated.
 - b) Key features of Kaizen include the following:
 - i) Improvements are based on many small changes rather than the radical changes that might arise from research and development.
 - ii) Ideas come from the workers themselves, so they are less likely to be radically different and therefore are easier to implement.
 - iii) Small improvements are less likely to require major capital investment than major process changes.
 - iv) The ideas come from the talents of the existing workforce, as opposed to using research, consultants, or equipment—any of which could be expensive.
 - v) All employees, including management, should continually seek ways to improve their own performance.
 - vi) Workers are encouraged to take ownership of their work and can help reinforce teamwork, thereby improving worker motivation.
 - c) The key features of Kaizen are the more tactical elements of CIP. The more strategic elements include deciding how to increase the value of the delivery process output to the customer (effectiveness) and how much flexibility is valuable in the process to meet changing needs.

- b. An **activity analysis** determines what is done, by whom, at what cost in time and other resources, and the value added by each activity.
 - 1) A value-added activity is necessary to remain in business. For example, a manufacturer would deem the conversion of raw materials into salable products a value-added activity.
 - a) Such an activity may be mandated (e.g., a regulatory requirement) or discretionary. The latter produces some changes not otherwise achievable that enables other activities to occur.
 - 2) A value-added cost is incurred to perform a value-added activity without waste. Most types of direct labor would be considered value-added cost because the costs are being incurred to directly produce the product.
 - 3) A nonvalue-added activity is unnecessary and should be eliminated. The act of generating nonsalable final products is a nonvalue-added activity.
 - a) An example of a nonvalue-added activity is where inventory has to be moved long distances from one work station to another in a production process. Similarly, inventory that has to wait in line before being processed is a waste. This is why just-in-time (JIT) inventory systems have proved popular: JIT eliminates much of the waste in a production process.
 - 4) A nonvalue-added cost is caused by a nonvalue-added activity or inefficient performance of a value-added activity. The costs of raw materials and direct labor expended on products that fail inspection would be considered nonvalue-added costs.
 - a) Thus, managing the causes of cost results in elimination of unnecessary activities as well as greater efficiency of activities.
- c. Financial and nonfinancial **measures of activity performance** address efficiency, quality, and time. The purpose is to assess how well activities meet customer demands.
 - 1) To satisfy customer needs and wants, activities should be efficient (a favorable input-to-output ratio) so that customers are willing to pay the prices charged.
 - 2) Activities should produce defect-free output (high quality), and that output should be produced in a timely manner (with less resource usage and in response to customer requirements).
- d. The selection of value-added activities in each place of the value chain reflects the firm's determination of its competitive advantage and its choice of competitive strategy.
 - 1) For example, different design strategies require different activities and costs. A firm might choose to be the low-cost producer of an undifferentiated product rather than compete on the basis of superior product quality.
- e. One aspect of process analysis is the management of time. Product development time is a crucial factor in the competitive equation. A company that is first in the market with a new product has obvious advantages.
 - 1) Reducing development time is also important because product life cycles are becoming shorter.
 - 2) Companies need to respond quickly and flexibly to new technology, changes in consumer tastes, and competitive challenges.

3. Business Process Reengineering (BPR)

- a. BPR is a complete rethinking of how business functions are performed to provide value to customers, that is, radical innovation instead of mere improvement, and a disregard for current jobs, hierarchies, and reporting relationships.
 - 1) Technological advances have increased the popularity of business process reengineering.
- b. A process is how something is accomplished in a firm. It is a set of activities directed toward the same objective. Reengineering is process innovation and core process redesign. Instead of improving existing procedures, it finds new ways of doing things. Thus, reengineering should be contrasted with process improvement, which consists of incremental but constant changes that improve efficiency.
 - 1) Accordingly, BPR techniques eliminate many traditional controls. They exploit modern technology to improve productivity and decrease the number of clerical workers. Thus, the emphasis is on developing controls that are automated and self-correcting and require minimal human intervention.
- c. The emphasis therefore shifts to monitoring internal control so management can determine when an operation may be out of control and corrective action is needed.
 - 1) Most BPR techniques also assume that humans will be motivated to work actively in improving operations when they are full participants in the process.
- d. Part of the BPR process involves looking at possible alternatives and determining the cost of those possible alternatives compared to the costs of maintaining the same processes.
 - 1) To do this, the management accountant must determine
 - a) The cost to reengineer the process, and
 - b) The expected savings.
 - 2) Some desirable alternatives may actually increase total costs.
 - a) For example, more expensive materials could lead to higher product quality. While this will increase total costs, it may be necessary to maintain/increase market share.
 - b) In situations like these, the management accountant would determine the savings attributed to possible alternatives compared to maintaining the same processes.
 - 3) After determining the cost and savings of BPR, the company must look at all of its goals and needs overall to determine if it should invest in a possible alternative.

EXAMPLE 11-1 BPR Cost and Savings

It currently takes 3 people working for 10 hours each at a rate of \$15 per hour to produce 10 units of Product P. For \$350, the company has the option of purchasing a machine that will allow the same 3 people to produce 10 units of Product P in 5 hours.

It currently costs \$450 (3 people × 10 hours × \$15 per hour) to produce 10 units of Product P. If the company purchases the machine, it will cost \$225 (3 people × 5 hours × \$15 per hour) to produce 10 units of Product P for a savings of \$225 (\$450 – \$225). Since this company will continually save \$225 every 10 hours, the company should probably purchase the machine.

4. Benchmarking

- a. The IMA's Statement on Management Accounting (SMA) *Effective Benchmarking*, issued in July 1995, describes techniques for improving the effectiveness of benchmarking, which is a means of helping companies with productivity management and business process reengineering.
 - 1) "Benchmarking involves continuously evaluating the practices of best-in-class organizations and adapting company processes to incorporate the best of these practices." It "analyzes and measures the key outputs of a business process or function against the best and also identifies the underlying key actions and root causes that contribute to the performance difference."
 - 2) Benchmarking is an ongoing process that entails quantitative and qualitative measurement of the difference between the company's performance of an activity and the performance by the best in the world. The benchmark organization need not be a competitor.
- b. The first phase in the benchmarking process is to select and prioritize benchmarking projects.
 - 1) An organization must understand its critical success factors and business environment to identify key business processes and drivers and to develop parameters defining what processes to benchmark.
 - 2) The criteria for selecting what to benchmark relate to the reasons for the existence of a process and its importance to the entity's mission, values, and strategy. These reasons relate in large part to satisfaction of end users or customer needs.
- c. The next phase is to organize benchmarking teams.
 - 1) A team organization is appropriate because it permits an equitable division of labor, participation by those responsible for implementing changes, and inclusion of a variety of functional expertise and work experience.
 - 2) Team members should have knowledge of the function to be benchmarked, respected positions in the company, good communication skills, teaming skills, motivation to innovate and to support cross-functional problem solving, and project management skills.

- d. The benchmarking team must thoroughly investigate and document internal processes.
 - 1) The organization should be seen as a series of processes, not as a fixed structure. A process is “a network of related and independent activities linked by the outputs they exchange.” One way to determine the primary characteristics of a process is to trace the path a request for a product or service takes through the organization.
 - 2) The benchmarking team must also develop a family of measures that are true indicators of process performance and a process taxonomy, that is, a set of process elements, measures, and phrases that describes the process to be benchmarked.
- e. Researching and identifying best-in-class performance is often the most difficult phase.
 - 1) The critical steps are setting up databases, choosing information-gathering methods (internal sources, external public domain sources, and original research are the possible approaches), formatting questionnaires (lists of questions prepared in advance), and selecting benchmarking partners.
- f. The data analysis phase entails identifying performance gaps, understanding the reasons they exist, and prioritizing the key activities that will facilitate the behavioral and process changes needed to implement the benchmarking study’s recommendations.
 - 1) Sophisticated statistical analysis and other methods may be needed when the study involves many variables, testing of assumptions, or presentation of quantified results.
- g. Leadership is most important in the implementation phase of the benchmarking process because the team must be able to justify its recommendations.
 - 1) Moreover, the process improvement teams must manage the implementation of approved changes.
- h. Benchmarking is a way for a company to learn its strengths and weaknesses by comparison to similar companies. Additional benefits of benchmarking include
 - 1) Best practices are identified and defined.
 - 2) Alternative solutions are evaluated.
 - 3) The competitive position is strengthened.
 - 4) The company goals are questioned.
 - 5) More people are held responsible for their performance.

5. Costs of Quality

- a. The IMA's Statement on Management Accounting *Managing Quality Improvements*, issued in 1993, describes four categories of costs of quality: prevention, appraisal, internal failure, and external failure. The organization should attempt to minimize its total cost of quality.
- b. Conformance costs include prevention and appraisal, which are both financial measures of internal performance.
 - 1) **Prevention** attempts to avoid defective output. These costs include preventive maintenance, employee training, review of equipment design, and evaluation of suppliers. Providing quality training to employees should reduce all types of quality costs.
 - 2) **Appraisal** encompasses such activities as statistical quality control programs, inspection, and testing.
- c. Nonconformance costs include costs of internal failure (a financial measure of internal performance) and external failure costs (a financial measure of customer satisfaction).
 - 1) **Internal failure** costs occur when defective products are detected before shipment.
 - a) Examples are scrap, rework, tooling changes, downtime, redesign of products or processes, lost output, reinspection and retesting, expediting of operations after delays, lost learning opportunities, and searching for and correcting problems.
 - 2) The costs of **external failure** or lost opportunity include lost profits from a decline in market share as dissatisfied customers make no repeat purchases, return products for refunds, cancel orders, and communicate their dissatisfaction to others.
 - a) Thus, external failure costs are incurred for customer service complaints; rejection, return, repair, or recall of products or services; warranty obligations; product liability claims; and customer losses.
 - i) Given the wide variety of causes of external failure, the best solution is to provide all employees with training about the importance of providing quality to customers.
 - b) Environmental costs are also external failure costs, e.g., fines for nonadherence to environmental law and loss of customer goodwill.
 - i) To minimize environmental damage and its resulting costs, the International Organization for Standardization has issued ISO 14000 standards to promote the reduction of environmental damage by an organization's products, services, and operations and to develop environmental auditing and performance evaluation systems.

6. Efficient accounting processes

- a. Improving accounting processes can increase a company's ability to minimize the costs of these processes while also maximizing their usefulness. Methods used to make accounting operations more efficient are described below.
- 1) **Process walk-throughs** involve following a transaction from the initial stage and observing every step in the financial reporting system until the transaction is recorded in the financial statements. Process walk-throughs help assess whether transactions are recorded correctly and whether the system catches errors, and they identify processes requiring improvement.
 - 2) **Process training** enables personnel to learn how to do their jobs more efficiently and creates a motivated, skilled, and effective workforce through which organizational goals are achieved.
 - 3) **Identification of waste and over-capacity** leads to more efficient accounting operations by eliminating unnecessary accounting transactions and freeing up accounting personnel to work on more urgent accounting issues.
 - 4) **Identifying the root cause of errors** uncovers issues related to accounting systems, errors by personnel, or issues with organizational procedures or instructions. Errors can then be decreased or eliminated by correcting problems associated with the process.
 - 5) **Reducing the accounting close cycle (fast close)** can be accomplished by successfully identifying the root cause of errors and correcting the problem; hence, speeding up the close process. Additionally, by conducting walk-throughs of various accounting processes and improving the processes wherever possible (i.e., processing of accounts payable, posting of accruals, submission of subsidiary results, etc.) will lead to reducing completion times.
 - 6) **Shared services** make accounting operations more efficient by having only one department provide accounting service for each department, division, and/or region. Shared services help standardize processes and accounting technology systems, making it easier to track and analyze costs across a company. Possible benefits include creation of best practices and more effective controls.

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CMA REVIEW

PART 1: FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS



E. Internal Controls (15% - Levels A, B, and C)

1. *Governance, risk, and compliance*

- a. Internal control structure and management philosophy (17.2)
- b. Internal control policies for safeguarding and assurance (17.1)
- c. Internal control risk (17.2)
- d. Corporate governance (17.1)
- e. External audit requirements (17.2)

2. *System controls and security measures*

- a. General accounting system controls (18.1-18.2)
- b. Application and transaction controls (18.2)
- c. Network controls (18.2-18.3)
- d. Backup controls (18.3)
- e. Business continuity planning (18.3)

STUDY UNIT SEVENTEEN

INTERNAL CONTROLS -- CORPORATE GOVERNANCE

17.1	Corporate Governance and Regulations Relating to Internal Control	1
17.2	Risk and Internal Control	12

Internal Controls

Management accountants are expected to have a thorough understanding of the risks inherent to, and the internal controls within, a business. Internal controls have always been a good idea in a well-run business, but with the passage of the Foreign Corrupt Practices Act in 1977, an effective internal control system became a legal requirement. The Sarbanes-Oxley Act of 2002 further enhanced the legal requirements for internal controls.

This study unit is the **first of two** on **internal controls**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The two study units are

Study Unit 17: Internal Controls -- Corporate Governance

Study Unit 18: Internal Controls -- Controls and Security Measures

If you are interested in reviewing more introductory or background material, go to www.glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

17.1 CORPORATE GOVERNANCE AND REGULATIONS RELATING TO INTERNAL CONTROL

1. Definition of Corporate Governance

- a. **Governance** is the combination of people, policies, procedures, and processes (including internal control) that help ensure that an entity effectively and efficiently directs its activities toward meeting the objectives of its stakeholders.
 - 1) Stakeholders are persons or entities who are affected by the activities of the entity. Among others, these include shareholders, employees, suppliers, customers, neighbors of the entity's facilities, and government regulators.
- b. Corporate governance can be either internal or external.
 - 1) Corporate charters and bylaws, boards of directors, and internal audit functions are internal.
 - 2) Laws, regulations, and the government regulators who enforce them are external.

2. Governance Practices

- a. Governance practices reflect the organization's unique culture and largely depend on it for effectiveness.
 - 1) The organizational culture
 - a) Sets values, objectives, and strategies;
 - b) Defines roles and behaviors;
 - c) Measures performance; and
 - d) Specifies accountability.
 - 2) Thus, the culture determines the degree of sensitivity to social responsibility.
- b. Governance practices may use various legal forms, structures, strategies, and procedures. They ensure that the organization
 - 1) Complies with society's legal and regulatory rules;
 - 2) Satisfies the generally accepted business norms, ethical principles, and social expectations of society;
 - 3) Provides overall benefit to society and enhances the interests of the specific stakeholders in both the long and short term; and
 - 4) Reports fully and truthfully to its stakeholders, including the public, to ensure accountability for its decisions, actions, and performances.
- c. Governance applies to all organizational activities. Thus, its processes provide overall direction for risk management activities.
 - 1) Internal control activities are in turn a key element of risk management. They implement the organization's risk management strategies.

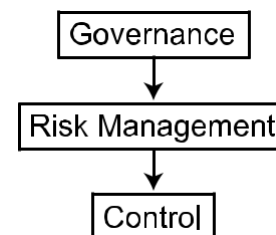


Figure 17-1

3. Governance in Corporations

- a. Unlike a sole proprietorship or a general partnership, a corporation is a legal entity created under authority of a **state statute** to carry out the purposes permitted by that statute and the articles of incorporation.
 - 1) The corporation ordinarily is treated as a **legal person** with rights and obligations separate from its owners and managers.
- b. Corporations are governed by **shareholders** (owners) who elect a board of directors and approve fundamental changes in the corporate structure.
 - 1) **Directors** establish corporate policies, adapt bylaws, and elect or appoint corporate **officers** who carry out the policies in the day-to-day management of the organization.
- c. Incorporation may be in any state. **Articles of incorporation** must be filed with the secretary of state or another designated official. The articles must include the following:
 - 1) Corporation's name (must differ from the name of any corporation authorized to do business in the state)
 - 2) Number of authorized shares of stock
 - 3) Street address of the corporation's initial registered office
 - 4) Name of the registered agent at that office
 - 5) Name and address of each incorporator

- d. **Bylaws** govern the internal structure and operation of the corporation.
 - 1) Initial bylaws are adopted by the incorporators or the board.
 - 2) They may contain any provision for managing the business and regulating the entity's affairs that does not conflict with the law or the articles.
- e. Personnel policies and procedures are integral to an efficient control environment. They influence the control consciousness of personnel.

4. Corporate Governance Process

- a. Governance has two major components: strategic direction and oversight.
 - 1) Strategic direction determines
 - a) The business model,
 - b) Overall objectives,
 - c) The approach to risk taking (including the risk appetite), and
 - d) The limits of organizational conduct.
 - 2) Oversight is the governance component with which internal auditing is most concerned. It is also the component to which risk management and control activities are most likely to be applied. The elements of oversight are
 - a) Risk management activities performed by senior management and risk owners and
 - b) Internal and external assurance activities.
- b. The **board** is the source of overall direction to, and the authority of, management. It also has the ultimate responsibility for oversight. Directors have a **fiduciary** duty to the corporation.
 - 1) Another responsibility is to identify stakeholders, whether directly involved with the business (employees, customers, and suppliers), indirectly involved (investors), or having influence over the business (regulators and competitors).
 - 2) The board must determine the expectations of stakeholders and the outcomes that are unacceptable.
 - 3) The most important function of the board's **audit committee** is to promote the independence of the external and internal auditors from undue influence by management. A related function is support for the recommendations made by auditors.
- c. **Management** performs day-to-day governance functions and determines
 - 1) What specific risks are to be managed,
 - 2) Who will be **risk owners** (managers responsible for specific day-to-day risks), and
 - 3) How specific risks will be managed.
- d. The functions of risk owners include
 - 1) Evaluating the adequacy of the design of risk management activities and the organization's ability to carry them out as designed;
 - 2) Determining whether risk management activities are operating as designed;
 - 3) Establishing monitoring activities; and
 - 4) Ensuring that information to be reported to senior management and the board is accurate, timely, and available.

- e. A **risk committee** may be created that
 - 1) Identifies key risks,
 - 2) Connects them to risk management processes,
 - 3) Delegates them to risk owners, and
 - 4) Considers whether tolerance levels delegated to risk owners are consistent with the organization's risk appetite.
- f. Governance expectations, including tolerance levels, must be periodically reevaluated by the board and senior management. The result may be changes in risk management activities.

5. Internal Control

- a. An organization establishes a system of internal control to help it manage many of the risks it faces. Statements on Auditing Standards define internal control as follows:
A process effected by those charged with governance, management, and other personnel that is designed to provide reasonable assurance about the achievement of the entity's objectives with regard to the reliability of financial reporting, effectiveness and efficiency of operations, and compliance with applicable laws and regulations. Internal control over safeguarding of assets against unauthorized acquisition, use, or disposition may include controls relating to financial reporting and operations objectives.
 - 1) The Committee of Sponsoring Organizations of the Treadway Commission's (COSO) definition of internal control is covered in item 6.b. in Subunit 17.2.
- b. One such risk is the risk of fraud. **Fraud** differs from error because it is intentional. It typically involves pressures or incentives to engage in wrongdoing and a perceived opportunity to do so. Examples are fraudulent financial reporting and misappropriation of assets.
 - 1) Internal controls are designed to prevent and detect fraud. However, because of the concealment aspects of fraudulent activity (e.g., collusion or falsification of documents), controls cannot give absolute assurance that material fraud will be prevented or detected.
- c. Controls must be cost beneficial. They should not have an adverse effect on efficiency or profitability. For example, the audit committee might review the internal auditor's work schedule.

6. Foreign Corrupt Practices Act

- a. The Foreign Corrupt Practices Act (FCPA), enacted in 1977, had its origin in the Watergate investigations. The FCPA is designed to prevent secret payments of corporate funds for purposes that Congress has determined to be contrary to public policy, such as bribery of foreign officials.
 - 1) The act amends the Securities Exchange Act of 1934 to prohibit a **domestic** concern engaged in interstate commerce, including any person acting on its behalf, whether or not doing business overseas and whether or not registered with the SEC, from offering or authorizing corrupt payments to any
 - a) Foreign official
 - b) Foreign political party or official thereof
 - c) Candidate for political office in a foreign country

- 2) Only political payments to **foreign government officials** are prohibited. Payments to foreign business owners or corporate officers are not addressed by the FCPA.
- 3) **Corrupt payments** are payments for the purpose of inducing the recipient to act or refrain from acting so that the domestic concern might obtain or retain business.
 - a) The FCPA prohibits a mere offer or promise of a bribe, even if it is not consummated.
 - b) The FCPA prohibits payment of anything of value, except for de minimis gifts (e.g., tokens of hospitality).
 - c) Payments are prohibited if the person making them knew or should have known that some or all of them would be used to influence a governmental official.
- 4) **Foreign officials** do not include clerical or ministerial employees.
 - a) EXAMPLE: Payments made to a clerk to expedite the processing of goods through customs may not be prohibited by the act.
 - b) Such payments are not prohibited as long as the recipient has no discretion in carrying out a governmental function.
 - c) Payments that are allowed under the written law of the foreign country are also not prohibited.
- 5) Regardless of whether they have foreign operations, all **public companies** must make and keep books, records, and accounts in reasonable detail that accurately and fairly reflect transactions and dispositions of assets.
- 6) All public companies registered under the 1934 act must devise and maintain a system of internal accounting control sufficient to provide **reasonable assurance** (not absolute assurance) that
 - a) Transactions are executed in accordance with management's general or specific authorization.
 - b) Transactions are recorded as necessary to
 - i) Permit preparation of financial statements in conformity with generally accepted accounting principles (GAAP) or any other criteria applicable to such statements and
 - ii) Maintain accountability for assets.
 - c) Access to assets is permitted only in accordance with management's general or specific authorization.
 - d) The recorded accountability for assets is compared with the existing assets at reasonable intervals, and appropriate action is taken with respect to any differences.
- 7) The penalties for an individual for each criminal violation of the corrupt practices provisions are a fine of up to \$100,000 or imprisonment for up to 5 years, or both. A corporation may be assessed a fine of up to \$2,000,000 for violation of the same section.
 - a) Fines imposed upon individuals may not be paid directly or indirectly by an employer.

- 8) The implications of the FCPA extend well beyond its anti-bribery provisions.
 - a) All American businesses and business people are involved. Management is particularly affected. The responsibility for internal control is not new, but the potential for civil and criminal liabilities represents an added burden.
 - b) The impact of the law and the threat its ambiguities pose may alter business operations. Management might decide to abandon direct selling operations in foreign countries in favor of the use of foreign agents in hopes that this might lessen their "reason to know."
- 9) A written code of ethics and conduct is a necessity. This code should be communicated and monitored by internal auditors for compliance.
 - a) The code might include an explanation of the FCPA and its penalties. A firm may require written representations from employees that they have read and understood the provisions of the code.
 - b) Written representations regarding compliance might also be requested at future times. Foreign agents should be made aware of the prohibitions of indirect payments.

7. Sarbanes-Oxley Act

- a. The Sarbanes-Oxley Act of 2002 was a response to numerous financial reporting scandals involving large public companies. The act contains provisions that impose new responsibilities on issuers (public companies) and their auditors. The act created the **Public Company Accounting Oversight Board (PCAOB)**, which establishes auditing standards for registered public accounting firms to apply in their audits of issuers.
 - 1) The act requires that each member of the issuer's **audit committee** be an **independent member** of the board of directors. To be independent, a director must not be affiliated with, or receive any compensation (other than for service on the board) from, the issuer. Additional requirements are as follows:
 - a) The audit committee must consist of **at least three** fully independent members.
 - b) At least **one** member of the audit committee must be a **financial expert**.
 - c) The audit committee must be directly responsible for appointing, compensating, and overseeing the work of the independent auditor. The independent auditor must report directly to the audit committee, not to management.
 - d) The audit committee also must establish procedures for
 - i) The receipt, retention, and treatment of complaints received regarding accounting, internal control, or auditing matters and
 - ii) The confidential, anonymous submission of employees' concerns regarding questionable accounting or auditing matters.
 - 2) **Prohibited nonaudit services. Section 201** of the act lists several activities that cannot be performed on behalf of audit clients. Item 8. on page 8 has a detailed explanation.
 - a) An accounting firm may perform non-audit services (e.g., tax services) for an audit client only if the activity is approved in advance by the audit committee of the client.

- 3) **Audit partner rotation. Section 203** of the act requires the lead auditor and the reviewing partner to be rotated off the audit so that the same individual is not supervising a client's audit for an extended period of time. Item 9. on the next page has a detailed explanation.
 - a) The lead audit partner cannot perform audit services for more than 5 consecutive fiscal years of the audit client.
- 4) **Corporate responsibility** of a public company. **Section 302** requires periodic statutory financial reports to include certain certifications. Item 11. beginning on page 9 has a detailed explanation.
- 5) **Internal control report. Section 404** of the act requires management to establish and document internal control procedures and to include in the annual report a report on the company's internal control over financial reporting.
 - a) This report is to include
 - i) A statement of management's responsibility for internal control;
 - ii) Management's assessment of the effectiveness of internal control as of the end of the most recent fiscal year;
 - iii) Identification of the framework used to evaluate the effectiveness of internal control (such as the report of the Committee of Sponsoring Organizations);
 - iv) A statement about whether significant changes in controls were made after their evaluation, including any corrective actions; and
 - v) A statement that the external auditor has issued an attestation report on management's assessment. Because of this requirement, two audit opinions are expressed: one on internal control and one on the financial statements.
 - b) The external auditor must attest to and report on management's assessment.
 - i) The auditor must evaluate whether the structure and procedures
 - (a) include records accurately and fairly reflecting the firm's transactions and
 - (b) provide reasonable assurance that transactions are recorded so as to permit statements to be prepared in accordance with GAAP.
 - ii) The auditor's report also must describe any material weaknesses in internal controls.
 - iii) The evaluation is not to be the subject of a separate engagement but be in conjunction with the audit of the financial statements.

8. Sarbanes-Oxley Section 201. SERVICES OUTSIDE THE SCOPE OF PRACTICE OF AUDITORS

(a) *PROHIBITED ACTIVITIES* - Section 10A of the Securities Exchange Act of 1934 is amended by adding the following: It shall be unlawful for a registered public accounting firm (and any associated person of that firm) that performs for any issuer any audit required by the rules of the Commission to provide to that issuer, contemporaneously with the audit, any non-audit service, including:

- (1) bookkeeping or other services related to the accounting records or financial statements of the audit client;
- (2) financial information systems design and implementation;
- (3) appraisal or valuation services, fairness opinions, or contribution-in-kind reports;
- (4) actuarial services;
- (5) internal audit outsourcing services;
- (6) management functions or human resources;
- (7) broker or dealer, investment adviser, or investment banking services;
- (8) legal services and expert services unrelated to the audit; and
- (9) any other services that the Board determines, by regulation, is impermissible.

PREAPPROVAL REQUIRED FOR NON-AUDIT SERVICES - A registered public accounting firm may engage in any non-audit service, including tax services, that is not described above, for an audit client only if the activity is approved in advance by the audit committee of the client.

(b) *EXEMPTION AUTHORITY* - The Board may, on a case-by-case basis, exempt any issuer, public accounting firm, or transaction from the prohibition provision to the extent that such exemption is necessary or appropriate in the public interest and is consistent with the protection of investors, and subject to review by the Commission.

9. Sarbanes-Oxley Section 203. AUDIT PARTNER ROTATION

- a. CPA firms must rotate audit partners so that the same individual is not supervising a client's audit for an extended period of time.
- (j) It shall be unlawful for a registered public accounting firm to provide audit services to an issuer if the lead (or coordinating) audit partner (having primary responsibility for the audit), or the audit partner responsible for reviewing the audit, has performed audit services for that issuer in each of the five previous fiscal years of that issuer.

10. Sarbanes-Oxley Section 204. AUDITOR REPORTS TO AUDIT COMMITTEES

- (a) Section 10A of the Securities Exchange Act of 1934 (15 U.S.C. 78j-1), as amended by this Act, is amended by adding at the end the following:
- (k) *REPORTS TO AUDIT COMMITTEES*. - Each registered public accounting firm that performs for any issuer any audit required by this title shall timely report to the audit committee of the issuer-
 - (1) all critical accounting policies and practices to be used;
 - (2) all alternative treatments of financial information within generally accepted accounting principles that have been discussed with management officials of the issuer, ramifications of the use of such alternative disclosures and treatments, and the treatment preferred by the registered public accounting firm; and
 - (3) other material written communications between the registered public accounting firm and the management of the issuer, such as any management letter or schedule of unadjusted differences.

11. Sarbanes-Oxley Section 302. CORPORATE RESPONSIBILITY FOR FINANCIAL REPORTS

- (a) *REGULATIONS REQUIRED* - The SEC shall require, for each company filing periodic reports under the Securities Exchange Act of 1934, that the principal executive officer or officers and the principal financial officer or officers, or persons performing similar functions, certify in each annual or quarterly report filed or submitted under the act that--
- (1) *the signing officer has reviewed the report;*
 - (2) *based on the officer's knowledge, the report does not contain any untrue statement of a material fact or omit to state a material fact necessary in order to make the statements made, in light of the circumstances under which such statements were made, not misleading;*
 - (3) *based on such officer's knowledge, the financial statements, and other financial information included in the report, fairly present in all material respects the financial condition and results of operations of the issuer as of, and for, the periods presented in the report;*
 - (4) *the signing officers--*
 - (A) *are responsible for establishing and maintaining internal controls;*
 - (B) *have designed such internal controls to ensure that material information relating to the issuer and its consolidated subsidiaries is made known to such officers by others within those entities, particularly during the period in which the periodic reports are being prepared;*
 - (C) *have evaluated the effectiveness of the issuer's internal controls as of a date within 90 days prior to the report; and*
 - (D) *have presented in the report their conclusions about the effectiveness of their internal controls based on their evaluation as of that date;*
 - (5) *the signing officers have disclosed to the issuer's auditors and the audit committee of the board of directors (or persons fulfilling the equivalent function)--*
 - (A) *all significant deficiencies in the design or operation of internal controls which could adversely affect the issuer's ability to record, process, summarize, and report financial data and have identified for the issuer's auditors any material weaknesses in internal controls; and*
 - (B) *any fraud, whether or not material, that involves management or other employees who have a significant role in the issuer's internal controls; and*
 - (6) *the signing officers have indicated in the report whether or not there were significant changes in internal controls or in other factors that could significantly affect internal controls subsequent to the date of their evaluation, including any corrective actions with regard to significant deficiencies and material weaknesses.*
- (b) *FOREIGN REINCORPORATIONS HAVE NO EFFECT*- Nothing in this section 302 shall be interpreted or applied in any way to allow any issuer to lessen the legal force of the statement required under this section 302, by an issuer having reincorporated or having engaged in any other transaction that resulted in the transfer of the corporate domicile or offices of the issuer from inside the United States to outside of the United States.

12. Sarbanes-Oxley Section 407. DISCLOSURE OF AUDIT COMMITTEE FINANCIAL EXPERT

- (a) *RULES DEFINING "FINANCIAL EXPERT"* - The Commission shall issue rules, as necessary or appropriate in the public interest and consistent with the protection of investors, to require each issuer, together with periodic reports required pursuant to sections 13(a) and 15(d) of the Securities Exchange Act of 1934, to disclose whether or not, and if not, the reasons therefor, the audit committee of that issuer is comprised of at least 1 member who is a financial expert, as such term is defined by the Commission.
- (b) *CONSIDERATIONS* - In defining the term "financial expert" for purposes of subsection (a), the Commission shall consider whether a person has, through education and experience as a public accountant or auditor or a principal financial officer, comptroller, or principal accounting officer of an issuer, or from a position involving the performance of similar functions
 - (1) an understanding of generally accepted accounting principles and financial statements;
 - (2) experience in
 - (A) the preparation or auditing of financial statements of generally comparable issuers; and
 - (B) the application of such principles in connection with the accounting for estimates, accruals, and reserves;
 - (3) experience with internal accounting controls; and
 - (4) an understanding of audit committee functions.

13. Audit Approaches

- a. The following are the four audit approaches:
 - 1) Substantive procedures (bottom-up)
 - 2) Balance sheet
 - 3) Systems-based
 - 4) Risk-based (top-down)
- b. The **substantive procedures approach** (vouching or direct verification) applies audit resources to large volumes of transactions and account balances without any particular focus on specified areas of the financial statements. A bottom-up approach is not risk-based and views all controls equally, therefore, testing a high number of controls instead of focusing on high-risk controls, processes, and transactions.
- c. Under the **balance sheet approach**, substantive procedures are performed on balance sheet accounts, with only limited procedures applied to income statement (profit and loss) accounts.
 - 1) The justification for this approach is that if the relevant management assertions for all balance sheet accounts are tested and verified, the income reported for the accounting period will not be materially misstated.
- d. The **systems-based approach** requires auditors to assess the effectiveness of the internal controls and then to perform substantive procedures primarily on accounts that are least likely to meet systems objectives. Reduced testing is appropriate for accounts that are likely to meet systems objectives.

- e. The **risk-based approach** directs audit resources to appropriate financial statement assertions based on the auditor's assessment of the risk of material misstatements (whether by fraud or error).
 - 1) Auditors are required to make risk assessments at the financial statement and relevant assertion levels based on an appropriate understanding of the entity and its environment, including internal controls.
 - a) Risks of material misstatement at the **financial statement level** relate pervasively to the financial statements as a whole and potentially affect many assertions.
 - b) **Assertions** are representations by management, made either explicitly or implicitly, about the recognition, measurement, presentation, and disclosure of information in the financial statements.
 - 2) This approach requires auditors to identify the key day-to-day risks faced by a business, consider the effect these risks could have on the financial statements, and plan their audit procedures accordingly.

14. Audit Opinions

- a. An external auditor may express four types of audit opinions in audit reports on financial statements.
 - 1) An **unmodified opinion** is the conclusion that the financial statements are presented fairly, in all material respects, in accordance with the framework.
 - 2) A **qualified opinion** is the conclusion that, except for the matter(s) described in the basis for qualified opinion paragraph, the financial statements are presented fairly, in all material respects, in accordance with the framework.
 - a) If the auditor has obtained sufficient appropriate audit evidence, misstatements are material but not pervasive.
 - b) If the auditor has not obtained sufficient appropriate audit evidence, the possible effects of undetected misstatements are material but not pervasive.
 - 3) An **adverse opinion** is the conclusion that, because of the significance of the matter(s) described in the basis for adverse opinion paragraph, the financial statements are not presented fairly.
 - a) The auditor has obtained sufficient appropriate audit evidence, but the misstatements are material and pervasive.
 - 4) A **disclaimer of opinion** is the conclusion that, because of the significance of the matter(s) described in the basis for disclaimer of opinion paragraph, the auditor has not been able to obtain sufficient appropriate audit evidence. Accordingly, the auditor does not express an opinion on the financial statements.
 - a) The possible effects of undetected misstatements are material and pervasive.

17.2 RISK AND INTERNAL CONTROL

1. The Assessment and Management of Risk

- a. Every organization faces risks, that is, unforeseen obstacles to the pursuit of its objectives. Risks take many forms and can originate from inside or outside the organization. Examples include the following:
 - 1) A hacker may break into a university's information systems, changing grades and awarding unearned degrees.
 - 2) The CEO may bribe a member of Congress to introduce legislation favorable to the firm's business.
 - 3) A foreign government may be overthrown in a coup d'etat, followed by the expropriation of the firm's assets in that country.
 - 4) An accounts payable clerk may establish fictitious vendors in the company's information systems and receive checks in payment for nonexistent goods or services.
 - 5) A spike in interest rates may make the firm's long-term capital projects unprofitable.
 - 6) The introduction of a new technology may make one of the firm's premier products obsolete.
- b. **Risk assessment** is the process whereby management identifies the organization's vulnerabilities.
 - 1) All systems of internal control involve tradeoffs between cost and benefit. For this reason, no system of internal control can be said to be "100% effective." Organizations accept the fact that risk can only be mitigated, not eliminated.
 - 2) **Risk management** is the ongoing process of designing and operating internal controls that mitigate the risks identified in the organization's risk assessment.
- c. Risk can be quantified as a combination of two factors: the severity of consequences and the likelihood of occurrence. The expected value of a loss due to a risk exposure can thus be stated numerically as the product of the two factors.
 - 1) Risk can also be assessed in qualitative terms.

EXAMPLE 17-1 Risk Assessment Factors

A company is assessing the risks of its systems being penetrated by hackers.

Event	Consequences	Likelihood
Minor penetration	Annoyance	90%
Unauthorized viewing of internal databases	Public embarrassment, Loss of customer confidence	8%
Unauthorized alteration of internal databases	PR crisis, Customer defection	2%

Although the occurrence of an annoyance-level incident is viewed as almost inevitable, the company will not find it worthwhile to institute the level of control necessary to prevent it.

By contrast, the occurrence of a disastrous-level incident is seen as remote, but the consequences are so severe that the company is willing to institute costly internal controls to ensure its prevention.

- d. In the AICPA's **audit risk** model, audit risk is defined as the risk that an auditor may express an inappropriate opinion on materially misstated financial statements. The model may be adapted to the system of internal control as follows:
- 1) **Inherent risk (IR)** is the susceptibility of one of the company's objectives to obstacles arising from the nature of the objective, assuming no related internal controls. For example, a uranium mine is inherently riskier than a shopping mall.
 - 2) **Control risk (CR)** is the risk that the controls put in place will fail to prevent an obstacle from interfering with the achievement of the objective. For example, a policy requiring two approvals for expenditures over a certain dollar amount could be bypassed by collusion.
 - 3) **Detection risk (DR)** is the risk that an obstacle to an objective will not be detected before a loss has occurred. For example, an embezzlement that continues for a year before detection is much costlier than one that is discovered after 1 month.
 - 4) **Total audit risk (TR)** may thus be stated as follows:

$$TR=IR \times CR \times DR$$

2. Objectives of Internal Control

- a. The three classes of objectives direct organizations to the different (but overlapping) elements of control.
 - 1) **Operations**
 - a) Operations objectives relate to **achieving the entity's mission**.
 - i) Appropriate objectives include improving (a) financial performance, (b) productivity, (c) quality, (d) innovation, and (e) customer satisfaction.
 - b) Operations objectives also include **safeguarding of assets**.
 - i) Objectives related to protecting and preserving assets assist in risk assessment and development of mitigating controls.
 - ii) Avoidance of waste, inefficiency, and bad business decisions relates to broader objectives than safeguarding of assets.
 - 2) **Reporting**
 - a) To make sound decisions, stakeholders must have reliable, timely, and transparent financial information.
 - b) Reports may be prepared for use by the organization and stakeholders.
 - c) Objectives may relate to
 - i) Financial and nonfinancial reporting
 - ii) Internal or external reporting
 - 3) **Compliance**
 - a) Entities are subject to laws, rules, and regulations that set minimum standards of conduct.
 - i) Examples include taxation, environmental protection, and employee relations.
 - ii) Compliance with **internal policies and procedures** is an operational matter.

4) The following is a useful memory aid for the COSO classes of objectives:

O	Operations
R	Reporting
C	Compliance

b. Linkage

- 1) The entity's overall objectives must connect its particular capabilities and prospects with the objectives of its business units.
- 2) As conditions change, the objectives (and related internal controls) of subunits must be altered to adapt to changes in the objectives of the entity as a whole.

c. Achievement of Objectives

- 1) An internal control system is more likely to provide reasonable assurance of achieving the reporting and compliance objectives than the operational objectives.
- 2) Reporting and compliance objectives are responses to standards established by external parties, such as regulators.
 - a) Thus, achieving these objectives depends on actions almost entirely within the entity's control.
- 3) However, operational effectiveness may not be within the entity's control because it is affected by human judgment and many external factors.

3. Roles and Responsibilities Regarding Internal Control

a. Responsible Parties

- 1) Board of Directors and Its Committees
 - a) The entity's commitment to integrity and ethical values is reflected in the board's selections for the chief executive officer (CEO) and senior vice president positions.
 - b) To be effective, board members should be objective, have a role regarding internal controls in an oversight only capacity, have knowledge of the organization's industry, and be willing to ask the relevant questions about management's decisions.
 - c) Important subcommittees of the board, in organizations of sufficient size and complexity, include the following:
 - i) Audit committee
 - ii) Compensation committee
 - iii) Nomination or governance committee
 - iv) Finance committee
 - v) Risk committee

2) Senior Management

- a) Senior management is responsible for establishing and maintaining the organization's system of internal control.
- b) The CEO also establishes the tone at the top. Organizations inevitably reflect the ethical values and control consciousness of the CEO.

3) Internal Auditors

- a) Although management is ultimately responsible for the design and operation of the system of internal controls, an organization's internal audit function has a consulting and advisory role.
- b) Internal audit also evaluates the adequacy and effectiveness of internal control in response to risks in the entity's oversight, operations, and information systems. For example, internal audit evaluates controls over the following:
 - i) Achievement of the organization's strategic objectives
 - ii) Reliability and integrity of financial and operational information
 - iii) Effectiveness and efficiency of operations and programs
 - iv) Safeguarding of assets
 - v) Compliance with laws, regulations, standards, policies, procedures, and contracts
- c) To remain independent in the conduct of these reviews, the internal audit function cannot be responsible for selecting and executing controls. This is solely the responsibility of management.

4) Other Entity Personnel

- a) Everyone in the organization must be aware that (s)he has a role to play in internal control, and every employee is expected to perform his or her appropriate control activities.
- b) In addition, all employees should understand that they are expected to inform those higher in the organization of instances of poor control.

b. External Parties

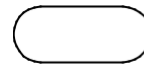
- 1) External Auditors
 - a) Traditionally, independent accountants have been required by their professional standards to consider the auditee's system of internal control as part of their audit of the entity's financial statements.
 - b) This is no longer simply a good practice self-imposed by the accounting profession. The PCAOB has made it a legal requirement for auditors of public companies to examine and report on internal control.
- 2) Legislators and Regulators
 - a) Congress passed the Foreign Corrupt Practices Act and the Sarbanes-Oxley Act, both of which set legal requirements with regard to internal control.
- 3) Parties Interacting with the Entity
 - a) The following are examples of control-related information received from outside sources:
 - i) A major customer informs management that a sales representative attempted to arrange a kickback scheme in return for the customer's business.
 - ii) A supplier reveals inventory and shipping problems by complaining about incomplete orders.
 - b) Parties without a supplier or customer relationship with the entity also may provide insight into the functioning of controls.
 - i) As a condition for granting a loan, a bank may require that certain ratios be kept above or below specified levels. As a result, management may need to pay closer attention to controls over cash and inventory levels than it did previously.
- 4) Financial Analysts, Bond Rating Agencies, and the News Media
 - a) The actions of these parties can inform an entity about how it is perceived by the world at large.
- 5) Outsourced Service Providers
 - a) Some of an entity's business functions, e.g., IT, human resources, or internal audit, may be performed by external service providers.
 - i) Accordingly, management must manage the risks of outsourcing. Thus, it must implement a means of evaluating the effectiveness of control over outsourced activities.

4. Flowcharting

a. Flowcharting is the representation of a process using pictorial symbols and is useful in understanding, evaluating, and documenting internal control and systems development. Flowcharts provide a visual of the various steps of a process from beginning to end. Flowcharts assist with identifying strengths and weaknesses in internal controls. A document flowchart effectively displays the document flows through various departments.

1) Flowcharting symbols have been standardized by both the American National Standards Institute (ANSI) and the International Organization for Standardization (ISO).

2) To the right are some standard flowcharting symbols representing process endpoints and connectors:



Starting or ending point



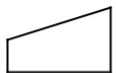
Connection between points on the same page



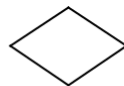
Connection between different pages of the flowchart

Figure 17-2

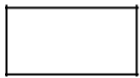
3) Below are some standard flowcharting symbols representing processes:



Keying operation



Decision point



Computer operation



Manual operation

Figure 17-3

4) Below are some standard flowcharting symbols representing input and output:



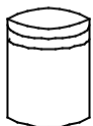
Generalized symbol for input or output used when the medium is not specified



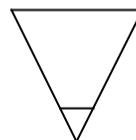
Display on a video terminal



Online storage (magnetic disk)



Database (magnetic disk)



Offline storage (file)

Figure 17-4

- b. Vertical flowcharts present successive steps in a top-to-bottom format. Before the advent of object-oriented programming, flowcharts were a very common tool for computer programmers to design the flow of a new system. Below is an example of a simple vertical flowchart.

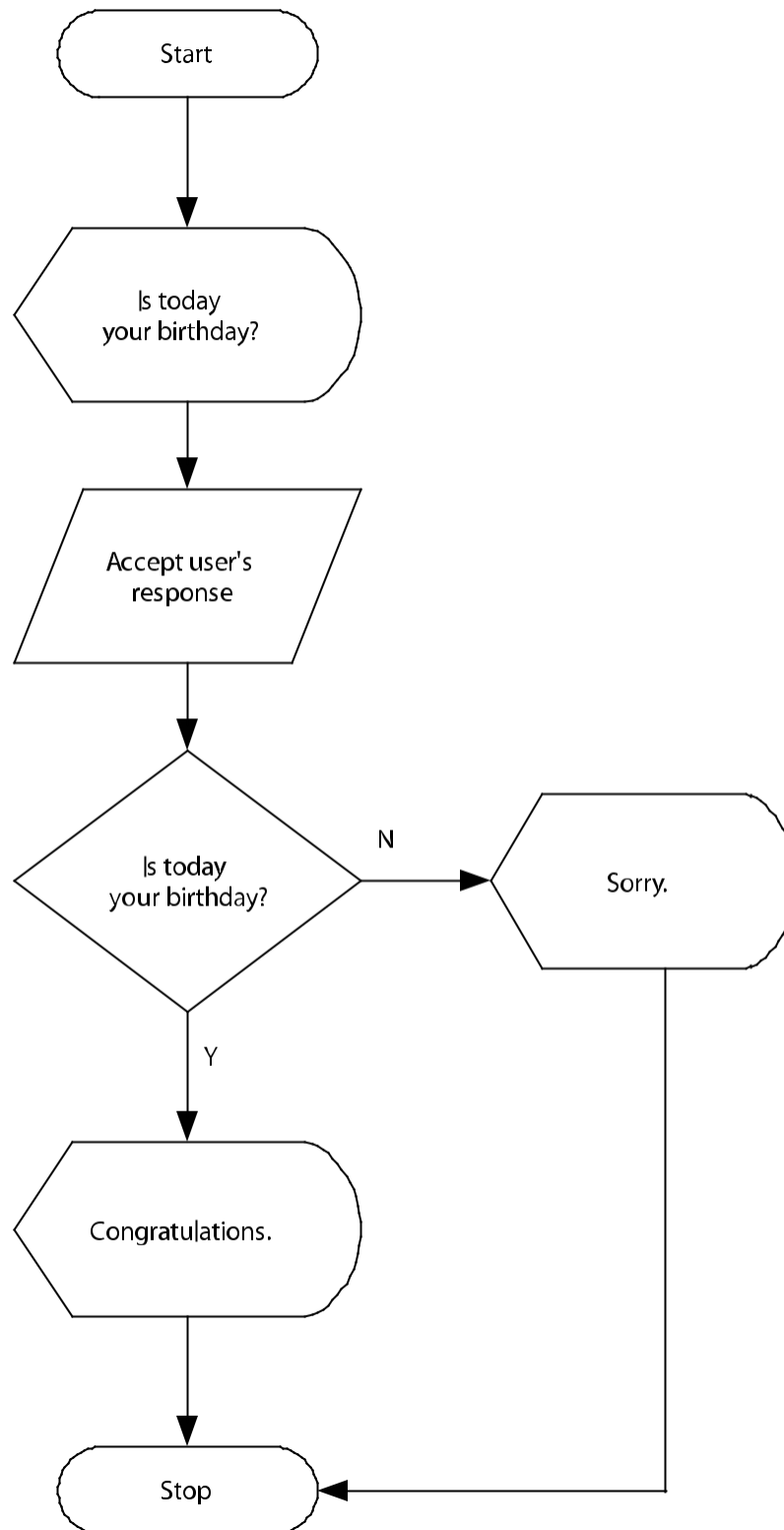


Figure 17-5

- c. Horizontal flowcharts, also called systems flowcharts, depict areas of responsibility (departments or functions) in vertical columns. Activities and documents flow back and forth between departments across the page.

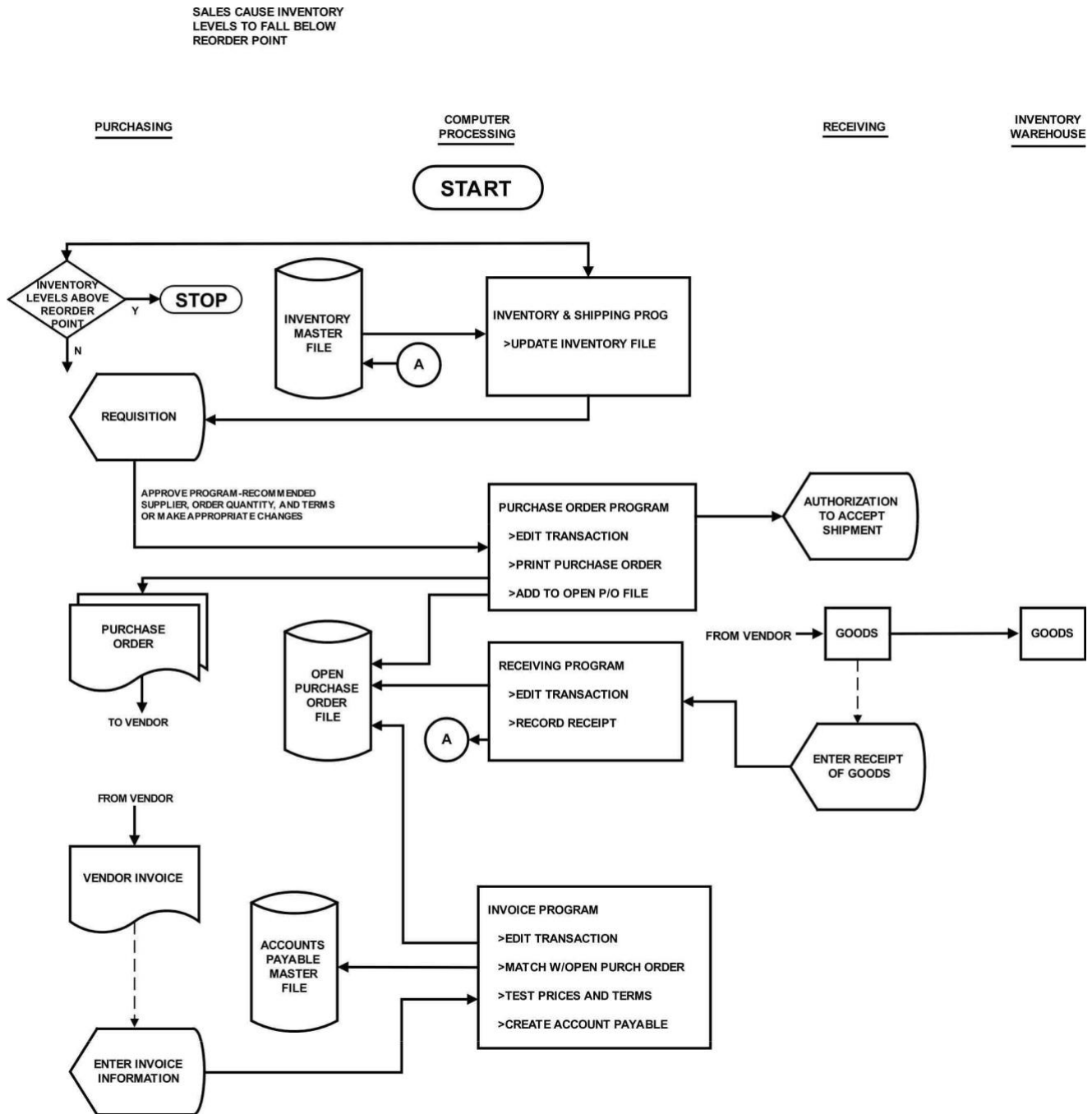


Figure 17-6

5. PCAOB Approach

- a. One of the requirements of the Sarbanes-Oxley Act is that the annual financial statement audit also address the firm's system of internal control over financial reporting.
 - 1) The PCAOB issued its Auditing Standard (AS) No. 2201, "An Audit of Internal Control Over Financial Reporting That Is Integrated with An Audit of Financial Statements," to provide guidance when these two audits are integrated.
 - 2) AS 2201 requires the external auditor to express an opinion on both the system of internal control over financial reporting and the fair presentation of financial statements.
 - a) The auditor's primary consideration when assessing a firm's internal controls over financial reporting is whether they affect the financial statement assertions.
 - b) Per AS 2201, the auditor should use a top-down (risk-based) approach to audit internal controls over financial reporting. Under the top-down approach, the auditor begins at the financial statement level, focusing on entity-level controls, and then works down to significant accounts and disclosures and their relevant assertions.
- b. AS 2201 focuses on the existence of material weaknesses in internal control:

Because a company's internal control cannot be considered effective if one or more material weaknesses exist, to form a basis for expressing an opinion, the auditor must plan and perform the audit to obtain competent evidence that is sufficient to obtain reasonable assurance about whether material weaknesses exist as of the date specified in management's assessment. A material weakness in internal control over financial reporting may exist even when financial statements are not materially misstated.

- 1) The AICPA's auditing standards define material weakness as follows:

A material weakness is a deficiency, or combination of deficiencies, in internal control that results in a reasonable possibility that a material misstatement of the financial statements will not be prevented or timely detected and corrected.
- 2) This financial reporting-oriented focus for internal controls stands in contrast with the broader view that internal controls are processes to aid the organization in achieving its goals.

6. COSO Framework

a. *Internal Control – Integrated Framework* is widely accepted as the standard for the design and operation of internal control systems.

- 1) However, regulatory or legal requirements may specify another control model or design.

BACKGROUND 17-1 Origin of COSO Framework

The Watergate investigations of 1973-74 revealed that U.S. companies were bribing government officials, politicians, and political parties in foreign countries. The result was the Foreign Corrupt Practices Act of 1977. The private sector also responded by forming the National Commission on Fraudulent Financial Reporting (NCFRR) in 1985. The NCFRR is known as the Treadway Commission because James C. Treadway was its first chair.

The Treadway Commission was originally sponsored and funded by five professional accounting organizations based in the United States. This group of five became known as the Committee of Sponsoring Organizations of the Treadway Commission (COSO). The Commission recommended that this group of five organizations cooperate in creating guidance for internal control. The result was *Internal Control – Integrated Framework*, published in 1992, which was modified in 1994 and again in 2013.

The executive summary is available at www.coso.org/documents/990025P-Executive-Summary-final-may20.pdf

b. The COSO model defines internal control as follows:

*Internal control is a process, effected by an entity's board of directors, management, and other personnel, designed to provide **reasonable** assurance (not absolute assurance) regarding the achievement of objectives relating to operations, reporting, and compliance.*

c. Thus, internal control is

- 1) Intended to **achieve** three classes of **objectives**
- 2) An **ongoing process**
- 3) Effected by **people** at **all** organizational **levels**, e.g., the board, management, and all other employees
- 4) Able to provide **reasonable**, but not absolute, **assurance**
- 5) **Adaptable** to an entity's structure

7. Components of Internal Control

a. Supporting the organization in its efforts to achieve objectives are the following five components of internal control:

- 1) Control environment
- 2) Risk assessment
- 3) Control activities
- 4) Information and communication
- 5) Monitoring

b. A useful memory aid for the COSO components of internal control is, "Controls stop **CRIME**."

C	Control activities
R	Risk assessment
I	Information and communication
M	Monitoring
E	Control environment

8. Control Environment

a. The control environment is a set of standards, processes, and structures that pervasively affects the system of internal control. Five principles relate to the control environment:

- 1) The organization demonstrates a commitment to **integrity and ethical values** by
 - a) Setting the **tone at the top**. Through words and actions, the board of directors and management communicate their attitude toward integrity and ethical values.
 - b) Establishing standards of conduct. The board and management create expectations that should be understood at all organizational levels and by outside service providers and business partners.
 - c) Evaluating the performance of individuals and teams based on the established standards of conduct.
 - d) Correcting deviations in a timely and consistent manner.
- 2) The board demonstrates independence from management and exercises **oversight** of internal control. The board
 - a) Establishes oversight responsibility. The board identifies and accepts its oversight responsibilities.
 - b) Applies relevant experience by defining, maintaining, and periodically evaluating the skills and expertise needed among its members to ask difficult questions of management and take appropriate actions.
 - c) Operates independently. The board includes enough members who are independent and objective in evaluations and decision making.
 - i) For example, in some jurisdictions, all members of the audit committee must be outside directors.
 - d) Provides oversight. The board is responsible for oversight of management's design, implementation, and conduct of internal control.
- 3) Management establishes, with board oversight, **structures, reporting lines, and appropriate authorities and responsibilities**. Management
 - a) Considers all structures of the entity. Variables considered in establishing organizational structures include the following:
 - i) Nature of the business
 - ii) Size and geographic scope of the entity
 - iii) Risks, some of them outsourced, and connections with outside service providers and partners
 - iv) Assignment of authority to different management levels
 - v) Definition of reporting lines
 - vi) Reporting requirements
 - b) Establishes and evaluates reporting lines. The trend in corporate governance has been to allow employees closer to day-to-day operations to make decisions.
 - c) Designs, assigns, and limits authorities and responsibilities.

- 4) The organization demonstrates a **commitment to attract, develop, and retain competent individuals** in alignment with objectives.
 - a) Policies and practices reflect expectations of competence. Internal control is strengthened when management specifies what competencies are needed for particular jobs.
 - b) The board and management evaluate competence and address shortcomings. Employees and outside service providers have the appropriate skills and knowledge to perform their jobs.
 - c) The organization attracts, develops, and retains individuals. The organization is committed to hiring individuals who are competent and have integrity. Ongoing training and mentoring are necessary to adapt employees to the control requirements of a changing environment.
 - d) Senior management and the board plan and prepare for succession.
- 5) The **organization holds individuals accountable** for their internal control responsibilities in pursuit of objectives. Management and the board
 - a) Enforce accountability through structures, authorities, and responsibilities
 - b) Establish performance measures, incentives, and rewards
 - c) Evaluate performance measures, incentives, and rewards for ongoing relevance
 - d) Consider excessive pressures
 - e) Evaluate performance and reward or discipline individuals
- b. The control environment reflects the attitude and actions of the board and management regarding the significance of control within the organization.
 - 1) Internal control policies and procedures help ensure that management directives to mitigate risks are carried out. They are intended to ensure that necessary actions are taken to address risks to achieve management's objectives.

9. Risk Assessment

- a. The risk assessment process encompasses an assessment of the risks themselves and the need to manage organizational change. This process is a basis for determining how the risks should be managed. Four principles relate to risk assessment:
 - 1) The organization **specifies objectives** with sufficient clarity to enable the identification and assessment of risks relating to five types of objectives.
 - a) Operations
 - b) External financial reporting
 - c) External nonfinancial reporting
 - d) Internal reporting
 - e) Compliance

- 2) The organization **identifies** risks to the achievement of its objectives across the entity and **analyzes** risks to determine the appropriate risk **response**. Management must focus carefully on risks at all levels of the entity and take the necessary actions to manage them.
 - a) **Risk response** considers how each risk should be managed; whether to accept, avoid, reduce, or share the risk; and the design of controls. Strategies for risk response include the following:
 - i) Risk **avoidance** ends the activity from which the risk arises. For example, the risk of having a pipeline sabotaged in an unstable region can be avoided by simply selling the pipeline.
 - ii) Risk **retention** accepts the risk of an activity. This term is a synonym for self-insurance.
 - iii) Risk **reduction** (mitigation) lowers the level of risk associated with an activity. For example, the risk of systems penetration can be reduced by maintaining a robust information security function within the organization.
 - iv) Risk **sharing** transfers some loss potential to another party. Examples are purchasing insurance, hedging, and entering into joint ventures.
 - v) Risk **exploitation** seeks risk to obtain a high return on investment.
- 3) The organization considers the potential for fraud in **assessing fraud risks** to the achievement of objectives. The organization must
 - a) Consider various types of fraud,
 - b) Assess incentives and pressures,
 - c) Assess opportunities, and
 - d) Assess attitudes and rationalizations.
- 4) The organization **identifies and assesses changes** that could significantly affect the system of internal control.
 - a) Significant changes could occur in an organization's external environment, business model, and leadership. Thus, internal controls must be adapted to the entity's changing circumstances.

10. Control Activities

- a. These policies and procedures help ensure that management directives and risk responses are carried out. Whether automated or manual, they are applied at various levels of the entity and stages of processes.
 - 1) They may be preventive or detective, and segregation of duties is usually present (discussed in more detail in Study Unit 18, Subunit 1).
 - 2) Three principles relate to control activities:
 - a) The organization **selects and develops control activities** that contribute to the mitigation of risks to the achievement of objectives to acceptable levels.
 - b) The organization selects and develops **general control activities** over technology to support the achievement of objectives.
 - c) The organization **deploys control activities** through **policies** that establish what is expected and **procedures** that put policies into action.

11. Information and Communication

- a. Information systems enable the organization to obtain, generate, use, and communicate information to (1) maintain accountability and (2) measure and review performance. Three principles relate to information and communication:
 - 1) The organization obtains or generates and uses **relevant, quality information** to support the functioning of internal control.
 - 2) The organization **internally communicates** information, including objectives and responsibilities for internal control, necessary to support the function of internal control.
 - 3) The organization **communicates with external parties** regarding matters affecting the functioning of internal control.

12. Monitoring Activities

- a. Control systems and the way controls are applied change over time. Monitoring is a process that assesses the quality of internal control performance over time to ensure that controls continue to meet the needs of the organization. The following are two principles related to monitoring activities:
 - 1) The organization selects, develops, and performs **ongoing or separate evaluations (or both)** to determine whether the components of internal control are present and functioning.
 - 2) The organization **evaluates and communicates control deficiencies** in a timely manner.

13. Relationship of Objectives, Components, and Organizational Structure

- a. The COSO model may be displayed as a cube with rows, slices, and columns. The rows are the five components, the slices are the three objectives, and the columns represent an entity's organizational structure.

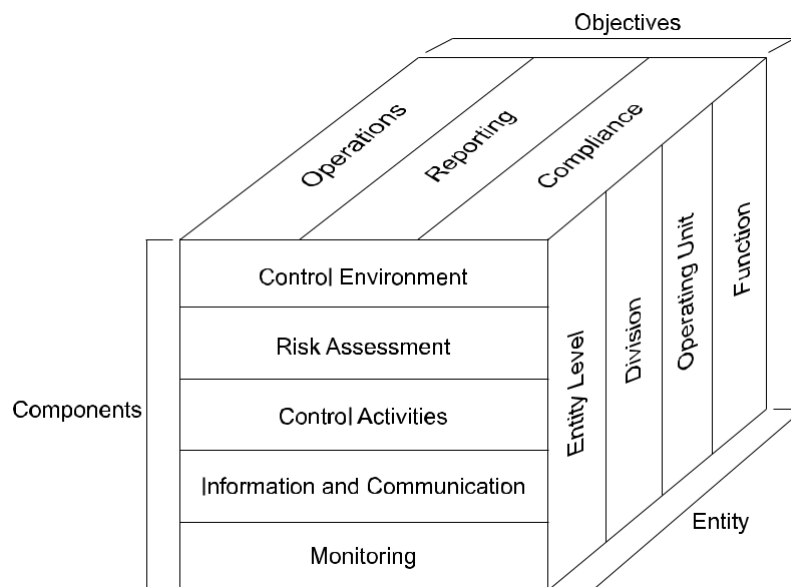


Figure 17-7

STUDY UNIT EIGHTEEN

INTERNAL CONTROLS -- CONTROLS AND SECURITY MEASURES

18.1	Control Procedures	1
18.2	Systems Controls and Information Security	11
18.3	Security Measures and Business Continuity Planning	21

This study unit is the **second of two** on **internal controls**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The two study units are

Study Unit 17: Internal Controls -- Corporate Governance

Study Unit 18: Internal Controls -- Controls and Security Measures

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.



When taking the exam, try to answer questions in consecutive order as much as possible. If you don't know an answer, make your best educated guess and mark it for review; you can go back to it if you have time. This helps you stay on track and prevents you from accidentally leaving any questions unanswered!

18.1 CONTROL PROCEDURES

1. The Control Process

- a. Control requires feedback on the results of organizational activities for the purposes of measurement and correction.
 - 1) The control process is a key element of management.
 - 2) Management is responsible for establishing goals and objectives, developing and implementing control procedures, and accomplishing desired results.
- b. The control process includes
 - 1) Establishing standards for the operation to be controlled,
 - 2) Measuring performance against the standards,
 - 3) Examining and analyzing deviations,
 - 4) Taking corrective action, and
 - 5) Reappraising the standards based on experience.
- c. An evaluation-reward system should be implemented to encourage compliance with the control system.
- d. The costs of internal control must not be greater than its benefits.
- e. No control system can eliminate all fraud. Experienced employees can often evade controls, and collusion between employees also can override many controls.

2. Types of Controls

a. Primary Controls

- 1) **Preventive controls** deter the occurrence of unwanted events. Thus, effective control stops problems before they occur.
 - a) Storing petty cash in a locked safe and segregation of duties are examples of this type of control.
 - b) IT examples include
 - i) Designing a database so that users cannot enter a letter in the field that stores a Social Security number
 - ii) Requiring the number of invoices in a batch to be entered before processing begins
 - iii) Establishing a formal security policy
 - iv) Using only clean and certified copies of software
 - v) Not using shareware software
 - vi) Checking new software with antivirus software
 - vii) Restricting access
 - viii) Educating users
 - ix) Using edit (field) checks to prevent certain types of incorrect data from entering a system
 - x) Preformatting a data entry screen so that certain fields must be filled before proceeding
- 2) **Detective controls** alert the proper people after an unwanted event. They are effective when detection occurs before material harm occurs. Examples of detective controls include
 - a) Automatic reporting to the accounts payable department of all rejected batches of invoices
 - b) Using hash totals [defined in Subunit 18.2, item 8.b.1)d)] to detect data entry errors and/or test for completeness
 - c) Installing burglar alarms
 - d) Examining system logs of actions that require scrutiny, such as repeated failed login attempts and the use of powerful utility programs
- 3) **Corrective controls** correct the negative effects of unwanted events. Examples of corrective controls include
 - a) Requiring that all cost variances over a certain amount are justified
 - b) Correcting errors reported on error listings
 - c) Isolating and removing viruses
 - d) Restarting from system crashes
- 4) **Directive controls** cause or encourage the occurrence of a desirable event.
 - a) Policy and procedure manuals are common examples.

b. Secondary Controls

- 1) **Compensatory (mitigative) controls** may reduce risk when the primary controls are ineffective. However, they do not, by themselves, reduce risk to an acceptable level.
 - a) An example is supervisory review when segregation of duties is not feasible.
- 2) **Complementary controls** work with other controls to reduce risk to an acceptable level.
 - a) For example, separating the functions of accounting for and custody of cash receipts is complemented by obtaining deposit slips validated by the bank.

c. Time-Based Classification

- 1) **Feedback controls** report information about completed activities. They permit improvement in future performance by learning from past mistakes. Thus, corrective action occurs after the fact. Inspection of completed goods is an example.
- 2) **Concurrent controls** adjust ongoing processes. These real-time controls monitor activities in the present to prevent them from deviating too far from standards. An example is close supervision of production-line workers.
- 3) **Feedforward controls** anticipate and prevent problems. These controls require a long-term perspective. Organizational policies and procedures are examples.

d. Financial vs. Operating Controls

- 1) **Financial controls** should be based on relevant established accounting principles.
 - a) Objectives of financial controls may include proper authorization; appropriate recordkeeping; safeguarding of assets; and compliance with laws, regulations, and contracts. These are sometimes called "accounting controls."
- 2) **Operating controls** apply to production and support activities. They are sometimes called "administrative controls."
 - a) Because they may lack established criteria or standards, they should be based on management principles and methods. They also should be designed with regard to the management functions of planning, organizing, directing, and controlling.
 - b) Documentary controls apply to documents, such as vendor invoices, purchase orders, and receiving reports, that evidence the performance of operating controls. A purchase order stating the quantity goes to the vendor and to accounts payable. The receiving department should be sent a blind copy (no price or quantity) so that a count is needed to prepare the receiving report. Payment for a purchase is authorized by accounts payable only after examination of the vendor's invoice, purchase order, and receiving report.

e. People-Based vs. System-Based Controls

- 1) **People-based controls** are dependent on the intervention of humans for their proper operation, for example, regular performance of bank reconciliations.
 - a) Checklists, such as lists of required procedures for month-end closing, can be valuable to ensure that people-based controls are executed when needed.
- 2) **System-based controls** are executed whenever needed with no human intervention.
 - a) An example is code in a computerized purchasing system that prevents any purchase order over a certain monetary threshold from being submitted to the vendor without managerial approval.
 - b) Other examples include control totals, reasonableness checks, and sequence tests.

f. **Broad Controls**

- 1) The two broad groupings of information systems control activities are **general controls** and **application controls**. These are examined in the next two subunits. General controls is the umbrella under which the IT function operates. They affect the organization's entire processing environment and commonly include controls over
 - a) Data center and network operations
 - b) Systems software acquisition, change, and maintenance
 - c) Access security
 - d) Application system acquisition, development, and maintenance

g. IT Administration Controls over Operations

- 1) A modern organization should recognize information technology as a separate function with its own set of management and technical skills. An organization that allows every functional area to acquire and administer its own systems in isolation is not serious about proper control.
- 2) Treating IT as a separate functional area of the organization involves the designation of a chief information officer (CIO) or chief technology officer (CTO) and the establishment of an information systems steering committee to set a coherent direction for the organization's systems and prioritize information technology projects.

h. Controls over software acquisition, change, and maintenance include

- 1) **Controls over systems software**, which ensure that operating systems, utilities, and database management systems are acquired and changed only under close supervision and that vendor updates are routinely installed.
- 2) **Controls over application software**, which ensure that programs used for transaction processing (e.g., payroll and accounts receivable) are cost-effective and stable.

i. **Hardware controls** are built into the equipment by the manufacturer. They ensure the proper internal handling of data as they are moved and stored.

- 1) They include parity checks, echo checks, read-after-write checks, and any other procedure built into the equipment to ensure data integrity.

3. Control Activities

- a. Control activities are designed and placed in operation to ensure that management's directives are executed. Thus, they should include the requisite steps to respond to the risks that threaten the attainment of organizational objectives.
 - 1) For this purpose, controls should be suitably designed to prevent or detect unfavorable conditions arising from particular risk exposures. They also should be placed in operation and operate effectively. If controls are not always enforced, they cannot operate effectively, no matter how effective their design.
 - 2) Control procedures are implemented to manage or limit risk in accordance with the entity's risk assessments whenever risk exposures exist that threaten loss of assets or misstatements of accounting or management information.
 - 3) Controls can be identified in the following areas:
 - a) Segregation of duties, including four basic functional responsibilities
 - b) Independent checks and verification
 - c) Safeguarding controls
 - d) Prenumbered forms
 - e) Specific document flow

4. Segregation of Duties



SUCCESS TIP

This subunit describes the types and provides examples of segregation of duties. As a CMA candidate, you must be able to identify and explain proper segregation of duties procedures. However, you should also be able to apply this concept to many different functions within the internal control process.

- a. The **segregation of accounting duties** is vital because a separation of functions (authorization, recording, and access to assets) may not be feasible in an IT environment. For example, a computer may print checks, record disbursements, and generate information for reconciling the account balance. These activities customarily are segregated in a manual system.
 - 1) Four types of functional responsibilities should be segregated:
 - a) Authority to execute transactions
 - i) For example, changes in payroll should be authorized by a department other than payroll. Thus, a personnel department should authorize any changes (to job status, pay rate, etc.) executed by a payroll employee.
 - b) Recordkeeping of the transactions
 - c) Custody of the assets affected by the transactions
 - d) Periodic reconciliation of the existing assets to recorded amounts

- 2) EXAMPLE: In the purchases-payables cycle,
 - a) Inventory control initiates purchase requisitions. The authority to execute and follow up on transactions with vendors is then vested in the purchasing department.
 - i) Specifications should be established for quality, nature, and amounts purchased.
 - b) Recordkeeping is done by accounts payable, not purchasing.
 - c) Custody of the assets involved is vested in the warehouse, not inventory control.
 - d) Goods received are approved by a separate receiving department.
 - e) Periodic reconciliation of the existing assets to recorded amounts is performed by inventory control, not the warehouse.
 - f) Purchasing is a high risk activity despite the stated controls. A purchasing-department employee may transact with parties related to the employee.
- 3) EXAMPLE: In the sales-receivables cycle (also known as the revenue cycle),
 - a) The authority to execute transactions is vested in the sales department, not the treasurer.
 - b) Recordkeeping is done by accounts receivable, not sales.
 - c) Custody of the assets involved is vested in the warehouse (in the case of the merchandise) and the treasurer (in the case of the cash).
 - d) Periodic reconciliation of the existing assets to recorded amounts is performed by the general ledger accounting group, not the treasurer.
 - e) Appropriate control over obsolete or spoiled materials requires that they be determined by an approved authority, separate from the regular user, to lack salability or regular usability.
 - f) Credit sales should be authorized prior to executing the sales.
- 4) EXAMPLE: In the payroll cycle,
 - a) The authority to execute transactions is vested in the human resources department, which authorizes the hiring and termination of employees and their rates of pay and deductions.
 - b) Recordkeeping is done by the payroll department.
 - c) Custody of the assets involved is vested in the treasurer.
 - d) Periodic reconciliation of the existing assets to recorded amounts is performed by the general ledger accounting group.
 - e) Supervisors in a department should approve employee time cards because they know whether hours were actually worked, but they should not distribute paychecks to that department's employees.

- 5) **EXAMPLE:** In the financing-treasury cycle,
- a) Transactions in debt and equity instruments should be properly authorized. The board of directors should review and approve all major funding and repayment proposals.
- 6) The following memory aid is for the functions that should be kept separate for proper segregation of duties:

A	Authorization
R	Recordkeeping
C	Custody
R	Reconciliation

- b. Despite best efforts to segregate certain duties, this control can be compromised through collusion among employees.

5. Independent Checks and Verification

- a. The reconciliation of recorded accountability with the assets must be performed by a part of the organization either (1) unconnected with the original transaction or (2) without custody of the assets involved.
 - 1) A comparison revealing that the assets do not agree with the recorded accountability provides evidence of unrecorded or improperly recorded transactions.
 - a) The converse, however, does not necessarily follow. For example, agreement of a cash count with the recorded balance does not provide evidence that all cash received has been properly recorded.
 - 2) The frequency of such comparisons for the purpose of safeguarding assets depends on the nature and amount of the assets involved and the cost of making the comparison.
 - a) For example, cash may be counted daily but materials inventory only annually.
- b. **EXAMPLE:** The general ledger group performs monthly reconciliations of bank statements to the company's records. This is an independent check on the work of the treasury function.
- c. Annual confirmations of accounts receivable balances sent to customers by an internal auditor are an effective means of verification. Similarly, loan balances can be confirmed with a bank.

6. Safeguarding Controls

- a. Safeguarding controls limit access to an organization's assets to authorized personnel. Access includes both direct physical access and indirect access through the preparation or processing of documents that authorize the use or disposition of assets.
 - 1) EXAMPLES:
 - a) A lockbox system for collecting cash receipts from customers
 - b) Daily, intact deposit of cash receipts after preparation and verification by two treasury employees
 - c) Approval of credit memos by the credit department, not sales
 - d) Writeoffs of uncollectible accounts by the supervisor of the credit department manager
 - e) Unescorted access to computer operations center prohibited to (1) all non-information systems personnel and (2) all non-operations information system personnel, such as developers
 - f) Online access to production application libraries prohibited to developers; online access to production databases prohibited to all users except the organizational "owners" of the data elements
 - g) Direct deposit of pay in lieu of distribution of physical paychecks; unclaimed paychecks held by the treasurer, not payroll; and maintenance of control over unused checks
 - h) Holding of securities in a safe deposit box; two employees always present when box is accessed
 - i) Physical measures taken to protect assets from natural disasters, e.g., floods, wind damage, earthquakes
 - j) Authorization of a payment voucher by accounts payable only after examining the supporting documents

7. Prenumbered Forms

- a. Sequentially prenumbered forms are the basis for a strong set of internal controls. Receiving reports in the warehouse and purchase orders in the sales department are common examples.
 - 1) When every hardcopy form is prenumbered, all can be accounted for; e.g., the date of their use and the person who filled them out can be ascertained. Any document in the sequence that is missing can be flagged for special scrutiny when it is processed.
 - a) During the periodic reconciliation, the verifying party can detect unrecorded and unauthorized transactions.

- 2) This functionality can be achieved even in a paperless environment. Applications can be coded to sequentially number initiated transactions, and proper review and approval can be verified online.
- 3) In addition to prenumbered forms, procedures ensuring that personnel do not receive documents inappropriate to their duties enhance internal control.
 - a) For example, documents authorizing the write-off of uncollectible receivables should not be routed to cashiers. These cashiers could later pocket the money if a written-off account was subsequently paid.

8. Specific Document Flow

- a. As an organization conducts business, documents and other evidence of business transactions should be created.
 - 1) Sequentially prenumbered forms increase the usability of this evidence.
- b. **Tracing and Vouching**
 - 1) **Tracing** follows a transaction forward from the triggering event to a resulting event, ensuring that the transaction was accounted for properly.
 - a) Tracing is used to gain assurance that a liability was properly accrued for all goods received.
 - 2) **Vouching** tracks a result backward to the originating event, ensuring that a recorded amount is properly supported.
 - a) Vouching is used to gain assurance that a receivable claimed is supported by a sale to a customer.

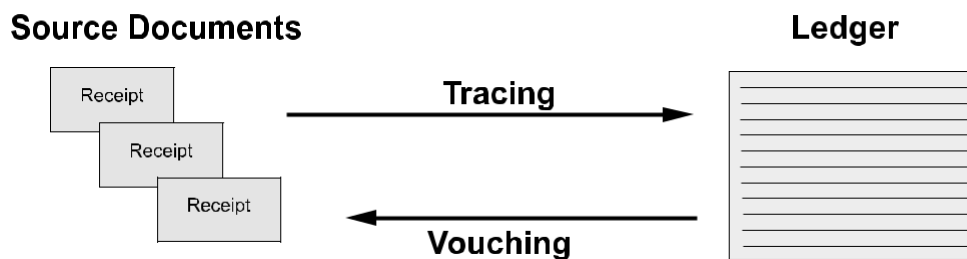


Figure 18-1

- c. By searching for missing records in the document flow, auditors can detect errors or fraud.

- d. The chart below describes some of the possible procedures and the information they provide.

Procedures	Information Provided
Trace shipping documents with sales invoices and journal entries. Account for the numerical sequence of sales orders, shipping documents, invoices, etc.	Sales and receivables were all accounted for.
Vouch sample of recorded sales to customer orders and shipping documents.	Sales occurred.
Trace cash receipts to specific accounts.	Accounts receivable are measured appropriately.
Vouch a sample of recorded cash receipts to accounts receivable and customer orders. Vouch a sample of recorded cash disbursements to approved vouchers.	Cash transactions occurred.
Vouch a sample of recorded purchases to documentation. Vouch a sample of recorded cost of sales to documentation.	Inventory transactions occurred.
Vouch a sample of recorded payables to documentation, e.g., requisitions, purchase orders, receiving reports, and approved invoices.	Purchases occurred.
Trace subsequent payments to recorded payables. Collect supporting documentation and search for unmatched documents to determine whether relevant documents have been lost, misplaced, or misfiled.	Accounts payable are all accounted for.

9. Compensating Controls

- a. Compensating controls replace the normal controls, such as segregation of duties, when the latter cannot feasibly be implemented.
 - 1) For example, in the finance and investment cycle, top management may authorize and execute investments and have access to the records, stock certificates, etc. The compensating control in this case is for at least two people to perform each function.
 - a) Providing oversight is an alternative to the performance of each function by at least two people. Thus, the board may authorize an investment with other functions (custody of stock certificates, management of the portfolio, and oversight of recordkeeping) performed by a top manager.
 - 2) Other compensating controls in the finance and investment cycle include
 - a) Periodic communications with the board,
 - b) Oversight by a committee of the board, and
 - c) Internal auditing's reconciliation of the securities portfolio with the recorded information.

18.2 SYSTEMS CONTROLS AND INFORMATION SECURITY

1. Information security encompasses not only computer hardware and software but all of an organization's information, regardless of what medium it resides on. Information security involves far more than just user IDs and passwords.
 - a. The importance of a broad definition of information security becomes clear in light of recent incidents of firms accidentally disposing of documents containing confidential customer information with their regular trash.
2. **Three Goals of Information Security**
 - a. **Availability** is the ability of the intended and authorized users to access computer resources to meet organizational goals.
 - b. **Confidentiality** is assurance of the secrecy of information that could adversely affect the organization if revealed to unauthorized persons.
 - c. **Integrity** is ensuring that data accurately reflect the business events underlying them and preventing the unauthorized or accidental modification of programs or data.
3. **Steps in Creating an Information Security Plan**
 - a. Threats to the organization's information, i.e., events that can potentially compromise an organization's information infrastructure, must be identified.
 - 1) Threats to confidentiality include the improper disposal of customer records. Threats to availability include viruses and denial-of-service attacks. Threats to integrity include employee errors and sabotage.
 - b. The risks that the identified threats entail must be determined.
 - 1) Risk analysis encompasses determining the likelihood of the identified threats and the level of damage that could potentially be done should the threats materialize.
 - 2) For example, an organization may conclude that, while the potential damage from sabotage is very high, its likelihood is quite low.
 - c. Controls that will compensate for the identified risks should be designed.
 - 1) Controls should be designed based on the combination of likelihood and potential damage determined in the risk analysis.
 - d. The new controls should be incorporated into a coherent, enterprise-wide information security plan.
 - 1) A plan should list the controls that will be put in place and how they will be enforced.
 - e. Policies for expectations of all persons, both employees and external users, with access to the organization's systems must be established.
 - 1) The most important policy is that which governs the information resources to which individuals have access and how the level of access will be tied to their job duties.
 - a) Carrying out such a policy requires the organization's systems to be able to tie data and program access to individual system IDs.
 - b) One provision of the policy should be for the immediate removal of access to the system for terminated employees.

4. Threats to Information Systems

- a. **Input manipulation** is an intrusion into a system by exploiting a vulnerability in a legitimate electronic portal, such as the input fields on a web page. An input field may call, for instance, for the user's address, but a knowledgeable hacker can implant HTML code in the input field that runs a system command giving him or her access to the organization's data.
- b. **Program alteration** is the deliberate changing of the processing routines of an application program. A famous, if fictitious, example is a piece of code that directs all amounts less than one cent to be directed to the malicious programmer's bank account.
- c. **Direct file alteration** is the deliberate changing of data in a database to the intruder's advantage. A common example is a hacker who uses unauthorized access to change his or her course grades while bypassing the normal audit trail.
- d. **Data theft** is the surreptitious copying of critical data elements from an organization's databases. Social Security and credit card numbers are common targets of this type of attack.
- e. **Sabotage** is the disruption of an organization's systems, not for personal gain, but simply for revenge or in the spirit of vandalism. Changing a company's website to include unflattering information that is not immediately noticeable is an example. Another example is a disgruntled programmer who injects a logic bomb into an application that will disrupt processing long after the programmer's departure from the company.
- f. **Malware** is short for malicious software. Basically, the term "malware" encompasses all harmful software, including all of those listed in item 4.
- g. **Viruses** are computer programs that propagate (replicate) themselves from one computer to another without the user's knowledge. Some are written only for the programmer's amusement and are relatively harmless. This type of virus may cause a clever or annoying message to appear on the user's screen. Others are malicious and can cause great inconvenience and even loss of data.
 - 1) One symptom is that programs load or execute more slowly.
 - 2) A common way of spreading a virus is by email attachments and downloads of public domain software.
 - 3) Countermeasures include access restriction, frequent change of passwords, and tests using virus detection software, although antivirus software may not detect new viruses or variants of known viruses.
- h. **Logic bombs** also destroy data but, unlike viruses, they remain on a single computer and do not replicate. Often they lie dormant until triggered by some occurrence, such as the arrival of a certain date.
- i. **Worms** are pieces of code that do not threaten the data on a computer (unlike viruses and logic bombs) but are destructive because of the rapidity with which they replicate themselves.
 - 1) A worm released onto the Internet will propagate from network to network, eventually overwhelming one or more servers with traffic.

- j. **Trojan horses** are voluntarily installed on a computer by the user because they are masquerading as programs the user wants.
 - 1) While the program may present the user with, for instance, an entertaining video game, it contains code that a hacker can activate to control the computer, retrieving sensitive data from it or using it to launch proxy attacks on other computers.
- k. **Back doors** are a means of obtaining access to a system while bypassing the usual password controls.
 - 1) IT personnel often deliberately design back doors into systems to allow system management during unusual circumstances.
 - 2) Hackers search for vulnerabilities in systems to exploit back doors for their own purposes.
- l. **Spyware** spies on a user without his or her knowledge and collects data, such as a history of keystrokes. Programs that capture keystrokes are called **keylogger** software.
- m. **Ransomware** holds a computer or files hostage and demands a ransom payment. Ransomware distributors do not really want to cause major trouble. They just want to take something hostage and get a quick payment from the computer user.
- n. **Theft** becomes increasingly problematic with the higher portability of laptop and palmtop computers. All organizations must establish policies for the proper physical protection of computing infrastructure assets.
- o. **Phishing** is the attempt to acquire sensitive information by pretending to be a trustworthy entity.

5. Systems Development Controls

- a. All information systems, automated or manual, perform four basic functions on information: input, processing, output, and storage.
 - 1) Proper management of the systems development process can enhance the accuracy, validity, safety, security, and adaptability of the controls over these functions.
- b. Effective systems development requires the setting of priorities. This can be achieved through a steering committee composed of managers from both the IT function and the end-user functions. The committee approves development projects, assigns resources, and reviews their progress.
 - 1) The steering committee also ensures that requests for new systems are aligned with the overall strategic plan of the organization.
 - 2) All newly developed systems should conform to established organizational standards for coding and documentation.
- c. The most critical separation of duties in an information system is between (1) computer operators, files, equipment, and production programs and (2) programmers and systems analysts.
 - 1) Also, neither a computer operator nor a systems analyst should be allowed to perform programming duties.
 - 2) Moreover, an operator should not have custody of files.

- d. Changes to existing systems should be subject to the same strict controls. Requests for changes should be initiated by an end user and authorized by management or the steering committee.
 - 1) All changes should be made to a working copy of the program. Production code should never be directly alterable by a programmer.
 - 2) All changes should be adequately tested before being placed in production. The test results should be demonstrated for and accepted by the user who requested the change.
 - a) Adequate testing must involve the use of incorrect data. The program must be able to appropriately handle data that do not conform to the ideal.
 - 3) The changed program code should be stored in a secure library during testing and while awaiting migration into production.
- e. Unauthorized program changes can be detected by code comparison. The version in use is compared electronically with an archived version known to be correct.

6. Physical Controls

- a. Physical controls limit physical access and environmental damage to computer equipment and important documents.
- b. **Access Controls**
 - 1) Access controls prevent improper use or manipulation of data files and programs. They ensure that only those persons with a bona fide purpose and authorization have access.
 - 2) No persons except operators should be allowed unmonitored access to the processing facility. This can be accomplished through the use of a guard desk, a keypad, or a magnetic card reader.
 - 3) **Passwords and ID numbers.** The use of passwords and identification numbers (for example, a PIN used for an ATM) is an effective control in an online system to prevent unauthorized access to files. Lists of authorized persons are maintained online. To avoid unauthorized access, the entity may combine
 - a) The entry of passwords or identification numbers;
 - b) A prearranged set of personal questions; and
 - c) The use of badges, magnetic cards, optically scanned cards, or biometric attributes.
 - 4) **Device authorization table.** This control grants access only to those physical devices that should logically need access. For example, because it is illogical for anyone to access the accounts receivable file from a manufacturing terminal, the device authorization table will deny access even when a valid password is used.
 - 5) **System access log.** This log records all uses and attempted uses of the system. The date and time, codes used, mode of access, data involved, and interventions by operators are recorded.
 - 6) **Encryption.** Encoding data before transmission over communication lines makes it more difficult for someone with access to the transmission to understand or modify its contents. Encryption technology converts data into a code.
 - a) Unauthorized users may still be able to access the data but, without the encryption key, will be unable to decode the information.

- 7) **Callback.** This feature requires the remote user to call, give identification, hang up, and wait for a call to an authorized number. This control ensures acceptance of data only from authorized modems. However, a call-forwarding device may thwart this control by transferring access from an authorized to an unauthorized number.
- 8) **Controlled disposal of documents.** One method of enforcing access restrictions is to destroy data when they are no longer in use. Thus, paper documents may be shredded, and magnetic media may be erased.
- 9) **Biometric technologies.** These are automated methods of establishing an individual's identity using physiological or behavioral traits. These characteristics include fingerprints, retina patterns, hand geometry, signature dynamics, speech, and keystroke dynamics.
- 10) **Automatic log-off.** The disconnection of inactive data terminals may prevent the viewing of sensitive data on an unattended work station.
- 11) **Security personnel.** An entity may hire security specialists. For example, (a) developing an information security policy for the entity, (b) commenting on security controls in new applications, and (c) monitoring and investigating unsuccessful access attempts are appropriate duties of the information security officer.

c. **Environmental Controls**

- 1) The processing facility should be equipped with both a cooling and heating system (to maintain a year-round constant level of temperature and humidity) and a fire suppression system.

7. **Logical Controls**

- a. Logical controls are established to limit system access (1) to authorized persons and (2) to authorized persons only to the extent necessary to perform their job duties. For example, database access lists should be reconciled with current payroll lists.
- b. **Authentication** is the act of ensuring that the person attempting to access the system is in fact who (s)he says (s)he is. The most widespread means of achieving this is through the use of IDs and passwords. The elements of user account management are as follows:
 - 1) Anyone attempting to access one of the organization's systems must supply a unique identifier (e.g., the person's name or other series of characters) and a password that is known only to that person and is not stored anywhere in the system in unencrypted format.
 - a) Not even information security personnel should be able to view unencrypted passwords. Security personnel can change passwords, but the policy should require that the user immediately change it to something secret.
 - 2) Password Optimization
 - a) Passwords should be difficult to guess. Ideally, passwords are at least eight characters long and contain both uppercase and lowercase letters as well as numerals.
 - b) A dialog can be designed to query the user for common names in his or her life (e.g., children, pets, sports teams) so that these words can be stored and never permitted by the system to be used as that person's password.
 - c) The system should force passwords to be changed periodically, e.g., every 90 days.

- 3) Password fatigue results when users must log on to several systems in the course of a day. Users are likely to write down their IDs and passwords in such cases, defeating the purpose of automated authentication.
 - a) Single sign-on can be the solution in well-managed systems environments. A single ID and password combination is required to allow a user access to all IT resources (s)he needs. A high level of maintenance and security awareness is required to make single sign-on successful.
 - b) The policy should prohibit employees from leaving their IDs and passwords written down in plain view.
- c. **Authorization** is the practice of ensuring that, once in the system, the user can only access those programs and data elements necessary to his or her job duties.
 - 1) In many cases, users should be able to view the contents of some data fields but not be able to change them.
 - 2) An example is an accounts receivable clerk who can view customers' credit limits but cannot change them. This same clerk can, however, change a customer's outstanding balance by entering or adjusting an invoice.
 - 3) To extend the example, only the head of the accounts receivable department should be able to execute the program that updates the accounts receivable master balance file. An individual clerk should have no such power.

8. Input, Processing, and Output Controls

- a. Operations is the part of an information system involved in daily processing of data and distribution of results. Key controls apply to input, processing, and output. **Input controls** provide reasonable assurance that data submitted for processing are (1) authorized, (2) complete, and (3) accurate. These controls vary depending on whether input is entered in online or batch mode.
 - 1) **Online** input controls can be used when data are keyed into an input screen.
 - a) Preformatting. The data entry screen mimics the old hardcopy document, forcing data entry in all necessary fields.
 - b) Edit (field) checks. The data entry screen prevents certain types of incorrect data from entering the system. For example, the system rejects any attempt to enter numerals in the Name field or letters in the Amount field. Dropdown menus can restrict the user's choices to only valid selections.
 - c) Limit (reasonableness) checks. Certain amounts can be restricted to appropriate ranges, such as hours worked < 20 per day, or invoices over \$100,000 requiring supervisor approval.
 - d) Check digits. An algorithm is applied to any kind of serial identifier to derive a check digit. During data entry, the check digit is recomputed by the system to ensure proper entry. Requiring the full number including the check digit to be keyed in all future data entry operations eliminates the possibility of dropped or transposed digits, etc.
 - i) However, if the check digit is a sum-of-digits, transposition errors will not be detected because the sum of transposed digits will not change. For example, the sum of serial identifier "1, 3, 6" is the same as the sum of "3, 6, 1."
 - e) Prompting. Asking questions of the user to ensure proper data entry.

- b. Program documentation is a control to ensure that programs are kept up to date and perform as intended.
 - 1) **Batch** input controls can be used when data are grouped for processing in "batches."
 - a) Management release. A batch is not released for processing until a manager reviews and approves it.
 - b) Record count. A batch is not released for processing unless the number of records in the batch, as reported by the system, matches the number calculated by the user.
 - c) Control total validation routines. A batch is not released for processing unless the sum of the dollar amounts of the individual items as reported by the system matches the amount calculated by the user.
 - d) Hash total. The arithmetic sum of a numeric field, which has no meaning by itself, can serve as a check that the same records that should have been processed were processed. An example is the sum of all Social Security numbers.
 - i) This number is much too unwieldy to be calculated by the user, but once it is calculated by the system, it can follow the batch through subsequent stages of processing.
- c. **Processing controls** provide reasonable assurance that (1) all data submitted for processing are processed and (2) only approved data are processed. These controls are built into the application code by programmers during the systems development process.
 - 1) Some processing controls repeat the steps performed by the input controls, such as limit checks and control totals.
 - 2) Validation. Identifiers are matched against master files to determine existence. For example, any accounts payable transaction in which the vendor number does not match a number on the vendor master file is rejected. Limit checks and field checks are other examples of input validation routines.
 - 3) Completeness. Any record with missing data is rejected.
 - 4) Arithmetic controls. Cross-footing compares an amount to the sum of its components. Zero-balance checking adds the debits and credits in a transaction or batch to ensure they sum to zero.
 - 5) Sequence check. Computer effort is expended most efficiently when data are processed in a logical order, such as by customer number. This check ensures the batch is sorted in this order before processing begins. Thus, it is not appropriate in a real-time environment.
 - 6) Run-to-run control totals. The controls associated with a given batch are checked after each stage of processing to ensure all transactions have been processed.
 - 7) Key integrity. A record's "key" is the group of values in designated fields that uniquely identify the record. No application process should be able to alter the data in these key fields.

- d. **Output controls** provide assurance that processing was complete and accurate.
 - 1) A complete audit trail should be generated by each process: batch number, time of submission, time of completion, number of records in batch, total dollars in batch, number of records rejected, total dollars rejected, etc.
 - a) The audit trail is immediately submitted to a reasonableness check by the user, who is most qualified to judge the adequacy of processing and the proper treatment of erroneous transactions.
 - 2) Error listings report all transactions rejected by the system. These should be corrected and resubmitted by the user.

9. **Computer-Assisted Audit Techniques (CAATs)**

- a. Certain controls relating to the input, processing, and output of data are internal to the computer system.
 - 1) They should be tested by procedures that are not traditionally performed in a manual environment.
 - 2) Such techniques have been characterized as auditing around the computer or auditing through the computer.
- b. **Auditing around the computer** is not appropriate when systems are sophisticated or the major controls are included in the computer programs. It may be appropriate for very simple systems that produce appropriate printed outputs.
 - 1) The auditor manually processes transactions and compares the results with the client's computer-processed results.
 - 2) Because only a small number of transactions can ordinarily be tested, the effectiveness of the tests of controls must be questioned.
 - 3) The computer is treated as a black box, and only inputs and outputs are evaluated.
- c. **Auditing through the computer**, i.e., concurrent auditing, uses the computer to test the processing logic and controls within the system and the records produced. Concurrent auditing techniques include the following:
 - 1) Processing test data
 - 2) Parallel simulation
 - 3) Generalized audit software
 - 4) Data extraction techniques
 - 5) Creation of an integrated test facility
 - 6) Programming embedded audit modules
- d. CAAT may be systems- or transaction-based or may provide automated methods for extracting and analyzing data.
- e. Test data consist of a set of dummy inputs containing both good and bad data elements. This approach subjects auditor-created data to the client's programs.
 - 1) The auditor can assess the controls embedded in the application by observing (a) whether the good data are correctly processed and (b) how well the system handles the bad input.
 - 2) Test data must never be mingled with real data, and test data must not be allowed to interfere with production processing. Monitoring by IT personnel is crucial when the auditor employs a test deck.

- f. Parallel simulation subjects client data to auditor-created programs.
 - 1) The goal is to determine whether the data are subjected to the processes that the client claims the application performs.
 - 2) Parallel simulation requires the auditor to have considerable technical knowledge. The auditor also must have extensive communications with client personnel to learn the designed functions of the application being imitated.
- g. Generalized audit software (GAS) packages allow the auditor to load a copy of the client's production data onto the auditor's own computer and perform various analytical procedures.
 - 1) The auditor can search for duplicate records, gaps in numerically sequenced records, high-monetary-amount transactions, suspect vendor numbers, etc. Control totals can be calculated, and balances can be stratified for receivables testing.
 - 2) The leading GAS packages are ACL (Audit Command Language) and IDEA (Interactive Data Extraction and Analysis).
 - 3) The hazards lie in ensuring that the data obtained are those required for the audit procedure being performed. Control totals and other methods are used for this purpose.
- h. Spreadsheet analysis. Electronic spreadsheets, such as Microsoft Excel, permit easy analysis of huge amounts of client data and the performance of what-if scenarios.
- i. Integrated test facility (ITF). In this approach, the auditor creates dummy files (a department, vendor, employee, or product) on the client's live production system. The objective is to determine whether a real-time system contains adequate controls.
 - 1) All transactions associated with the dummy files are processed by the live system, and the auditor can observe the results.
 - 2) Use of an ITF requires great care to ensure that no transactions associated with the dummy files are included in production reports and output files.
- j. An embedded audit module is an integral part of an application system. It is designed to identify and report actual transactions and other information that meet criteria having audit significance.
 - 1) An advantage is that it permits continuous monitoring of online, real-time systems.
 - 2) A disadvantage is that audit hooks must be programmed into the operating system and applications programs to permit insertion of audit modules.
- k. Application tracing uses a feature of the programming language in which the application was written.
 - 1) Tracing aids computer programmers in following the step-by-step operation of a computer program's source code. It can be used by auditors for the same purpose.
- l. System mapping is similar to application tracing. But mapping is performed by another computer program instead of by the auditor. A mapping program searches for unused code. For example, fraud might be committed by unexecuted program code that is activated if an unscrupulous programmer becomes dissatisfied or is terminated.

10. Storage Controls

- a. Program documentation. Complete, up-to-date documentation of all programs and associated operating procedures is necessary for efficient operation of an information system. Maintenance of programs is important to provide for continuity and consistency of data processing services to users.
- b. Dual write routines. The data can be stored on two separate physical devices (usually magnetic hard drives) so that a mishap to one does not destroy the organization's data set.
 - 1) Especially important in this regard is the technology known as RAID (redundant array of independent discs), a data storage virtualization technology that combines multiple physical disk drive components into a single logical unit for the purposes of data redundancy, performance, or both.
 - 2) **Spooling** refers to sending data to an intermediate storage that is accessible by the peripheral device when needed. For example, print files are spooled to an intermediary storage for the printer to access when it is ready to print. This differs from a dual write routine in that data are spooled to a separate physical device to shorten processing time rather than for the purpose of data protection.
- c. Validity checks. Hardware that transmits or receives data compares the bits in each byte to the permissible combinations to determine whether they constitute a valid structure.
- d. Physical controls. Mounting hard drives in physically secure rooms and storing portable media in locked storage areas are vital to preventing the compromise of confidential data.
- e. Snapshot copies of often-used data files are sometimes made to allow a file to be used while it is being backed up. However, the risk is that such a file may not be current.
- f. **Cloud computing.** Organizations are relieved of the need to manage the storage of both applications and data because all the software and data they need are located on the Internet.
 - 1) Cloud computing is defined as a standardized IT capability (services, software, infrastructure) delivered via the Internet in a pay-per-use, self-service way.
 - 2) Advantages of cloud computing include lower infrastructure investments and maintenance costs, increased mobility, and lower personnel and utility costs.
 - 3) Disadvantages of cloud computing include less control than over an internal IT department, more difficulty ensuring data security and privacy, and less compatibility with existing tools and software.

18.3 SECURITY MEASURES AND BUSINESS CONTINUITY PLANNING

1. Inherent Risks of the Internet

- a. Password Attacks
 - 1) A **brute-force** attack uses password cracking software to try large numbers of letter and number combinations to access a network. A simple variation is the use of password cracking software that tries all the words in a dictionary.
 - 2) Passwords also may be compromised by Trojan horses, IP spoofing, and packet sniffers. **Spoofing** is identity misrepresentation in cyberspace (e.g., using a false website to obtain visitor information). **Sniffing** is the use of software to eavesdrop on information sent by a user to the host computer of a website.
- b. A man-in-the-middle attack takes advantage of network packet sniffing and routing and transport protocols.
 - 1) These attacks may be used to steal data, obtain access to the network during a rightful user's active session, analyze the traffic on the network to learn about its operations and users, insert new data or modify the data being transmitted, and deny service.
 - 2) Encryption is the effective response to man-in-the-middle attacks. The encrypted data will be useless to the attacker unless it can be decrypted.
- c. A **denial-of-service** (DoS) attack is an attempt to overload an organization's network with so many messages that it cannot function (i.e., induce a system crash).
 - 1) A distributed denial-of-service (DDoS) attack comes from multiple sources, for example, the machines of several innocent parties infected by Trojan horses. When activated, these programs send messages to the target and leave the connection open.
 - 2) A DoS attack may establish as many network connections as possible to exclude other users, thus overloading primary memory or corrupting file systems.

2. Use of Data Encryption

- a. Encryption technology converts data into a code. Unauthorized users may still be able to access the data, but without the encryption key, they cannot decode it.
 - 1) Encryption technology may be either hardware- or software-based. Two major types of encryption software exist.
- b. **Public-key**, or asymmetric, encryption is the more secure of the two because it requires two keys: The public key for coding messages is widely known, but the private key for decoding messages is kept secret by the recipient.
 - 1) The parties who wish to transmit coded messages must use algorithmically-related pairs of public and private keys.
 - 2) The sender uses the recipient's public key, obtained from a directory, to encode the message, and transmits the message to the recipient. The recipient then uses the public key and the related private (secret) key to decode the message.
 - 3) Neither party knows the other's private key. The related public key and private key pair is issued by a certificate authority (a third-party fiduciary, e.g., VeriSign or Thawte). However, the private key is issued only to one party.
 - a) RSA, named for its developers (Rivest, Shamir, and Adelman), is the most commonly used public-key method.

- c. **Private-key**, or symmetric, encryption is less secure because it requires only a single key for each pair of parties that want to send each other coded messages.
 - 1) Data Encryption Standard (DES), a shared private-key method developed by the U.S. government, was the most prevalent secret-key method. It is based on numbers with 56 binary digits.
 - 2) Triple DES was designed as a replacement for DES and is based on the DES algorithm. It uses an effective key strength of 168.
 - 3) The Advanced Encryption Standard (AES) is a symmetric key algorithm designed as a replacement for Triple DES. It is used worldwide by U.S. government organizations to protect sensitive information.

3. Firewalls

- a. A firewall is a combination of hardware and software that separates an internal network from an external network, such as the Internet, and prevents passage of traffic deemed suspicious. Two principal types of firewalls are network firewalls and application firewalls.
 - 1) Firewall systems ordinarily produce reports on organization-wide Internet use, exception reports for unusual usage patterns, and system penetration-attempt reports. These reports are very helpful as a method of continuous monitoring, or logging, of the system.
- b. **Network firewalls** regulate traffic to an entire network, such as an organization's LAN.
 - 1) The firewall examines each query and, depending on the rules set up by the network security administrator, denies entry to the network based on the source, destination, or other data in the header.
 - 2) Queries from a particular source address that repeatedly fail to gain access to the network might indicate a penetration attempt. The firewall can notify network security personnel who can then investigate.
- c. Vulnerability and penetration testing work in unison to identify computer system security weaknesses that require fixes.
 - 1) **Vulnerability testing** (commonly referred to as vulnerability scanning) identifies weaknesses in the IT infrastructure (e.g., operating systems, outdated applications, missing patches, obsolete protocols and certificates, physical access, or open ports).
 - a) Vulnerability testing provides information on
 - i) Targets for penetration testing and
 - ii) How to mitigate identified vulnerabilities.

- 2) **Penetration testing** involves using tools and techniques commonly used by hackers to gain access to an application, system, or network by circumventing its security features.
 - a) Generally, penetration tests focus on multiple vulnerabilities on one or more systems that can be used to gain more extensive access rather than focusing on only a single vulnerability.
 - b) Penetration testing determines
 - i) How well the system tolerates real-world-style attack patterns,
 - ii) The experience level an attacker needs to successfully access the system,
 - iii) The countermeasures required to protect against attacks, and
 - iv) The ability to detect attacks and respond appropriately.
- d. **Application firewalls** regulate traffic to a specified application, such as an email or file transfer application.
 - 1) An application firewall is based on proxy server technology. The firewall becomes a proxy, or intermediary, between the computer actually sending the packet and the application in question.
 - a) This arrangement allows for a high level of security over the application but at the cost of slowing down communications.
 - 2) Since an application firewall only provides security for a single application, it is not a substitute for a network firewall.
- e. A firewall alone is not an adequate defense against computer viruses. Specialized antivirus software is also necessary.

4. **Routine Backup and Offsite Rotation**

- a. An organization's data is more valuable than its hardware.
 - 1) Hardware can be replaced for a price, but each organization's data bundle is unique and is indispensable to carrying on business. If it is ever destroyed, it cannot be replaced. For this reason, periodic backup and rotation are essential.
- b. The offsite location must be temperature and humidity controlled and guarded against physical intrusion.
 - 1) Just as important, it must be geographically remote enough from the site of the organization's main operations that it would not be affected by the same natural disaster. It does the organization no good to have sound backup procedures if the files are not accessible or have been destroyed.
 - a) An alternative is not to keep all files in one location.
- c. A typical backup routine involves duplicating all data files and application programs once a month. (Application files must be backed up as well as data files because programs can change too.)
 - 1) Incremental changes, that is, only those data elements and programs that have changed since the last full monthly backup, are backed up every week and kept at the main processing center. (Transporting the weekly backups to the offsite location is generally not cost-effective.)
- d. In case of an interruption of normal processing, the organization's systems can be restored such that, at most, 3 weeks of business information is lost.
 - 1) This is not an ideal situation, but it is preferable to a complete loss of a company's files, which could essentially put it out of business.

5. Business Continuity Planning

- a. A comprehensive computer security plan should be developed to provide reasonable assurance of safeguarding physical facilities and hardware and the privacy and integrity of data.
 - 1) Such a plan assists management in ensuring that benefits exceed costs.
- b. Because absolute assurance of computer security cannot be provided, continuity (disaster) planning is necessary.
 - 1) Disaster recovery is the process of resuming normal information processing operations after the occurrence of a major interruption.
 - 2) Business continuity planning is the continuation of business by other means during the period in which computer processing is unavailable or less than normal.
- c. Two major types of disasters must be planned for: those in which the data center is physically available and those in which it is not.
 - 1) Examples of the first type of disaster are power failure, random intrusions such as viruses, and deliberate intrusions, such as hacking incidents.
 - a) The organization's physical facilities are sound, but immediate action is required to keep normal processing going.
 - 2) The second type of disaster is much more serious. This type is caused by disasters, such as floods, fires, hurricanes, earthquakes, etc.
 - a) An occurrence of this type necessitates the existence of an alternate processing facility.
- d. Dealing with Specific Types of Disasters
 - 1) Power failures can be guarded against by the purchase of backup electrical generators.
 - a) These can be programmed to automatically begin running as soon as a dip in the level of electric current is detected.
 - b) This is a widespread practice in settings such as hospitals where 24-hour system availability is crucial.
 - 2) Attacks such as viruses and denials-of-service call for a completely different response.
 - a) The system must be brought down "gracefully" to halt the spread of the infection. The IT staff must be well trained in the nature of the latest virus threats to know how to isolate the damage and bring the system back to full operation.
 - b) Antivirus software will stop most attacks. But this software may not detect certain viruses. One sign that a virus has infected a computer is that a program takes longer than usual to load or execute.

- 3) The most extreme disaster happens when the organization's main facility becomes uninhabitable because of flood, fire, earthquake, etc. It is to prepare for these cases that organizations contract for alternate processing facilities. The disaster recovery plan should include instructions for appointing a recovery team when a disaster occurs.
 - a) An alternate processing facility is a physical location maintained by an outside contractor for the express purpose of providing processing facilities for customers in case of disaster.
 - b) The recovery center, like the offsite storage location for backup files, must be far enough away that it will likely be unaffected by the same natural disaster that forced the abandonment of the main facility.
 - i) Usually, companies contract for backup facilities in another city.
 - c) Once the determination is made that processing is no longer possible at the principal site, the backup files are retrieved from the secure storage location and taken to the recovery center.
 - i) The first step in restoration is to set up the operating system.
 - d) Recovery centers can take many forms. Organizations determine which facility is best by calculating the tradeoff between the cost of the contract and the cost of downtime.
 - i) A **hot site** is a fully operational processing facility that is immediately available. A flying-start site is a hot site with the latest data and software that permit startup within a few minutes or even a few seconds.



**AUTHOR'S
NOTE**

An example of a hot site is a **mirrored data center**. A mirrored data center is one that runs parallel to a company's main center of operations, usually when system downtime would be unacceptable. For example, downtime in the systems of e-commerce companies may have substantial impacts. A mirrored data center involves automatic backups to a data storage facility that is geographically remote from the primary data center. Special software is used to instantly copy information from the primary data center to the secondary data center so that the secondary data center can immediately take the place of the primary data center.

- ii) A **warm site** is a facility with limited hardware, such as communications and networking equipment, already installed but lacking the necessary servers and client terminals.
- iii) A **cold site** is a shell facility lacking most infrastructure but readily available for the quick installation of hardware.

APPENDIX G

PRODUCTION CYCLES

Below and on the following page are descriptions of job-order costing and process costing. They include sequential T-accounts illustrating the flow of costs through the production cycle.

Job-Order Costing

Job-order costing is used when each end product is unique. Because the end products are few, tracking their costs is relatively simple. Manufacturers that use job-order costing include construction and shipbuilding. Service industries include software design and plumbing.

Cost Flows

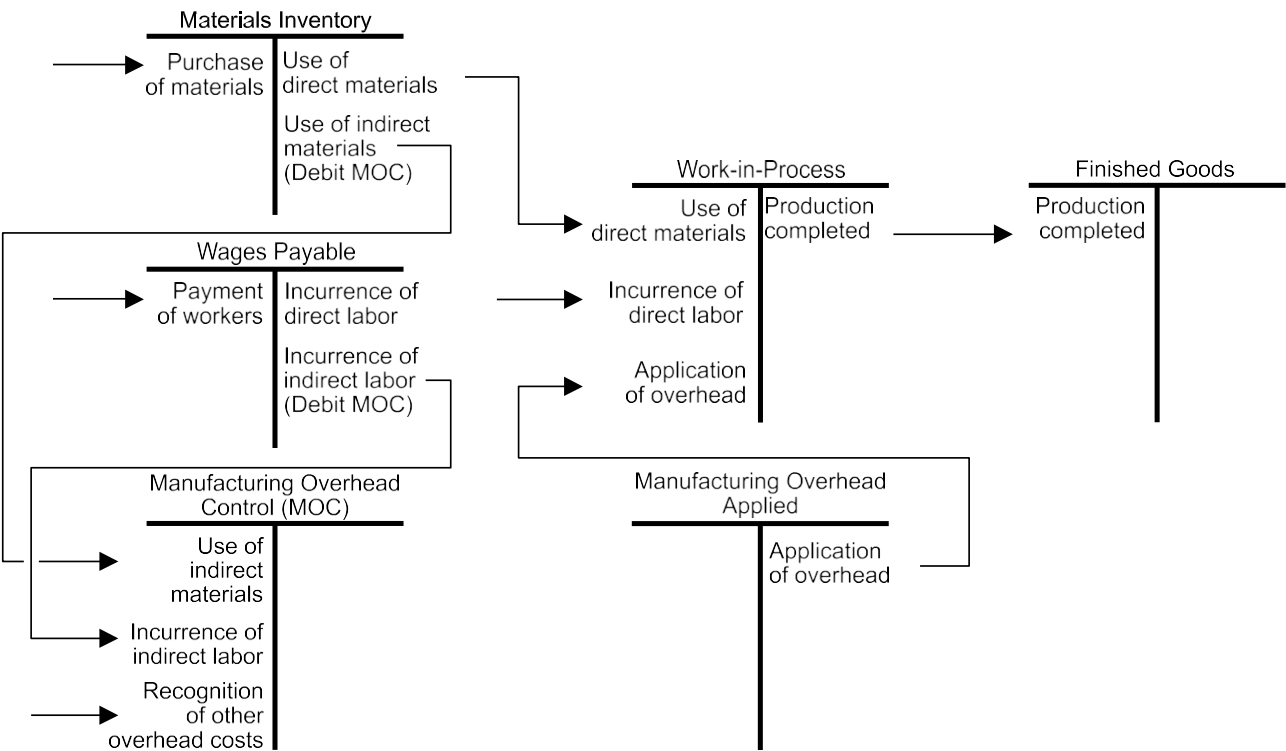


Figure G-1

Process Costing

Process cost accounting assigns costs to inventoriable goods or services. It applies to relatively homogeneous products that are mass produced on a continuous basis (e.g., petroleum products, thread, and computer monitors).

Process Cost Flows

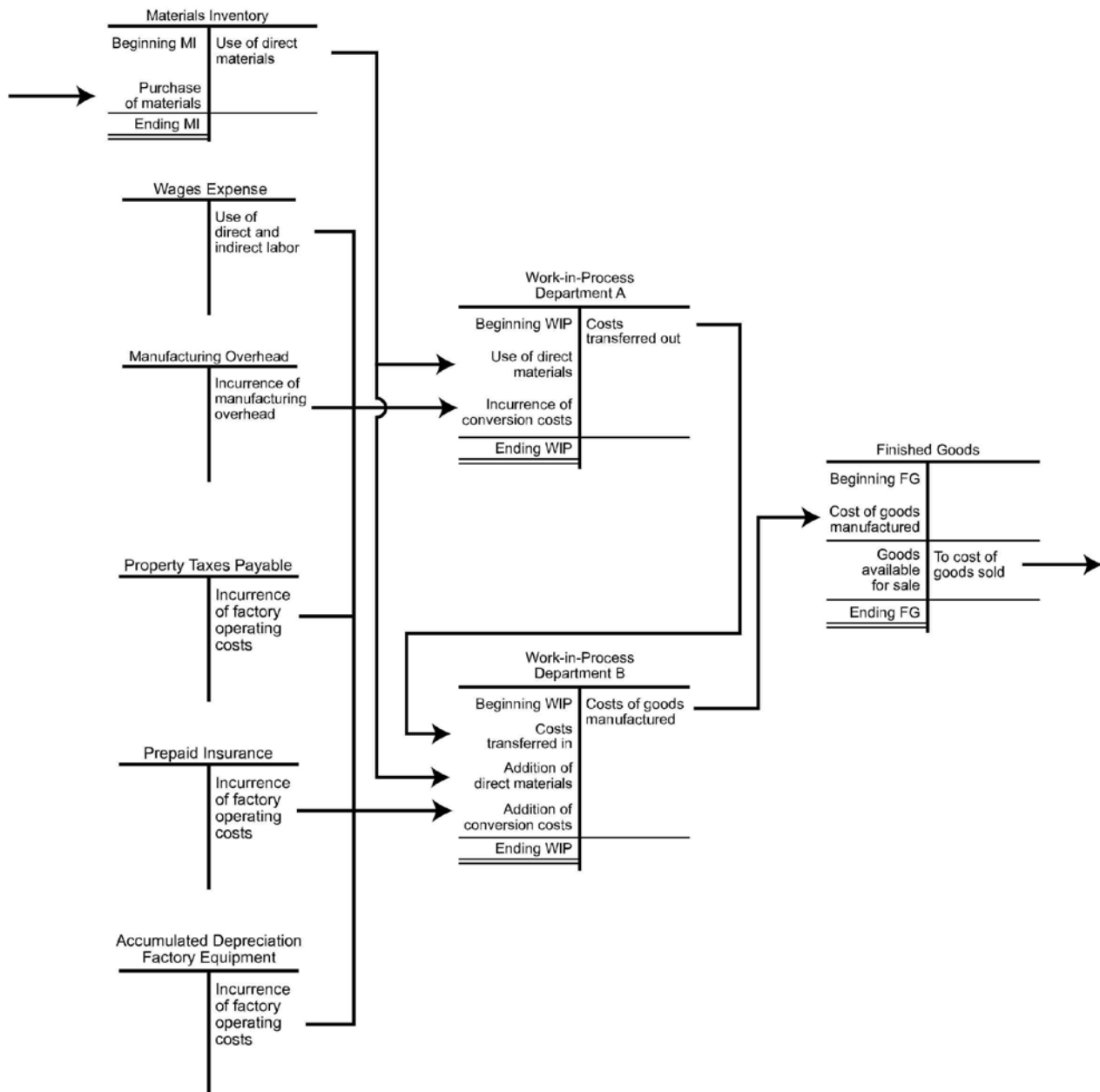


Figure G-2

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**2021
EDITION**

CMA REVIEW

PART 1:

FINANCIAL PLANNING, PERFORMANCE, AND ANALYTICS



F. Technology and Analytics (15% - Levels A, B, and C)

1. *Information systems*

- a. Accounting information systems (19.1)
- b. Enterprise resource planning systems (10.2)
- c. Enterprise performance management systems (19.1)

2. *Data governance*

- a. Data policies and procedures (19.2)
- b. Life cycle of data (19.2)
- c. Controls against security breaches (18.2)

3. *Technology-enabled finance transformation* (20.1)

- a. System development life cycle
- b. Process automation
- c. Innovative applications

4. *Data analytics* (20.2)

- a. Business intelligence
- b. Data mining
- c. Analytic tools
- d. Data visualization

STUDY UNIT NINETEEN

INFORMATION SYSTEMS AND DATA GOVERNANCE

19.1	Accounting Information Systems	2
19.2	Data Governance and Risk	8
19.3	COSO Framework -- Internal Control for Data Governance	15

Technology and Analytics

Data analytics, sometimes called analysis of **big data**, is a growing phenomenon in accounting. Big data refers to sets of data that are too large to be evaluated by traditional means of analysis. Thus, different software is needed than what is commonly used by managerial accountants and auditors. The challenges of big data include how to capture the data, how to store it, how to analyze it, and how to visualize what has been analyzed. A prerequisite to working with big data is a working knowledge of accounting information systems.

The area of technology and analytics was recently added to the CMA exam, and candidates will be expected to know everything from definitions to how to interpret analytical cases. Study Units 19 and 20 have been carefully developed to break down these concepts in an easy-to-understand manner.

This study unit is the **first of two** on **technology and analytics**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The two study units are

Study Unit 19: Information Systems and Data Governance

Study Unit 20: Systems Development and Data Analytics

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

19.1 ACCOUNTING INFORMATION SYSTEMS

1. A management information system (MIS) typically receives input from a transaction processing system, aggregates it, then reports it in a format useful to middle management in running the business. An information system is an essential part of a company's value chain. For this reason, MISs are often classified by function or activity, such as the following:
 - a. Accounting: general ledger, accounts receivable, purchasing and accounts payable, payroll processing, fixed asset management, and tax accounting; other aspects of accounting information systems are described in item 3. below.
 - 1) These single-function systems have been expanded to cover complete cycles within the company's operations, such as the revenue-to-cash cycle (which covers sales, receivables, and cash), the production cycle, and the purchases and payables cycle.
 - b. Finance: capital budgeting, operational budgeting, and cash management.
 - c. Manufacturing: production planning, cost control, and quality control.
 - d. Logistics: inventory management and transportation planning.
 - e. Marketing: sales analysis and forecasting.
 - f. Human resources: projecting payroll, projecting benefits obligations, employment-level planning, and employee evaluation tracking.
2. These single-function systems, often called stovepipe systems because of their limited focus, are gradually being replaced by integrated systems that link multiple business activities across the enterprise.
 - a. The most comprehensive integrated system is termed an enterprise resource planning (ERP) system.
3. An accounting information system (AIS) is a subsystem of an MIS that processes routine, highly structured financial and transaction data relevant to financial as well as managerial accounting. An AIS is composed of the **general ledger/financial reporting system (GL/FRS)**, **management reporting system (MRS)**, and **transaction processing system (TPS)**.
 - a. The **GL/FRS** is concerned with transactions with external parties (e.g., customers, suppliers, governments, owners, and creditors) reflected in financial statements prepared in conformity with GAAP.
 - 1) The GL/FRS typically produces the income statement, balance sheet, statement of cash flow, and other reports required by management.

- b. The **MRS** provides information useful to decision making and internal management, including the internal activities recorded in the cost accounting system and the preparation of related reports and analysis.
 - 1) The MRS typically produces production reports, pro forma financial statements, budgets, cost-volume-profit analysis, and other internal reports.
- c. The **TPS** is a system that performs the routine transactions necessary to conduct business. A transaction is a single discrete event that can be stored in an information system.
 - 1) Each transaction must succeed or fail as a complete unit; it can never be only partially complete.
 - 2) Transaction processing is designed to maintain a system's integrity.

4. Accounting Cycles

- a. The accounting process can be described in terms of five cycles:
 - 1) Sales to customers on credit and recognition of receivables
 - 2) Collection of cash from customer receivables
 - 3) Purchases on credit and recognition of payables
 - 4) Payment (disbursement) of cash to satisfy trade payables (includes both expense and capitalized transactions)
 - 5) Payment of employees for work performed and allocation of costs
- b. Appendix F contains five flowcharts and accompanying tables depicting the steps in the cycles and the controls in each step for an organization large enough to have an optimal segregation of duties.
 - 1) In small- and medium-sized organizations, some duties must be combined. The internal auditor must assess whether organizational segregation of duties is adequate.



**AUTHOR'S
NOTE**

Except for manual checks and remittance advices, the flowcharts presented do not assume use of either a paper-based or an electronic system. Each document symbol represents a business activity or control, whether manual or computerized.

5. Enterprise Resource Planning (ERP)

- a. Study Unit 10, Subunit 2, has detailed coverage of traditional and current ERP systems.

6. Data, Databases, and Database Management System (DBMS)

- a. A **database** is an organized collection of data in a computer system.
- b. Data in the database are integrated to eliminate redundancy of data items. A single integrated system allows for improved data accessibility.
 - 1) When systems within the organization are not integrated, they not only may contain different data but also may define and update data in inconsistent ways. Thus, determining the location of data and ensuring their consistency are more difficult.
- c. A DBMS is an integrated set of computer programs that (1) create the database; (2) maintain the elements; (3) safeguard the data from loss or destruction; and (4) make the data available to applications, programs, and inquiries.
 - 1) The DBMS allows programmers and designers to work independently of the technical structure of the database.
 - a) Before the development of DBMSs, designing and coding programs that used databases was extremely time-consuming (and therefore expensive) because programmers had to know the exact contents and characteristics of the data in every database.
 - b) DBMSs provide a common language for referring to databases, easing the design and coding of application programs.
 - c) A DBMS includes security features. Thus, a specified user's access may be limited to certain data fields or logical views depending on the individual's assigned duties.
 - d) DB2 (IBM), Oracle (Oracle Corp.), SQL Server (Microsoft), and Access (Microsoft) are examples of DBMSs.
- d. The term "database" often includes the DBMS.
- e. Databases and the associated DBMSs permit efficient storage and retrieval of data for formal system applications.
 - 1) They also permit increased access to data (e.g., to answer inquiries for data not contained in formal system outputs) as well as updating of files by transaction processing.
 - 2) These increased capabilities, however, result in increased cost because they require
 - a) The use of sophisticated hardware (direct-access devices)
 - b) Sophisticated software (the DBMS)
 - c) Highly trained technical personnel (database administrator, staff)
 - d) Increased security controls

7. Two Early Database Structures

- a. Storing all related data on one storage device creates security problems.
 - 1) Should hardware or software malfunctions occur, or unauthorized access be achieved, the results could be disastrous.

- 2) Greater emphasis on security is required to provide backup and restrict access to the database.
 - a) For example, the system may employ **dual logging**, that is, use of two transaction logs written simultaneously on separate storage media.
 - b) It may also use a snapshot technique to capture data values before and after transaction processing.
 - c) The files that store these values can be used to reconstruct the database in the event of data loss or corruption.

8. Relational Database Structure

- a. A relational structure organizes data in a conceptual arrangement.
 - 1) An individual data item is called a field or column (e.g., name, date, amount).
 - a) Related fields are brought together in a record or row (e.g., for a single sales transaction).
 - b) Multiple records make up a file or table (e.g., sales).
 - c) Tables can be joined or linked based on common fields rather than on high-overhead pointers or linked lists as in other database structures.
 - d) Every record in a table has a field (or group of fields) designated as the key. The value (or combination of values) in the key uniquely identifies each record.
- b. Note that in a relational structure, each data element is stored as few times as necessary. This is accomplished through the process of normalization. Normalization prevents inconsistent deletion, insertion, and updating of data items.
 - 1) The relational structure requires careful planning, but it is easy to maintain and processes queries efficiently.
- c. The three basic operations in the relational model are selecting, joining, and projecting.
 - 1) Selecting creates a subset of records that meet certain criteria.
 - 2) Joining is the combining of relational tables based on a common field or combination of fields.
 - 3) Projecting results in the requested subset of columns from the table. This operation creates a new table containing only the required information.
- d. Two features that make the relational data structure stand out are cardinality and referential integrity.
 - 1) **Cardinality** refers to how close a given data element is to being unique.
 - a) A data element that can only exist once in a given table has high cardinality.
 - b) A data element that is not unique in a given table but that has a restricted range of possible values is said to have normal cardinality.
 - c) A data element that has a very small range of values is said to have low cardinality. A field that can contain only male/female or true/false is an example.
 - 2) **Referential integrity** means that for a record to be entered in a given table, there must already be a record in some other table(s).

- e. The tremendous advantage of a relational data structure is that searching for records is greatly facilitated.
 - 1) For example, a user can specify a customer and see all the parts that customer has ordered, or the user can specify a part and see all the customers who have ordered it. Such queries were extremely resource-intensive, if not impossible, under older data structures.
- f. A distributed database is stored in two or more physical sites using either replication or partitioning.
 - 1) The replication or snapshot technique makes duplicates to be stored at multiple locations.
 - a) Changes are periodically copied and sent to each location. If a database is small, storing multiple copies may be cheaper than retrieving records from a central site.
 - 2) Fragmentation or partitioning stores specific records where they are most needed.
 - a) For example, a financial institution may store a particular customer's data at the branch where (s)he usually transacts his or her business. If the customer executes a transaction at another branch, the pertinent data are retrieved via communication lines.

9. Additional Terminology

- a. The **database administrator (DBA)** is the individual who has overall responsibility for developing and maintaining the database and for establishing controls to protect its integrity.
 - 1) Thus, only the DBA should be able to update data dictionaries. In small systems, the DBA may perform some functions of a DBMS. In larger applications, the DBA uses a DBMS as a primary tool.
 - 2) The responsibility for creating, maintaining, securing, restricting access to, and redefining and restructuring the database belongs to the DBA.
- b. The **data dictionary** is a file that describes both the physical and logical characteristics of every data element in a database.
 - 1) The data dictionary includes, for example, the name of the data element (e.g., employee name, part number), the amount of disk space required to store the data element (in bytes), and what kind of data is allowed in the data element (e.g., alphabetic, numeric).
 - 2) Thus, the data dictionary contains the size, format, usage, meaning, and ownership of every data element. This greatly simplifies the programming process.
- c. Data from a relational database can be displayed in graphs and reports, changed, and otherwise controlled using a program called **Query Management Facility (QMF)**.
- d. The **schema** is a description of the overall logical structure of the database using data-definition language, which is the connection between the logical and physical structures of the database.
 - 1) A **subschema** describes a particular user's (application's) view of a part of the database using data definition language.

- e. The **database mapping facility** is software that is used to evaluate and document the structure of the database.
- f. An **object-oriented database** is a response to the need to store not only numbers and characters but also graphics and multimedia applications.
 - 1) Translating these data into tables and rows is difficult. However, in an object-oriented database, they can be stored, along with the procedures acting on them, within an object.
- g. Advanced database systems provide for **online analytical processing (OLAP)**, also called multidimensional data analysis, which is the ability to analyze large amounts of data from numerous perspectives.
 - 1) OLAP is an integral part of the data warehouse concept.
 - 2) Using OLAP, users can compare data in many dimensions, such as sales by product, sales by geography, and sales by salesperson.
 - 3) The following technologies are replacing OLAP:
 - a) **In-memory analytics** is a process that queries data stored directly in the computer's memory (i.e., RAM) as opposed to querying data located on the computer's internal or external storage devices (e.g., hard drives).
 - b) **Search engine technology** stores data at a document/transaction level, and data is not pre-aggregated like it would be when contained in an OLAP or in-memory technology application. Users are able to have full access to their raw data and create the aggregations themselves.
- h. A **data warehouse** contains not only current operating data but also historical information from throughout the organization. Thus, data from all operational systems are integrated, consolidated, and standardized in an organization-wide database into which data are copied periodically. These data are maintained on one platform and can be read but not changed.
 - 1) **Data cleansing** cleans up data in a database that is incorrect, incomplete, or duplicated before loading it into the database. It improves the quality of data. The need for data cleansing increases when multiple data sources are integrated.
 - 2) Data mining is facilitated by a data warehouse. **Data mining**, which is covered thoroughly in Study Unit 20, Subunit 2, is the process of analyzing data from different perspectives and summarizing it into useful information. Data mining software ordinarily is used.
 - a) For example, data mining software can help find abnormal patterns and unforeseen correlations among the data.
 - b) Internal auditors can use data mining techniques to detect fraud.
 - 3) A **data mart** is a subset of the data warehouse that can be tailored to user data requirements.
 - a) Because data marts are developed for specific groups of users, they are often created for individual business lines or departments.

10. **Enterprise performance management (EPM)** (also known as corporate performance management or business performance management) is associated with
- Business intelligence and ERP.
 - Budgeting and forecasting
 - Financial reporting
 - Analyzing performance results and identifying ways to improve

19.2 DATA GOVERNANCE AND RISK

1. Overview

- Organizations need to be aware of the unique risks associated with a computer-based business information system.
- IT security (or **cybersecurity**) is information security applied to computer hardware, software, and computer networks.
- Safe computing can be achieved by using carefully crafted policies and procedures in conjunction with antivirus and access control software.
- The most comprehensive indicator of an information system's compliance with prescribed procedures is the control the system has over the data. This includes the capacity and complexity of the system, as well as the accessibility of the data to the end-user.

2. System Availability

- The ability to make use of any computer-based system is dependent on
 - An uninterrupted flow of electricity
 - Protection of computer hardware from environmental hazards (e.g., fire and water)
 - Protection of software and data files from unauthorized alteration
 - Preservation of functioning communications channels between devices

3. Volatile Transaction Trails

- In any computer-based environment, a complete trail useful for audit purposes might exist for only a short time or in only computer-readable form. In online and real-time systems, data are entered directly into the computer, eliminating portions of the audit trail traditionally provided by source documents.

4. **Decreased Human Involvement**

- a. Because employees who enter transactions may never see the final results, the potential for detecting errors is reduced. Also, output from a computer system often carries a mystique of infallibility, reducing the incentive of system users to closely examine reports and transaction logs.

5. **Uniform Processing of Transactions**

- a. Computer processing uniformly subjects like transactions to the same processing instructions, thereby virtually eliminating clerical error. Thus, it permits consistent application of predefined business rules and the performance of complex calculations in high volume.
- b. However, programming errors (or other similar systematic errors in either the hardware or software) will result in all like transactions being processed incorrectly.

6. **Unauthorized Access**

- a. When accounting records were kept in pen-and-ink format, physical access to them was the only way to carry out an alteration. Once they are computer-based, however, access may be gained by parties both internal and external to the organization.
- b. Security measures, such as firewalls and user ID-and-password combinations, are vital to maintaining security over data in an automated environment.

7. **Data Vulnerability**

- a. Destruction of hardware devices or units of storage media could have disastrous consequences if they contain the only copies of crucial data files or application programs.
- b. For this reason, it is vital that an organization's computer files be duplicated and stored offsite periodically.

8. **Reduced Segregation of Duties**

- a. Many functions once performed by separate individuals may be combined in an automated environment.
 - 1) Receiving cash, issuing a receipt to the payor, preparing the deposit slip, and preparing the journal entry may once have been performed by separate individuals.
 - 2) In a computer-based system, the receipt, deposit slip, and journal entry may be automatically generated by the computer. If the same employee who receives the cash is also responsible for entering the relevant data into the system, the potential for fraud or error is increased.

9. Reduced Individual Authorization of Transactions

- a. Certain transactions may be initiated automatically by a computer-based system. This is becoming more widespread as an increasing number of business processes become automated.
 - 1) For example, an enterprise resource planning system at a manufacturing concern may automatically generate a purchase order when raw materials inventory reaches a certain level. If the company shares an EDI system with the vendor, the purchase order may be sent to the vendor electronically without any human intervention.
- b. This reduced level of oversight for individual transactions requires careful coding to ensure that computer programs accurately reflect management's goals for business processes.
 - 1) Independent verification of transactions is an important compensating control in the absence of segregation of duties and reduced individual authorization. A third party performs the verification to ensure that the transactions were appropriately processed.

10. Malicious Software (Malware)

- a. Malware is a term describing any program code that enters a computer system that has the potential to degrade that system. Common forms of malware are discussed in Study Unit 18, Subunit 2.

11. Overview of COBIT

- a. COBIT is the best-known control and governance framework that addresses information technology.
 - 1) In its original version, COBIT was focused on controls for specific IT processes.
 - 2) Over the years, information technology has gradually pervaded every facet of the organization's operations. IT can no longer be viewed as a function distinct from other aspects of the organization.
 - a) The evolution of COBIT has reflected this change in the nature of IT within the organization.

BACKGROUND 19-1 COBIT

When originally published in 1996, COBIT was an acronym for Control Objectives for Information and Related Technology. COBIT 2019, the most recent version, was published in 2018 by ISACA (formerly known as the Information Systems Audit and Control Association) and is available at www.isaca.org/resources/cobit.

12. Information Criteria

- a. **Effectiveness** deals with information's relevance to the business process and receipt in a timely, correct, consistent, and usable manner.
- b. **Efficiency** concerns the provision of information through the optimal (most productive and economical) use of resources.
- c. **Confidentiality** concerns the protection of sensitive information from unauthorized disclosure. This includes safeguarding of resources.
- d. **Integrity** relates to the accuracy of information and the completeness of data, as well as to the validity of the conclusions reached using that data, as assumed by the users and their organizational values and expectations.
- e. **Availability** relates to information being available when required by the organization both now and in the future.
- f. **Compliance** deals with complying with the laws, regulations, and contractual agreements to which the organization may be subject. This includes both external laws and regulations and internal policies and procedures.
- g. **Reliability** relates to the usefulness of data. This includes whether the information is suitable for management's use in the operations of the entity and allows for management to exercise its governance responsibilities.

13. Data Governance Focus Areas

- a. **Strategic alignment** is dedicated to ensuring the linkage of business plans with IT plans while defining, maintaining, and validating the IT value proposition. Furthermore, IT operations must be aligned with enterprise operations.
- b. **Value delivery** is concerned with implementing the value proposition throughout the delivery cycle. It ensures that IT delivers the promised benefits, while concentrating on optimizing costs and proving the intrinsic value of IT.
- c. **Resource management** deals with the optimal investment in, and the proper management of, critical IT resources.
- d. **IT risk** is the risk associated with the use, ownership, operation, involvement, influence, and adaption of IT within an organization.
- e. **Risk management** includes risk awareness by senior management, knowledge of compliance requirements, transparency about risks to the organization, and embedding risk management responsibilities into the entity's structure.
 - 1) Establishing and maintaining controls are management functions. These are not under the purview of stockholders, the internal auditors, or the IT department (although internal auditors may evaluate the systems of internal controls).
- f. **Performance measurement** tracks and monitors the entire process, including strategy implementation, the completion of projects, resource usage and cost measures, process performance, and service delivery.

14. COBIT 2019

- a. COBIT 2019 expands on COBIT 5's key principles for a **governance system** applicable to IT governance to include six governance system principles and three **governance framework** principles. A governance system is the rules, practices, and processes that direct and regulate an entity. A governance framework is the structure upon which the governance system is built.
 - 1) The six principles for a **governance system** are summarized as follows:
 - a) Provide **stakeholder value**. Achieving value requires a strategy and a governance system.
 - b) Take a **holistic** approach. Create synergies among the components interconnected in the system.
 - i) Governance system **components** were called “enablers” under COBIT 5.
 - ii) Components can be **generic** (components applied in principle to any circumstances) or **variant** (components designed for a given purpose or context in a focus area).
 - c) Utilize a **dynamic** governance system. The governance system must be dynamic when dealing with a change in design factors (e.g., personnel, infrastructure, applications, etc.) and must be accompanied by consideration of its systemic effects.
 - d) Keep governance **distinct** from management. Governance tasks should be differentiated from management tasks.
 - e) Tailor governance to **enterprise needs**. The governance system must be designed to meet an organization's requirements.
 - i) **Design factors** affect the blueprint of a governance system.
 - ii) Design factors include, but are not limited to, threat landscape, technology adoption strategy, and enterprise strategy and goals.
 - f) Achieve **end-to-end** enterprise coverage. The emphasis is not solely on the IT function but on all information, processes, and technology that contribute to organizational goal achievement.

15. Data Life Cycle

- a. A life cycle consists of phases, and each phase has its own characteristics. The following are seven phases of the data life cycle.
 - 1) **Data Capture**
 - a) The first phase is to obtain the data. It is the act of creating information that does not yet exist within the organization.
 - b) The primary ways in which data can be captured include
 - i) **Utilizing data**. The organization obtains already existing data outside the organization, for example, it may buy or otherwise acquire data from an outside entity.
 - ii) **Data entry**. New data values can be created within the organization.
 - iii) **Signal reception**. The organization acquires data that has already been created by control systems within the organization.

2) Data Maintenance

- a) The second stage of the life cycle, after the data have been captured, can be defined as supplying data to the points at which data synthesis and data usage occur.
- b) Data maintenance involves processing the data without yet deriving any value from it for the enterprise. It often involves cleansing and enrichment of the data.

3) Data Synthesis and Analytics

- a) Data synthesis and analytics can be defined as the creation of data values by using inductive logic. It typically involves using other data as inputs.
- b) Analytics uses modeling, such as risk modeling, actuarial modeling, and modeling for investment decisions.

4) Data Usage

- a) The next phase is to use the data in support of the enterprise. This stage can be defined as the application of data to tasks the enterprise needs to run and manage itself.
- b) This typically includes activities outside the regular data life cycle.
- c) Data usage may have data governance challenges. For example, is it legal to use the data in the ways management wants to use it?
 - i) There may be regulatory or contractual constraints on how data may be used; the role of data governance is to ensure that legalities are observed.

5) Data Publication

- a) Data publication refers to the sending of data to a location outside the organization, for example, in monthly statements to customers.

6) Data Archival

- a) A single data value may be used for quite some time, but eventually the end of its lifetime usefulness will be encountered. At that point, the organization should archive the data value.
- b) Data archival involves copying data to a location where it is stored in case it is ever needed again. At the same time, the data is removed from all active environments.
- c) In summary, a data archive is a location where data is stored, but there is no regular maintenance and probably little usage.

7) Data Purging

- a) At the end of the data life cycle, every copy of a data item is removed from the enterprise. This is typically done from the archive.
- b) Disposing of all copies of data is often difficult, which results in a data governance challenge in this phase because it may be difficult to prove that a full purge has actually occurred.

16. Record Retention Policy

- a. It is important for every organization to have a formal record retention policy (sometimes referred to as a records management policy), which provides for the retention and periodic destruction of documents and other records.
- b. Keeping and maintaining too many records, or storing them longer than needed, can create unneeded costs for an organization.
 - 1) For example, if a company becomes involved in a legal dispute, the task of going through a voluminous amount of records to produce relevant material can become an enormous and costly challenge. However, that same legal dispute may be based on old records, so there is a fine balance between keeping records too long and not keeping them long enough.
- c. For some types of documents, there are minimum retention periods imposed by law, such as those related to taxes and employees, and certain industries may be governed further by particular regulations.
 - 1) Even the minimum legal retention periods sometimes have uncertain descriptions.
 - a) For instance, tax records must be kept at least 7 years, unless there is suspected fraud, in which case, there is no minimum. Thus, it is sometimes better to keep tax records longer than the minimum period in case the taxpayer needs to prove lack of fraud at some unforeseen point in the future.
- d. If the records needed for a particular legal case have been destroyed pursuant to a well-established records retention policy that is routinely complied with, often the courts may assume the organization complied with its duty.
 - 1) However, as illustrated by the Arthur Andersen & Co. shredding incident at Enron, once litigation is likely, it is too late to establish a records retention policy or to begin complying with a policy that was not followed in the past.

19.3 COSO FRAMEWORK -- INTERNAL CONTROL FOR DATA GOVERNANCE



SUCCESS TIP

COSO requirements are covered in Study Unit 17, Subunit 2, with more detailed coverage of internal controls. However, the COSO material on data governance is highlighted in the section of the CMA exam testing data analytics. Thus, candidates should develop a thorough understanding of the COSO framework. Review the COSO coverage in Study Unit 17, Subunit 2, before continuing this study unit.

1. Overview

- a. Effective corporate governance relies heavily on effective systems of (1) internal control and (2) enterprise risk management. The Committee of Sponsoring Organizations of the Treadway Commission (COSO) has established a widely accepted framework for each system.
- b. Although the COSO *Internal Control – Integrated Framework* is widely accepted as the standard for the design and operation of internal control systems, regulatory or legal requirements may specify another control framework or design.
- c. The COSO framework consists primarily of a definition of internal control, categories of objectives, components and related principles, and requirements of an effective system of internal control.

2. Relationship of Objectives, Components, and Organizational Structure

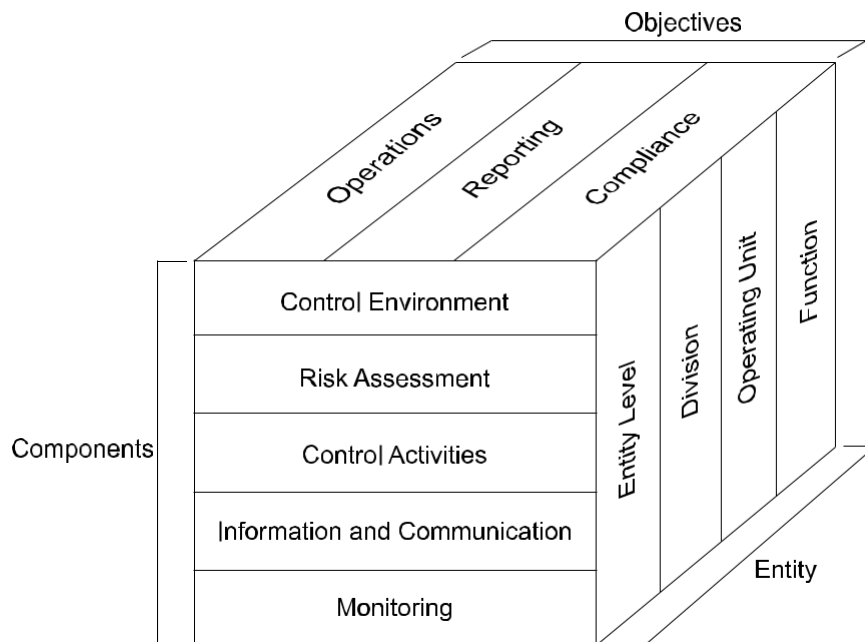


Figure 19-1

- a. The COSO model may be represented by a cube with rows, slices, and columns. (1) The rows are the five components, (2) the slices are the three objectives, and (3) the columns are an entity's organizational structure.

3. Requirements for Effective Internal Control

- a. A system of internal control is effective if it provides **reasonable assurance** of achieving an entity's objectives relating to operations, reporting, and compliance. Such a system reduces the risk(s) of not achieving those objectives to an **acceptable level**.
- b. An effective system of internal control requires that each of the five components of internal control, and relevant principles, is present and functioning.
 - 1) **Present** refers to whether the components and relevant principles exist in the design and implementation of the system of internal control.
 - 2) **Functioning** refers to whether the components and relevant principles continue to exist in the operation of the system of internal control.
 - 3) The five components should operate together in an integrated manner.
 - a) **Operating together** refers to whether all five components collectively reduce the risk of not achieving an objective to an acceptable level.

4. Other Considerations

- a. The use of **judgment** is required in designing, implementing, and conducting internal control and assessing its effectiveness.
- b. The use of **outsourced service providers** for certain business processes does not relieve the organization of its responsibility for the system of internal control.
- c. Although **technology innovation** creates opportunities and risks, the principles in the COSO framework do not change.
- d. The **organization's size** may affect how it implements internal control.
 - 1) Senior management of smaller organizations typically have a wider span of control and greater direct interaction with personnel than senior management of larger organizations. Larger organizations may need to rely on more formal mechanisms of control (e.g., written reports, formal meetings, or conference calls).
 - 2) Larger organizations have more resources than smaller organizations. Consequently, a smaller organization may have to outsource all, or parts, of its internal audit function or incur higher costs relative to larger organizations because of the lack of economies of scale.

5. Roles and Responsibilities Regarding Internal Control

a. Internal Parties

- 1) Board of Directors
 - a) Responsible for **overseeing** the system of internal control.
 - b) Defines expectations about integrity, ethical values, transparency, and accountability through its authority to select and terminate the CEO.
 - c) Often performs certain duties through committees, which include the
 - i) Audit Committee
 - ii) Compensation Committee
 - iii) Nomination/Governance Committee
 - iv) Risk Committee
 - v) Finance Committee

- 2) Senior Management
 - a) Sets the **tone at the top** and has primary responsibility for establishing proper ethical culture.
 - b) Sets objectives.
 - c) Has **overall responsibility** for designing, implementing, and operating an effective system of internal control.
 - i) Maintains oversight and control over the entity's risks.
 - ii) Guides the development and performance of control activities at the entity level.
 - iii) Assigns responsibility for establishing more specific internal controls at the different levels of the entity.
 - iv) Communicates expectations.
 - v) Evaluates control deficiencies.
- 3) Operational Management
 - a) Provides the **first line of defense** for effective management of risk and control.
 - b) Develops and implements control and risk management processes.
- 4) Business-Enabling Functions
 - a) Provide the **second line of defense** for effective management of risk and control.
 - b) These functions support the entity through specialized skills and include various risk management and compliance functions.
 - c) Additionally, these functions are typically responsible for the ongoing monitoring of control and risk.
- 5) Internal Auditors
 - a) Provide the **third line of defense** for effective management of risk and control.
 - b) Evaluate the adequacy and effectiveness of controls in responding to risks in the entity's oversight, operations, and information systems.
 - c) To remain independent, the internal audit activity cannot be responsible for selecting and executing controls.
- 6) Other Entity Personnel
 - a) Everyone in the organization is expected to (1) competently perform his or her appropriate control activities and (2) inform those higher in the organization about ineffective control.

b. External Parties

- 1) External Auditors
 - a) Independent accountants have been required to consider the auditee's system of internal control as part of their audit of the financial statements.
 - b) The PCAOB also requires auditors of public companies to examine and report on internal control.
- 2) Legislators and Regulators
 - a) The Foreign Corrupt Practices Act and the Sarbanes-Oxley Act set legal requirements regarding internal control.
- 3) Parties Interacting with the Entity
 - a) The following are examples of control-related information received from outside sources:
 - i) A major customer informs management that a sales representative attempted to arrange a kickback scheme in return for the customer's business.
 - ii) A supplier reveals inventory and shipping problems by complaining about incomplete orders.
 - b) Parties without a supplier or customer relationship with the entity also may provide insight into the functioning of controls.
 - i) As a condition for granting a loan, a bank may require that certain ratios be kept above or below specified levels. As a result, management may need to pay closer attention to controls over cash and inventory levels than it did previously.
- 4) Financial Analysts, Bond Rating Agencies, and the News Media
 - a) The actions of these parties can inform an entity about how it is perceived by the world at large.
- 5) Outsourced Service Providers
 - a) Some of an entity's business functions, e.g., IT, human resources, or internal audit, may be performed by external service providers.
 - i) Accordingly, the entity must understand the activities and controls related to the outsourced services and how the provider's internal control system affects the entity's system of internal control.

6. Limitations

- a. Internal control only provides **reasonable assurance** of achieving objectives. It cannot provide absolute assurance because any system of internal control has the following **inherent limitations**:
 - 1) Established objectives must be **suitable** for internal control.
 - a) For example, if an entity establishes unrealistic objectives, the system of internal controls will be ineffective.
 - 2) Human **judgment** is faulty, and controls may fail because of simple errors or mistakes.
 - 3) Controls may fail due to **breakdowns** (e.g., employee misunderstanding, carelessness, or fatigue).
 - 4) **Management** may inappropriately **override** internal controls, e.g., to fraudulently achieve revenue projections or hide liabilities.
 - 5) Manual or automated controls can be circumvented by **collusion**.
 - 6) **External events** are beyond an organization's control.

STUDY UNIT TWENTY

SYSTEMS DEVELOPMENT AND DATA ANALYTICS

20.1	Technology-Enabled Finance Transformation	1
20.2	Analytics and Big Data	14

This study unit is the **second of two** on **technology and analytics**. The relative weight assigned to this major topic in Part 1 of the exam is **15%**. The two study units are

Study Unit 19: Information Systems and Data Governance

Study Unit 20: Systems Development and Data Analytics

If you are interested in reviewing more introductory or background material, go to www.Glem.com/CMAIntroVideos for a list of suggested third-party overviews of this topic. The following Glem outline material is more than sufficient to help you pass the CMA exam. Any additional introductory or background material is for your personal enrichment.

20.1 TECHNOLOGY-ENABLED FINANCE TRANSFORMATION

1. Systems Development Life Cycle (SDLC)

- a. The SDLC approach is the traditional methodology applied to the development of large, highly structured application systems. A major advantage of the life-cycle approach is enhanced management and control of the development process. SDLC consists of the following five steps:
 - 1) **Systems strategy** requires understanding the organization's needs.
 - 2) **Project initiation** is the process by which systems proposals are assessed.
 - 3) **In-house development** is generally chosen for unique information needs.
 - 4) **Commercial packages** are generally chosen for common needs rather than developing a new system from scratch.
 - 5) **Maintenance and support** involves ensuring the system accommodates changing user needs.
- b. Portions of the phases can overlap.
- c. Note that the feedback gathered during the maintenance of a system provides information for developing the next generation of systems, hence the name **life cycle**.

2. The **phases and component steps of a traditional SDLC** can be described as follows:

a. **Initiation, Feasibility, and Planning**

- 1) The SDLC begins by recognizing there is a need for a new system, gaining an understanding of the situation to determine whether it is feasible to create a solution, and formulating a plan.

b. **Requirements Analysis and Definition**

- 1) A formal proposal for a new system is submitted to the IT steering committee, describing the need for the application and the business function(s) that it will affect.
- 2) Feasibility studies are conducted to determine
 - a) What technology the new system will require
 - b) What economic resources must be committed to the new system
 - c) How the new system will affect current operations
- 3) The steering committee gives its go-ahead for the project.

c. **System Conceptual Design**

- 1) Logical design consists of mapping the flow and storage of the data elements that will be used by the new system and the new program modules that will constitute the new system.
 - a) Data flow diagrams, system interface diagrams, and structured flowcharts are commonly used in this step.
 - b) Some data elements may already be stored in existing databases. Good logical design ensures that they are not duplicated.
- 2) Physical design involves planning the specific interactions of the new program code and data elements with the hardware platform (existing or planned for purchase) on which the new system will operate.
- 3) Systems analysts are heavily involved in these two steps.

d. **Building and Development**

- 1) The actual program code and database structures that will be used in the new system are written.
- 2) Hardware is acquired, and physical infrastructure is assembled.

e. **Testing and Quality Control**

- 1) Testing is performed during system development with the intent of identifying errors or other defects. The job of testing is an iterative process because when one error is corrected, it can illuminate other errors or even create new ones. Testing determines whether the system
 - a) Meets the requirements that guided its design and development
 - b) Responds correctly to all kinds of inputs
 - c) Performs its functions within an acceptable time
 - d) Achieves the general result its stakeholders desire

- 2) Although the number of possible tests to apply is almost limitless, developers cannot test everything. All testing uses strategy to select tests that are feasible for the available time and resources.
 - a) Combinatorial test design identifies the minimum number of tests needed to get the coverage developers want.
- 3) The following are various methods available to test systems:
 - a) **Static testing** examines the program's code and its associated documentation through reviews, walkthroughs, or inspections but does not require the program to be executed.
 - b) **Dynamic testing** involves executing programmed code with a given set of test cases.
 - c) **White-box testing** tests the internal structures or workings of a program, as opposed to the functionality exposed to the end-user.
 - d) **Black-box testing** focuses on how well the software works. It treats the software as if it were a "black box" and examines functionality without any knowledge of the source code supporting the program.
 - e) **Gray-box testing** goes further than a black-box test. A gray-box evaluation involves having knowledge of the program's internal data structures and algorithms for the purpose of designing tests while simultaneously performing those tests at the user, or black-box, level.
- 4) There are four levels of tests:
 - a) The first level is known as **unit testing** and refers to tests that verify
 - i) The functionality or workability of a specific section of code and
 - ii) The means of handling data that are passed between multiple units or subsystems components.
 - b) **Integration testing** is any type of software testing that seeks to verify the interfaces between components against a software design. Integration testing works to expose defects in the interfaces and interactions between integrated components (modules).
 - c) **System testing**, or end-to-end testing, tests a completely integrated system to verify that the system meets its requirements.
 - d) **Acceptance testing** is conducted to determine whether the system meets the organization's needs and is ready for release.

f. Acceptance, Installation, and Implementation

- 1) User acceptance testing is the final step before placing the system in live operation.
 - a) IT must demonstrate to the users who submitted the original request that the system performs the desired functions.
 - b) Once the users are satisfied with the new system, they acknowledge formal acceptance and implementation begins.
- 2) Four strategies for converting to the new system can be used.
 - a) With **parallel** operation, the old and new systems both run at full capacity for a given period.
 - i) This strategy is the safest because the old system is still producing output (in case there are major problems with the new system), but it is also the most expensive and time-consuming.
 - b) With **direct changeover** (direct cutover) conversion, the old system is shut down and the new one takes over processing at once.
 - i) This is the least expensive and time-consuming strategy, but it is also the riskiest because the new system cannot be reverted to the original.
 - c) Under **pilot** conversion, one branch, department, or division at a time is fully converted to the new system.
 - i) Experience gained from each installation is used to benefit the next one. One disadvantage of this strategy is extension of the conversion time.
 - d) In some cases, **phased** conversion is possible. Under this strategy, one function of the new system at a time is placed in operation.
 - i) For instance, if the new system is an integrated accounting application, accounts receivable could be installed, then accounts payable, cash management, materials handling, etc.
 - ii) The advantage of this strategy is that the users are able to learn one part of the system at a time.
- 3) Training and documentation are critical.
 - a) The users must feel comfortable with the new system and have plenty of guidance available, either in hard copy or online.
 - b) Documentation consists of more than just operations manuals for the users. Layouts of the program code and database structures must also be available for the programmers who must modify and maintain the system.

g. Operations and Maintenance

- 1) After a system becomes operational, it should be monitored to ensure ongoing performance and continuous improvement.
- 2) Systems follow-up or post-audit evaluation is a subsequent review of the efficiency and effectiveness of the system after it has operated for a substantial time (e.g., 1 year).

3. Program Change Control

- a. Over the life of an application, users are constantly asking for changes. The process of managing these changes is referred to as systems maintenance, and the relevant controls are called **program change controls**.
- b. Once a change to a system has been approved, the programmer should save a copy of the production program in a test area of the computer, sometimes called a "sandbox."
 - 1) Only in emergencies, and then only under close supervision, should a change be made directly to the production version of a computer program.
 - 2) The IT function must be able to revert back to a prior version immediately if unexpected results are encountered during an emergency change.
- c. The programmer makes the necessary changes to this copy of the program's source code.
- d. The programmer transforms the changed program into a form the computer can execute. The resulting machine-ready program is referred to as object code, or more precisely, executable code.
- e. Once the programmer has the executable version of the changed program, (s)he tests it to see if it performs the new task as expected.
 - 1) This testing process absolutely must not be run against production data. A special set of test data must be available for running test programs.
- f. The programmer demonstrates the new functionality for the user who made the request.
 - 1) Either the user accepts the new program or the programmer goes back and makes further changes.
- g. Once the program is in a form acceptable to the user, the programmer moves it to a holding area.
 - 1) Programmers (except in emergencies) should never be able to put programs directly into production.
- h. The programmer's supervisor reviews the new program, approves it, and authorizes its move into production, which is generally carried out by operations personnel.
 - 1) The compensating control is that operators generally lack the programming knowledge to put fraudulent code into production.

4. End-User vs. Centralized Computing

- a. End-user computing (EUC) involves user-created or user-acquired systems that are maintained and operated outside of traditional information systems controls.
 - 1) Certain environmental control risks are more likely in EUC. They include copyright violations that occur when unauthorized copies of software are made or when software is installed on multiple computers.
 - 2) Unauthorized access to application programs and related data is another concern. EUC lacks physical access controls, application-level controls, and other controls found in mainframe or networked environments.
 - 3) Moreover, EUC may not have adequate backup, recovery, and contingency planning. The result may be an inability to recreate the system or its data.

- b. Program development, documentation, and maintenance also may lack the centralized control found in larger systems.
 - 1) The risk of allowing end-users to develop their own applications is decentralization of control. These applications may not be reviewed by independent outside systems analysts and are not created using a formal development methodology. They also may not be subject to appropriate standards, controls, and quality assurance procedures.
 - 2) When end-users create their own applications and files, private information systems in which data are largely uncontrolled may proliferate. Systems may contain the same information, but EUC applications may update and define the data in different ways. Thus, determining the location of data and ensuring data consistency become more difficult.
 - 3) EUC applications should contain controls that allow users to rely on the information produced. Identification of applications is more difficult than in a traditional centralized computing environment because few people know about and use them. There are three steps that should be considered:
 - a) The first step is to discover the existence of controls and their intended functions.
 - i) One approach is to take an organization-wide inventory of major EUC applications.
 - ii) An alternative is for the auditors and the primary user (a function or department) to review major EUC applications.
 - b) The second step is risk assessment.
 - c) The third step is to review the controls included in the applications chosen in the risk assessment.
- c. In a personal computer setting, the user is often the programmer and operator. Thus, the protections provided by segregation of duties are eliminated.
- d. The audit trail is diminished because of the lack of history files, incomplete printed output, etc.
- e. In general, available security features for stand-alone machines are limited compared with those in a network.
- f. Responsibility for the control of EUC exists at the organizational, departmental, and individual user levels. The end-user is directly responsible for security of equipment. Acquisition of hardware and software, taking equipment inventories, and strategic planning of EUC are organizational- and departmental-level responsibilities.

5. Business Process Analysis and Design

- a. The organizational needs assessment is a detailed process of study and evaluation of how information systems can be deployed to help the organization meet its goals. The steps in the assessment are as follows:
 - 1) Determine whether current systems support organizational goals.
 - 2) Determine the needs unmet by current systems.
 - 3) Determine the capacity of current systems to accommodate projected growth.
 - 4) Propose a path for information systems deployment to achieve organizational goals within budgetary constraints.
- b. A business process is a flow of actions performed on goods and/or information to accomplish a discrete objective.
 - 1) Examples include hiring a new employee, recruiting a new customer, and filling a customer order.
- c. Some business processes are contained entirely within a single functional area; e.g., hiring a new employee is performed by the human resources function.
 - 1) Other processes cross functional boundaries. Filling a customer order requires the participation of the sales department, the warehouse, and accounts receivable.
- d. In the early days of automated system deployment, hardware and software were very expensive. Systems tended to be designed to serve a single process or even a single functional area. Tremendous gains in processing power and storage capacity have made integrated systems, i.e., those that combine multiple processes, the norm.
 - 1) The most advanced of these are enterprise resource planning (ERP) systems.
- e. The automation of a process, or the acquisition of an integrated system, presents the organization with an opportunity for business process reengineering.
 - 1) **Business process reengineering** involves a complete rethinking of how business functions are performed to provide value to customers, that is, radical innovation instead of mere improvement and a disregard for current jobs, hierarchies, and reporting relationships.

6. Business Process Design and Participants' Roles

- a. The everyday functioning of a business process affects multiple stakeholder groups.
 - 1) Input from each group should be considered in the design of the process. However, some stakeholders will be more active participants.
- b. End-users are generally the drivers of a new or redesigned process.
 - 1) For example, the customers of a multi-division business may have open accounts with several of the divisions. Whenever a customer calls, the customer relations department would like the most up-to-date customer balance information for all divisions to be accessible at once.
 - 2) Although the motivation for the new process begins with the customer service department, personnel in various divisions as well as the central IT function will be affected.
- c. Because IT pervades every aspect of operations in a modern organization, the **IT steering committee** must study each request for a new process and either approve or deny it.
 - 1) Typical members of the steering committee include the chief information officer and the head of systems development from the IT function. Executive management from each division is also represented.
 - 2) The committee members have an understanding of the interactions of the organization's current systems and how they will affect and be affected by new or redesigned business processes.
- d. Once a new process or system has been approved, a project team is assembled, consisting of representatives of the end-users who requested it and the IT personnel who will design and build the software components that will support it.
- e. Upper management supports process design by making sufficient resources available to ensure successful implementation of the new process.
- f. If the new process or system crosses organizational boundaries, as is the case with electronic data interchange (EDI) systems, external parties, such as representatives of the vendor or customer businesses, are participants.

7. Robotic Process Automation (RPA)

- a. RPA is a form of machine learning technology that enables a computer to acquire knowledge and mimic the actions of the person(s) using it to perform a task.
 - 1) RPA systems capture data through a graphical interface and process the data used for repetitive tasks.
- b. Benefits of RPA include, but are not limited to, the following:
 - 1) Robots can perform continuously without needing to take time off.
 - 2) It eliminates the element of human error and improves efficiency (subsequent to successful programming and implementation).
 - 3) Although initial implementation requires capital investment, long-term costs associated with RPA are considerably lower than labor costs.

8. Artificial Intelligence (AI)

- a. AI is computer software designed to perceive, reason, and understand.
 - 1) Historically, computer software works through a series of if/then conditions in which every operation has exactly two possible outcomes (yes/no, on/off, true/false, one/zero).
 - a) Human reasoning, on the other hand, is extremely complex, based on deduction, induction, intuition, emotion, and biochemistry, resulting in a range of possible outcomes.
 - 2) AI attempts to imitate human decision making, which hinges on this combination of knowledge and intuition (i.e., remembering relationships between variables based on experience).
 - 3) The advantages of AI in a business environment are that IT systems
 - a) Can work 24 hours a day;
 - b) Will not become ill, die, or be hired away; and
 - c) Are extremely fast processors of data, especially if numerous rules (procedures) must be evaluated.
- b. There are several types of AI:
 - 1) **Neural networks** are a collection of processing elements working together to process information much like the human brain, including learning from previous situations and generalizing concepts.
 - 2) **Case-based reasoning systems** use a process similar to that used by humans to learn from previous, similar experiences.
 - 3) **Rule-based expert systems** function on the basis of set rules to arrive at an answer. These cannot be changed by the system itself. They must be changed by an outside source (i.e., the computer programmer).
 - 4) **Intelligent agents** are programs that apply a built-in or learned knowledge base to execute a specific, repetitive, and predictable task, for example, showing a computer user how to perform a task or searching websites for particular financial information.
 - 5) An **expert system** is an interactive system that attempts to imitate the reasoning of a human expert in a given field. It is useful for addressing unstructured problems when there is a local shortage of human experts.

- c. The following are ways AI can be utilized to improve the efficiency and effectiveness of processing accounting data:
- 1) Automate data entry and analysis
 - a) Information obtained from document images (e.g., expense reports, vendor invoices, customer payments, etc.) is automatically assigned to applicable accounting general ledger accounts.
 - i) For example, a payee deposits a check at an ATM, the ATM takes a picture of the check data (e.g., payor's bank account data, dollar amount, etc.) and funds are extracted from, and deposited into, the payor's and payee's bank account, respectively, and recorded.
 - b) By continuously repeating the data entry process, the computer eventually learns to identify variances and anomalies; thus, corrections can be made earlier and the integrity of accounting reports is strengthened.
 - 2) Reduce fraud
 - a) Generally, auditors are capable of reviewing only a small fraction of disbursement documents (e.g., vendor invoices, contract payments, employee expense reports, etc.) manually, whereas AI has the capability of reviewing the entire population of documents.
 - i) Because AI has the capability to learn by analyzing vast amounts of data, it is more efficient and effective at identifying fraud.
 - 3) Strengthen expenditure disbursement policies
 - a) The entire population of scanned disbursement documentation, such as (1) purchase requisitions and purchase orders, (2) vendor invoices, (3) credit card activity, and (4) employee expense reports (including supporting documents), can be analyzed to identify disbursements not in compliance with policy (i.e., disallowed vendors and travel extras).

9. Cloud Computing

- a. Cloud computing ("the cloud") is a popular term relating to on-demand access to resources that are accessed on the Internet and shared by others.
- b. Advantages of using cloud computing include fast access to software, a reduced need for investment in IT infrastructure, and the ability to use "pay as you go" services.
- c. IT security in the cloud is potentially more difficult due to the convenience and ease of access to sensitive data provided by cloud computing services.

- d. There are three primary categories of cloud services:
- 1) **Infrastructure-as-a-Service (IaaS)**
 - 2) **Platform-as-a-Service (PaaS)**
 - a) PaaS is a form of cloud computing wherein a third-party provides both hardware and software tools. These tools are typically for application development.
 - 3) **Software-as-a-Service (SaaS)**
 - a) SaaS is a software distribution model in which customers can subscribe to web-based applications. An example is a cloud storage service like Dropbox, which enables customers to store, share, and synchronize files across devices.
 - b) Benefits of SaaS include
 - i) Users do not have to buy, maintain, support, or update as much computer hardware or software. SaaS products are maintained and updated by the third-party host.
 - ii) Users can subscribe for only the amount of time the service is needed.
 - iii) Users can choose only the basic service or add select features.
 - iv) Users can access the software from anywhere, allowing for work at home or on site with the client.
 - c) SaaS can also have disadvantages.
 - i) Users must rely on outsiders to keep the software running and maintain a secure environment for the company's data.
 - ii) Providers can have service disruptions or experience a security breach, the latter of which can release the user's data to a fraud perpetrator.
- e. Cloud computing has benefited from the rise of smartphones and tablets.
- 1) Because these devices have limited memory, personal data (e.g., pictures, contacts, etc.) may be stored on the cloud to be retrieved later so that available memory can be used for application software.
 - 2) Generally, smart machine technology characteristics include, but are not limited to, (a) learning and operating on their own, (b) adapting their behavior based on experience (learning), and (c) generating unanticipated results.
- f. Analysts must have a detailed understanding of the available data and possess some sense of the answers they are looking for. The data are only as valuable as the business outcomes they make possible. The way businesses make use of data allows full recognition of the data's true value and their potential to improve decision-making capabilities and measure them against the results of positive business outcomes.

10. Blockchain

- a. In the late 15th century, the double-entry system and the general ledger were revolutionary concepts that changed commerce. The invention of the double-entry system allowed third parties to track the finances of a business. The firm maintains a private ledger of its transactions while a trusted independent party (i.e., the auditor) confirms the ledger's accuracy.
- b. A blockchain is an innovative technology that has the potential to change accounting.
 - 1) A **blockchain** is a type of digital database (or ledger) that provides proof of who owns what at any moment in time because each transaction has been added to the ledger.
 - a) The ledger is a type of **distributed ledger** that is encrypted, public, and shared widely for anyone to view. Every transaction makes the ledger grow larger. Data is not centralized; therefore, there is no central administrator.
 - 2) The term "blockchain" derives from the nature of a ledger. It contains the history of every transaction and forms a **chain**. When a change of ownership occurs, a new "block" is linked to the original chain to form a new chain.
 - a) A **block** is the current part of a blockchain. It records some or all of the recent transactions and becomes a permanent database in the blockchain.
 - b) A blockchain is primarily used to verify financial transactions within digital currencies (i.e., cryptocurrency transactions), though it is possible to digitize, code, and insert practically any document into the blockchain.
 - i) **Cryptocurrency** is a digital asset designed to be a medium of exchange using cryptography (encryption) to secure the transactions, control the creation of additional units of the currency, and verify the transfer of funds. Bitcoin is a type of cryptocurrency.
 - c) Owning a **bitcoin**, which uses a blockchain, indicates that the owner has a piece of information (or a block) within the blockchain ledger.
 - d) The blockchain allows each bitcoin owner to transfer a quantity of currency directly to another party connected to the same network without the need for a financial institution to mediate the process.
 - 3) The essence of a blockchain is that many third parties confirm that each new block is legitimate and part of the new complete chain. Each time a proposed transaction is entered in the form of a new block, **miners** authenticate the transaction by agreeing on what the current version of the blockchain should be.
 - a) In the example of Bitcoin, the miners are paid with some new bitcoin for verifying the new chain.

- 4) A key element of a blockchain is a **consensus mechanism**. It is a cryptographic process that takes control of the ledger from one party (i.e., the firm) and allows it to be examined and maintained by multiple independent entities. No centralized organization controls the chain. The official chain is agreed upon by a majority of the participating miners.
 - a) For example, blockchains and consensus mechanisms are similar to Google Docs. Users in a Google Doc can edit documents at the same time, and the most updated versions are always available. All users of the document must agree to any changes made.
- 5) In accounting, a company's ledger can be edited secretly, and the changes may not necessarily be obvious. But, using blockchain technology, editing of a chain is immediately obvious because the majority of third parties have a different chain. Thus, if anyone tries to alter a previous transaction, the public can see that the ledger has been tampered with and is presumably wrong. In the future, blockchain technology could potentially remove bookkeeping from private firms and turn it over to the public.
 - a) Falsifying records using a blockchain requires changing over 50% of the peer chains in the ecosystem. In the case of Bitcoin, which has over 10,000 miners at any moment in time, a bad actor would have to control many computers to alter a chain.

11. Smart Contracts

- a. Smart contracts are computerized transaction protocols that execute the terms of a contract. A smart contract is a collection of code and data (sometimes referred to as function and state) that is deployed using cryptographically signed transactions on a blockchain network.
- b. The general objectives of smart contract design are to satisfy common contractual conditions (such as payment terms, liens, confidentiality, and even enforcement), minimize exceptions (both malicious and accidental), and minimize the need for trusted intermediaries (e.g., lawyers).
- c. The code, being on a blockchain, is also tamper evident and tamper resistant and therefore can be used as a trusted third party.
- d. A smart contract can perform calculations, store information, expose properties to reflect a publicly exposed state, and, if appropriate, automatically send funds to other accounts.
- e. Smart contracts must be deterministic (i.e., given an input, they must always produce the same output based on that input).

20.2 ANALYTICS AND BIG DATA

1. Business Intelligence (BI)

- a. BI is associated with the collection of applications, tools, and best practices that transform data into actionable information for (1) managerial control, (2) strategic planning, and (3) making better decisions.
 - 1) BI gives upper management information about the present status of the organization and how to steer it in the intended direction.
 - 2) BI gives an executive immediate information about an organization's critical success factors.
- b. BI tools rely on advanced software to search vast amounts of data obtained from both within and outside the organization to discover patterns, trends, and relationships.
- c. BI tools display information about the organization as bar graphs, pie charts, columnar reports, or any other format considered appropriate to upper management's decision making.
 - 1) These displays are sometimes grouped into what is termed a **digital dashboard**.
 - a) Stock price trends, sales by region and date, on-time delivery performance, instantaneous cash balances, and profitability by customer are possible metrics.

2. Data Analytics (DA)

- a. DA involves qualitative and quantitative methodologies and procedures to retrieve data out of data sources and then inspect the data (in accordance with predetermined requirements) based on data type to facilitate the decision-making process.
 - 1) A **data type** specifies the type of value and the applicable mathematical, relational and non-relational, or logical operation methodologies that can be applied without resulting in an error.
- b. For-profit entities, not-for-profit entities, and government agencies (federal, state, and local) utilize DA to reach a conclusion based on evidence and reasoning to make well-supported decisions and formulate strong business models.
 - 1) Organizations can also use business analytics to rule out proposed strategic plans and models that would not be beneficial or work for the organization.
- c. Management utilizes DA to evaluate operational, financial, and other data to identify any deviations from the norm (e.g., anomaly detection, potential risks) and opportunities for enhancement or advancement.

d. Data analytics contains five stages as follows:

1) **Define questions**

- a) Goals and objectives that the organization is trying to achieve should be identified. Key performance indicators (KPIs) must be identified to assist with measuring whether an organization is progressing towards its goals and objectives. Examples of KPIs include
 - i) Current ratio
 - ii) Net profit margin
 - iii) Budget variance
 - iv) Debt to equity ratio
 - v) Payment error rate
- b) Clearly defined goals and objectives assist the IT team with selecting the most appropriate technology source to use for the analysis.
- c) Early adoption of goals and KPIs helps keep the analysis on course and avoid worthless analysis.

2) **Obtain relevant data** (commonly referred to as **information discovery**)

- a) Access to every piece of data available allows for
 - i) Valuable analysis,
 - ii) More precise correlations,
 - iii) Construction of meaningful analysis models and forecasts, and
 - iv) Identification of actionable insights.

3) **Clean/normalize data**

- a) Cleaning data consists of, but is not limited to, flushing out useless information and identifying missing data.
- b) Data governance assists with ensuring data are accurate and usable.
- c) Normalizing data involves storing each data element as few times as necessary. It results in a reduction of data and strengthened data integrity for use of a specific purpose.
 - i) **Integrity** is ensuring that data accurately reflect the business events underlying them and that any anomalies are rectified.

4) Analyze data

- a) As the data collected are analyzed, a determination can be made as to whether the data are the exact data needed. The determination includes, but is not limited to,
 - i) Assessing whether additional data are needed,
 - ii) Collecting new and/or different data,
 - iii) Revising the original question, and
 - iv) Formulating additional questions.
- b) Data analytics methods include the following application types:
 - i) **Descriptive analysis** is the most basic and most used method. It concentrates on reporting actual results.
 - ii) **Diagnostic analysis** provides insight into the reason certain results occurred.
 - iii) **Predictive analysis** involves applying assumptions to data and predicting future results. Item 9. on page 23 contains a more detailed discussion of predictive analysis.
 - iv) **Prescriptive analysis** concentrates on what an organization needs to do in order for the predicted future results to actually occur.
 - v) **Anomaly detection** is used to identify unusual patterns or deviations from the norm or expected results.
 - vi) **Network analysis** consists of analyzing network data and statistics to find patterns.
 - vii) **Text analysis** involves the utilization of text mining and natural language algorithms to find patterns in unstructured text.
- c) Personnel generally will select data to trace to supporting source documentation, such as invoices, contracts, and payments, and perform the following additional procedures:
 - i) Review and confirm the details of the data selected.
 - ii) Analyze the findings and determine compliance or non-compliance with policy.
 - iii) Analyze the findings for accuracy.
 - iv) Identify internal controls requiring enhancement or, if no controls exist, assist with the creation of a control.

5) **Communicate results**

- a) The primary purpose of communicating results with appropriate management is to ensure the accuracy of the information used, conclusions, and recommendations.
- b) Data visualization or graphic illustrations (e.g., charts, graphs, network analysis, etc.), written repetition (e.g., summaries), and itemized lists (bulleted or numbered) are good ways of emphasizing information.
 - i) Using visual aids to support a discussion of major points results in the most retention of information.
- c) Generally, language should be fact-based and neutral.
 - i) Using too strong a word or a word inappropriate for the particular recipient may induce an unwanted response. Thus, high-connotation language should be chosen carefully to appeal to the specific recipient.

3. **Data Mining**

- a. Data mining is the search for unexpected relationships among data. It combines information technology and statistics with a goal of analyzing data from different perspectives and summarizing it into useful information.
 - 1) The classic example of the use of data mining was the discovery by convenience stores that diapers and beer often appeared on the same sales transaction in the late evening.
 - 2) Another example is the process Netflix uses to make decisions about which new shows to acquire and produce. Data scientists at Netflix gather information on trends in what and how consumers are watching, then draw conclusions from those trends on what consumers are most likely to watch in the future.
- b. Common tasks performed during data mining to achieve the goals of revealing patterns and providing insights include
 - 1) Detecting anomalies to identify data that deviates from what is expected. Outlier data can help an organization find fraud, software issues, or other unexpected occurrences.
 - 2) Finding relationships between variables to sort data into clusters that differ from known structures. Clustering can be challenging because discovering new group structures may take multiple rounds of trial and error.
 - 3) Classifying data within the boundaries of known structures that have been generalized to accommodate new data sets.
 - 4) Performing regression analysis, including both linear regression and multiple regression, to model the relationships among the data as accurately as possible.
 - 5) Summarizing the data so it can be presented to those who would make use of the information. Summarizations can be visual representations of the data or documents containing aggregated information.

- c. Data mining is an iterative process by means of continuous review and rethinking (i.e., enhancements of previous analysis) while working to obtain the desired results. Once a finding has been made, further data mining may lead to even more conclusions.
- d. Some people claim data mining is a science because a computer system is used to find solutions, while others claim that data mining is an art because distinguishing meaningful patterns from the computer output is highly dependent on the judgment, expertise, and intuition of the analyst.
 - 1) The first step in data mining is finding an anomaly, such as an outlier, cluster, unexpected change, or a deviation from what was expected. This involves identifying unusual data records, which might be interesting, or perhaps data errors that could require further investigation.
 - 2) The next step is to find relationships between variables and then cluster (group) those relationships in some manner. Clustering is not always easy because the group structure is not known in advance.
 - 3) Once the data have been clustered, the next step is to generalize the relationships so that the demonstrated structure applies to new data as well as the original database.
 - 4) Regression analysis, including both linear regression and multiple regression, attempts to find a quantitative function or equation that models the data with the least error, that is, to estimate the relationships among the data or datasets.
 - 5) The final step in data mining provides a representation of the data set, including visualizations and reports.
- e. Query tools can be used to assist with the retrieval of data during a data mining exercise.
 - 1) Structured query language (SQL) is a language used to access and manipulate data within relational database management systems (RDBMSs). Queries are constructed and executed using a set of commands to create, update, and retrieve information. Queries can be used to join multiple tables to manipulate or retrieve joined records with a single command.

4. Simple Regression

- a. Regression analysis is the process of deriving a linear equation that describes the relationship between two variables.
 - 1) Simple regression is used for one independent variable. Multiple regression is used for more than one.
- b. The simple regression equation is the algebraic formula for a straight line.

$$y = a + bx$$

If: y = the dependent variable
 a = the y intercept
 b = the slope of the regression line
 x = the independent variable

- 1) The best straight line that fits a set of data points is derived using calculus.

- c. Regression analysis is particularly valuable for quantifying risk in financial risk management as well as for budgeting and cost accounting purposes.
 - 1) One extremely common application of simple regression in a business setting is the estimation of a mixed cost function, one with a fixed component and a variable component.
 - 2) The y-axis intercept is the fixed portion, and the slope of the regression line is the variable portion.

EXAMPLE 20-1 Linear Regression Analysis of Manufacturing Costs

A firm has performed a linear regression analysis and determined that total manufacturing costs (y) consist of fixed costs of \$420,000 and variable costs of \$32 per unit of output. This relationship can be stated mathematically as follows:

$$y = \$420,000 + \$32x$$

If the firm is planning to produce 12,000 units of output, its forecast for total manufacturing costs is \$804,000 [$\$420,000 + (\$32 \times 12,000)$].

5. Multiple Regression

- a. Multiple regression is used when there is more than one independent variable. For example, sales might depend on the amount of advertising, the number of salespeople in the field, the price of the product, and perhaps other independent variables. Multiple regression will show how much each of the independent variables contributes to the overall level of sales.
 - 1) Thus, multiple regression allows a firm to identify many factors (independent variables) and weigh each one according to its influence on the overall outcome.
- b. The **coefficient of determination (r^2)**, or the coefficient of correlation squared, is a measure of the fit between the independent and dependent variables.
 - 1) The coefficient of determination is the proportion of the total variation in the dependent variable that is accounted for by the independent variable.
 - 2) The value of r^2 ranges from 0 to 1. The closer the value of r^2 to 1, the more useful the independent variable (x) is in explaining or predicting the variation in the dependent variable (y).

EXAMPLE 20-2 Coefficient of Determination

A car dealership determines that new car sales are a function of disposable income with a coefficient of correlation of .8. Thus, 64% ($.8^2$) of the variation of new car sales from the average can be explained by changes in disposable income.

6. Standard Error

- a. The standard error measures how well the linear equation represents the data. It is the vertical distance between the data points in a scatter diagram and the regression line.
 - 1) The closer the data points to the regression line, the lower the standard error.

7. Aspects of Regression Analysis

- a. The linear relationship established for x and y is valid only across the relevant range, the range from the highest to the smallest measures in the data set. The user must identify the relevant range and ensure that projections lie within it.
- b. Regression analysis assumes that past relationships are a basis for valid projections.
- c. Regression does not determine causality.
 - 1) Although x and y move together, the apparent relationship may be caused by some other factor. For example, car wash sales volume and sunny weather are strongly correlated, but car wash sales do not cause sunny weather.
- d. The **goodness-of-fit test** assists with determining whether the sample is representative of the population (validates assumptions).
 - 1) The **confidence level** is the percentage of times that a sample is expected to be representative of the population; i.e., a confidence level of 95% should result in representative samples 95% of the time.
 - a) A person selecting an item at random from a normally distributed population can be, for example, 95% confident that the value of the item is within 1.96 standard deviations of the mean and 99% confident that it will fall within 2.57 standard deviations of the mean.
 - 2) A **confidence interval** for a given confidence level is the range around a sample value that is expected to contain the true population value. It is constructed using the confidence coefficient for the number of standard deviations (based on the normal distribution) for the confidence level chosen.
 - a) If repeated random samples are drawn from a normally distributed population and the auditor specifies a 95% confidence level, the probability is that 95% of the confidence intervals constructed around the sample results will contain the population value.
- e. Advantages of regression analysis include the following:
 - 1) It uses data efficiently.
 - 2) Good results can be obtained with relatively small sets of data.
 - 3) The theory associated with linear regression is well-understood and allows for construction of different types of easily-interpretable statistical intervals for predictions, calibrations, and optimizations.
- f. The disadvantages or limitations of regression analysis include the following:
 - 1) Outputs of regression can lie outside the relevant range.
 - 2) The shapes that linear models can assume over long ranges are limited.
 - 3) Sometimes the extrapolation properties will be poor.
 - 4) The regression line may be ultrasensitive to outliers.

8. Various Approaches to Quantify or Understand Risk

a. Sensitivity Analysis

- 1) Sensitivity analysis uses trial-and-error to determine the effects of changes in variables or assumptions on final results. It is useful in deciding whether expending additional resources to obtain better forecasts is justified.
- 2) The trial-and-error method inherent in sensitivity analysis is greatly facilitated by the use of computer software.
 - a) A major use of sensitivity analysis is in capital budgeting. Small changes in interest rates or payoffs can make a significant difference in the profitability of a project.
- 3) Sensitivity analysis is limited due to
 - a) The use of assumptions instead of facts
 - b) The consideration of variables individually as opposed to all together

b. Simulation

- 1) This method is a sophisticated refinement of probability theory and sensitivity analysis. The computer is used to generate many examples of results based on various assumptions.
- 2) Project simulation is frequently expensive. Unless a project is exceptionally large and expensive, full-scale simulation is usually not worthwhile.

c. Monte Carlo Simulation

- 1) This method often is used in simulation to generate the individual values for a random variable. The performance of a quantitative model under uncertainty may be investigated by randomly selecting values for each of the variables in the model (based on the probability distribution of each variable) and then calculating the value of the solution.
- 2) Performing this process many times produces the distribution of results from the model.

d. Delphi Approach

- 1) The Delphi approach solicits opinions from experts, summarizes the opinions, and feeds the summaries back to the experts (without revealing participants to each other).
 - a) The process is repeated until the opinions converge on an optimal solution.

e. Time Series Analysis

- 1) Time series analysis (also called trend analysis) is the process of projecting future trends based on past experience. It is a regression model in which the independent variable is time.
- 2) A **seasonal** pattern often exists when a time series is influenced by seasonal factors (e.g., the quarter of the year, the month, or the day of the week).
 - a) For example, a ski resort will typically show a seasonal pattern for its revenues.
 - b) Seasonality is normally always of a known period. Thus, seasonal time series are sometimes called periodic time series.
 - c) Similarly, a manufacturer of lawn mowers will have greater production during the winter and spring than during the summer and fall months.
- 3) A **cyclical pattern** exists when data points show rises and falls that are not of a fixed or seasonal pattern. The duration of these cyclical fluctuations is usually at least a couple of years.
 - a) An example is business cycles, which typically last several years, but the length of the current cycle is never known in advance.
 - b) Cyclical behavior is often confused with seasonal behavior. However, if the fluctuations are not of a fixed period, they are cyclical; if the period is unchanging and associated with some aspect of the calendar, the pattern is seasonal.
 - c) Normally, the average length of cycles is longer than the length of a seasonal pattern, and the magnitude of the depth and rise of cycles tends to be more variable than the magnitude of seasonal patterns, the latter of which occur at least annually.
- 4) An irregular pattern exists when random or unplanned factors occur. A time series does not occur with irregular information.
 - a) For instance, a ski resort's revenues may generally be expected to be high during winter; however, the resort may experience an unplanned increase in revenues in March or April due to an unexpected snow storm.
- 5) Benefits of time series analysis include
 - a) The ability to project expected upcoming activities
 - i) For example, sales of snow jackets are higher during winter than summer.
 - b) Identifying components or elements that impact the anticipated occurrence in the future
 - c) Analyzing current results against activity in the past to determine trend shifts
- 6) Limitations of time series analysis include the following:
 - a) Components or elements that play a role in projections may become less reliable over time.
 - b) The conclusions resulting from the analysis could be misleading.
 - i) For example, higher sales of snow jackets could be the result of an increase in population or could be increasing at a decreasing rate.
 - c) Predictability of applicable components or elements in the past are assumed to remain unchanged for the future.

9. Predictive Analytics

- a. Predictive analytics is technology that uses data (historical, current, and predicted), statistical algorithms, and machine learning techniques to draw insights on such trends as consumer or industry behavior patterns. Additionally, recommendations can be made based on underlying assumptions to data and predicting future results.
- b. The likelihood of predicted scenarios can be assessed by calculating probabilities to illustrate their feasibilities.
 - 1) Assumptions (e.g., product type, units sold, unit costs, seasonal product status, etc.) must be true to life to ensure the accuracy and quality of the results of the analysis.
- c. Recommendations can then be made of what would be required in order to achieve each predicted scenario.
- d. A common use of predictive analytics in the retail sector occurs when a customer selects an item to purchase online and prepares to finalize the transaction; the web page then displays additional products other customers purchased in conjunction with the initial item.
 - 1) For example, a customer purchases a table saw and, upon checking out, the web page displays images of saw blades, eye-protection goggles, and gloves that other customers bought along with the table saw.

10. Exploratory Data Analysis (EDA)

- a. EDA is an approach to data analysis that differs from traditional modeling and analytic techniques by encouraging the data itself to reveal its underlying structure rather than prematurely applying a hypothesis or statistical method. It is often used as a first step or pre-step in the data analysis process.
 - 1) The main role of EDA is to open-mindedly explore in order to gain some new, often unsuspected, insight into the data.
- b. EDA seeks to
 - 1) Maximize insight into a data set,
 - 2) Uncover an underlying structure,
 - 3) Identify important variables,
 - 4) Check for outliers and anomalies,
 - 5) Test underlying assumptions,
 - 6) Develop parsimonious models, and
 - 7) Determine optimal factors.
- c. EDA often uses visual and graphical tools, such as histograms and scatter plots.

11. What-If Analysis

- a. What-if analysis is a process of determining the effects on outcomes in a model through changes in scenarios.
- b. For example, an analyst could build a model that projects potential profits for
 - 1) Every 50, 100, or 150 units sold;
 - 2) A decrease in distribution costs by 5% for every 50 units sold; or
 - 3) An increase in sales labor hours by 5% for every 50 units sold.
- c. **Goal seeking** occurs when the decision maker has a specific outcome in mind and needs to determine how it can be achieved.

12. Big Data

- a. Big data is an evolving term that describes any voluminous amount of structured, semi-structured, or unstructured data that has the potential to be mined for information to reveal relationships and dependencies or to predict outcomes and behaviors.
 - 1) **Structured data** refers to data that are highly organized into predefined groupings and are typically maintained in relational databases. The data are predefined such that each item falls into a specific anticipated data type (e.g., string, float, integer, date, boolean) that can easily be sorted and searched by computer programs. For example, sales data are mostly structured.
 - 2) **Semi-structured data** refers to data that are not as highly organized as structured data but still have some identifying information that can be used for organization by computer programs. With certain processes, semi-structured data can be stored in relational databases, which can be handled in the same way as structured data. For example, XML and XBRL data can be converted and stored in relational databases for analysis.
 - 3) **Unstructured data** refers to information that has little or no predefined organizational structure. This lack of organization makes such data much more difficult for computer programs to search, sort, and analyze. For example, audios, videos, and images are data types that are difficult for computer programs to analyze.
- b. Big data includes information collected from numerous sources, including popular social media sites, Internet-enabled devices, machine data, videos, and even voice recordings. The information collected is converted from low-density data into high-density data (data that has value).
- c. Data are processed with analytic and algorithmic tools to reveal meaningful information.
- d. Data functions as a strategic asset when processing the data results in information (end product) that is relevant, high-quality, and timely. The information is then provided to decision makers who use sound judgment to
 - 1) Make well-informed and timely decisions
 - 2) Identify valuable trends and patterns
 - 3) Identify demand opportunities for products or services
 - 4) Formulate plans based on identified trends and patterns in order to enhance results

13. Big Data is often characterized by **The “4 Vs”** -- volume, variety, velocity, and veracity.

- a. **Volume.** This term is used to describe the large amount of data captured over time.
 - 1) Depending on the amount of data, the number of servers required could range from a single server to thousands of servers.
 - 2) Real-time sensors used in the Internet of Things (IoT) have become one of the top sources of data.
 - a) IoT is a system of interrelated computing devices that are provided with real-time sensors and have the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. Therefore, IoT submits large amounts of data that contribute to a high volume of captured data.

- b. **Variety.** Data exist in a wide variety of file types.
 - 1) Structured data file types are generally maintained by Structured Query Language (SQL). Files of this type are used for managing relational databases and performing various operations on the data in them.
 - 2) Unstructured data file types (i.e., streaming data from sensors, text, audio, images, and videos) are maintained by non-relational databases (i.e., NoSQL).
- c. **Velocity.** This term refers to the speed at which big data must be analyzed.
 - 1) Analysts must have a detailed understanding of the available data and possess some sense of what answer(s) they are looking for. The ability to look for these answers makes data analytics as much an art as a science.
 - 2) The computing power required to quickly process large volumes and varieties of data can overwhelm a single server or even multiple servers. Organizations must apply adequate computing power to big data tasks to achieve the desired velocity.
 - 3) Businesses are hesitant to invest in an extensive server and storage infrastructure that might only be used occasionally to complete big data tasks. As a result, cloud computing has emerged as a primary source for hosting big data projects.
- d. **Veracity.** This term refers to the trustworthiness of the data.
 - 1) The user wants to know that the data are representative of the population. Collecting large amounts of statistics and numbers is of little value if they cannot be relied upon and used.
- e. **The “5th V.”** More and more businesses are using big data because of the **value** of the information resulting from the culmination of analyzing large information flows and identifying opportunities for improvement.
 - 1) Use of big data is only as valuable as the business outcomes it makes possible. It is how businesses make use of data that allows full recognition of its true value and potential to improve decision-making capabilities and enhance positive business outcomes.
 - a) **Volume-based value.** The more data businesses have on customers, both recent and historical, the greater the insights. This leads to better decisions about acquiring, retaining, increasing, and managing customer relationships.
 - b) **Variety-based value.** In the digital era, the capability to acquire and analyze varied data is extremely valuable. Varied data enable businesses to develop and personalize customer platforms, which increase engagement and awareness of customer needs and expectations.
 - c) **Velocity-based value.** The faster businesses process data, the more time they will have to ask the right questions and seek answers. Quick analysis capabilities provide businesses with the right data in time to achieve their customer relationship management goals. For example, learning that a woman is pregnant based on her buying habits is only valuable if a business can develop an advertising program instantly to sell the woman pre-natal care items.

14. Uses

- a. Big data uses inductive statistics and combines them with concepts from nonlinear system identification (i.e., output is not directly proportional to the input) to deduce inferences from large sets of data, which reveals relationships and interdependencies. Alternatively, big data can be used to predict outcomes and behaviors.
 - 1) Big data can be used to identify opportunities for
 - a) Cost reductions,
 - b) Time reductions,
 - c) New product development and optimized offerings, and
 - d) New customers.
 - 2) Big data analytics tools can complete missing pieces through **data fusion**, which is the process of integrating data and knowledge representing the same object into a consistent and useful representation of the population.

15. Key Technologies

- a. **Data management.** Data need to be high-quality and complete before they can be reliably analyzed. Thus, businesses need to
 - 1) Establish processes that build and maintain standards for high-quality data
 - 2) Establish a master data management program
- b. **Data mining** examines large amounts of data to discover patterns.
 - 1) It sifts through repetitive noise in data, pinpoints what is relevant, uses that information to assess likely outcomes, and then accelerates the pace of informed decision making.
- c. **Hadoop** is an open-source, Java-based software framework that stores large amounts of data and runs applications on clusters of commodity hardware.
 - 1) It uses a distributed computing approach.
- d. **In-memory analytics** analyzes data from system memory instead of secondary storage. This approach
 - 1) Derives immediate results by removing data preparation and analytical processing delays and
 - 2) Enables iterative and interactive analytic scenarios more efficiently.
- e. **Predictive analytics** uses a combination of data, statistical algorithms, and machine-learning techniques (such as expert systems) to identify the probability of future outcomes based on historical data.
- f. **Text mining** analyzes text data from the Web, comment fields, books, and other text-based sources through the use of machine learning or natural language processing technology.
 - 1) It can be used to identify new topics and term relationships.

16. Limitations of Big Data (Including Other Data Analytics)

- a. Data analytics is not suitable for all applications and analyses. Situations in which data analytics may be unsuitable include but are not limited to those in which
 - 1) **User-level data results are incomplete.** Generally, the data available to an organization are restricted to data obtained from individuals who visited the organization's Internet resources (i.e., website) or viewed the organization's advertisements on the Internet.
 - a) The data are only representative of the target market; thus, untapped markets could potentially exist, the data of which are not being captured.
 - 2) **Providing the answer to why the analysis results are what they are is difficult.** Data are processed by (a) separating data into groups and applying analytic methods or (b) analyzing data directly using algorithmic methods. Both processes can result in forecasting outcomes and providing guidance, but algorithmic analytic methods tend to make it more difficult for non-technical people to justify choosing a certain course of action over another course of action (e.g., spending more capital expenditures on product Z over product P).
 - 3) **Data are subject to useless information (commonly known as noise).** A single incorrect or useless variable can corrupt the results and require additional labor hours to work with the data in order to attain meaningful results.
 - 4) **User-level data results require interpretation prior to use.** Generally, collected data are converted from text format to data visualizations. Data visualizations assist with identifying trends and correlations that run the risk of going undetected in text-based data.

17. Data Visualization

- a. Data visualization refers to any effort to help readers or other users understand the significance of data by placing it in a visual context, such as a graph or chart. Trends, correlations, or other arrangements that might often be overlooked in text-based data can be illustrated and thus recognized more easily with data visualization software. Visualizations are for the benefit of decision makers; the computers themselves do not use or need visualizations for decision making.
- b. Data visualization tools may provide interactive capabilities that allow users to manipulate data or drill into the data for what-if-type analysis.
 - 1) Visualizations facilitate visual comparisons between data elements, which helps decision makers identify patterns, deviations from patterns, and outliers in the analysis stage. This is accomplished in an efficient manner.
 - 2) Visualizations are usually understood by a wider audience because they reduce the message to its core components and use minimal jargon.
 - 3) Visualization options include histograms, boxplots, scatter plots, dot plots, tables, dashboards, bar charts, pie charts, line charts, and bubble charts. Additionally, adding colors and shapes assists with more effectively communicating results by decreasing the burden of decision makers interpreting vast amounts of information and instead concentrating on the more vital information.
 - a) Business intelligence tools include the capability of grouping the various visualization options onto a **digital dashboard**. The dashboard allows the user to pre-select the chart types he or she will use for assistance with decision making.

18. Statistical Control Charts

- a. Statistical control charts are graphic aids for monitoring the status of any process subject to acceptable or unacceptable variations during repeated operations.
 - 1) They also have applications of direct interest to auditors and accountants, for example, (a) unit cost of production, (b) direct labor hours used, (c) ratio of actual expenses to budgeted expenses, (d) number of calls by sales personnel, or (e) total accounts receivable.

EXAMPLE 20-3 Statistical Control Chart

The chart below depicts 2 weeks of production by a manufacturer who produces a single precision part each day. To be usable, the part can vary from the standard by no more than ± 0.1 millimeter.

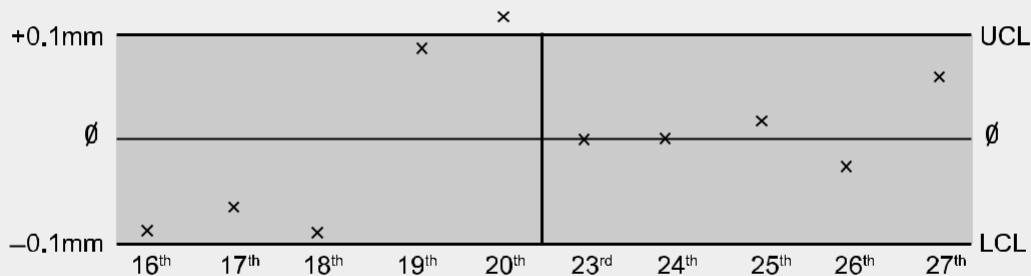


Figure 20-1

The part produced on the 20th had to be scrapped, and changes were made to the equipment to return the process to the controlled state for the following week's production.

19. Pareto Diagrams

- a. A Pareto diagram is a bar chart that assists managers in quality control analysis.

EXAMPLE 20-4 Pareto Diagram

The chief administrative officer wants to know which departments are generating the most travel vouchers that have to be returned to the submitter because of incomplete documentation.

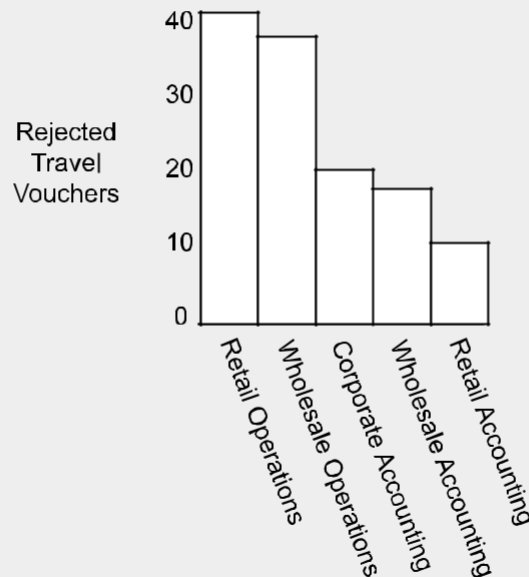


Figure 20-2

20. Histograms

- a. A histogram is similar to a Pareto diagram. The major distinction is that histograms display a continuum for the independent variable.

EXAMPLE 20-5 Histogram

The CAO wants to know the amount of a typical rejected travel voucher.



Figure 20-3

21. Fishbone Diagrams

- a. A fishbone diagram (also called a cause-and-effect diagram) is a total quality management process improvement method that is useful in studying causation (why the actual and desired situations differ).

EXAMPLE 20-6 Fishbone Diagram

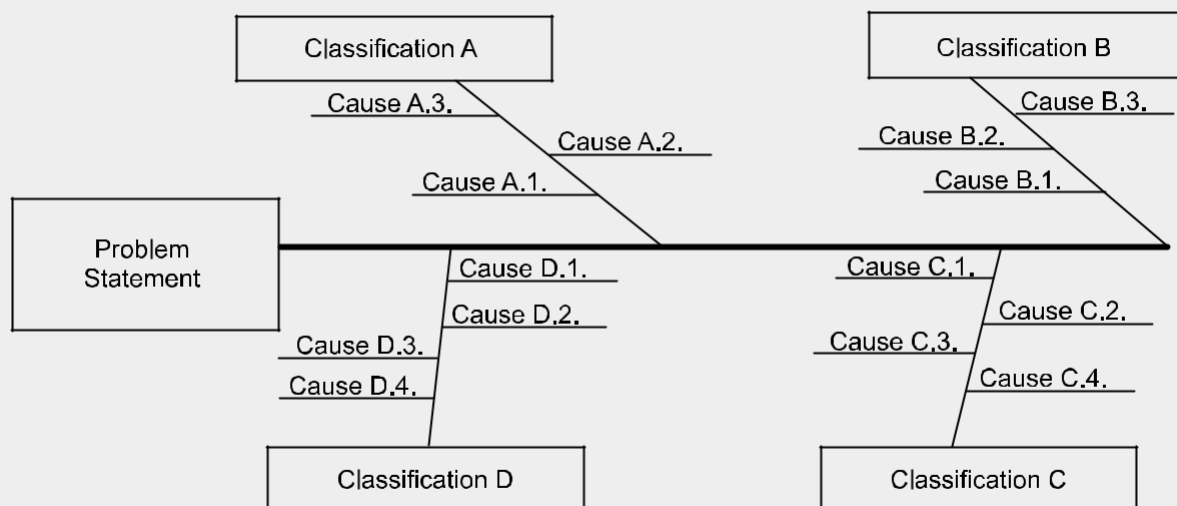
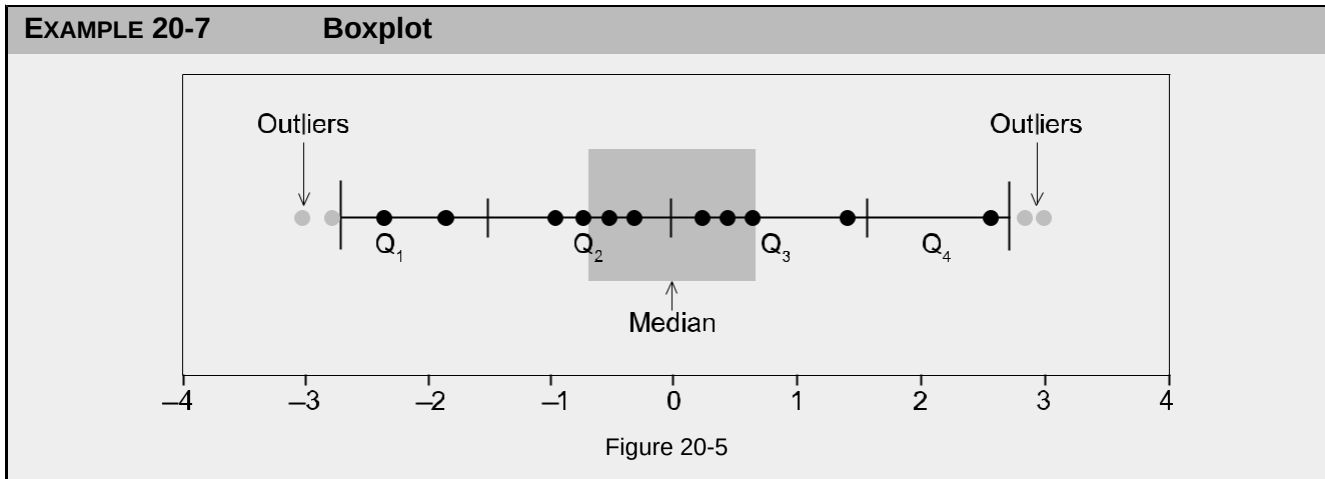


Figure 20-4

22. Boxplots

- a. A boxplot provides a visual to illustrate how tight or spread out data is and whether any outliers exist by displaying the data distribution in quartiles.



23. Tables

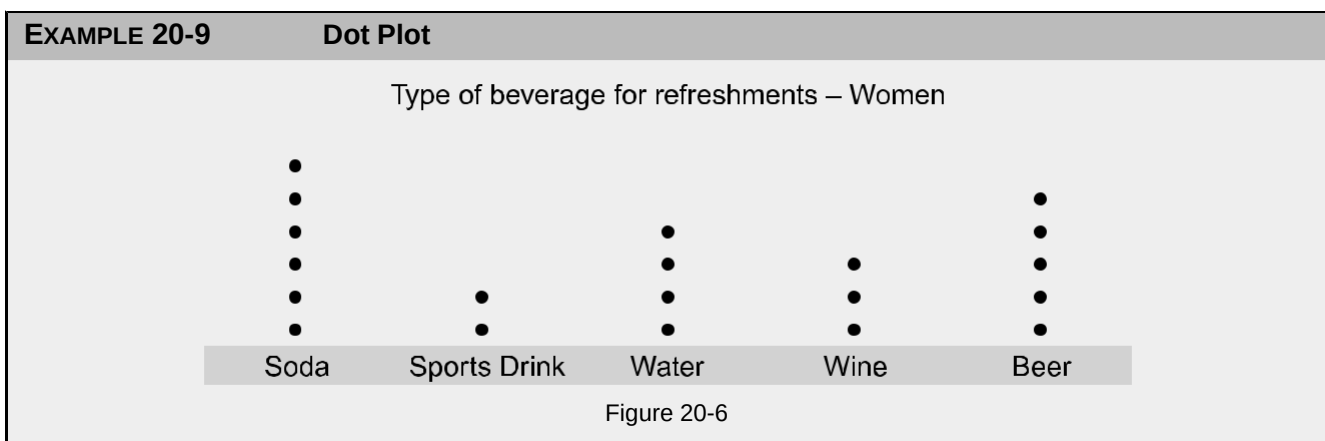
- a. A table illustrates data in rows and columns.

EXAMPLE 20-8 Table

	Type of beverage for refreshments				
Beverage	Soda	Sports Drink	Water	Wine	Beer
Women	6	2	4	3	5
Men	5	1	3	5	6

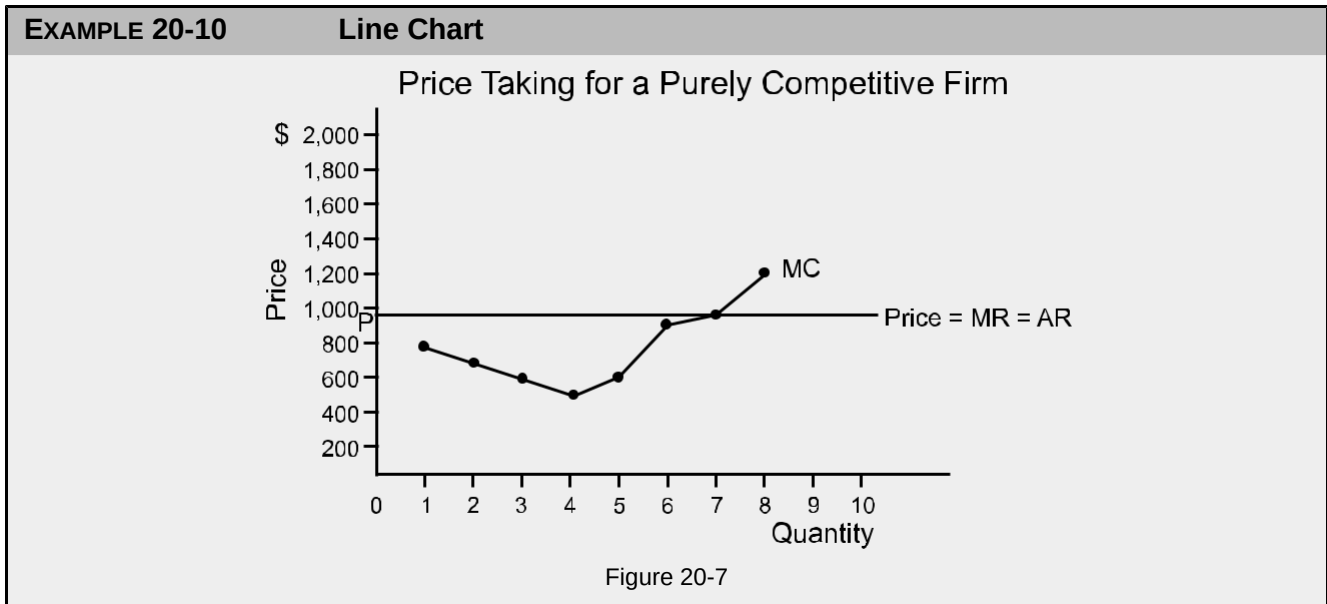
24. Dot Plots

- a. A dot plot (similar to a histogram) is a graph that illustrates the distribution for each factor using dots.



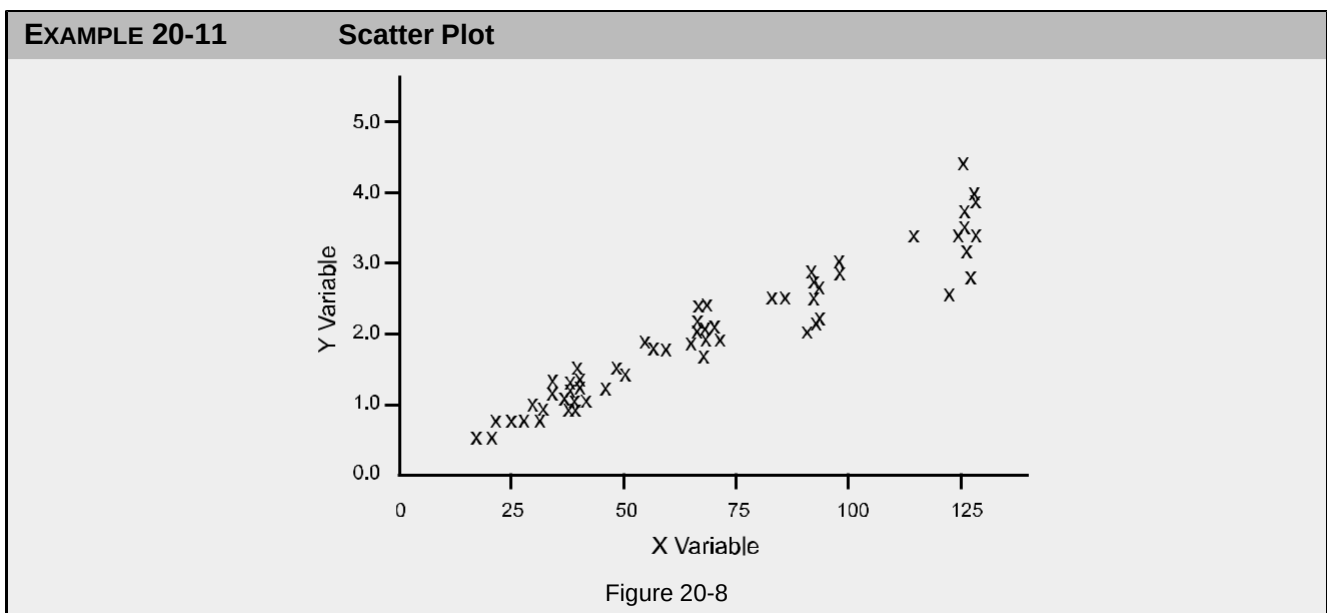
25. Line Charts

- a. A line chart illustrates information as a series of data points (commonly known as markers) connected by a straight line. It is effective for identifying patterns.



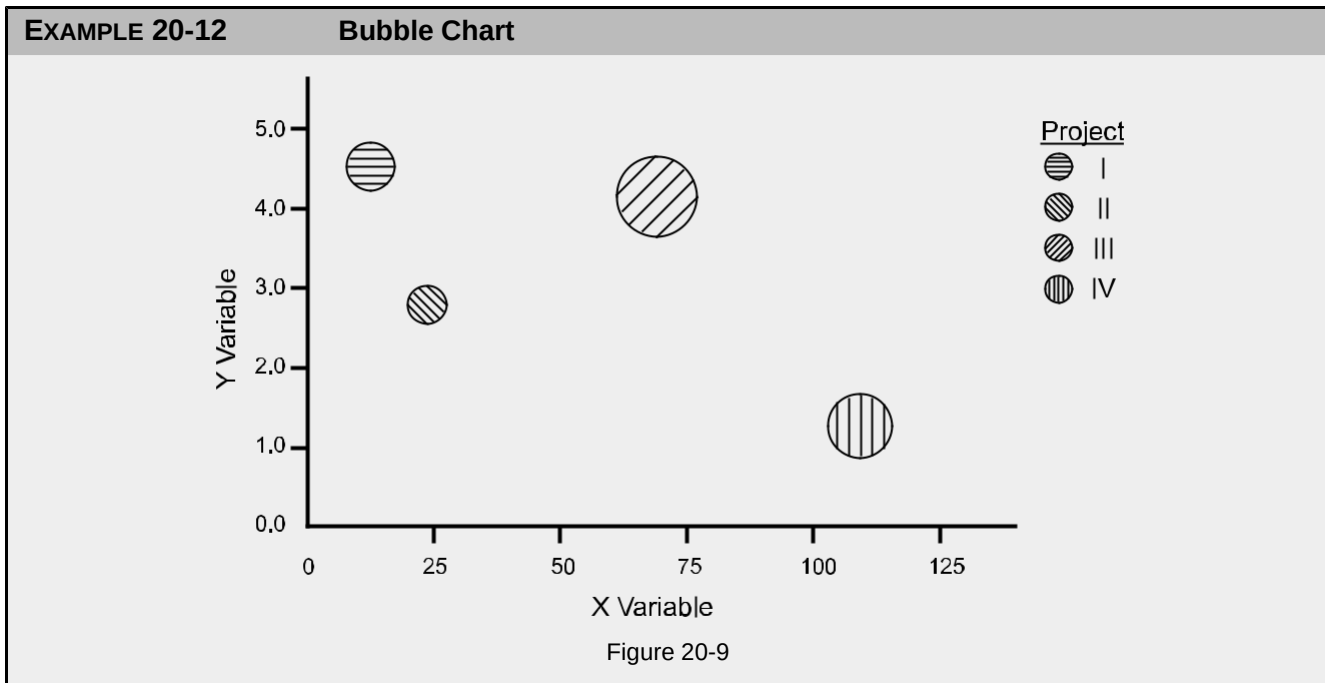
26. Scatter Plots

- a. A scatter plot (similar to a line chart) utilizes the horizontal (variable x) and vertical (variable y) axes to illustrate the correlation among two variables.
- 1) A positive correlation is identified by the data points forming a relatively straight line with the x and y values increasing and decreasing together.
 - 2) A negative correlation (or inverse correlation) is identified by the data points forming a relatively straight line, where the value on one axis increases as the value of the other axis decreases.
 - 3) A lack of correlation is identified by the scattering of data and the general absence of a straight line among the plotted data.



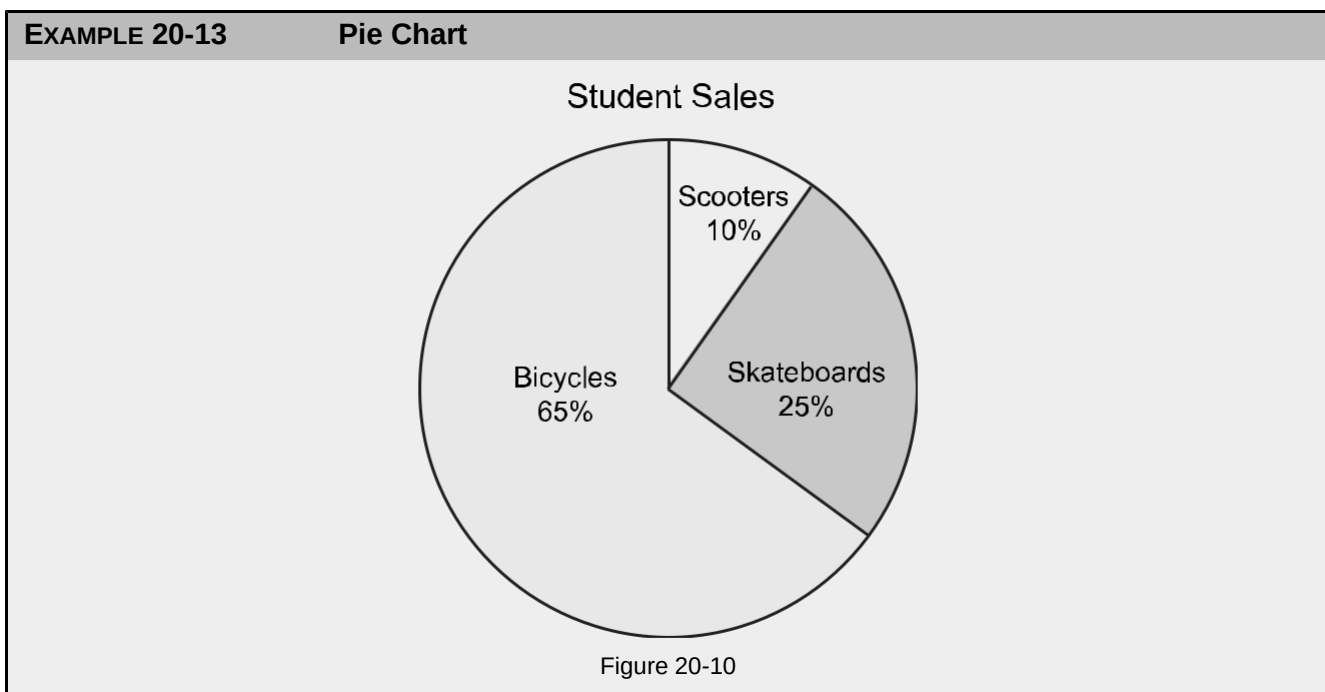
27. Bubble Charts

- a. A bubble chart is similar to a scatter plot; however, a bubble chart incorporates a third variable consisting of a grouping of data points represented by the size of the bubble.



28. Pie Charts

- a. A pie chart is a graph in the shape of a circle with each “pie slice” representing relative sections of a whole population size.
- 1) A limitation of pie charts is that too many slices makes comparability extremely difficult, which makes them less precise than other charts.



29. Limitations of Visualizations

- a. Visualizations can be misleading. The coder who creates an image can influence the viewer's interpretation of that image.
- b. Visualization tools can be manipulated to present a biased picture.

EXAMPLE 20-14 Biased Chart Scales

An increase in expenses from \$500,000 to \$501,000 is hardly noticeable on a chart that shows expenses on a scale from \$0 to \$1 million. However, if the chart shows expenses on a scale from \$499,000 to \$502,000, the expense line will show a steep vertical tendency.

In the first instance, the coder may have wanted to show that expenses had been flat, while in the second case, the coder may have wanted to show that expenses had increased tremendously.

30. Best Practices for Visualization Tools

- a. To avoid distortion in the communication of complex information, best practice guidelines for chart and graph design include, but are not limited to, the following:
 - 1) Utilize the full axis by ensuring the axis begins with zero and does not skip values.
 - 2) Do not overload the design with too much information.
 - 3) Limit the amount of colors used to ensure differences among data points stand out.
 - a) Avoid using variant shades of the same color (e.g., use red and not light red, soft red, medium red, etc.).
 - 4) Ensure the design does not require the user to interpret relationships among data.
 - a) Display calculations to avoid misinterpretation.
 - 5) Ask others for feedback on the design before finalizing the chart. Keep in mind that what might be clear to one person may actually be confusing to others.

APPENDIX F

ACCOUNTING CYCLES

On the following pages are five flowcharts and accompanying tables describing the steps in five basic accounting cycles and the controls in each step for an organization large enough to have an optimal segregation of duties.

NOTE: Except for manual checks and remittance advices, the flowcharts presented do not assume the use of either a paper-based or an electronic system. Each document symbol represents a business activity or control, whether manual or computerized.

NOTE: In the diagrams that follow, documents that originate outside the organization are separated by a thick border.

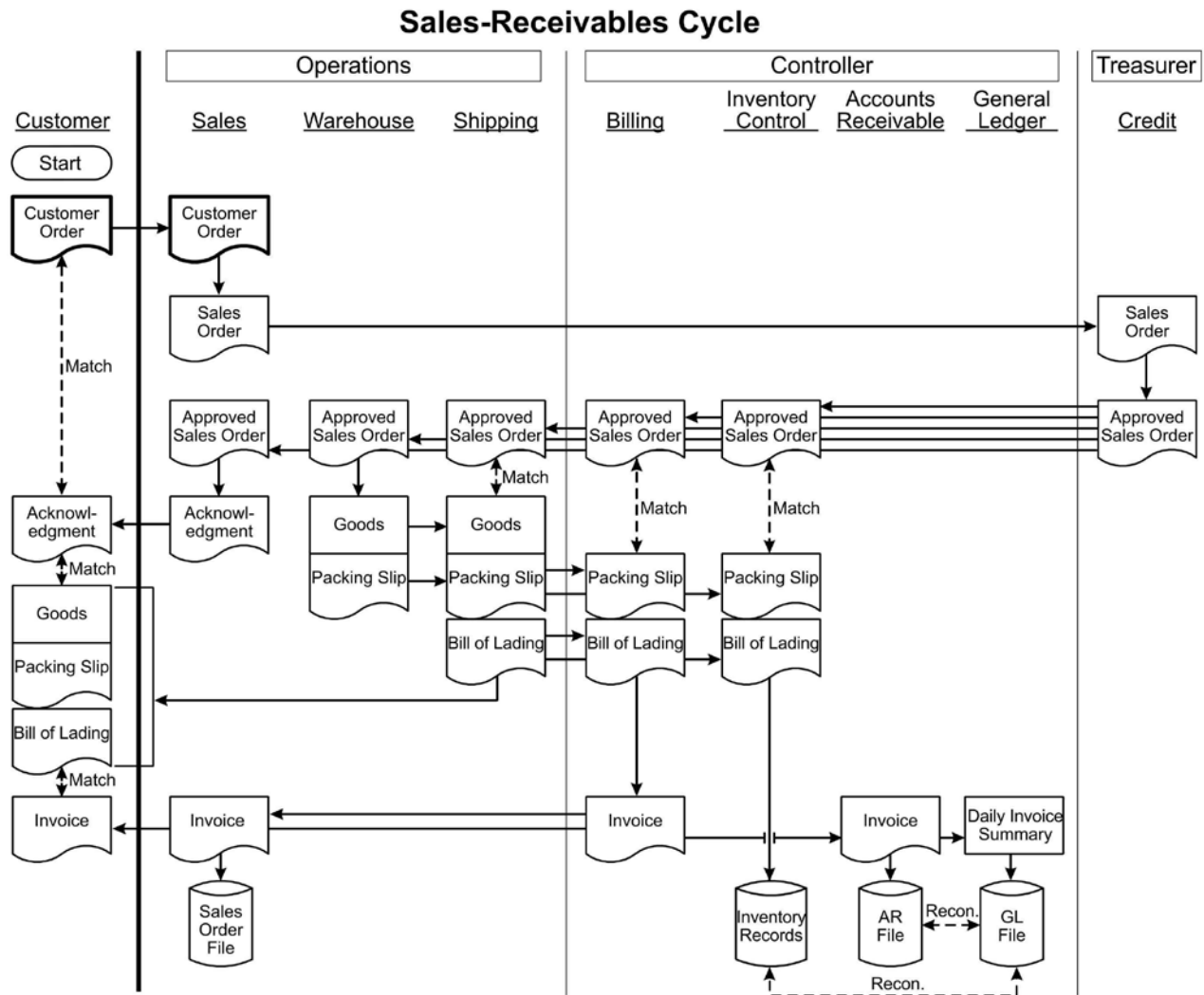


Figure F-1

Sales-Receivables Cycle

Function:	Authorization					Custody			Recording		
Department:	Customer	Sales	Credit	Billing	Shipping	Warehouse			Inventory Control	Accounts Receivable	General Ledger

Step	Business Activity	Embedded Control
1	Sales receives a customer order for merchandise, prepares a sales order , and forwards it to Credit.	Reconciling sequentially numbered sales orders helps ensure that customer orders are legitimate.
2	Credit performs a credit check on the customer. If the customer is creditworthy, Credit approves the sales order .	Ensures that goods are shipped only to actual customers and that the account is unlikely to become delinquent.
3	Credit sends the approved sales order to Sales, Warehouse, Shipping, Billing, and Inventory Control.	Notifies these departments that a legitimate sale has been made.
4	Upon receipt of an approved sales order , Sales sends an acknowledgment to the customer.	The customer's expectation of receiving goods reduces the chances of misrouting or misappropriation.
5	Upon receipt of an approved sales order , Warehouse pulls the merchandise, prepares a packing slip , and forwards both to Shipping.	Ensures that merchandise is removed from Warehouse only as part of a legitimate sale.
6	Shipping verifies that the goods received from Warehouse match the approved sales order , prepares a bill of lading , and sends the shipment to the customer.	Ensures that the correct goods are shipped.
7	Shipping forwards the packing slip and bill of lading to Inventory Control and Billing.	Notifies these departments that the goods have actually been shipped.
8	Upon receipt of the packing slip and bill of lading , Inventory Control matches them with the approved sales order and updates the inventory system.	Ensures that inventory unit counts are updated once the goods have actually been shipped. Updating inventory and GL files separately provides an additional accounting control when they are periodically reconciled.
9	Upon receipt of the packing slip and bill of lading , Billing matches them with the approved sales order , prepares an invoice , and sends it to the customer. If the invoice is paper-based, a remittance advice is included for use in the cash receipts cycle.	Ensures that customers are billed for all goods, and only those goods, that were actually shipped. Reconciling sequentially numbered invoice transactions helps prevent misappropriation of goods.
10	Sales receives the invoice from Billing and updates the sales order file.	Prevents double shipment of completed orders and allows follow-up of partially filled orders.
11	Accounts Receivable receives the invoice from Billing and posts a journal entry to the AR file.	Ensures that customer accounts are kept current.
12	Accounts Receivable prepares a summary of all invoices for the day and forwards it to General Ledger for posting of the total to the GL file.	Updating AR and GL files separately provides an additional accounting control when they are periodically reconciled.

Cash Receipts Cycle

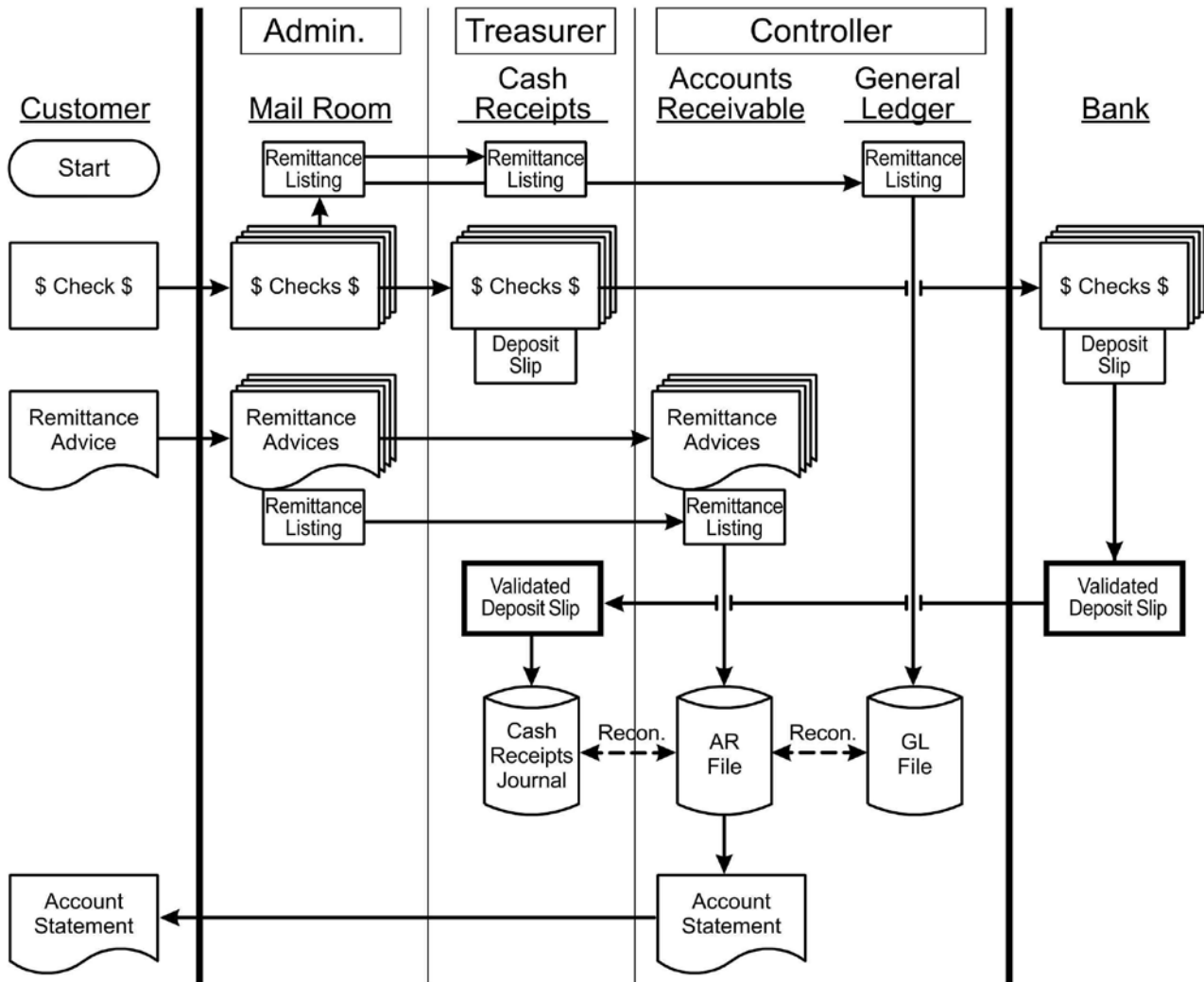


Figure F-2

Cash Receipts Cycle

Function:	Authorization		Custody		Recording	
Department:	Customer	Bank	Mail Room	Cash Receipts	Accounts Receivable	General Ledger

Step	Business Activity	Embedded Control
1	Mail Room opens customer mail. Two clerks are present at all times. Customer checks are immediately endorsed "For Deposit Only." Remittance advices are separated.	Reduces risk of misappropriation by a single employee.
2	Mail Room prepares a remittance listing of all checks received during the day and forwards it with the checks to Cash Receipts.	Remittance listing provides a control total for later reconciliation.
3	Cash Receipts prepares a deposit slip and deposits checks in Bank. Bank validates the deposit slip .	Bank provides independent evidence that the full amount was deposited.
4	Upon receipt of the validated deposit slip , Cash Receipts posts a journal entry to the cash receipts journal.	Ensures that the cash receipts journal is updated for the amount actually deposited.
5	Mail Room also sends the remittance listing to General Ledger for posting of the total to the GL file.	Updating AR and GL files separately provides an additional accounting control when they are periodically reconciled.
6	Mail Room also sends the remittance listing and remittance advices to Accounts Receivable for updating of customer accounts.	Ensures that customer accounts are kept current.
7	Accounts Receivable periodically sends account statements to customers showing all sales and payment activity.	Customers will complain about mistaken billings or missing payments.

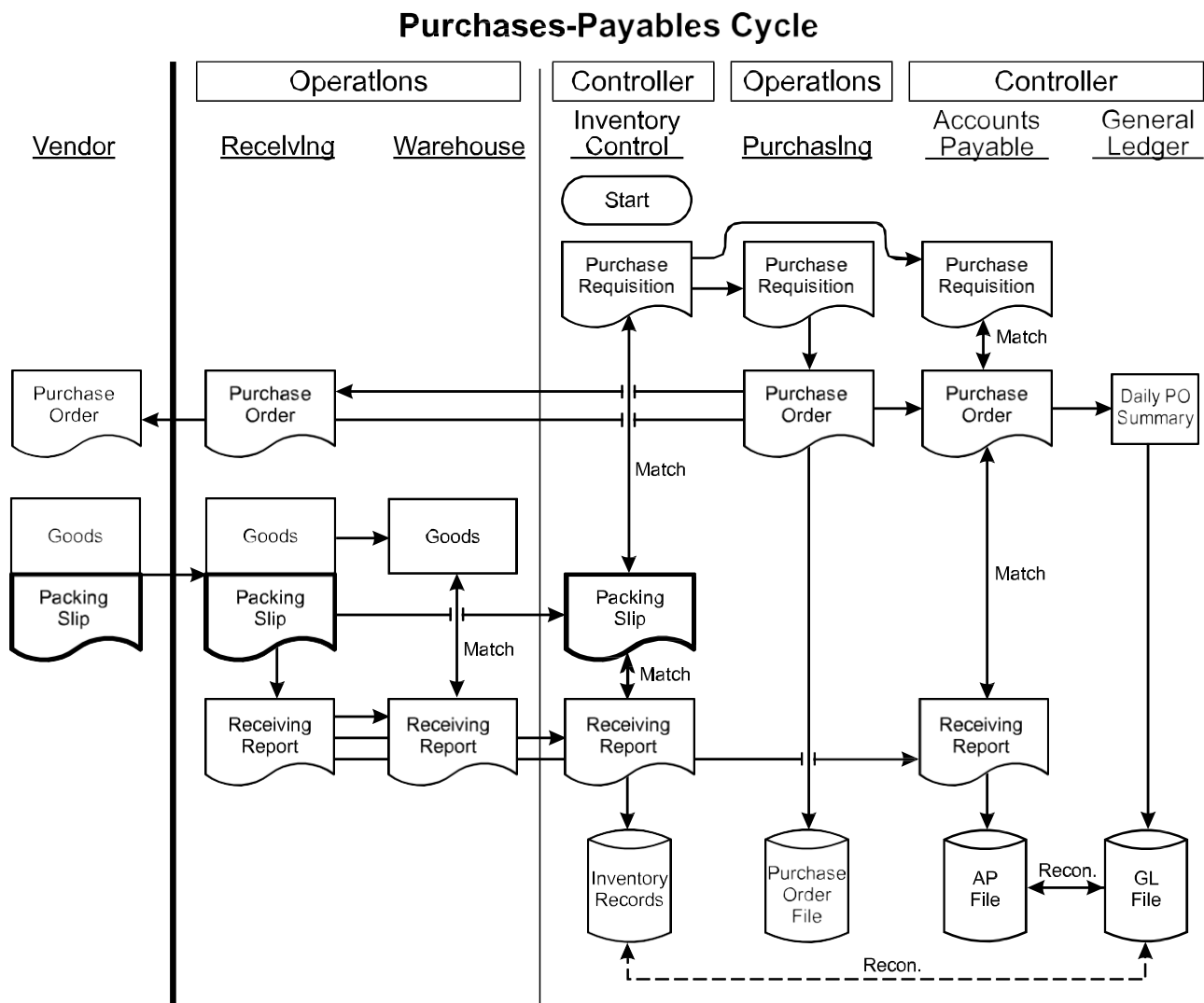


Figure F-3

Purchases-Payables Cycle

Function:	Authorization		Custody		Recording	
Department:	Inventory Control	Purchasing	Vendor	Receiving	Warehouse	Accounts Payable General Ledger

Step	Business Activity	Embedded Control
1	Inventory Control prepares a purchase requisition when inventory approaches the reorder point and sends it to Purchasing and Accounts Payable.	Predetermined inventory levels trigger authorization to initiate purchase transaction.
2	Purchasing locates authorized vendor in vendor file, prepares a purchase order , and updates the purchase order file.	Ensures that goods are bought only from vendors who have been preapproved for reliability. Reconciling sequentially numbered purchase orders helps ensure that customer orders are legitimate.
3	Purchasing sends the purchase order to Vendor, Receiving, and Accounts Payable. Receiving's copy has blank quantities.	Vendor prepares merchandise for shipment. Receiving is put on notice to expect shipment. Accounts Payable is put on notice that liability to this vendor is about to increase.
4	Accounts Payable prepares a summary of all purchase orders issued that day and forwards it to General Ledger for posting of the total to the GL file.	Updating AP and GL files separately provides an additional accounting control when they are periodically reconciled.
5	Goods arrive at Receiving with a packing slip .	Because quantities are blank on Receiving's copy of the purchase order, employees cannot assume the order is correct as received and must count items.
6	Receiving prepares a receiving report and forwards it with the goods to Warehouse.	Detects discrepancies between the vendor packing slip and actual goods received.
7	Warehouse verifies that goods received match those listed on the receiving report .	Detects any loss or damage between Receiving and Warehouse.
8	Receiving sends the receiving report and packing slip to Inventory Control for matching with the purchase requisition and updating of inventory records.	Ensures that inventory records are current. Updating inventory and GL files separately provides an additional accounting control when they are periodically reconciled.
9	Receiving also sends the receiving report to Accounts Payable for matching with the purchase order and purchase requisition and updating of the AP file.	Ensures that vendor accounts are current.

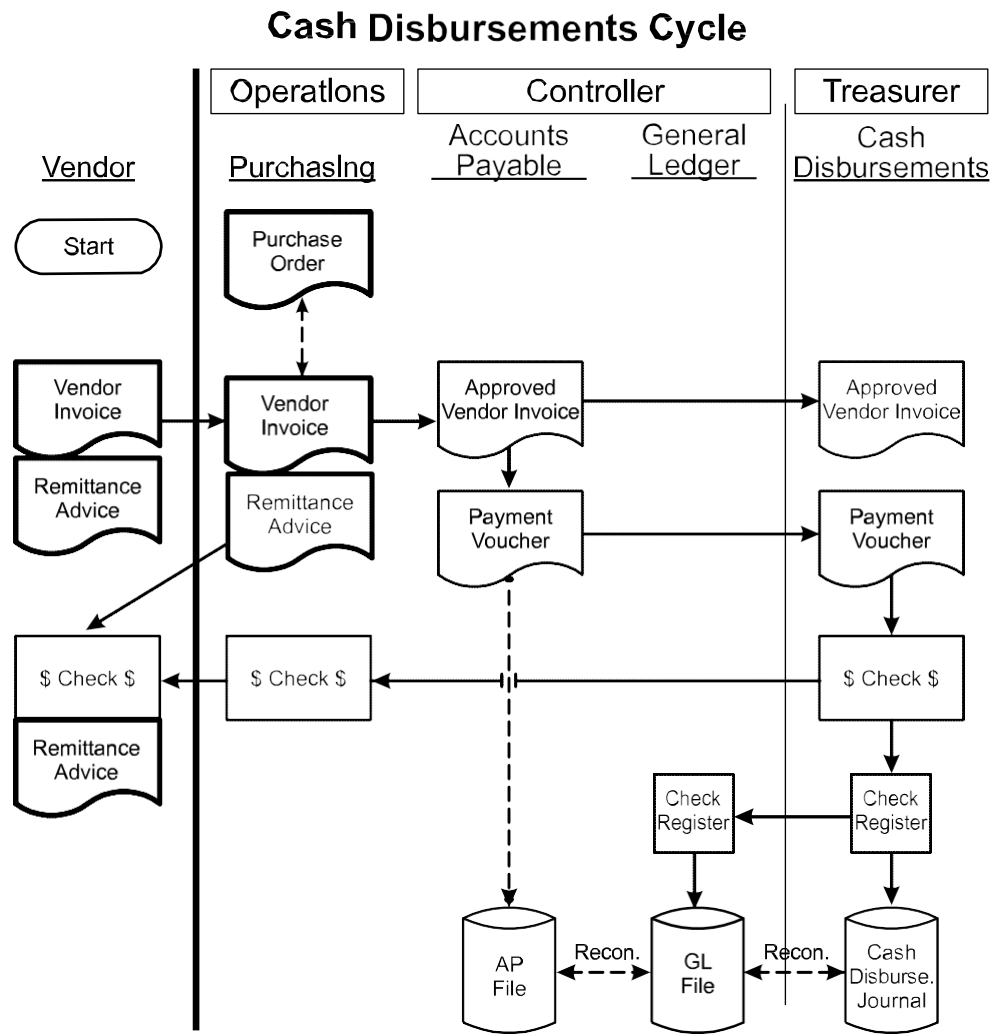


Figure F-4

Cash Disbursements Cycle

Function:	Authorization		Custody	Recording	
Department:	Vendor	Purchasing	Cash Disbursements	Accounts Payable	General Ledger

Step	Business Activity	Embedded Control
1	Purchasing receives a vendor invoice and remittance advice . The remittance advice is separated and filed. The invoice is matched with the purchase order and approved for payment. The purchase order is marked as closed, and the approval is forwarded to Accounts Payable.	Ensures that vendors are timely paid for goods received and that Purchasing can follow up on partially filled orders.
2	Accounts Payable matches the approved vendor invoice with the AP file and issues a payment voucher to Cash Disbursements.	Ensures that the invoice is for goods actually received and that duplicate payment cannot be made.
3	Upon receipt of a payment voucher with an approved vendor invoice , Cash Disbursements issues a check and forwards it to Purchasing.	Ensures that payments are made only when goods have actually been received.
4	Purchasing sends the remittance advice with the check to Vendor.	Settles liability to Vendor.
5	Cash Disbursements prepares a check register of all checks issued during the day and posts a journal entry to the cash disbursements journal.	Ensures that the cash disbursements journal is updated for the total of checks requested.
6	The check register is also forwarded to General Ledger for posting of the total to the GL file.	Updating AP and GL files separately provides an additional accounting control when they are periodically reconciled.

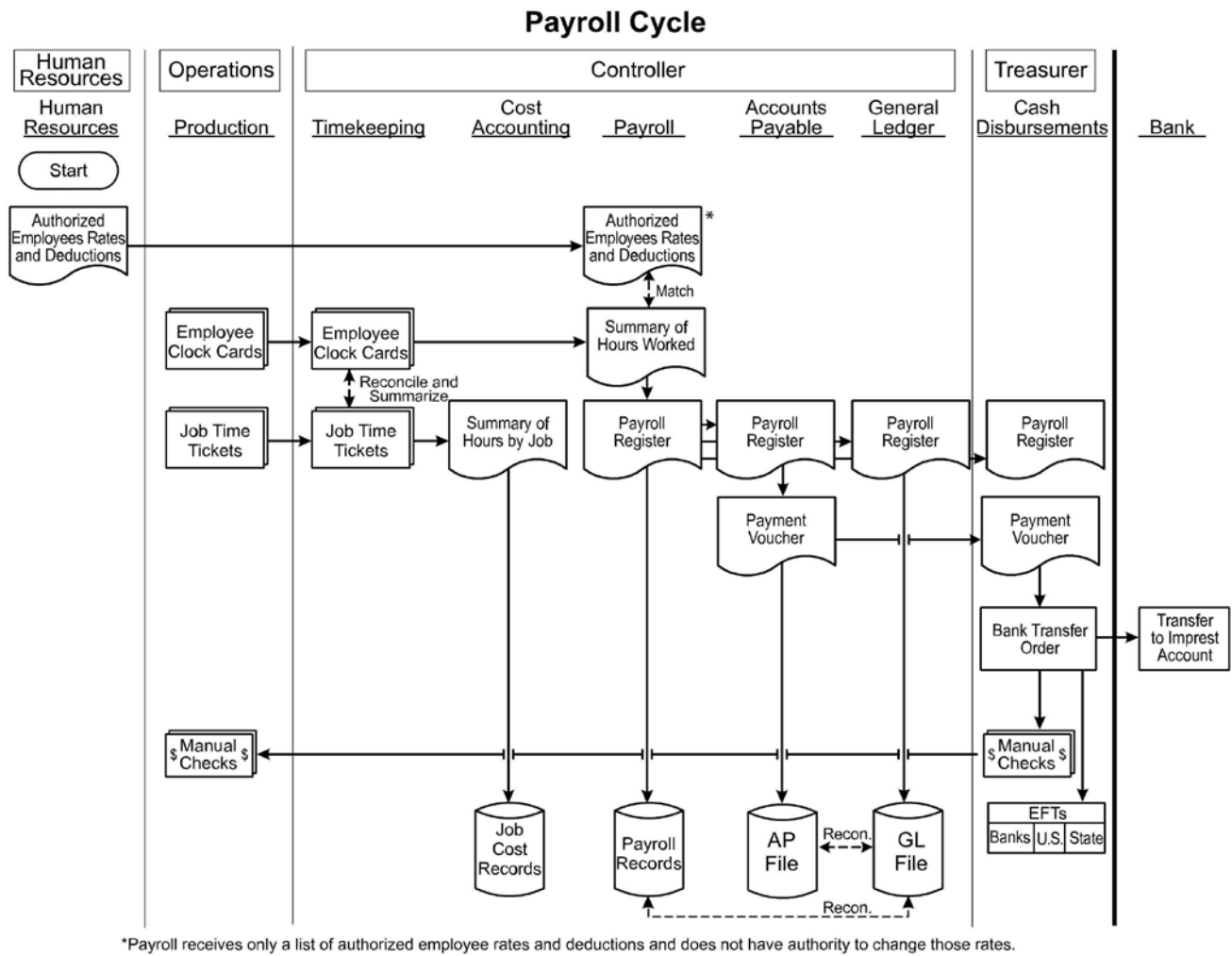


Figure F-5

Payroll Cycle

Function:	Authorization		Custody		Recording				
Department:	Human Resources	Production	Cash Disbursements	Bank	Time-keeping	Cost Accounting	Payroll	Accounts Payable	General Ledger

Step	Business Activity	Embedded Control
1	Human Resources sends a list of authorized employees , pay rates, and deductions to Payroll.	Ensures that only actual persons are carried on the payroll and that rates of pay and withholding amounts are accurate.
2	Employees register the start and end times of their workdays on clock cards .	Mechanically or electronically captures employee work hours.
3	Employees record time worked on various tasks on job time tickets .	Allows accumulation of labor costs by job as well as tracking of direct and indirect labor.
4	At the end of the pay period, a production supervisor approves clock cards and job time tickets and forwards them to Timekeeping.	Ensures that employees worked only authorized hours.
5	Timekeeping reconciles the clock cards and job time tickets .	Ensures that employees are paid only for actual hours worked.
6	Timekeeping prepares a summary of hours worked by job and forwards it to Cost Accounting for updating of the job records.	Ensures that direct labor costs are appropriately assigned to jobs.
7	Timekeeping prepares a summary of hours worked by employee and forwards it to Payroll. Payroll matches it with the authorized employee list , prepares a payroll register , and updates the payroll records.	Ensures that employees are paid the proper amount.
8	Accounts Payable receives the payroll register from Payroll, prepares a payment voucher , and forwards it along with the payroll register to Cash Disbursements.	Ensures that a payable is accrued. Authorizes the movement of cash into the payroll imprest account.
9	Accounts Payable also forwards the payroll register to General Ledger for posting of the total to the GL file.	Updating AP and GL files separately provides an additional accounting control when they are periodically reconciled.
10	Cash Disbursements compares the payment voucher with the payroll register total and initiates appropriate bank transfers .	Ensures that the correct amount is transferred to the payroll imprest account and governmental authorities.
11	Cash Disbursements executes three bank transfers .	Use of an imprest payroll account allows idle funds to be invested and funds related to uncashed checks to be isolated. In the U.S., federal taxes withheld are transferred to the U.S. Treasury. In the U.S., state taxes withheld are transferred to the state government.
12	Employees paid by manual check are given checks by Treasury personnel, not by Payroll or their supervisors.	Ensures that Payroll or supervisory personnel cannot perpetrate fraud through the creation of fictitious employees.